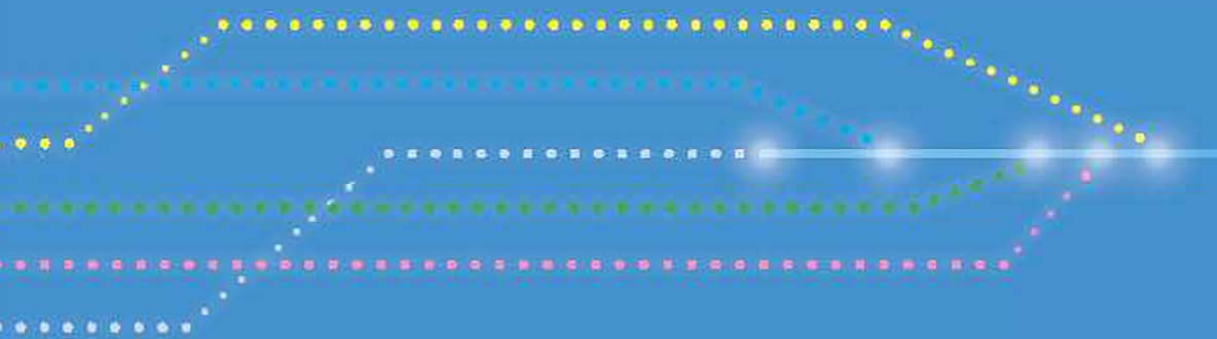


Handbook on Supply and Use Tables and Input Output-Tables with Extensions and Applications



United Nations

Department of Economic and Social Affairs
Statistics Division

Studies in Methods

Series F No.74, Rev.1

Handbook of National Accounting

Handbook on Supply and Use Tables and Input-Output Tables with Extensions and Applications

Edited white cover version



United Nations
New York 2018

Department of Economic and Social Affairs

The Department of Economic and Social Affairs of the United Nations is a vital interface between global policies in the economic, social and environmental spheres and national action. The Department works in three main interlinked areas: (i) it compiles, generates and analyses a wide range of economic, social and environmental data and information on which United Nations Member States draw to review common problems and to take stock of policy options; (ii) it facilitates the negotiations of Member States in many intergovernmental bodies on joint courses of action to address ongoing or emerging global challenges; and (iii) it advises interested Governments on the ways and means of translating policy frameworks developed in United Nations conferences and summits into programmes at the country level and, through technical assistance, helps build national capacities.

Note

The designations employed and the presentation of the material in the present publication do not imply the expression of any opinion whatsoever on the part of the United Nations concerning the legal status of any country or of its authorities, or the delimitations of its frontiers. The term “country” as used in this report also refers, as appropriate, to territories or areas. The designations of country groups are intended solely for statistical or analytical convenience and do not necessarily express a judgement about the stage reached by a particular country, territory or area in the development process. Mention of the names of firms and commercial products does not imply endorsement by the United Nations. The symbols of United Nations documents are composed of capital letters and numbers.

ST/ESA/STAT/SER.F/74/Rev.1

UNITED NATIONS PUBLICATION

Sales No.:

ISBN: 978-92-1-1

eISBN: 978-92-1-0

Copyright © United Nations, 2018

All rights reserved

Preface and acknowledgements

The present *Handbook on Supply and Use Tables and Input-Output Tables with Extensions and Applications* has been prepared as part of a series of handbooks on national accounting in support of the implementation of the System of National Accounts 2008 (2008 SNA). The objective of this Handbook is to provide step-by-step guidance for the compilation of supply and use tables (SUTs) and input-output tables (IOTs) and an overview of the possible extensions of SUTs and IOTs which increase their usefulness as analytical tools.

Preparation of the Handbook started as an update of the 1999 United Nations publication entitled *Handbook of National Accounting: Input-Output Table Compilation and Analysis*,¹ with the aim of incorporating changes in the underlying international economic accounting standards, most notably the 2008 SNA, and classifications; extending the scope of the Handbook to provide fuller coverage of SUTs; and providing practical compilation guidance for countries with advanced and less advanced statistical systems. In this process, however, the Handbook has also evolved to include an innovative approach to the compilation of SUTs and IOTs in the following three main areas: first, the underlying use of an integrated approach to statistics; second, the use of a business model for the compilation of SUTs and IOTs linking the various parts through the compilation scheme known as the “H-Approach”; and, third, the mainstreaming of environmental considerations.

The Handbook builds on the experience, practices and guidance available at national and regional level, including the Eurostat *Manual of Supply, Use and Input-Output Tables* (Eurostat, 2008). It provides a consistent worked example of SUTs and IOTs, which runs throughout the chapters (as far as practically possible) in order to facilitate understanding of the various compilation steps. It also provides examples of best practices to illustrate certain aspects of the compilation of SUTs, along with clear recommendations, principles and guidelines in order to ensure best practice.

For the preparation and drafting of the Handbook, an editorial board was established in May 2013, comprising 12 members and the United Nations Statistics Division. The editorial board members were leading international experts, including members of the International Input-Output Association, with decades of accumulated knowledge and experience from different regions and from different institutions, such as national statistical offices, central banks, international organizations and the academic community.

An editor (Sanjiv Mahajan, Office for National Statistics, United Kingdom) was appointed to lead the work of the editorial board and coordinate the contributions of experts for the various

¹ ST/ESA/STAT/SER.F/74, Sales No. E.99.XVII.9.

chapters. Initial drafts of the chapters were prepared by members of the editorial board, including the editor. These were further refined and aligned by the editor in liaison with respective members of the board and the United Nations Statistics Division into a coherent set of chapters. This was achieved through many bilateral electronic communications between the editor and chapter authors, a face-to-face meeting of all board members in New York in May 2014, and a final editorial board review prior to a global consultation.

The Handbook is therefore the outcome of a collaborative team effort led by the editor in liaison with the United Nations Statistics Division and the editorial board. This team comprises the following:

- Sanjiv Mahajan, editor Office for National Statistics, United Kingdom
- Joerg Beutel Konstanz University of Applied Sciences, Germany
- Simon Guerrero Central Bank of Chile
- Satoshi Inomata Institute of Developing Economies, Japan External Trade Organization
- Soren Larsen Statistics Denmark
- Brian Moyer Bureau of Economic Analysis, United States of America
- Isabelle Remond-Tiedrez European Commission, Eurostat
- José M. Rueda-Cantuche European Commission, Joint Research Centre
- Liv Hobbelstad Simpson Norway
- Bent Thage Denmark
- Catherine Van Rompaey Statistics Canada
- Piet Verbiest Statistics Netherlands
- Ilaria Di Matteo United Nations Statistics Division

The editorial board members contributed initial draft chapters and a detail review of all the chapters in the various rounds of consultation. Substantive contributions on specific topics, including initial draft chapters, were provided by the editorial board members as follows: Joerg Beutel (transforming SUTs into IOTs, compiling physical SUTs (PSUTs) and environmentally extended IOTs (EE-IOTs), extension of SUTs and IOTs and modelling applications of IOTs); Simon Guerrero (examples of country practices); Satoshi Inomata (multi-country SUTs and IOTs); Soren Larsen (compiling the use table); Brian Moyer (compiling the import use table and domestic use table); José M. Rueda-Cantuche (transforming SUTs into IOTs and projecting SUTs and IOTs); Liv Hobbelstad Simpson (guidance for countries with limited statistical resources and examples of country practices); Bent Thage (classification of industries and products, compiling the supply table, use table, valuation matrices, import use table and domestic use table, and transforming SUTs into IOTs); Catherine Van Rompaey (regional SUTs); and Piet Verbiest (compiling SUTs in volume terms and balancing). The editor also provided substantive contributions to these topics, initial draft chapters and all other topics in the Handbook, and brought

all the material together through numerous iterations with editorial board members reflecting many changes and improvements.

The contributions by the editor and the members of the editorial board and their commitment to the Handbook are very much acknowledged and appreciated. The following specific contributions are also acknowledged: Joerg Beutel, in formatting and standardizing tables, charts, boxes and figures throughout the Handbook; Ilaria Di Matteo, in reorganizing the chapters and ensuring overall coherence and consistency of the Handbook; and Erwin Kolleritsch (Statistics Austria), in kindly providing and checking much of the empirical data supporting the SUTs and IOTs in parts two and three of the Handbook.

The Handbook also benefited from specific inputs provided by Issam Alsammak (Statistics Canada), Gary Brown (Office for National Statistics, United Kingdom), Andrew Cadogan (Australian Bureau of Statistics), Duncan Elliot (Office for National Statistics, United Kingdom), Antonio F. Amores (European Commission Joint Research Centre), Ziad Ghanem (Statistics Canada), Manfred Lenzen (University of Sydney, Australia), Bo Meng (Institute of Developing Economies, Japan External Trade Organization), Louis de Mesnard (University of Bourgogne, France), Carol Moylan and Tom Howells (Bureau of Economic Analysis, United States), Jan Oosterhaven (University of Groningen, Netherlands), Ole Gravgard Pedersen (Statistics Denmark), Xesús Pereira (University of Santiago de Compostela, Spain), Joao Rodrigues (Technical University of Lisbon, Portugal), Jaroslav Sixta (Czech Statistical Office), Silke Stapel-Weber (European Commission, Eurostat), Umed Temursho (European Commission Joint Research Centre), Norihiko Yamano and Nadim Ahmad (OECD), and Herman Smith, Julian Chow, Gulab Singh, Benson Sim and Alessandra Alfieri (United Nations Statistics Division).

Feedback was also received from participants at various meetings and conferences, most notably the annual International Input-Output Association (2014, 2015 and 2016) and various regional national accounts meetings. The Handbook has benefited greatly from the numerous useful comments and suggestions made by national statistical offices, central banks, regional commissions, academic associations and international organizations, and also by the Intersecretariat Working Group on National Accounts during the global consultation in the period August to October 2017.

The Handbook was prepared under the supervision of Herman Smith and the overall responsibility of Ivo Havinga, both of the United Nations Statistics Division.

Contents

PREFACE AND ACKNOWLEDGEMENTS.....	I
CONTENTS	V
ABBREVIATIONS.....	XVII
PART ONE.....	1
CHAPTER 1. INTRODUCTION.....	3
A. BACKGROUND.....	3
B. USES OF SUTs AND IOTs.....	5
C. SYSTEM OF NATIONAL ACCOUNTS.....	7
D. OBJECTIVES OF THIS HANDBOOK	12
E. STRUCTURE OF THE HANDBOOK.....	15
PART TWO	21
CHAPTER 2. OVERVIEW OF THE SUPPLY AND USE TABLES AND INPUT-OUTPUT TABLES ...	23
A. INTRODUCTION.....	23
B. OVERVIEW OF SUTs	23
C. OVERVIEW OF IOTs	35
D. STRUCTURE OF SUTs AND IOTs: BASIC ELEMENTS.....	38
E. COMPILING SUTs AS AN INTEGRAL PART OF THE NATIONAL ACCOUNTS	57
CHAPTER 3. BUSINESS PROCESSES AND PRODUCTION STAGES.....	69
A. INTRODUCTION.....	69
B. INSTITUTIONAL ARRANGEMENTS.....	70
C. OVERVIEW OF THE GENERIC STATISTICAL BUSINESS PROCESS MODEL	71
D. OVERALL STRATEGY FOR THE COMPILATION OF SUTs AND IOTs.....	75
ANNEX A TO CHAPTER 3: EXAMPLES OF INSTITUTIONAL ARRANGEMENTS IN COUNTRIES....	90
A. CENTRALIZED PRODUCTION OF ECONOMIC STATISTICS: CANADA	90
B. CENTRALIZED PRODUCTION OF ECONOMIC STATISTICS: NORWAY	91
C. CENTRALIZED PRODUCTION OF ECONOMIC STATISTICS: UNITED KINGDOM	92
D. DECENTRALIZED PRODUCTION OF ECONOMIC STATISTICS: CHILE	93
E. DECENTRALIZED PRODUCTION OF ECONOMIC STATISTICS: UNITED STATES OF AMERICA.....	95
CHAPTER 4. SPECIFY NEEDS, DESIGN, BUILD AND COLLECT STAGE.....	99
A. INTRODUCTION.....	99
B. SPECIFY NEEDS, DESIGN AND BUILD PHASES	99
C. COLLECT PHASE	120
PART THREE	129
CHAPTER 5. COMPILING THE SUPPLY TABLE	131
A. INTRODUCTION.....	131
B. STRUCTURE OF THE SUPPLY TABLE	131
C. DOMESTIC OUTPUT.....	135

D. IMPORTS OF GOODS AND SERVICES	145
ANNEX A TO CHAPTER 5: SAMPLE QUESTIONNAIRE COLLECTING SALES OF GOODS AND SERVICES, INVENTORIES OF GOODS AND TRADE-RELATED DATA	153
CHAPTER 6. COMPILING THE USE TABLE	157
A. INTRODUCTION.....	157
B. STRUCTURE OF THE USE TABLE	157
C. INTERMEDIATE CONSUMPTION PART OF THE USE TABLE.....	162
D. GVA PART OF THE USE TABLE.....	170
E. FINAL CONSUMPTION EXPENDITURE PART OF THE USE TABLE	173
F. GROSS CAPITAL FORMATION PART OF THE USE TABLE.....	182
G. EXPORTS	196
ANNEX A TO CHAPTER 6: SAMPLE QUESTIONNAIRE COLLECTING PURCHASES OF GOODS AND SERVICES FOR INTERMEDIATE CONSUMPTION	198
ANNEX B TO CHAPTER 6: IMPACT OF CAPITALIZING THE COSTS OF RESEARCH AND DEVELOPMENT IN SUTS AND IOTS	201
A. RESEARCH AND DEVELOPMENT AS FIXED CAPITAL FORMATION	201
B. IMPLICATIONS OF VALUATION OF OUTPUT AS SUM OF COSTS.....	202
C. OWN-ACCOUNT RESEARCH AND DEVELOPMENT AS PRINCIPAL OR SECONDARY OUTPUT	203
D. BALANCING SUPPLY AND USE OF RESEARCH AND DEVELOPMENT SERVICES.....	205
CHAPTER 7. COMPILING THE VALUATION MATRICES.....	207
A. INTRODUCTION.....	207
B. VALUATION OF PRODUCT FLOWS.....	207
C. TRADE MARGINS	216
D. TRANSPORT MARGINS	229
E. TAXES ON PRODUCTS AND SUBSIDIES ON PRODUCTS	236
ANNEX A TO CHAPTER 7. EXAMPLE FOR DERIVING TRADE MARGINS IN SUTS BASED ON SURVEY DATA	241
A. SUPPLY TABLE.....	243
B. USE TABLE	245
CHAPTER 8. COMPILING THE IMPORTS USE TABLE AND DOMESTIC USE TABLE	249
A. INTRODUCTION.....	249
B. STRUCTURE OF THE IMPORTS USE TABLE AND DOMESTIC USE TABLE	250
C. COMPILATION OF THE IMPORTS USE TABLE	255
D. SPECIFIC ISSUES IN THE COMPILATION OF THE IMPORTS USE TABLE	259
E. ENHANCEMENTS TO THE IMPORTS USE TABLE FOR ANALYTICAL USES	268
CHAPTER 9. COMPILING SUTS IN VOLUME TERMS	271
A. INTRODUCTION.....	271
B. RECOGNITION OF ALTERNATIVE APPROACHES.....	272
C. OVERVIEW OF THE STEPS IN THE H-APPROACH WITH A FOCUS ON VOLUMES	273
D. PRICE AND VOLUME INDICATORS IN THEORY.....	284
E. PRICE AND VOLUME INDICATORS IN PRACTICE	285
F. INPUT-OUTPUT TABLES IN VOLUME TERMS.....	301

CHAPTER 10.	LINKING THE SUPPLY AND USE TABLES TO THE INSTITUTIONAL SECTOR	
ACCOUNTS	303	
A.	INTRODUCTION.....	303
B.	INSTITUTIONAL SECTORS AND SUBSECTORS	304
C.	TABLE LINKING SUTs AND INSTITUTIONAL SECTOR ACCOUNTS.....	307
D.	COMPILATION METHODS	315
CHAPTER 11.	BALANCING THE SUPPLY AND USE TABLES	319
A.	INTRODUCTION.....	319
B.	OVERVIEW OF THE SYSTEM AND BASIC IDENTITIES	320
C.	BALANCING.....	324
D.	STEP-BY-STEP PROCEDURE FOR SIMULTANEOUS BALANCING	332
E.	ALTERNATIVE BALANCING METHODS.....	336
F.	EXTENDING THE BALANCING OF SUTs TO INCLUDE INSTITUTIONAL SECTOR ACCOUNTS, IOTs, PSUTs AND EE-IOTs.....	338
G.	PRACTICAL ASPECTS OF BALANCING	342
ANNEX A TO CHAPTER 11.	BALANCING SUPPLY AND USE TABLES	350
CHAPTER 12.	TRANSFORMING THE SUPPLY AND USE TABLES INTO INPUT-OUTPUT TABLES	
	369	
A.	INTRODUCTION.....	369
B.	OVERVIEW OF THE RELATIONSHIP BETWEEN IOTs AND SUTs	369
C.	CONVERSION OF SUTs TO IOTs.....	374
D.	INPUT-OUTPUT FRAMEWORK	379
E.	EMPIRICAL APPLICATION OF THE TRANSFORMATION MODELS	400
ANNEX A TO CHAPTER 12.	MATHEMATICAL DERIVATION OF DIFFERENT IOTs	411
A.	PRODUCT-BY-PRODUCT IOTs AND INDUSTRY-BY-INDUSTRY IOTs	411
B.	PRODUCT-BY-PRODUCT IOTs.....	412
C.	INDUSTRY-BY-INDUSTRY IOTs	414
D.	USE OF A HYBRID TECHNOLOGY ASSUMPTION FOR PRODUCT-BY-PRODUCT IOTs	416
ANNEX B TO CHAPTER 12.	CLASSICAL CAUSES AND TREATMENT OF NEGATIVE CELL ENTRIES	
IN THE PRODUCT TECHNOLOGY.....	419	
A.	CLASSICAL CAUSES OF NEGATIVE ELEMENTS IN THE PRODUCT TECHNOLOGY	419
B.	OVERALL STRATEGY FOR REMOVING NEGATIVES.....	421
C.	SPECIFIC APPROACHES TO DEALING WITH NEGATIVES.....	421
ANNEX C TO CHAPTER 12.	EXAMPLES OF REVIEWS OF APPROACHES TO THE TREATMENT OF	
SECONDARY PRODUCTS	427	
CHAPTER 13.	COMPILING PHYSICAL SUPPLY AND USE TABLES AND ENVIRONMENTALLY	
EXTENDED INPUT-OUTPUT TABLES	429	
A.	INTRODUCTION.....	429
B.	OVERVIEW OF PSUTs	430
C.	COMPILATION OF PSUTs	442
D.	ENVIRONMENTAL EXTENDED IOTs.....	449
E.	COMPILATION OF EE-IOTs	453
F.	COUNTRY EXAMPLES	454

CHAPTER 14. SUPPLY AND USE TABLES AND QUARTERLY NATIONAL ACCOUNTS	467
A. INTRODUCTION.....	467
B. QUARTERLY NATIONAL ACCOUNTS.....	468
C. SUTS AND QUARTERLY NATIONAL ACCOUNTS.....	474
CHAPTER 15. DISSEMINATING SUPPLY, USE AND INPUT-OUTPUT TABLES	487
A. INTRODUCTION.....	487
B. USER IDENTIFICATION	487
C. DISSEMINATION STRATEGY	488
D. COMMUNICATIONS OF SUTs AND IOTs WITH USERS	493
E. DISSEMINATION FORMAT FOR SUTs AND IOTs.....	494
F. STATISTICAL DATA AND METADATA EXCHANGE INITIATIVE.....	497
PART FOUR.....	499
CHAPTER 16. REGIONAL SUPPLY AND USE TABLES	501
A. INTRODUCTION.....	501
B. ISSUES ARISING IN AND METHODS FOR THE COMPILATION OF REGIONAL SUTs AND IOTs.....	501
C. EXAMPLE OF BOTTOM-UP METHODS FOR REGIONAL SUTs: CANADIAN EXPERIENCE	504
CHAPTER 17. MULTI-COUNTRY SUPPLY AND USE TABLES AND INPUT-OUTPUT TABLES.....	521
A. INTRODUCTION.....	521
B. OVERVIEW OF MULTI-COUNTRY SUTs AND IOTs AND MAIN COMPILATION ISSUES	522
C. COMPILATION PROCEDURE.....	530
D. MULTI-COUNTRY INPUT-OUTPUT DATABASE INITIATIVES	541
E. WAY AHEAD.....	543
CHAPTER 18. PROJECTING SUPPLY, USE AND INPUT-OUTPUT TABLES.....	549
A. INTRODUCTION.....	549
B. SITUATIONS NEEDING PROJECTION METHODS	549
C. GENERAL APPROACHES TO PROJECTION FROM A HISTORICAL PERSPECTIVE	551
D. NUMERICAL EXAMPLES.....	567
E. CRITERIA TO CONSIDER WHEN CHOOSING A METHOD	580
CHAPTER 19. EXTENSIONS OF SUTs AND IOTs AS PART OF SATELLITE SYSTEMS.....	583
A. INTRODUCTION.....	583
B. OVERVIEW OF POSSIBLE EXTENSIONS.....	584
C. SOCIAL ACCOUNTING MATRIX	591
D. EXTENDED INPUT-OUTPUT TABLES.....	597
E. OTHER EXAMPLES OF SATELLITE SYSTEMS.....	601
CHAPTER 20. MODELLING APPLICATIONS OF IOTs	603
A. INTRODUCTION.....	603
B. NUMERICAL EXAMPLE OF IOTs AS A STARTING POINT	604
C. DISTINCTION BETWEEN PRICE, VOLUME, QUANTITY, QUALITY AND PHYSICAL UNITS	605
D. INPUT COEFFICIENTS	609
E. OUTPUT COEFFICIENTS.....	611
F. QUANTITY MODEL OF INPUT-OUTPUT ANALYSIS	612
G. PRICE MODEL OF INPUT-OUTPUT ANALYSIS.....	616

H.	INPUT-OUTPUT MODELS WITH INPUT AND OUTPUT COEFFICIENTS	622
I.	CENTRAL MODEL OF INPUT-OUTPUT ANALYSIS	623
J.	INDICATORS	626
K.	MULTIPLIERS.....	629
L.	INTER-INDUSTRIAL LINKAGE ANALYSIS	636
CHAPTER 21. EXAMPLES OF COMPILATION PRACTICES.....		641
A.	INTRODUCTION.....	641
B.	BASIC CONSIDERATIONS FOR THE COMPILATION OF NATIONAL ACCOUNTS AND SUTs	642
C.	EFFECT ON GDP OF INTEGRATING SUTs IN THE NATIONAL ACCOUNTS FOR MALAWI.....	655
D.	DEVELOPMENT OF THE APPLICATION OF THE INPUT-OUTPUT FRAMEWORK IN THE CZECH REPUBLIC.....	659
E.	CONTINUAL CHANGE, DEVELOPMENT AND IMPROVEMENT IN CHILE	663
REFERENCES		675
ADDITIONAL READING		695
Box 1.1	Evolution of the SUTs and IOTs within the national accounts.....	13
Box 2.1	Numerical example of the SUTs system	28
Box 2.2	Numerical example showing a use table split between consumption of domestic production and imports	29
Box 2.3	SUTs and product-by-product IOTs.....	37
Box 2.4	SUTs and industry-by-industry IOTs.....	38
Box 2.5	Three approaches to measuring GDP.....	41
Box 2.6	Other classifications of products	46
Box 2.7	SNA recommendations on partitioning of vertically and horizontally integrated enterprises	53
Box 2.8	Overview of the valuation in SUTs and IOTs.....	57
Box 2.9	Calculation of output for market and non-market producers	60
Box 2.10	Example of derivation of GDP from balanced SUTs.....	64
Box 3.1	Examples of the main recommendations, principles and guidelines provided in this Handbook.....	85
Box 4.1	Example of in-house custom-built software: Statistics Netherlands.....	100
Box 4.2	ERETES	101
Box 4.3	Data sources generally used	121
Box 5.1	Redefinitions	144
Box 5.2	Consistency issues with the CIF/FOB adjustment	150
Box 6.1	Example of a calculation of the values of an input column	168
Box 6.2	Classification of individual consumption according to purpose	174
Box 6.3:	Non-durable, semi-durable and durable goods	176
Box 6.4	Classification of the purposes of NPISHs	179
Box 6.5	Classification of functions of government	181
Box 6.6	Gross fixed capital formation by type of asset.....	183

Box 7.1 Compilation process for trade margins	221
Box 7.2 Examples of transport costs which do not form transport margins	231
Box 7.3 Options to consider where no data exists on transport margins	234
Box 8.1 Standard services components of BPM 6.....	257
Box 9.1 Treatment of newly introduced and disappearing taxes and subsidies	297
Box 10.1 Compilation methods used to link SUTs to the institutional sector accounts.....	315
Box 11.1 Example of discrepancies balanced in current prices and in volume terms.....	331
Box 11.2 Example of simultaneous balancing comparing volume indices	331
Box 11.3 Methods used for automated balancing SUTs.....	345
Box 12.1 Clarification of IOTs terminology.....	373
Box 12.2 Input-output framework for domestic output and imports	381
Box 12.3 Basic transformations of SUTs to IOTs	383
Box 13.1 Selected reference material	444
Box 15.1 Fundamental Principles of Official Statistics.....	488
Box 15.2 Reference metadata in the SDMX metadata structure for SUTs and IOTs	493
Box 17.1 Background papers of each database initiative	541
Box 17.2 Overview of the main features of the various databases	543
Box 18.1 Methods for projection of SUTs and IOTs.....	554
Box 18.2 SUTs and IOTs for Austria, 2005 and 2006.....	568
Box 18.3 Results using the GRAS Method	570
Box 18.4 Flow diagram of the GRAS method.....	571
Box 18.5 Results using the SUT-RAS method.....	574
Box 18.6 Flow diagram of the SUT-RAS method.....	575
Box 18.7 Results using the SUT-Euro method.....	578
Box 18.8 Flow diagram of the SUT-EURO method.....	580
Box 19.1 Measurement performance and social progress: overview of Stiglitz-Sen-Fitoussi Commission 2009 report.....	588
Box 20.1 Quantities, prices, values and volumes in IOTs	607
Box 20.2 Quantity input-output model	620
Box 20.3 Price input-output model	621
Box 20.4 Multipliers in the input-output model	631
Box 21.1 Material product system and Phare projects	660
 Figure 1.1 Overview of the links between SUTs and the SNA framework.....	 9
Figure 2.1: Graphical overview of supply and use tables	27
Figure 2.2 Schematic overview of the compilation of SUTs and IOTs: H-Approach.....	30
Figure 2.3 System of national accounts in matrix form.....	41
Figure 2.4 Overview of SUTs and IOTs as part of the SNA compilation	59
Figure 3.1 Phases of the GSBPM	72
Figure 3.2 Simplified business processing model for compiling SUTs, IOTs, and PSUTs	74

Figure 3.3 Structure of the SUTs and the links covered in this Handbook	77
Figure 3.4 Compilation of SUTs and IOTs in current prices and in volume terms.....	79
Figure 3.5 Evolution of compiling SUTs and IOTs in the first three years.....	80
Figure 3.6 First year of compilation	82
Figure 3.7 Second year of compilation	84
Figure 4.1 Overview of SUTs and IOTs as part of the SNA compilation.....	123
Figure 5.1 Link between valuation matrices in the supply table and the use table.....	135
Figure 6.1 Three-dimensional view of SUTs.....	162
Figure 7.1 Schematic representation of the valuation matrices in the SUTs.....	210
Figure 7.2 Alternative distribution channels of goods.....	227
Figure 9.1 Overview of the compilation schematic layout linking SUTs in current prices and in volume terms.....	274
Figure 9.2 Link between SUTs in current prices and in volume terms	278
Figure 10.1 Links between the industry accounts and the institutional sector accounts	304
Figure 10.2 Link between the SUTs and institutional sector accounts.....	309
Figure 11.1 Simplified SUTs system.....	321
Figure 11.2 Six-pack.....	325
Figure 11.3 Overview of the SUTs balancing framework for simultaneous balancing.....	333
Figure 11.4 Sources of feedback loops emanating from the balancing process	342
Figure 12.1 Transformation of SUTs into IOTs	372
Figure 12.2 Basic transformation models	378
Figure 13.1 Physical flows of natural inputs, products and residuals.....	431
Figure 13.2 Overview of the compilation system for PSUTs.....	446
Figure 13.3 Key feedback loops in producing and balancing the PSUTs and environmental extended IOTs.....	449
Figure 13.4 Danish SUTs framework extended with physical flows	456
Figure 13.5 From source data to PSUTs.....	458
Figure 14.1 Quarterly GDP production (output) aggregate: data availability and estimation in the United Kingdom.....	470
Figure 14.2 Quarterly GDP expenditure components: data availability and estimation in the United Kingdom	471
Figure 15.1 Release calendar covering SUTs, IOTs and national accounts: Statistics Denmark	490
Figure 15.2 Measuring United Kingdom GDP and SUTs: revision policy	491
Figure 17.1 Schematic representation of multi-country SUTs (three-country case)	523
Figure 17.2 Schematic representation of multi-country IOTs (three country case)	524
Figure 17.3 System of multi-country SUTs and its conceptual correspondence to a national SUTs framework.....	532
Figure 17.4 Splitting the import matrix by country of origin	535
Figure 17.5 Converting valuation scheme	536

Figure 17.6 Formation of the export vector to rest of the world.....	538
Figure 17.7 Transformation to multi-country IOTs.....	540
Figure 21.1: Illustration of a database for the product-flow method used in smaller countries .	649
Figure 21.2 Supply and use table.....	670
Table 2.1: Simplified structure of the supply table.....	24
Table 2.2: Simplified structure of the use table	24
Table 2.3: Supply and use tables framework	25
Table 2.4 Schematic view of the physical supply table	34
Table 2.5 Schematic view of the physical use table	34
Table 2.6 Simplified IOT (product by product).....	36
Table 2.7 Links between the use table and functional classifications	51
Table 2.8 Simplified table linking the SUTs to the institutional sector accounts.....	65
Table 4.1 Examples of the size of published and internal working level SUTs and IOTs	107
Table 5.1 Numerical example of a supply table at basic prices.....	132
Table 5.2 Supply table at basic prices, including a transformation into purchasers' prices	133
Table 5.3 Data adjustment for external trade of goods and services	148
Table 5.4 CIF and FOB adjustment row	149
Table 6.1 Use table at purchasers' prices.....	158
Table 6.2 Intermediate consumption of selected inputs into "Manufacture of rubber and plastic products"	169
Table 6.3 Sample product balance for "Gelatine and gelatine derivatives".....	170
Table 6.4 Categories of final consumption expenditure	173
Table 6.5 Table linking final expenditures by purpose (COICOP) and product (CPC).....	175
Table 6.6 Final consumption expenditure of households (by COICOP headings).....	176
Table 6.7 Table linking final consumption expenditures of NPISHs by purpose (COPNI) and by product (CPC)	180
Table 6.8 Table linking final consumption expenditure of general government by COFOG and CPC	182
Table 6.9 Categories of gross capital formation	183
Table 6.10 Table linking gross fixed capital formation by industries, assets and products	185
Table 6.11 Gross fixed capital formation by investing industry.....	187
Table 6.12 Table linking change in inventories industries, assets and products	192
Table 7.1 Supply table at basic prices, including a transformation into purchasers' prices	212
Table 7.2 Use table at purchasers' prices.....	213
Table 7.3 Use-side valuation matrices	214
Table 7.4 Use table at basic prices.....	216
Table 7.5 Trade turnover and trade margins for wholesale and retail trade margins	224
Table 8.1 Structure of the imports use table	251
Table 8.2 Numerical example of the imports use table	251

Table 8.3 Structure of the domestic use table	252
Table 8.4 Numerical example of a domestic use table	252
Table 8.5 Example of an input table for imports at basic prices.....	253
Table 8.6 Input-output table for domestic output at basic prices.....	254
Table 8.7 Input-output table for domestic output at basic prices, net exports with adjustment items	255
Table 8.8 Processing within the country	261
Table 8.9 Goods sent abroad for processing	262
Table 8.10 Industry 1 – Alternative input structures	264
Table 9.1 Supply table in current prices and in volume terms.....	280
Table 9.2 Use table in current prices and in volume terms.....	281
Table 9.3 Gross domestic product in current prices and in volume terms.....	283
Table 10.1 Summary of institutional sectors and subsectors	306
Table 10.2 Main features of the SUT approach and the institutional sector approach	307
Table 10.3 Numerical example showing the table linking the SUTs and institutional sector accounts.....	310
Table 10.4 Goods and services for the whole economy	312
Table 10.5 Production account for the whole economy	313
Table 10.6 Link between GDP and industry GVA	313
Table 10.7 Generation of income account for the whole economy	314
Table 12.1 Product-by-product IOT at basic prices.....	370
Table 12.2 Numerical example of rectangular SUTs for the transformation	375
Table 12.3 Numerical example of square SUTs for the transformation	376
Table 12.4 Transformation matrix for the product technology assumption	386
Table 12.5 Product-by-product IOTs based on product technology	387
Table 12.6 Transformation matrix for industry technology assumption	388
Table 12.7 Product-by-product IOTs based on industry technology	388
Table 12.8 Matrix for hybrid technology.....	389
Table 12.9 Transformation matrix for hybrid technology assumption	390
Table 12.10 IOTs based on the hybrid technology assumption.....	390
Table 12.11 Transformation matrix for the fixed industry sales structure assumption	391
Table 12.12 IOTs based on the fixed industry sales structure assumption	392
Table 12.13 Transformation matrix for the fixed product sales structure assumption for rectangular SUTs	394
Table 12.14 IOTs based on the fixed product sales structure assumption derived from rectangular SUTs	394
Table 12.15 Transformation matrix for the fixed product sales structure assumption for square SUTs	395
Table 12.16 IOTs based on the fixed product sales structure assumption for square SUTs.....	396

Table 12.17 Absolute deviation of IOTs based on rectangular SUTs less IOTs based on square SUTs for Model D	397
Table 12.18 Alternative presentations of product-by-product IOTs.....	399
Table 12.19 Empirical example of product-by-product IOTs	402
Table 12.20 Empirical example of industry-by-industry IOTs.....	404
Table 13.1 General PSUT	432
Table 13.2 Classes of natural input.....	434
Table 13.3 Typical components for groups of residuals.....	436
Table 13.4 List of individual components of SEEA physical PSUTs	443
Table 13.5 Common national data sources and links to SEEA component accounts.....	443
Table 13.6 Single region IOT with environmental data	450
Table 13.7 Single region IOT in hybrid units.....	452
Table 13.8 Industry-by -industry IOTs (upper block of Table 13.6)	453
Table 13.9 Environmental data by industry (lower block of Table 13.6).....	453
Table 13.10 PSUTs in Denmark	460
Table 13.11 SUTs for the Netherlands, 2010	462
Table 13.12 PSUTs for the Netherlands, 2010	463
Table 14.1 Balancing supply and use of products	477
Table 16.1 SUTs framework for interregional SUTs.....	507
Table 16.2 Interregional and international trade flows by province and territory, 2010	510
Table 17.1 Adjustment targets for national tables of selected countries in the Asian international input-output table for the year 2000.....	530
Table 18.1 Categorization of methods	561
Table 19.1 Structure of a social accounting matrix	594
Table 19.2 Numerical example of a social accounting matrix.....	595
Table 19.3 Extended IOT with satellite systems	599
Table 20.1 IOT at basic prices	605
Table 20.2 Input coefficients of IOTs.....	610
Table 20.3 Output coefficients of IOTs	612
Table 20.4 Input coefficients for domestic intermediate consumption.....	613
Table 20.5 Leontief matrix	614
Table 20.6 Leontief inverse	615
Table 20.7 Quantity input-output model based on monetary data.....	616
Table 20.8 Price input-output model based on monetary data.....	619
Table 20.9 Emission model.....	625
Table 20.10 Input indicators for production activities per unit of output.....	628
Table 20.11 Output multipliers (Leontief inverse)	629
Table 20.12 Multipliers for products	632
Table 20.13 Input content of final use by category	635
Table 20.14 Backward linkages.....	637

Table 20.15 Forward linkages.....	637
Table 20.16 Forward and backward linkages	638
Table 20.17 Normalized forward and backward linkages	638
Table 21.1 Historical benchmark exercises	664
Table 21.2 Annual business surveys	666
Table 21.3 Administrative records.....	667
Table 21.4 SUTs for tobacco products, year 2008, current prices.....	672
Table 21.5 SUT for cleaning and toiletry products, year 2008, current prices	672
Table 21.6 IOTs for domestic output at basic prices, 2008	673

Abbreviations

AIOT	Asian International Input-Output Tables
ANZSIC	Australian and New Zealand Standard Industrial Classification
ARIMA	autoregressive integrated moving average
BEC	classification by broad economic categories
BPM	Balance of Payments and International Investment Position Manual
BTDIxE	Bilateral Trade Database by Industry and End-Use
CH ₄	methane
CIF	cost, insurance and freight
CO ₂	carbon dioxide
COFOG	Classification of the Functions of Government
COICOP	Classification of the Purposes of Non-profit Institutions Serving Households
COPNI	Classification of the Purposes of Non-profit Institutions Serving Households
COPP	Classification of the Outlays of Producers According to Purpose
CPA	Classification of Products by Activity
CPC	Central Product Classification
CPI	consumer price index
CRAS	cell-corrected RAS method
CREEA	compiling and refining of economic and environmental accounts
CSPI	corporate services price index
EBOPS	Extended Balance of Payments Services Classification
ECE	Economic Commission for Europe

EE-IOT	environmentally extended input-output tables
EPI	export price index
ERETES	equilibre ressources-emplois et tableau entrées-sorties
ESA	European system of accounts
EU	European Union
FAO	Food and Agriculture Organization of the United Nations
FBS	finance and business services
FI	fixed industry
FIGARO	full international and global accounts for research in input-output analysis
FISIM	financial intermediation services indirectly measured
FOB	free on board
FP	fixed product
GDP	gross domestic product
GENESIS	online databank of the Federal Statistical Office of Germany
GNI	gross national income
GRAS	generalized RAS
GSBPM	Generic Statistical Business Process Model
GTAP	global trade analysis project
GTAP-MRIO	multi-region input-output table based on the global trade analysis project database
GVA	gross value added
HS	Harmonized Commodity Description and Coding System
ICIO	inter-country input-output
ICPIs	intermediate consumption price indices

ICT	information and communications technology
ILO	International Labour Organization
IMF	International Monetary Fund
IMTS	International Merchandise Trade Statistics: Concepts and Definitions
INSEE	Institut national de la statistique et des études économiques
IOTs	input-output tables
IPCC	Intergovernmental Panel on Climate Change
IPIs	import price indices
ISIC	International Standard Industrial Classification of All Economic Activities
KLEMS	Integrated Industry-Level Production Account
KRAS	Konfliktfreies RAS
MPS	Material Product System
MRIO	multi-region input-output
N ₂ O	nitrous oxide
NACE	Statistical Classification of Economic Activities in the European Community
NAICS	North American Industry Classification System
NMVO	non-methane volatile organic compounds
NO _x	mono-nitrogen oxides
NPISH	non-profit institution serving households
OECD	Organization for Economic Cooperation and Development
PIOT	physical input-output tables
PPP	purchasing power parities
PSUT	physical supply and use table
PYPs	previous years' prices
R&D	research and development

RAS	ranking and scaling data reconciliation method
RPI	retail prices index
SAM	social accounting matrix
SDMX	Statistical Data and Metadata Exchange
SEEA	System of Environmental and Economic Accounts
SESAME	system of economic and social accounting matrices and extensions
SITC	Standard International Trade Classification
SNA	System of National Accounts
SO ₂	sulphur dioxide
SUT-Euro	Euro method for SUTs
SUTs	supply and use tables
TEC	trade by enterprise characteristics data
TiVA	trade in value added
TLS	taxes less subsidies on products
TRAS	three-stage RAS
TTM	trade and transport margins
UNSD	United Nations Statistics Division
UNWTO	World Tourism Organization
VAT	value added taxes
WCO	World Customs Organization
WIOD	World Input-Output Database
WPI	wholesale price index
WTO	World Trade Organization

Part one

Chapter 1. Introduction

A. Background

1.1. The supply and use tables (SUTs) are an integral part of the System of National Accounts 2008 (2008 SNA) forming the central framework for the compilation of a single and coherent estimate of gross domestic product (GDP) integrating all the components of production, income and expenditure approaches, and providing key links to other parts of the SNA framework.

1.2. In their simplest form, the SUTs describe how products (goods and services) are brought into an economy (either as a result of domestic production or imports from other countries) in the supply table, and how those same products (intermediate consumption; final consumption by household, non-profit institutions serving households, and general government; gross capital formation; and exports) are used in the use table.

1.3. The SUTs also provide the link between components of gross value added (GVA), industry inputs and outputs. Although typically they show only the industry dimension, SUTs can also be formulated to show the role of different institutional sectors (for example, non-financial corporations, government, and others) providing an important linking mechanism to the different accounts of the SNA framework (the goods and services account, production account, generation of income account and the capital account).

1.4. Importantly, and by design, these interlinkages facilitate data confrontation and the examination of the consistency of data on goods and services obtained from different statistical sources, such as business surveys, household surveys and administrative data within a single detailed framework. As such, they provide a powerful mechanism for feedback on the quality and coherency of primary data sources.

1.5. The SUTs do not just provide a framework to ensure the best quality estimates of GDP and its components: they are also an important analytical resource in their own right, showing the interaction between producers and consumers. When measured in volume terms, the SUTs provide the basis for a rich stream of analyses, notably in the field of structural analysis, and in particular productivity, where in recent years SUTs have been widely accepted as an important tool for

KLEMS-type² productivity measures. Just as important is their growing use as the basis for deriving the input-output tables (IOTs).

1.6. In many respects, the IOTs, which show the links between final uses and intermediate uses of goods and services defined according to industry outputs (industry-by-industry tables) or according to product outputs (product-by-product tables) predate the SUTs. The IOTs also show separately the consumption of domestically produced and imported goods and services. The widespread availability of SUTs has meant, however, that the SUTs form the starting point for constructing IOTs and, in turn, an entire swathe of related analytical products and indicators, such as the Leontief inverse and other type of analyses, including output multipliers, employment multipliers, and others.

1.7. The SUTs and IOTs are compiled by many countries in the course of producing their core national accounts, thereby improving the coherency and consistency of their national account estimates. The ability to readily create IOTs from SUTs (as shown in chapter 12) has helped to reinforce the momentum behind the evolution, role and use of SUTs.

1.8. SUTs and IOTs have received much attention in recent years. This is because their analytical properties allow for a much wider set of analyses, not only of the national economy and the regions within a nation but also of the interlinkages between economies at the global level and also of environmental impacts.

1.9. Further momentum has been generated for the role of SUTs and IOTs in step with the rapidly growing impact of globalization and the international fragmentation of production. For a full understanding of international interdependencies and their impact on important policy areas, such as trade, competitiveness and sustainable development, there is increasing need to view production and consumption through a global value chain lens. In other words, multi-country and regional SUTs and IOTs have become essential tools to inform policy and policymakers. Over the past five years, a number of efforts have been made by the international statistics community to meet these needs, such as the trade in value added database prepared by the Organization for Economic Cooperation and Development (OECD) and the World Trade Organization (WTO), and other comparable databases such as the World Input-Output Database (WIOD) and the Handbook on Accounting for Global Value Chains prepared by the Expert Group on International Trade and Economic Globalization Statistics.

1.10. Given these developments and, in particular, the heightened importance of SUTs and IOTs, the timing of the present Handbook is important and highly relevant. The present chapter provides a general introduction to the various issues considered in greater detail in the various chapters that follow. Section B of this introductory chapter provides a general overview of the roles and uses of

² KLEMS is an industry-level growth and productivity research project, based on the analysis of capital (K), labour (L), energy (E), materials (M) and service (S) inputs.

SUTs and IOTs. Section C covers the SNA and its links to SUTs and IOTs. Section D covers the objectives of the Handbook and its new features compared to previous manuals on the subject. Lastly, section E briefly outlines the structure and content of the Handbook.

B. Uses of SUTs and IOTs

1.11. The uses of SUTs and IOTs are multiple and their statistical and analytical importance has increased with time and in response to new and emerging issues, such as globalization and sustainable development, with its three pillars of social, economic and environmental development. Where possible, the analytical uses of SUTs and IOTs are presented below in parallel. As SUTs form the basis for the compilation of IOTs, the uses of the two types of tables are treated in the same way in this section.

1.12. As mentioned above, the SUTs combine in a single framework the three approaches to measuring GDP, namely, the production approach, the income approach and the expenditure approach. All three approaches are based on sets of data with various levels of detail and a range of different sources. Combining the data in a single statistical framework compels compilers to use harmonized and unique classifications of producers, users and income receivers, together with harmonized and unique classifications and definitions of products and income categories. Under these conditions, corresponding data can be related and compared in an organized manner. Combining the three data sets provides an opportunity to analyse the causes of discrepancies, make necessary adjustments and fill data gaps when necessary.

1.13. An important objective of national accounts is to estimate year-to-year and quarter-to-quarter changes in a number of macroeconomic variables. When dealing with production, use and the generation of value added, it is important to divide the current price changes into volume changes (representing what is termed “real” growth) and price changes. When SUTs are compiled simultaneously in current prices and in volume measures (as recommended in this Handbook, using what is known as the “H-Approach”), there are considerable advantages in the overall quality and consistency of the information provided. During the entire statistical process – from the processing and analysis of the source data through to, and including, the balancing of the SUTs – data in current prices and deflated data are obtained simultaneously and consistently with each other.

1.14. In addition to annual national accounts, SUTs can be used in the compilation of quarterly national accounts. This may range from the compilation and balancing of quarterly SUTs to the mere use of the SUTs framework to highlight possible discrepancies between quarterly product supply and use. The annual estimates of GVA can, for example, be used as weights in the quarterly estimate of GDP in volume terms to reflect the most recent period. In addition, SUTs can provide weighting schemes for price and volume indices.

1.15. The SUTs and IOTs serve also as the basis for compiling a range of accounts –regional, environmental, labour, tourism, etc. The clear links of these satellite systems with both the SUTs

and the IOTs ensure the consistency of the satellite systems with the concepts and methods of the core national accounts and allow for feedback loops with the SUTs during the compilation and balancing process of the frameworks involved. For instance, the SUTs can support the compilation of regional accounts by including clear links to variables like regional GVA. When physical environmental flows are linked to the SUTs and IOTs in the environmental accounts, they provide feedback loops to the compilation of SUTs by contrasting physical and monetary measures of the supply and use of products. When SUTs are linked to labour and capital, they can be used for productivity analyses that link economic growth to the use of intermediate inputs. Lastly, social accounting matrices elaborate the linkages between SUTs and sector accounts. They capture transactions and transfers between all economic agents in the accounting system and measures effects of macroeconomic policies on distribution.

1.16. The SUTs and IOTs also provide the basis for different types of analytical uses at micro and macro levels (see, for example, United Nations, 2002; Mahajan, 2004a; and Mahajan, 2006). Various examples are included in the list of additional reading at the end of this Handbook. Examples include the following:

- Economic analyses: export shares, import penetration, concentration ratios, links between prices and costs, links between energy production, consumption and emissions, etc.
- Impact and policy analyses: sensitivity analyses, analyses of the impacts of taxation changes, price changes, introduction of a minimum wage, specific economic crisis, earthquakes, etc., analyses of consumption and demand-based accounting and analyses of air emissions, material flows, energy, water, etc.
- Industrial and sectoral analyses: changes over time to specific sectors, such as information and communications technology (ICT), oil and gas, food, sport, creative arts, tourism, health, etc., and, more recently, analyses covering the digital economy, sharing economy, collaborative economy and also product-specific global value chains.
- Local government type investment planning: construction projects, shopping centres, new motorways, rural planning, etc.
- Base structures for modelling: computable general equilibrium models, environmental analyses, supply-side-based models, etc.

1.17. The role of SUTs and IOTs in understanding global value chains is of particular importance, given the interconnected nature of today's global economy. SUTs constitute the centrepiece of the internationally compatible accounting framework for a systematic and detailed description of the economy, its various components on the supply and use side and its relations to other economies. The construction of international SUTs and IOTs makes it possible, in combination with trade statistics, to follow the trade in value added and to understand who ultimately benefits from the trade of finished goods in terms of value added, employment, and other factors. The compilation of international or global SUTs and IOTs tables poses a number of

compilation challenges (including, for example, the recording of goods sent abroad for processing and the recording of the production abroad and merchanting operations affecting SUTs and IOTs) and relies on the availability of national SUTs and IOTs on a comparable basis.

1.18. In addition, the inclusion of the environmental dimension in the SUTs and IOTs further enhances the usefulness of these tables by allowing the integration and consistency of the economic and environmental information and an understanding of the interlinkages between the economy and the environment. Incorporating environmental considerations as part of the regular compilation of SUTs improves the quality, coherence and consistency of the related outputs and the process provides powerful feedback loops for identifying improvements.

C. System of National Accounts

1.19. The SNA provides an internationally compatible framework for a systematic and detailed description of a total economy (namely, that of a region, country or group of countries), its components and its relations with other total economies. The 2008 SNA (United Nations, European Commission, IMF, OECD and World Bank, 2009) is the latest version of the SNA, which was adopted by the United Nations Statistical Commission in 2008.

1.20. The SNA describes the basic features of the accounting system in terms of concepts, principles, statistical units and their groupings, etc. The SNA gives an overview of the sequence of accounts, the balancing items associated with each account, a brief description of key aggregates and the role of SUTs and the input-output framework. The key accounting sequence includes the following stages: production of goods and services, transactions relating to products (goods and services) and also to non-produced assets, transactions which distribute and redistribute income and wealth, financial transactions and balance sheets.

1.21. The SNA framework also draws in other aspects, such as price and volume measurement, population, labour market measures, regional accounts and various specific conceptual issues. Figure 1.1 provides an overview of how SUTs and IOTs fit within the SNA framework. In particular, it shows which accounts in the SNA sequence of accounts are more directly linked with SUTs and IOTs, namely, production accounts, generation of income accounts, use of disposable income accounts and capital accounts.

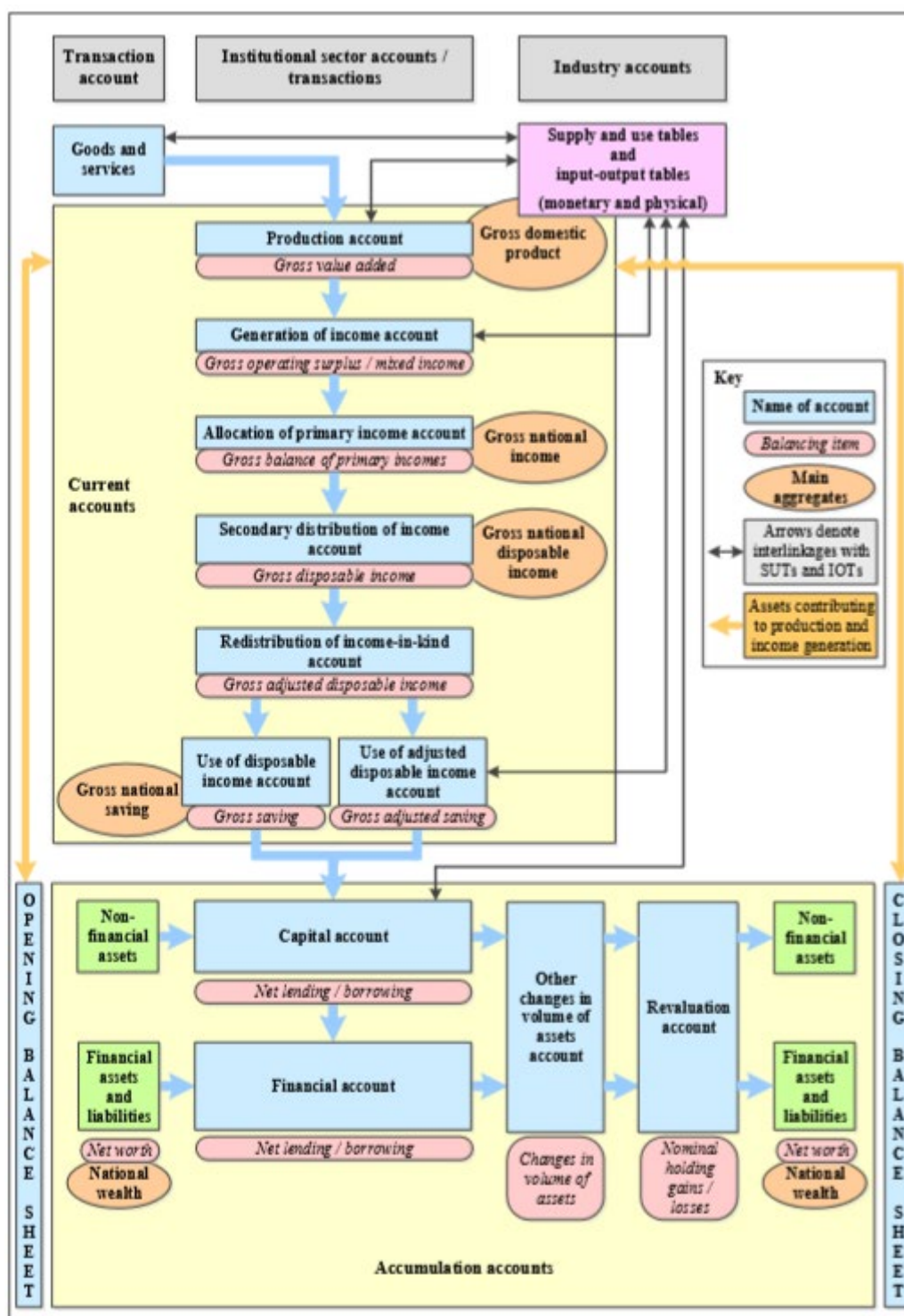
1.22. Producing annual SUTs simultaneously both in current prices and in volume terms, not only ensures consistency for price volume measures, it also allows for the estimation of the volume of GVA through what is termed “double deflation”, which is recommended in the 2008 SNA.

1.23. As noted above, the SUTs are an integral part of the SNA, determining a single estimate of GDP both in current prices and in volume terms and linked to the institutional sector accounts. For example, the goods and services account for the total economy can be directly compiled from the SUTs through appropriate aggregation. In addition, by using the breakdown of GVA by

components in the use table, the production account and the generation of income account can easily be compiled from the SUTs and linked to the institutional sectors.

1.24. Another important aspect linking the SUTs and the institutional sector accounts is the statistical unit. The SNA uses two types of units and two corresponding ways of subdividing the economy, which are quite different and serve separate analytical purposes. The units can be classified to an industry for use in the SUTs and to an institutional sector for use in the institutional sector accounts.

Figure 1.1 Overview of the links between SUTs and the SNA framework



Compiled by Sanjiv Mahajan, May 2014

1.25. The first purpose of describing production, income, expenditure and financial flows, and balance sheets, is served by grouping institutional units into institutional sectors on the basis of

their principal functions, behaviour and objectives. The SNA enables a complete set of flow accounts and balance sheets to be compiled for each sector and subsector, and also for the total economy. The five institutional sectors distinguished in the SNA are the following:

- Non-financial corporations
- Financial corporations
- General government
- Households
- Non-profit institutions serving households

1.26. The SNA also describes the transactions between these five institutional sectors and the rest of the world. These institutional sectors can be further split into subsectors, for example, general government can be split into central government and local government.

1.27. The second purpose of describing processes of production and for input-output analysis is served by the grouping of local kind-of-activity units (or establishments) into industries, on the basis of their type of activity. An activity is characterized by an input of products, a production process and an output of products.

1.28. In order to ensure consistency between SUTs and the institutional sector accounts, a link table is compiled as an integrated part of the system. In this link table, a cross-classification of output, intermediate consumption, components of value added (and other possible variables of industries) between the industries and the institutional sectors is shown. Thus, this table links the main macroeconomic variables from the SUTs to the institutional sector accounts, providing a picture of local kind-of-activity units and one based on institutional units. As both types of units are classified differently, the link table also provides a picture of the relations of output, intermediate consumption, value added, and other variables, originating in the different industries and institutional sectors.

1.29. The SUTs – consistent with the national accounts – are normally produced in connection with the final or benchmarked versions of the macroeconomic data some two or three years after first preliminary results of the national accounts are published. The SUTs, however, should play a more vital role at the heart of national accounts in the production of preliminary annual or even quarterly accounts. Once the SUTs compilation system is in place on an annual basis, the statistical benefits are significant.

1.30. SUTs can play various roles in the national accounts. One, for example, is to update SUTs – often in a more aggregated version – from the previous year with information available for the preliminary estimates in order to have a complete set of SUTs available that are consistent with the national accounts. This procedure is a good method for revealing inconsistencies in the aggregated preliminary national accounts. Another role of SUTs could arise from new information

in a situation in which new, and more, detailed information on total supply and exports is available at an early stage, the structure and relationships in the SUTs of the previous year could be used to project SUTs for domestic output and imports.

1.31. The compilation of SUTs was in the past associated solely with the construction of IOTs. The SUTs were therefore seen as an intermediate step in the compilation of IOTs. This meant, effectively, that the SUTs were only compiled out after the compilation of the national accounts had been completed. This approach, in fact, has significant limitations because the independently calculated national accounts aggregates had to be kept unchanged despite inconsistencies identified through the SUTs system.

1.32. SUTs are now seen as more than just as a step in the construction of IOTs: it is the SUTs that provide the ideal framework guaranteeing the coherency and consistency of supply and use of products in the system in current prices, and in volume terms, thereby improving the quality of the national accounts, and in turn the key economic aggregates.

1.33. The compilation of SUTs is thus recommended as part of the regular annual compilation of national accounts. The annual compilation of SUTs is also one of the recommended data sets used in assessing the scope of implementation of the 2008 SNA.³ The compilation of SUTs on a quarterly basis can also play a role in improving the quality and coherence of quarterly national accounts (the role played by SUTs in quarterly national accounts is further elaborated in chapter 14).

1.34. The approach to the compilation of SUTs as an integral component of the production of national accounts may be formulated in general terms as follows:

- SUTs are produced as a central element of the compilation of national accounts with a view to providing a key link to various parts of the SNA framework.
- SUTs provide a statistical framework representing the most efficient means of incorporating all basic data – aggregated or detailed – covering the components of the three approaches to measuring GDP, and linking to the institutional sector accounts in a systematic way.
- SUTs effectively ensure the consistency and reconciliation of results at a detailed level and thereby improve the overall quality of the national accounts.
- SUTs are compiled and balanced in both current prices and in volume terms.
- SUTs are produced annually or even, if possible, on a quarterly basis, the ideal option.

³ See table 2 of the 2011 report of the Intersecretariat Working Group on National Accounts to the United Nations Statistical Commission at its forty-second session (E/CN.3/2011/6), available at: <http://unstats.un.org/unsd/statcom/doc11/2011-6-NationalAccounts-E.pdf>.

- SUTs can provide a powerful feedback mechanism on the coherency and consistency of source data, such as business surveys, and on the classification of units on the business register.

1.35. When balanced, SUTs provide a coherent, consistent and wholly integrated suite of statistics for a single period (for example, a year), which include:

- A single estimate of GDP in current prices and in volume terms, which is underpinned with components of the production, income (only in current prices) and expenditure approaches to measuring GDP
- Detailed goods and services account in current prices and in volume terms (not by institutional sector)
- Production accounts by industry and by institutional sector in current prices and in volume terms.
- Generation of income accounts by industry and by institutional sector (both in current prices only)
- Link to the use of disposable income account through the flows of final consumption expenditures and capital account through gross capital formation (and its components) balanced via SUTs.

1.36. These guidelines should form part of the strategic tools used to improve the quality of the national accounts.

D. Objectives of this Handbook

1.37. The theoretical development of IOTs has a long history. Box 1.1 provides a description of the evolution of both IOTs and SUTs within the context of national accounts. The United Nations Statistics Division has followed the theoretical development and the practical work of national statistics offices on IOTs and SUTs from the outset. Starting in 1996, it has prepared a number of publications, under the guidance of the United Nations Statistical Commission, such as those listed among the references under United Nations, 1966, 1973 and 1999, to share practices, update the methodology in line with the updates of the SNA, and provide guidance on the compilation of IOTs.

1.38. This Handbook continues those efforts in cooperation with other international organizations and experts, providing practical and step-by-step guidance for the compilation of SUTs and IOTs based on the latest international statistical standards set out in the 2008 SNA and the sixth edition of the International Monetary Fund (IMF) *Balance of Payments and International Investment Position Manual* (BPM 6) (IMF, 2009).

Box 1.1 Evolution of the SUTs and IOTs within the national accounts

The national accounting system is continuously evolving across the various domains to reflect developments in and improvements to the quality of economic statistics and the evolution of economies, with a view to providing a relevant measurement of the economy. Over the past four centuries many significant contributions have been provided by people from various disciplines and countries, resulting in the system as it stands today and how it relates to SUTs and IOTs. Below is a short description of this evolution.

Wassily Leontief (1905–1999), who is often referred to as the pioneer of input-output-based economics, made the first of many key contributions with the publication of his article “Quantitative input and output relations in the economic system of the United States”. This article discussed the construction of an economic transactions table that Leontief had based on the *tableau économique*, proposed by François Quesnay in 1758.

The framework was developed and applied as an economic tool with the construction of the first IOTs for the United States covering the years 1919 and 1929 published in 1936. Later, Leontief developed the first input-output-based model, which was based on theories developed by Leon Walras published in 1874 and 1877. Leontief’s pioneering work was recognized by the award to him of the Nobel Prize in Economics in 1973. As a result, input-output analysis has become a major tool in developing quantitative economics as a science.

The role of SUTs and IOTs has evolved within national accounts. The 1953 SNA (United Nations, 1953) included no reference to SUTs or IOTs. The 1968 SNA (United Nations, 1968), however, presented the integration of an input-output framework into the integrated economic accounts of the SNA. The conceptual development of the integrated economic accounts of the SNA earned Richard Stone the Nobel Prize in Economic Science in 1984, for having made fundamental contributions to the development of the SNA and having thereby greatly improved the basis for empirical economic analysis.

Alongside Leontief and Stone, other Noble laureates include Ragnar Frisch and Jan Tinbergen in 1969, Paul Samuelson in 1970, Simon Kuznets in 1971, John Hicks in 1972 and James Meade in 1977, who have all contributed to the foundations of the measurements used in today’s SNA and the interlinkages between various sectors and activities in an economy.

The latest evolution of SUTs was recognized in the 1993 SNA (United Nations, CEC, IMF, OECD and World Bank, 1993), chapter XV of which covered both SUTs and IOTs. With the latest version of the SNA, the 2008 SNA, the role and applications of SUTs have been further enhanced, and this in turn will help to meet many analytical needs, as reflected in chapters 14 and 28 of the 2008 SNA.

1.39. This Handbook may, therefore, be viewed as an update of the United Nations *Handbook of Input-Output Table Compilation and Analysis* (United Nations, 1999). In response, however, to the ever-increasing importance of SUTs in their own right, this Handbook extends the scope of the previous publication by providing a more detailed description and compilation guidance for SUTs. As stated in the 2008 SNA, “only supply and use tables provide a sufficiently rigorous framework to eliminate discrepancies in the measured flows of goods and services throughout the economy to ensure the alternative measures of GDP converge to the same value” (2008 SNA, para. 14.15).

1.40. The Handbook builds on the experience, practices and guidance available at national and regional level, such as the *Eurostat Manual of Supply, Use and Input-Output Tables* (Eurostat, 2008). At the same time, however, it provides an innovative approach to the compilations of SUTs and IOTs in the following three main areas:

- Underlying use of an integrated approach to statistics

- Use of a business model for the compilation of SUTs and IOTs linking the various parts through an H-Approach compilation scheme
- Mainstreaming of environmental considerations, through the inclusion of the environmental focus of chapter 13 at the core of part three of the Handbook

1.41. The compilation guidance provided in this Handbook relies on an integrated statistics approach whereby the production of statistics in the various domains is not seen in isolation but as part of an integrated process using common concepts, definitions, business registers and frames, statistical units, estimation methods and data sources to improve the consistency of the statistics compiled, to reduce the respondent burden, and potentially to reduce the statistical agency costs. In particular, the consistency of the basic economic information that feeds into the national accounts and the SUTs with the classifications, concepts and definitions of the 2008 SNA greatly reduces the discrepancies across data from different sources, thus facilitating their reconciliation as part of the integration process. The integrated statistics approach is described in the *Guidelines on Integrated Economic Statistics* (United Nations, 2013).

1.42. This Handbook follows the Generic Statistical Business Process Model (GSBPM) (UNECE, 2013) to describe the production of statistics in a general and process-oriented manner. The underlying concepts and principles of the GSBPM have been followed in describing the business process and stages of the statistical production processes underpinning the compilation of SUTs and IOTs. Chapter 3 of this Handbook describes these links in more detail in the context of SUTs and IOTs. In addition, the chapters in parts two and three of the Handbook are linked to the different parts of these stages of the statistical production process.

1.43. With the adoption of the *System of Environmental-Economic Accounting* (SEEA) (United Nations, European Commission, FAO, IMF, OECD and World Bank, 2014) by the United Nations Statistical Commission, the extension of SUTs and IOTs to include environmental flows in monetary and physical terms has become an internationally agreed standard. Including environmental consideration from the outset in the compilation of SUTs brings a number of advantages. It facilitates the integration and reconciliation of the information, it enhances the quality of the information, and it significantly increases the uses to which the tabulations may be put.

1.44. In line with the United Nations Statistical Commission,⁴ this Handbook recommends the annual compilation of SUTs. In addition, the Handbook promotes the compilation of these tables as an integral part of the compilation of national accounts in order to ensure full consistency of the basic data and also of the macroeconomic estimates that are derived from the accounts.

⁴ See the 2011 report of the Intersecretariat Working Group on National Accounts to the United Nations Statistical Commission at its forty-second session (E/CN.3/2011/6), available online at: <http://unstats.un.org/unsd/statcom/doc11/2011-6-NationalAccounts-E.pdf>

1.45. The Handbook provides a consistent numerical example of SUTs and IOTs that runs throughout the chapters (as far as practically possible), in order to facilitate understanding of the various compilation steps. It also provides examples of best practices to illustrate certain aspects of the compilation of SUTs. It should be noted that, in the numerical examples provided in the Handbook, the numbers may not add up exactly to the totals because of rounding.

1.46. The target audience for this Handbook mainly includes compilers of SUTs and IOTs with a basic knowledge and understanding of the SNA. Since, however, the Handbook provides an overview of the whole statistical production process, managers or staff in charge of the programme of national accounts and of economic and environmental accounts would also benefit from the Handbook in gaining an overall understanding of the requirements for the compilation of SUTs and IOTs. Lastly, analytical users may also benefit from perusing the Handbook, as it would provide them with a better understanding of the compilation steps, thus increasing the analytical applications of SUTs and IOTs.

E. Structure of the Handbook

1.47. The Handbook consists of the following four main parts:

- Part one, providing an introduction to the Handbook, set out in chapter 1.
- Part two, describing the overview of SUTs and IOTs, the fundamental building blocks required, cross-cutting issues and the main stages of the GSBPM – namely, the design, building and collection phases. Part two comprises chapters 2–4.
- Part three, describing the compilation, balancing and dissemination phases of SUTs and IOTs. This also includes the physical SUTs (PSUTs), environmentally extended IOTs (EE-IOTs) and the SUTs links to the quarterly national accounts. Part three comprises chapters 5–15.
- Part four, providing examples of the extensions and applications of SUTs and IOTs. Part four comprises chapters 16–21.

Part one

1.48. As indicated above, chapter 1 provides an introduction to the Handbook; it describes the importance of SUTs and IOTs for statistical purposes (for example, compilation of annual and quarterly national accounts, etc.), for policymaking and for analytical purposes (for example, economic forecasting, assessing the impact of globalization). It also provides a general description of the SNA and where the SUTs fit within the SNA framework. This chapter also describes the overall approach of the Handbook (including in comparison to previous handbooks) to the compilation of SUTs and IOTs and provides a general outline of its contents.

Part two

1.49. Chapter 2 provides a conceptual overview of SUTs and IOTs and describes the basic elements determining their structure and compilation. These include the accounting principles of the SNA, the classifications of economic activities and products, the choice of the statistical units and how they affect SUTs and IOTs, and the valuation in SUTs and IOTs. The chapter identifies the advantages of compiling SUTs as an integral part of the national accounts and how the SUTs are used to obtain consistent estimates of GDP. It also describes in more detail the extended perspective adopted in this Handbook to SUTs and IOTs, incorporating an environmental dimension which makes possible an integrated overview of the framework from the very outset.

1.50. Chapter 3 provides an overview of the different phases that constitute the statistical production process of SUTs and IOTs, based on the stages of the GSBPM and in line with the United Nations Guidelines on Integrated Economic Statistics (United Nations, 2013). This chapter also provides an overview of the different institutional set-ups in countries which may have an impact on the compilation process. The compilation phases specific to SUTs and IOTs are presented in this chapter together with a link to the relevant chapters of the Handbook.

1.51. Chapter 4 covers specific phases of the GSBPM, namely the specify needs, design, build and collect phases. It provides a description of the elements that should be considered and carefully evaluated at the beginning of the compilation process, such as the level of detail of the industry and products in the tables, the compilation schedule, the revision policy, resources, typical data sources, and others. These and other issues are covered in this chapter, thus providing a foundation for the compilation of SUTs and IOTs.

Part three

1.52. Chapter 5 describes the conceptual and practical aspects of the compilation of the supply table and how the so-called “unbalanced” supply table is put together from the typical data sources for SUTs, such as business surveys, administrative data and others.

1.53. Chapter 6 describes the conceptual and practical aspects of the compilation of the use table. As in chapter 5, this chapter shows how an unbalanced use table is constructed on the basis of typical data sources.

1.54. Chapter 7 describes how to compile the valuation matrices necessary to bridge the different valuation concepts of the product flows. This chapter covers the main concepts and methodologies of compiling matrices for trade margins, transport margins, taxes on products and subsidies on products.

1.55. Chapter 8 describes the structure of the imports use table and the domestic use table and the steps necessary to disaggregate the use table into an imports use table and a domestic use table. Historically, the compilation of these tables was largely viewed as an intermediate – though not

essential – step in the compilation of IOTs. That said, however, the imports use table and the domestic use table are becoming increasingly important in their own right for analytical purposes.

1.56. Chapter 9 covers the compilation of SUTs in volume terms. It follows the recommendation that SUTs should simultaneously be compiled in current prices and in volume terms. The compilation of SUTs in volume terms can start after the SUTs have been compiled in current prices (although the current price tables do not need to be balanced) but there is need for a simultaneous presentation of volume and price indices.

1.57. Chapter 10 describes the importance of linking SUTs and the institutional sector accounts, which involve data by industry that are to be subdivided according to the institutional sectors to which the units within each industry are assigned. The chapter provides guidance on how to compile the cross-tabulation between industries and institutional sectors and presents various approaches that may be followed in establishing the link between the SUTs and the institutional accounts. It also identified certain issues that may arise in the compilation of the linking table.

1.58. Chapter 11 describes the manual and automated balancing procedures of SUTs in both current prices and in volume terms. This is important for full consistency of the detailed information. The various checks related to product, industry and macro identities, benchmarking with national accounts, and comparison with previous SUTs, if available, are explained. It is recommended that SUTs should be produced and balanced simultaneously at basic and purchasers' prices and also for domestic and imported products, all of which should be both in current prices and in volume terms. A further dimension, and challenge to be surmounted, is the need to cover both annual SUTs and, if possible, quarterly SUTs.

1.59. The sequence of chapters represents the preferred scenario for the compilation of SUTs. Different approaches may be followed, however. An increasing number of countries have achieved the preferred scenario. This scenario for the compilation of SUTs and IOTs may be seen as ambitious but it can be realized through gradual improvements in source data, production processes and the information technology environment.

1.60. Chapter 12 provides an overview of the IOTs (product by product and industry by industry) and describes the methods and the underlying assumptions for transforming SUTs into IOTs. The compilation of IOTs is quite different in nature from that of SUTs and relies on the availability of SUTs. The compilation of IOTs is considered more as an analytical step than a compilation process and, for this purpose, is viewed as a transition from statistics to modelling.

1.61. Chapter 13 describes the structure of the SUTs in physical units where additional rows and columns are added to show flows from the environment to the economy and vice versa. This chapter also describes typical data sources for the compilation of these tables and examples of specific issues in which the SEEA and the SNA differ (for example, the treatment of international flows and the treatment of goods for processing), and shows how standard economic IOTs in monetary units may be extended to include information on the environment in physical units in

the environmentally extended IOTs. Physical IOTs are also an extension of the SUTs framework, extended to take into account environmental considerations. They consist of a transformation of the PSUTs. Given certain conceptual and practical issues in the compilation of physical IOTs, however, the focus of the 2012 SEEA – and thus of this chapter – has shifted towards the compilation of environmentally extended IOTs rather than physical IOTs. Examples of two country practices are also presented in this chapter.

1.62. Chapter 14 provides an overview of how SUTs may be used to improve the quarterly national accounts. Since there are various scenarios that can be used in practice, this chapter focuses only on three main situations which illustrate the use of SUTs to various degrees in the compilation of the quarterly national accounts.

1.63. Data dissemination is an important activity for any statistical production process, as it provides users with a range of statistics produced to internationally agreed guidelines. Presenting SUTs and IOTs to users in a clear, transparent and user-friendly manner is thus an important task for the statisticians. Chapter 15 provides an overview of the elements that should be considered when disseminating SUTs and IOTs, such as the identification of users' needs in order to tailor dissemination to the main types of users of SUTs and IOTs, the importance of having a dissemination strategy and the elements that should be covered in the strategy. Reference to the Statistical Data and Metadata Exchange (SDMX) for SUTs and IOTs is also provided in this chapter.

Part four

1.64. Chapter 16 describes the methods for compiling regional (subnational) SUTs and the main compilation issues, such as the disaggregation of the information at subnational level, among others. Different issues and challenges are covered through a bottom-up and top-down compilation approach.

1.65. Although the focus of this Handbook is mainly on the compilation of national SUTs and national IOTs, there is a growing demand for these instruments to capture the structure and mechanism of the cross-border fragmentation of production activities. Chapter 17 provides an overview of multi-country SUTs and IOTs, the main compilation issues, and a simplified compilation procedure. This chapter also reviews current international initiatives in this area.

1.66. Chapter 18 deals with the projections of SUTs and IOTs. Many users also require comparable input-output products that are comparable in terms of frequencies and timeliness. For example, some countries produce quarterly SUTs, some countries produce annual SUTs and some countries produce SUTs on a less regular basis. Consequently, a variety of methods, techniques and approaches exist for the projection of SUTs and IOTs and for dealing with data gaps. These techniques also can help producers, for example, to deal with periods between benchmarked years. This chapter examines various methods and techniques used, along with a range of literature available on how to surmount the problem of incomplete data, thus allowing the estimation and

projection of IOTs. The chapter also presents a numerical example for three methods: the generalized RAS, the SUT-RAS and the Euro methods.

1.67. Chapter 19 describes the main extensions of supply, use and input-output tables as part of a satellite system which is routinely used for economic analysis. Several examples of the disaggregation of the use table and various satellite accounts are reviewed, including such extensions as social accounting matrices, extended IOTs and other satellite systems.

1.68. Chapter 20 describes the different types of input-output models and provides a broad overview illustrating the benefits and the approaches used. The traditional quantity model and price model of input-output analysis are presented for monetary IOTs and physical IOTs. Input and output coefficients, the Leontief inverse, price and quantity models, indicators, multipliers and inter-industrial linkages were developed for an empirical extended IOT, with extensions for gross fixed capital formation, capital stock, employment, energy, air emissions, waste, sewage and water.

1.69. Chapter 21 provides examples of compilation practices from various countries with different statistical systems. In general, the compilation practices can vary greatly depending on the resources available, the statistical infrastructure, registers, surveys, methodologies and other factors. This chapter provides guidance for countries with limited statistical resources and illustrates differences and challenges in the compilation of SUTs and IOTs.

Part two

Chapter 2. Overview of the supply and use tables and input-output tables

A. Introduction

2.1. Before providing step-by-step guidance on the compilation of SUTs and IOTs, it is important to ensure a general understanding of the two sets of tables. The main objective of this chapter is, therefore, to provide an overview of SUTs and IOTs, which may be found in sections B and C, respectively. Section D introduces the fundamental elements of the SUTs and IOTs, such as the underlying classifications, the statistical units and the valuation methods. Some of these elements are discussed in more detail in subsequent chapters. Lastly, section E elaborates on the importance of compiling SUTs as an integral part of the national accounts.

2.2. Although this chapter covers a wide range of challenges and issues that must be tackled when planning and building a new system of SUTs and IOTs, all aspects may not be achievable in countries with limited resources. It is worth recognizing that a system may be established with a moderate level of ambition using available data, even if these are incomplete. Nonetheless, it is preferable to have a SUTs-type environment for reconciliation of the various statistical sources, rather than only an unbalanced series of national accounts aggregates.

B. Overview of SUTs

2.3. The SUTs describe the whole economy by industry (for example, the motor vehicle industry) and by product (for example, sports goods). The tables show links between components of GVA, industry inputs and outputs, and product supply and use. The SUTs link different institutional sectors of the economy (for example, non-financial corporations) together with details of imports and exports of goods and services, final consumption expenditure of government, household and non-profit institutions serving households (referred to as NPISHs), and capital formation.

2.4. As their name suggests, SUTs consist of two interlinked tables: the supply table and the use table. The supply table shows the supply of goods and services by type of product and by type of industry, distinguishing between supply by domestic industries and imports of goods and services. In other words, the supply table provides information on the output (by product) generated by economic activities and the imports (by product) from abroad. The totals in the last column represent the total supply by products and the totals in the bottom row represent the total output by economic activity and total imports. A simplified supply table is presented in Table 2.1 below.

Table 2.1: Simplified structure of the supply table

Industries Products	Industries				Imports	Total
	Agriculture, forestry, etc.	Mining and quarrying	...	Services		
Agriculture, forestry, etc. Ores and minerals, etc. ... Services	Output by product by industry				Imports by product	Total supply by product
Total	Total output by industry				Total imports	Total supply

2.5. The second table is the use table, which provides information on the uses of the different products. The use table shows the use of goods and services by type of product and by type of use, in other words, as intermediate consumption by industry, final consumption, gross capital formation or exports. Furthermore, the table shows the components of gross value added by industry – namely, compensation of employees, other taxes less subsidies on production, consumption of fixed capital and net operating surplus. While the totals by row represent the total uses by product, the total by column represent the total output by economic activity, total final consumption, total gross fixed capital formation and total exports. Table 2.2 below shows the simplified structure of the use table.

Table 2.2: Simplified structure of the use table

Industries Products	Industries				Final uses			Total
	Agriculture, forestry, etc.	Mining and quarrying	...	Services	Final consumption	Gross capital formation	Exports	
Agriculture, forestry, etc. Ores and minerals, etc. ... Services	Intermediate consumption by product and by industry				Final uses by product and by category			Total use by product
Value added	Value added by component and by industry							Value added
Total	Total output by industry				Total final uses by category			

 Empty cells by definition

2.6. The classification of products, in practice, is often more detailed than the classification of industries, thus generating rectangular SUTs. For example, the output of the dairy industry is separately shown in the SUTs for the products of processed milk, butter, yoghurt, cheese and so forth, and not as only one aggregate product for all dairy products.

2.7. There are three basic identities that hold between the supply table and the use table. The first identity corresponds to the fundamental identity in national accounts, whereby for each economic activity the following holds:

$$\text{Identity (1)} \quad \text{Output} = \text{Intermediate consumption} + \text{GVA}$$

2.8. The second identity is that the total supply by product is equal to the total use by product. This means that the amount of products available for use in an economy must have been supplied either by domestic production or by imports, and the same amount of products entering an economy in an accounting period must be used for intermediate consumption, final consumption, capital formation or exports. This means that, for each product (or group of products):

$$\text{Identity (2)} \quad \text{Output} + \text{Imports} = \text{Intermediate consumption} + \text{Final consumption} + \text{Capital formation} + \text{Exports}$$

2.9. Another important identity which is also key when linking the production and income approaches to calculating GDP and the industry and institutional sector dimension through the SUTs is the following:

$$\text{Identity (3)} \quad \text{For each industry, the GVA using the production approach equals the GVA estimate using the income approach.}$$

2.10. These identities are fundamental in the balancing process that is carried out when compiling SUTs both in current prices and in volume terms, all through a time series dimension.

2.11. Once balanced, the supply table and the use table can be integrated into a single matrix – often referred to as the SUTs framework, which is shown in Table 2.3 below. This table clearly shows the two basic identities linking the SUTs. The total supply by product (left part of the bottom row of Table 2.3) equals the total use by product (top part of the last column of Table 2.3) and the total outputs by industry are identical in both SUTs (the middle part of the bottom row equals the middle part of the last column). The schematic view of SUTs in Table 2.3 also serves as the underlying matrix for projection methods (see chapter 18).

Table 2.3: Supply and use tables framework

		Products				Industries				Final uses			Total
		Agriculture, forestry, etc.	Ores and minerals, etc.	...	Services	Agriculture, forestry, etc.	Mining and quarrying	...	Services	Final consumption	Gross capital formation	Exports	
Products	Agriculture, forestry, etc.					Intermediate consumption by product and by industry				Final uses by product and by category			Total use by product
	Ores and minerals, etc.												
	Services												
Industries	Agriculture, forestry, etc.	Output by product by industry											Total output by industry
	Mining and quarrying												
	...												
	Services												
Value added		Value added by component and by industry											Value added
Imports		Total imports by product											Total imports
Total		Total supply by product				Total output by industry				Total final uses by category			

Empty cells by definition

2.12. SUTs thus bring together the components of each of the three approaches to measuring GDP– namely, the production, income and expenditure approaches:

(a) Production approach:

$$\text{GDP} = \text{Output (at basic prices)} - \text{Intermediate consumption} + \text{Taxes less subsidies on products}$$

(b) Income approach:

$$\text{GDP} = \text{Compensation of employees} + \text{Gross operating surplus} + \text{Other taxes less subsidies on production} + \text{Taxes less subsidies on products}$$

(c) Expenditure approach:

$$\text{GDP} = \text{Final consumption} + \text{Gross capital formation} + \text{Exports} - \text{Imports}$$

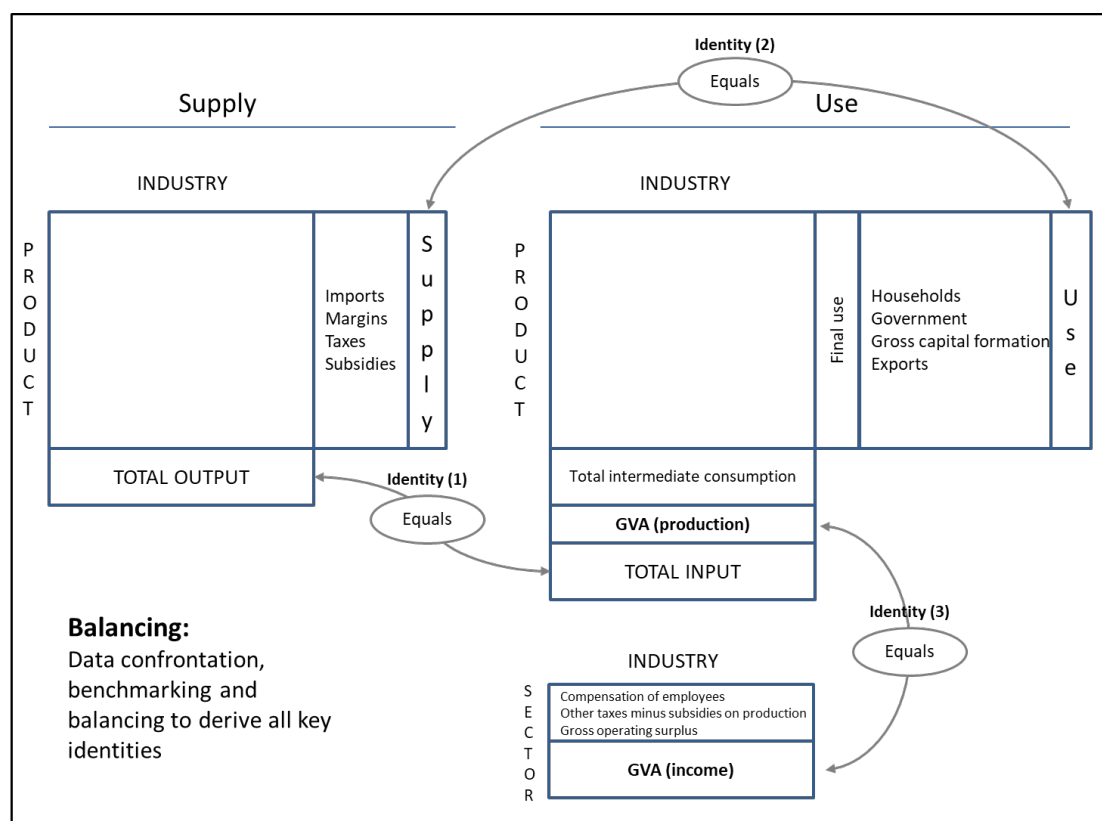
2.13. When balanced, SUTs show, by definition, a single estimate of GDP both in current prices and in volume terms. This underlines the importance of the recommendation that SUTs be compiled as part of the annual regular compilation of the national accounts, as they ensure the consistency and coherence of the national accounts components, namely, goods and services account, production account (by industry and by institutional sector) and generation of income account (by industry and by institutional sector), and make it possible to derive a single estimate of GDP. The institutional sector links are covered in more detail in chapter 10.

2.14. The SUTs also have links to other accounts, such as disposable income accounts (covering variables like household final consumption expenditure) and accumulation accounts (covering variables like gross fixed capital formation as part of the capital account).

2.15. Producing annual SUTs simultaneously both in current prices and in volume terms (preferably, when two successive years of current price SUTs are available) ensures coherence and consistency for both price and volume measures. In addition, this approach allows for the estimation of the volume of GVA through what may be termed “double deflation”, where GVA is derived by deducting intermediate consumption in volume terms from total output in volume terms. This can be achieved on the basis of an individual unit, industry, institutional sector, and for the whole economy.

2.16. SUTs can also be compiled on a quarterly basis to derive official estimates of quarterly GDP. Developing quarterly SUTs may be highly demanding in terms of resources, time and data availability but have the advantage of significantly improving the quality of the estimate of quarterly GDP.

2.17. Figure 2.1 provides a graphical overview of the SUTs, explicitly identifying the main identities that are ensured in balanced SUTs. Box 2.1 presents a numerical example of balanced SUTs.

Figure 2.1: Graphical overview of supply and use tables

2.18. The use table records the intermediate consumption and final uses by type of product but it does not distinguish between the consumption of domestically produced goods and services and that of imported goods and services. Although such a split is not a necessary condition for the creation of balanced SUTs in current prices, it is a key step linking SUTs and IOTs. The disaggregation of the use table into two tables, the domestic use table and the imports use table, is shown in Box 2.2, with a numerical example.

2.19. The compilation of the imports use table is necessary to ensure good quality volume estimates (in particular, GVA by industry) and these tables are becoming increasingly important owing to the growing impact of globalization and the need to measure global value chains and trade in value added.

Box 2.1 Numerical example of the SUTs system

Supply table						
		Industries			Imports	Total supply
		Agriculture	Manufacturing and construction	Services		
Products	Agriculture	270	30	50	20	370
	Manufacturing	6	380	87	42	515
	Construction	4	50	13	8	75
	Trade, transport and communication	10	15	210	7	242
	Finance and business services	6	17	240	11	274
	Other services	4	8	100	12	124
Total		300	500	700	100	1 600

Use table								
		Industries			Final use			Total use
		Agriculture	Manufacturing and construction	Services	Final consumption expenditure	Gross capital formation	Exports	
Products	Agriculture	34	59	93	131	21	32	370
	Manufacturing	97	107	57	122	73	59	515
	Construction	9	12	4	17	30	3	75
	Trade, transport and communication	42	24	11	140	20	5	242
	Finance and business services	14	53	42	116	31	18	274
	Other services	14	35	22	35	10	8	124
Taxes less subsidies on products		4	5	12	52	6	1	80
GVA		86	205	459				750
Total		300	500	700	613	191	126	2 430

Supply and use tables framework														
		Products						Industries			Final use			Total
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	Agriculture	Manufacturing and construction	Services	Final consumption expenditure	Gross capital formation	Exports	
Products	Agriculture							34	59	93	131	21	32	370
	Manufacturing							97	107	57	122	73	59	515
	Construction							9	12	4	17	30	3	75
	Trade, transport and communication							42	24	11	140	20	5	242
	Finance and business services							14	53	42	116	31	18	274
	Other services							14	35	22	35	10	8	124
Industries	Agriculture	270	6	4	10	6	4							300
	Manufacturing and construction	30	380	50	15	17	8							500
	Services	50	87	13	210	240	100							700
	Taxes less subsidies on products							4	5	12	52	6	1	80
	GVA							86	205	459				750
	Imports	20	42	8	7	11	12							100
Total		370	515	75	242	274	124	300	500	700	613	191	126	4 030

= Zero by definition

The **supply table** shows the supply of goods and services by product and by type of supplier, distinguishing supply by domestic industries and imports of goods and services. The domestic output of industries is shown by products. The vector of imports comprises the nation’s total imports of goods and services by product.

The **use table** shows the use of goods and services by product and by type of use, i.e. as intermediate consumption by industry, final consumption expenditure, gross capital formation and exports of goods and services. The intermediate uses and final uses reflect the consumption of domestically produced goods and services and also of imported goods and services. Furthermore, although the table is shown in summary form, it should be noted that there are components underlying the headings, for example, GVA can be split between compensation of employees, other taxes less subsidies on production, consumption of fixed capital and net operating surplus.

Note that, for illustrative purposes, it is assumed that the SUTs presented here are compiled on a consistent valuation basis.

The **supply table** shows the supply of goods and services by product and by type of supplier, distinguishing supply by domestic industries and imports of goods and services. The domestic output of industries is shown by products. The vector of imports comprises the nation's total imports of goods and services by product.

The **use table** shows the use of goods and services by product and by type of use, i.e. as intermediate consumption by industry, final consumption expenditure, gross capital formation and exports of goods and services. The intermediate uses and final uses reflect the consumption of domestically produced goods and services and also of imported goods and services. Furthermore, although the table is shown in summary form, it should be noted that there are components underlying the headings, for example, GVA can be split between compensation of employees, other taxes less subsidies on production, consumption of fixed capital and net operating surplus.

Note that, for illustrative purposes, it is assumed that the SUTs presented here are compiled on a consistent valuation basis.

2.20. Once the imports use table is constructed, the domestic use table can be estimated by subtracting the imports use table from the use table. The imports use table and the domestic use table form the basis for the construction of input imports tables and domestic IOTs, respectively. More detail may be found in chapters 8 and 12.

Box 2.2 Numerical example showing a use table split between consumption of domestic production and imports

Supply table							
	Industries			Output	Imports	Total supply	
	Agriculture	Manufact. and construction	Services				
Products							
Agriculture	270	30	50	350	20	370	
Manufacturing	6	380	87	473	42	515	
Construction	4	50	13	67	8	75	
Trade, transport and communication	10	15	210	235	7	242	
Finance and business services	6	17	240	263	11	274	
Other services	4	8	100	112	12	124	
Total	300	500	700	1 500	100	1 600	

Use table							
	Industries			Final use			Total use
	Agriculture	Manufact. and construction	Services	Final consumption expenditure	Gross capital formation	Exports	
Products							
Agriculture	34	59	93	131	21	32	370
Manufacturing	97	107	57	122	73	59	515
Construction	9	12	4	17	30	3	75
Trade, transport and communication	42	24	11	140	20	5	242
Finance and business services	14	53	42	116	31	18	274
Other services	14	35	22	35	10	8	124
Taxes less subsidies on products	4	5	12	52	6	1	80
GVA	86	205	459				750
Total	300	500	700	613	191	126	2 430

Domestic use table							
	Industries			Final use			Total use
	Agriculture	Manufact. and construction	Services	Final consumption expenditure	Gross capital formation	Exports	
Products							
Agriculture	30	50	90	130	20	30	350
Manufacturing	85	90	51	120	70	57	473
Construction	5	10	3	16	30	3	67
Trade, transport and communication	40	20	10	140	20	5	235
Finance and business services	10	50	40	115	30	18	263
Other services	10	30	20	35	10	7	112
Taxes less subsidies on products	4	5	12	52	6	1	80
GVA	86	205	459				750
Output	300	500	700	613	191	126	2 430

Imports use table							
	Industries			Final use			Total
	Agriculture	Manufact. and construction	Services	Final consumption expenditure	Gross capital formation	Exports	
Products							
Agriculture	4	9	3	1	1	2	20
Manufacturing	12	17	6	2	3	2	42
Construction	4	2	1	1			8
Trade, transport and communication	2	4	1				7
Finance and business services	4	3	2	1	1		11
Other services	4	5	2			1	12
Total	30	40	15	5	5	5	100

Empty cells by definition

The **domestic use table** is derived by subtracting the imports use table from the total use table shown in box 2.1.

The imports of goods and services are then shown separately as a new row denoted as “Imports” in the domestic use table. The domestic use table shows the input requirements of industries in terms of domestic intermediates, imported intermediates and primary inputs (GVA). It also shows the use of domestic output of products for intermediate uses and final uses.

The **imports use table** includes information on the use of imported products for intermediate consumption and final uses and the column totals, which match the estimates shown in the “Imports” row.

1. Supply and use tables in current prices and in volume terms: H-Approach

2.21. The SUTs framework not only constrains the current value estimates of supply and use of products to balance exactly, it also provides a means of ensuring that the corresponding volume estimates in previous years’ prices are balanced and that the series of prices implied by the existence of one table in current prices and one in volume terms are strictly consistent. In general, the best way to ensure mutual consistency is to prepare the SUTs in current values and in volume terms at the same time (2008 SNA, para. 14.136).

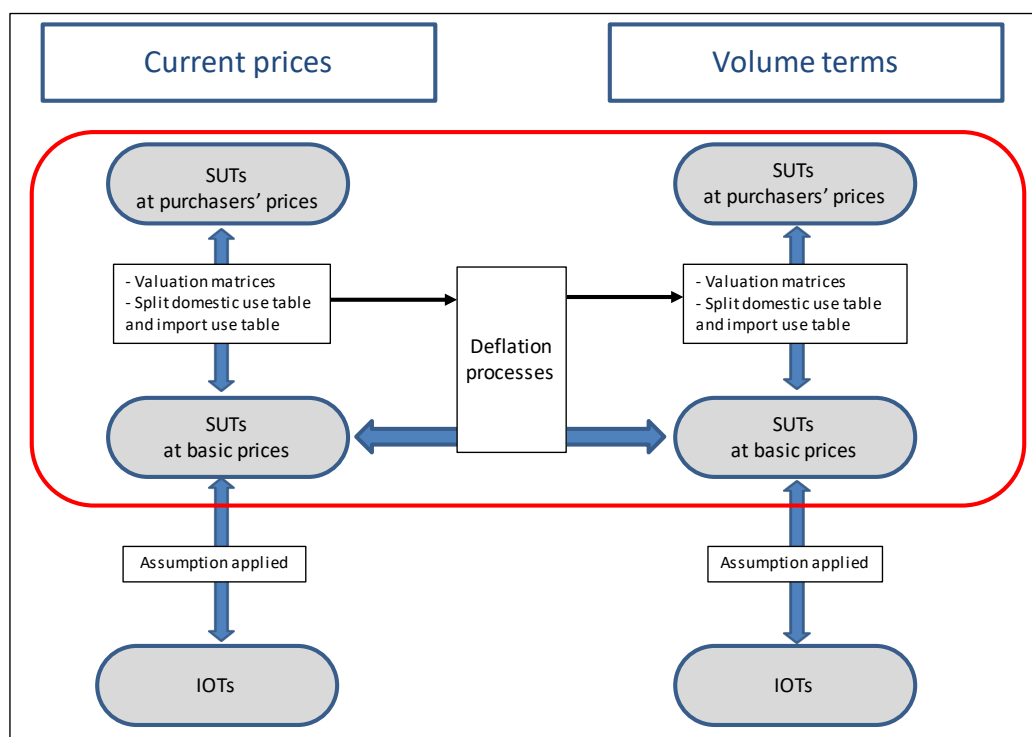
2.22. The compilation and balancing of SUTs in current prices and in volume terms for a sequence of years also helps to balance the changes in volumes, values and prices in the best possible way (the key condition for the attainment of this outcome is that SUTs are available in current prices both for the current year and for the previous year). This approach ensures a high degree of quality in terms of coherence and consistency over time – and recommended as the best

approach for the production of SUTs. Producing annual SUTs simultaneously both in current prices and in volume terms also allows estimation of the volume of GVA through double deflation, whereby GVA in previous years' prices is derived by deducting intermediate consumption in previous years' prices from total output in previous years' prices. Then, the change in volume of GVA between each pair of consecutive years is given by the change of GVA in previous years' prices compared to GVA of the previous year at current prices of that year.

2.23. The SUTs at purchasers' prices and at basic prices in current prices and in volume terms can be compiled and balanced sequentially or simultaneously. In both cases, powerful feedback loops covering quality in terms of consistency and coherence are available. More details are provided in chapters 9 and 11.

2.24. Figure 2.2 shows an overview of the H-Approach for an integrated compilation of SUTs (and IOTs) in current prices and volume terms. The H-Approach is the recommended compilation approach, which brings together the compilation of SUTs in current prices and volume terms, the valuation at basic prices, producers' prices and purchasers' prices, and the links with the compilation of IOTs. The matrices covering other taxes on production, other subsidies on production, trade margins and transport margins are the valuation matrices which link between basic prices, producers' prices and purchasers' prices.

Figure 2.2 Schematic overview of the compilation of SUTs and IOTs: H-Approach



2.25. The diagram in Figure 2.2 may be visualized as the letter “H” with the left vertical arm representing SUTs and IOTs in current prices and the right vertical arm representing the SUTs and IOTs in previous years’ prices. The horizontal transition represents the deflation process using, for example, a combination of prices, volume indicators and rates of the previous year applied to the volumes.

2.26. The SUTs in current prices are decomposed into the component parts (imports and the valuation matrices on the left-hand side of the H-Approach), each of which is deflated separately as appropriate (the join in the middle), and then added back to get to a purchasers’ prices valuation in previous years’ prices (the right-hand side). This means that basic prices play the dominant role in the process, and the initial compilation flow is from top left to middle left and that of deflation to middle right, and then to top right.

2.27. When final use deflators are deemed to be better for final use components at purchasers’ prices, then the H-Approach allows for the use of higher quality deflators, which are perhaps more appropriate. In these types of examples, it is possible to work with purchasers’ prices where the data are believed to be more reliable, making appropriate adjustments, working from top right to middle right, then onto middle left and to top left, and still ensuring a balance at each stage. Similarly, if high quality volume indicators are available, then this can better inform, for example, the step between the middle left and middle right with adjustments as appropriate.

2.28. For balancing purposes, areas such as value added tax (VAT) on products and changes in inventories may be separated out as balanced matrices (whereby the impact on production, income and expenditure is equal and the matrices are in balance) to avoid any balancing adjustments, but this may overcomplicate the system.

2.29. It is important to note that the scheme presented in Figure 2.2 should not be taken as one to be implemented as a whole. In practice, for example, if a country wants to focus exclusively on the compilation of annual SUTs, the focus should be on the compilation steps of the SUTs within the bold line box in Figure 2.2 which are recommended in order to achieve balanced SUTs in current and in volume terms. If, however, a country wants to compile SUTs and IOTs, all the steps in Figure 2.2 should be completed in order to guarantee important feedback loops and enhance the quality of the tabulations.

2.30. When planning for the compilation of SUTs, it is useful to keep in mind the compilation approach in Figure 2.2, since it is naturally linked to the production of time series of SUTs (and IOTs) both in current prices and in previous years’ prices using chain-linked volumes. More detail of this may be found in chapter 9. Although SUTs in volume terms for one period can be compiled using SUTs in current prices for one period and deflators, the preferred approach contains a time-series dimension and the following principles:

- To compile SUTs in volume terms, the following are needed:

- Balanced set of SUTs at purchasers' prices in current prices for the present year and the previous year
- Deflators for each flow
- Previous year's SUTs in current prices of that year are needed to derive volume estimates. The SUTs in figure 2.2 are balanced for illustrative purposes. In reality, however, they may be unbalanced and an iterative balancing process may be necessary. This allows for the first SUTs in previous years' prices to be compiled.
- Each transition stage is created in a balanced format which provides a much easier systematic process and build. This means that each of the transition matrices covering taxes, subsidies, trade and transport margins and import of goods and services will be balanced, in other words, the supply-side row total will equal the use-side row total.
- For some of the variables, like household final consumption expenditure, there are already present deflation approaches using consumer price indices to generate the corresponding estimates in previous years' prices. These estimates are likely to differ from those generated with the H-Approach but would also feature in the reconciliation and balancing process of the estimates, and form an example of working from right to left through balanced adjustments.

2.31. The H-Approach provides a transparent, coherent and consistent approach for the compilation and balancing of SUTs. For example, balancing adjustments to one part of the SUTs can be assessed in terms of their impact on other areas of the SUTs, and also in terms of time series.

2. PSUTs

2.32. The SUTs described in the previous sections are part of the 2008 SNA framework. As such, they reflect the production boundary of the SNA and they are compiled in monetary units. The tables can, however, be extended to include the environment as providers of natural inputs into the economy and as absorbers of residuals from the economy. The extension of these tables and, more generally, of the accounting framework of the SNA to include environmental considerations is carried out in the 2012 SEEA. The SEEA enables the analyses of the interaction between the environment and the economy, such as an assessment of the use of natural resources, the generation of waste by the economy and waste flows into the environment.

2.33. The SEEA central framework comprises a sequence of accounts – namely, the SUTs in monetary and physical units, the asset accounts in physical and monetary units, and environmental activity accounts and related flows. The present Handbook covers the SUTs of the SEEA and, in particular, since the monetary SUTs of the SEEA are the same as the SNA, the Handbook focuses on the PSUTs of the SEEA. For additional information on the complete set of accounts of the

SEEA, the reader is directed to the publication United Nations, European Commission, FAO, IMF, OECD and World Bank (2014).

2.34. PSUTs are used to assess how an economy supplies and uses energy, water, materials, and also their changes in production and consumption patterns over time, and therefore, in combination with data from monetary SUTs, changes in productivity and intensity in the use of natural inputs and the release of residuals can be examined. The structure of PSUTs is based on the monetary SUTs with extensions to incorporate a column for the environment and rows for natural inputs and residuals.

2.35. Table 2.4 and Table 2.5 provide the simplified structure of the physical supply table and use table, respectively. In order to address specific environmental domains (for example, accounting for water, energy, timber and other resources), these tables are compiled for a disaggregation of products and industries which is relevant for the particular environmental domain of interest. In the case of energy, for example, the products of interest that can be explicitly shown in the table include coal, peat and peat products, natural gas and others. The industries of interest include the main suppliers of energy products (for example, electricity generation, manufacture of gas and other products) and the main users of energy products (for example, manufacturing, transportation and others). These tables are compiled in monetary units – within the SNA context – and in physical units as shown in Table 2.4 and Table 2.5.

2.36. In the PSUTs, the SUTs of the SNA are augmented to include a block of rows for “natural inputs” and a block of rows for “residuals”. The block for natural inputs is used to describe the flows from the environment to the economy; in other words, this block describes the extraction of natural inputs (for example, water, energy resources and others) from their location in the environment as a part of economic production processes or that are directly used in production. Natural inputs may be, first, natural resource inputs, such as mineral and energy resources or timber resources; second, inputs from renewable energy sources, such as solar energy captured by economic units; or, third, other natural inputs such as inputs from soil (for example, soil nutrients) and inputs from air (for example, oxygen absorbed in combustion processes) (2012 SEEA, para. 2.89). When an industry, for example, extracts water as part of the economic production process, this is recorded in the block of natural inputs in the use table in Table 2.5. It is assumed that the environment provides (that is, supplies) all the natural inputs that are used into the economic production process.

2.37. The blocks for “residuals” represent the flows of solid, liquid and gaseous materials, and energy, that are discarded, discharged or emitted to the environment (for example, emission to air) by establishments and households through processes of production, consumption or accumulation but that may also flow within the economy, as is the case when, for example, solid waste is collected as part of a waste collection scheme (2012 SEEA, para. 2.92).

2.38. The block for residuals in the supply table (Table 2.4) represents the flows of waste from the economy to the environment and thus it includes the generation and disposal of waste during economic production activities (generation of waste by industries) and generated during final consumption (mainly by households). While the block of residuals in the use table (Table 2.5) shows, for example, the collection and treatment of waste and other residuals by economic activities, the accumulation of waste in controlled landfills and the residuals flows direct to the environment.

Table 2.4 Schematic view of the physical supply table

Industries		Industries				Imports	Final consumption	Gross capital formation/ Accumulation	Environment	Total
		Agriculture, forestry, etc.	Mining and quarrying	...	Services					
Natural inputs	Mineral and energy resources								Flows from the environment	Total supply by natural inputs
	Water									
Products	Agriculture, forestry, etc.	Output by product by industry				Imports by product				Total supply by product
	Ores and minerals; etc.									
Residuals	Solid waste	Residuals generated by industry					Residuals generated by final consumption	Residuals from scrapping and demolition of produced assets		Total supply by residuals
	Wastewater									
...										

 Empty cells by definition

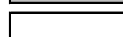
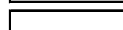
 Cells may contain relevant flows

Table 2.5 Schematic view of the physical use table

Industries		Industries				Exports	Final consumption	Gross capital formation/Accumulation	Environment	Total
		Agriculture, forestry, etc.	Mining and quarrying	...	Services					
Natural inputs	Mineral and energy resources	Extraction of Natural inputs								Total use by natural inputs
	Water									
Products	Agriculture, forestry, etc.	Intermediate consumption by product and by industry				Final uses by product and by category				Total use by product
	Ores and minerals; etc.									
Residuals	Solid waste	Collection and treatment of waste and other residuals						Accumulation of waste in controlled landfilled	Residual flows direct to the environment	Total use by residuals
	Wastewater									
	...									

 Empty cells by definition

 Cells may contain relevant flows

2.39. The supply and use identity applies to both physical and monetary flows. For each product measured in physical terms (for example, cubic metres of timber), the quantity of output and imports (total supply of products) must equal the quantity of intermediate consumption, household final consumption, gross capital formation and exports (total use of products). The equality between supply and use also applies to the total supply and use of natural inputs and the total supply and use of residuals. In addition to the supply and use identity, the PSUTs incorporate the

input-output identity, implying that the total flows into the economy either are returned to the environment or accumulate in the economy.

C. Overview of IOTs

2.40. For many analytical purposes, a transformation from a pair of SUTs into a single IOT where total input (row totals) and total output (column totals) are equal brings considerable advantages. IOTs have algebraic properties that make them particularly suitable for analyses that enable estimates of the effects of changing relative prices, labour and capital requirements in the face of changing output levels, the consequences of changing patterns of demand and so on. They may also be used as the basis for an expanded version that may be used to estimate the demands made by the economy on the environment (2008 SNA, para. 28.35).

2.41. An IOT is essentially derived from the use table, where either the columns representing industries are replaced by products or where the rows representing the products are replaced by industries through a transformation process which involves a range of assumptions. The resulting intermediate consumption matrix is then square, showing products in both rows and columns or industries in both. In both cases, the row totals for the complete matrix match the column totals for the complete matrix, product-by-product matrix or industry-by-industry matrix as the case may be (2008 SNA, para. 28.32). Of course, the classifications in the IOTs coincide with those in the SUTs, as the former is a transformation of the latter.

2.42. It is recommended that the IOTs be derived from SUTs. The IOTs derived from the SUTs further describe the interrelationships between industries and products, along with the sale and purchase relationships between producers and consumers within an economy. They can be produced to illustrate flows between the sales and purchases (final and intermediate) of industry outputs (referred to as industry-by-industry tables) or to illustrate the sales and purchases (final and intermediate) of product outputs (referred to as product-by-product tables).

2.43. The derivation of IOTs from the system of SUTs may also reveal inconsistencies and weaknesses in the SUTs. This is made possible by the powerful quality-related feedback loop from the IOTs to the SUTs and vice versa.

2.44. Table 2.6 provides a simplified IOT where the columns of the original use table referring to industry-based structures are transformed into product-based structures. The relations between output and input are now relations between products and primary inputs necessary to produce outputs in similar production units. Primary inputs are inputs that are not outputs of other industries. They include the imports of goods and services and the components of GVA, such as compensation of employees, and others. They are necessary to the production process but are not produced anywhere in the domestic economy.

Table 2.6 Simplified IOT (product by product)

Products Products	Products				Final uses			Total
	Agriculture, forestry, etc.	Ores and minerals; etc.	...	Services	Final consumption	Gross capital formation	Exports	
Agriculture, forestry, etc. Ores and minerals; etc. ... Services	Intermediate consumption by product				Final uses by product and by category			Total use by product
Imports	Intermediate consumption of imported products				Final use of imported products			
Value added	Value added by component							Value added
Total	Total supply				Total final uses by category			

 Empty cells by definition

2.45. For the transformation of SUTs into IOTs, various assumptions have to be made or adjustments are required based on industry-by-industry or product-by-product assumptions:

- *Product-by-product IOTs*: these may be compiled using either the product technology assumption (whereby each product is produced in its own specific way, irrespective of the industry where it is produced) or the industry technology assumption (whereby each industry has its own specific means of production, irrespective of its product mix).
- *Industry-by-industry IOTs*: these may be compiled using either the fixed industry sales structure assumption (whereby each industry has its own specific sales structure, irrespective of its product mix) or the fixed product sales structure assumption (whereby each product has its own specific sales structure, irrespective of the industry where it is produced).

A mixture of both assumptions can also be applied by implementing a hybrid technology assumption. The correct use and understanding of the terminology, transformation process and assumptions applied are covered in more detail in chapter 12 of this Handbook.

2.46. The selection of the appropriate type of IOT – product-by-product or industry-by-industry – depends on a number of statistical and practical considerations. For example, industry-by-industry IOTs are closer to statistical sources and actual market transactions. Product-by-product IOTs are believed to be more similar in terms of cost structure and production activities. This view has been somewhat weakened, however, by changes introduced in the 2008 SNA, such as the strict implementation of changes in economic ownership.

2.47. In the IOTs, two identities of the SUTs system are reduced to a single type of identity. It is typical for IOTs that, for each product or industry, the input equals output and total input equal total output.

2.48. The figures of total output and total input by product are the same as total supply and total use by product of the SUTs – this holds for product-by-product IOTs. The industry-based structures are transformed into product-based structures. In this transformation, the final use data

are left unchanged. The transformation only rearranges the data on the basis of the production matrix of the intermediate use table by applying certain analytical assumptions to the relations between primary and secondary outputs.

2.49. In general, and for analytical purposes, it is recommended to separate the use table into the use table for domestic output and the imports use table. More detail on the compilation of the domestic use table and the imports use table may be found in chapter 8.

2.50. Box 2.3 and Box 2.4 show a simplified numerical example of a sequence of tables – based on the SUTs shown in Box 2.1 and Box 2.2 – necessary for compiling product-by-product IOTs and industry-by-industry IOTs, respectively.

Box 2.3 SUTs and product-by-product IOTs

Supply table							
	Industries			Domestic supply	Imports	Total supply	
	Agriculture	Manufacturing and construction	Services				
Products							
Agriculture	270	30	50	350	20	370	
Manufacturing and construction	10	430	100	540	50	590	
Services	20	40	550	610	30	640	
Total	300	500	700	1500	100	1600	

Use table							
	Industries			Final use			Total use
	Agriculture	Manufacturing and construction	Services	Final consumption expenditure	Gross capital formation	Exports	
Products							
Agriculture	34	59	93	131	21	32	370
Manufacturing and construction	106	119	61	139	103	62	590
Services	70	112	75	291	61	31	640
Taxes less subsidies on products	4	5	12	52	6	1	80
GVA	86	205	459				750
Total	300	500	700	613	191	126	

Use table for domestic output							
	Industries			Final use			Total
	Agriculture	Manufacturing and construction	Services	Final consumption expenditure	Gross capital formation	Exports	
Products							
Agriculture	30	50	90	130	20	30	350
Manufacturing and construction	90	100	54	136	100	60	540
Services	60	100	70	290	60	30	610
Imports	30	40	15	5	5	5	100
Taxes less subsidies on products	4	5	12	52	6	1	80
GVA	86	205	459				750
Total	300	500	700	613	191	126	

Imports use table							
	Industries			Final use			Total
	Agriculture	Manufacturing and construction	Services	Final consumption expenditure	Gross capital formation	Exports	
Products							
Agriculture	4	9	3	1	1	2	20
Manufacturing and construction	16	19	7	3	3	2	50
Services	10	12	5	1	1	1	30
Total	30	40	15	5	5	5	100

Input-output table for domestic output (product by product)							
	Products			Final use			Output
	Agriculture	Manufacturing and construction	Services	Final consumption expenditure	Gross capital formation	Exports	
Products							
Agriculture	34.08	52.23	83.69	130.00	20.00	30.00	350.00
Manufacturing and construction	113.17	111.84	18.99	136.00	100.00	60.00	540.00
Services	73.23	114.19	42.58	290.00	60.00	30.00	610.00
Imports	37.73	46.07	1.20	5.00	5.00	5.00	100.00
Taxes less subsidies on products	4.58	4.83	11.59	52.00	6.00	1.00	80.00
GVA	87.21	210.84	451.95				750.00
Total	350.00	540.00	610.00	613.00	191.00	126.00	

Input table for imports (product by product)							
	Products			Final use			Total
	Agriculture	Manufacturing and construction	Services	Final consumption expenditure	Gross capital formation	Exports	
Products							
Agriculture	4.91	10.75	0.34	1.00	1.00	2.00	20.00
Manufacturing and construction	20.22	21.68	0.11	3.00	3.00	2.00	50.00
Services	12.60	13.65	0.74	1.00	1.00	1.00	30.00
Total	37.73	46.07	1.20	5.00	5.00	5.00	100.00

In product-by-product IOTs, all inputs are allocated to similar production units. They are derived from the SUTs system on the basis of analytical assumptions (see chapter 12 for further detail on the derivation of IOTs).

Product-by-product IOTs are further away from statistical sources than industry-by-industry IOTs.

Box 2.4 SUTs and industry-by-industry IOTs

Supply table							
Products	Industries			Domestic supply	Imports	Total supply	
	Agriculture	Manufacturing and construction	Services				
Agriculture	270	30	50	350	20	370	
Manufacturing and construction	10	430	100	540	50	590	
Services	20	40	550	610	30	640	
Total	300	500	700	1500	100	1600	

Use table							
Products	Industries			Final use			Total use
	Agriculture	Manufacturing and construction	Services	Final consumption	Gross capital formation	Exports	
Agriculture	34	59	93	131	21	32	370
Manufacturing and construction	106	119	61	139	103	62	590
Services	70	112	75	291	61	31	640
Taxes less subsidies on products	4	5	12	52	6	1	80
GVA	86	205	459				750
Total	300	500	700	613	191	126	

■ = Zero by definition

Use table for domestic output							
Products	Industries			Final use			Total
	Agriculture	Manufacturing and construction	Services	Final consumption expenditure	Gross capital formation	Exports	
Agriculture	30	50	90	130	20	30	350
Manufacturing and construction	90	100	54	136	100	60	540
Services	60	100	70	290	60	30	610
Imports	30	40	15	5	5	5	100
Taxes less subsidies on products	4	5	12	52	6	1	80
GVA	86	205	459				750
Total	300	500	700	613	191	126	

Imports use table						
Products	Industries			Final use		
	Agriculture	Manufacturing and construction	Services	Final consumption expenditure	Gross capital formation	Exports
Agriculture	4	9	3	1	1	2
Manufacturing and construction	16	19	7	3	3	2
Services	10	12	5	1	1	1
Total	30	40	15	5	5	5

Input-output table for domestic output (industry by industry)							
Industries	Industries			Final use			Total
	Agriculture	Manufacturing and construction	Services	Final consumption expenditure	Gross capital formation	Exports	
Agriculture	26.78	43.70	72.72	112.31	19.25	25.24	300.00
Manufacturing and construction	78.17	90.47	55.30	138.46	85.28	52.32	500.00
Services	75.05	115.83	85.97	305.23	75.47	42.45	700.00
Imports	30.00	40.00	15.00	5.00	5.00	5.00	100.00
Taxes less subsidies on products	4.00	5.00	12.00	52.00	6.00	1.00	80.00
GVA	86.00	205.00	459.00				750.00
Total	300.00	500.00	700.00	613.00	191.00	126.00	

Input table for imports (industry by industry)							
Industries	Industries			Final use			Total
	Agriculture	Manufacturing and construction	Services	Final consumption expenditure	Gross capital formation	Exports	
Agriculture	3.71	7.69	2.61	0.86	0.86	1.61	17.34
Manufacturing and construction	13.74	16.69	6.16	2.54	2.54	1.83	43.50
Services	12.55	15.62	6.23	1.60	1.60	1.56	39.17
Total	30.00	40.00	15.00	5.00	5.00	5.00	100.00

In industry-by-industry IOTs, inputs are allocated to industries. They are derived from the SUTs system on the basis of pragmatic assumptions. The intermediate input of industries consists of output of industries rather than products (of industry adjusted products) (see chapter 12 for details on the derivation of IOTs).

Industry-by-industry IOTs are closer to statistical sources and actual observations than product-by-product IOTs.

D. Structure of SUTs and IOTs: basic elements

2.51. Defining the structure of SUTs and IOTs is a principal first step and depends on a number of basic elements which form the backbone of these tabulations. These elements include:

- Principles of the accounting system underlying the SNA applied to SUTs and IOTs
- Classification of economic activities and its level of detail
- Classification of products and its level of detail

- Choice of the statistical units
- Valuation

2.52. Considerations on these elements reflect the specific context in which the tables are compiled, which include the analytical objectives, the data availability, the economic structure of the country, and others. Each of these elements is described below.

1. Principles of the accounting system underlying the SNA applied to SUTs and IOTs

2.53. The accounting system underlying the SNA is derived from broad bookkeeping principles and is applied to the structure and links in the SUTs and IOTs. There are three bookkeeping principles underlying the SNA accounting system:

- Vertical double-entry bookkeeping, also known as double-entry bookkeeping
- Horizontal double-entry bookkeeping
- Quadruple-entry bookkeeping

2.54. The main characteristic of vertical double-entry bookkeeping is that each transaction leads to at least two entries, traditionally referred to as a credit entry and a debit entry. This principle ensures that the total of all credit entries and all debit entries for all transactions are equal, thus permitting a check on consistency of accounts for a single unit. Each transaction requires two entries.

2.55. The concept of horizontal double-entry bookkeeping is useful for compiling accounts that reflect the mutual economic relationships between different institutional units in a consistent manner. It implies that, if unit A provides something to unit B, the accounts of both A and B show the transaction for the same amount: as a payment in A's account and as a receipt in B's account. Horizontal double-entry bookkeeping ensures the consistency of recording for each transaction category by counterparties. For example, dividends payable throughout the economy should be equal to dividends receivable throughout the economy once transactions with the rest of the world are taken into account.

2.56. The simultaneous application of vertical and horizontal double-entry bookkeeping results in quadruple-entry bookkeeping – which forms the accounting system underlying the SNA. It deals in a coherent manner with multiple transactors or groups of transactors, each of which satisfies vertical double-entry bookkeeping requirements. A single transaction between two counterparties thus gives rise to four entries. In contrast to business bookkeeping, national accounts deal with interactions among a multitude of units in parallel, and thus require special care from a consistency point of view.

2.57. An account records and displays all the flows and stocks for a given aspect of economic life. In each account, the sum of resources is equal to the sum of uses with a balancing item to

ensure this equality. Normally the balancing item will be an economic measure which is itself of interest.

2.58. The accounts can be built up for different areas of the economy by employing a system of economic accounts which highlight, for example, production, income and financial transactions. In many cases, these accounts can be elaborated and set out for different institutional units and groups of units (or institutional sectors). Usually a balancing item has to be introduced between the total resources and total uses of these units or sectors and, when summed across the whole economy, these balancing items constitute significant aggregates.

2.59. The accounting structure is uniform throughout the system and applies to all units in the economy, whether they are institutional units, subsectors, sectors or the whole economy, although some accounts (or transactions) may not be relevant for some institutional sectors.

2.60. The national accounting system uses two types of units and two corresponding ways of subdividing the economy, which are quite different and serve separate analytical purposes:

- The first purpose, namely that of describing production, income, expenditure and financial flows, and balance sheets, is served by grouping institutional units into institutional sectors on the basis of their principal functions, behaviour and objectives. The national accounting system enables a complete set of flow accounts and balance sheets to be compiled for each sector, and subsector, and also for the total economy.
- The second purpose, namely that of describing processes of production and for input-output analysis, is served by the system grouping establishments into industries on the basis of their type of activity. An activity is characterized by an input of products, a production process and an output of products.

2.61. Figure 2.3 shows in matrix form an overview of the structure of the SNA. The degree of subdivisions of the columns and rows using the relevant classifications determines the degree of detail of the accounts. The shaded rows and columns for goods and services and production by industry indicate those parts of the system relevant for the compilation of SUTs and IOTs, and clearly indicate that SUTs are at the core of the national accounts system.

2.62. The three approaches to measuring GDP (production, income and expenditure) are shown in Box 2.5 and can be derived from the data in Figure 2.3 generating a single estimate of GDP.

2.63. All the aggregate components and detailed components are included in the SUTs and IOTs-related part of the system.

Figure 2.3 System of national accounts in matrix form

millions of euros

Expenditures Assets Receipts Liabilities		Opening balance	Goods and services	Production by industry	Income and consumption	Accumulation	Rest of the world	Closing balance	Total excl. balance
		1	2	3	4	5	6	7	8
Opening balance	1					Assets of domestic sectors (real and financial)	Financial assets of the rest of the world		
Goods and services	2			303 492 Intermediate consumption	226 258 Final consumption	74 612 Gross capital formation	165 648 Exports of goods and services		770 009
Production by industry	3		578 360 Output at basic prices						578 360
Income and consumption	4		33 778 Taxes less subsidies on products	274 868 Value added at basic prices			38 023 Primary incomes and current transfers from rest of the world		346 670
Accumulation	5	Liabilities of domestic sectors			79 669 Gross savings		- 5 057 Deficit on the balance of payments	Liabilities of domestic sectors	74 612
Rest of the world	6	Financial liabilities of the rest of the world	157 871 Imports of goods and services		40 743 Primary incomes and current transfers to the rest of the world			Financial liabilities of the rest of the world	198 614
Closing balance	7					Assets of domestic sectors (real and financial)	Financial assets of the rest of the world		
Total excl. balance	8		770 009	578 360	346 670	74 612	198 614		

Table based on 2011 figures from Austria

Box 2.5 Three approaches to measuring GDP

Production approach millions of euros		Income approach millions of euros		Expenditure approach millions of euros	
Variable	Value	Variable	Value	Variable	Value
Output at basic prices	578 360	Compensation of employees	144 343	Final consumption	226 258
- Intermediate consumption	- 303 492	+ Other taxes less subsidies on production	4 858	+ Gross capital formation	74 612
= Gross value added at basic prices	274 868	+ Consumption of fixed capital	53 469	+ Exports of goods and services	165 648
+ Taxes less subsidies on products	33 778	+ Net operating surplus	72 198	- Imports of goods and services	- 157 871
= Gross domestic product	308 647	= Gross value added at basic prices	274 868	= Gross domestic product	308 647
		+ Taxes less subsidies on products	33 778		
		= Gross domestic product	308 647		

Table based on 2011 figures from Austria

2. Classification of economic activities

2.64. The International Standard Industrial Classification of All Economic Activities (ISIC) is the international reference classification of economic activities (also referred to as “industries”). The fourth revision, ISIC Rev. 4, was issued by the United Nations in 2008 (United Nations, 2008). Its main purpose is to provide a set of activity categories that can be used for collecting and presenting internationally comparable statistics by economic activity.

2.65. In general, the scope of ISIC covers productive activity, that is, all economic activities within the production boundary as described in the SNA (with one exception for activities in Class 9820– “Undifferentiated service-producing activities of private households for own use”). The classification is used to classify statistical units such as establishments or enterprises, according to the economic activity in which they mainly engage. All categories at each level of the classification are mutually exclusive. ISIC Rev. 4 is the reference classification of production activities of the 2008 SNA.

2.66. The structure of ISIC consists of 21 sections, 88 divisions, 238 groups and 419 classes. The principles and criteria used to define and delineate the categories are based on the inputs of goods, services and factors of production, the process and technology of production, the characteristics of outputs, and the use to which the outputs are put. At the class level of the classification, preference has been given to the process and technology of production in defining individual ISIC classes, in particular in the classes related to services. The list of products that defines a class is called the principal products of that class. At the division and group levels, characteristics of outputs and the use to which outputs are put become more important for the creation of analytically useful aggregation categories.

2.67. At national and regional levels, there may be need for recourse to a level of detail that reflects specific national and regional circumstances. It is important, however, that these classifications are compatible with ISIC Rev. 4 at an aggregated level of detail. At its thirty-seventh session, the United Nations Statistical Commission recommended that countries adapt their national classifications in a way that allows them to report data at least at the two-digit level of ISIC Rev. 4 without loss of information.⁵ Examples of regional classifications include the industrial classification used in the European Union, the second revision of the Statistical Classification of Economic Activities in the European Community (NACE Rev. 2), which is identical with ISIC Rev. 4 up to the two-digit level (divisions) of the classification. At lower levels, NACE has provided more detail suitable for the European users of the classification. The additional detail can always be aggregated to ISIC categories at the three-digit and four-digit levels, within the same structure. The North American Industry Classification System (NAICS), although it has a substantially different structure from ISIC, has been designed in a way that

⁵ See *Official Records of the Economic and Social Council, 2002, Supplement No.4 (E/2006/24)*, chapter I, para. 3, item 37/105 (b).

statistical data collected according to NAICS can be re-aggregated into the two-digit divisions of ISIC Rev. 4. The Australian and New Zealand Standard Industrial Classification (ANZSIC) was revised in 2006. Its structure broadly follows the ISIC structure, so that categories at the division and more detailed levels can be aggregated into the two-digit categories of ISIC.

2.68. An economic unit may engage in a variety of production activities. The classification of the economic unit is made in accordance with the importance of the production activities. In this regard, the activities of an economic unit are subdivided into principal activity, secondary activity and ancillary activities. The principal activity of an economic entity is the activity that contributes most to the value added of the entity, as determined by what is known as the “top-down method”. This method follows a hierarchical order, starting with the identification of the relevant category at the highest level (section) and progressing down through the levels of the classification to the lowest level (classes). The effect of this top-down method is such that the principal activity need not account for 50 per cent or more of the total value added of an entity or even that its generated value added exceed that of all other activities carried out by the unit, although, in the majority of cases, it will do so (United Nations, 2008, para. 57).

2.69. In practice, it is often impossible to obtain information on the GVA of the different activities performed and the activity classification has to be determined by using substitute criteria, such as employment and turnover.

2.70. Products resulting from a principal activity are either principal products or by-products. By-products are products that are necessarily produced together with principal products, for example, hides produced when producing meat by slaughtering animals. Since normal patterns of horizontal integration have been taken into account when defining the ISIC classification, such commonly integrated activities are usually included in the same ISIC class, even though the outputs of the activities have quite different characteristics. Thus ISIC class 1010 “processing and preserving of meat” also includes hides, skins, wool and feathers originating from slaughtered animals (United Nations, 2008, paras. 57 and 120).

2.71. A secondary activity is a separate activity the products of which are ultimately intended for third parties and that is not the principal activity of the entity in question. The outputs of secondary activities are called secondary products, including any by-products associated with these outputs. Most economic entities produce at least some secondary products.

2.72. Traditionally, the existence of by-products has been seen as creating problems in input-output analysis as they would disturb supply-and-use relationships. Additional demand for the principal product would therefore also result in more output of the by-product, without there automatically being any additional demand for that increased output. In the case of more complex production processes than meat and hides, for example in the chemical and electronic industries, it will, however, be very difficult or even, in the absence of special technical insight, impossible for the compilers of SUTs to identify by-products separately.

2.73. A distinction is made between principal and secondary activities on the one hand and ancillary activities on the other. All economic units require some basic, routine services to support their production activities. When they are provided in-house, they are called ancillary activities. In general, an ancillary activity is a supporting activity undertaken within an enterprise in order to create the conditions within which the principal and secondary activities can be carried out (2008 SNA, para. 5.36). Ancillary activities typically produce services that are commonly found as inputs into almost any kind of economic activity. These outputs are always services and intended for intermediate consumption within the same entity. They include, for example, the maintenance of records, files or accounts in written form or on computers; the purchase of materials and equipment; the provision of electronic and traditional written communication facilities; the hiring, training, managing and paying of employees; the storing of materials or equipment; warehousing; the provision of security and surveillance, and others.

2.74. Some of these activities are found in every economic entity. The output of an ancillary activity is not explicitly recognized and recorded separately in the SNA. It follows that the use of this output is also not recorded. All the inputs consumed by an ancillary activity – materials, labour, consumption of fixed capital, and so forth – are treated as inputs into the principal or secondary activity that it supports.

2.75. The following activities are not to be considered ancillary: producing goods or services as part of gross fixed capital formation and research and development activities, which are considered to be part of gross fixed capital formation in the 2008 SNA. These items will therefore appear as either principal or secondary output. Goods that become embodied in the output of the principal or secondary activities are not outputs of ancillary activities.

2.76. More details on principal products, secondary products and ancillary products specific to the construction of the supply table may be found in chapter 5.

(a) Considerations for the compilation of SUTs and IOTs

2.77. In SUTs and IOTs, industries should be classified according to ISIC Rev. 4. The major advantage of using established international industrial classifications is that comparability with other types of economic statistics and the national accounts is not compromised. The choice is therefore not which industrial classifications should be used in the SUTs and IOTs but rather the level of detail.

2.78. At the working level, it is recommended to use the most detailed level of classification of industry, taking into consideration user needs, the availability of data and the level of detail used in the national accounts. Furthermore, certain compilation aspects also influence the choice of working level, such as the distinction between industries which are allowed to deduct VAT and those that are not, the distinction between market and non-market producers, and the explicit identification of certain industry subdivisions that are relevant for the compilation of the trade and transport margin matrices. In addition, the link between SUTs and the institutional sector accounts

should be reflected. These considerations are further elaborated in chapters 4–7. In general, the level of detail in the published and disseminated SUTs differs from that at the working level: SUTs tend to be published at a more aggregated level of detail which takes into account users' needs and confidentiality.

3. Classification of products

2.79. The international reference classification of products is the Central Product Classification (CPC). The latest revision, CPC Version 2.1, was issued by the United Nations in 2015 (United Nations, 2015). The primary purpose of CPC is to classify all goods and services that are the result of production in the economy. CPC presents categories for all products that can be the object of domestic or international transactions or that can be entered into stocks. It includes products that are the output of economic activity, including transportable goods, non-transportable goods and services. CPC in general follows the definition of products within the SNA.

2.80. The importance of the industrial origin of goods and services was underscored by the attempt to group into one CPC subclass mainly the products that are the output of a single ISIC class. Through their linkage to the criterion of industrial origin, the input structure, technology and organization of production characteristics of products are also reflected in the structure of CPC. The criterion of industrial origin of products is one of the classification principles applied by ISIC.

2.81. CPC was developed primarily to enhance harmonization among various fields of economic and related statistics and to strengthen the role of national accounts as an instrument for the coordination of economic statistics. It provides a basis for transforming basic statistics from their original classifications into a standard classification for analytical use. As a general purpose classification, CPC provides less detail than other specific classification systems in areas or applications for which such systems are available, for example the Harmonized Commodity Description and Coding System (Harmonized System). The Harmonized System⁶ codes provide building blocks for the part dealing with transportable goods and take into account the basic categories of economic supply and use of products as specified in the SNA such as intermediate consumption, final consumption, capital formation, and imports and exports.

2.82. CPC is a system of categories that are both exhaustive and mutually exclusive. This means that if a product does not fit into one CPC category, it must automatically fit into another. In CPC Version 2.1, in total there are 10 sections, 71 divisions, 329 groups, 1,299 classes and 2,887 subclasses. Each subclass in sections 0–4 of CPC is defined as the equivalent of one heading or subheading or the aggregation of several headings or subheadings of the Harmonized System, since the Harmonized System is a detailed classification of transportable goods that is widely accepted for use in international trade statistics by virtually all countries. Other classifications of

⁶ The Harmonized Commodity Description and Coding System is the classification used for international trade statistics.

products may be used at country and regional level, however, these classifications are in general broadly consistent with CPC Version 2.1. The Classification of Products by Activity (CPA) is based on NACE – and therefore follows a different aggregation structure than CPC – and detailed categories that are mostly aligned with CPC. Exceptions exist for areas where CPC deviates from the Harmonized System, since CPA maintains a closer link with the Combined Nomenclature, which is the European version of that classification.

2.83. CPC and ISIC are both general purpose classifications, with ISIC representing the activity side. Each subclass of CPC consists of goods or services that are predominantly produced in a specific class of ISIC. The relationship between industries and their products is complex, however, and changes over time, and it should be noted that there has been no intention of establishing a one-to-one correspondence between CPC and ISIC. Such an effort is considered neither practical nor desirable as it might lead to an inadequate description of CPC categories, in particular at the higher levels.

2.84. The classification of a product in the service part of CPC does not automatically imply that the product cannot be a principal output of a goods producing industry. Thus the two CPC divisions: (87) Maintenance, repair and installation (except construction) services; and (88) Manufacturing services on physical inputs owned by others, both appear in the business and production services section of CPC but the units carrying out these activities on a fee or contract basis are classified in the same ISIC category as units producing the same goods or services for their own account. The correspondence table between CPC Version 2.1 and ISIC Rev. 4 (see the United Nations Statistics Division classification website at: <http://unstats.un.org/unsd/class/default.asp>) shows that 125 subclasses of CPC division 88 are defined to correspond to 125 manufacturing industry classes of ISIC. This implies that these manufacturing services are the principal output (and not as might have been expected, the secondary output) of the corresponding manufacturing activities. In other words, there are no service industries producing these services. This example shows the need to ensure that services of these kinds are correctly entered into the domestic output matrix, requiring a considerable number of products.

2.85. Box 2.6 shows other classification of products, such as the Harmonized System, the Standard International Trade Classification (SITC), the Classification by Broad Economic Categories (BEC) and the Extended Balance of Payments Services Classification (EBOPS) and how they relate to CPC. The basis for grouping products in the SUTs (and IOTs) is thus most commonly an aggregation of CPC sections, divisions or groups (2008 SNA, para. 14.22).

Box 2.6 Other classifications of products

The Harmonized Commodity Description and Coding System, generally referred to as the Harmonized System or simply HS, is a multipurpose international product nomenclature developed by the World Customs Organization (WCO). The system is used by more than 200 countries and economies as a basis for their collection of external trade statistics for goods. It is also extensively used by governments, international

organizations and the private sector for many other purposes, such as internal taxes, trade policies, monitoring of controlled goods, rules of origin, freight tariffs, transport statistics, price monitoring, quota controls, compilation of national accounts, and economic research and analysis.

As a result of the intensive world-wide use and degree of detail that the Harmonized System provides, it is a fundamental classification system and provides a key link for the definitions of all other classifications of goods (including the goods part of CPC), as well as for the definition of the classes of ISIC. The latest version now available is HS 2012.

The Harmonized System explanatory notes are a part of a commodity database giving the Harmonized System classification of more than 200,000 products actually traded internationally. This high number of background products also makes it evident that, at the level of external trade statistics (usually 5,000–10,000 products, as most countries apply further subdivisions of the 5,000 Harmonized System codes), there will be no homogeneous products, and of course, even less so at the much higher level of aggregation applied in a SUTs system.

SITC and BEC are both classifications of goods defined in terms of the Harmonized System, and also primarily used in relation to external trade data. SITC distinguishes around 3,000 products at its most detailed level. It is primarily used as an alternative to the Harmonized System publication level of external trade statistics, and there will usually be no advantage in applying it in SUTs, rather than using the Harmonized System directly. Following the breakdown of products according to BEC (food, materials, fuel, capital goods, transport equipment, consumer goods), these groupings may be used as a reference when deciding on uses of some products but this breakdown is not applicable as the main product classification in the SUTs system. Furthermore, BEC is not an international standard classification in the same sense as the Harmonized System or SITC.

The relationship between CPC and SITC is similar to that between CPC and the Harmonized System, since SITC also uses the subheadings of the Harmonized System as building blocks to create product groupings that are more suitable for the economic analysis of trade. BEC is related to CPC through its close correlation with SITC and is designed to serve as a means for converting external trade data compiled by use of the SITC into end-use categories that are meaningful within the SNA framework. It is generally possible to rearrange entire CPC subclasses into BEC categories through the correspondences between CPC and SITC, and between SITC and BEC.

EBOPS 2010 (United Nations, European Commission, IMF, OECD and WTO, 2011) is a classification of trade in services that was developed to provide further breakdowns of the BPM 6 classification so as to meet a number of user requirements, including the provision of information required under the General Agreement on Trade in Services. It builds upon the BPM 6 classification of services. In BPM 6, 12 main service categories are identified and broken down into a list of standard and supplementary components. EBOPS 2010 consists of a further breakdown of these components into more detailed sub-items. EBOPS 2010 also includes several supplementary items for the recording of useful additional information regarding services transactions in various sectors such as, travel and tourism or insurance services. Like the BPM 6 services classification, EBOPS 2010 is primarily a product-based classification. Items of these classifications may be described in terms of CPC. Correspondences cannot, however, be established in the areas of travel, construction, and government goods and services, n.i.e., which focus on the mode of consumption of goods and services or the status of the transactor, rather than on the type of product consumed. A detailed correspondence between EBOPS 2010 and CPC Version 2, may be found online at <http://unstats.un.org/unsd/tradeserv/TFSITS/msits2010/annexes.htm>.

(a) Considerations for the compilation of products in the SUTs and IOTs

2.86. Based on the description of the product classifications and their level of detail, it is obvious that products in the SUTs, even when a high level of detail is applied, as is the case with 2,000 products, will nonetheless represent aggregated groups of products when compared with the detail applied in basic statistics, and even more so when compared with the real-world variety of products. Accordingly, analogies to the notion of homogeneous products, which are often assumed in standard economic theory, will in general be inappropriate, as there can be no homogeneous products or production processes at the SUTs or IOTs level of aggregation. Many economies usually consist of hundreds of thousands of producing units, of which virtually no two are completely identical, and there are millions of different products and even more production processes. It is therefore important to realize that national accounts and SUTs record economic transactions, and not physically identifiable products or related technical production processes, which will in general be outside the sphere of official statistics.

2.87. Even very detailed basic statistics already represent highly aggregated data when compared to the number of real-world products. As previously mentioned, the HS contains descriptions of 200,000 products. Statistics on products are collected at a maximum detail of, say, 10,000 products, and only in selected areas such as external trade statistics and output from manufacturing industries. Furthermore, products that are identical in a physical sense may be different in an economic sense when they are sold at different prices to different purchasers. This may, for example, happen because of the way transportation costs are invoiced. The concept of basic prices is defined specifically to include this possibility. Statistics on the breakdown of products for intermediate consumption will often give less detail than production statistics and may sometimes be collected from enterprise units rather than establishment units, and in most cases the statistical coverage of purchases is irregular or limited to certain industries, for example, mainly manufacturing industries, but even in this case the compilers of SUTs may be confronted with the task of further aggregation.

2.88. To gain a better understanding of the level of aggregation, it is useful to consider the product definitions required when selecting items and collecting prices for compiling price indices such as consumer price indices and producer price indices. Each item must be defined more precisely than by just referring to even the most detailed product classification. The same applies when collecting prices for use in the International Comparison Programme. Official statistics have in these cases to make selections from a product universe at a much more detailed level than 10,000 product groups, in order to compile a sound price index. This places the notion of “homogeneity”, as often applied in connection with SUTs and IOTs, in perspective (see 2008 SNA, para. 14.144).

2.89. As a result, the term “homogenous” in the context of the SUTs system usually means “mutually exclusive”. As outlined above, international activity and product classifications aim at mutually exclusive classification criteria. Yet within any group of products fulfilling this criterion, there may be considerable “non-homogeneity” depending on the level of aggregation. The

classification of products in this (mutually exclusive) sense is statistically possible at any level of aggregation but a product in the SUTs will usually represent a basket of products, and the contents of the basket will, furthermore, vary from one cell to another along the rows of both the supply table and the use table. For the classification of producer units into industries the same “mutually exclusive” conditions basically hold, although the situation is somewhat different as the statistically observed input structures will usually represent a mixture of intermediate consumption structures for many individual products, some of which will also be produced in other industries. As a consequence, industries producing mutually exclusive products can only be derived on certain assumptions which do not in general form part of the compilation of the SUTs. Redefinitions (see chapter 5) may be seen as an exception.

4. Other classifications relevant for SUTs and IOTs

2.90. The SUTs system distinguishes a large number of products and industries. Final uses, however, often distinguish only final consumption, gross capital formation and exports at a very aggregate level. The functional classifications help to support the compilation of SUTs and allow for a wide range of other analyses. It is mainly the disaggregated SUTs which allow us to identify the different purposes of expenditure on a product basis.

2.91. The SNA uses special classifications to analyse consumption and other outlays according to the purpose for which the expenditure is undertaken. Such functional classifications and associated detail – the Classifications of Expenditure According to Purpose (United Nations, 2000a) – can be found in chapter 29 of 2008 SNA, on satellite accounts and other extensions. These classifications include, in particular, the classification of the functions of government (COFOG); the classification of individual consumption according to purpose (COICOP), the classification of the purposes of non-profit institutions serving households (COPNI) and the classification of the outlays of producers according to purpose (COPP). The main purpose of these classifications is to provide more detailed statistics for a wide variety of analytical uses. In 2018, the United Nations issued the first revision of COICOP, COICOP 2018 (United Nations, 2018), to reflect users’ need for more detail and several other issues that required a revision of the classification.

2.92. For the SUTs, it is recommended that the lower-level detail be produced in the form of disaggregated matrices as subsystems feeding into the central compilation of SUTs in current prices and in volume terms. As a result, the correspondence between categories of these functional classifications and CPC makes it possible to bring the basic data into the use tables.

2.93. The correspondence table between categories of CPC and COICOP has been established and available on the United Nations Statistics Division classification website at <http://unstats.un.org/unsd/class/default.asp>. When making decisions on the details of the product classification to be applied in SUTs, the possibility of establishing transformation tables to COICOP at group levels or class levels and to make use of the reverse transformation, from the

products surveyed in the household budget surveys to the products of the SUTs, should be taken into consideration. These transformation matrices are keys to the use of the consumer price index (CPI) in the volume estimates, as sub-indices of the CPI will usually be based on the COICOP classification. In addition, for the purposes of household budget surveys and purchasing power parities (PPPs), COICOP is applied at a more detailed level, including as many as 300 or more subclasses.

2.94. Table 2.7 shows the types of links and extensions. Some of the key areas are covered in this section but more detail in terms of compilation is provided in chapter 6.

2.95. COPP provides detailed information on outlays of producers for current production, infrastructure research and development, environmental protection, marketing and human resource development. It should be noted that COPP is included here more for completeness of presentation of the functional classifications. COPP is not widely used and does not fit well in the SUTs framework, as its outlays include wages and other types of costs in addition to intermediate consumption. In principle, COPP applies to all producers, whether market or non-market or for own final use.

Table 2.7 Links between the use table and functional classifications

Use table

		INDUSTRIES				FINAL USE					Total
		Agriculture	..	Other services	Total	Final consumption expenditure			Gross capital formation	Exports	
						Households	NPISH	General government			
Products	Agriculture										
	Other services										
Value added											
Total											

Final consumption expenditure by households (COICOP 2018)

		COICOP					
		Food and non-alcoholic beverages	Alcoholic beverages, tobacco and narcotics	Clothing and footwear	...	Personal care, social protection and miscellaneous goods and services	Total
Products	Agriculture						
	Other services						
Total							

Final consumption expenditure by NPISH (COPNI)

		COPNI					
		Housing	Health	Recreation and culture	...	Services n.e.c.	Total
Products	Agriculture						
	Other services						
Total							

Final consumption expenditure by government (COFOG)

		COFOG					
		General public services	Defence	Public order and safety	...	Social protection	Total
Products	Agriculture						
	Other services						
Total							

5. Statistical units

2.96. In general, the same statistical unit is the basis for compiling the use table and for compiling the supply table. Different choices of statistical units are available for the compiler and it is important to have a clear understanding of the impact of the choice of different statistical units has on the SUTs and on the IOTs.

2.97. Different types of statistical units may be defined (for example, enterprise group, local unit, kind-of-activity unit, and others). For SUTs, however, the focus is on two specific statistical units: enterprises and establishments (local kind-of-activity units).

2.98. An enterprise is defined as the view of an institutional unit as a producer of goods and services (where institutional units are economic entities that have autonomy of decision making and have clear links with the legal units). An establishment is an enterprise or part of an enterprise that is situated in a single location and in which only a single productive activity is carried out or in which the principal productive activity accounts for most of the value added.

2.99. The impact of globalization and the way in which multi-national businesses control and operate their activities pose a number of challenges, including the basis of the statistical unit for measurement of national activity versus global activity. Following the recommendations of the 2008 SNA, however, the establishment is the unit that is more suitable for the analysis of production in which the technology of production plays an important role. The establishment is therefore the recommended unit for the compilation of the production part of the national accounts and therefore the compilation of SUTs. This means, as a rule, that multi-product enterprises must be partitioned into smaller and more uniform units with regard to their kind of production, if possible.

2.100. Trying to collect data on sub-establishment production processes as part of the input-output compilation is an approach that has no natural limitation and that will, apart from the costs, almost invariably become skewed by the specific knowledge and insight that the compilers happen to possess and lead to non-transparent and uneven compilation processes.

2.101. In practice, the extent of partitioning enterprises into establishments varies across countries, depending on whether the creation of establishments is based on a relatively modest breakdown of institutional units or whether, alternatively, the starting point is a register of all local producer units. The latter case follows the formal definitions set out in the 2008 SNA and would lead to a purer activity classification than the former. Recommendations for partitioning vertical and horizontal integrated enterprises are briefly outlined in Box 2.7.

Box 2.7 SNA recommendations on partitioning of vertically and horizontally integrated enterprises

A **horizontally integrated enterprise** is one in which several different kinds of activities that produce different kinds of goods or services for sale on the market are carried out simultaneously, using the same factors of production (2008 SNA, para. 5.21).

Horizontal integration occurs when an activity results in end products with different characteristics. This could theoretically be interpreted as activities carried out simultaneously using the same factors of production. In this case, it will not be possible to separate them statistically into different processes, assign them to different units or generally provide separate data for these activities. Another example would be the production of electricity through a waste-incineration process. The activity of waste disposal and the activity of electricity production cannot be separated in this case.

Within the SNA, a separate establishment should be identified for each different kind of activity wherever possible (2008 SNA, para. 5.22).

A **vertically integrated enterprise** is one in which different stages of production, which are usually carried out by different enterprises, are carried out in succession by different parts of the same enterprise (2008 SNA, para. 5.23). Vertical integration of activities occurs wherever the different stages of production are carried out in succession by the same unit and the output of one process serves as the input to the next process. Examples of common vertical integration include tree-felling and subsequent on-site sawmilling, mining of metal ores and manufacture of basic iron and steel, operation of a clay pit combined with a brickworks or production of synthetic fibres in a textile mill.

While the procedure for the treatment of vertically integrated activities could be applied to any unit, it should be noted that the SNA recommends that, when a vertically integrated enterprise spans two or more sections of ISIC, at least one establishment must be distinguished within each section. With such a treatment, activities of units engaged in vertically integrated activities will not cross section boundaries of ISIC (2008 SNA, para. 5.26). If this approach has not already been followed in basic statistics, the compilers of SUTs will exceptionally have to deal with individual producer units.

6. Valuation in the SUTs

2.102. More than one set of prices may be used to value outputs and inputs depending on how taxes and subsidies on products, and also transport charges and trade margins, are recorded. The 2008 SNA distinguishes three main valuation concepts of the flows of goods and services: basic prices, producers' prices and purchasers' prices.

2.103. The valuation of the data for the use table (for example, intermediate and final consumption) is different from the valuation of the data for the production side of the supply table. In fact, the valuation of use table is based on the actual price paid by the users for the goods and services (i.e., purchasers' price) while the valuation of the production data in the supply table is based on output at basic prices – this in line with the 2008 SNA.

2.104. In order to balance the SUTs, the same valuation should be used. For this purpose, specific matrices have to be compiled for trade and transport margins and taxes and subsidies on products. The compilation of these valuation matrices is an important component of the compilation of SUTs

and IOTs. Chapter 7 provides a detailed description of the compilation steps for the valuation matrices and the compilation issues.

2.105. An overview of the three different valuations – basic prices, producers’ prices and purchasers’ prices – is provided below. They differ as a result of the treatment of taxes on products less subsidies on products, and trade and transport margins.

(a) Basic prices

2.106. Basic prices are the preferred method in the 2008 SNA for valuing output in the accounts. This price basis reflects the amount receivable by the producer from the purchaser for a unit of goods or services, minus any taxes payable, and plus any subsidy receivable on that unit as a consequence of production or sale (for instance, the cost of production).

2.107. The value of output at basic prices reflects the sum of intermediate consumption of goods and services at purchasers’ prices, compensation of employees, return to capital for market producers’ own capital formation, and other taxes less subsidies on production. Other taxes on production include items such as property taxes and business rates, business licences, motor vehicle licenses, mission permits issued by governments under cap-and-trade schemes, and others. Basic prices exclude any transport charges invoiced separately by the producer. When a valuation at basic prices is definitely not feasible, then a proxy as close as possible to basic prices should be used.

2.108. The basic price valuation is the preferred valuation for the construction of IOTs which in turn are used in constructing structural models of the economy or modelling particular features of economic behaviour. When compiling the IOTs, it is therefore necessary also to value the purchases by products at basic prices, a process which is further explained in chapter 7.

(b) Producers’ prices

2.109. Producers’ prices may be thought of as the prices of goods and services “at the factory gate”, so to speak. This valuation includes all taxes on production and taxes on products, for example excise duties. Producers’ prices relate to basic prices as follow:

Producers’ prices	<i>equals</i>	basic prices
	<i>plus</i>	taxes on products (excluding invoiced VAT)
	<i>less</i>	subsidies on products.

2.110. Although the producers’ price valuation is valid and noted in the 2008 SNA, it not recommended for use in the 2008 SNA. At the same time, this valuation still forms the basis for some business survey data. Accordingly, if relevant, specific steps are needed to change data based on business survey to basic prices, as appropriate, for use in national accounts and SUTs.

(c) Purchasers' prices

2.111. Purchasers' prices are those prices payable by the purchaser and include transport costs, trade margins and taxes (unless the taxes are deductible by the purchaser). Purchasers' prices are defined as follows:

Purchasers' prices *equals* producers' prices
 plus any non-deductible VAT or similar tax payable by the purchaser
 plus transport costs paid separately by the purchaser and not included in the producers' price.
 plus trade margins.

2.112. Where taxes and subsidies on products and other taxes and subsidies on production are concerned, some short definitions are provided below:

- Taxes on products include, in particular, value added taxes, taxes and duties on imports, and taxes on products such as stamp taxes on the sale of petrol, diesel, alcoholic beverages and tobacco.
- Subsidies on products include import subsidies and other subsidies on products.
- Other taxes on production consist of all taxes that enterprises incur as a result of engaging in production, independently of the quantity or value of the goods and services produced or sold. These may be payable on the land, fixed assets, business and property rates or labour employed in the production process or on certain activities or transactions.
- Other subsidies on production consist of subsidies which resident producer units may receive as a consequence of engaging in production, including in particular subsidies on payroll or work force, subsidies to reduce pollution and grants for interest relief.

2.113. In the use table, transactions are recorded at purchasers' prices. In the supply table, domestic production is recorded at basic prices and imports by type of product at cost, insurance and freight (CIF) prices. In the SNA and the balance of payments, total imports of goods are valued at free on board (FOB) prices. Further details on these connections and the adjustments required may be found in chapter 5, section D. Accordingly, additional columns are included in the supply table in order to complete the valuation gap between total use and total supply of products. These include information on trade and transport margins, taxes on products and subsidies on products.

(d) Value added tax

2.114. VAT is a wide-ranging tax usually designed to cover most or all goods and services. In some countries, VAT may replace most other forms of taxes on products but it may also be levied in addition to certain other taxes on products, such as excise duties on tobacco, alcoholic beverages

or fuel oils. VAT is a tax on products collected in stages by enterprises. Producers are required to charge certain percentage rates of VAT on the goods or services that they sell. VAT is shown separately on the sellers' invoices so that purchasers know the amounts that they have paid. Producers are generally not required, however, to remit to the government the full amounts of the VAT invoiced to their customers because they are permitted to deduct the VAT that they themselves have paid on goods and services purchased for their own intermediate consumption, resale or gross fixed capital formation.

2.115. Deductible VAT is the VAT payable on purchases of goods or services intended for intermediate consumption, gross fixed capital formation or for resale that producers are permitted to deduct from their own VAT liability to the government in respect of VAT invoiced to their customers. Non-deductible VAT is VAT payable by purchasers that is not deductible from their own VAT liability, if any.

2.116. The SNA requires that the net system of recording VAT should be followed. In the net system, outputs of goods and services are valued excluding invoiced VAT; imports are similarly valued excluding invoiced VAT; and purchases of goods and services are recorded including non-deductible VAT.

(e) Valuation in SUTs and IOTs

2.117. Box 2.8 presents an overview of the valuation in the compilation of SUTs and IOTs in a simplified numerical example. this overview underlines the different valuation layers: the supply table at basic prices including the transformation into purchasers' prices is considered with the use table at purchasers' prices (total supply equals total use). In a second step, valuation matrices are compiled – one for the trade and transport margins and the other for the taxes less subsidies on products – in order to transform the use table from purchasers' prices to basic prices. In this way, the supply table at basic price can be considered in relation to the use table at basic prices (total supply at basic prices equals total use at basic prices). The use table at basic prices is further split between the domestic use table and imports use table at basic prices. The SUTs at basic prices are the starting point for the compilation of IOTs, which are compiled at basic prices.

Box 2.8 Overview of the valuation in SUTs and IOTs

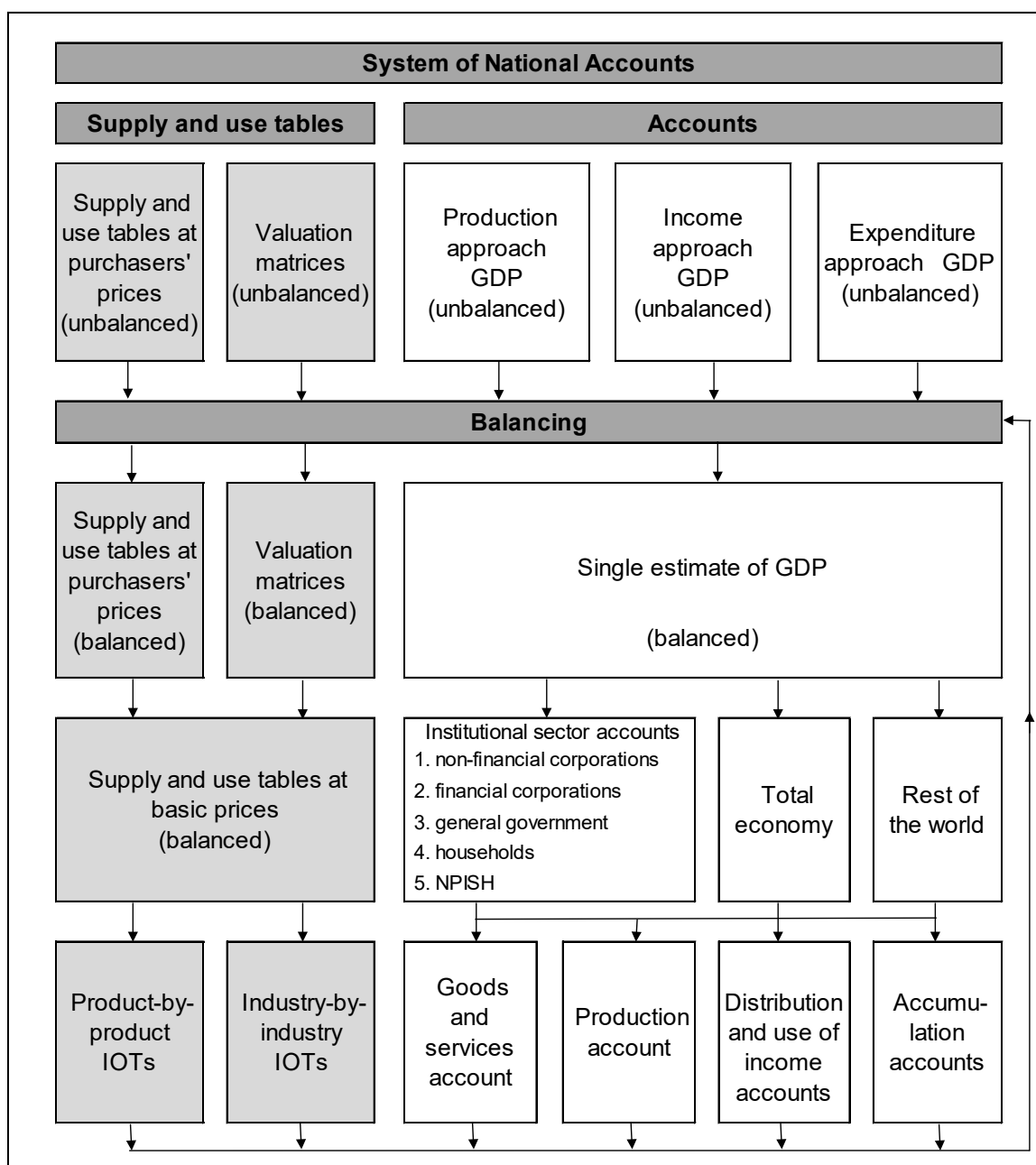
BALANCED SUPPLY AND USE SYSTEM									
Supply table at basic prices including a transformation into purchasers' prices					Use table at purchasers' prices				
	Industries			Output at basic prices	Imports	Supply at basic prices	Trade and transport margins	Taxes less subsidies on	Total supply at purchasers' prices
	Agriculture	Manuf. and constr.	Services						
Products									
Agriculture	270	30	50	350	20	370	28	20	418
Manufacturing	6	380	87	473	42	515	80	21	616
Construction	4	50	13	67	8	75	22	4	101
Trade, transport and communication	10	15	210	235	7	242	- 130	13	125
Finance and business services	6	17	240	263	11	274		15	289
Other services	4	8	100	112	12	124			131
Total	300	500	700	1 500	100	1 600		80	1 680
COMPILATION OF VALUATION TABLES									
Trade and transport margins									
	Industries			Final consumption expenditure	Gross capital formation	Exports	Total		
	Agriculture	Manuf. and constr.	Services						
Products									
Agriculture	3	5	4	13	2	1			26
Manufacturing	16	14	5	33	10	2			80
Construction	3	4	1	5	8	1			22
Trade, transport and communication	- 22	- 23	- 10	- 51	- 20	- 4			- 130
Finance and business services									
Other services									
Total									
Taxes less subsidies on products									
	Industries			Final consumption expenditure	Gross capital formation	Exports	Total		
	Agriculture	Manuf. and constr.	Services						
Products									
Agriculture	1	1	6	11	1				20
Manufacturing	2	2	2	12	2	1			4
Construction			1	2	1				
Trade, transport and communication	1	1	1	9	1				13
Finance and business services			1	12	1				15
Other services									
Total	4	5	12	52	6	1			80
TRANSFORMATION OF SUPPLY AND USE TABLES TO BASIC PRICES									
Supply table at basic prices					Use table at basic prices				
	Industries			Output at basic prices	Imports	Supply at basic prices	Final consumption expenditure	Gross capital formation	Exports
	Agriculture	Manuf. and constr.	Services						
Products									
Agriculture	270	30	50	350	20	370	34	59	93
Manufacturing	6	380	87	473	42	515	97	107	57
Construction	4	50	13	67	8	75	9	12	4
Trade, transport and communication	10	15	210	235	7	242	42	24	11
Finance and business services	6	17	240	263	11	274	14	53	42
Other services	4	8	100	112	12	124	14	35	22
Total	300	500	700	1 500	100	1 600	4	5	12
COMPILATION OF SEPARATE USE TABLES FOR DOMESTIC OUTPUT AND IMPORTS									
Imports use table at basic prices					Domestic use table for domestic output at basic prices				
	Industries			Final cons. exp.	Gross capital form.	Exports	Total		
	Agriculture	Manuf. and constr.	Services						
Products									
Agriculture	4	9	3	1	1	2			20
Manufacturing	12	17	6	2	3	2			42
Construction	4	2	1	1					8
Trade, transport and communication	2	4	1						7
Finance and business services	4	3	2	1	1				11
Other services	4	5	2			1			12
Total	30	40	15	5	5	5			100
Domestic use table for domestic output at basic prices									
	Industries			Final consumption expenditure	Gross capital formation	Exports	Total use at basic prices		
	Agriculture	Manuf. and constr.	Services						
Products									
Agriculture	30	50	90	130	20	30			350
Manufacturing	85	90	51	120	70	57			473
Construction	5	10	3	16	30	3			67
Trade, transport and communication	40	20	10	140	20	5			235
Finance and business services	10	50	40	115	30	18			263
Other services	10	30	20	35	10	7			112
Imports	30	40	15	5	5	5			100
Taxes less subsidies on products	4	5	12	52	6	1			80
GVA	86	205	459	613	191	126			750
Total	300	500	700	613	191	126			

Empty cells by definition

E. Compiling SUTs as an integral part of the national accounts

2.118. As mentioned before, the compilation of SUTs should be seen as an integral part of the compilation of the national accounts. Figure 2.4 provides a general overview of how the compilation of SUTs and IOTs fits within the compilation of national accounts conforming to the same statistical standards (for example, 2008 SNA, BPM 6, 2012 SEEA, IMF Government Finance Statistics, and others), and using the same basic sources generally used for the compilation of national accounts.

2.119. One important feature of the approach outlined in Figure 2.4 is the level at which the traditional annual and quarterly balancing process of the national accounts and balance of payments system takes place. Balanced macroeconomic data can be derived at a more aggregated level by applying the production, income and expenditure approaches. A recommended, better quality option is that the system be balanced at the same time for the institutional sector accounts and the SUTs at a lower-level disaggregation of products and industries. In many countries, the annual and quarterly estimates of GDP are obtained from the production, income and expenditure approaches and reconciled using SUTs. Some countries have a long tradition and much experience in using detailed production data based on establishments (local kind-of-activity units) as the statistical unit for compiling GDP estimates following the production approach.

Figure 2.4 Overview of SUTs and IOTs as part of the SNA compilation

1. Different approaches to measuring GDP

2.120. The three approaches to measuring GDP form the basis of estimating GDP both quarterly and annually. The use of three different methods which, as far as possible, use independent sources of information avoids sole reliance on one source and is conducive to greater confidence in the overall estimation process. This in turn also underpins not only the quality of the key aggregates but also of the underlying details. The SUTs combine the three approaches in a consistent manner.

(a) Production approach

2.121. The production approach looks at the contribution of each economic unit by estimating the value of their output less the value of goods and services used up in the production process to produce their output, this is also known as GVA. Using the production approach:

$$\begin{array}{lcl} \text{GVA at basic prices} & \text{equals} & \text{output at basic prices} \\ & \text{less} & \text{intermediate consumption at purchasers' prices} \end{array}$$

and then,

$$\begin{array}{lcl} \text{GDP} & \text{equals} & \text{GVA at basic prices} \\ & \text{plus} & \text{taxes on products} \\ & \text{less} & \text{subsidies on products} \end{array}$$

GDP is also the balancing item of the production account for the whole economy.

2.122. The distinction between market and non-market producers (see 2008 SNA, para. 6.133, for the definitions) is important for the determinants of both total output and gross value added which is covered in this section. While the output of market producers is determined from the revenue side, the output of non-market producers is calculated as the costs of all inputs including labour cost and consumption of fixed capital. Box 2.9 provides an overview of the calculation of output for market and non-market producers.

2.123. The estimate of output for producing units in the non-market sector is derived by summing their costs, for example, intermediate consumption, compensation of employees, other taxes (less subsidies) on production and consumption of fixed capital. GVA is the sum of compensation of employees, other taxes (less subsidies) on production and consumption of fixed capital.

2.124. The production approach to measuring GDP, and the estimates of GVA, can be implemented by using an industry dimension or by an institutional sector dimension. GVA is the variable used when producing labour productivity estimates and also output per worker uses GVA as the output measure.

Box 2.9 Calculation of output for market and non-market producers

Market producers and producers for own final use		
Total output	<i>equals</i>	total sales of goods and services (as invoiced, excluding VAT)
(at basic prices)	<i>plus</i>	changes in inventories of work-in-progress and finished goods
	<i>plus</i>	output produced for own use, for example research and development (R&D), computer software and construction (also known as own account capital formation) and household production of agriculture products for own use
	<i>less</i>	purchases of goods or services for resale without further processing (thereby only including the gross margin within output)
	<i>plus</i>	income earned in kind
	<i>less</i>	any taxes on products
	<i>plus</i>	any subsidies on products

Total intermediate consumption (at purchasers' prices)	<i>equals</i>	total purchases of goods and services for use as inputs to the production process (excluding employment costs and fixed capital formation)
	<i>less</i>	changes in inventories of materials and fuels
	<i>less</i>	any purchased or bought-in R&D, computer software (treated as capital expenditure, assuming this is included in the purchases in the first place)
	<i>plus</i>	Financial intermediation services indirectly measured (FISIM)
	<i>plus</i>	any imputed insurance premium supplements
	<i>less</i>	any payments to employees such as income earned-in-kind
Gross value added (at basic prices)	<i>equals</i>	total output (at basic prices)
	<i>less</i>	total intermediate consumption (at purchasers' prices)
Non-market producers		
Total output (at basic prices)	<i>equals</i>	total intermediate consumption (at purchasers' prices)
	<i>plus</i>	compensation of employees (labour costs)
	<i>plus</i>	imputed charge for consumption of fixed capital (sometimes called depreciation)
	<i>plus</i>	other taxes on production and imports
	<i>less</i>	other subsidies on production
Gross value added (at basic prices)	<i>equals</i>	compensation of employees (labour costs)
	<i>plus</i>	imputed charge for consumption of fixed capital (depreciation)
	<i>plus</i>	other taxes on production and imports
	<i>less</i>	other subsidies on production
Final consumption expenditure (at purchasers' prices)	<i>equals</i>	total intermediate consumption at purchasers' prices
	<i>plus</i>	gross value added at basic prices
	<i>equals</i>	total output at basic prices
	<i>less</i>	market output
	<i>less</i>	payment for non-market output
	<i>less</i>	output produced for own final use
	<i>equals</i>	non-market output

(b) Income approach

2.125. Using the income approach, GDP is obtained by adding together the income components that make up value added. GDP by income approach covers only the income generated within the domestic economy:

GDP	<i>equals</i>	compensation of employees
	<i>plus</i>	gross operating surplus and gross mixed income
	<i>plus</i>	other taxes less subsidies on production
	<i>plus</i>	taxes on products and imports.
	<i>less</i>	subsidies on products

The above income approach provides estimates of GDP market prices.

2.126. As its name suggests, the income approach adds up all income earned by resident individuals or corporations in the production of goods and services and is therefore the sum of uses in the generation of income account for the total economy (or alternatively the sum of primary incomes distributed by resident producer units).

2.127. The income approach to measuring GDP can be analysed either by industry, by institutional sector or by type of factor income. The type of factor income approach is often linked to the source data and allows for the incorporation of various administrative data sources. These include, for example, generating direct estimates of mixed income (using labour force data and administrative data) and gross trading profit and loss (using company accounts data) as complementary estimates and not as residuals.

2.128. Based on factor incomes, to estimate gross operating surplus the following categories are added:

- Self-employment income (mixed income and quasi-corporations)
- Gross trading profits of private financial corporations
- Gross trading profits of private non-financial corporations
- Gross trading surplus of public corporations (financial and non-financial)
- Rental income
- Non-market consumption of fixed capital

and the following categories are deducted:

- Holding gains and losses on inventories
- Intermediate consumption of financial intermediation services indirectly measured (referred to as FISIM)

2.129. Producing all three dimensions in a single, integrated SUTs framework provides a natural link between the production account and generation of income account, both by industry and by institutional sector. This approach also ensures a high degree of consistency and coherency across the accounts.

2.130. It should be noted that the income approach to measuring GDP cannot be used to calculate chained linked volume measures directly because it is not possible to separate income components into prices and quantities in the same way as for goods and services. However, a chained linked volume measure of the income based total can be obtained indirectly. The expenditure based GDP deflator at market prices (also known as the index of total home costs) can be used to deflate the current market price income based total estimate to provide a chained linked volume measure of the total income component of GDP for balancing purposes.

(c) Expenditure approach

2.131. In the expenditure approach, GDP is obtained by adding the final expenditures or uses by consumers and producers of goods and services produced within the domestic economy. The total is obtained from the sum of final consumption expenditure on goods and services by households,

NPISHs and government, gross capital formation (gross fixed capital formation on tangible and intangible fixed assets, changes in inventories and acquisitions less disposals of valuables) and net exports of goods and services.

2.132. Using the expenditure approach:

GDP	<i>equals</i>	Final consumption expenditure (households, NPISHs and government)
	<i>plus</i>	gross fixed capital formation
	<i>plus</i>	change in inventories
	<i>plus</i>	acquisitions less disposals of valuables
	<i>plus</i>	exports
	<i>less</i>	imports

2.133. The data for these categories are estimated from a wide variety of sources, including business surveys, expenditure surveys, the government's internal accounting system, surveys of traders and the administrative documents used in importing and exporting goods.

2.134. To avoid double counting in this approach, it is important to classify consumption expenditures as either final or intermediate. Final consumption expenditure involves the consumption of goods purchased by or for the ultimate consumer or user. These expenditures are final because the goods are no longer part of the economic flow or being traded in the market place. Intermediate consumption, on the other hand, is consumption of goods and services that are used or consumed in the production process. Gross capital formation is treated separately from intermediate consumption as the goods (or services) involved are not used up within the production process in an accounting period, except for depreciating over time.

2.135. Exports include all sales to non-residents, and exports of both goods and services have to be regarded as final consumption expenditure, since they are final as far as the domestic economy is concerned. Imports of goods and services are deducted because they are not part of the production of the domestic economy but produced in another economy.

2.136. The expenditure approach to measuring GDP is also used to estimate chain-linked volume measures of GDP. The chained-linked volume measure shows the change in GDP after the effects of inflation have been removed.

2.137. Box 2.10 shows a numerical example of how a single estimate of GDP can be derived from a balanced SUTs system by extracting the components of the production, income and expenditure approaches to measuring GDP from the supply table and use table.

Box 2.10 Example of derivation of GDP from balanced SUTs

Supply table at basic prices, including a transformation into purchasers' prices															Millions of euros			
		INDUSTRIES							Imports	Total supply at basic prices	VALUATION							
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	Total			Wholesale trade margins	Retail trade margins	Transport margins	Taxes on products	Subsidies on products	Total	Total supply at purchasers' prices	
		(1)	(2)	(3)	(4)	(5)	(6)	(7)			(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
Products	Agriculture	(1)	8 782					8 782	3 271	12 052	1 052	873	274	386	- 107	2 479	14 532	
	Manufacturing	(2)	796	182 982	643	1 808	133	44	186 405	124 590	310 995	29 777	19 061	2 540	21 041	- 49	72 370	383 364
	Construction	(3)	83	961	43 060	734	255	179	45 272	563	45 835				1 542		1 542	47 377
	Trade	(4)	1	4 773	311	54 204	640	257	60 187	600	60 787	- 31 301	- 21 040		586		- 51 755	9 032
	Transport	(5)	13	465	66	25 538	128	125	26 335	8 150	34 485			- 2 800	628	- 448	- 2 620	31 865
	Communication	(6)	160	1 781	139	43 912	1 253	982	48 228	6 234	54 463	472	1 021	9	3 592	- 34	5 059	59 522
	Finance and business services	(7)	29	8 902	698	7 588	106 909	3 381	127 508	7 061	134 569			- 22	4 865		4 842	139 411
	Other services	(8)	3	85	13	1 053	143	74 346	75 643	824	76 467			85	1 777		1 861	78 329
Total		(9)	9 867	199 950	44 931	134 837	109 461	79 314	578 360	151 293	729 653	0	0	0	34 416	- 638	33 778	763 431
Adjustments	CF/FOB adjustments on imports	(10)							- 97	- 97								- 97
	Direct purchases abroad by residents	(11)							6 675	6 675								6 675
	Total	(12)	9 867	199 950	44 931	134 837	109 461	79 314	578 360	157 871	736 230				34 416	- 638	33 778	770 009
Total of which:																		
	Market output	(13)	9 763	195 916	41 462	127 401	88 330	18 116	480 989		480 989							480 989
	Output for own final use	(14)	104	4 029	3 468	2 134	19 890	2 670	32 295		32 295							32 295
	Non-market output	(15)	0	4	0	5 302	1 241	58 528	65 075		65 075							65 075

Use table at purchasers' prices															Millions of euros		
		INDUSTRIES							FINAL USE							Total use at purchasers' prices	
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	Total	Final consumption expenditure			Gross fixed capital formation	Changes in valuables	Changes in inventories	Exports		Total
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	Households	NPISH	General government	(11)	(12)	(13)	(14)		(15)
Products	Agriculture	(1)	2 583	6 570	16	371	34	49	3 595			180		- 27	1 161	4 909	14 532
	Manufacturing	(2)	2 205	107 190	12 441	16 874	6 015	8 797	71 438		3 180	26 756	2 183	3 034	123 252	229 842	383 364
	Construction	(3)	105	2 440	9 528	2 446	3 907	1 604	20 029	1 667		25 155		- 38	563	27 348	47 377
	Trade	(4)	33	1 883	119	2 240	259	308	4 842			67	45		753	4 189	9 032
	Transport	(5)	14	4 386	267	8 399	822	321	14 208	5 833		3 370			8 453	17 656	31 865
	Communication	(6)	34	2 563	299	9 359	5 919	1 833	20 008	26 444		121	5 976		67	6 905	39 514
	Finance and business services	(7)	457	13 578	4 736	20 359	29 166	9 134	73 470	38 838		1 006	11 170		- 178	11 145	61 981
	Other services	(8)	8	382	59	1 171	415	1 794	3 829	14 923	5 416	53 373	113	107	1	567	74 500
Total at purchasers' prices		(9)	5 440	138 991	27 466	61 219	46 538	23 839	303 492	166 063	5 416	61 050	69 418	2 335	2 859	152 800	459 939
Adjustments	CF/FOB adjustments on exports	(10)														- 97	- 97
	Direct purchases abroad by residents	(11)							6 675								6 675
	Purchases on the domestic territory by non-residents	(12)							- 12 945							12 945	
Total at purchasers' prices		(13)	5 440	138 991	27 466	61 219	46 538	23 839	303 492	159 792	5 416	61 050	69 418	2 335	2 859	165 648	466 517
VALUE ADDED	Compensation of employees	(14)	551	30 679	10 239	37 906	22 997	41 971	144 343								
	Other taxes less subsidies on production	(15)	- 1 627	1 077	546	1 755	2 004	1 103	4 858								
	Consumption of fixed capital	(16)	1 845	12 750	1 542	10 917	18 934	7 480	53 469								
	Net operating surplus	(17)	3 658	16 453	5 138	23 040	18 989	4 921	72 198								
	Gross operating surplus	(18)	5 503	29 203	6 680	33 957	37 923	12 401	125 667								
	GVA	(19)	4 427	60 959	17 465	73 618	62 923	55 475	274 868								
	Total input at basic prices	(20)	9 867	199 950	44 931	134 837	109 461	79 314	578 360								

Table based on 2011 figures from Austria

The box below shows how a single estimate of GDP at market prices can be derived from the above balanced SUTs system by extracting the components of the production, income and expenditure approaches to measuring GDP from either the supply table or the use table.

Calculation of gross domestic product

Production approach			Income approach			Expenditure approach			Millions of euros	
Total output at basic prices	578 360		Compensation of employees	144 343		Final consumption expenditure by Households		159 792		
- Intermediate consumption at purchasers' prices	- 303 492		+ Other taxes less subsidies on production	4 858		+ Final consumption expenditure by NPISH		5 416		
			+ Consumption of fixed capital	53 469		+ Final consumption expenditure by General government		61 050		
			+ Net operating surplus	72 198		+ Gross fixed capital formation		69 418		
= GVA	274 868		= GVA	274 868		+ Changes in valuables		2 335		
						+ Changes in inventories		2 859		
+ Taxes less subsidies on products	33 778		+ Taxes less subsidies on products	33 778		+ Exports		165 648		
						- Imports		- 157 871		
= GDP	308 647		= GDP	308 647		= GDP		308 647		

Table based on 2011 figures from Austria

2. Linking SUTs to the institutional sector accounts

2.138. It is important to link the SUTs to the institutional sector accounts in order to have a complete, consistent and integrated set of accounts, as highlighted in Figure 1.1. The SNA uses two types of units and two ways to subdivide the economy. Both are quite different and serve different analytical purposes. In order to describe production, income, expenditure and financial

flows, and balance sheets, the SNA uses institutional units which, on the basis of their principal functions, behaviour and objectives, are grouped into institutional sectors like non-financial corporations and financial corporations. For the institutional units, the full set of accounts is covered in the system.

2.139. A simplified version of a table covering the main institutional sectors is shown in Table 2.8. Further details may be found in chapter 10, on linking the institutional sector accounts to the SUTs.

Table 2.8 Simplified table linking the SUTs to the institutional sector accounts

INSTITUTIONAL SECTORS	INDUSTRIES				Total
	1	2	...	n	
1. Non-financial corporations					
Total output					
Market output					
Output for own final use					
Non-market output					
Intermediate consumption					
GVA at basic prices					
Compensation of employees					
Other net taxes on production and imports					
Consumption of fixed capital					
Operating surplus, net					
Gross fixed capital formation					
2. Financial corporations					
Total output					
:					
Gross fixed capital formation					
3. General government					
Total output					
:					
Gross fixed capital formation					
4. Households					
Total output					
:					
Gross fixed capital formation					
5. Non-profit institutions serving households					
Total output					
:					
Gross fixed capital formation					
6. Total					
Total output					
:					
Gross fixed capital formation					

2.140. When describing the processes of production (and input-output analyses), the system uses the establishment as the statistical unit and groups it into industries on the basis of its principal activity. For the establishment, only a limited set of accounts is feasible, namely those accounts of the SUTs framework.

2.141. In order to show the relationships between the accounts of the production processes and the accounts of the institutional units, a link table can be compiled as an integrated part of the system. In this link table, a cross-classification of output, intermediate consumption, components

of GVA (and possible other variables of industries) between the industries and the institutional sectors is shown. This link table should help to ensure consistency of data compiled on the basis of establishment and on the basis of institutional units. As both units are classified differently, the link table also provides a picture of the relationships between output, intermediate consumption, GVA, and other variables, originating in the different industries and institutional sectors.

3. Benefits of compiling SUTs as an integral part of the national accounts

2.142. There are a number of advantages of producing SUTs as an integral part of the national accounts and this approach is therefore recommended in this Handbook.

2.143. From a methodological point of view, the following may be identified:

- SUTs provide the ideal framework for integrating the components of the three approaches to measuring GDP both in current prices and in volume terms.
- When statistical discrepancies exist amongst the macroeconomic aggregates, it is less clear where adjustments could be applied. Through their detailed examination of the supply and use of products, however, the SUTs provide a powerful approach to identifying which areas could be adjusted.
- SUTs allow for the data confrontation of different primary sources by bringing them together into a single framework, and facilitate efforts to prioritize how resources could be allocated to seek quality improvements.
- Where statistical information is incomplete or contradictory, as may happen with gross fixed capital formation or household final consumption expenditure, alternative estimates can be made in a transparent way using the SUTs framework, ensuring consistency and coherence.
- SUTs provide a full framework for establishing the connection between the various valuation concepts in national accounts, from basic prices through to purchasers' prices.
- SUTs form the ideal framework for estimating GVA through double deflation and GDP in volume terms, while also ensuring coherence of deflation across the different areas.

2.144. In terms of practical benefits:

- SUTs offer new options to incorporate all existing information, including from primary sources, on a consistent basis. This is also true for information that is only periodically available, as well as a framework for making reliable estimates, including plausible restrictions and identities.
- When SUTs are produced as an integral part of the national accounts, it is relatively easy to compile IOTs. These IOTs derived from SUTs will be fully compatible and consistent with all figures from the national accounts, adding to the credibility and analytical usefulness of both products.

- SUTs that are consistent with the national accounts are normally produced in connection with benchmarked macroeconomic data some two or three years after the initial preliminary results of the national accounts are published. SUTs can also play a vital role in the production of preliminary annual or even quarterly accounts.

2.145. Once the SUTs system is in place on an annual basis, the benefits are significant and can take various forms:

- SUTs from the previous year can be updated with information available for the preliminary year in order to produce a complete set of SUTs (albeit at a more aggregated level) that are consistent with the preliminary figures. This procedure is a good method for revealing inconsistencies in the aggregated preliminary figures at an early stage.
- SUTs can be used to incorporate new information; for example, when new detailed information on total supply and exports is available earlier, then the structure of SUTs of the previous year could be used to project SUTs for domestic output and imports.

Chapter 3. Business processes and production stages

A. Introduction

3.1. The compilation of monetary and physical SUTs and thus IOTs is viewed as part of a statistical production process which starts from the identification of the objectives and users' needs to the dissemination of the tabulations and the evaluation of the production process. The various stages of compilation of SUTs (and IOTs) presented in this Handbook follow those of the GSBPM (United Nations Economic Commission for Europe, 2013). The GSBPM explicitly identifies and organizes the compilation steps and the interdependencies between them in a generic statistical business process. Thus it provides a useful flexible framework by which to describe the compilation process for SUTs and IOTs.

3.2. It should be mentioned that country practices in the compilation of SUTs and IOTs vary considerably, since they are specific to the particular context in which they take place. For example, they depend on the specific institutional arrangements of the statistical system, the statistical legal framework, the legal, political, regional and taxation arrangements, the statistical units, the business registers, the range of processes, publication schedules, revision policies, resources, data availability, confidentiality, and also the final outputs. Despite the great variability in country practices, there are common steps in the compilation of SUTs and IOTs. In the description of the compilation stages of the GSBPM in this Handbook, these common steps that are flexible and applicable to all countries are identified.

3.3. There is an overarching framework within which the statistical production process takes place and this should be taken into consideration in the design of the compilation process and also in the actual compilation of SUTs and IOTs. This includes the statistical institutional arrangement in the country and the data and metadata quality framework.

3.4. The objective of this chapter is to provide an overview of the compilation steps for SUTs and IOTs. Section B presents an overview of the different institutional set-ups in various countries. Section C outlines the GSBPM compilation stages that relate to SUTs and IOTs and, lastly, section D provides a schematic summary of the compilation steps and their links with the relevant chapters of the Handbook and a summary of the main recommendations, principles and guidelines for the compilation of SUTs, IOTs, PSUTs and EE-IOTs that are covered by this Handbook. Annex A to chapter 3 provides examples of institutional arrangements for the compilation of economic statistics in countries.

B. Institutional arrangements

3.5. The institutional arrangements are generally understood as a set of agreements between the agencies involved regarding the division of responsibilities in the collection, processing, compilation and dissemination of data. They are fundamental to an effective statistical system and essential for the management of an integrated economic statistics programme. The functions and responsibilities of the lead statistical agency in the country can be carried out more efficiently if it is supported in its role by institutional arrangements such as advisory committees, relationship meetings, memorandums of understanding, service-level agreements, technical cooperation and a legal framework that protects the confidentiality and integrity of the data while allowing for the sharing of data between partner statistical agencies (United Nations, 2013, para. 3.23).

3.6. Apart from the legal framework and other factors, institutional arrangements depend on the kind of national statistical system that exists in a given country, namely, whether it is centralized or decentralized. A national statistical service is considered centralized if the management and operations of the statistical programmes are predominantly the responsibility of a single autonomous government agency, and decentralized if the statistical programmes are managed and operated under the authority of several government departments. Under this arrangement, a particular agency is usually entrusted with the responsibility of coordinating the statistical activities of the various departments.

3.7. In economic statistics, countries have different institutional arrangements under which such bodies as, for example, the national statistical office and the central bank have different roles and responsibilities. Countries often follow a decentralized approach, under which the collection of economic statistics is split across different institutions within the country, so that, for example, the national accounts (non-financial accounts) are compiled by the national statistics office, the balance of payments and financial accounts are compiled by the central bank, and the government finance statistics covering the public sector are compiled by the finance ministry.

3.8. When countries are considering either building or redesigning their systems or changing the roles and responsibilities of the various institutions involved, the undertaking should be approached with the aim of producing integrated economic accounts throughout the entire statistical production process. The motivation for integrated economic statistics comes from the benefits that such data sets provide for coordinated national and global policy initiatives in an increasingly interconnected world. The integration of economic statistics involves the use of common concepts, definitions, estimation methods and data sources for statistical reconciliation, helping to improve the coherence and consistency of a wide range of economic statistics and to reduce the respondent burden and overall costs. Integration therefore is not specific either to the type of statistical system (centralized versus decentralized) or to the level of development of the statistical system. This approach has the following prerequisites:

- Adoption of the conceptual framework of the SNA and the SEEA as the umbrella framework for organizing economic statistics
- Alignment of the interdependencies of the components of the statistical production process (statistical units, classifications and others)
- Establishment of enabling institutional arrangements for statistical integration (United Nations, 2013, para. 2.5)

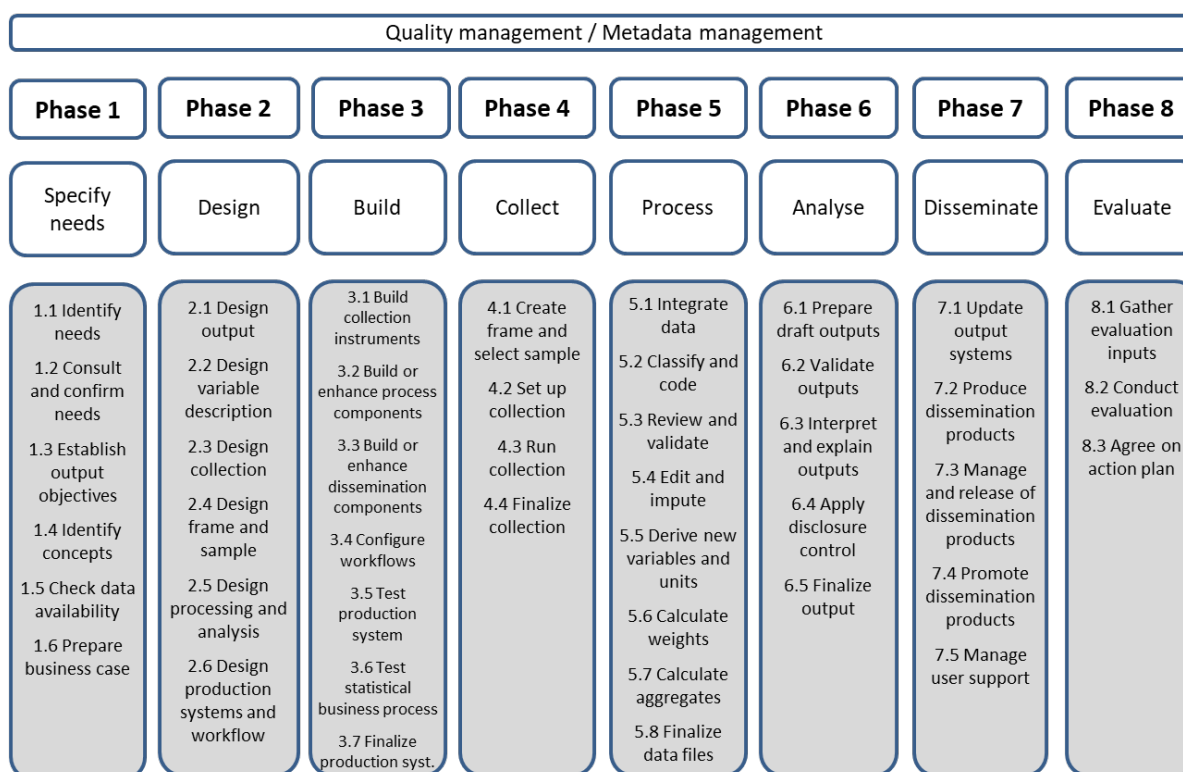
3.9. Examples of institutional arrangements in different countries may be found in the annex to this chapter. In general, it could be said that, beyond being conducive to the coherence and consistency of official economic statistics, a centralized arrangement may provide more comparability and harmonization both within the statistical system and with the statistical system of other countries. Although the transition to an integrated system may incur large investment costs, it would generate great benefits in terms of improved quality and reduced costs in both the short term and the long term.

3.10. The roles and responsibilities of the various institutions in countries evolve over time and aspects of the historical evolution of these arrangements are reflected in the country examples covered in this chapter. One such example is that of Finland, where, in 2014, the compilation of the balance of payments was transferred from the country's central bank to its statistics office. Finland now follows the practice of other countries such as Denmark, Ireland, Luxembourg, Malta, Norway and the United Kingdom, where the balance of payments is compiled alongside the national accounts within the statistical office and not in the central bank.

C. Overview of the Generic Statistical Business Process Model

3.11. The GSBPM describes, and defines, the set of business processes needed to produce official statistics. It provides a standard framework and harmonized terminology to help statistical organizations to modernize their statistical production processes, and to share methods and components. The GSBPM can also be used for integrating data and metadata standards, as a template for process documentation, for harmonizing statistical computing infrastructures, and to provide a framework for process quality assessment and improvement. The GSBPM is a reference model that can be used in a flexible manner to describe, document, organize and communicate the statistical production process in question.

3.12. The GSBPM consists of a sequence of eight phases: (1) Specify needs; (2) Design; (3) Build; (4) Collect; (5) Process; (6) Analyse; (7) Disseminate; and (8) Evaluate. An overview of the phases, together with the sub-elements of each phase, may be seen in Figure 3.1.

Figure 3.1 Phases of the GSBPM

3.13. The GSBPM is not a rigid framework in which all steps must be followed in a strict order; rather, it helps to identify the possible steps in the statistical business process, and the interdependencies between them. Although presentation of the GSBPM follows the logical sequence of steps in most statistical business processes (for example, business surveys), in different circumstances the elements of the model may occur in a different order. In addition, in compiling SUTs and IOTs, some sub processes will be revisited a number of times, forming iterative loops, in particular within the “Process” and “Analyse” phases.

3.14. This section focuses on the business processes in national accounts, in particular, the compilation of SUTs, PSUTs and IOTs. The business process and stages of production covered in this chapter therefore reflect the application of the underlying GSBPM. As a result, Figure 3.2 provides an overview of a simplified business processing model specific for the compilation of SUTs, PSUTs and IOTs.

3.15. In the compilation of SUTs and IOTs, the sequential stages in the compilation of the GSBPM may be summarized as follows, and as presented in Figure 3.2:

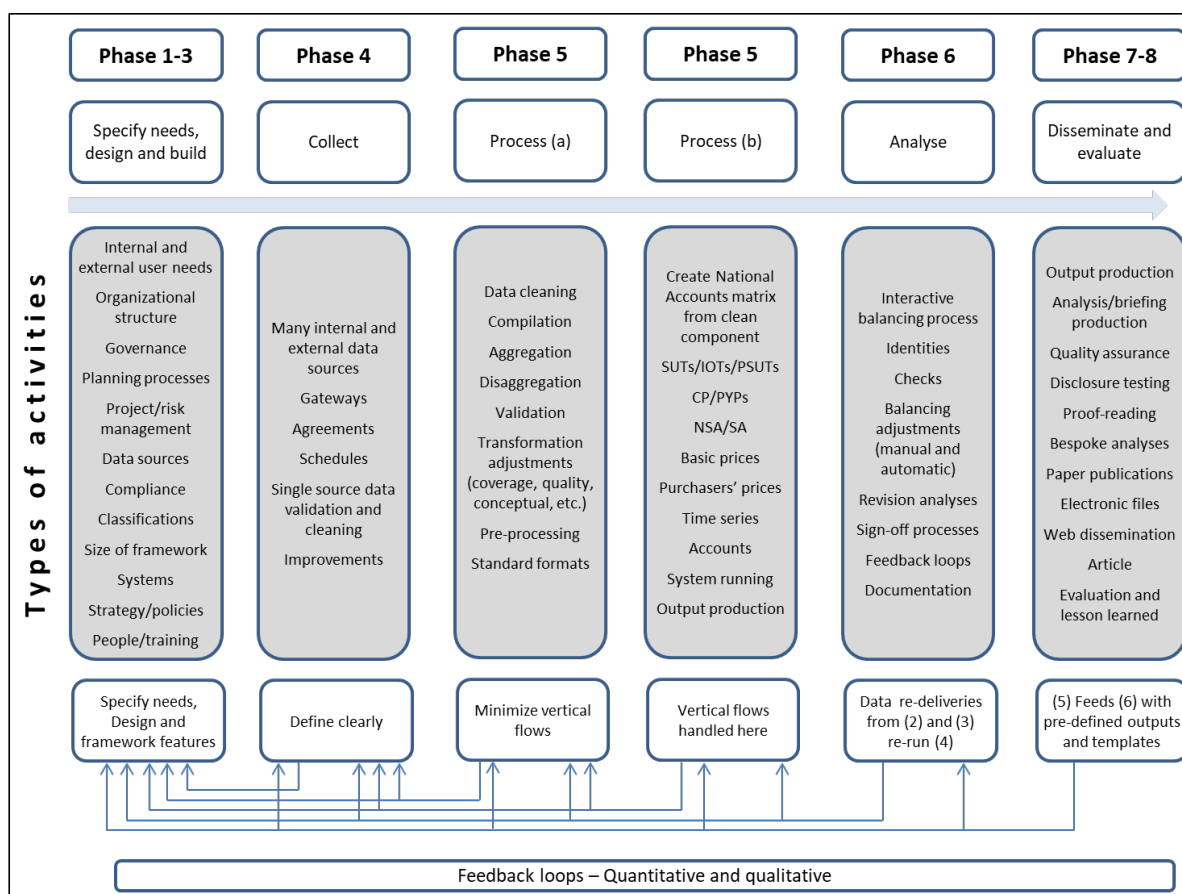
- Phases 1–3: Specify needs, design and build. This stage includes tasks related to the phases: 1 – “Specify needs”, 2 – “Design”, and 3 – “Build” of the GSBPM set out in Figure 3.1. It

covers all the pre-collection activities of setting up the system, including identifying users' needs, designing the system, determining the size of the SUTs and IOTs, and other tasks.

- Phase 4: Collect. This relates to the activities of data gathering from various sources. In general, the compilers of SUTs (and IOTs) rely on data already collected for the purposes of national accounts which have often already been adjusted to fit into the national accounting framework.
- Phase 5: Process. This stage corresponds to a number of activities related to the data cleaning, adjustments and transformation that are needed in order to start putting the data into an unbalanced SUT. This stage is very important in the compilation of SUTs and IOTs and is therefore separated into two steps in Figure 3.2. The first corresponds to all the activities necessary to put the data in the initial unbalanced SUTs. This involves data cleaning, pre-processing, aggregation and disaggregation of the basic data and any other adjustment to the basic data to fit into the national accounts concepts of the SUTs. The second step in this phase corresponds to all the activities of setting up an initial (unbalanced) set of SUTs at purchasers' and basic prices and in current prices and volume terms.
- Phase 6: Analyse. This stage corresponds mainly to the activities of balancing (manual and automated) SUTs and IOTs and the feedback loop to the source data to resolve inconsistencies. As a result, there is a continuous loop between this and the previous phase, making it possible to achieve balanced SUTs and IOTs. In this stage, the final output of the compilation process is prepared, validated and finalized.
- Phases 7 and 8: Disseminate and evaluate. This stage refers to the activities related to the dissemination of the output tables, which include the preparation of printed publications, press releases and websites, the promotion of dissemination products and other tasks, as well as the activities related to the evaluation of the production process and also of the output in the light of internal or external feedback.

3.16. The grey boxes for each stage in Figure 3.2 include examples of the types of functions undertaken. They are listed in no particular order of importance and are linked with one another.

3.17. The broad approach is to move and process data from left to right, with minimal backward loops, even though effective feedback loops are critical at each phase, and the incorporation of new, or improved, data deliveries are unavoidable. Good data version control at each stage is needed, enabling the generation of a wide-range of outputs, articles and analyses such as revision analysis.

Figure 3.2 Simplified business processing model for compiling SUTs, IOTs, and PSUTs

3.18. Each phase should be viewed as cumulative, even when allowing for the iterative nature of the balancing process. The incorporation of balancing adjustments should be viewed as a cumulative step and not as creating a loop.

3.19. It is important to prepare proper documentation throughout the various compilation stages and in particular during the stage of balancing and adjustment. The steps and links between the source data through to the balanced data should be recorded and documented separately and reviewed in subsequent balancing exercises to investigate source data incoherence, bias and other factors. For example, moving from the original source data (such as business survey data, administrative-based data, company accounts-based data, and other types of data) to the validated 2008 SNA data, a number of adjustments may need to be made in such areas as the following:

- Coverage (including exhaustiveness) adjustments
- Conceptual adjustments
- Quality adjustments

- Balancing and coherence adjustments

3.20. In general, for all the stages of the compilation process, it is important also to have in place the following:

- Data version control – for example, data storage, conventions allowing easy access and revision analyses, and also pre and post automated balancing analyses
- Clear controls and disciplines – for example, read and write access for each stage, setting out operational standards, change controls and testing, and other measures
- Appropriate staffing – for example, ensuring that all staff are trained and skilled to undertake the different functions and ensuring that sufficient staff are in place for each phase
- Clear organizational structure of the staff involved – for example, clear roles and responsibilities of staff, as staff members can have more than one role in more than one phase

D. Overall strategy for the compilation of SUTs and IOTs

3.21. Within the stages of the overall production process presented in the previous section, the “Process” and “Analyse” phases (5 and 6) have a particular structure in the compilation of SUTs and IOTs. This section provides an overview of the steps that are generally undertaken to construct SUTs and IOTs after the data have been gathered. In addition, since the compilation of SUTs and IOTs is not seen as a one-time exercise but as part of a continuous programme, this section also provides the strategy for compiling SUTs and IOTs in current prices and in previous years’ prices for the first year of compilation and the subsequent years.

3.22. The first step in compiling SUTs and IOTs is to populate the various separate parts of the supply table and use table (as shown in Figure 3.3) with the available data. This leads to the construction of unbalanced SUTs which are then subjected to a balancing process to reconcile all the entries.

3.23. The steps that are generally used by countries to construct an unbalanced version of the SUTs are presented below:

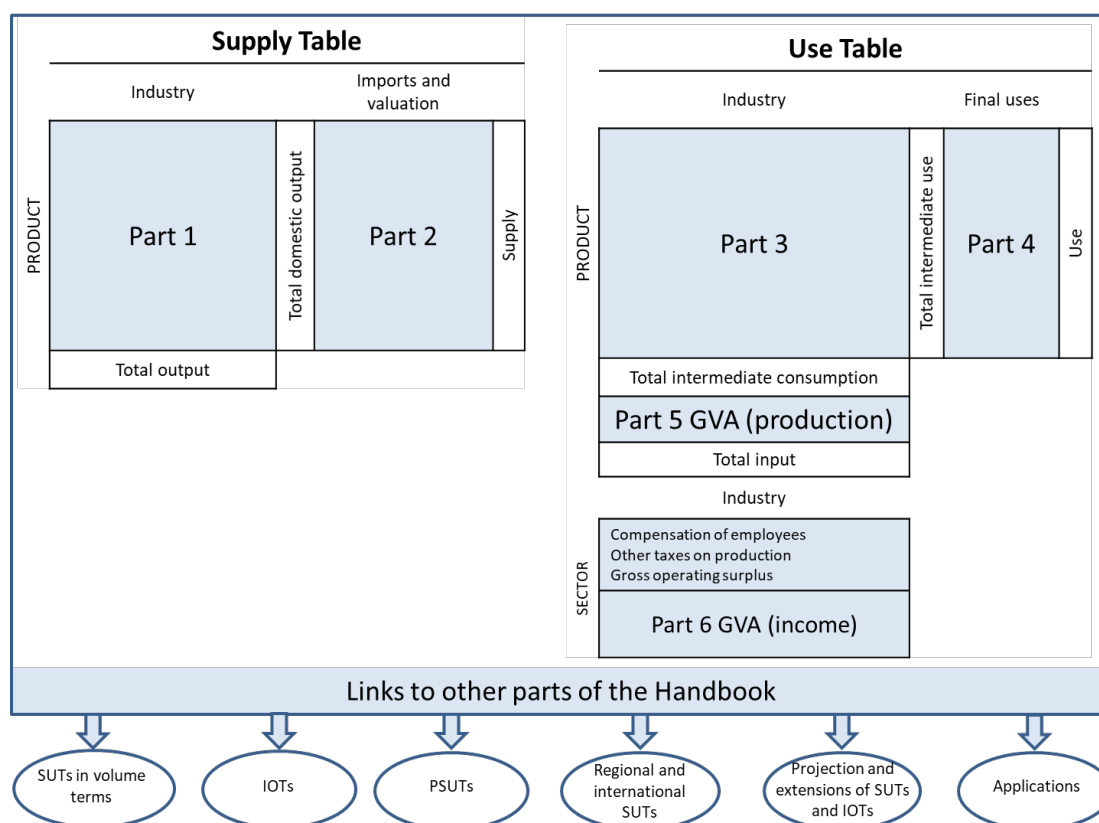
- *Step 1 – construction of the supply table:* This consists of filling the available data into an initial unbalanced supply table, which covers domestic output by product (part 1 in Figure 3.3) and the imports of goods and services and the valuation matrices comprising information on taxes less subsidies on products, trade margins and transport margins (part 2 in Figure 3.3). These valuation matrices allow the transformation of total supply of products at basic prices (formed by summing the domestic output and the imports) to total supply of products at purchasers’ prices. The construction of this initial unbalanced supply table is presented in chapter 5 of this Handbook.

- *Step 2 – construction of the use table:* In a manner similar to step 1, this step consists of filling the use table with the available data, which cover: the intermediate consumption at purchasers' prices (showing the input requirements of goods and services for the production of the domestic output of each industry shown in the supply table – part 3 in Figure 3.3); final uses at purchasers' prices and for each category, such as final consumption and gross fixed capital formation, for which separate compilation steps will be needed (part 4 in Figure 3.3); and production-based GVA at basic prices shown by industry (part 5 in Figure 3.3). This compilation step is covered in chapter 6 of this Handbook.
- *Step 3 – compilation of the valuation matrices:* These matrices are essential to the preparation of SUTs at basic prices. They expand the valuation columns in part 2 of the supply table in Figure 3.3 into corresponding matrices for intermediate consumption and final consumption of the use table. This compilation step is described in chapter 7 of this Handbook.
- *Step 4 – compilation of the imports use table and domestic use table at basic prices:* This step is essential to increasing the analytical uses of SUTs by distinguishing the use of imported and domestic products. This compilation step is presented in chapter 8 of this Handbook.
- *Step 5 – compilation of the SUTs in volume terms (previous years' prices):* When balanced both in current prices and in volume terms, the SUTs ensure coherent and consistent deflation of the components of the production and expenditure approaches to measuring GDP as well as coherent and consistent estimates of price and volume indices. This requires that SUTs are compiled in volume term at this stage of the compilation process. The compilation of SUTs in volume terms is described in chapter 9 of this Handbook.
- *Step 6 – Linking SUTs with the institutional sector accounts:* Linking SUTs and the institutional sector accounts is an important step in the compilation of SUTs, ensuring the full integration and consistency of SUTs with the national accounts. This link is provided by compiling a linking table between the sectors and industries (part 6 in Figure 3.3). The compilation of the linking tables is presented in chapter 10 of this Handbook.

3.24. These six steps above are generally followed in that order; however, there is a significant level of interdependency in the compilation process. For example, trade and transport margins and taxes less subsidies on products are necessary for the transformation of the use table from purchasers' prices to basic prices and also for conversion of the supply of products at basic prices to purchasers' prices in the supply table, to enable the balancing of products at purchasers' prices. This information may partly be derived from estimates based on the use tables and linked to estimates from the supply table at basic prices.

3.25. On the other hand, estimates of certain final uses may require basic supply side information if, for example, the product flow⁷ method is being applied. Nevertheless, allowing for interdependencies in the compilation of these tables, it is vital that the tables are viewed and accepted as primary estimates.

Figure 3.3 Structure of the SUTs and the links covered in this Handbook



3.26. Once these six steps are completed, the result is unbalanced SUTs at purchasers' prices and basic prices. This represents the start of a balancing procedure which is an iterative procedure integrating the following aspects:

- Balancing of SUTs at purchasers' prices
- Compilation of valuation matrices
- Transformation of SUTs into basic prices

⁷ Following the terminology used in 2008 SNA (para. 14.2), in this Handbook the expressions "product balance" and "product flow" methods are used in preference to "commodity balance" and "commodity flow method", as reflecting the more recent usage of the word "product" in place of "commodity". It is noted, however, that the change in terminology does not indicate a change in methodology.

- Compilation of separate use tables for use of domestic output and use of imports of goods and services
- Balancing of SUTs at purchasers' prices and at basic prices
- Balancing the production-based GVA and income-based GVA, providing the link to the institutional sector accounts

All of the above should be balanced with time series in mind to ensure consistent movements of levels and growth rates.

3.27. One of the key reasons why the SUTs are balanced first at purchasers' prices is to reflect as closely as possible the basis of the survey data that is being fed into the use table. The intermediate uses and final uses, for example, are collected close to the economic reality of the prices paid by purchasers, in other words, the purchasers' prices. In addition, no valuation issues exist with such variables as compensation of employees and other taxes less subsidies on production.

3.28. These aspects should, however, be viewed alongside the domestic output part of the supply table reflecting data collected from producers whereby the output is valued at basic prices. Thus a balance between the two is needed.

3.29. For the SUTs balanced at purchasers' prices, the two key identities are:

- Total supply of products at purchasers' prices equals total uses of products at purchasers' prices.
- Total output of industries at basic prices equals total input of industries at basic prices.

3.30. Balancing is not just necessary in order to achieve the above identities but also makes it possible to trace inconsistencies of basic data and estimation methods. Ideally, the balancing of the SUTs system should be done both in current prices and in volume terms simultaneously. In fact, balancing in this manner means that the process is not complete until the transformation into basic prices and the separation of the use of domestically produced products from the use of imported goods and services have been achieved, as these are key steps in producing the SUTs in volume terms. These steps are in practice interrelated and provide a powerful feedback loop in terms of quality and validity of the various component estimates.

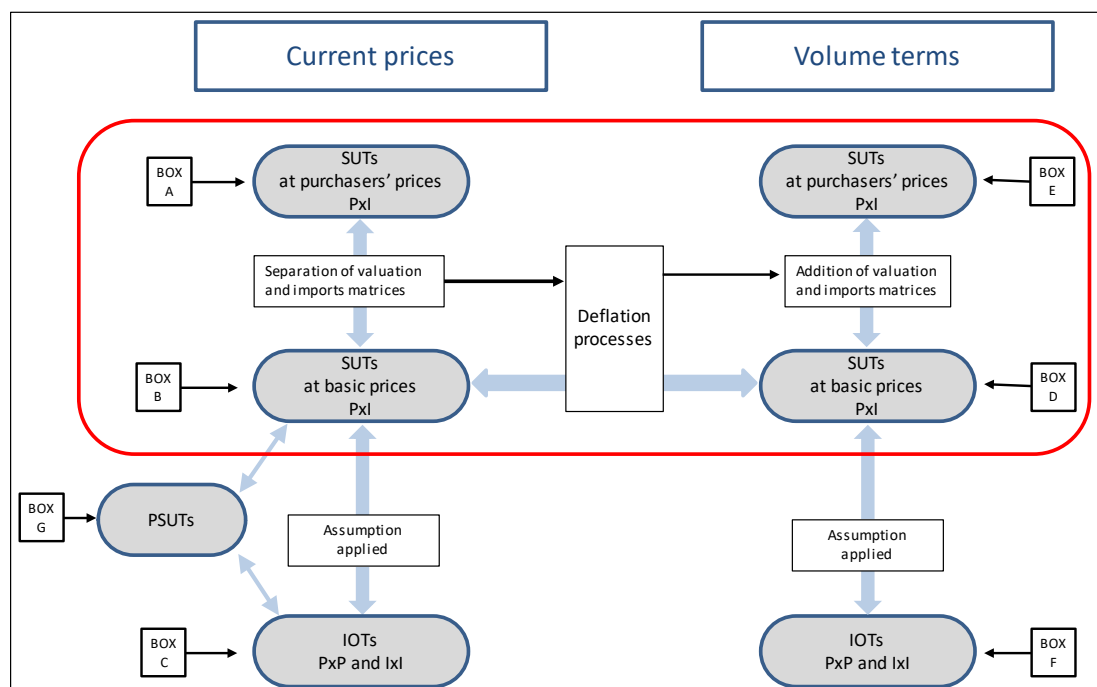
1. Compilation of SUTs in current prices and volume terms

3.31. The SUTs framework in Figure 3.3, when treated in summary form, can be combined with the H-Approach to show a simplified version of the compilation schematic configuration when the SUTs are compiled in current prices and in volume terms. Figure 3.4 illustrates the sequence of steps involved in the compilation of SUTs, PSUTs, and IOTs. The inner box outlined in red focuses on the compilation of SUTs. Thus, countries that intend to compile only monetary SUTs can focus on the steps within the red box and follow the compilation sequence indicated by the arrows in the

figure. The order of compilation of PSUTs and IOTs in the figure does not imply a compilation sequence for these tabulations. The compilation of these tables reflects the country's priority.

3.32. The simplified illustration provided in Figure 3.4 of the compilation of SUTs, PSUTs and IOTs can be seen in relation, on the one hand, to phase 4, "Process", when it comes to compiling unbalanced SUTs and PSUTs and, on the other hand, to phase 5, "Analyse", when it comes to compiling balanced SUTs, PSUTs, and IOTs.

Figure 3.4 Compilation of SUTs and IOTs in current prices and in volume terms



PxP: product-by-product

PxI: product-by-industry

IxI: industry-by-industry

3.33. In compiling the seven boxes, in the sequence from box A to box G, a further dimension of their evolution needs to be reflected. In year 1 of the compilation process, boxes A, B and C representing current prices are produced in that sequence covering the economy for year (T) for SUTs and IOTs, and box G covering PSUTs which are linked to the outputs of boxes B and C.

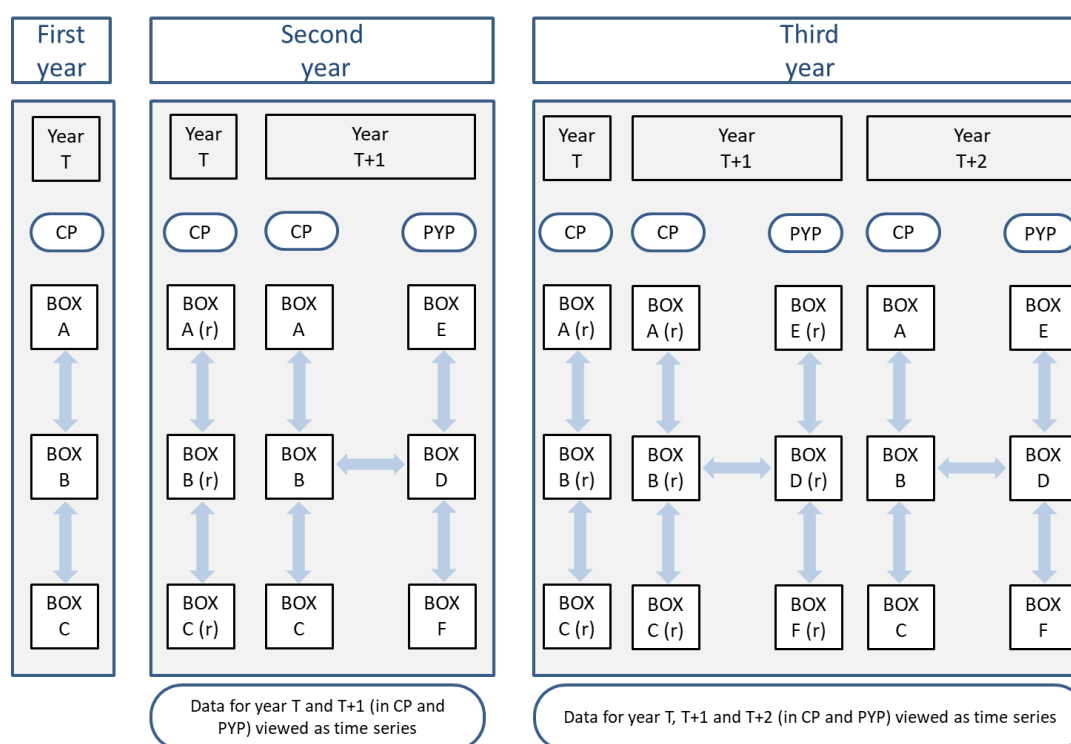
3.34. As mentioned in chapter 2, SUTs in volume terms for one period can be compiled using SUTs in current prices for one period and deflators. The preferred approach, however, includes a time-series dimension and boxes D, E and F representing the previous years' prices should not be compiled in year 1 as there are no SUTs in current prices for the previous year (T-1).

3.35. It is therefore essential to have two consecutive years of SUTs in current prices to enable the production of the first year of SUTs in previous years' prices. If the SUTs are produced less frequently, say, every five years, it is much more difficult to produce SUTs in volume terms.

3.36. In year 2, boxes A, B and C are produced covering the economy for year (T+1) together with any revisions to the data for boxes A, B and C for the year (T). In addition, the first set of SUTs in previous years' prices can be compiled for year (T+1). In each year thereafter, the process will extend the availability of SUTs by an extra year in current prices and previous years' prices, while also incorporating any revisions to SUTs for earlier periods to ensure consistent time series.

3.37. Figure 3.5 provides a summary of the evolution dimension for the first three years. As time passes, different challenges will evolve, such as the need to retain an ever-increasing number of years of SUTs on a consistent basis, the need for a revisions policy, data version and vintage control, managing the production of consistent levels and growth rates, the organizational arrangement of resources which may not increase each year, among others. It is thus important to ensure that this process is properly planned and managed from the start.

Figure 3.5 Evolution of compiling SUTs and IOTs in the first three years



3.38. Based on the overall strategy for the compilation of SUTs and IOTs, it is possible to provide step-by-step guidance. This is provided for the first year of compilation, and then subsequent years

of compilation, as there are some additional steps that need to be considered in order to ensure a fully consistent time series of SUTs in current prices and in previous years' prices.

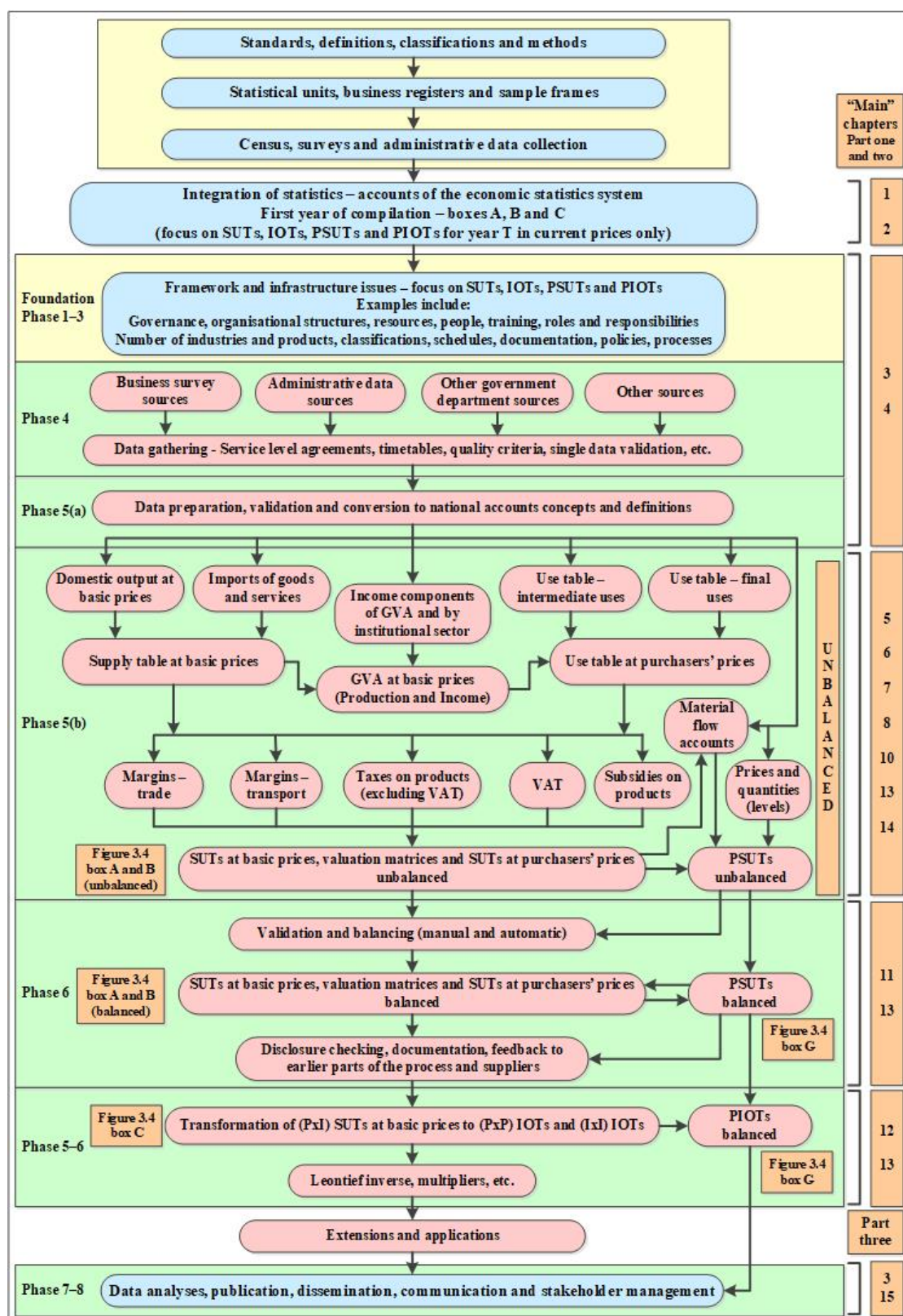
(a) Step-by-step summary for the first year of compilation

3.39. Figure 3.6 provides a description of the various phases in the compilation of SUTs and IOTs for year 1 of the compilation, together with the reference to the relevant chapters of the Handbook. The figure also contains references to boxes A–G of Figure 3.4 and the compilation stages of Figure 3.2. Although boxes A, B and C form key outputs, there are various intermediate stages and intermediate outputs as denoted in the separate stages in Figure 3.6.

3.40. Before starting the first year of compilation, as illustrated in Figure 3.6, compilers should ensure that the overall framework is well in place, comprising: standards, definitions, classifications and methods, statistical units, business registers and sample frames, census, survey and administrative data collection.

3.41. The following key features of Figure 3.6 should be noted:

- It is consistent with Figure 3.5 and covers the simplified business process model for compiling SUTs and IOTs. Each block of work in Figure 3.6 is split according to the type of work as indicated in the six different stages in Figure 3.2.
- It follows the underlying principles and features of the GSBPM.
- The flow of work is kept as logical and sequential as possible and follows the H-Approach as covered earlier. As mentioned earlier in this chapter, however, the compilation of SUTs and IOTs includes several interrelated processes and dependencies which must be reflected and retained. Furthermore, in some cases, there is more than one approach available, for compiling trade margins using a supply-side approach or use-side approach or both.
- The flows in Figure 3.6 do not present backward loops, although effective feedback loops are critical at each phase and improve the process. For example, the compilation and balancing of PSUTs provides an important feedback loop to the compilation of monetary SUTs, thus enhancing the quality of physical and monetary SUTs.
- Integrated links bring the PSUTs together with such input data as the prices and quantities (levels) and material flow accounts alongside the SUTs and IOTs.
- Each phase of work is also linked to the main chapters in parts A and B of this Handbook, providing much more detail on the compilation – these links provide the key chapters but not all references.
- The same approach has not been applied to part C (Extensions and applications) of the Handbook as there are many variations and options.

Figure 3.6 First year of compilation


(b) Step-by-step summary for the subsequent years of compilation

3.42. Having completed the year 1 results, Figure 3.7 provides a detailed stage of production for year 2 with the focus on the SUTs in previous years' prices, which, as mentioned, can only be compiled when SUTs have been compiled for the current year and the previous year.

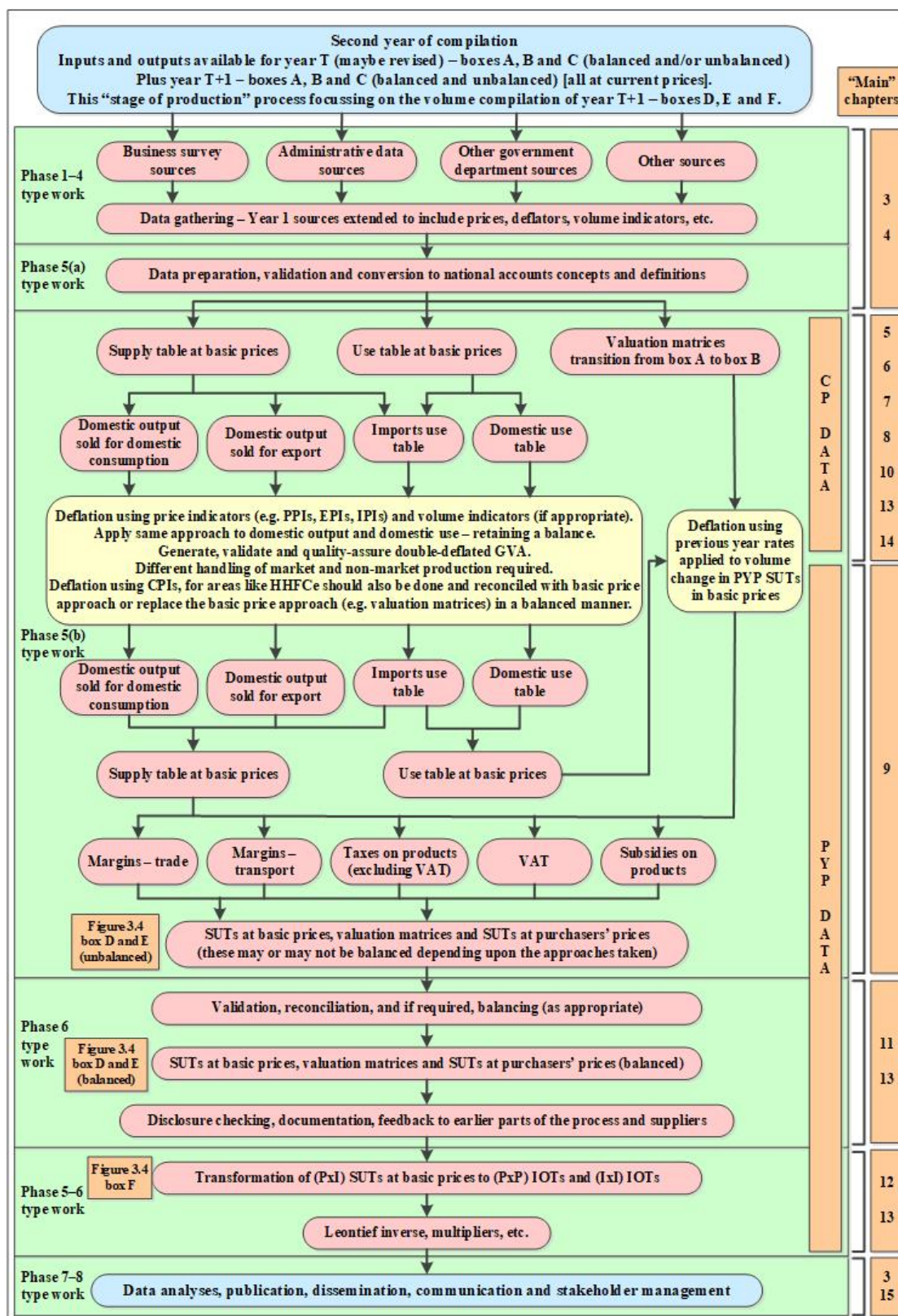
3.43. In compiling SUTs in previous years' prices for the first time, there may be several years of SUTs in current prices already available. If so, then the compilation is just an extended version of the process indicated in Figure 3.7, and it is better in terms of quality and consistency as there is a time series dimension in place immediately.

3.44. Other features to note in Figure 3.7 are the following:

- The focus of the steps in Figure 3.7 is the right-hand side of the H-Approach of Figure 3.4 and builds on the detail available from the left-hand side – assuming the left-hand side products are available.
- The deflation approach follows the underlying H-Approach and is covered in chapter 9, on compiling supply and use tables in volume terms.
- Other approaches are available but this is the recommended approach.
- The compilation of IOTs in volume terms is not essential but is not resource-intensive if all the other parts are available.

3.45. An additional feature is also achieved whereby GVA in volume terms is arrived at using the SNA recommended approach, namely, double deflation. However, the results from this approach need additional quality assurance against other indicators. This is to ensure that the quality of the GVA estimate in volume terms (and, in turn, GDP) is not reduced as a consequence of the errors in either the current price estimates of output and intermediate consumption or inappropriate deflation of these two variables.

Figure 3.7 Second year of compilation



2. Summary of the main recommendations, principles and guidelines of the Handbook

3.46. Box 3.1 provides a list of the main recommendations, principles and guidelines relevant for the compilation of SUTs, IOTs, PSUTs (and EE-IOTs) and related products presented in this Handbook, covering various aspects such as the organizational or institutional environment, compilation, data strategy and requirements, and balancing.

3.47. The recommendations and guidelines presented in box 3.1 may be viewed as aspirational as they provide the model scenario for the compilation and dissemination of SUTs, IOTs and related products. Countries can gradually implement the recommendations and guidelines in accordance with their specific situation in terms of such factors as data availability, resource constraints, legal framework and others, and in line with their priorities.

Box 3.1 Examples of the main recommendations, principles and guidelines provided in this Handbook

A. Organizational or institutional environment

1. The organization of the economic statistics system should follow an integrated economic statistics approach. The use of the GSBPM to organize the statistical production process would facilitate the compilation of SUTs, IOTs and related products.
2. National accounts should have very close links with all its suppliers, in particular, the business register, business surveys and administrative sources.
3. The compilation of the various components of the SNA framework should be coordinated and integrated in terms of production processes, such as production schedules, feedback loops, coherence, and other features:
 - National accounts (including balance of payments and monetary financial statistics, government finance statistics)
 - SUTs and IOTs together with PSUTs and EE-IOTs
 - Environmental-economic accounts to be closely linked with the compilation of SUTs
 - Regional accounts
 - Prices
 - Labour market statistics
4. The compilation of SUTs and IOTs should be performed as part of the regular compilation of the national accounts and within the core national accounts. This would have the following effects:
 - Leads to better quality, coherence and consistency of national accounts, balance of payments and related statistics
 - Creates effective and powerful data quality and coherence feedback loops, which in turn help to address structural issues and biases and to prioritize resources to targeted improvements
5. The final estimates of the national accounts aggregates should be derived from the balanced SUTs framework and not the other way around. For example, the SUTs based estimates should be not confined to predetermined estimates or already published estimates.

6. The compilation of SUTs and IOTs should reflect stakeholder interests. This can be achieved by organizing regular meetings with data suppliers and users together with other regular stakeholders.

7. Appropriate internal governance should be exercised to ensure accountability and guidance, supported by programme, project and process management, including risk management, under a framework reflecting:

- Schedules, timetables and customer-supplier service-level agreements set in place to ensure a regular supply of source data, briefings and evaluation reviews
- Various standards and policies, such as revision policy, confidentiality and disclosure controls, and others
- Staff recruitment, retention and skill development

8. Skill development needs to take into consideration the following types of training requirements:

- National accounts – Technical skill focus covering national accounts concepts, methods, processes and guidance, and also such functions as developments, compilation, coordination, balancing, analyses and dissemination
- Systems – IT systems, programming, data management (standards and principles), data dissemination, also covering website management, and other systems, including the role of dedicated IT professionals supporting economic statistics
- Management – staff management, effective leadership, communication, and other aspects

9. For effective and sustainable production of SUTs and IOTs, it is important to have sufficient computing capacity in place that includes:

- Robust, reliant, structured, quick and well-documented systems
- Database software and hardware, speed, structure, flexibility, statistical functionality, data management and links to web dissemination

10. It is important that the statistical production process is well documented and kept up to date, reflecting:

- Operational, methodological, system, metadata and recording-specific issues, adjustments, etc. for each quarterly or annual exercise

11. The compilation of SUTs and IOTs is to be performed with due consideration for the costs and resources available and also other criteria such as data availability, data quality and time.

B. Compilation

1. SUTs (and IOTs) should be compiled annually and, if possible, on a quarterly basis, following the H-Approach for the production of SUTs and IOTs in current prices and in previous years' prices (including valuation and imports matrices). The application of the H-Approach makes it possible for the volume of GVA to be estimated using a double deflation method and also ensures greater coherence, linking SUTs to various other parts of the SNA framework.

2. SUTs should be produced first, then IOTs derived from the SUTs, using additional information and assumptions.

3. Rectangular SUTs should be compiled with more products than those provided for by industries:

- The greater the detail, better the quality – while more detail will increase the burden on business, systems and resources, it can improve the quality of balancing
- Improved matching between prices and values, thereby ensuring better quality of the data in volume terms
- Compilation (and balancing) should be undertaken at the greatest level of detail available – time, quality and resources permitting. Due to confidentiality-type criteria, however, the level of publication may – or will – be aggregated to a higher level

4. Standard international statistical classifications (such as ISIC, CPC, COICOP, and others) should be used at appropriate detailed levels to ensure international comparability. Within these classifications, greater granularity may be desired for specific economies.
5. Consistent statistical units should be used throughout the process, from the business register and business surveys through to the SUTs.
6. Ideally, SUTs and PSUTs (and EE-IOTs, as appropriate) should be based on sound and complete data sources, reflecting:
 - Common concepts, definitions and classifications
 - Comprehensive and up-to-date statistical business register
 - Wide range of (preferably annual) regular business surveys (including structural detail), household surveys, administrative data, prices, and other sources
 - Benchmarking and reconciliation – preferably conducted annually, reflecting rapidly changing economies (minimizing the use of fixed factor or stability assumptions)
 - Incorporation of labour and capital information ensuring improved coherence for productivity estimates
 - Appropriate choice of index number formulae and base year
7. All the data building blocks should be recorded separately, namely, source data, coverage adjustments (including exhaustiveness), conceptual adjustments, quality adjustments, balancing adjustments, and others
8. A table should be compiled linking the SUTs and the institutional sector accounts, including:
 - Goods and services
 - Production accounts by industry and by institutional sector
 - Generation of income accounts by industry and by institutional sector
 - Parts of the use of disposable income account (such as household final consumption expenditure) and parts of the capital account – by industry and by institutional sector (such as gross capital formation and its components)
9. Use of the bottom-up approach should be preferred in the compilation of regional SUTs, which should be reconciled with national SUTs.
10. For the derivation of IOTs, the following should be the methods most frequently used:
 - Model A (product-by-product) IOTs using the product technology assumption
 - Model D (industry-by-industry) IOTs using the fixed product sales structure assumption
 - Hybrid – mix of technologies usually chosen to avoid having any negatives
11. Comprehensive documentation should be prepared on operational methods and methodology, including appropriate metadata and revision analysis.
12. Efforts should be made to keep up to date with, and contribute to, internationally evolving and agreed changes to concepts, methods and systems developments.

C. Data strategy and requirements

1. SUTs are data-hungry and a range of timely, comprehensive, consistent and coherent data sources are needed. The data strategy should reflect a range of aspects.
2. These should include data-handling aspects such as:
 - Data collection (for example, questionnaire design, electronic data capture, receipt of all the data that a company can provide, etc.)
 - Data processing, data editing, metadata and data warehousing

- Data quality frameworks
- Data dissemination and use of SDMX standards

3. The following structural and data-collection issues should be reflected:

- Comprehensive and up-to-date statistical business register used as the sampling frame for all business surveys
- Use of as many data sources as possible, censuses, business and household surveys, administrative data, company accounts, regulatory accounts, company websites, and other sources
- An international business unit handling all aspects of multinational enterprise groups, from profiling the business structures to data collection and data reconciliation and feeding coherent data through to the various statistical domains. In addition, the need to develop links and share data with other national statistics offices and national central banks for statistical purposes only.
- Frequency of information – monthly, quarterly, annually or five-yearly. The more regular, the better the information reflects rapidly changing industry structures of sales and inputs, changing patterns of household consumption, impact of globalization on trade flows, and other factors
- Sufficient, appropriate and relevant, price indices matching the current price values for deflation or use of suitable volume only indicators where price information may be unavailable
- Strategy for handling, and reviewing, areas where data may be missing

4. The following more general needs should be included:

- Need to minimize the burden on business
- Need to have confidentiality and disclosure testing processes

D. Balancing

1. SUTs should be balanced in current prices and in volume terms, thus leading to:

- A single estimate of GDP incorporating the components of production, income and expenditure approaches to measuring GDP
- Volume estimates of GVA through double deflation
- Balance between supply of products and use of products and between industry inputs and industry outputs

2. The balancing process should simultaneously encompass:

- SUTs at basic prices and at purchasers' prices
- SUTs in current prices and in volume terms (preferably, previous years' prices)
- SUTs links to IOTs, PSUTs and EE-IOTs (as appropriate)
- Link with the institutional sector accounts

3. Balancing should strongly promote integration of the following:

- Goods and services, production account, generation of income account, parts of the capital account and use of disposable income account
- Incorporation of PSUTs and EE-IOTs (as appropriate)
- Productivity estimates (labour, capital and multifactor)

4. Simultaneous balancing should be preferred to sequential balancing. If this is not possible, sequential balancing (first in current prices, then in volume terms), with quick and effective feedback loops, should be considered as an alternative.

5. The organization of the balancing function can be set up in different ways across teams. A centralized balancing approach should, however, be preferred to the decentralized balancing arrangement whereby the balancing of the various elements related to SUTs and IOTs (such as current and constant prices for a single year and for a time series, links with

productivity, regional accounts, among others) is carried out at the same time and within the same unit in order to ensure the full consistency of all SUTs-related products.

6. The production and balancing of SUTs should enable the identification of source data incoherence. Mechanisms should be developed to provide feedback to data suppliers and help prioritize areas for improvement and allocation of resources.

7. An annual review and evaluation of the balancing adjustments should be carried out, to identify and address any evolving biases.

Annex A to chapter 3: Examples of institutional arrangements in countries

A3.1 This annex presents examples of institutional arrangements in selected countries. The examples cover the cases of centralized and decentralized statistical systems.

A. Centralized production of economic statistics: Canada

A3.2 As a centralized national statistics office, Statistics Canada is accorded the legal mandate to collect and disseminate a broad range of statistics by a federal act (the Statistics Act). Provisions in the Act also serve to protect data confidentiality and assure political neutrality and an arms-length relationship with policymakers.

A3.3 Users are regularly consulted, and the office, through various channels, ensures that priority requirements are established and met. These channels include national advisory committees, federal-provincial consultations and regular bilateral meetings with key policy partners such as the federal finance department and the Central Bank.

A3.4 Statistics Canada produces a full suite of macroeconomic accounts, including:

- National accounts (including financial and wealth accounts)
- Balance of payments
- Government finance statistics
- Productivity measures
- Environmental accounts (natural resource stocks, along with physical flows of energy use, greenhouse gas emissions and water use)
- Selected satellite accounts covering tourism, culture and pensions

A3.5 The compilation processes are integrated to assure data coherence across components of the Canadian macroeconomic accounts, and regional SUTs serve as the integrating mechanism for the production dimensions. The integration is achieved through annual benchmarking and reconciliation processes with current price measures of GDP income and expenditure, real GDP by industry and labour and multifactor productivity. Data coherence is a requirement for key policy applications, such as the input to fiscal formulas for the sharing of sales tax revenues among the Federal Government and provincial jurisdictions or to formulas for equalizing fiscal capacity among Canadian provinces.

A3.6 Economic surveys, along with labour market data, price statistics and international trade statistics, are produced within Statistics Canada, ensuring the alignment of priority-setting for feeder programmes to the macroeconomic accounts.

A3.7 All business surveys are linked to a central business register maintained through regular updates from administrative files. The survey content is harmonized, as are approaches to collection, processing and estimation within an integrated business statistics programme framework. The use of administrative data is optimized throughout all stages of the process, and continuous access to required files is assured through formalized arrangements with data providers, such as the Canada Revenue Agency and national and provincial regulatory authorities.

A3.8 In recent years, Statistics Canada has made significant progress towards implementing consistent classification standards across all feeder programmes, thereby facilitating the compilation of SUTs. The North American Industry Classification System (NAICS) serves as the basis of industry statistics and North American Product Classification (NAPCS) the basis of statistics on products. Continuing efforts are being made to ensure compliance and to coordinate input from the macroeconomic accounts and feeder programme areas into the development of updated standards.

B. Centralized production of economic statistics: Norway

A3.9 Statistics Norway has overall responsibility for official statistics in Norway, and also for the conduct of extensive research and analysis activities. Statistics Norway reports to the Ministry of Finance, with the support of the Statistics Act of 1989. Statistics Norway is a professional, autonomous organization with the mandate to determine what it publishes, and when and how the publishing shall take place.

A3.10 Statistics Norway is responsible for the production and maintenance of the business register, along with the business surveys using samples drawn from this register.

A3.11 The Department of National Accounts and Industry Statistics comprises nine divisions, with the following responsibilities:

- National accounts
- Primary industry statistics
- Manufacturing and research and development statistics
- Construction and service statistics
- Transport, tourism and ICT statistics
- Energy statistics
- Natural resources and environmental statistics

- Accounting statistics
- Business register

A3.12 The division for national accounts is responsible for the following:

- Quarterly and annual national accounts (including SUTs, IOTs and regional accounts)
- Quarterly and annual non-financial accounts
- Quarterly balance of payments

The balance of payments has been an integral part of the national accounts since the 1950s. Satellite accounts are also prepared by the division for national accounts but not on a regular basis.

A3.13 The Department of Prices, Financial and External Trade Statistics comprises six divisions, with the following responsibilities:

- Financial market statistics (including financial accounts)
- Public finance statistics
- Financial corporations
- External trade statistics (in liaison with the Customs Department)
- Price statistics
- Development cooperation

Banking statistics were originally under the responsibility of Statistics Norway and were then moved to Bank of Norway before being moved back to Statistics Norway.

C. Centralized production of economic statistics: United Kingdom

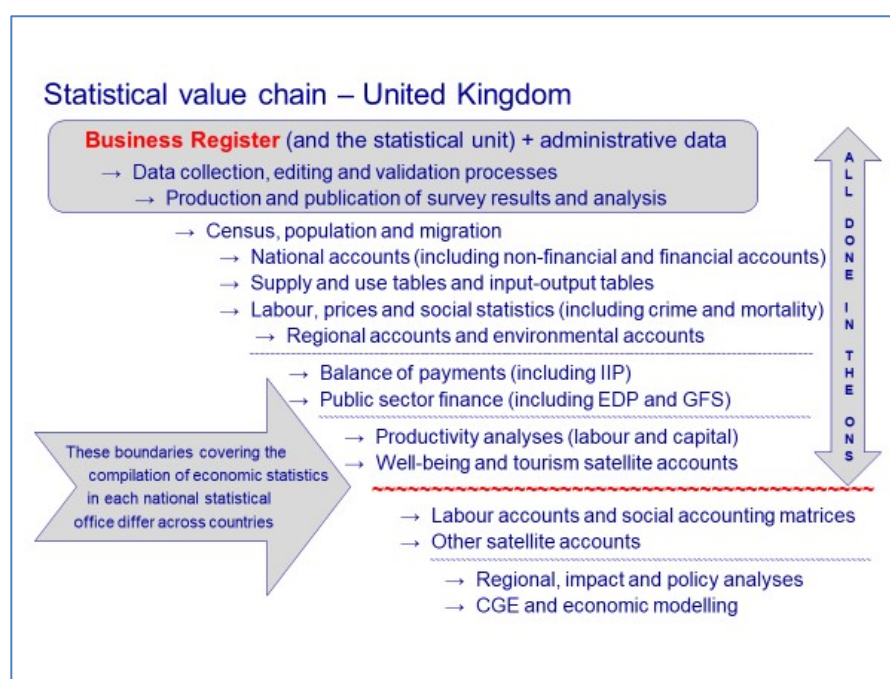
A3.14 The structure of the United Kingdom statistical system has evolved over many decades, helped by several reorganizations of statistical departments and changes in legislation, consolidating the responsibility for almost all economic statistics under the Office for National Statistics and the Government Statistical Service. The United Kingdom system continues to evolve, for example, by developing better links and access to administrative data.

A3.15 Currently, the United Kingdom has in place resources, systems and processes for producing detailed, integrated and timely quarterly and annual economic accounts. The Office for National Statistics, as an independent statistical body with a central role, is wholly responsible for the compilation of the national accounts, balance of payments, public sector finance statistics, labour market statistics and price statistics. The compilation of SUTs is central to the annual national accounts system. The Office also produces regional accounts, environmental accounts and IOTs (Mahajan, 2016).

A3.16 The Office for National Statistics is one of the few national statistics offices with such centralized responsibility and coverage of economic statistics – this has only been the case since the late 1980s. Furthermore, since 2011, all the above economic statistics are being produced at the same location.

A3.17 The independent status of the Office for National Statistics is supported by national legislation, pursuant to which it reports to the United Kingdom Statistics Authority. The Statistics Authority, which was established on 1 April 2008 under the Statistics and Registration Service Act 2007, is a non-ministerial department overseen by Parliament and not by a government minister.

Figure 3A.1: Integrated process of compiling national accounts and balance of payments United Kingdom



Compiled by Sanjiv Mahajan March 2014

D. Decentralized production of economic statistics: Chile

A3.18 Economic statistics in Chile are mainly produced by three institutions:

- Central Bank of Chile, which is responsible for the compilation of most of the macroeconomics statistics, namely, national accounts (non-financial and financial accounts), balance of payments and international investment position
- Finance Ministry, which produces the government finance statistics
- National Statistics Office, which undertakes the data collection covering economic and business surveys and the compilation of price indicators, labour market indicators and socio-demographic data

A3.19 The organizational structure described above allows the Bank to achieve a high level of consistency between external statistics and national accounts in terms of both the statistics themselves and the methodology.

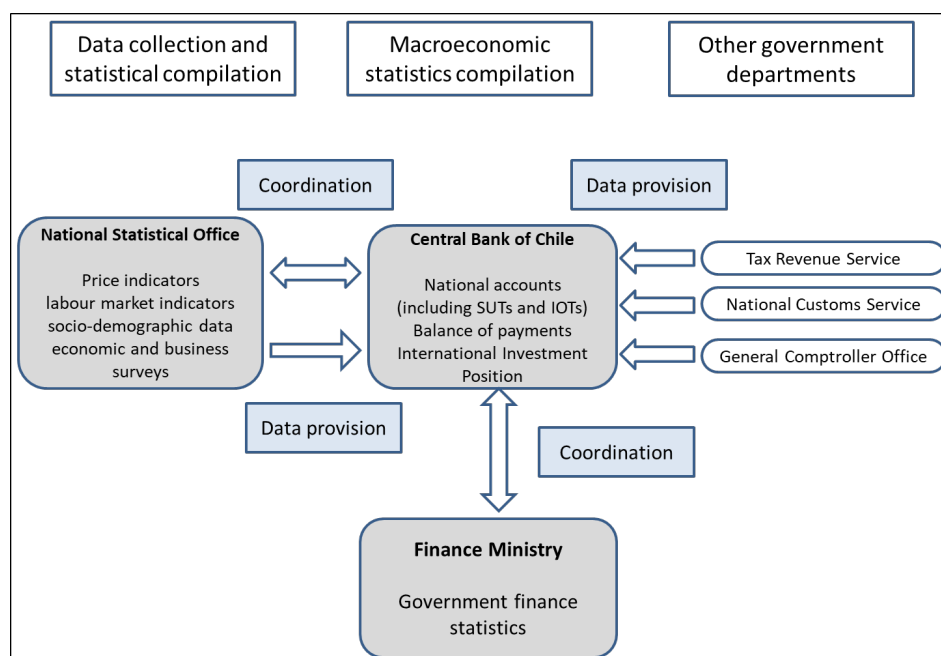
A3.20 In the production of macroeconomics statistics, the Bank uses a significant amount of data, provided mainly by the Tax Revenue Service, the National Customs Service, the General Comptroller's Office and the National Statistics Office, the latter being the main provider of statistics for national accounts compilation. Dependency on the statistics from the National Statistics Office entails a high degree of coordination between both institutions. To this end, a framework agreement is in place to ensure that the requirements and conditions for the provision of statistical products are met. In addition, a committee with members from both institutions regularly meets to coordinate issues related to data collection and the specific needs of national accounts.

A3.21 There are strong links between the Central Bank of Chile and the National Statistics Office, buttressed by a continuous programme to improve the cooperation and the quality of the links and the data flows between the customer and supplier.

A3.22 Other salient features of the Chilean system include the following:

- The SUTs and IOTs are compiled within the national accounts in the Central Bank of Chile.
- Where the balance of payments is concerned, the Central Bank of Chile collects the data on international trade in services to supplement the data on the international trade in goods collected and provided by the National Customs Service.
- The Central Bank of Chile also produces regional GDP figures on an annual basis.
- The national statistics office produces a business register which, in turn, is employed by the Central Bank of Chile after making improvements and modifications.
- Although the environmental accounts are not produced for Chile, various efforts have been undertaken by the Minister of the Environment to produce a range of environmental indicators.

A3.23 The diagram below shows the components of the statistical system in Chile.

Figure 3A.2 Statistical system in Chile

E. Decentralized production of economic statistics: United States of America

A3.24 The United States of America has a highly decentralized statistical system, under which responsibility for producing a substantial portion of official federal economic statistics is divided among 13 agencies that have statistical work as their principal mission.

A3.25 There are also numerous other entities that are considered part of the statistical system in the United States but statistical work is not their principal mission. Most of the country's primary economic indicators are produced by one of three main federal statistical agencies, while the United States Census Bureau conducts economic censuses and surveys. The three main agencies and their responsibilities are the following:

- **Bureau of Economic Analysis:** this body relies primarily on data generated by other agencies to compile the national accounts (non-financial accounts) and the balance of payments.
- **Federal Reserve Board (the United States central bank):** this compiles the financial accounts and government finance statistics.
- **Bureau of Labor Statistics:** this body prepares the labour market statistics and price statistics.

A3.26 The Bureau of Economic Analysis also undertakes a number of business surveys. At the same time, however, most of the statistics used by the Bureau in preparing GDP and input-output

accounts come from non-Bureau sources, including other statistical offices. The Census Bureau provides most of the other expenditure components of GDP and output and intermediate purchases within the input-output framework. For the period 1997–2017, the Bureau of Economic Analysis produced SUTs at basic prices with a transformation to purchasers' prices, make/use tables at purchasers' prices for benchmark years and at producers' prices annually, together with IOTs. Annual make/use tables at producers' prices at a more aggregated level of detail are also available for the period 1947–1996.

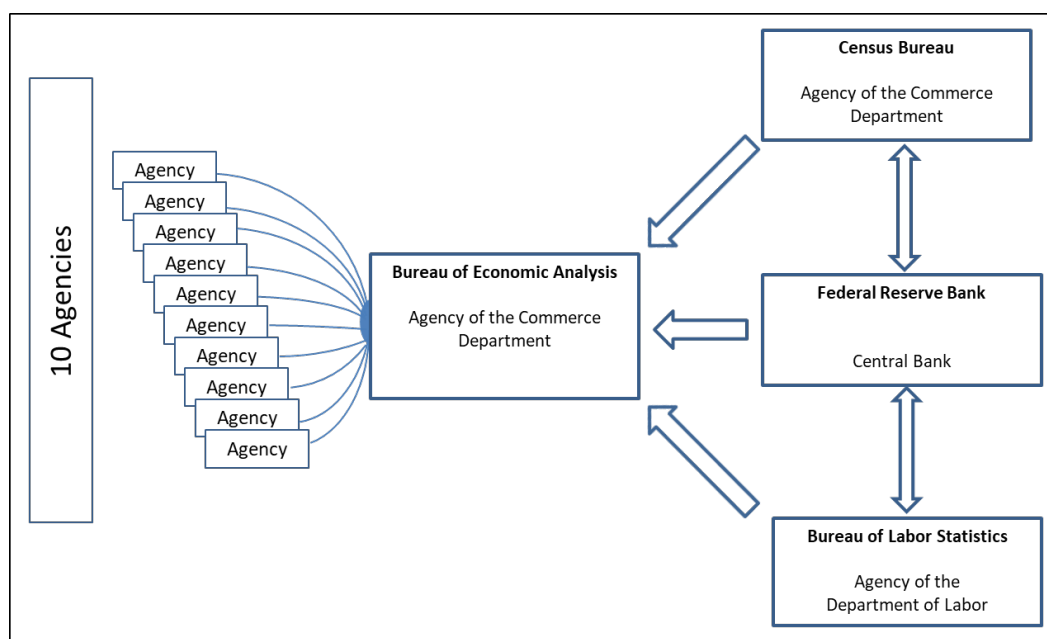
A3.27 The agencies each produce and maintain their own business register, often created using different sources, as detailed below:

- The Census Bureau's business establishment list is compiled mainly from federal tax forms and used as the primary sampling frame for the five-year economic censuses and many of the economic surveys.
- The Bureau of Labor Statistics business establishment list is based on information collected in connection with the joint federal and state unemployment insurance programme and used by Bureau establishment surveys, including the producer price index (PPI) survey.

Figure 3A.3 Overview of the products produced by the main agencies

Bureau of Labor Statistics Agency of the Department of Labor	Bureau of Economic Analysis Agency of the Commerce Department	Census Bureau Agency of the Commerce Department	Federal Reserve Bank Central Bank
Consumer price index Employment projections Producer price index Employment rates State and local Employment Consumers' expenditures Productivity	Balance of payments International investment position GDP and GDP by industry GDP by state and metropolitan area Corporate profits Input-Output tables Personal income and outlays Personal consumption expenditures Regional I-O modelling system State and local employment Business surveys (e.g. FDI)	Economic census – 5 yearly Decennial census – 10 yearly American community survey Median income Population International trade in goods Monthly, quarterly and annual economic indicators	Financial accounts Consumer credit and other household finance statistics Industrial production and capacity utilization Money stock measures Interest rates Bank assets and liabilities Bank structure data Business finance statistics Exchange rates and international data

Figure 3A.4 Overview of the statistical system structure



A3.28 Data sharing between agencies, as practised in the United States, may significantly improve data harmonization and eventually lead to savings. In recent years, the Bureau of Economic Analysis has signed memorandums of understanding with other agencies and ministries to facilitate the exchange of data, including confidential data.

A3.29 Although there may be differences in the concepts of statistics employed by different agencies and in their statistical coverage (for example, productivity statistics are published in the United States by the Bureau of Labor Statistics and then used by the Bureau of Economic Analysis in its measurement of national accounts), the confrontation of the data themselves or of the data processing steps raises the data validation to another level and enhances the quality of statistics.

Chapter 4. Specify needs, design, build and collect stage

A. Introduction

4.1. The objective of this chapter is to describe the activities specific to the compilation of SUTs and IOTs which take place during the phases of the GSBPM relating to the specification of needs, designing, building and collecting, referred to together as the “specify needs, design, build and collect stage”. The chapter will also explore specific elements that need to be considered when compiling SUTs and IOTs. These phases of the GSBPM is of particular importance in the compilation process because of its direct implications for all the remaining phases and their elements need to be continuously reviewed and corresponding adjustments made to the process.

4.2. The following section of the chapter focuses on the “specify needs”, “design” and “build” phases of this stage. The level of detail of the industry and products in the tables must be carefully evaluated at the beginning of the compilation process, together with other elements such as the compilation schedule, the revision policy and others. Section C moves on to the collect phase and describes the main data sources used for SUTs and IOTs.

B. Specify needs, design and build phases

1. Specify needs

4.3. The identification of user’s needs is a fundamental step in the compilation of any statistics, as it aims to identify what statistics need to be compiled, in which format, when and for what purpose. All these elements affect the planning of the compilation process of SUTs and IOTs since they have implications, for example, for the choice of the level of industry and product detail of the SUTs. Thus an assessment of the objectives of these tabulations has to take place during the specify needs phase of the statistical production process and this assessment must be regularly reviewed in the light of feedback from users, to ensure the relevance of the compiled SUTs and IOTs. During this phase, consultation with relevant stakeholders, through meetings, workshops and surveys, is of key importance.

4.4. Other elements of this phase include the identification of the statistical outputs that are required to meet the user needs and checking the data availability to see if existing data sources can meet the user requirements, if there are alternative data sources that would be more suitable for the specific statistics, and if there are data gaps to fill.

4.5. There are many different users – policymakers, analysts, researchers and others – and uses – planning, modelling, monitoring and so forth – of SUTs and IOTs. It is therefore important to maintain links with the users to ensure that their needs are met through a single efficient compilation process. For example, in the context of growing concerns about the environment, if a specific environmental topic, such as water, energy, fish, or forests, is considered as a key user need to be addressed, it is important to develop and design a compilation process which includes these elements from the outset, rather than adjusting *ex post facto* the statistical output through modelling based on various assumptions.

4.6. The compilation of any statistical output depends to a large extent on the availability of appropriate infrastructure for information technology and human resources. The technology has changed enormously over the last fifty years. Statistics which were once compiled with calculators are now processed in seconds by modern computers, laptops or even smartphones. This rapid development has facilitated the work of statisticians and improved the timeliness of their statistical outputs. When compiling SUTs and IOTs, a variety of software, databases and custom-designed platforms is available and can be adapted to the specific compilation process in any given country.

4.7. It is therefore important to have a clear understanding of the information technology requirements necessary for all the phases of the compilation process. In practice, more than one software package is often required and the ones selected are those which best meet the diverse functional requirements of the specific phase and which are able to communicate with one another in an easy, effective and efficient manner. If the links between the software packages are cumbersome or time-consuming, alternatives should be sought. Many national statistical offices have developed in-house tailored software to meet their needs – this has advantages but may entail greater overhead maintenance and training requirements.

4.8. Box 4.1 and Box 4.2 provide examples of custom-made software produced and maintained by, in the first case, Statistics Netherlands and, in the second, the Institut national de la statistique et des études économiques (INSEE) – the French national statistical office – and Eurostat – the statistical office of the European Union.

Box 4.1 Example of in-house custom-built software: Statistics Netherlands

Statistics Netherlands has a long-standing tradition of compiling SUTs both in current prices and in volume terms, and also IOTs, and therefore has extensive experience in tackling the various challenges associated with the use of computer systems to produce and maintain long-run series of SUTs and IOTs.

Statistics Netherlands has been publishing SUTs since 1990 (relating to the benchmark revision for the year 1987) and IOTs since the 1950s (for years starting from 1948).

Statistics Netherlands has always used in-house custom-built software for the compilation of the balanced SUTs and IOTs. There are two separate software tools which are both updated continuously on an SQL database. These are combined with a graphical user interface in Visual Basic for Applications (VBA) code, enabling data to be accessed and adjusted by the national accounts staff. Both systems can handle quarterly and annual data.

The first application includes tools for the transformation of source data to national accounts definitions and standards, both in current prices and in volume terms.

The second application includes tools for simultaneous balancing of the SUTs in current prices and in volume terms, for compiling the valuation matrices and for transforming the SUTs to industry-by-industry IOTs.

The present systems are based on a major overhaul carried out in 1995 and have been continuously updated. A key rebuild took place in 2004/2005, reflecting a new programming language and also including new features.

Box 4.2 ERETES

ERETES – the acronym for “équilibre ressources-emplois et tableau entrées-sorties” (supply-use balances and input-output tables) – is a computer system designed to assist national accountants to compile the SUTs and integrated economic accounts (including sector accounts), which complies with the principles and guidelines set out by the SNA. ERETES was developed by INSEE and Eurostat. In 2014, it was being used by several countries in Africa and Latin America and the Caribbean. Further countries are expected to adopt the system. The ERETES system is available in French, Spanish and English.

Although the objective of ERETES is to generate SUTs and the integrated economic accounts, it can also be used by countries that have limited data resources. The minimum data requirements are an enterprise survey and a household budget survey, foreign trade statistics, government accounts, balance of payments and banking statistics. With these data, ERETES can help countries to generate estimates of GDP in current prices. If sufficient price and volume indices are available, then estimates of GDP in volume terms can also be generated. The compilation of SUTs would also require information on intermediate consumption and trade and transport margins.

One key advantage of ERETES over other computer systems is that it is supported by a permanent secretariat that can call on a group of multilingual national accountants and information technology experts with extensive experience of applying the system in a number of developing countries. ERETES is regularly updated and improved. ERETES is available at: <http://www.ernetes.net/EN/index.htm>.

4.9. Another successful example of custom-built software produced by one national statistical office, and then made available for use by other countries under specific terms and conditions, may be seen in the Norwegian software SNA-NT. Here, Statistics Norway provided both the software and the associated human resources for training in the use of the applications by, for example, Malawi, the Czech Republic and Slovakia.

4.10. When choosing the software and hardware to support the compilation of SUTs and IOTs as part of the national accounts, consideration should be given to various criteria such as the database environment, in particular its flexibility and structure, its statistical functionality and diagnostic tools required, the necessary availability of mathematical functions such as matrix calculations, the resources and costs, the training programme, compatibility with data suppliers, data management, and the data dissemination platform envisaged.

4.11. Across the world a range of different products are used to this end, such as Oracle, JAVA programming, MATLAB, SPSS, SQL, SAS, Excel and custom-made software built to meet specific requirements. For example, the use of Excel or input systems such as an Oracle database could provide an effective solution in the “collect” and “process” phases, for preparing and validating the data, while SAS could be used for the further processing of SUTs and IOTs. Tools such as Excel, an output system and web-tools for dissemination may offer the best means of validating and balancing, and also of analysing and disseminating the data.

4.12. Skilled and trained human resources are also a fundamental pillar for the compilation of SUTs and IOTs. It is thus important to recruit and retain skilled and effective staff and develop and use internal and external training opportunities on the theoretical and practical aspects of the compilation of economic statistics.

4.13. An important step in this phase is the preparation of a document summarizing the findings of all the activities mentioned above – namely: needs identification; establishment of output objectives; checking of data availability and information technology requirements, and others – in the form of a business case, with a view to securing approval to implement the new or modified statistical business process. Such a business case would need to conform to the requirements of the approval body but would typically include elements such as a description of the “as-is” business process (if it already exists), with information on how the current statistics are produced, highlighting any inefficiencies and issues to be addressed; the proposed “to-be” solution (with clear improvements and benefits); detailing how the statistical business process will be developed to produce the new or revised statistics; an assessment of costs and benefits, and any external constraints.

2. Design and build phases

4.14. The design phase comprises all the activities undertaken to define the statistical output and the concepts, methods, collection instruments and operational processes necessary. Accordingly, this phase includes all the design elements needed to define or refine the statistical output identified in the previous phase, all relevant metadata, ready for use later in the statistical business process, and quality assurance procedures.

4.15. These activities make substantial use of international and national standards, in order to reduce the length and cost of the design process and enhance comparability and usability of outputs. Organizations are also encouraged to reuse or adapt design elements from existing processes. In addition, outputs of design processes may form the basis for future standards at the national and international levels.

4.16. The design of the statistical output for SUTs and IOTs consists of the size and layout of the tables; the breakdown of industries and products; disclosure control methods; processes governing access to any confidential information; and the identification of the statistical variables needed, which is then linked to the data collection phase.

4.17. In the build phase, the production solution is put together and tested to the point where it is ready for use in a live environment. This phase is broken down into several activities, which include a review of data sources; the configuration of the workflow from data collection through to dissemination; and testing of the statistical business process. For statistical outputs produced on a regular basis, the design and build phases usually occur for the first iteration, and following a review or a change in methodology or technology, rather than for every iteration.

4.18. During the design and build phases a number of specific issues in the compilation of SUTs and IOTs should be considered. These include, for example, the choice of the level of detail of industries and products of SUTs and IOTs at working level and in the dissemination phase; how to handle confidentiality throughout the compilation process; the schedule for the compilation and dissemination of SUTs and IOTs; the revision policy and analysis; the resources required to sustain the compilation; the benchmarking; and the choice of the index formula and the base year. In addition, it is important to create and maintain documentation for all phases of the compilation process to serve as a quality control measure of the process. All these activities in the design and build phases are further elaborated in the following sections.

(a) Level of industry, product detail and size of SUTs and IOTs

4.19. The level of industry and product detail of the published and disseminated SUTs and IOTs greatly depends on the objectives of the tabulations and their uses. The industries and products explicitly identified in the disseminated tables reflect to a great extent the users' needs and the specific policy concern of interest. For example, if a specific environmental domain is of interest, such as energy, specific industries and products are likely to be explicitly identified in the disseminated tables in order to address the specific domain. The ultimate level of aggregation of the disseminated SUTs and IOTs has an impact on – and at the same time is affected by – the data availability and the data collection, compilation and balancing procedures.

4.20. The number of products in the SUTs is usually higher than the number of industries, thus showing more than one primary product for each industry, causing the SUTs to be rectangular. Their size and shape will have appropriate implications for IOTs, the physical tables and other related products analyses (for example, productivity).

4.21. The level of detail of industries and products at the working level is generally very disaggregated and the recommendation is to work with the most detailed level of aggregation taking into consideration the constraints posed by the available data, resources, and burden on business. Various aspects need to be considered, including the user needs, the availability of data, and the level of detail used in national accounts. For example, compilation aspects that influence the level of detail (since they facilitate the compilation and validation of the data at the working level) include:

- Distinction between industries which are allowed to deduct VAT and those that are not allowed to deduct VAT, and different VAT rates per product and categories exempt from

VAT to facilitate the compilation of the valuation matrices in particular for taxes and subsidies.

- Distinction between market and non-market activities to facilitate the understanding of the input structure of GVA.
- Subdivisions of industries according to institutional sectors which have an impact on the linking table between SUTs and the institutional sector accounts.
- Separate identification of certain industry and product subdivisions to facilitate the compilation of the trade and transport margin matrices, such as section G and its divisions of ISIC Rev. 4.
- Links to international industry, product and functional classifications.
- Links between output structures, input structures, price indices and values for deflation, along with the availability of price statistics to generate estimates in volume terms.
- Links between domestic supply and exports, and also intermediate consumption and imports, to study the input structure of the industries.
- Links to the environmental accounts.
- Enterprises that trade internationally and have links to global value chains (see extended SUTs and the OECD trade by enterprise characteristics (TEC) database).
- Availability and quality of source data.
- Size and value of output and value added of the industry: if the industry is too large and heterogeneous then it should be further broken down. The same can apply to products.
- Benchmarking (annual as opposed to five-yearly) using comprehensive sources and censuses, since the level of detail in benchmarking years is much larger than that during regular annual compilation of SUTs and IOTs.
- Annual chain-linking the volume estimates.
- Staff resources, time schedules for production and publication, confidentiality and system infrastructure.

4.22. An appropriate choice of the level of industry and product detail in the SUTs at working level will facilitate the compilation and the search for causes of inconsistencies. For many products, it is possible, by their nature, to identify the industry in which they are used. For example, fertilizers are mainly used in agriculture, crude oil in oil refineries, concrete in construction, and so forth. For some products, it may also be possible to identify whether they are used as

intermediate consumption or final consumption. Haircut services, for example, can be assumed to be mainly consumed by households and thus recorded in household final consumption expenditure. The more detailed the classification of products used in SUTs, the easier it is to use expert knowledge to supplement surveys in allocating products to different uses.

4.23. Linking and matching products to valuation-related elements (taxes, subsidies, trade and transport margins) make the composition of transactions more transparent and clear-cut and significantly facilitates their analysis.

4.24. If the products are more detailed, it will also mean that the number of users of a certain product is greatly reduced. Where there is only one producer and one user of a product, the search for the cause of inconsistencies would only require investigating two source statistics. When a product has 20 users, for example, the search becomes more complicated.

4.25. Questions relating to the layout of SUTs tables also arise for the final use part of the SUTs. It may be useful to integrate the functional classifications in the final consumption data, showing final consumption by products and by consumption purpose. On the other hand, it may be better to keep such detail and transformation separate, but this integration will still be needed in some form or other in the compilation of SUTs and the other accounts of the system.

4.26. In the dissemination phase, the size and breakdown between industries and products shown in the SUTs (and thus in the IOTs) mainly reflect the user's needs and the objective of the tabulations, taking into account confidentiality considerations. Other presentational considerations include, for example, the size and relative value of output and value added for the industries and the size and value of supply for the products. Industries or products that are not economically significant or relevant for a particular economy may be aggregated together, while a more detail breakdown may be shown for economic activities that contribute substantially to GDP, in order more effectively to analyse the cost structures and the interdependencies with other economic activities.

4.27. It should be mentioned that, when compiling consistent annual, or quarterly, SUTs, the stability of the level of detail of the applied classifications is also important, as many ratios and proportions will usually be taken as a starting point in the estimates for the following year.

4.28. A higher degree of product detail also supports the use of certain estimation methods, for example the product flow method of compiling national accounts (namely, balancing the supply and use of products) by taking into account the relevant differentiation concerning product tax rates, margin rates and homogeneity in prices. Moreover, it is much easier to distribute disaggregated products and services across industries and final use categories with the product flow method than at a higher aggregate level. Detailed product accounts also help in the balancing procedure, as it is easier to explore and detect the causes of imbalances if the basis is determined by homogeneous single products rather than aggregate groups of products. The work on a detailed product level certainly increases the data quality but has resource and systems implications. At

higher levels of aggregation, problems of imbalance might not even be detected at all and therefore not remedied.

4.29. Table 4.1 provides an example of the size of SUTs and IOTs compiled by selected countries. It is worth noting that the internal working level of the industry and product detail used for compilation and balancing is much higher than which is actually published. For example, in the United States, the working level in producing the SUTs is over 800 industries, whereas in Denmark, there are around 2,350 products at the working level but the SUTs are published only at a level of 64 products by 64 industries.

4.30. It is important to distinguish between the detail required for the compilation and balancing work at the working level as opposed to the information required for the publication. The in-house operating detail should be the same or, as in most cases, in greater detail in terms of number of industries and products than that allowed for disclosure by the publication. For example, many countries distinguish hundreds or even thousands of products but do not publish at these levels, as a great deal of confidential information would thereby be released. It should be noted, however, that countries often allow people outside the national statistical office to have access to more detailed data, albeit confidential and under signed agreements, for analytical purposes.

Table 4.1 Examples of the size of published and internal working level SUTs and IOTs

Country ^{(1),(2)}	National Supply and Use Tables					National Input-Output Tables			
	Current prices				Volume terms or PYPs	Current prices			
	Internal working / compilation levels		Published levels		Do you produce such tables: Y (yes) N (no) or P (plan to)	Internal working / compilation levels		Published levels	
	Number of products	Number of industries	Number of products	Number of industries		Number of industries / products		Number of industries / products	
						P x P Tables	I x I Tables	P x P Tables	I x I Tables
Argentina ⁽³⁾	271	162	271	162	N	n/a	124	n/a	124
Australia ⁽⁴⁾	301	67	n/a	n/a	Y	n/a	114	n/a	114
Austria	573	135	74	74	P	74	n/a	74	n/a
Belgium ⁽⁵⁾	355	135	64	64	P	135	n/a	64	n/a
Brunei Darussalam ⁽⁶⁾	324	74	74	74	N	74	74	74	74
Canada	490	230	490	230	P	n/a	230	n/a	230
Chile	275	160	180	111	Y	n/a	111	n/a	111
Columbia	369	61	61	61	Y	61	61	61	61
Costa Rica	183	146	183	138	Y	183	136	183	128
Czech Republic	252	120	88	88	Y	184	184	82	82
Denmark ⁽⁷⁾	2 350	117	117	117	Y	n/a	117	n/a	117
Estonia	247	98	64	64	Y	64	n/a	64	n/a
Finland	776	179	64	64	Y	n/a	179	n/a	64
Germany	86	63	85	63	P	73	n/a	72	n/a
Hungary ⁽⁸⁾	820	242	64	64	Y	88	88	64	64
Iceland ⁽⁹⁾	561	142	n/a	n/a	P	n/a	n/a	n/a	n/a
India ⁽¹⁰⁾	142	126	140	66	N	130	n/a	130	n/a
Indonesia ⁽¹¹⁾	244	81	n/a	n/a	P	251	n/a	185	n/a
Ireland ⁽¹²⁾	82	82	58	58	Y	82	n/a	58	n/a
Kuwait ⁽¹³⁾	43	43	n/a	n/a	N	43	43	n/a	n/a
Mexico ⁽¹⁴⁾	819	814	262	262	P	814	262	814	262
Netherlands	614	128	85	76	Y	n/a	128	n/a	76
New Zealand ⁽¹⁵⁾	299	118	201	106	P	n/a	106	n/a	106
Norway	860	156	64	64	P	n/a	156	n/a	64
Republic of Korea	1 851	328	384	328	N	1 851	n/a	384	n/a
Saudi Arabia ⁽¹⁶⁾	59	59	18	18	N	59	59	n/a	n/a
Serbia	216	88	n/a	n/a	N	n/a	n/a	n/a	n/a
Singapore	71	71	71	71	N	n/a	71	n/a	71
Slovakia	290	88	64	64	Y	88	n/a	64	n/a
Slovenia ⁽¹⁷⁾	350	230	64	64	Y	64	n/a	64	n/a
South Africa	104	293	104	62	N	n/a	50	n/a	50
Sweden ⁽¹⁸⁾	403	97	62	64	P	62	n/a	62	n/a
United Kingdom ⁽¹⁹⁾	112	112	112	112	P	112	n/a	112	n/a
United Republic of Tanzania ⁽²⁰⁾	250	59	250	59	P	n/a	n/a	n/a	n/a
United States of America ⁽²¹⁾	4 988	819	73	71	P	73	71	n/a	n/a
Submissions to European Commission reflect EU Member States ⁽²²⁾	64	64	64	64	P	64	64	64	64

Key: PYP – previous year's prices
P x P – product-by-product
I x I – industry-by-industry

- (1) The above table was compiled by Sanjiv Mahajan (Office for National Statistics, United Kingdom) at the time of preparation of this Handbook. The numbers are indicative for the reference year 2014 (as at December 2016) unless denoted otherwise.
- (2) Other differences will exist in terms of comparability, given that the tables are compiled on different bases, in terms of the frequency of the tables, classifications used, the SNA version, latest reference year, latest benchmark year, valuation of SUTs (basic prices or producers' prices or purchasers' prices), assumptions underpinning the IOTs, etc.
- (3) For Argentina the SUTs reference year is 2004 and the IOTs reference year is 1997.
- (4) For Australia the IOTs working level operates at 1268 products and 114 industries.

- (5) For Belgium the IOTs reference year is 2010.
- (6) For Brunei Darussalam the reference year for SUTs and IOTs is 2005.
- (7) For Denmark the number of products can vary from year to year. Detailed data can be made available outside Statistics Denmark for analytical purposes.
- (8) For Hungary the IOTs reference year is 2010 (compiled five-yearly).
- (9) For Iceland the SUTs are produced but not yet published. The plans are to publish 64 products x 64 industries.
- (10) For India the SUTs reference year is 2012/13 and the IOTs reference year is 2007/08. Industry outputs are published in full whereas industry uses are aggregated to higher levels.
- (11) For Indonesia the SUTs and IOTs reference year is 2010.
- (12) For Ireland the SUTs reference year is 2013 and the IOTs reference year is 2011.
- (13) For Kuwait the SUTs and IOTs reference year is 2010.
- (14) For Mexico the reference is year 2008, the PxP IOTs reference year is 2008 (814 products), updated version for 2012 (259 products), and the IxI IOTs reference year is 2012.
- (15) For New Zealand the production of SUTs in volume terms is being planned but there are no plans for publication.
- (16) For Saudi Arabia the SUTs and IOTs internal working levels reference year is 2011 and the published SUTs reference year is 2013.
- (17) For Slovenia the SUTs reference year is 2013 and the IOTs reference year is 2010. SUTs and IOTs will be published in 2017, both with a reference year of 2014.
- (18) For Sweden the SUTs in PYPs are produced but not yet published. A version is, however, submitted to the Commission (Eurostat) and the plan is to publish at some stage.
- (19) In the United Kingdom the IOTs PxP tables use the hybrid assumption approach
- (20) For the United Republic of Tanzania the SUTs reference year is 2007. IOTs for the reference year 2007 are being finalized.
- (21) For the United States, in benchmark years, the SUTs are published at the 389 product and 389 industry level of detail. While IOTs themselves are not currently published, the IxI Leontief Inverse tables calculated from IOTs are published at the 71 industry level of detail, and PxP tables at the 73 product level of detail.
- (22) All EU Member States are expected to provide:
 - Annual SUTs both in current prices and PYPs as well as five-yearly IOTs under the ESA 2010 Transmission Programme after the expiry of any National Derogations.
 - The SUTs and IOTs are to be supplied using 64 products and 64 industries.

4.31. Confidentiality is a fundamental principle of official statistics (see United Nations, 2013b). It ensures that individual data collected by statistical agencies for statistical compilation, whether they refer to natural or legal persons, are to be strictly confidential and used exclusively for statistical purposes. It is important therefore that procedures are put in place to ensure the confidentiality of the information disseminated.

4.32. Countries may apply different criteria to decide whether specific data may be disclosed or not. This decision is likely to be driven by the legislation in place underpinning the collection of data from businesses. The decision is normally influenced by the number of enterprises observed in an industry or by whether the data can be disclosed through a process of deduction. One solution would be to choose a higher aggregation level with a sufficient number of enterprises in an industry to overcome any disclosure problems. There may not always be an easy solution for some industries or the products to which they should be allocated. The price is a loss of information due to aggregation, resulting in the increased heterogeneity of the SUTs system. Other methods might therefore be explored or combined, such as creating or redefining products.

4.33. Cases where there are one or two dominant producers in an industry, such as mining, extraction of crude oil, sugar, pharmaceuticals and others, pose a different challenge. In these cases, it is recommended, when necessary, that specific permission is sought from the business when their data are publicly available from other public sources, for example published company

annual reports and accounts. If permission is not granted, then suppression of the relevant cells in the SUTs should be considered. The aggregation of industries and products with non-disclosive industries and products should, however, be avoided, as this results in the loss of useful details for non-disclosive industries and products.

(b) Schedule of the work programme

4.34. Aligning the monthly, quarterly and annual timetables covering data collection processes, compilation processes, data supply, validation, balancing and publication for all the accounts and outputs is a key step in ensuring coherence and consistency. This should include the compilation of SUTs and other input-output-related products, as appropriate.

4.35. The overall process needs therefore to be split into well-defined blocks of work with clearly defined processes and linkages between the processes, in such a way that they all fit within realistic schedules (including contingency planning and risk management) with clear roles and responsibilities for the staff and management involved. The governance of the programme should be clear, with regular monitoring and meetings scheduled at key junctures, for example, linked to key milestones in the process. The project management of the process should ensure that dedicated resources are attributable to this support function.

4.36. The schedule needs to incorporate deadlines of both data providers and data users, together with various internal intermediate deadlines. For annual business surveys, the time lag from changing the questionnaire to incorporating the new results and publication in SUTs and national accounts could be around three years; thus it is important to retain schedules, which are regularly reviewed and reflect the incorporation of continual data improvements.

4.37. In general, it is useful to put in place service-level agreements with data providers and also with data users. Agreements with data providers would cover the types of data to be provided, the quality criteria, briefings, schedules and the format in which the data will be delivered. Important elements to consider in such service-level agreements include:

- Clear ownership – senior representatives from both the supplier side and the customer side
- Reasons for data requirement
- Publication of results including disclosure requirements
- Process of data delivery by data provider (for example, format)
- What data are required (need to specify the variables needed)
- Timing of data deliveries
- Required briefing to accompany the data
- Handling of customer queries

- Quality (covering such criteria as consistency, credibility, revisions, precision and communication)
- Methodological notes supporting the data (for example, sources, methods, coverage, and others)
- Development, improvements and consultation
- Arrangements for review of the process, among others

4.38. Service-level agreements with data users, on the other side, would cover, for example, users' deadlines (such as those linked to policy agendas, research schedules, and other factors) to be reflected in the statistical production schedule. For example, the finance ministry or the central bank or both may require structural updates on the economy to fit with their policy review. Such user requirements may form part of regular annual schedules for the producers of SUTs.

4.39. In general, there should be a regular review, possibly on an annual basis, of all aspects of the process (including timetables, data quality, implementation of future changes, and other considerations) with both data providers and data users, to ensure continuous improvement and to guide changes as necessary.

4.40. The schedule for the compilation of SUTs naturally depends on the periodicity and frequency of SUTs and IOTs. In general, it is recommended that SUTs be compiled annually, in line with the United Nations Statistical Commission recommendations on the scope of the implementation of the 2008 System of National Accounts.⁸ While it is recommended that a benchmark system of SUTs based on specific survey results be compiled every five years, rapid changes in the economy, the external impacts of globalization, the increasing rate of change of technology and its impact, new products, new industries, the impact of digitization and other factors may affect the production process of SUTs and the periodicity of benchmarked tables, with the result that an annual benchmark process may be preferred.

4.41. Schedules setting the frequency of SUTs compilation may be assessed against the revision policy and the uses of intra-annual sources (such as quarterly and monthly short-term indicators), with a view to incorporating if necessary the revision guidelines and indicators policy.

(c) Revision policy and analysis

4.42. Revisions to time series data are an important part of the production process. Changes to published data can occur for many different reasons. For example, forecast data may be replaced by survey data, the reclassification of industries, methodological changes to the way in which data are estimated or just through the correction of errors. Changes due to the correction of errors should be identified as corrections and distinguished from revisions that are more commonly associated

⁸ See the recommended data set in the report of the Intersecretariat Working Group on National Accounts (E/CN.3/2011/6), available at <http://unstats.un.org/unsd/statcom/doc11/2011-6-NationalAccounts-E.pdf>.

with improving estimates as more information is gathered over time (Mahajan, 2015). There is a conflict between the release of timely estimates and that of accurate estimates. If statistical offices and central banks waited to publish the most accurate data possible, given the nature of data collection, there would be a large time lag between the date to which the data refer and the date of publication.

4.43. Bringing together data for the purposes of compiling SUTs and other input-output-related products integrated within the national accounts involves aligning production timetables (quarterly or annual) and schedules, and also revision policies. Ideally, a highly effective revision policy to ensure that revisions are implemented in a coordinated and coherent fashion across the accounts should cover the national accounts, balance of payments and government finance statistics, should include the SUTs and IOTs, and should also extend to primary source data and other domains, such as regional accounts and environmental accounts.

4.44. The revision policy should reflect appropriate criteria to assess each revision, including when best to implement the changes, for example, in a quarterly exercise (for example, for short-term revisions) or in an annual exercise (for example, for long-period revisions). This will have an impact on how to compile SUTs and IOTs and the revision guidelines should be operated flexibly, reflecting such issues as economic significance and practical aspects such as their impact on resources and systems.

4.45. The usual guidelines applied to all the accounts cover:

- Revisions to the latest quarters for an incomplete year (these can affect annual SUTs)
- Revisions to past recent quarters since the last full benchmark year
- Revisions made through the annual process to recent years, say, between three and five completed years
- Revisions to a longer period, sometimes viewed as a major revision. Many earlier years may be revised depending on whether they meet certain criteria such as those relating to methodological improvements, correction of errors and economic significance

4.46. Any changes to back data will also have an impact on the monthly and quarterly seasonally adjusted estimates.

4.47. In some countries, occasional or major revisions of national accounts are usually carried out every five years and require more resources if the revision is implemented on the basis of a large SUTs system. A revision at a more aggregated level is always easier and less demanding.

4.48. The various revision practices have different pros and cons:

- Regular revisions to SUTs and national accounts help to preserve good quality levels and growth rates. Some users do not welcome regular revisions, however, whereas others do not welcome the “big bang” approach to revisions.
- Five-yearly revision exercises may mean more significant changes for a number of years until the next revision window.
- Revisions only at the aggregated level may provide problems with the detail and provide discontinuities in the long time series of annual SUTs
- Revisions also at the level of IOTs ensures that SUTs and IOTs remain consistent.

4.49. Revisions applied only to part of the accounts and not all the relevant outputs would generate incoherence across different outputs and not help users, for example, in some countries, SUTs may be revised but the IOTs are not revised, thus the links between the two products are out of line. This situation should be avoided.

4.50. Analysis of revisions can provide information about the reliability of estimates and how they change between the first estimate and final estimate as well as a source for the identification of any biases, (Mahajan, 2004b). Note that revision analysis does not give information on the accuracy of an estimate, for example, the final estimate may not be accurate in terms of sampling error or non-sampling error.

4.51. The knowledge of the source and the reasons for the revisions is key and helps producers, and users, to better understand the data. Sometimes with major revisions going back in time, understanding the changes can be quite complicated, such as changes in definitions, classifications and data. For analytical users of SUTs and IOTs, it is important to understand the reasons and impact, especially when there is a mix of revisions, such as, for example, with the introduction of 2008 SNA, necessitating the revision of SUTs and IOTs going back 30–40 years or more. The forthcoming handbook on backcasting methodology (United Nations, forthcoming) will provide more information on this topic.

(d) Resources

4.52. When considering the resources and human requirements for the full integration of SUTs and IOTs (and also physical SUTs) in the compilation of national accounts and balance of payments, it is essential to distinguish between the first compilation of the SUTs, the recurrent production of SUTs and the process surrounding major backward revisions.

4.53. A substantial amount of resources is required to build up an integrated SUTs framework for the first time. This work involves establishing all the planning, conceptual and methodological work, data collection needs, requirements of all the individual industry and product balances, the development of appropriate techniques for incorporating the primary sources and new software for handling the SUTs system and the necessary training and investment in staff and systems. The

investment may lead to considerable changes in working procedures towards a better integration of activities from data collection through to publication, following an integrated statistics approach.

4.54. The resources needed to establish an integrated SUTs framework should, however, be viewed against the way in which the development will evolve, for example, the level of integration and the organization of roles and responsibilities across the statistical domains. The implementation of the recommendations and guidelines provided in this Handbook can be carried out in a gradual manner taking into account countries' specific situation in terms of the resources available and their national priorities. Countries' practices may vary considerably, among such scenarios as the following:

- Only annual SUTs in current prices
- Annual SUTs in both current prices and in volume terms
- Annual SUTs complemented with a quarterly SUTs system
- Annual and quarterly SUTs in current prices and in volume terms
- SUTs at basic prices or at purchasers' prices
- Benchmarking annually or, say, at least every five years
- Links between SUTs and IOTs:
 - SUTs only
 - SUTs and IOTs, both in current prices (possibly also in volume terms)
 - Only IOTs and no SUTs – this is clearly the least favoured scenario but it has a historical legacy in some countries

4.55. Developing a new SUTs and national accounts system poses different challenges involving the change of existing production systems while maintaining business-as-usual activities.

4.56. Where periodicity is concerned, at least every five years a benchmark system of SUTs should be compiled which is based on more exhaustive specific survey and administrative data results. As described earlier in this chapter, however, when considering schedules for SUTs (and IOTs), with rapidly changing and developing economies, the impact of globalization and digitization, and other factors, it is recommended that the development of new SUTs systems should reflect an annual benchmarking and reconciliation process. This will also help to avoid significant revisions and distortions to the levels of data. Together with annual chain-linking, this will mean better quality measurement of the growth rates of GDP in volume terms, in particular for the more recent periods. Appropriate techniques have been developed and the trends of

structural change (for example, composition of outputs and intermediate consumption) during the previous years can be used to take forward the structures in SUTs, if no new structural information is available, for example, beyond benchmark years.

4.57. The experience from countries which have integrated the SUTs framework into the compilation of their national accounts suggests that the resources needed are similar to the resources for those countries following a more traditional approach with a separate SUTs compilation. It is therefore recommended that SUTs are compiled as an integrated process and a regular part of the compilation of the national accounts. The fact that IOTs are produced with relatively few additional resources if a SUTs system is in place militates in favour of an integrated and regular approach for the compilation of IOTs.

(e) Benchmarking, extrapolation using indicators, price and volume information

4.58. Planning a new system of SUTs is generally linked to a benchmark year for which the most important areas of the economy are covered by censuses and surveys. This is especially important when policy decisions are based on the levels of the figures, for example, the level of gross national income (GNI). Some detailed data sources are collected at more or less regular intervals and will not all be available for the benchmark year. Hence, figures that do not relate to the actual year will need to be corrected for the changes that have taken place between the reference period of the data and the period for which they are being used. This can be done using indicators for value or volume and price indicators.

4.59. When a balanced benchmark SUT exists, the compilation of SUTs for following (or previous) years will usually be considerably easier. It is possible to use information on the structures of the benchmark table to fill the gaps between those cells for which no new source data are available or to extrapolate as appropriate. For example, input structures will change over time and information on new input structures can then be used as it becomes available, replacing the extrapolated information. Taking into account that the other estimated structures are subject to uncertainty, it may be sufficient to review them at intervals of a few years, one at a time. Even when such structures are reused from the previous year, they will change over time as a result of balancing the SUTs. This could happen even more rapidly with the impact of globalization and the development of new products.

(f) Features of GDP in volume terms

4.60. When looking at the change in the economy over time, the main concern is often whether more goods and services are actually being produced now than at some time in the past. With productivity, however, the point of interest is whether this output is increasing relative to the inputs.

4.61. Over time, changes in current price GDP show changes in the monetary value of the components of GDP. These changes in value can reflect changes in both price and volume, making

it difficult to establish how much of an increase in the series is due either to increased activity in the economy (volume change) or to an increase in the price level. For productivity measures, only volume changes are used. It is therefore useful to measure GDP in volume terms (preferably in previous years' prices), meaning that price effects are excluded, including in current prices. In most cases, the revaluation of current price data to remove price effects (known as deflation) is carried out by using price indices such as component series of the retail prices index or producer price index, to deflate current price series at a detailed level of disaggregation.

4.62. At the international level, constant price estimates and chain-linked volume measures are two common measures for volume change of GDP. Under the constant price method, a certain year is selected as the base year; constant price volume estimates of GDP for subsequent years are the aggregation of its components computed by multiplying the price of each component in the base year by its volume in the current year, and the real growth is derived from the comparison of constant price volume estimates at different years.

4.63. Under the chain-linked volume measures method, the annually reweighted chain-linking approach is adopted to compile the volume measures of GDP and its components. First, the volume estimates of major components of GDP in the current year are revalued at preceding year prices, which in practice are calculated by deflating the current price values of subcomponents by the relevant price indices. Second, the short-term volume indices for different years, calculated by dividing the volume estimate of GDP from the initial step by the current price GDP in the previous year, are chain-linked to a selected reference year in order to obtain a continuous time series of the chain volume indices of GDP and its components. The chain-linked volume index series can be converted into the chained monetary series by multiplying the chain-linked volume index by the current price value in the reference year.

4.64. For some series, price indices for particular goods and services are used to deflate the current price series, such as components of the following:

- Consumer price index (CPI)
- Retail price index (RPI)
- Producer price index (PPI)
- Corporate services price index (CSPI)
- Import prices
- Export prices

4.65. The process known as “double deflation” is the preferred method for the estimation of GVA in volume terms. This is achieved by deflating the value of output and the value of intermediate inputs separately to get corresponding volume measures, and then subtracting the

latter from the former. This double deflation approach means that an industry's total output is deflated by the price of its primary and secondary output, while each intermediate input is deflated by its own price index.

4.66. This is in contrast to the single deflation method, whereby GVA in current prices is deflated directly using an output-based deflator to arrive at GVA estimates in volume terms. The single indicator volume estimates can also be derived in other ways, for example, by deflating output with output price indices, assuming a constant GVA to total output ratios from the base year, or using volume indicators directly. This direct price deflation of GVA is not recommended by the SNA when using a single indicator method.

4.67. The SUTs offer a major advantage and a natural framework that enables double deflation to be applied in a coherent and consistent manner across the national accounts.

4.68. Chain-linked volume measure series are expressed as index numbers in which the series are simply scaled proportionately to a value of 100 in the reference year. These index numbers are volume indices of the so-called "base weighted" or "Laspeyres" form.

4.69. Aggregate price indices are of the "Paasche" or "current-weighted" form. They are generally calculated indirectly by dividing the current price value by the corresponding chained volume measure and multiplying by 100. Examples are the GDP deflator and the households' consumption deflator.

4.70. Value indices are calculated by scaling current price values proportionately to a value of 100 in the reference year. By definition, such a value index, if divided by the corresponding volume index and multiplied by 100, will give the corresponding price index.

4.71. From the point of view of production, GDP at market prices is at best estimated with reference to annually compiled SUTs both in current prices and in previous years' prices. The SUTs are compiled in previous years' prices in order to achieve an accurate breakdown of value changes in subsequent years according to volume and price changes.

4.72. The base-year table provides the specific weights for each industry and product, used in the index formulae by which the price data are aggregated.

4.73. The great statistical benefit of a system based on previous years' prices is the fact that the weights in the index formulae are always up-to-date, thus reflecting the structure of the recent past, and in turn optimizing the quality of the GDP growth rates in volume terms for more recent periods.

(g) Choice of index number formulae

4.74. In order to calculate price and volume measures, a number of methodological choices have to be made, for example:

- Which index number formulae will be applied
- Whether a fixed base year or an annually changing base year will be applied

4.75. Different index formulae may be applied using different weighting schemes. It is beyond the scope of this manual to discuss in depth the theoretical and practical considerations relating to this choice. Chapter 15 of the 2008 SNA and Eurostat (2016) provide much more detail on the choice of index formulae.

4.76. Economic theory suggests that an index formula that assigns equal weight to the current year and the base year is to be preferred. This is one of the reasons why the SNA has a preference, albeit not a strong one, for the so-called “superlative” indices, like Tornqvist and Fisher.

4.77. Although the superlative indices have a number of attractions, it should be noted they also have notable disadvantages:

- Superlative indices are demanding in their data requirements and will increase the work burden significantly.
- Superlative indices are less intuitive than Laspeyres and Paasche indices.
- Superlative indices are not additively consistent, which is a serious constraint when applied in an accounting framework.
- Values change does not always equal volume change times price change.

4.78. From a practical point of view, a number of requirements can be imposed on the index numbers:

- The applied index formulae should be a good approximation of the changes obtained by the superlative indices.
- A change in value must be divided into a price change and a volume change without a residual.
- Values in volume terms for aggregates should equal the sum of values in volume terms of constituent parts, applying the same index formulae.
- In addition, it is sensible for the index formulae to be relatively straightforward and easy to interpret for users.

4.79. The last three requirements can only be met with the application of the Laspeyres volume index formula and the Paasche price index formula. The formulae underpinning these indices are shown below

Laspeyres volume index

$$L_q = \frac{\sum p_0 q_t}{\sum p_0 q_0} = \frac{\sum p_0 q_0 \frac{q_t}{q_0}}{\sum p_0 q_0}$$

Paasche price index

$$P_p = \frac{\sum p_t q_t}{\sum p_0 q_t} = \frac{\sum p_t q_t}{\sum p_t q_t / \frac{p_t}{p_0}}$$

where p is the price and q is the quantity.

4.80. The characteristic for the Laspeyres volume index is that the volume changes of individual goods are weighted with the value of the transaction concerned in the base year..

4.81. The characteristic for the Paasche price index is that the price changes of individual goods are weighted with their value of the transaction concerned in the current year. The deflated values derived with this index formula combination can easily be explained as values in prices of the base year.

4.82. It can be easily shown that the decomposition of value changes, in terms of volume and price changes, do not have a residual.

$$P_v = \frac{\sum p_t q_t}{\sum p_0 q_0} = L_q \times P_p$$

4.83. The deflation of values in current prices by using a Paasche price index gives:

$$P_v/P_p = \sum p_t q_t / \frac{\sum p_t q_t}{\sum p_0 q_t} = \sum p_0 q_t$$

4.84. This illustrates that the deflated aggregate equals the sum of deflated components, which means that additivity in the SUTs for volume estimates is assured. The use of Laspeyres volume indices and Paasche price indices ensures that the current price identities also hold in volume terms. This also means that, after balancing, for every product, the total supply equals total use, and for every industry, the total output equals total intermediate consumption plus gross value added and, in volume terms, this is:

Total supply in volume terms *equals* Total use in volume terms

Total output in volume terms *equals* Total intermediate consumption in volume terms

plus Total value added in volume terms

(h) Choice of the base year

4.85. By applying the Laspeyres volume index number formula, the volume changes are weighted with the values of the concerning transaction in a base year. The question arises which year should be chosen as the base year. Generally speaking, there is a choice between a fixed base year and a changing base year.

4.86. With the method of fixed weights for a series of years, the weights are derived from a single year in the past. An advantage of this method is that, with long time series of values at prices of the base year, the deflated components of aggregates add up exactly to the deflated aggregate. A serious disadvantage, however, is that volume changes of aggregates are calculated with outdated weights. This disadvantage is especially severe when relative prices change rapidly and, as a result, economic growth can be significantly misrepresented. In addition, the disappearance of products (such as vinyl records, cathode ray tube televisions, and so forth) or the appearance of new products (such as mobile telephony, pharmaceutical tablets, iPads, and the like) can lead to notable distortion in the estimates of economic growth. Even a fixed base year has to be changed every five years, and then all previously published real growth rates will be revised – this change is often not welcomed by users.

4.87. Applying a changing base year means that the weights are updated every year and are usually derived from the previous year. Since those weights are more up-to-date, a better approximation of the volume changes is obtained than with the method of using fixed weights. The time series can be obtained by multiplying separately estimated year-to-year volume indices – this is called “chaining”.

4.88. An important advantage of the chaining method is that the above-mentioned misrepresentation of growth rates is avoided. In fact, chain-linked volume measures provide a more reliable measure of volume growth, provided individual prices and quantities tend to increase or decrease steadily over time. They also have a key disadvantage, however, in that when the time series are in prices of the previous year, the deflated components of an aggregate then no longer add up exactly to the deflated aggregate. As a result, mathematical discrepancies will appear that cannot be removed without distorting the underlying actual volume and price movements.

(i) Documentation

4.89. As the compilation of SUTs is a complex process, a thorough documentation of the basic data and the methods used, the problems encountered, the solutions applied, and the results achieved is highly recommended. Such an annual (or quarterly, if appropriate) inventory is not only worthwhile for purposes of publication but also for internal use in the compilation process itself and future exercises. When SUTs have to be balanced, information on, in particular, the sources and methods of estimation for each single supply and use element is needed to evaluate and analyse industry and product imbalances. The documentation helps to evaluate the data quality and outline the strategy for balancing. The balancing steps should also be documented, of course, in order to avoid repeating changes and the destruction of already balanced data.

4.90. Documentation of the various compilation steps can also point to missing data issues and problems of basic data quality. It is important that such findings are used as feedback to primary statistics and to identify priorities in improving the compilation methodology. A documentation

system for the SUTs should be seen in the frame of the overall documentation for the system of national accounts data.

4.91. Producing a balanced set of SUTs for several years is like solving a puzzle. At first, all macroeconomic data, survey results, census results and other valid economic data on supply and use of products in the economy have to be collected. In a second step, missing data have to be estimated on the basis of harmonized methodologies and documented procedures. In the third and final stage, the balancing of the SUTs system generates a consistent set of macroeconomic variables in current prices and in previous years' prices. Thus the documentation of all stages of the compilation of hard data in terms of sources and soft data in terms of estimates and adjustments, ideally for each cell of the SUTs system, is key. In turn, a quality assessment using clear criteria for each cell can also be developed over time.

4.92. Links between survey data and final national accounts data should be maintained in the system, in particular separately recording documentation on the survey data, coverage adjustments, conceptual and valuation adjustments, quality adjustments and balancing and coherence adjustments. Analyses of these types of adjustments over time and over successive exercises can also help to highlight any biases, incoherence in data sources, and other shortcomings and, in turn, help in developing priorities and strategies for further improvements and investment.

C. Collect phase

4.93. The collect phase of the GSBPM (see figure 3.2) consists of all the activities concerning the collection and gathering of all necessary information (data and metadata), using different collection modes (including extractions from statistical, administrative and other non-statistical registers and databases), and their storage in an appropriate structured environment for further processing. While it may include the validation of dataset formats, it does not include any transformations of the data themselves, as these are all effected in the “process” phase. For statistical outputs produced regularly, this phase occurs in each iteration.

4.94. Generally, the compilation of SUTs and IOTs relies on the data sources used for calculating GDP according to the production, income and expenditure approaches. An overview of the range of data sources generally used is provided in Box 4.3.

4.95. It should be noted that this situation is reversed in some countries, such as Chile and Japan, where the requirements of SUTs (and IOTs) are first clearly defined and then the surveys (regular and ad hoc) are undertaken to collect data to meet these requirements. This may be ideal for SUTs (and IOTs), but it may also be more costly.

4.96. The compilation of SUTs is data demanding and in principle could be based on many sources that could help to populate the SUTs. It is strongly recommended, however, for purposes of timeliness, that more data sources should be regularly available, and that these are also based on official statistics, preferably using the same business register sampling frame. It is also worth

noting that many sources can provide data feeding into both the supply table and the use table for the respective industries and products, as appropriate. For example, annual structural surveys provide details on both sales and purchases.

Box 4.3 Data sources generally used

Monthly, quarterly, annual, regular and ad hoc business surveys based on the business register sampling frame:

- Monthly business surveys covering production, distribution, construction and service industries and financial information such as employment and turnover
- Monthly surveys collecting price-related details such as producer (input and output) prices covering all industries (for example, manufacturing and services), retail and consumer price indices, import prices, export prices, and earnings and wage prices
- Quarterly business surveys covering areas like capital expenditure, inventories and profits
- Annual business surveys covering structural business statistics for each industry:
 - Detailed information covering such variables as turnover, employment, purchases, capital expenditure, inventories, holding gains, taxes and subsidies, and others
 - Detailed sales by type of product
 - Detailed purchase data by type of product, covering intermediate-type products and capital-type product separately
- Range of quarterly and annual surveys covering financial services, in particular financial assets and liabilities, international trade in services and foreign and direct investment
- Economic censuses every three to five years.

Household-based surveys:

- Living costs and food surveys – collecting details on expenditure by households
- International passenger surveys – expenditure by residents abroad and non-residents' expenditure in the domestic economy
- Labour force surveys – collecting details on employment including hours worked

Administrative data:

- Census – population estimates for grossing purposes for use in household-based surveys
- Pay and profits data relating to tax and employment records from collecting government departments, along with data covering self-employed incomes
- VAT payments data and turnover subject to VAT by industry (and by product where differential rates exist) from the tax collecting departments
- Imports and exports of goods data collected by customs

Other government departments:

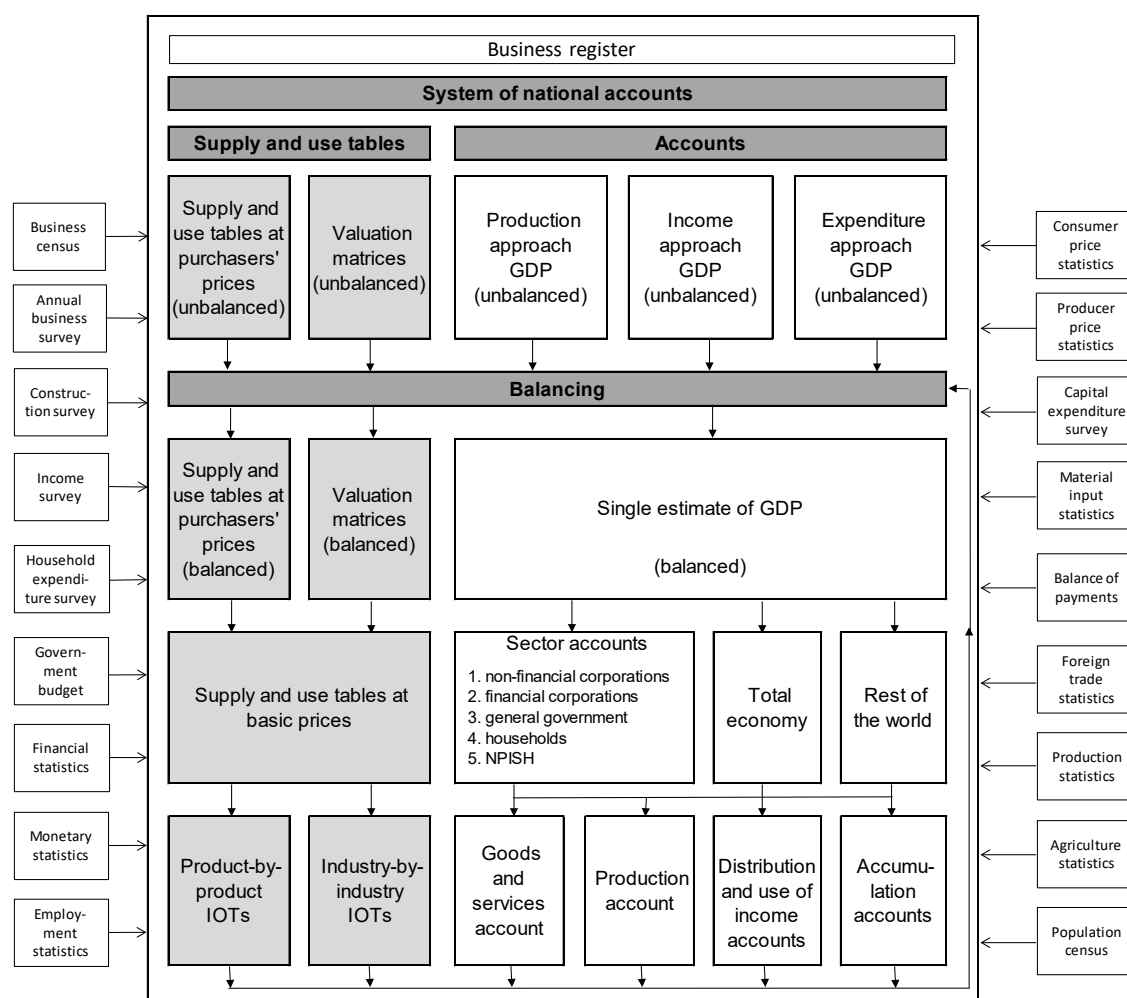
- Agriculture industry data from the agricultural ministry
- Banking industry data from the central bank
- Data on government incomes and expenditures from the finance ministry – expenditure will need to be split between the individual consumption and collective consumption categories. This source can also provide the full range of taxes receivable and subsidies payable

Other sources:

- Company annual report and accounts (in general, the direct use is limited, however these are often used for public utility companies like electricity, gas, water, postal services, telecommunications and others, in many countries)
- Regulatory bodies' accounts
- Insurance data from the insurance industry regulators
- Airline data from the airline industry regulators
- Financial details from company websites which supplement the company annual report and accounts

4.97. The compilation of SUTs, whether carried out as an integral part of the compilation of national accounts (as recommended in this Handbook) or as a separate compilation from that of the national accounts, is based to a large extent on the same data sources as those used for the national accounts. Figure 4.1 expands on figure 2.3 in chapter 2 to show the typical data sources used for the compilation of national accounts, which also feed into the compilation of SUTs and IOTs. Often the same source can provide data feeding into more than one of the approaches to measuring GDP. For example, the agricultural data from agricultural departments often feeds into all three approaches to measuring GDP, thereby ensuring natural consistency and coherence of the data used in SUTs.

4.98. In general, there is a strong correlation with the level of detail used within the SUTs and the quality of the product balances and the aggregates. The more disaggregated the level of industries and products, the higher the degree of matching of individual products in terms of allocation of uses, prices and other characteristics and hence the better the quality. If the product level is too aggregated, the individual products may be too broad and heterogeneous, and therefore of lesser quality.

Figure 4.1 Overview of SUTs and IOTs as part of the SNA compilation

1. Typical data sources

(a) Structural surveys

4.99. Many countries rely on annual structural type surveys which fit in naturally to the needs of the SUTs framework. One of the advantages of such a set-up is that it allows the collection of a range of variables from a single source – the statistical unit – thereby ensuring the consistency and coherence of each variable and across variables. Thus, the employment data will be consistent with the turnover data and, in turn, with the derived GVA data.

4.100. Where annual structural surveys exist, data for a number of key variables are collected with more detail supplemented from other surveys (for example, detailed purchases data). These variables can include:

- Sales of goods and services (and an appropriate product breakdown)
- Purchases of goods and services (and an appropriate product breakdown)
- Purchases and sales of goods for resale without any further processing (this helps to produce the trade margins by type of product)
- Changes in inventories (a split between materials and fuels, work-in-progress and finished goods not sold is required, as the first category affects intermediate consumption and the latter two categories affect output)
- Capital expenditure (and an appropriate product breakdown)
- Employment costs (and an appropriate breakdown)
- Taxes on products and production (covering business rates, excise duties and others)
- Subsidies on products and production (covering agriculture, transport and others)
- Areas such as research and development, and international trade in services, which link to other related surveys

4.101. Often data compiled to respond to statistical regulations are used in the compilation of SUTs. For example, in the European Union, statistical regulations for member States cover various business statistics derived from monthly, quarterly and annual surveys providing short-term indicators and structural business statistics. Data from many of these sources also feed into the compilation of SUTs. The Handbook on the Design and Implementation of Business Surveys (Eurostat, 1998), provides extensive detail on the conduct of such surveys.

(b) Administrative data

4.102. Administrative data constitute a key data source for the compilation of both quarterly and annual national accounts and, in some countries, administrative data may be the main data source. The use of administrative data is growing under the influence of a number of factors, such as good coverage, declining resource impact on national statistical offices and lessening of the response burden.

4.103. Administrative data have statistical strengths and weaknesses vis-à-vis sample surveys. Apart from the low cost of obtaining administrative data, their major strength is that they commonly have complete – or nearly complete – coverage of the fields to which they relate. As a result, there are no sampling errors and any non-sampling errors, such as those arising from an out-of-date business register and inadequate new business provisions, are scarce and minor.

4.104. The weaknesses of administrative data arise from the fact that they are by-products of administrative systems which are not generally designed to meet the needs of the national accounts. Examples of these weaknesses include the following:

- Available data do not meet national accounting definitions (for example, wages rather than compensation of employees, or a measure of depreciation that differs from the national accounting concept of consumption of fixed capital).
- Purchases registered in the VAT system will usually include both purchases for intermediate consumption and for gross capital formation.
- Data are not recorded on an accrual basis (for example, exports and imports from customs are recorded as they cross the customs frontier and not when they change ownership).
- Data are incomplete (for example, movement of oil rigs in and out of territorial waters are excluded from customs data).
- Data may not be disaggregated in a desirable way (for example, government expenditures may not distinguish between wages and intermediate consumption or new motor vehicle registrations may not distinguish between household and business use).
- Administrative data may be untimely (for example, company tax data).
- Administrative data may undergo change as a result of a change in policy.

(c) Business and company accounts-based statistics

4.105. The values of outputs, inputs, gross capital formation and other elements have their counterparts in business accounts or government accounts, but the concepts used in business accounting often do not consistently follow national accounts definitions. Further details on this issue may be found in United Nations (2000b) and Mahajan (2013). A few examples are listed below:

- Differences between concepts: financial intermediation services indirectly measured (referred to as FISIM), insurance services, and others.
- Change in inventories: different valuations, correction for holding gains and losses, and others.
- Distinction between intermediate consumption and capital formation: acquisitions of machinery and equipment included in current expenses, and others.
- Distinction between intermediate consumption and compensation of employees: fringe benefits, links to own account production, and others.
- Use in business accounts of a time frame that does not follow the calendar year: often accounts that are closest to the calendar year can be used as an approximation for the annual

values but it may be appropriate to correct figures for some big enterprises that have accounts that cover other periods or may have large seasonal patterns, such as the gas supply industry.

- Use in government accounts of a fiscal year that differs from the calendar year. It can be misleading to use the information for the nearest fiscal year as if it were identical to calendar year data. Annual data can be calculated by weighting together data for the two fiscal years that overlap the actual year. A better method is, however, to use quarterly or monthly information to split the data from each fiscal year into the shares belonging to different calendar years where such information exists.

(d) VAT-based statistics

4.106. The VAT system usually provides statistics on those units that are covered by the business register, which will also use VAT-registered businesses as a source. The business register generally includes the units that collect VAT on all or a part of their turnover and those that can deduct VAT on all or part of their purchases (capital or current).

4.107. The VAT-based statistics may exist in published form but, even when not published, they can usually be obtained from the authorities that are responsible for collection of VAT, with appropriate service level and confidentiality agreements in place. Typically, this source will contain information on VAT-liable, zero-rated and VAT-exempt turnover and also on deductible and non-deductible purchases with a classification by industries.

4.108. The VAT-based statistics tend to be available on both a quarterly and an annual basis, and are usually available shortly after the reference period. A correction based on the final dates of payment may be necessary if the statistics show payments instead of accruals.

4.109. There are pitfalls to bear in mind when using VAT-based statistics for national accounts purposes:

- The concepts used in VAT-based statistics are different from those used in national accounts. As VAT-based turnover covers sales of products from own production, sales of traded goods and sales of used capital equipment, the VAT purchases cover purchases of goods and services for use as inputs, goods intended for resale and also purchases of capital equipment. Before these figures can be used in national accounts, the VAT purchases must be split into the different shares based on details payments.
- VAT-based statistics do not show figures for units with activities that are not VAT-liable and they may not include units with turnover below certain thresholds. Such thresholds differ from country to country but will often exclude a significant share of the smaller enterprises. The informal economy can – by its very nature – be assumed not to be included in the VAT-based register sources.

- The industry classification of the VAT-based units may not be the same as the classification used in national accounts.
- For practical reasons, the units accepted in VAT-based statistics may be enterprises, kind-of-activity units, establishments, or even conglomerates of enterprises which are allowed a joint registration for payment of VAT.

4.110. Despite such caveats, the VAT-based statistics may still be the most reliable data source for the size of some industries that are poorly covered by other sources. The figures from VAT-based statistics should, however, be seen as the minimal size of the industries in question. It can be necessary to add values for units below the threshold values and VAT-exempt units, including those operating in the informal and hidden economy.

(e) Missing data

4.111. If certain data are not available in the official statistical system, the first option would be to check whether such data are available outside official statistics. One such example is when intermediate data on advertising costs are not available as a separate item in the business surveys. In this case, one possible source could be data from relevant trade associations observing the advertising market. Despite the fact that the data are often not sufficiently comprehensive, or the classifications differ from the official classifications, so that the data do not fully conform to the required concepts, these data certainly give a good indication of the advertising market across the various industries.

4.112. There may be a full set of data observations for a period, and then again for a later period. Various modelling techniques exist for generating estimates for the intervening periods to populate the SUTs, such as the basic Holt-Winters approach (Holt, 1957; Winters, 1960). When balancing SUTs, however, these estimates should be treated as being of much lower quality than the more reliable and up-to-date estimates.

4.113. Furthermore, if no specific data are available, the expert advice of chambers of commerce, trade associations, research institutes or other similar organizations could be useful.

4.114. In certain industries, a single company or a few companies are the big players in that market. Those could also be specifically contacted for expert input or to request some of their internal data on a confidential basis. For example, telecommunication companies may provide their revenue data by type of customer; supermarket chains may be asked to provide data on their sales by products; major railway companies may provide data on the goods transported, and so on.

4.115. Annual company reports and accounts, publications of regulatory bodies and trade associations, and internet company websites are very useful sources of financial data for businesses and households.

4.116. Certain estimates can be based on the identities and coherence of the SUTs framework. This holds true for the application of the product flow method, where detailed supply data are used to estimate certain use data. The product flow method basically applies fixed allocations which will need to be reviewed each year. The method should be applied with great caution in populating SUTs and will depend on the level of product detail. The collection of primary data from various sources with data confrontation provides the best approach to the populating of SUTs and the attainment of good quality results.

(f) Exhaustiveness: methods of grossing up

4.117. Statistical sources usually exclude units with employment or turnover below certain thresholds, while national accounts data should include estimates for these missing units. The methods used for grossing up will typically be based on an estimate of employment in the excluded units, and assumptions on output, input, capital formation and other factors by employee. These assumptions should as far as possible reflect the conditions in comparable units but when the small units are not covered by source data, the grossing up procedure will necessarily add to the uncertainty of the estimated totals. It can also be expected that the structures of outputs and inputs of small units are somewhat different from those found in the units covered by collected data.

4.118. There is considerable interest in the phenomenon of the non-observed economy. This term is used to describe activities that, for one reason or another, are not captured in regular statistical enquiries, because they are underground, illegal or informal, consist in household production for own final use, or simply because of deficiencies in the basic data collection system. Guidance in this regard is provided in the handbook on measuring the non-observed economy (OECD, ILO and Interstate Statistical Commission, 2002) and in the manual on the informal sector (ILO, 2013).

4.119. In countries where a significant share of total output and input is found in the informal economy, it can be appropriate to conduct specific surveys of this activity. With a view to confronting data on the supply and use of labour, useful information on this subject may actually be found in population censuses, household budget surveys or labour force surveys. In this respect, the SUTs framework, in which available source data are combined and balanced, offers the most promising means of arriving at exhaustive estimates of economic activity.

Part three

Chapter 5. Compiling the supply table

A. Introduction

5.1. The first step in the compilation of SUTs and IOTs is the construction of an initial and unbalanced version of the supply table. The values entered into the tables should reflect, as far as possible, all available knowledge and data on the product structure of each column, although many values may need to be changed when the SUTs system is balanced. This applies to estimated totals and also to the values of supply of specific products.

5.2. Before balancing takes place, the estimates for domestic supply, imports of goods and services should be checked for credibility, and if necessary adjusted as appropriate. These will then form the starting point for the balancing process.

5.3. This chapter focuses on the steps and data sources needed to compile this initial, unbalanced version of the supply table. Section B provides a more detailed overview of the structure of the supply table. Section C focuses on the compilation of the domestic output table and the necessary compilation steps and section D focuses on the compilation of the imports of goods and services. Annex A to chapter 5 provides an example of a questionnaire on the collection of sales of goods and services, inventories of goods and trade-related data.

B. Structure of the supply table

5.4. The supply table shows the supply of goods and services for a given period of time by type of product of an economy and distinguishes between the output of domestic industries and imports by type of product. The supply table is generally compiled first at basic prices, reflecting the valuation of the data sources. As illustrated in Table 5.1, the supply table at basic prices consists of two main parts: domestic output and imports of goods and services.

5.5. The domestic output matrix contains information on the supply of products by the different industries. The column for the imports of goods and services contains information on the total imports by product. The matrices for domestic output and imports of goods and services have the same row structure, defined by product categories. This structure allows for the horizontal aggregation of all the elements and the transition from total supply of products at basic prices to total supply at purchasers' prices.

Table 5.1 Numerical example of a supply table at basic prices

Millions of euros

		INDUSTRIES							Imports	Total supply at basic prices
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	Output at basic prices		
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
PRODUCTS	Agriculture (1)	8 782						8 782	3 271	12 052
	Manufacturing (2)	796	182 982	643	1 808	133	44	186 405	124 590	310 995
	Construction (3)	83	961	43 060	734	255	179	45 272	563	45 835
	Trade (4)	1	4 773	311	54 204	640	257	60 187	600	60 787
	Transport (5)	13	465	66	25 538	128	125	26 335	8 150	34 485
	Communication (6)	160	1 781	139	43 912	1 253	982	48 228	6 234	54 463
	Finance and business services (7)	29	8 902	698	7 588	106 909	3 381	127 508	7 061	134 569
	Other services (8)	3	85	13	1 053	143	74 346	75 643	824	76 467
	Total (9)	9 867	199 950	44 931	134 837	109 461	79 314	578 360	151 293	729 653
Adjustments	CIF/FOB adjustments on imports (10)								- 97	- 97
	Direct purchases abroad by residents (11)								6 675	6 675
	Total (12)	9 867	199 950	44 931	134 837	109 461	79 314	578 360	157 871	736 230
Total of which:										
	Market output (13)	9 763	195 916	41 462	127 401	88 330	18 116	480 989		
	Output for own final use (14)	104	4 029	3 468	2 134	19 890	2 670	32 295		
	Non-market output (15)	0	4	0	5 302	1 241	58 528	65 075		

Table based on 2011 figures from Austria

5.6. The supply table at basic prices is then transformed into the supply table at purchasers' prices, through the addition of valuation adjustments represented by valuation matrices covering trade margins, transport margins, taxes on products and subsidies on products. Table 5.2 shows the valuation adjustments which are added to the columns of the supply table at basic prices to arrive at the total supply of each product at purchasers' prices.

5.7. The first step in the compilation of an initial version of the supply table therefore involves the compilation of data for total domestic output at basic prices and imports valued at CIF prices aggregated to total supply at basic prices. The second step involves the compilation of trade and transport margins, taxes on products less subsidies on products which are used to convert the total supply of products at basic prices to the total supply of products at purchasers' prices.

5.8. The data in the domestic output matrix are valued at basic prices, which is the amount receivable by the producer from the purchaser for a unit of a good or service produced as output minus any tax payable, plus any subsidy receivable by the producer as a consequence of its production or sale. The value of output of goods excludes any transport charges invoiced separately by the producer.

5.9. Data on imports by product from foreign trade statistics are usually valued at CIF prices. In the 2008 SNA and BPM 6, however, in which total imports of goods are valued FOB, an extra row has to be added for the CIF/FOB adjustments on imports, in order to reconcile the different valuations. These adjustments are shown in row (10) of Table 5.1 and explained in detail in section

D. In addition, a further adjustment is added in the supply table to account for direct purchases made abroad by residents: this is shown in row (11) in Table 5.1.

5.10. These adjustments in the supply table (rows (10) and (11) in Table 5.1 and Table 5.2) have corresponding entries in the use table (rows (10) and (11) in table 6.1) under the columns for exports and final consumption expenditures by households. It should be noted that some countries do not show these estimates in separate rows but consolidate the values across the product groups in the respective columns, thereby providing a different product balance.

Table 5.2 Supply table at basic prices, including a transformation into purchasers' prices

Millions of euros											
			INDUSTRIES						Output at basic prices	Imports	Total supply at basic prices
			Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services			
PRODUCTS	Agriculture	(1)	8 782	0	0	0	0	0	8 782	3 271	12 052
	Manufacturing	(2)	796	182 982	643	1 808	133	44	186 405	124 590	310 995
	Construction	(3)	83	961	43 060	734	255	179	45 272	563	45 835
	Trade	(4)	1	4 773	311	54 204	640	257	60 187	600	60 787
	Transport	(5)	13	465	66	25 538	128	125	26 335	8 150	34 485
	Communication	(6)	160	1 781	139	43 912	1 253	982	48 228	6 234	54 463
	Finance and business services	(7)	29	8 902	698	7 588	106 909	3 381	127 508	7 061	134 569
	Other services	(8)	3	85	13	1 053	143	74 346	75 643	824	76 467
	Total	(9)	9 867	199 950	44 931	134 837	109 461	79 314	578 360	151 293	729 653
Adjustments	CIF/FOB adjustments on imports	(10)								- 97	- 97
	Direct purchases abroad by residents	(11)								6 675	6 675
	Total	(12)	9 867	199 950	44 931	134 837	109 461	79 314	578 360	157 871	736 230
Total of which:											
	Market output	(13)	9 763	195 916	41 462	127 401	88 330	18 116	480 989		
	Output for own final use	(14)	104	4 029	3 468	2 134	19 890	2 670	32 295		
	Non-market output	(15)	0	4	0	5 302	1 241	58 528	65 075		

		Total supply at basic prices	VALUATION MATRICES						Total supply at purchasers' prices		
			Trade margins	Transport margins	VAT	Taxes on products	Subsidies on products	Total			
										(10)	(11)
PRODUCTS	Agriculture	(1)	12 052	1 926	274	329	57	- 107	2 479	14 532	
	Manufacturing	(2)	310 995	48 838	2 540	13 175	7 866	- 49	72 370	383 364	
	Construction	(3)	45 835	0	0	1 529	13	0	1 542	47 377	
	Trade	(4)	60 787	- 52 341	0	575	11	0	- 51 755	9 032	
	Transport	(5)	34 485	0	- 2 800	558	71	- 448	- 2 620	31 865	
	Communication	(6)	54 463	1 493	9	3 375	217	- 34	5 059	59 522	
	Finance and business services	(7)	134 569	0	- 22	2 706	2 159	0	4 842	139 411	
	Other services	(8)	76 467	85	0	1 201	576	0	1 861	78 329	
	Total	(9)	729 653	0	0	23 447	10 969	- 638	33 778	763 431	
Adjustments	CIF/FOB adjustments on imports	(10)	- 97							- 97	- 97
	Direct purchases abroad by residents	(11)	6 675							6 675	6 675
	Total	(12)	736 230	0	0	23 447	10 969	- 638	40 356	770 009	
Total of which:											
	Market output	(13)									
	Output for own final use	(14)									
	Non-market output	(15)									

Table based on 2011 figures from Austria

5.11. A distinction may be made in the SUTs between the three types of production: market output; output for own final use; and non-market output. In the domestic output matrix, however, these three categories of production are usually grouped together in the relevant industries and shown in three supplementary rows for each industry. Thus government services are distributed in the system according to the various activities in which the government is engaged: for example, public administration services, education services, health services, recreation services, social welfare services, and others, but are shown together with the corresponding market producers. For example, health services provided by market and non-market producers (within the same industry) are shown as a total. Furthermore, for some industries, the supplementary rows are useful for the link with the institutional sector accounts.

5.12. Although the supplementary rows make it possible to split output by industry into the three categories of output, there is no product dimension. Ideally, each industry should be shown separately (also reflecting different structures and links between the output and the inputs) or additional analyses produced for the user.

5.13. Imports of goods and services are classified by type of product. Since this table is designed to show the total supply by type of products, the valuation of imports of goods should be compatible with the valuation of the domestic production of goods. Imports by type of product are therefore valued at CIF prices which are comparable with the domestic output at basic prices. Although total imports of goods are valued at FOB prices, it is not easy to move the imports of goods by type of product to CIF prices, as this depends upon the source data and the person providing the transportation (see 2008 SNA, para. 14.77).

5.14. The addition of the two components – production and imports – gives the total supply of products at basic prices.

5.15. The supply table at purchasers' prices is obtained by adding various valuation matrices (earned on both domestic output and imports) to the total supply at basic prices, thus enabling movement from one valuation to another. The valuation matrices include:

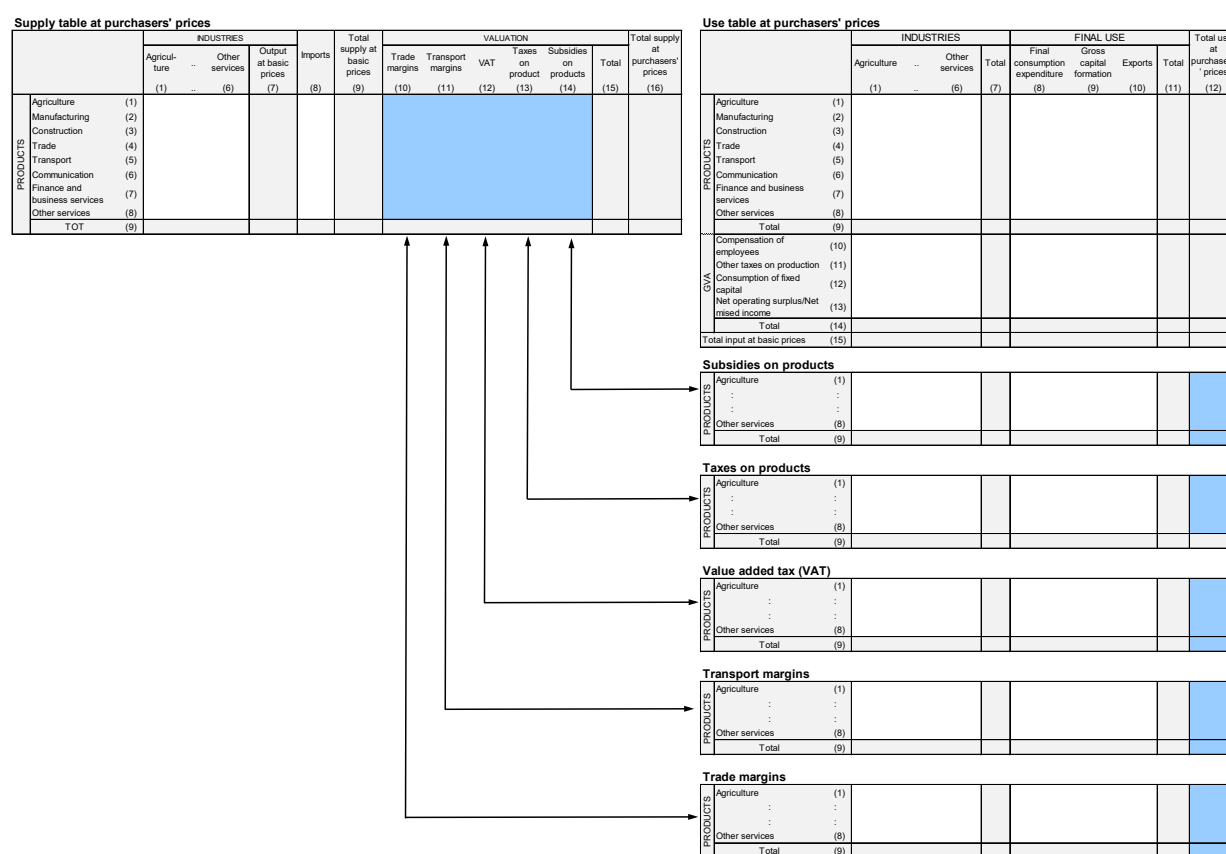
- Trade margins
- Transport margins
- Taxes on products (with non-deductible VAT treated separately from other taxes on products)
- Subsidies on products (which are deducted)

5.16. It should be noted that, when the supply table is shown with the final column summing to purchasers' prices, it is referred to as the supply table at purchasers' prices. This is actually just the supply table at basic prices with the addition of the valuation columns. The production and import sections of the supply table have not been changed and remain valued at basic prices.

5.17. The task of compiling SUTs is a highly integrated process. This is particularly true for the estimation of the valuation vectors and matrices, where it is often necessary to rely also on estimates from the use table side in order to obtain the valuation vectors entered into the supply table. Figure 5.1 provides an overview of how the valuation matrix in the supply table is linked to a sequence of valuation matrices in the use table. This figure also demonstrates the interconnections between the valuation matrices linking the supply table and use table. The estimation of the valuation matrices, considering both the supply table and the use table, is dealt with in chapter 7.

5.18. The rest of this chapter focuses on the compilation of the supply table at basic prices.

Figure 5.1 Link between valuation matrices in the supply table and the use table



C. Domestic output

1. Structure of the domestic output table

5.19. The first and most elaborated part of the supply table is the domestic output matrix. This records data on the production of the economy classified along two dimensions: the rows represent the type of products (based on CPC Version 2.1) and the columns represent the different industry groupings (based on ISIC Rev. 4). Thus the rows show the domestic output matrix, a single product by producing industry, and the columns show all the products produced by a single industry.

Although this is consistent with CPC and ISIC, countries may however use different and more detailed classifications, for example, those which reflect country-specific activities.

5.20. The domestic output matrix reflects the principal and secondary products of industries including by-products. It is the principal activity of the statistical unit that determines its classification to a specific industry. In special cases where the domestic output matrix is square (the number of products being equal to the number of industries), and the sequence of products arranged to reflect the sequence of the industries (based on their principal activities), the principal activity of an industry is reported on the diagonal of the matrix, while the secondary activities of an industry are listed as off-diagonal entries.

5.21. In practice, however, it is common to have more products than industries. For this reason, the production part of the supply table is usually a rectangular matrix with more rows than columns, as shown in Table 5.1. This reflects the fact that it may be of more interest to specify, for example, different kinds of agricultural crops, in the case of agriculture, and of less interest or practical use to distinguish farms specializing in each possible type of crop. In this case, all the crops would still form the principal output of agriculture, whereas, for example, the production of wine or the construction of buildings for own use would be treated as secondary outputs of the industry. The greater the level of product detail, the more scattered the entries will be around the principal products. In these cases, it is not possible to observe directly the distinction of principal products versus secondary products or production in the rectangular domestic output matrix.

5.22. Annex A to this chapter provides an extract of a survey questionnaire collecting data on sales of goods and services by type of product, in addition to other variables by product, such as opening stocks (inventories), closing stocks (inventories) and trade margins.

5.23. Even though the industry concept is already being applied in the national accounts, the existing level of detail or precise delimitation should not be viewed as a constraint when compiling SUTs and, in particular, when compiling benchmark tables. On the other hand, the way in which statistical units are defined and classified in the business register and covered in basic statistics represents a real constraint on the possible choices concerning industries in the SUTs. Even though industries may, in the process of compiling SUTs, be redefined to some extent or otherwise modified in terms of their basic statistics, the options are much more limited than the range of choices available when it comes to deciding what product classification should be applied.

5.24. The choice of the level of detail for industries and products to be used in the SUTs must be based on a thorough examination of the available statistics and considerations concerning the advantages of using product details in balancing, in estimating margins and taxes on products by uses, final uses by purpose, in volume estimates and in other applications. The general recommendation, however, is to work with as much detail as possible, as any aggregation of basic statistics will also entail a loss of information that could at some stage have contributed to the overall quality of the balanced SUTs (on this issue, see also chapter 4).

5.25. It is also necessary to clarify any user requirements about the format of the final table, including international reporting. In general, it would be an advantage to work at a more detailed level than that warranted by current uses, in order to extract the maximum information from available data sources and to be prepared for the emergence of new uses and for transformations necessary to comply with future changes to economic activity and product classifications.

2. Primary statistics and data sources

5.26. The structure of economic entities varies from small enterprises engaged in one or a few activities that are undertaken either at, or from, a single geographical location to large and complex enterprises engaged in many different activities. These enterprises may be horizontally or vertically integrated, and their activities may be undertaken either at, or from, many geographical locations. The way in which producer units are defined, measured statistically, broken down or aggregated is of fundamental importance when compiling SUTs.

5.27. In practice, compilers of SUTs will not deal with individual economic units but only with the aggregates of units in the form of industries, usually based on current business statistics by economic activity. To arrive at a full understanding of the role of these statistics in the compilation of SUTs, it is necessary to assess the delimitation of units that influence the properties of the industries.

5.28. The most important prerequisite for the collection of basic statistics is the business register and the types of economic units that it holds. Ideally, business registers will contain two types of unit: enterprise units and establishments.

5.29. Usually the enterprises form the core units of the business register, as they are easier to identify and track on a current basis because of their legal status. The number of establishments created depends on the register policy adopted (in other words, how many enterprises are partitioned into establishments). Different geographical locations of the production units will be one of the main criteria for subdividing an enterprise into several establishments.

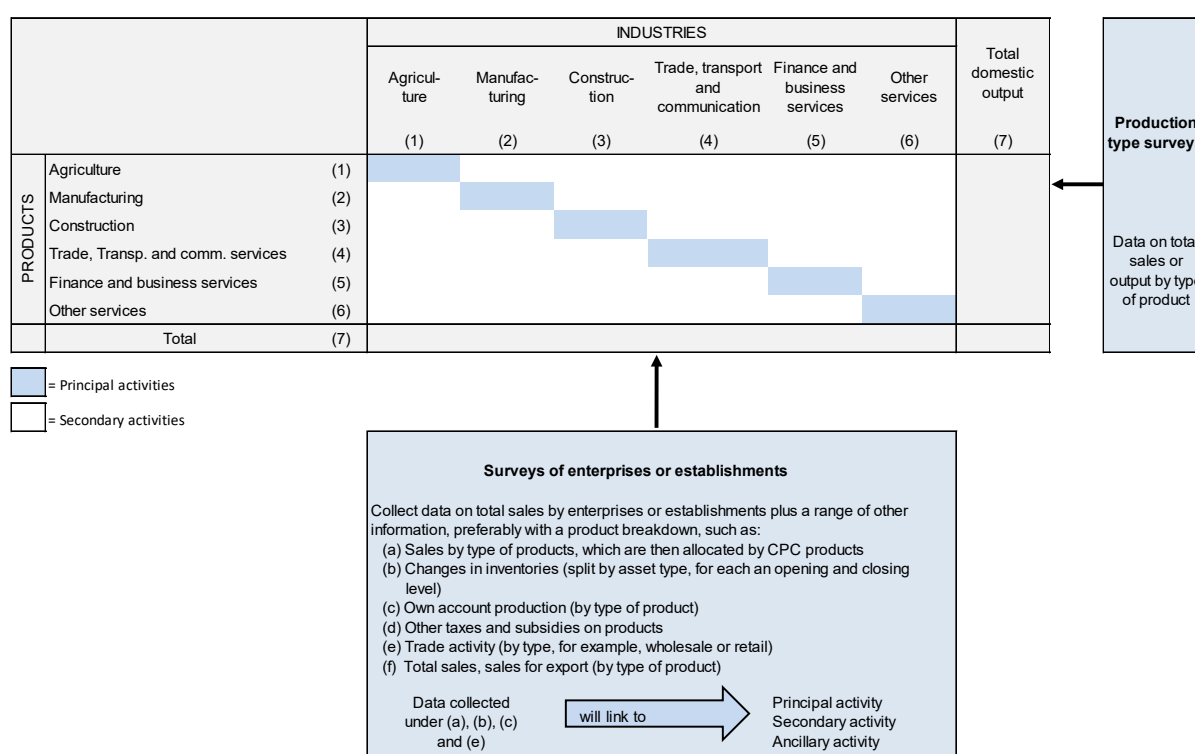
5.30. In the collection of basic statistics, the enterprise will usually serve as the collection entity and, to the extent that the enterprise is made up of several establishments, the enterprise will be requested to report a separate range of statistics for each of those establishments. This has implications for both the supply table and the use table, as some types of costs can only be reported at the enterprise level, whereas it is possible to report all regular production costs for the individual establishments.

5.31. As the large majority of enterprises are small or medium in size, and tend to engage in one kind of activity only, the enterprise and the establishment units may be identical in these cases. Large enterprises, however, which often contribute the bulk of the production of an economy, will often cover different kinds of economic activity and therefore, in formal terms, be made up of several establishment units.

5.32. It is important to note that many primary sources, such as enterprise surveys and production surveys, which are used to collect data feeding into the domestic output matrix, often also collect data at the same time through the same survey questionnaire but feed these into the use table (for example, data on the industries' input structures and gross fixed capital formation). This approach provides the data feeding into the SUTs with a high degree of coherence and consistency.

5.33. The estimates of the domestic output matrix are usually based on two main types of information sources: enterprise surveys and production surveys. Additional information, such as administrative sources, company accounts and others, will also be used. Figure 5.2 provides a simplified view of the different types of information used in compiling the domestic output matrix.

Figure 5.2 Different types of information used in compiling the production matrix



5.34. Starting with the enterprise survey, the principal objective is to supply information on the main structural characteristics of the different economic activities. The basic unit of this type of survey tends to be the enterprise. From this source, it is possible to estimate the total production by activity, starting with its private accounting business systems. On the other hand, the production surveys allow the estimation of the total production by type of product.

5.35. Combining both sources of information, enterprise statistics and production statistics, it is possible to combine the data and obtain the production by type of product, by principal activities of the enterprise and by principal activities of the establishment that belongs to this enterprise. Consequently, the principal production and the secondary production of a product can be identified, primarily for industrial products. In many cases, lack of information makes it necessary

to use reasonable assumptions about what products are constitute the industries' secondary production.

5.36. Many enterprises may perform some construction work, for example, own-account gross capital formation and minor maintenance and repair work. Enterprises in the manufacturing and service industries are often involved in either wholesale trade or retail trade or even both. Many service industry enterprises may provide retail trade services as a secondary activity. Lastly, activities involving the rental of real estate and leasing of equipment are often secondary activities.

5.37. In basic statistics, output by products will usually be available for goods-producing industries such as agriculture, mining and manufacturing industries – at least for enterprises and establishments above a certain threshold – and similarly imports and exports of goods will be covered in great detail by external trade statistics. For service industries, a breakdown of output by individual kinds of services (as defined in the CPC classification) is less common, although in recent years many countries have developed such statistics. If there is a lack of product statistics for services, the output by the most detailed service activities of ISIC may be used as proxy, assuming that all output consists of the services characteristic for that particular industry. Concerning product breakdowns, construction will be placed between these two extremes.

5.38. For manufacturing units below a certain threshold, output statistics by products will usually be missing, whereas total output will be estimated based on either business surveys or administrative records. Working at the most detailed activity level available, output from these small units can be broken down into the products of the system, for example, by assuming that the composition by product is identical to that which has been observed for the smallest category of those units for which output statistics by product exist. During the balancing process, this assumption may be modified and the output redistributed by product.

5.39. The products recorded in the domestic output part of the supply table should be output valued at basic prices at the time it is completed. For manufacturing industries, it is usually only sales by product that are given in the surveys and adjustments would need to be made for change in inventories of finished goods and work-in-progress, in order to move from sales to output.

5.40. When information exists about opening and closing inventories by industry, it can be assumed that the composition by product is identical to the sales by product. The change in inventories of finished products and work-in-progress by product can be derived by applying relevant price indices and assumptions about inventory valuation principles used by enterprises. As the reliability of these data by product is limited, however, and as, by definition, these data should always be identical on the supply side and the use side, there is no need to adjust the sales figures or enter them into the final use category of changes of inventories at this stage. In the system, it is the actual sales figures that are relevant for the distribution by users, and the estimated data for change in inventories can therefore be simply imposed on the system after the balancing has been completed. This is, however, not the case for change in inventories of materials and fuels

(recorded in the intermediate use part of the use table) and in trade. Changes in inventories of agricultural products and mining products will usually have to be included in the system from the outset, as the output data will often refer not to sales but to actual output.

3. Principal and secondary products

5.41. The distinction between principal and secondary production has traditionally played a prominent role in I-O literature, as the existence of secondary production requires certain assumptions for IOTs to be compiled. It should be noted, however, that a match between products and industries (determining in which industry a product is the principal output) is really only necessary in those cases where the chosen techniques for deriving IOTs as a starting point requires the SUTs to be aggregated to square tables, cases where the sequence of the aggregated products is made comparable to the sequence of industries. With regard to other techniques for the compilation of IOTs and for the purpose of the SUTs, there is no need to match products and industries. It should thus be noted that, when industry-by-industry IOTs are derived on the assumption of fixed product sales structures, there is no need first to aggregate the rectangular SUTs.

5.42. When necessary, the match between product and principal producer can be derived either theoretically (by identifying for each product the principal producer according to the ISIC definitions of the principal products of each industry – the correspondence keys are available on the United Nations Statistics Division classification website at <http://unstats.un.org/unsd/class/default.asp>) or empirically (established by observation of the actual domestic output matrix, the industry being the main producer of each product).

5.43. In principle, the empirical match will be the most precise in the sense that it depicts the production relationships as they actually exist in this particular economy. The theoretical match may be the preferred approach when considering time series and international comparisons. These matching methods also demonstrate that the product classification applied when compiling SUTs can be chosen completely independently of whether or not it may subsequently be necessary to derive square SUTs.

5.44. When the domestic output matrix is aggregated to a square matrix and arranged so that the entries for the primary products fall on the diagonal, the off-diagonal elements show the extent of secondary production. This refers to that part of a product which is produced by industries other than the one where it principally belongs either formally according to the industrial classification (theoretical aggregation key), or according to the industry which is actually the main producer (empirical aggregation key).

5.45. As the secondary production observed in the domestic output matrix depends on the level of aggregation both of products and of industries, secondary production does not possess any observable characteristics of its own. The elusive character of the concept of secondary production makes it difficult to justify that a product should be of particular interest statistically just because

it is produced in two or more industries at a certain level of industry or product aggregation. When the industry and product classifications to be used in SUTs have been decided (inclusive of possible redefinitions), the principal versus secondary distinction plays no role in the subsequent elaboration and balancing within the SUTs framework.

5.46. For most countries, the domestic output matrix is characterized by showing secondary production almost exclusively for manufacturing industries, whereas for most other industries, practically all the production is found on the diagonal elements (or in the rectangular table – in what is known as the “diagonal field”). There are three main reasons for this:

- Basic statistics for manufacturing industries have traditionally included detailed product statistics and thus make identification of secondary production possible.
- For service industries, the diagonal structure is simply due to the fact that, more often than not, very limited detail has been collected on the type of product breakdown of these service activities. Thus, the total output from establishments (or even enterprises) must be assumed to be the primary output of the industries to which the units are classified in the business register.
- The activities of industries such as agriculture, construction and trade are often defined in a purer form (the industries covering all their principal products, and only those) in the national accounts and SUTs than in the business register. In this case, all secondary agricultural, construction and trade activities in other industries would have been transferred to their main activities. Alternatively, data for some activities may have been constructed in such a way that from the outset no secondary production exists, for example, agricultural activity is measured as the sum of all agricultural products, construction activity as the value of new construction and repairs, and so forth.

5.47. When the rectangular SUTs have been balanced, there may be a need to aggregate them into a square system either for dissemination purposes or for compiling IOTs by methods that require square SUTs. In a square system, the number of product groups must be identical to the number of industries, and furthermore, the products aggregated in such a way that the resulting product groups can be given corresponding industry names, indicating the industry of which they are principal products. If aggregation is made solely for dissemination purposes, the product aggregation could also be carried out to, for example, higher levels of CPC – which, as mentioned above, have no direct correspondence with ISIC defined industries.

4. Ancillary activities

5.48. When the production of an enterprise takes place in two or more different establishments, certain ancillary activities may be carried out centrally for the collective benefit of all the establishments. If, in such a case, a producer unit is undertaking purely ancillary activities and is statistically observable, in that separable accounts for the production that it undertakes are readily

available, or if it is in a geographically different location from the establishments that it serves, it may be desirable and useful to consider it as a separate unit and to allocate it to the industrial classification corresponding to its principal activity. Another exception is the case when some products are used both for own ancillary use and also supplied to another unit (see 2008 SNA, para. 14.33). In the 2008 SNA, however, it is recommended that statisticians do not make extraordinary efforts to create separate establishments for these activities artificially in cases where suitable basic data are not available.

5.49. The fact that establishments, even at a detailed level, are classified to the same activity, does not mean that they are in all respects identical. Each establishment has its own unique institutional and organizational characteristics, which may influence the composition of its purchases as much as the underlying technical production processes. Two establishments producing identical products may have quite different input structures, depending on the degree of reliance on purchased semi-fabricated products, outsourcing of certain activities (see also chapter 8, section D, on goods sent abroad for processing), whether it owns the capital equipment and buildings it uses rather than leasing or renting them, and so forth, and in general, on the degree of vertical integration of the various stages of the production processes. There is no way that these institutional characteristics inherent in the original source data should be eliminated from the SUTs (or subsequently from the IOTs) nor does the SNA expect the compilers to try to do so.

5.50. Institutional arrangements of production not only differ between establishments classified in the same industry but also across countries and over time. It is obvious that there are serious limitations to the view that the SUTs (and the IOTs) portray the technical characteristics of a production system. From a statistical point of view, the achievable elimination of institutional arrangements is obtained by using establishments as production units (with the possible additional partitioning of vertically integrated enterprises as discussed above), given that the establishments are designed with this purpose in mind and there are no official statistics providing production structures below this level.

5.51. In some countries, the recommended establishment unit approach may not be achievable in practice, since, for legal, practical or historic reasons, statistics are only collected for enterprise units. Even though compilers of the SUTs may in this situation try to break down the most important multi-activity enterprises into their constituent establishment units, there is in general no feasible alternative to working with the existing data. In this case it is still possible to compile SUTs, although the overall picture of the productive system will become less precise and to some extent blurred, which will also have an adverse effect on the resulting SUTs and IOTs.

5.52. It should, however, be recognized that the important objectives of compiling the SUTs may also be achieved when the data are based on enterprise units (see SNA 2008, para. 14.21), although some product-flow and common-sense procedures may be more difficult to apply because of the less stringent definition of industries, as the composition of output from the enterprise units will also be crossing the borderlines between sections of ISIC. For these reasons, an enterprise-based

approach will in general require more thorough coverage of statistical source data. It should also be borne in mind that there are no automatic methods available that can disentangle this dataset and transform it into SUTs or IOTs with analytical properties comparable with those resulting from SUTs based on establishment-type units. Depending on the specific circumstances, it may in such cases be decided to compile SUTs alone and not IOTs.

5. Redefinitions

5.53. Redefinitions refer to adjustments made to the source data relative to the way in which they are obtained from the primary statistics, in order to obtain “purer” industries, so to speak, for use in the SUTs. This is an exception to the previously mentioned rule that SUTs compilers should not attempt to create their own versions of basic statistics. That would not be cost-effective nor would it be conducive to the comparability of SUTs with other economic statistics or on an international scale. In practice, deviations from the way in which enterprises or establishments are defined in the business register and reflected in primary statistics should be limited in scope.

5.54. Such redefinitions may be seen as implementing the SNA recommendation to partition vertically or horizontally integrated enterprises or establishments that have production in two or more sections of ISIC Rev. 4 (2008 SNA, paras. 5.52–5.54). Redefinitions are generally carried out manually, using product-specific input structures based on specific insight into the activities, leading to results that will come as no surprise nor give rise to negative elements, as might have been the case had more automatic methods been applied. By reducing secondary production, redefinitions facilitate the subsequent compilation of IOTs, and compilers of SUTs should be aware of how the choice of compilation techniques will affect the subsequent calculation of the IOTs.

5.55. Redefinitions (more background is provided in Box 5.1) are usually concentrated on a few major activities, such as agriculture, energy, construction and trade, or a few major enterprises, such as mining operations. Redefinitions affect all those activities from which secondary output is being removed. For some activities, redefinition-type adjustments may have been carried out already in the source data, as in the following cases:

- The European Union System of Agricultural Accounts requires that all agricultural activity is covered by these accounts and there are very limited possibilities for the retention of non-agricultural secondary production within the system definition of agriculture.
- All rented dwellings are usually grouped together in one single industry (together with owner-occupied dwellings) independently of the activity of the actual owner.
- Trade activities outside the trade industries (trade as secondary activity), by definition, have already been separately identified when compiling the national accounts, as only the trade margins, and not the gross turnover of the traded products, should be counted as output, and may have been grouped together with trade as primary activity.

- Construction activities are also frequently redefined to form one single, “pure” construction activity, often because total output has been defined by adding up the values of specific types of construction output rather than output from building establishments, or alternatively inputs have been determined from the supply of construction materials. Any of these approaches will also facilitate the distribution of building materials for intermediate consumption.

Box 5.1 Redefinitions

In Miller and Blair (2009), on page 141, redefinitions are defined as: “Factoring out the amount of secondary products produced as well as the inputs used in that production and reassigning both to the industry for which the product is classified as primary”.

In addition, a distinction is made between *specific redefinition* and *mechanical redefinition* (page 215), where the former is the “by hand” procedure and the latter refers to the various mathematical procedures that can be applied to eliminate secondary production when producing IOTs from SUTs (covered in chapter 12 of Miller and Blair).

The specific redefinition or two-step process emerges from the practice in several countries. It is explained in detail for the United States in Guo J. and others (2002). The Bureau of Economic Analysis paper was presented at the fourteenth International Conference on Input-Output Techniques, held in Montreal, Canada, in 2002. The article also analyses the differences between the resulting tables when redefinitions are not applied (case 1), and when they are applied (case 2).

The redefinition method is also used in Canada and Denmark, whereas the industry-by-industry IOTs in Norway are more of the case 1 type, in that they retain the micro-macro link to a maximum degree.

The industry-by-industry IOTs of the Netherlands seems to fit somewhere between case 1 and case 2.

In France, the first step (redéfinition) is based on enterprise units and is carried out to an extent that the supply table becomes diagonal. The use tables thereby also form the IOTs, and the second step (compiling the IOTs) becomes superfluous.

5.56. Although the redefinitions serve the purpose of creating purer activities and thus facilitate I-O analysis, their main purpose is to arrive at an activity classification that is applicable for use in the national accounts, and thus conducive to the integrated compilation of SUTs and national accounts. Three different situations can be distinguished:

- Case 1: no redefinitions take place in the national accounts, the SUTs and the industry-by-industry IOTs.
- Case 2: redefinitions have been carried out for all national accounts data and in the SUTs prior to the calculation of the industry-by-industry IOTs.
- Case 3: redefinitions are not carried out when the current national accounts are compiled but applied when the SUTs and the industry-by-industry IOTs are compiled.

5.57. In the first two cases, the consistency and comparability of the current national accounts (tables by industry), and of the SUTs and IOTs classifications are upheld, but that is not so in the third case. Ideally, the choice of redefinitions should already be introduced in the general

classification of industries used in national accounts. Not only will the manually prepared redefinitions be more precise at these earlier stages, but they will also facilitate the balancing of the system as the need to make a large number of small input entries to many cells of the use table will be obviated.

D. Imports of goods and services

1. General description and definition

5.58. The second part of the supply table covers the total imports of goods and services. In national accounts, imports refer to transactions that occur when there are changes of economic ownership of goods between residents and non-residents, whether or not there are corresponding physical movements of goods across frontiers.

5.59. International merchandise trade statistics represent the main source of data for imports of goods. International standards are specified in International Merchandise Trade Statistics: Concepts and Definitions (IMTS 2010) (United Nations, 2011). For imports of services, the main source of data is either the details available in the balance of payments statistics or specialized statistics on international trade in services (for example, business surveys), according to the international standards given in the 2010 Manual on Statistics of International Trade in Services (United Nations, European Commission, IMF, OECD and WTO, 2011) in connection with product classifications.

5.60. Some differences exist, however, between the concepts used in international trade statistics and the 2008 SNA and BPM 6, and adjustments therefore need to be made to the basic statistics in order that they can be used in the SUTs. The BPM 6 identifies sources of difference between the IMTS and the 2008 SNA and BPM 6 concepts of imports that may occur in countries. In this regard, it recommended that a standard reconciliation table be compiled to assist users in understanding these differences.

5.61. One major difference is the valuation used to record imports of goods. While IMTS 2010 uses a CIF valuation for imports, the 2008 SNA and BPM 6 use a uniform FOB valuation for both exports and imports of goods. The 2008 SNA states, in paragraph 3.149:

Imports and exports of goods are recorded in the SNA at border values. Total imports and exports of goods are valued free-on-board (FOB, that is, at the exporter's customs frontier). As it may not be possible to obtain FOB values for detailed product breakdowns, the tables containing details on foreign trade show imports of goods valued at the importer's customs frontier (CIF, that is, cost, insurance and freight), supplemented with global adjustments to FOB values. CIF values include the insurance and freight charges incurred between the exporter's frontier and that of the importer. The value on the commercial invoice may of course differ from both of these.

5.62. The adjustments for the FOB and CIF valuation of imports are described in more detail in the next section.

5.63. Another difference is the time of recording. In the 2008 SNA and BPM 6, the time of recording of imports and exports corresponds to the time that ownership of the goods is transferred. By contrast, IMTS follows the timing of customs processing. While this timing is often an acceptable approximation to the change of economic ownership principle, adjustments may be needed in some cases, such as for items with large values or goods sent on consignment (that is, dispatched before they are sold). It should be noted that, in the case of goods sent abroad for processing with no change of economic ownership, the values of goods movements are included in the IMTS-based recording but are to be excluded from the ownership-based recording in the national accounts and balance of payments. It is recommended, however, that the values of goods movements be entered as supplementary items in the balance of payments, to indicate the nature of these arrangements.

5.64. Other adjustments to IMTS may be needed to bring estimates into line with the change of economic ownership of goods, either generally or because of the particular coverage of each country. Possible examples include:

- Merchanting
- Non-monetary gold
- Goods entering or leaving the territory illegally
- Goods procured in ports by carriers
- Goods moving physically but where no change of economic ownership has taken place such as in cases of operating leases

5.65. To maintain consistency with BPM 6, the 2008 SNA introduced new treatment relating to merchanting and goods sent abroad for processing. Merchanting is a process whereby a unit in economy X purchases goods from economy Y for sale in economy Z (sometimes within economy Y itself). The goods legally change ownership but do not physically enter the economy where the owner is resident. By convention, the purchases of the goods intended for resale is shown as negative exports. When the goods are sold, they are shown as positive exports. When the purchase and sale take place in the same period, the difference is shown as an addition to exports. If the purchase takes place in an accounting period, the negative export is offset by an increase in inventories of goods for resale, even though those goods are held abroad.

5.66. The surplus on this item in the foreign trade statistics is by its nature a trade margin and should be included in the output of the industry. In the main, this activity takes place in the trade industry. In exceptional cases, this may lead to an overall deficit on the item in the foreign trade statistics but the trade margin would usually still remain positive (the deficit added to changes in inventories). As indicated, trade margins from merchanting activity primarily occur in the trade

industry but can occur in many other industries, unless all trade is redefined to the trade industry. Given that business statistics provide source data as a starting point for the compilation of SUTs, merchanting activity can then appear in various industries, for example, oil companies and pharmaceutical companies.

5.67. The new treatment of goods sent abroad for processing is dealt with in more detail in chapter 8 of this Handbook.

5.68. A special category within imports is the direct purchases abroad by residents. This item covers all purchases of goods and services made by residents while travelling abroad for business or pleasure. Two categories must be distinguished because they require different treatments:

- Expenditure by resident business travellers: this item refers to intermediate consumption of several industries to which the travellers belong (in the use table) and imports of services (in the supply table).
- Expenditure by other resident travellers on personal trips: this expenditure is recorded in final consumption expenditures by households (in the use table) and imports of services (in the supply table).

5.69. Imports broken down by products in the SUTs do not include direct purchases abroad by residents. Consequently, these must be included in an adjustment row to obtain the overall value of imports (row (11) in Table 5.1).

5.70. In tables 5.1 and 5.2, the estimates for CIF/FOB adjustments on imports and the direct purchases abroad by residents are shown separately in the rows. It should be noted, however, that some countries do not show these estimates in separate rows but consolidate the values in the product groups in the respective columns. This situation in turn leads to different product balances but does not change the imports aggregate total. This is often due to the coverage of the data sources and, in these cases, appropriate adjustments should be applied to extract the corresponding entries to generate the separate entries.

5.71. Goods procured in ports by carriers may be included in a similar adjustment row. It should also be noted that imports and exports of ships and aircraft may have to be given special attention, as these transactions may follow special recording procedures in the external trade statistics that are not consistent with the way in which output or gross fixed capital formation should be recorded in the national accounts.

5.72. Imports of goods and services in SUTs are dealt with in more detail in chapter 8.

2. Valuation for imports: CIF and FOB valuation

5.73. In the 2008 SNA and BPM 6, the total imports of goods are valued FOB. The data on imports by detailed products from the foreign trade statistics used in the SUTs are usually available at CIF prices, however, following the International Merchandise Trade Statistics (United Nations,

2011). To reconcile the different valuations used for total imports of goods and for the product components of imports, two types of adjustments are needed. These adjustments are presented below.

(a) Data adjustment

5.74. The first type of adjustment must be made to the data of the balance of payments prior to entering data from this data source into the SUTs system. This adjustment is necessary in order to ensure as a starting point a consistent set of data for imports and exports of goods and services that can be balanced across the SUTs. This adjustment is illustrated in Table 5.3.

5.75. The starting point is the account for the rest of the world, as shown in columns (1) and (2) of Table 5.3 (which mirrors the balance of payments according to BPM 6), where only the entries for goods and services are shown and where imports of goods are valued FOB (372 in column (2) of Table 5.3). This is the value of the goods at the point of exit from the exporter's economy, including transport charges and trade margins up to the border point. The CIF value of imports (382 in column (6) of Table 5.3) of goods measures the value of a good delivered at the point of entry into the importing economy. The difference between the two values (10 in column (4) of Table 5.3) is made up of the costs of transportation, insurance and other expenditures between the point of exit of the exporter's country and the point of entry into the importer's country.

5.76. The services linked to the difference between the FOB and CIF values can be delivered by either resident producers or non-resident producers. To the extent that non-resident producers are involved, the BPM 6 imports of services must be reduced with their services (7 in column (4) of Table 5.3) to avoid double counting, as these services are now included in the CIF value of the imported goods. Adjustment for the services delivered by resident producers (3 in column (3) of Table 5.3) is a bit trickier, as a service that, according to the BPM 6 definition, is a purely domestic transaction will now appear as an import of services included in the CIF value of imported goods. As this import originates from resident producers, it is necessary to introduce a balancing service export of the same value.

Table 5.3 Data adjustment for external trade of goods and services

	SNA/BPM balance of payments		Introducing imports CIF		SUTs balance of payments	
	Uses (FOB) (1)	Resources (FOB) (2)	Uses (3)	Resources (4)	Uses (CIF) (5)	Resources (CIF) (6)
Imports of goods		372		10		382
Exports of goods	462				462	
Imports of services		84		-7		77
Exports of services	78		3		81	
Total	540	456	3	3	543	459
Balance		84		0		84

Note: In practice there will be a further breakdown of both goods and, in particular, services in the balance of payments, and therefore also for the adjustments in columns (3) and (4).

5.77. In Table 5.3, all data adjustments are shown in columns (3) and (4), and the resulting “SUTs Balance of Payments” in columns (5) and (6). It is noted that the balance of the adjustment items is zero, and consequently, the surplus on the transactions in goods and services – 84 in column (2) – is identical in the two alternative ways of presenting the external transactions.

5.78. The “SUTs balance of payments” represents the framework of source data for external trade for SUTs with the appropriate product breakdowns. The composition by specific services making up the CIF and FOB difference will usually be available from the working tables of the balance of payments compilers, as their starting point for the FOB recording of imports will usually have been imports of goods from the external trade statistics valued at CIF. Regular surveys may also have been carried out to illuminate the CIF and FOB difference and the related service structure.

5.79. It is important to underline that the above data adjustment is not the CIF and FOB adjustment often seen as a separate row in SUTs or IOTs. The data adjustment must be made before starting compiling SUTs. At the detailed product level, the supply and use of the individual services are adjusted so that they can meaningfully be balanced under the CIF valuation of goods, and these data adjustments will not be separately identifiable in the completed SUTs.

(b) CIF and FOB adjustment row

5.80. The CIF and FOB adjustment is an ex post facto adjustment made at the macro level to the totals for exports and imports of goods and services to derive the corresponding totals found in the SNA (the goods and services account and the rest of the world account).

5.81. In principle, the purpose of this adjustment is to demonstrate that the data in SUTs are consistent with the rest of the national accounts and to avoid the double-counting of CIF-type services provided by residents. The CIF and FOB adjustment row has no balancing or other methodological functions in the SUTs, and it may be omitted from the SUTs and also the IOTs if there is no special need to maintain the exact conceptual relationship to the national accounts.

5.82. Table 5.4 illustrates the place and content of the CIF and FOB adjustment row in SUTs, albeit limited here to external trade data.

Table 5.4 CIF and FOB adjustment row

	Supply Table Imports		Use Table Exports	
	Goods	Services	Goods	Services
SUT total	382	77	462	81
CIF/FOB adjustment	-10	7		-3
BOP total	372	84	462	78

5.83. The “SUT total” row includes the totals for imports and exports of goods and services in the balanced SUTs system, consistent with the “SUTs Balance of Payments” in Table 5.3.

5.84. In order to obtain totals for the external transactions identical to those found in the rest of the national accounts (and the balance of payments), the adjustments shown in the “CIF/FOB adjustment” row of Table 5.4 are introduced. These adjustments mirror those that were made as data adjustments in Table 5.3. The two types of adjustments have quite different roles, however:

- Those in Table 5.3 relate in principle to columns of the SUTs and must necessarily be carried out prior to the balancing, and there is no way to avoid this adjustment.
- On the other hand, the CIF/FOB adjustment in Table 5.4 is a kind of “memo” row of the SUTs that can be added ex post facto, or even omitted if there is no need to demonstrate consistency with the national accounts.

5.85. It should be noted that, if goods and services are lumped together in SUTs, the CIF/FOB adjustment row will only include the adjustment item, -3, for both imports and exports.

5.86. From a bookkeeping perspective, the data adjustment for exports of services (3 in Table 5.3) could alternatively be recorded as a negative import, even though this action entails a less logical explanation of how the domestic output of services are disposed of and also requires the existing imports of those services to be sufficient to prevent a negative net result.

5.87. With this approach, there would be adjustments in Table 5.3 for imports only, showing identical numerical changes for goods and services, respectively. The CIF/FOB adjustment row in Table 5.4 would in this case have entries only for imports (-10 for goods and +10 for services), and if imports were not shown separately for goods and for services, the CIF/FOB adjustment row would be empty. Further details covering issues of consistency in the SNA are provided in Box 5.2.

Box 5.2 Consistency issues with the CIF/FOB adjustment

The CIF/FOB adjustment is dealt with in both the *Eurostat Manual of Supply, Use and Input-Output Tables* (Eurostat, 2008) and in the supply and use table chapters 14 and 28 of the 2008 SNA.

In the numerical example in Eurostat (2008) (pages 60, 70 and 122), external trade in goods and services are lumped together and, as explained above, for this case, the CIF/FOB adjustment row therefore contains identical negative adjustments for imports and exports. In the more extensively elaborated numerical example (pages 113–115), where comparisons are also made to the treatment in the 1993 SNA, ex ante data adjustments are mixed up with the ex post CIF/FOB adjustment in a complicated manner.

The CIF/FOB adjustment table has both a column and a row dimension, and the final outcome is incorrect because the ordinary exports of services linked to exports of goods are included in the adjustment.

The exposition of the CIF/FOB adjustment in the 2008 SNA is unclear because it starts out from the assumption that the SUTs have been balanced using inconsistent data, namely imports of goods valued CIF, and services as defined as in the BPM 6 based on imports being valued FOB. This shortcoming can obviously not be remedied by ex post adjustments to columns and rows, as any new column data would require a new balancing of the SUTs.

The final outcome in the 2008 SNA are SUTs with a CIF/FOB column in the supply table (table 14.12 of the 2008 SNA), in addition to the CIF/FOB adjustment row with adjustments for imports only (the resident producers' delivery of services linked to imports CIF being treated as negative imports).

If the CIF/FOB adjustment column in the supply table (table 14.12 of the 2008 SNA) is added to the column for imports of services, services as defined in the SUTs balance of payments are obtained, so that in principle this could be taken to indicate the ex ante data adjustment. This is not easily understood from the exposition, however, and, to compound the lack of clarity, the CIF/FOB adjustment is being distributed by user (in table 14.15 of the 2008 SNA), a step for which there is no explanation.

Annex A to chapter 5: Sample questionnaire collecting sales of goods and services, inventories of goods and trade-related data

A5.1 The extract shown in figure A5.1 is from a business survey questionnaire from the Statistical Office of Serbia. Data are collected for each industry and by product covering the following areas:

- Sales of goods produced by the enterprise
- Closing stocks of products and work-in-progress
- Sales of merchandise
- Trade margin
- Closing stocks of goods for resale

A5.2 Additional tables collecting data on the sales of industrial and non-industrial services provide the full coverage of goods and services needed to calculate the industry totals. An extract of these tables may be seen in figure A5.2. These data make it possible to calculate the industry output by product and trade margins required to populate the domestic output part of the supply table and the trade margins column, as shown in Table 5.2. In some countries, opening stock values are also collected.

Figure A5.1 Extract of questionnaire covering sales of goods, inventories of goods and trade activity

No.	Code	Product description	Sales of goods produced by the enterprise (group of accounts 61)	Closing stocks of products and work in progress (group of accounts 10 and 11)	Sales of merchandise (group of accounts 60)	Trade margins amount rate %	Closing stocks of goods for resale (group of accounts 13)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1000		TOTAL					
		AGRICULTURAL PRODUCTS, RAW AND UNPROCESSED PRODUCTS OF PLANT AND ANIMAL ORIGIN					
1001	01.11.1–01.11.4	Cereals, all kinds (except rice), cereal seeds					
1002	01.11.6, 01.11.7	Green leguminous vegetables (beans, peas, lentils and other)					
1003	01.11.8	Soya beans, groundnuts (row) and cotton seed					
1004	01.11.9	Other oil seeds—sunflower, sesame, flax, etc.					
1005	01.11.12	Rice, not husked					
1006	01.13 except 01.13.7	Vegetables, raw and seeds					
1007	01.13.7	Sugar beet and sugar beet seed					
1008	01.13.8	Mushrooms and truffles					
1009	01.15	Unprocessed, raw tobacco					
1010	01.16	Fibre crops (flax, cotton, hemp and other, used in textile industry)					
1011	01.19.1	Forage crops and vegetative matter for livestock feeding unprocessed form					
1012	01.19.2	Flower and flower seeds					
1013	01.21	Grapes					
1014	01.22, 01.23	Tropical and subtropical fruits, all kinds (including citrus, figs, etc.)					
1015	01.24, 01.25 except 01.25.3	Other fruits, tree and bush fruits, except nuts (apples, pears, cherries, berries, etc.)					
1016	01.25.3	Nuts (almonds, hazelnut, walnuts, etc.)					
1017	01.26	Olives, coconuts (raw, unprocessed)					
1018	01.27	Coffee beans, tea leaves, cocoa beans, not roasted					
1019	01.28	Spices, aromatic, drug and pharmaceutical crops					
1020	01.11.5, 01.14, 01.19.3, 01.29, 01.3	Vegetables and fruit seeds, other seeds; grass, unprocessed straw and other residues of cereals; seeds for trees and seedlings; planting materials, sugar cane and other raw, unprocessed and untreated products of plant origin not elsewhere					
1021	01.4. except 01.45.3 & 01.49.3	Live animals and animal products (unprocessed milk, eggs, natural honey; seeds and embryos of animals, except raw skins, shorn wool and skins, etc.)					
1022	01.45.3, 01.49.3	Raw fur skins, shorn wool, skins (excluding products of slaughterhouses and industrial meat production, see 1036)					
1023	01.49, part	Other agricultural animal origin products, raw, unprocessed and untreated, not elsewhere classified					
1024	01.7	Hunting and trapping products, raw, unprocessed					
		PRODUCTS OF FORESTRY					
1025	02.2	Wood in the rough—logs, fuel wood and other raw products of forestry, odds and ends included					
1026	02.1, 02.3	Forest trees and seeds, wild growing edible products; natural cork, varnish, balsams and other natural gums and resins and other raw products of forestry not elsewhere classified					
		FISH AND OTHER FISHING PRODUCTS, UNPROCESSED AND UNTREATED					
1027	03	Fish, sea food and other fishing products; aquaculture products (raw, unprocessed and untreated)					
		MINING AND QUARRYING PRODUCTS, UNPROCESSED; CRUDE AND NATURAL GAS					
1028	05.1, 05.2	Coals, hard coal and lignite (coal for heating included)					
1029	06.1	Crude petroleum, bituminous or oil shale and tar sands. Note, petroleum products entered in row 1082					
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
		MANUFACTURING INDUSTRY PRODUCTS					
		Food products and other processed products of plant and animal origin; used as reproduced material					
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
		Production of electricity and manufactured gas (excluding natural gas extraction and petrol gases in refineries); trade and distribution of electricity and manufactured gas					
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
		Construction					
1162	41, part	Development of building projects					
1163	41, part	Construction works of residential and non-residential buildings					
1164	42	Construction and construction works of civil engineering					
1165	43	Specialized constructions works					

Figure A5.2 Extract of questionnaire covering sales of industrial and non-industrial services

No.	CPA code	Code and service description	Sales of services (groups of accounts 61 and 65, part)
(1)	(2)	(3)	(4)
2000		TOTAL	
		Support services directly linked with the production of goods and services	
2001	01.6 part	Support agricultural services to crop production	
2002	01.6 part	Support services to animal production (animal farming ; veterinary services excluded (row 2059)	
2003	02.10.2, 02.4	Support services to forestry (cultivation and logging of trees, excluded)	
2004	09	Mining support services, services to petroleum and natural gas extraction	
2005	13.3	Textile finishing services—bleaching, dyeing, printing etc.	
2006	16.10.9	Drying, impregnation or chemical treatment services of timber and product of wood; support services in the processing of wood and wood products not elsewhere classified	
2007	25.5	Forging, pressing, stamping and roll-forming services of metal	
2008	25.6	Treatment and coating services of metals; machining	
2009	24.5	Casting services of metal and steel	
		Subcontracted services in industry and construction, trade services and other intermediation commissions. Note: enter only the value of the services, value of materials of goods excluded	
2010	14, part	Subcontracted operations in textile industry (excluding value of materials)	
2011	15, part	Subcontracted operations in footwear and leather production industry (excluding value of materials)	
2012	16, part	Subcontracted operations in production of processed wood and wood products (value of materials, excluded)	
2013	25, part	Subcontracted operations as part of machine industry—processing and finishing materials services (value of materials, excluded)	
2014	—	Other subcontracted operations in production of goods of other enterprises (value of materials, excluded), please specify	
2015	46.1	Trade commissions	
2016	—	Other intermediation commissions please specify	
		Repair, maintenance, installation services; conversion, reconstruction and fitting out of transport equipment	
2017	33.1	Repair and maintenance services of fabricated metal products, machinery and equipment, except motor vehicles	
2018	95.1	Repair services of computers and communication equipment	
2019	95.2	Repair services of personal and household goods	
2020	45.2	Maintenance and repair services of motor vehicles	
2021	33.2	Installation services of industrial machinery and equipment	
2022	29.20.4, 29.20.5	Reconditioning, assembly, fitting out and bodywork services of motor vehicles, except installation, maintenance and repair services	
2023	30.11.9, 30.20.9, 30.30.6	Conversion, reconstruction and fitting out services of other transport equipment, except installation, maintenance and repair services	
		Transportation services Note include transportation equipment rental services with driver and removal services	
2024	49.1 and 49.3	Land transport services—passengers, taxi include	
2025	49.2 and 49.4	Land transport services—freight	
2026	50.1	Water transport services—passengers	
2027	50.2	Water transport services—freight	
2028	51.1	Air transport services—passengers	
2029	51.2	Air transport services—freight	
2030	52.2	Support services for transportation (loading, unloading, hauling, towing, parking service, etc., transportation excluded)	
		Other services	
2031	18	Printing services and services related to printing (newspaper printing, pre-press, binding and related services, reproduction services of recorded media)	
2032	35.30	Steam, hot water, air conditioning supply services	
2033	36	Natural water, water treatment, supply and distribution services	
2034	37	Sewerage services, removal and treatment services	
2035	38	Waste collection, treatment and disposal services	
⋮	⋮	⋮	⋮
		Donations and state subsidies (group accounts 64), lease of intangible assets and income from fees and charges. Note: 2089 and 2090 positions are not entered	
2086		Donations and other unconditioned transfers in cash or in kind by resident legal and natural persons (account 640 and 641)	
2087	—	Donations and other unconditioned transfers in cash or in kind by foreign legal and natural persons (account 640 and 641)	
2088	—	Subsidies, grants, donations and transfers of state and local government bodies (account 640 and 641)	
2089	—	Income from fees for usage of public non-produced assets (this is filled out only by budgetary units—account 741500)	
2090	—	Income from administrative and legal fees (this is filled out only by budgetary units—account 742200)	

Chapter 6. Compiling the use table

A. Introduction

6.1. This chapter primarily deals with the construction of an initial, unbalanced version of the use table. The values entered into the tables should, as far as possible, reflect all available knowledge and data on the product structure of each column, although many values may need to be changed when the SUTs system is balanced. This applies to estimated totals and also to the values for specific products. Before balancing takes place, the estimates for intermediate consumption, final uses and GVA, and the components of GVA (if available at this stage of the process), should be checked for credibility and, if necessary, adjusted as appropriate. These will then form the starting point for the balancing process.

6.2. In section B, this chapter provides an overview of the structure of the use table and describes the main blocks of the table. Section C focuses on the intermediate consumption; section D on the GVA part of the table; section E on the final consumption expenditure; section F on the gross capital formation; and section G on the exports of goods and services. The chapter has two annexes: annex A set out an example of a questionnaire for the collection of data on the purchase of goods and services for intermediate consumption and annex B provides a description of the impact of the change in treatment of research and development according to the 2008 SNA.

B. Structure of the use table

6.3. The use table shows the use of goods and services by product and by type of use for intermediate consumption by industry, final consumption expenditure, gross capital formation and exports. The use table also shows the components of GVA by industry for compensation of employees, other taxes less subsidies on production, consumption of fixed capital, and net operating surplus and net mixed income. Table 6.1 illustrates the structure of the use table.

6.4. The use table has two main objectives:

- The columns show the cost structure of each industry and the product structure of each type of final use.
- The rows show the distribution of each product and primary input (labour and capital) by uses.

6.5. It is customary to compile the use table, at least initially, at purchasers' prices. This valuation relates most closely to the basis of the data collected via business and household surveys and is known by the purchasers of the products.

Table 6.1 Use table at purchasers' prices

Millions of euros

		INDUSTRIES							FINAL USE								Total use at purchasers' prices	
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	Total	Final consumption expenditure			Gross fixed capital formation	Changes in valuations	Changes in inventories	Exports	Total		
									Households	NPISH	General government							
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	
PRODUCTS	Agriculture	(1)	2 583	6 570	16	371	34	49	9 623	3 595			180		- 27	1 161	4 909	14 532
	Manufacturing	(2)	2 205	107 190	12 441	16 874	6 015	8 797	153 522	71 438		3 180	26 756	2 183	3 034	123 252	229 842	383 364
	Construction	(3)	105	2 440	9 528	2 446	3 907	1 604	20 029	1 667			25 155		- 38	563	27 348	47 377
	Trade	(4)	33	1 883	119	2 240	259	308	4 842	3 325			67	45		753	4 189	9 032
	Transport	(5)	14	4 386	267	8 399	822	321	14 208	5 833		3 370				8 453	17 656	31 865
	Communication	(6)	34	2 563	299	9 359	5 919	1 833	20 008	26 444		121	5 976		67	6 905	39 514	59 522
	Finance and business services	(7)	457	13 578	4 736	20 359	29 166	9 134	77 430	38 838		1 006	11 170		- 178	11 145	61 981	139 411
	Other services	(8)	8	382	59	1 171	415	1 794	3 829	14 923	5 416	53 373	113	107	1	567	74 500	78 329
	Total at purchasers' prices before adjustments	(9)	5 440	138 991	27 466	61 219	46 538	23 839	303 492	166 063	5 416	61 050	69 418	2 335	2 859	152 800	459 939	763 431
Adjustments	CI/FOB adjustments on exports	(10)														- 97	- 97	- 97
	Direct purchases abroad by residents	(11)								6 675							6 675	6 675
	Purchases in the domestic territory by non-residents	(12)								- 12 945						12 945		
	Total at purchasers' prices	(13)	5 440	138 991	27 466	61 219	46 538	23 839	303 492	159 792	5 416	61 050	69 418	2 335	2 859	165 648	466 517	770 009
GVA	Compensation of employees	(14)	551	30 679	10 239	37 906	22 997	41 971	144 343									
	Other taxes less subsidies on production	(15)	- 1 627	1 077	546	1 755	2 004	1 103	4 858									
	Consumption of fixed capital	(16)	1 845	12 750	1 542	10 917	18 934	7 480	53 469									
	Net operating surplus/net mixed income	(17)	3 658	16 453	5 138	23 040	18 989	4 921	72 198									
	Gross operating surplus/gross mixed income	(18)	5 503	29 203	6 680	33 957	37 923	12 401	125 667									
	GVA	(19)	4 427	60 959	17 465	73 618	62 923	55 475	274 868									
	Total input at basic prices	(20)	9 867	199 950	44 931	134 837	109 461	79 314	578 360									

Empty cell by construction

Table based on 2011 figures from Austria

6.6. The upper part of the use table (rows (1)–(9) in Table 6.1) shows how the use of goods and services is distributed as intermediate consumption by industry, final consumption expenditure, gross capital formation and exports. The rows of this part of the table correspond to the same rows of the supply table. Each row in the upper part of the SUTs represents a product balance for each product.

6.7. In the lower left part of the use table (rows (14)–(19) in Table 6.1), the components of GVA are shown below intermediate consumption for each industry. If the industry output is given and the intermediate consumption of products determined in the use table, GVA of an industry can be estimated, in the first instance, as a residual variable. If, however, the income measure components of GVA (compensation of employees, other net taxes on production, consumption of fixed capital) are known, the residual value is net operating surplus and net mixed income. Net operating surplus can also be directly estimated using business accounts providing an alternative for data confrontation with the residual approach – the linking between business accounts and national accounts is covered in chapter 2. For each industry, the sum of intermediate consumption at purchasers' prices and GVA at basic prices will equal the value of output at basic prices shown as column totals in the supply table.

6.8. The columns of the use table cover the following categories:

- Industries (columns (1)–(6) in Table 6.1). Intermediate consumption will be shown with the same breakdown (number of columns) as for the industries’ domestic production in the supply table.
- Final consumption expenditure (columns (8)–(10) in Table 6.1), which consists of final consumption of households, NPISHs and general government, The latter is typically broken down into individual and collective consumption.
- Gross capital formation (columns (11)–(13) in Table 6.1), which is broken down in its components of total value of the gross fixed capital formation, changes in inventories and acquisitions less disposals of valuables. This can be further broken down by types of assets or industries.
- Exports of goods and services (column (14) in Table 6.1), which may be shown as a single column or as goods and services in separate columns. These can be further broken down as columns for export of domestically produced products and re-exports. Exports can also be broken down by countries or geographical or market groupings of countries.

6.9. As described in chapter 4, the actual number of rows and columns in the SUTs will depend, among other aspects, on resources and availability of source data.

6.10. The use table also includes a number of rows (in particular, rows (10)–(12) in Table 6.1) which contain adjustments. In particular, these rows contain the adjustment for the valuation of exports (CIF/FOB adjustments on exports), direct purchases abroad by residents, and purchase in the domestic territory by non-residents. In the SUTs, total imports and exports are valued FOB. Data on detailed flows of imports from foreign trade statistics are valued, however, at CIF prices. To reconcile the different valuations used for total imports FOB and the imported products CIF, a total CIF/FOB adjustment row on imports is added to the supply table. The same negative entries are shown in the CIF/FOB adjustment row for exports. More details on the CIF/FOB issues may be found in chapter 5.

6.11. The adjustments for direct purchases abroad by residents and purchases in the domestic territory by non-residents have to be made because the final consumption expenditure of households, as broken down by product, includes direct purchases by non-residents in the domestic territory and these must be treated as exports. Similarly, direct purchases by residents abroad must be treated as imports and thus included in the total final consumption expenditure of households.

6.12. The purchases by residents abroad are treated as both imports and final consumption expenditure of households. Thus an appropriate positive amount has to be entered in the imports column of the supply table and, at the same time, as a positive entry in the column of final consumption expenditure of households in the use table. The purchases in the domestic territory by non-residents are treated as exports and deducted from households’ final consumption expenditure. Thus the corresponding amount entered in the exports column with a positive value

is deducted in the column of final consumption expenditure of households. The balance of the row is zero.

6.13. The SUTs provide a framework enabling the supply and use of all products to be balanced, and the total outputs and inputs of each industry to be balanced. In this stage of the compilation process, however, it is recommended that the SUTs be populated using data based on the best possible data sources before the balancing process takes place. Preferably, regular (quarterly and annual) business surveys based on a high-quality comprehensive business register, along with household-based surveys and the use of administrative data, should be used.

6.14. A simple method for drawing up product balances – traditionally referred to as the “product flow” method – is to distribute the value available for domestic use based on the characteristics of each product. This may work well for products that have specific uses, for instance, as input in a particular industry or as gross fixed capital formation. Most of the products in the SUTs are, however, broad categories of goods or services that may have several different uses. Estimates of the use side that are solely based on the supply of specific products can, however, be used where information from the use side is not available.

6.15. For compiling the use table, two general options are available: the input approach and the output approach. In the input approach, the cost structures of industries and input structures of final use categories are compiled on the basis of specific survey results, while in the output approach the allocation of goods and services is determined with the product flow method. As the input approach is based on collected data, it is the recommended approach for populating the intermediate use part of the use table. The output approach is an alternative, providing a cross-check and forming the basis of the balancing process.

6.16. There is no absolute rule determining whether to give priority to columns or rows of a use table. This depends on basic surveys and the specific country practices of national accounts, and also on such indicators as quality and coverage of the data. It is, however, recommended that the compilation process be started by column, because data received from basic sources will then be fully reflected. At the same time, this method is consistent with the institutional approach to identifying the input structures for industries by intermediate consumption and GVA, and for the categories of final use (consumption, gross capital formation, exports) by product. A distinction must be made, however, between population of the tables with a tendency by column and the balancing of the tables with a stronger row dimension.

6.17. The prime objective of the use table is to identify the cost structures of industries and the input structure of final uses. The input approach can be implemented if survey results are available which identify the main cost structures – the survey approach to the collection of input data is recommended. The main types of sources for the input approach are the establishment survey, the consumer expenditure survey, the government expenditure survey, and the capital expenditure

survey. At the same time, the use table identifies the use of products and primary inputs. The main sources of the output approach are production statistics and foreign trade statistics.

6.18. The output approach (product flow method) is highly dependent on survey results from production statistics and foreign trade statistics.

6.19. Only in the absence of any surveys on the cost structure of industries or any information from input methods should the product flow method be considered. The product flow method can be useful for compiling the rows in a first stage, even if later on they are changed during the balancing process. If product flows are compiled at a very detailed level, it will be possible to break down intermediate consumption between industries even in the absence of complete and direct information on cost structures. Only in the case of specific products, for example, ships, military aircraft, nuclear fuels, and others, should the product flow method be preferred.

6.20. The output approach is often used to compile use tables. It is a valuable complement of the input approach. The product flow method facilitates identifying the output structure of goods and services. The more homogeneous goods and services are, the easier it will be to allocate the use in specific industries or categories of final uses. The product flow method is widely applied for rectangular systems of products and industries in which the number of products is much larger than the number of industries. Provided that survey results are available, the input approach is the best option to identify cost structures. The product flow method is a valuable cross-check for the input approach. It makes possible the identification of a more refined structure of intermediate and final uses in terms of specific products. The product flow method is also a powerful tool when it comes to balancing the system.

6.21. Nevertheless, the first stage is always to compile the totals of industries in terms of output, intermediates and GVA. This takes place in the production accounts of the system. Then the categories of final uses are added which were derived from specific surveys and statistics and product flow accounts.

1. Three-dimensional presentation of the SUTs

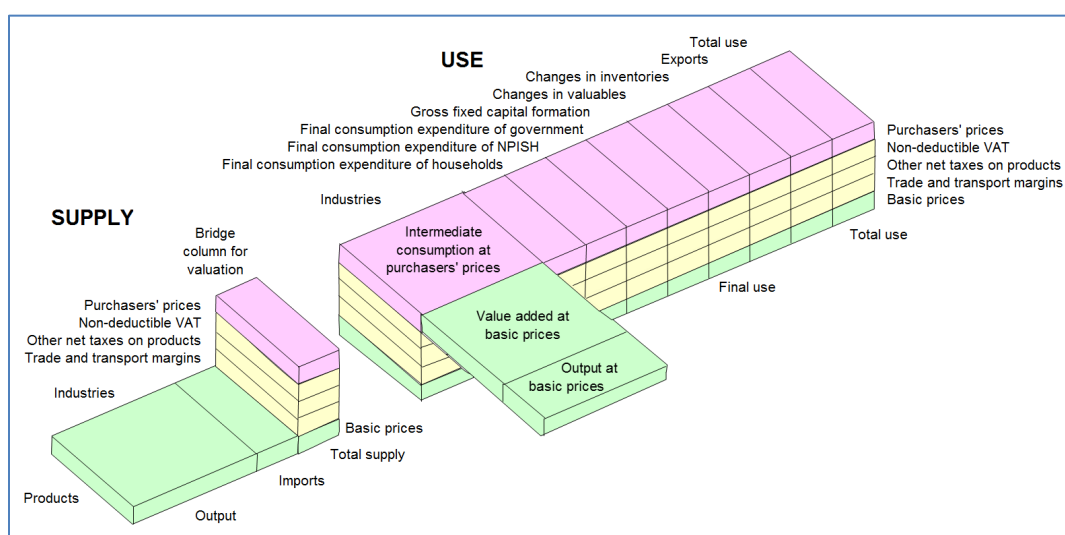
6.22. Trade and transport margins, taxes less subsidies on products (except VAT) and non-deductible VAT must be distributed by products in the valuation table (sometimes referred to as the “bridge column”), shown as part of the supply by products at purchasers’ prices. It is recommended that, where the supply values from the different levels of the valuation table are distributed by uses in “valuation layers”, the matrices should have the same size and format as the upper part of the use table at purchasers’ prices.

6.23. Figure 6.1 shows how the layers can be shown, stacked one above the other, in a three-dimensional representation of the use table. In this way, it will be possible to look at each product balance as a vertical slice, going from left to right, from the SUTs, where the supply is shown at basic prices, while the uses can be seen as a table showing how purchasers’ prices are transformed

into basic prices by removal of trade and transport margins, net taxes on products excluding VAT and non-deductible VAT.

6.24. The exact distribution of trade and transport margins and net taxes on products by uses cannot usually be observed in the data sources and need therefore to be estimated on the basis of whatever data are available, and also on common-sense assumptions. Further details of this approach may be found in chapter 7. Hence the establishment of the full three-dimensional system will require some additional resources. It can, however, be very useful as a tool for keeping track of the use of trade and transport margins and net taxes on products by uses, which are also needed for estimation of SUTs in volume terms and IOTs (both in current prices and in volume terms).

Figure 6.1 Three-dimensional view of SUTs



C. Intermediate consumption part of the use table

6.25. This section describes how to put together an initial unbalanced version of the intermediate consumption part of the use table and the data sources that can be used.

6.26. Intermediate consumption consists of the value of the goods and services consumed as inputs by a process of production, excluding fixed assets whose consumption is recorded as consumption of fixed capital (2008 SNA, para. 6.213). It thus includes all non-durable goods and services with an expected life of less than one year which are used up in the process of production by industries, thus excluding any goods purchased for resale without any further processing. The bought and not-consumed goods are entered in changes in inventories. Goods paid for by employers for the benefit of their staff can be regarded as remuneration in kind entered in compensation of employees.

6.27. The compilation methods for intermediate consumption vary across countries, depending on the data sources available. The recommended approach is to have regular (for example, annual)

data collection on input structures, an approach that is even more advisable with globalization and new technologies contributing to rapid structural change. The most commonly used approach, however, consists in starting with total intermediate consumption by industry – namely, the “Total” row (row (9) in Table 6.1). Then there is a balancing process with the amounts which are available for intermediate use of the various products (see the following sections). Lastly, an equilibrium is obtained between the sum of the “Total” row and the sum of the “Total” column.

6.28. Nevertheless, in some countries, in particular those in which accounts rely on data sources on enterprises (institutional approach), intermediate consumption may be initially known at a high level. Thus the main problem consists in distributing this total intermediate consumption among industries.

6.29. Intermediate consumption can be broken down by industries according to ISIC and by institutional sectors. As for the supply table, it is also possible to distinguish between market producers, non-market producers and producers for own final use and between producers in the formal and the informal economy. The columns in the use table correspond to the same classifications used in the supply table. When estimating the input structure, however, the number of columns does not necessarily need to be identical, although it is recommended that, where appropriate, the industry columns headings in the supply table and the use table are the same.

6.30. The separated columns of an ISIC category (for example, market and non-market) can have significantly different input structures. Trade margins on their inputs may also differ because of the use of different trading channels and discounts and taxation rules may vary, such as those relating to the deductibility of VAT.

6.31. When a distinction is made between market producers, non-market producers and producers for own final use, the inputs in each of these categories of units could be shown as separate submatrices. This would, however, leave many input columns empty or almost empty. A practical solution could be to separate only market and non-market producers within the same ISIC category into different columns where both have a significant size, for example, in the case of health and education services. Similarly, production on own account or informal activity may be shown in separate columns, if it is of special interest, for instance within agriculture, construction or trade.

1. Initial, unbalanced version of the intermediate consumption part of the use table

6.32. The information used for the construction of an initial set of estimates for the intermediate consumption part of the use table will be drawn from various source data.

6.33. For some industries, the source data may consist of a complete picture of outputs and inputs by products. These estimates will typically combine information on physical volumes and prices and may also use information from accounting data. The estimates may be carried out outside the actual SUTs compilation process. A similar situation obtains for the grossing up of survey results

to cover the entire industries in question, for example, agriculture, forestry and fishing. From the perspective of the national accounts and SUTs compilation, the values of all inputs are assumed to be already grossed up and ready to be entered into the SUTs framework.

6.34. It is important to ensure consistency between estimated outputs, inputs and change in inventories within each industry. Products delivered between units within the industry should appear with the same value for sales and purchases of the industry, except for the costs of change of ownership.

6.35. If different data sources are used to draw up the output and input sides of an industry, it is important to make sure that the units behind the data are defined in a similar way. Otherwise there is a danger that the industry's GVA could be overestimated or underestimated. It is also important to ensure that all the inputs are covered and that less important inputs are not missing from the input structure.

6.36. As mentioned before, the most common method used in filling the initial and unbalanced version of the intermediate consumption part of the use table is, first, to estimate the values for total intermediate consumption by industry; second, to enter in the table known values of intermediate consumption by product and by industry when available; and, third, to use additional information on cost structures to estimate all other values in this part of the use table.

(a) Total intermediate consumption by industry

6.37. The data sources used to populate the use table so as to obtain the value of total intermediate consumption by industry (see chapter 4 for more detail on this) include statistical surveys, accounts-based statistics and VAT-based statistics. It should be noted that the same source for the industry providing data feeding into the supply table often also forms the source for the data feeding into the use table.

6.38. For some industries, all inputs are provided by the data supplier and total intermediate consumption is calculated as the sum of these inputs. For most industries, however, total intermediate consumption needs to be estimated from annual and/or quarterly business surveys based on information from business accounts, government accounts and other sources, such as annual reports and financial statements of the economic units themselves.

(b) Inputs by products for each industry

Known values for specific cells, rows or columns

6.39. Some values for cells of the intermediate consumption part of the use table may be known from source data, or they may already have been calculated within the framework of independent subsystems, such as the use of energy, FISIM or insurance services by industries. For some industries, the entire columns of initial inputs can be drawn up in subsystems from which they can be transferred to the SUTs framework. Entire rows and columns may be filled in this way. The

source data used to calculate such values may not necessarily be available at purchasers' prices and they may need to be converted using the best possible assumption on trade and transport margins, net taxes on products and non-deductible VAT.

Input structures

6.40. Surveys on cost structures and other information on the input structure could be used for estimating the input structure in the intermediate consumption part of the use table.

6.41. Ideally, detailed purchases of goods and services should be collected annually covering all industries through surveys of cost structures. These annual surveys typically capture various aspects such as:

- Year-on-year structural changes
- Technological innovation and change
- Contracting out
- Company restructuring, mergers and takeovers
- Non-consolidation of businesses and industries
- Technological product and import substitution
- Economies of scale
- Inventory control
- Price changes

6.42. The collection of regular purchases details also makes possible the better measurement of sudden economic change, due for example to the swine flu crisis (affecting agricultural and slaughtering industries), storm damage (affecting construction and insurance industries), growth of Internet activity, or other natural disasters which may affect the supply and price of certain intermediate products with an impact on the production chains and other consequences.

6.43. Annual surveys of cost structure would also considerably facilitate the production of good quality SUTs in volume measures, which has a key annual focus. Annex A to this chapter provides an example of the type of costs and inventories questions, by type of product covering all goods and services that could be included in a questionnaire.

6.44. The information from surveys based on business accounts may contain some useful subtotals, for example, purchases of goods or indirect production costs, but it would usually be insufficient to provide a full breakdown by product and by industry. Similarly, government

accounts may contain supplementary information on the input of products, but generally not with sufficient detail for full input structures.

6.45. Known totals and subtotals must be supplemented by other information. Surveys on the use of raw-materials and services are typically used to serve the needs of national accounts, specifically the use table. The coverage of such surveys has traditionally been limited to manufacturing and other industries that have a significant input of raw materials. The cost structures of construction, distribution and service industries should also be surveyed, however. The level of detail should preferably make it possible to aggregate the collected data into the products used in the SUTs but, for practical reasons, it may equally be the product categories from surveys that need to be split into more detailed categories used in the SUTs – the latter approach generates lower quality estimates. A survey may only exist for earlier or later years than the year for which the SUTs are being compiled, in which case the revaluation of the values to prices of the year in question should be considered.

6.46. Surveys of cost structures may not have total coverage. They may exclude units below a threshold or they may be based on small samples. When results are grossed up to industry totals, uncertainty is added to the figures. Input structures of small establishments that are not covered by the survey will probably be unlike the structures found in the survey. Furthermore, surveys may collect data for enterprises rather than establishments. Enterprise-based data will include inputs of establishments classified in other industries, and therefore reflect some inputs which should not be included here but in another industry.

6.47. Cost structures typically also include some acquisition of capital equipment originally treated as current expenditure in business accounts.

6.48. If annual surveys are not available, the use of older SUTs, where a more detailed survey may have been conducted, can be used to form an initial structural base for a more recent period. Where such periodic surveys exist, different processes can be applied to generate more recent time series and meaningful data using more recent control totals applied to old structures.

6.49. Lastly, respondents do not always know all their inputs, and categories whose descriptions begin with “Other” or finish with “not elsewhere classified (n.e.c.)” will usually be overstated in such surveys. It may be a good idea to apply some common-sense corrections to the survey data before they are used to create input structures. The grossed-up values of inputs by products calculated from surveys are probably, despite their inaccuracy, often the best possible initial estimates of input structures, but they should be used with some caution.

6.50. The availability of statistics on cost structures can vary greatly among countries. For example, there are countries where surveys on cost structure are conducted yearly for all industries; there are countries where only manufacturing industries are surveyed annually and most other industries are surveyed with regular intervals; and there are countries where such surveys are scarce, outdated or altogether missing.

6.51. It is advisable that the need for new surveys is considered well in advance when new benchmark SUTs are planned, as it may take several years to plan and implement and obtain useable results from such surveys. Where no statistics are available, the possible existence of alternative data sources should also be investigated.

6.52. In some countries, it is common for annual reports and accounts of enterprises to include detailed descriptions of the use of inputs, for example, supporting purchase ledger details. The information is, however, typically shown in a rather unsystematic way. To make this detail useful, the data must be categorized in a way that corresponds to the product classification used in SUTs. A similar approach can be used for inputs in general government, if the accounts for central and local government contain detailed information that can be used to create input structures.

6.53. If nothing is known about inputs for some industries, it may be possible to use input structures from other industries that are assumed to be similar, but with some modifications based on expert opinions. As a last resort, input structures from neighbouring countries or similar activities in other countries could be considered, in particular, if those structures are based on actual source data, and account should also be taken of possible differences in the extent of processing and other factors. The initial inputs that are based on these kinds of approaches are uncertain, and they are more likely to be adjusted – within reasonable limits – in the balancing process.

6.54. In countries where the informal economy forms a considerable share of output in specific industries, the structure of inputs used in informal units can be expected to differ from the structure found in surveys of formal businesses. Within agriculture and related activities, the input in informal units may be covered by agricultural statistics. Otherwise it may be appropriate to make a separate estimation of the input structure in the informal economy. Information on the use of inputs in informal activities may be found in household budget surveys or labour force surveys, as mentioned above, if not in special studies of the informal activities. Such data may give an incomplete picture of the outputs and inputs and it may be necessary to add some supplementary assumptions based on expert knowledge before they can be used in SUTs.

(c) Putting together known values and information on product structure

6.55. Information on what are referred to as the “known” values and information on the input structure can be put together in order to fill an initial version of the values of intermediate consumption by product of an industry. Box 6.1 provides a numerical example of how all the available information can be combined.

Box 6.1 Example of a calculation of the values of an input column

In this example the total input, 2,500, is supposed to be known. A survey-based input structure sums to 100 per cent.

The values of input of products 2 and 5 are known as 150 and 300 respectively. The known values are treated as predetermined and subtracted from the total value that is going to be distributed by products.

The residual total value of inputs, $2,500 - 450 = 2,050$, is now distributed proportionally with the “survey-based input structure” excluding the “known values”.

Finally, the known and the calculated input values are added together to form the complete initial column of intermediate consumption for the industry in question.

Input in an industry Products	Survey based input structure	Known (predetermined) values	Survey-based input structure excl. known values	Inputs estimated from survey-based structure	Result: Input column
	%	Value	%	Value	Value
	(1)	(2)	(3) = (1) * 100.0/80.0	(4) = (3) * (2500-450)/100	(5) = (2) + (4)
Product 1	16.0	0	20.0	410	410
Product 2	6.0	150	0.0	0	150
Product 3	7.0	0	8.8	179	179
Product 4	44.0	0	55.0	1 128	1 128
Product 5	14.0	300	0.0	0	300
Product 6	8.0	0	10.0	205	205
Product 7	5.0	0	6.3	128	128
Total	100.0	450	100.0	2 050	2 500

(d) Grossed-up data versus the data collected by surveys

6.56. Putting together information from different kinds of sources in the SUTs framework generally will result in an unbalanced set of SUTs. The total uses of a product will therefore differ from the total supply for most products. In the final, balanced version of the use table such differences are removed either by manual adjustments or by automatic methods.

6.57. The corrections that are necessary to remove the differences between the first estimates of supply and use should as far as possible retain those values that are considered to be reliable statistics. For this purpose it is useful to be able to distinguish between:

- Inputs by product that have actually been reported as primary statistics by respondents and values that are found in annual reports, government accounts or other reliable sources
- Initial inputs by product that are the result of grossing up to the estimates for total input in each industry

6.58. In most cases, the difference between the two will represent the value that can be removed during the balancing process when total initial use exceeds total initial supply for a product. It can be useful if the reliable parts of the input values are shown together with the grossed-up values in the tables presented to the people working on the manual balancing of SUTs.

6.59. An example, of part of an industry input structure (column) with supplementary reliable values at purchasers' prices in the rightmost column is shown in Table 6.2. Columns (6) and (7) show values from the two approaches, which in turn should be investigated to achieve a plausible industry input structure and, in turn, an agreed estimate for each product. Apart from wood-wood, all the other products show significant differences, which call into question the reliability of the data feeding into the SUTs for the input structure for this industry.

Table 6.2 Intermediate consumption of selected inputs into "Manufacture of rubber and plastic products"

	Transaction	Basic price	Margins	Net taxes on products (excl. VAT)	VAT	Purchasers' price	Reported values
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Wood-wood	..	..	..	..	..	..	..
Wood in logs or roughly cut	P.2	22	0	0	0	22	20
Plywood, laminated wood	P.2	5 625	607	0	0	6 232	2 340
Packaging material, wood	P.2	10 286	1 024	0	0	11 310	10 539
Other wood products	P.2	20 085	580	0	0	20 665	12 816
Paper in rolls and sheets	P.2	20 352	1 854	0	0	22 206	20 353
...	..	3 027	46	0	0	3 073	1 329
...	..	..	..	..	..	..	..

P.2: Intermediate consumption.

6.60. An example showing part of a product balance (row) with supplementary reliable values in the rightmost column is shown in Table 6.3. Here the reported values on the supply side refer to basic prices, while the reported values on the use side refer to purchasers' prices. Although the basic price estimates are in balance (allowing for rounding differences), columns (6) and (7) show significant differences at purchasers' prices between the estimated value and the reported value.

Table 6.3 Sample product balance for "Gelatine and gelatine derivatives"**Supply**

	Transaction	Basic price	Reported values
	(1)	(2)	(3)
Other food products, n.e.c.	P.1	12	12
Paints and soap, etc.	P.1	230 779	230 779
Imports	P.1	136 245	136 244
Gelatine and gelatine derivatives	Total	367 036	367 035

Use

	Transaction	Basic price	Margins	Net taxes on products (excl. VAT)	VAT	Purchasers' price	Reported values
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Meat products	P.2	14 029	4 749			18 778	3 498
Fish products	P.2	932	31			963	
Dairy products	P.2	9 925	7 958			17 883	
Bakery products	P.2	133	110			243	134
Other food prod. n.e.c.	P.2	109 577	22 735			132 312	67 143
Paints and soap, etc.	P.2	34 113	7 170			41 283	26 765
Pharmaceuticals, medicine	P.2	64 658	9 516			74 174	7 315
Rubber and plastic products	P.2	17 812	3 754			21 566	10 436
	..	..	..	..	..	..	..
Change in inventories, materials	P.52	736	141			877	
Change in inventories, goods for resale	P.52	736	141			877	
Exports of domestic production	P.6	98 487	357			98 844	97 197
Re-exports	P.6	10 287	35			10 322	10 150
Gelatine and gelatine derivatives	Total	367 035	56 697				222 638

P.1: Output

P.2: Intermediate consumption

P.6: Exports of goods and services

P.52: Changes in inventories

D. GVA part of the use table

6.61. Once the intermediate consumption part of the use table has been estimated, it is possible to calculate the GVA for each industry. The GVA at basic prices is estimated as total output at basic prices from the supply table minus total intermediate consumption at purchasers' prices from the upper part of the use table. The GVA can be broken down into the following components:

- Compensation of employees
- Other taxes less subsidies on production
- Gross operating surplus and gross mixed income

6.62. Gross operating surplus can be further split into net operating surplus and consumption of fixed capital on gross operating surplus. In addition, gross mixed income can be further split into net mixed income and consumption of fixed capital on gross mixed income. If information is available, these breakdowns could be shown in the Use table as follows:

Gross value added

Compensation of employees

Other taxes on production

Other subsidies on production

Gross operating surplus

Consumption of fixed capital on gross operating surplus

Net operating surplus

Gross mixed income

Consumption of fixed capital on mixed income

Net mixed income

6.63. Each of the categories of GVA is described below, together with the relevant data sources.

1. Compensation of employees

6.64. Compensation of employees is defined as the total remuneration, in cash or in kind, payable by an enterprise to an employee in return for work done by the latter during the accounting period (2008 SNA, para. 7.5). Compensation of employees has two main components: wages and salaries payable in cash or in kind; and social insurance contributions payable by employers (actual and imputed) (2008 SNA, para. 7.42). Generally, statistics drawn from business accounts and government accounts show values for wages and salaries and possibly also other costs related to employment. In both cases, there may be conceptual differences from the national accounts concepts, due, for example, to the different treatment of particular issues such as fringe benefits, employers actual and imputed social contributions, and other factors. In addition, the information usually needs to be grossed up to cover the part of each industry that is not covered by the statistics (for example, non-exhaustive business register) and values that are only available for enterprises which would need to be distributed by establishments before use.

6.65. Information from tax-collecting authorities can provide data on compensation of employees that will also cover industries not fully covered by accounts statistics or surveys. This information may contain a distribution by industries that can be more or less consistent with the industry classification used in other statistical sources. If a business register is used to classify data from the various sources, it is likely that the figures from various sources will be classified in the same way. As the structure of many economic units will change over time, the same units may nevertheless be classified differently in different data sources. It should be borne in mind that data collected for administrative purposes may refer to units that are neither enterprises nor establishments, thus for the purposes of national accounts further alignment may be necessary for consistency.

6.66. In some countries, data for compensation of employees in a number of industries is estimated based on labour force surveys, household budget surveys or occasional industrial censuses in combination with population censuses. This is of particular relevance to cases where a considerable share of economic activity takes place in the informal economy.

2. Other taxes less subsidies on production

6.67. The other component of GVA at basic prices consists of other taxes less subsidies on production, which may be shown separately. Other taxes on production consist of all taxes except taxes on products that enterprises incur as a result of engaging in production (2008 SNA, para. 7.97). Similarly, other subsidies on production consist of subsidies except subsidies on products that resident enterprises may receive as a consequence of engaging in production. There are different types of taxes and subsidies on production. They may include taxes or subsidies on payroll or workforce, subsidies to reduce pollution, recurrent taxes on land, buildings or other structures, and others.

6.68. Generally, government accounts contain information on the total taxes and subsidy on production for each type of taxes and subsidies as they cover the various taxes and subsidies covered by the legislation.

6.69. The distribution of the taxes and subsidies by industries may be available from source data, but when this is not the case, the total taxes and subsidies on production have to be allocated to relevant industries (proportionally to the items to which they relate). The amounts should be shown at an accrual basis.

3. Operating surplus and mixed income

6.70. The value of “gross operating surplus and gross mixed income” is obtained as a residual when compensation of employees and other taxes and subsidies are subtracted from GVA at basic prices by industry. Estimates of these industry totals are usually available early in the process of compilation of the use table and should be checked for credibility before balancing SUTs. As shown in chapter 2, however, direct estimates of mixed income and gross operating surplus can be estimated using administrative sources in addition to company accounts. In doing so, a complementary estimate can provide data confrontation with the residual estimate.

6.71. The consumption of fixed capital on operating surplus and mixed income is usually based on the perpetual inventory model method. The corresponding values for net operating surplus and net mixed income are usually calculated in one of the final steps in the compilation of SUTs as the calculation of consumption of fixed capital requires finalized data on gross fixed capital formation broken down by industries, by institutional sector and by type of asset.

E. Final consumption expenditure part of the use table

6.72. Final consumption expenditure is the amount of expenditure on consumption goods and services (2008 SNA, para. 9.7). Final consumption expenditure can be disaggregated between individual consumption expenditure and collective consumption expenditures. The first consists of expenditure on individual consumption goods or services that are acquired by a household and used to satisfy the needs or wants of members of that household. The latter consists of the expenditures for collective consumption services which are services provided simultaneously to all members of the community or to all members of a particular section of the community, such as all households living in a particular region.

6.73. Final consumption expenditure is disaggregated in the use table in final consumption expenditure by households, NPISHs or general government. Table 6.4 shows the structure of the final consumption expenditure in the use table.

Table 6.4 Categories of final consumption expenditure

Products	FINAL CONSUMPTION			
	Households	NPISHs	General government	Total
Product 1				
Product 2				
⋮				
Product N				
Total				

6.74. The manner in which the submatrix of the use table is compiled, showing the use of products for final consumption, is similar for each of the three types of final consumer (households, NPISHs and general government) but starts from a different classification for each of them, reflecting the way (and the basic functional classifications) in which the basic data are collected.

1. Household final consumption expenditure

6.75. Household final consumption expenditure consists of the expenditure, including expenditure whose value must be estimated indirectly, incurred by resident households on individual consumption goods and services, including those sold at prices that are not economically significant and including consumption goods and services acquired abroad (2008 SNA, para. 9.113).

6.76. Information on consumption by households usually starts from household surveys. In these surveys, household expenditure is classified according to COICOP, as shown in Box 6.2. It is therefore recommended to underpin the compilation of this part of the use table with a table linking data on the final consumption expenditures by purpose and by products. This will greatly improve the quality of the data and will ensure that the different analyses of household final consumption expenditure are consistent and coherent with the balanced SUTs. It will also ensure homogeneity in the deflation process, thus ensuring better quality volume data.

Box 6.2 Classification of individual consumption according to purpose

COICOP is an integral part of the SNA, but it is also intended for use in three other statistical areas: household budget surveys, consumer price indices and international comparisons of GDP and its component expenditures. The purposes defined in COICOP are based on the classifications of consumer expenditures which national statistical offices have developed for their own use to serve a variety of analytic applications. Although COICOP is not strictly linked to any particular model of consumer behaviour, the classification is designed to broadly reflect differences in income elasticities. In 2018 the United Nations issued the first revision of COICOP, COICOP 2018 (United Nations, 2018), to reflect users' need for more detail and several other issues that required a revision of the classification. There are 15 divisions in COICOP 2018:

- 01 - Food and non-alcoholic beverages
- 02 - Alcoholic beverages, tobacco and narcotics
- 03 - Clothing and footwear
- 04 - Housing, water, electricity, gas and other fuels
- 05 - Furnishings, household equipment and routine household maintenance
- 06 - Health
- 07 - Transport
- 08 - Information and communication
- 09 - Recreation, sport and culture
- 10 - Education services
- 11 - Restaurants and accommodation services
- 12 - Insurance and financial services
- 13 - Personal care, social protection and miscellaneous goods and services
- 14 - Individual consumption expenditure of non-profit institutions serving households (NPISHs)
- 15 - Individual consumption expenditure of general government

The first 13 divisions add up to total individual consumption expenditure of private households. The last two identify those parts of consumption expenditure by non-profit institutions serving households and general government that are treated as social transfers in kind. Together all 15 items represent actual final consumption by households.

6.77. Table 6.5 shows the table that is used to cross-classify data of final consumption expenditures by purpose (COICOP classes) and by products (CPC classes). The list of products in this table is the same as that used for the supply table and the intermediate consumption part of the use table. Of course, the higher the level of disaggregation, the better the quality and precision of allocation to products and the greater the homogeneity for deflation purposes.

6.78. The compilation of Table 6.5 relies on different data sources and is often based on household budget surveys which directly collect details of expenditure on goods and services by

households and also on retail trade surveys (although some adjustments are needed). These data sources are discussed in the following sections.

Table 6.5 Table linking final expenditures by purpose (COICOP) and product (CPC)

Consumption COICOP groups	Household consumption (COICOP)				
	Food	Non-alcoholic beverages	...	Other services	Total
Products	01.1	01.2	...	13.9	01.1-13.9
Product 1					
Product 2					
:					
Product N					
Total					

6.79. Some adjustments may be needed to ensure that the final consumption expenditures by households reflect the final consumption expenditures of resident households. This means that, if the starting point is the final expenditures that took place in the territory by households, adjustments are needed to remove the expenditures in the country by non-resident households and include the final expenditures of resident households abroad. When distributions by COICOP groups of these adjustment items are unknown, they can be placed in one or two supplementary columns with a positive and a negative value as appropriate.

6.80. Very few products are exclusively used for household final consumption expenditure. For example, some domestic supply of typical consumer-related goods (for example, food) is used as intermediate consumption in restaurants, transport services and government institutions, as individual consumption of NPISHs and general government or – in the case of durables – as fixed capital formation. Accordingly, the amounts spent by households (this is considered as final consumption expenditures by households) cannot be determined from the supply of such products without knowledge of the other uses.

6.81. For some products, full information on household final consumption expenditure can be provided by subsystems established outside the SUTs framework. The underlying data may have the form of physical volume and price information, such as energy. Administrative sources can provide a rich source of detail, for example, covering purchases of motor vehicles, school fees and other outlays such as education, prescription medicine, and other expenses on health services.

6.82. Table 6.6 provides a numerical example of the link between household final consumption expenditure of households in the use table and COICOP by product breakdown. In this example, the largest expenditure items are housing, followed by transport, restaurants and food.

Table 6.6 Final consumption expenditure of households (by COICOP headings)

COICOP	CLASSIFICATION OF INDIVIDUAL CONSUMPTION BY PURPOSE (COICOP)												Total use at purchasers' prices
	Food and non-alcoholic beverages COICOP 01	Alcoholic beverages, tobacco and narcotics COICOP 02	Clothing and footwear COICOP 03	Housing, water, electricity, gas and other fuels COICOP 04	Furnishings, household equipment and routine household maintenance COICOP 05	Health COICOP 06	Transport COICOP 07	Communication COICOP 08	Recreation and culture COICOP 09	Education COICOP 10	Restaurants and hotels COICOP 11	Miscellaneous goods and services COICOP 12	
PRODUCTS	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Agricultural products (1)	2 240	60	0	434	0	0	0	0	860	0	0	0	3 595
Manufactured products (2)	14 016	5 537	9 749	7 090	10 198	2 254	12 809	2 548	2 767	0	0	4 470	71 438
Construction (3)	0	0	0	1 667	0	0	0	0	0	0	0	0	1 667
Trade, transport, communication (4)	0	0	0	0	0	0	7 827	3 361	4 407	0	20 008	0	35 602
Financial and business services (5)	0	0	0	26 218	747	0	1 339	0	2 590	0	0	7 944	38 838
Other services (6)	0	0	194	77	212	3 730	235	133	4 204	1 221	0	4 918	14 923
Total (7)	16 257	5 597	9 943	35 487	11 157	5 984	22 209	6 041	14 827	1 221	20 008	17 332	166 063

Table based on 2011 figures from Austria

Note: the products “Trade”, “Transport” and “Communication” of table 6.1 are presented together in table 6.6. At the time of drafting, this table was compiled in accordance with COICOP (United Nations, 2000a).

6.83. COICOP also distinguishes household final consumption expenditure according to the following product classes: services (S), non-durables (ND), semi-durables (SD) and durables (D). This supplementary classification provides data for other analytic applications, such as assessing household stocks of goods and the cyclical variation in consumer demand. For example, in dealing with the stock of capital goods held by households, goods in COICOP classes that are identified as durables provide the basic elements for such estimates. Box 6.3 provides a description of durable, semi-durable, non-durable goods and services.

Box 6.3: Non-durable, semi-durable and durable goods

Non-durable goods are defined as goods that can be used only once, for example: food, non-alcoholic beverages, alcoholic beverages, tobacco, materials for the maintenance and repair of dwellings, pharmaceuticals, fuels, energy, garden plants, flowers, pets, newspapers and stationery.

Semi-durable goods differ from durable goods in that their expected lifetime of use, although more than one year, is significantly shorter, and their purchase price is typically less than that of durable goods. For example: clothing, footwear, household textiles, motor vehicle spare parts, recording media, games, toys, books, and electrical appliances for personal care.

Durable goods are those goods which can be used repeatedly or continuously over a period of more than a year, for example: furniture and furnishings, carpets, major tools, vehicles, telephone equipment, computers, photographic equipment, jewellery, clocks and watches.

Services include cleaning and hire of clothing, actual and imputed housing rental, repair services, domestic services, outpatient and hospital services, transport services, post and telecommunication services, recreational and cultural services, education, catering, accommodation, hairdressing, insurance and financial services.

(a) Household budget surveys

6.84. Household budget surveys provide a good source for expenses classified by purpose and by product according to internationally agreed standards. Such surveys may also include

information on a wider range of household activities and living conditions, and sometimes may be named “living standard surveys”. Household budget surveys are generally used to compile or update the weights for the basket of goods used in the CPIs and to collect information on household income, possession of assets, type and equipment of dwellings, outlays for repair and maintenance, production for own use and other informal economic activity. This supplementary information is often useful for national accounts and PPPs and may actually have been collected with this purpose in mind, and also for satellite accounts such as social accounting matrices. Household budget surveys can contain information relating to consumption of goods produced for own consumption and services from owner-occupied dwellings which in some countries may not be available from other data sources.

6.85. Household expenditure surveys frequently use COICOP as the basis for the collection of household expenditure information. The results are reallocated to products classified by CPC and then used to estimate the vector of household final consumption expenditure by product for the SUTs.

6.86. Household budget surveys often provide good initial estimates feeding into household final consumption expenditure. Attention should be paid, however, to the coverage of the survey in order to ensure that the survey results can be used for final consumption expenditures by households. For example, some household budget surveys may not cover the year for which the SUTs are being compiled, in which case the survey data should be referenced to prices of the actual year and, if possible, corrected for the development in volumes from the surveyed period. In addition, since household final consumption expenditure refers to the total resident population, the statistician must ensure that the results of the household budget surveys are grossed up to cover the total population.

6.87. If no household budget survey exists for the SUTs reference year, an alternative approach may be to use the structure of expenditure from the last household budget survey and prorate this structure to the estimate for total expenditure of the reference year. This clearly assumes a fixed basket of spending on goods and services (this does not allow for relative price changes or changes due to new products or products not anymore consumed). The balancing process will, however, generate changes to this structure, for example reflecting changes from the supply side.

(b) Retail trade surveys

6.88. Household final consumption is linked to turnover of retail trade after adjusting for sales to businesses and non-residents. Consumers buy most of their goods from retail outlets. Retail trade statistics provide data on turnover broken down into product groups. Retail trade statistics, however, do not include imputed transactions, such as imputed rentals of owner-occupied dwellings and FISIM, which are included in household final consumption expenditure. These are

compiled using different sources and affect other parts of the national accounts, not just household final consumption expenditure.

6.89. Retail sales statistics that are used to feed into household final consumption expenditure need to undergo appropriate adjustments. For example, although consumers purchase most of their goods and services from retail outlets, they also purchase goods and services from units not classified to the retail industry, for example, directly from manufacturers and service industries. On the other hand, sales by retail outlets are not all consumed by resident household consumers but purchases from these retail outlets are also made by non-residents (for example, visitors) and these are categorized as exports made by businesses and, in turn, treated as intermediate consumption.

6.90. Accordingly, the turnover of retail trade and some services disaggregated by detailed industries can provide valuable information on household final consumption expenditure exclusively by broad categories of products and there is no one-to-one correspondence between retail trade turnover by industries and household final consumption expenditure by COICOP groups.

6.91. It should also be borne in mind that informal activities may contribute significantly to household final consumption expenditure. In the cases like retail trade, it would be particularly useful to extrapolate the structure of the column totals from already existing SUTs to form initial estimates for subsequent years.

(c) Products subject to regulations, taxes or subsidies

6.92. It is often possible to obtain detailed data on products that are subject to regulation, taxation or subsidization, since this information is available from the responsible authorities.

6.93. Motor vehicles, alcohol and tobacco are typical examples of products subject to regulation. It may, for example, be possible to use information on motor vehicle registration to determine household final consumption expenditure of motor vehicles. Information on the use of alcohol or tobacco (for example, related to duties paid) could be used to determine consumption of products that are used for household final consumption expenditure, taking into account that these products could also be used for hospitality by businesses or as an input used by restaurants, in which case they would be treated as intermediate consumption and thus excluded from final consumption expenditures by households.

2. Final consumption expenditure of NPISHs

6.94. Final consumption expenditure of NPISHs consists of the expenditure, including expenditure whose value must be estimated indirectly, incurred by resident NPISHs on individual consumption goods and services and possibly on collective consumption services (2008 SNA, para. 9.115).

6.95. Similar to the final consumption expenditure of households, it is useful to cross-classify the consumption expenditures of NPISHs by products (according to CPC) and by purpose. The reference classification of these expenditures by purpose is COPNI (United Nations, 2000a), see Box 6.4. by convention, all consumption expenditures of NPISHs are treated as individual consumption (see 2008 SNA, para. 9.107). Thus, all consumption expenditures of NPISHs are described in COPNI, and also in division 13 of COICOP.

6.96. Table 6.7 shows the matrix that links the final consumption expenditures by NPISHs by purpose (according to COPNI) and by product (according to CPC).

6.97. In some countries, NPISHs may produce most of the services within education, health or social protection, while in other countries such services may mainly be produced by general government and private enterprises. It may be appropriate to provide separate columns for COPNI divisions in the use table. Where the activity of NPISHs is negligible, its detailed breakdown can be considered to be less relevant.

6.98. Various data sources will be used to cover the details of NPISHs by industry and by product. It is recommended, for example, that a business survey is used based on a sample of NPISHs selected from the business register. The grossing-up methodology for a sample survey would need to reflect that NPISHs are non-market bodies and not market bodies, in other words, output is the sum of costs and not related to turnover. Other sources may also include company accounts, regulatory bodies and collective group accounts covering, say, a group of trade unions or religious bodies.

Box 6.4 Classification of the purposes of NPISHs

The main use of COPNI is to classify expenditures by NPISHs in a manner consistent with the purposes of the individual consumption expenditures of households and general government in order to obtain the SNA aggregate of actual final consumption of households.

COPNI can also be used to facilitate international comparisons of the activities of NPISHs. In many countries, activities of these institutions are an important complement to government activities in terms of supplying education, health and social protection services to the population. In some countries, NPISHs are also becoming prominent in non-traditional areas, such as environmental protection, the protection of human rights and the defence of minority groups.

Nine divisions are distinguished in COPNI:

- 01 – Housing
- 02 – Health
- 03 – Recreation and culture
- 04 – Education
- 05 – Social protection
- 06 – Religion

07 – Political parties, labour and professional organizations

08 – Environmental protection

09 – Services n.e.c.

Note that the nine divisions of COPNI correspond to division 14 of COICOP 2018, which classifies the individual consumption expenditure of NPISHs (see box 6.2).

Table 6.7 Table linking final consumption expenditures of NPISHs by purpose (COPNI) and by product (CPC)

COPNI Division	Final consumption of NPISHs (COPNI)				
	Housing	Health	...	Other services n.e.c.	Total
Products	01	02	...	09	01-09
Product 1					
Product 2					
:					
Product N					
Total					

3. Final consumption expenditure of general government

6.99. General government final consumption expenditure consists of expenditure, including expenditure whose value must be estimated indirectly, incurred by general government on both individual consumption goods and services and collective consumption services (2008 SNA, para. 9.114).

6.100. Final consumption expenditures of general government can be classified in several ways. For example:

- According to whether the goods or services have been produced by market or non-market producers.
- According to whether the expenditures are on collective services or individual goods or services.
- By function or purpose according to the Classification of the Functions of Government (COFOG).
- By type of good or service according to CPC.

6.101. The column of final consumption of general government in the use table is usually underpinned with a matrix linking the final consumption of general government by product and by purpose. The reference classification of final consumption of general government by purpose is

COFOG (United Nations, 2000a) (see Box 6.5). Data from government accounts are usually classified by COFOG groups. This classification may be more or less detailed but should in most cases make it possible to distinguish between individual consumption and collective consumption. Individual consumption corresponds to group 15 of COICOP.

Box 6.5 Classification of functions of government

A major use of COFOG is to identify consumption expenditures that benefit individual households and that are transferred to division 15 of COICOP 2018 in order to derive the SNA aggregate of actual final consumption of households (or actual individual consumption). The divisions, groups and classes covering these expenditures are clearly indicated in the classification. COFOG also permits trends in government outlays on particular functions or purposes to be examined over time.

COFOG is used in the analysis and presentation of statistics on government finance. COFOG consists of ten divisions:

- 01 – General public services
- 02 – Defence
- 03 – Public order and safety
- 04 – Economic affairs
- 05 – Environmental protection
- 06 – Housing and community amenities
- 07 – Health
- 08 – Recreation, culture and religion
- 09 – Education
- 10 – Social protection

Each class of COFOG is clearly identified as collective services or individual services. In general, all of classes 01–06 are collective services, as are sections 07.5 and 07.6 of health, sections 08.3–08.6 of recreation, culture and religion, sections 09.7 and 09.8 of education, and sections 10.8 and 10.9 of social protection. These sections cover expenditures on general administration, regulation, research that is not recorded as capital formation and so on. The remaining sections of health, recreation, culture and religion, education and social protection (which dominate each of the classes) are individual services (2008 SNA, para. 9.100).

6.102. Table 6.8 shows the table linking consumption expenditures of the general government by purpose (COFOG) and by product (CPC). An alternative would be to have column headings in terms of industries (ISIC) and the row headings by product (CPC).

6.103. The transformation of the data based on type of function or purpose to industry is very important, for example, as many regulatory and administrative services are classified to the public administration and defence function or purpose but should be classified to industries such as health and education. Again the level of detail is determined by its importance in the country. A split by level of government could depict different characteristics in terms of COFOG categories, inputs and outputs.

Table 6.8 Table linking final consumption expenditure of general government by COFOG and CPC

Collective services						Individual services					
Divisions (COFOG)	Collective consumption general government (COFOG)					Divisions (COFOG)	Individual consumption general government (COFOG)				Total
	General public services	Defence	...	Social protection	Total		Health	Recreation culture and religion	Education	Social protection	
Products	01	02	...	10		Products	07*	08*	09*	10*	
Product 1						Product 1					
Product 2						Product 2					
:						:					
Product N						Product N					
Total						Total					

6.104. Individual consumption of general government consists of two parts which may be shown as separate columns in the use table:

- Goods and services produced by general government as a non-market producer
- Goods and services purchased by general government from market producers for onward transmission to households either free or at prices that are not economically significant. These goods and services are not included in the output of general government.

6.105. Data sources for general government primarily rely on central government and local government administrative data, and are mainly provided by the finance ministries and local government bodies. These are often supplemented with specific survey data such as very detailed local government expenditure. Furthermore, some estimates are based on models such as the perpetual inventory model estimating the consumption of fixed capital for both the central government and local government sectors. All these sources may use a COFOG basis or an industry basis – in both cases, they will need a CPC product breakdown.

F. Gross capital formation part of the use table

6.106. Gross fixed capital formation is measured by the total value of a producer's acquisitions, less disposals, of fixed assets during the accounting period plus certain specified expenditure on services that adds to the value of non-produced assets (2008 SNA, para. 10.32). Gross capital formation is measured by the total value of the gross fixed capital formation, changes in inventories and acquisitions less disposals of valuables (2008 SNA, para. 10.31). In the use table, GCF is usually at least broken down into three separate columns to display its components separately as shown in Table 6.9. These are discussed in the following sections.

Table 6.9 Categories of gross capital formation

	Gross capital formation			
	Gross fixed capital formation	Changes in inventories	Acquisitions less disposals of valuables	Total
Products				
Product 1				
Product 2				
:				
Product N				
Total				

1. Gross fixed capital formation

6.107. Gross fixed capital formation is measured by the total value of a producer's acquisitions, less disposals, of fixed assets during the accounting period plus certain specified expenditure on services that adds to the value of non-produced assets (2008 SNA, para. 10.32). Fixed assets are produced assets that are used repeatedly or continuously in production processes for more than one year (2008 SNA, para. 10.11). Gross fixed capital formation is also described as capital investment, fixed investment or capital expenditure.

6.108. One approach to the compilation of this part of the use table is the demand-based approach, which requires detailed information on investment. Under this approach a matrix is compiled linking gross fixed capital formation by industries (according to ISIC), by type of asset (see Box 6.6) and by products (according to CPC). This matrix is often referred to as the "investment matrix". In order to develop such a matrix by industry, and by institutional sector, for each type of asset there should be an allocation to the appropriate product. In some cases, there may be a one-to-one relationship between the asset and product but in the case of machinery, for example, there are many one-to-many relationships.

Box 6.6 Gross fixed capital formation by type of asset

Gross fixed capital formation is usually shown by type of asset. The types of assets distinguished in the 2008 SNA are the following (see 2008 SNA, chapter 10, table 10.2).

Gross fixed capital formation by type of asset:

Dwellings

Other buildings and structures

Buildings other than dwellings

Other structures

Land improvements

Machinery and equipment

Transport equipment

ICT equipment

Other machinery and equipment

Weapons systems
Cultivated biological resources
Animal resources yielding repeat products
Tree, crop and plant resources yielding repeat products
Costs of ownership transfer on non-produced assets
Intellectual property products
Research and development
Mineral exploration and evaluation
Computer software and databases
Computer software
Databases
Entertainment, literary or artistic originals
Other intellectual property products
The output and capital formation of research and development are particularly difficult to measure. In theory, the value of the output of research and development is equal to the value of discounted future benefits that businesses get from their research and development investment. These future benefits are difficult to estimate. Furthermore, most research and development is produced on an own-account basis. For that reason, the sum-of-costs approach to the valuation of output will usually be applied. More detail on the impact of capitalizing research and development costs on SUTs and IOTs is provided in annex B to this chapter.

6.109. Table 6.10 shows the structure of the matrix linking gross fixed capital formation by type of asset, by product and by industry. Typically this matrix is based on surveys of capital expenditure, which tend to focus on institutional sectors, industries and assets; supplementary data sources are often also needed, however, such as, for example, specifically designed surveys which collect and provide investment detail by type of product. Detailed information on investments is of particular importance for gross fixed capital formation, which tends to be an erratic series and cannot be modelled easily: for instance, not all businesses buy vehicles every year, and the length of their use will vary across, and within, industries.

Table 6.10 Table linking gross fixed capital formation by industries, assets and products

6.110. In countries where such level of detail data is not available, the product flow approach may be applied, using assumptions linking the output of a product to the destination of the product in terms of its purpose. This is a less optimal approach but allows output and demand to be matched.

6.111. It is important to note that gross fixed capital formation – like other product-based variables in the use table – is valued at purchasers’ prices. It is recorded, however, by including any non-deductible VAT and excluding any deductible VAT. This will have an impact on those industries and products to which exemption applies and will be consistent with the valuation of the intermediate inputs for the corresponding industries.

6.112. In its simplest version, the use table may show gross fixed capital formation as a single column, and this would also fulfil the requirements for compiling SUTs and some users' needs. The single column approach may be preferred if information on gross fixed capital formation is

missing or incomplete. It should, however, be possible to distinguish between broad groups of assets based on the product classification used in the SUTs framework.

6.113. The quality of the product breakdown is greatly enhanced, however, by the greater level of detail linking industries, institutional sectors, assets and products. The disaggregation of gross fixed capital formation by industries and institutional sectors is also needed to calculate the consumption of fixed capital by industry, and in turn, the value of non-market producers' output. The breakdown can be done by columns that correspond fully to the columns for output and intermediate consumption. It may, however, be feasible to limit the number of columns in such a breakdown if precise information from source data is lacking.

6.114. If all combinations of gross fixed capital formation by types, industries and institutional sectors of a detailed matrix were shown as columns in the use table, these columns would completely dominate the presentation. Furthermore, a disproportionate share of the resources needed to balance the SUTs might be required to distribute products between the gross fixed capital formation columns, and finalization of the SUTs might be delayed unnecessarily.

6.115. One practical solution could be to show columns for a few broad categories of industries. Another could be to show only columns for different types of capital formation within the SUTs framework. The breakdown by industries could instead take place outside the central SUTs framework in a subsystem of investment matrices. Here, gross fixed capital formation by product from the final balanced version of the use table could be allocated to specific industries and institutional sectors as a separate process.

6.116. Estimates of gross fixed capital formation by industries have, however, an important role in the preparation of initial column totals for gross fixed capital formation for the use table. Furthermore, a preliminary version of gross fixed capital formation by products and industries can provide the starting point for the gross fixed capital formation columns of the use table.

6.117. Table 6.11 illustrates the gross fixed capital formation by industry and product link to the gross fixed capital formation column in the use table. In essence, this identifies the producers of capital goods in the rows and the investing industries in the columns.

Table 6.11 Gross fixed capital formation by investing industry

INDUSTRIES	INVESTING INDUSTRIES						Total at purchasers' prices
	Agriculture	Manufacturing	Construction	Trade, transport, communication	Financial and business services	Other services	
PRODUCTS	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Agricultural products (1)	128	2	0	4	36	10	180
Manufactured products (2)	1 223	7 225	664	5 893	9 124	2 626	26 756
Construction (3)	828	1 752	224	3 822	15 995	2 534	25 155
Trade, transport, communication (4)	12	1 206	189	2 280	1 672	684	6 043
Financial and business services (5)	124	4 331	82	1 324	3 065	2 245	11 170
Other services (6)	0	0	0	0	0	113	113
Total (7)	2 314	14 516	1 160	13 323	29 892	8 212	69 418

Table based on 2011 figures from Austria

Note: the products "Trade", "Transport" and "Communication" of table 6.1 are presented together in table 6.11

6.118. The gross fixed capital formation matrix has further roles to play. The calculation of capital stock data and the calculation of a valuation matrix for non-deductible VAT require an assessment of gross fixed capital formation by product (by producing industry) and investor (by investing industry). In the investment matrix, the user concept of capital – and not the owner concept of capital – should be reflected.

6.119. The assessment of consumption of fixed capital as a component of GVA should be based on empirical capital stock data. The human-made capital stock is derived from cumulative investment of the past in buildings, machinery and transport equipment, based on the actual lifetime of a capital good and allowing for retirements and obsolescence by using the perpetual investor method (see OECD, 2009).

6.120. Consumption of fixed capital is calculated at the current replacement cost of the net capital stock. The net capital stock is defined as the financial value of the gross capital stock still in use. Clearly, it will be easier to estimate these types of matrices on the basis of a rectangular SUTs system with many disaggregated homogeneous products.

(a) Sources for gross fixed capital formation

6.121. For industries that are covered by business statistics, the source data will usually include information on purchases and sales of capital equipment. It is normally possible to distinguish between gross fixed capital formation in buildings, structures, vehicles and machinery and equipment. Investment in intellectual property products is usually also shown but it is not necessarily classified by categories that can be used for national accounts. For example, these sources will often include purchased software and may also show a value for larger software projects but not all projects produced on own account. The measurement process is further complicated with the inclusion of acquisitions of patents, franchises and goodwill and the fact that research and development may or may not be wholly or partially capitalized in the source data and not necessarily recorded in line with the requirements of national accounts.

6.122. The distinction between intermediate consumption and gross fixed capital formation in business statistics sourced from business accounts may be different from that recorded in the national account. For example, items such as purchased computer software may be recorded as current expenditure in business accounts but as gross fixed capital formation in the national accounts, thus an equal and opposite adjustment is required. There are other conceptual differences between company accounts and national accounts, such as the following:

- Business accounts may not fully show gross fixed capital formation according to economic ownership. Some financially leased assets may be included in investment by their legal owners.
- Sales of existing assets should be treated as a negative gross fixed capital formation valued at the actual prices obtained by the seller. When the sale takes place between two resident producers, the positive and negative investment will cancel out for the economy as a whole, except for costs of change of ownership. In business accounts, the figures for disposals of assets will often be shown at historical cost while the corresponding cumulated depreciation is shown as a separate item. The difference is the bookkeeping value of the sold asset. If the actual price obtained differs from this value, the residual is included in secondary income, and should be reviewed and adjusted for as appropriate. In practice, it can usually be assumed that the difference between the negative gross fixed capital formation and the bookkeeping value of the sold asset is insignificant but there may be important exceptions where figures from company accounts are misleading.
- In business accounts, it is common practice to treat minor or regular purchases of equipment as current expenses. Such acquisitions may not always be identifiable in the accounts. For some big corporations, the threshold for classifying purchases as investment can actually be high, say, \$10,000 or more, but practices may vary between countries owing to differences in legislation and taxation rules.
- Own account production of capital goods may be capitalized in business accounts. Even if this is not the case, the accounts may contain information on the value of own account production. The value shown may not, however, be at basic prices as it may only include the direct cost of raw materials and wages and salaries attributed to its production, in which case a correction for indirect costs and gross operating surplus may be appropriate.
- Own account production of intellectual property products may not be directly identifiable in company accounts. Production of software, databases, research and development and literary, artistic or entertainment originals may sometimes have been capitalized as intangible assets in company accounts but may be treated as current expenses. For some intangibles, gross fixed capital formation may be covered by business surveys. In the absence of further detail, it is recommended that gross fixed capital formation is estimated based on the wages and salaries paid for this kind of work with an appropriate mark-up for other

expenses and typical gross operating surplus. More details on this matter may be found in the Handbook on Deriving Capital Measures of Intellectual Property Products (OECD, 2010).

6.123. Information on gross fixed capital formation within general government can usually be found in government accounts. Many of the above issues are also relevant to investment within general government. Accounts of central and local government will usually include a level of detail that reveals the distinction between intermediate consumption and gross fixed capital formation. Extra-budgetary units within general government may, on the other hand, provide less information on the nature of their costs as may also be the case for NPISHs.

6.124. Special care should be taken when projects are partially or wholly financed by capital transfers from outside, for instance from international organizations. In such cases, the accounts may show values that are net of financing from outside. In national accounts, gross fixed capital formation should record the full value of such projects.

6.125. The value of investment covered by business accounts will often give an incomplete picture of total gross fixed capital formation because some industries are only partially covered or lack information on some types of investment. The initial estimates of gross fixed capital formation may be prepared within an investment matrix framework that shows investment by industries, institutional sectors, types and products. In the event that uncertain data and so-called “guestimates” are used in many cells within such a framework, the framework can indicate the cells that are badly covered by source data but the cells should definitely contain values.

(b) Gross fixed capital formation by products

6.126. It is recommended that regular business surveys should be used as the key source for gross fixed capital formation by product, in particular since gross fixed capital formation is an erratic time series and cannot be modelled easily. If possible, these surveys should be linked to those collecting details on purchases of goods and services for intermediate consumption, in order to avoid double-counting or missing expenditure. Surveys of the product structure of gross fixed capital formation may exist but may not cover all industries. As for intermediate consumption, it may be possible to find detailed information in the annual reports of enterprises.

6.127. Government accounts contain information providing much more detail than a simple distribution by main types of investment. This information can, including for investment products, typically appear in an unsystematic form and it will need to be coded by product categories in SUTs before it can be used.

6.128. For many industries available information on the product dimension of investment is limited to a few categories or even non-existent. Initial estimates will therefore require some common-sense decisions.

6.129. In the end, the product structure of gross fixed capital formation will to a large extent be determined by the availability of investment products within the SUTs framework. In the simplest case, where other information is unavailable, gross fixed capital formation will alone be determined by the supply of typical investment products that are largely not used for other purposes.

6.130. Very few products are used exclusively for gross fixed capital formation. In most cases, the distribution between intermediate consumption, household final consumption expenditure and gross fixed capital formation that can be estimated using information from business surveys and company accounts is uncertain, in particular if the rows of the SUTs represent broad categories of products. The disaggregation of products can reduce this uncertainty, when the necessary source data are available. For some products, the distribution may be based on other types of information, as in the following examples:

- If an official register of motor vehicles is accessible, it may be possible to identify the changes from year to year in the numbers of different types of cars and lorries by age, size and ownership. Combined with typical prices for the various groups of vehicles, the register may be used to estimate household final consumption expenditure together with gross fixed capital formation by industry of new and used vehicles. Registers may also provide information to help distribute registration taxes by purpose.
- Similarly, it may be possible to use register information to follow the capital stock, and change in capital stock for other types of transport equipment.
- Official registers of buildings may also include information on type, size, use and owners and may be used for the distribution of buildings by purpose. For dwellings and private commercial buildings that are not completely covered by company accounts or business surveys, register information can be used in the estimation of the value of investment.
- Information on investment in buildings, transport equipment or specific types of machinery and investment may be collected by business surveys. Some countries carry out quarterly and annual business surveys covering gross fixed capital formation and underlying details. On the other hand, some countries occasionally carry out comprehensive industrial censuses that may contain information on the use of capital equipment for units that are not covered by other kinds of statistics.
- Household budget surveys can include information on investment in new dwellings and capital repairs. In countries with a large informal economy, household budget surveys may be the most important source for the estimation of investment in buildings, machinery and equipment in small farms (which may also, however, be included in agricultural censuses), small retail trade and repair workshops.

6.131. Furthermore, the product structure of investment tends to be more volatile than the structure of input cost structures. Nevertheless and despite its uncertainties, an initial version of investment by products estimated from the uses side is usually preferred to an estimated distribution based solely on the supply of specific products.

2. Changes in inventories

6.132. In order to define changes in inventories, it is useful to define first what is covered by inventories. Inventories are produced assets, consisting of goods and services, that came into existence in the current period or in an earlier period and that are held for sale, use in production or other use at a later date (2008, SNA, para. 10.12). Changes in inventories are measured by the value of the entries into inventories less the value of withdrawals and less the value of any recurrent losses of goods held in inventories during the accounting period. Some of these acquisitions and disposals are attributable to actual purchases or sales but others reflect transactions that are internal to the enterprise (2008 SNA, para. 10.118).

6.133. The change in inventories column in the use table should be underpinned with a matrix linking, as column headings, the classification of industries (for example, ISIC), and, as row headings, the product grouping (for example, CPC) as appearing in the SUTs for each type of asset. Changes in inventories can be analysed by industry and by types of assets, which need to be linked via CPC in the SUTs as illustrated in Table 6.12.

Table 6.12 Table linking change in inventories industries, assets and products

Changes in inventories used for correction from sales and purchases to output and intermediate consumption

ASSET TYPE	Changes in inventories					Total
	Materials and supplies	Finished goods	Work-in-progress	Goods for resale, wholesale	Goods for resale, retail	
PRODUCTS						
1						
2						

ASSET TYPE	Changes in inventories, Materials and supplies					Total
	Agriculture	Forestry and logging	Fishing and aquaculture	...	Activities of households as employers	
PRODUCTS	ISIC 01	ISIC 02	ISIC 03	...	ISIC 97	
1						
2						

ASSET TYPE	Changes in inventories, Finished goods					Total
	Agriculture	Forestry and logging	Fishing and aquaculture	...	Activities of households as employers	
PRODUCTS	ISIC 01	ISIC 02	ISIC 03	...	ISIC 97	
1						
2						

ASSET TYPE	Changes in inventories, Work in progress					Total
	Agriculture	Forestry and logging	Fishing and aquaculture	...	Activities of households as employers	
PRODUCTS	ISIC 01	ISIC 02	ISIC 03	...	ISIC 97	
1						
2						

ASSET TYPE	Changes in inventories, Wholesale trade					Total
	Agriculture	Forestry and logging	Fishing and aquaculture	...	Activities of households as employers	
PRODUCTS	ISIC 01	ISIC 02	ISIC 03	...	ISIC 97	
1						
2						

ASSET TYPE	Changes in inventories, Retail trade					Total
	Agriculture	Forestry and logging	Fishing and aquaculture	...	Activities of households as employers	
PRODUCTS	ISIC 01	ISIC 02	ISIC 03	...	ISIC 97	
1						
2						

6.134. The change in inventories should separately distinguish the following types of assets:

- Materials and supplies, which consist of all products that an enterprise holds in inventory with the intention of using them as intermediate inputs into production.
- Work-in-progress, which consists of output produced by an enterprise that is not yet sufficiently processed to be in a state in which it is normally supplied to other institutional units.
- Finished goods, which consist of goods produced as outputs that their producer does not intend to process further before supplying them to other institutional units.

- Military inventories, which consist of single-use items, such as ammunition, missiles, rockets, bombs and other items, delivered by weapons or weapons systems. Some single-use items such as ballistic missiles highly destructive may be classified as fixed assets in the sense that they provide deterrence services against aggressors.
- Goods for resale, which are goods acquired by enterprises, such as wholesalers or retailers, for the purpose of reselling them to their customers.

6.135. Although change in inventories for all asset types appear in the final uses part of the use table, they also play a role across other parts of the supply table and use table:

- For each industry, intermediate consumption can be calculated as purchases of goods and services less change in inventories of materials, fuels and raw materials.
- For each industry, output can be calculated as sales plus change in inventories of work-in-progress and finished goods. Producers of services may actually have inventories of work-in-progress in the form of projects lasting for more than one accounting period, such as films, advertising campaigns, legal contracts, and so forth.

6.136. The output value in trade can be calculated as sales less purchases less change in inventories of goods purchased for resale without any further processing. If a distinction between wholesale and retail trade is made in SUTs, then the calculation of the output value for each of these industries requires separate values for the change in inventories of goods for resale.

6.137. For estimating output and intermediate consumption, changes in inventories are usually estimated by industry and institutional sector. These detailed breakdowns are, however, seldom shown in the use table.

6.138. It must be noted that the changes in inventories reported by the survey respondents should be adjusted to comply with national accounts definitions: thus the changes in inventories should not include any holding gains or losses. Source data may be presented as output and intermediate consumption instead of sales and purchases. In practice, the data collected may nevertheless contain values of sales and purchases. These data may be replaced by a correction that uses inventories that appear in the company accounts. If such changes include any holding gains or losses, the correct treatment is to remove the original correction for inventory changes and replace it by a correction that uses inventory changes according to national accounts definitions.

6.139. Statistics based on company accounts usually contain information on inventory changes and also on stocks of inventories. The value shown for change in inventories will usually include holding gains and losses and will be misleading if significant price changes take place during the year. A correction for holding gains and losses can usually be based on the nominal values of opening and closing stocks, or the book value levels. It will require the availability of adequate information on price changes during the year. The correction can typically be carried out by the

inflation or deflation of opening and closing stocks to the average prices of the year before calculation of the difference. To calculate constant price values of stock of inventories (and changes in current prices), these prices values should be broken down by products for which price indices or volume indicators can be found.

6.140. Annual company accounts may not contain information of opening stocks, in which case it might be necessary to use values of closing stocks from the previous year as a measure of opening stocks. As coverage changes and establishments are reclassified, the observed differences between closing stocks and opening stocks may then need some adjustments at the industry level.

6.141. Given the links between sales, purchases and inventories, along with other variables, it is recommended that the data on inventories are collected via the same survey questionnaires to ensure coherence across the variables being collected with the same source and at the same point in time.

6.142. The Eurostat-OECD Compilation Guide on Inventories (Eurostat and OECD, 2017) provides more detail relating to compilation issues and guidance.

(a) Estimation of changes in inventories by product

6.143. To fill in the columns for changes in inventories, the totals for this item must be broken down into changes in inventories for the products used in the SUTs and must be completed as part of generating the estimates of output and intermediate consumption.

6.144. For some products, the values of changes in inventories may be calculated based on knowledge of physical opening stocks and closing stocks and information on the development in prices, for example, for crops and livestock in agriculture and for energy products. The calculation of changes in inventories should be an integral part of subsystems used to provide the complete product balances for such products.

6.145. The inventories of most industries include a broad selection of products that are usually not known from statistical sources. The totals of opening stocks and closing stocks used to calculate changes in inventories need to be distributed by products based on assumptions on the product structure for each total. The following are typical examples:

- Inventories of finished goods and work-in-progress can be distributed proportionally with those outputs of each industry that can appear in inventories of goods. Caution will be needed on service products.
- Inventories of raw materials and fuels can be distributed proportionally with the use of inputs of each industry that can appear in inventories.

- Inventories of goods for resale can be distributed using various proxies for the product structure, for example, output or input in specific industries, household final consumption expenditure in specific COICOP groups or gross fixed capital formation of specific types.
- Values of inventories of specific products may already be known from other calculations. Such values can be retained as predetermined values.
- Some products, for instance electricity, are not likely to appear in inventories. Services should only appear in inventories as work-in-progress.

6.146. Calculating the distribution of changes in inventories by products in this way is of course uncertain and it may be adjusted during the balancing of SUTs. There is also a danger that errors in source data will not be detected if changes in inventories are used as balancing items or not subjected to adequate quality assurance. The total of changes in inventories should, at least, only be adjusted in exceptional cases, where this is believed to be the most realistic solution to a balancing problem.

6.147. It should be noted that any balancing adjustment to the asset composition of changes in inventories, or to the total changes in inventories, will affect different parts of the supply table and use table and will have the same total impact on each of the production, income and expenditure approaches to measuring GDP. This is because the various components of changes in inventories feed into the estimation of output and intermediate consumption and, in turn, GVA and final uses, thus the impact of any adjustment will be equal on production, income and expenditure. It is more likely that quality adjustments will be made to source data, where such changes will, and should, change the intermediate consumption and output and, in turn, GVA of the corresponding industries.

3. Acquisitions less disposals of valuables

6.148. Valuables are produced goods of considerable value that are not used primarily for purposes of production or consumption but are held as stores of value over time (2008 SNA, para. 10.13). Valuables include precious metals and stones, antiques and other art objects and other valuable items. Acquisitions less disposals of valuables capture these alternative forms of investments. Capital formation in valuables usually needs to be based on the domestic supply of specific goods, that is, imports less exports plus the margin. Valuables are by nature difficult to distribute across industries (based on establishments) as they share certain properties with financial assets, and the industry breakdown does not reflect those valuables held by, for example, households.

6.149. Trade data by detailed product provide a good source for identifying such items. Whereas the margin or fee type data by detailed product may be collected via questionnaires sent, for example, to auctioneers. For SUTs, the product breakdown is key: thus the relevance of imports

less exports by type of product. In terms of the purchaser, this is less important, as the purchaser could be industry, government or households.

G. Exports

6.150. Exports are shown in the use table by product. Depending on the specific user's need, an additional breakdown by column could be provided to show exports by destination. It should be noted that the treatment, issues and sources of data applied to imports of goods and services are also applicable to exports of goods and services. More detail on the imports of goods and services may be found in chapters 4, 5 and 8, including such issues as the new treatment of goods sent abroad for processing.

6.151. In the use table, exports are valued FOB at the point of exit from the exporter's economy. This value includes the cost of transport from the exporter's premises to the border of the exporting economy. FOB price includes:

- Value of goods at basic prices
- Trade and transport services to the border
- Taxes minus subsidies on products; there is no VAT on exports

6.152. Since data on exports on goods are collected on a FOB basis, no further transformation is needed.

1. Data sources

6.153. Most countries have comprehensive foreign trade statistics for goods. Data are generally collected according to the Harmonized System, valued at FOB and often available with a high level of detail, say by six-digit or eight-digit Harmonized System codes. Thus the only adjustment needed to the basic data is the conversion between Harmonized System codes and CPC. It is usually possible to convert the data from the Harmonized System classification using a correspondence table from the United Nations website (<http://unstats.un.org/unsd/class/default.asp>), supplemented by a conversion table that defines the SUTs products as aggregates of CPC classifications.

6.154. Various adjustments would be needed to move the foreign trade statistics on to a balance of payments basis in line with BPM 6, such as the change in economic ownership and the difference crossing the border.

6.155. Foreign trade statistics will also usually include the distribution of exports of goods by countries for all products.

6.156. Enterprises with exports below certain threshold values can be allowed to report their foreign trade without a distribution by products, for example with survey-based external trade

statistics. In this case, the values of exports by products will need to be grossed up to cover total exports. The difference between grossed-up and reported values is uncertain, and may need to be corrected during the balancing of SUTs. As for inputs, information on the reported values can be shown together with the grossed-up values in the tables presented to those working on the manual balancing of SUTs before any automated balancing process.

6.157. The main source for data on exports of services is the balance of payments-based data and the sources used to produce these data. The classification according to the EBOPS 2010 (United Nations *et al.* 2011 and IMF, 2009) will usually provide sufficient detail for conversion into the classification used in the SUTs. Failing this, access may be available to statistics that show imports and exports by industries. A conversion into detailed SUTs-based products can be established based on the coding in balance of payments and the information on the industry classification of exporting units.

Annex A to chapter 6: Sample questionnaire collecting purchases of goods and services for intermediate consumption

A6.1 The extract shown in Figure A6.1 is from a business survey questionnaire from the Statistical Office of Serbia. The data are collected for each industry and by product covering the following:

- Cost of materials
- Closing stocks of materials and fuels

A6.2 Full coverage of goods and services consumed as intermediate consumption to calculate the industry totals is achieved via further tables collecting data on the costs of industrial and non-industrial services, an extract from which is shown in figure A6.2. These data make it possible to calculate the intermediate consumption by product required to populate the intermediate use part of the use table, as shown in table 6.1.

Figure A6.1 Extract from questionnaire covering costs and closing stocks of raw materials and other material inputs

No.	Code	Product description	Cost of materials (group of account 5.1)	Closing stocks (group of accounts 10)
1	2	3	4	5
3000		TOTAL		
		AGRICULTURAL PRODUCTS, RAW AND UNPROCESSED PRODUCTS OF PLANT AND ANIMAL ORIGIN		
3001	01.11.1 - 01.11.4	Cereals, all kinds (except rice), cereal seeds		
3002	01.11.6 - 01.11.7	Green leguminous vegetables (beans, peas, lentils and other)		
3003	01.11.8	Soya beans, groundnuts (row) and cotton seed		
3004	01.11.9	Other oil seeds - sunflower, sesame, lin etc.		
3005	01.11.12	Rice, not husked		
3006	01.13 except 01.13.7	Vegetables, raw		
3007	01.13.7	Sugar beet and sugar beet seed		
3008	01.13.8	Mushrooms and truffles		
3009	01.15	Unmanufactured tobacco		
3010	01.16	Fibre crops (lin, cotton and other fibre crops, used in textile industry)		
3011	01.19.1	Forage crops and vegetative matter for livestock feeding unprocessed form		
3012	01.19.2	Flowers and flower seeds		
3013	01.21	Grapes		
3014	01.22 - 01.23	Tropical and subtropical fruits (citrus, figs etc.)		
3015	01.24, 01.25 except 01.25.3	Other fruits, tree and bush fruits, except nuts (apples, pears, cherries, berries etc.)		
3016	01.25.3	Nuts (almonds, hazelnut, walnuts etc.)		
3017	01.26	Olives, coconuts (row, unprocessed)		
3018	01.27	Coffee beans, tea leaves, cocoa beans, not roasted		
3019	01.28	Spices, aromatic, drug and pharmaceutical crops		
3020	01.11.5, 01.14, 01.19.3, 01.29, 01.3	Vegetables and fruit seeds, other seeds, grass, unprocessed straw and other residues of cereals, seeds for trees and seedlings, planting materials, sugar cane and other raw, unprocessed and untreated products of plant origin n.e.c.		
3021	01.4. except 01.45.3 & 01.49.3	Live animals and raw animal products (unprocessed milk, eggs, natural honey, except raw skins, shorn wool and skins, see line 3022, etc.)		
3022	01.45.3, 01.49.3	Raw fur skins, shorn wool, skins (excluding products of slaughterhouses and industrial meat production, see 1036)		
3023	01.49. part	Other animal products, raw, unprocessed and untreated		
3024	01.7	Hunting and trapping products, raw		
		PRODUCTS OF FORESTRY		
3025	02.2	Wood in the rough - logs, fuel wood and other raw products of forestry		
3026	02.1, 02.3	Forest trees and seeds, wild growing edible products; natural cork, varnish, balsams and other raw products of forestry n.e.c.		
		FISH AND OTHER FISHING PRODUCTS, UNPROCESSED AND UNTREATED		
3027	03	Fish and other fishing products; aquaculture products (raw, unprocessed and untreated)		
		MINING AND QUARRYING PRODUCTS, UNPROCESSED		
3028	05.1, 05.2	Coals, hard coal and lignite		
3029	06.1	Crude petroleum, bituminous or oil shale and tar sands. Note petroleum products - fuels are entered in the row 3118		
3030	06.2	Natural gas, processed (Manufactured gas distributed through mains, heating gas and petroleum gases from refineries should be reported in rows 3116 and 3119)		
3031	07.1	Iron ores		
3032	07.2	Other metal ores		
3033	08.1	Stone, sand, and clay and other raw materials for construction, industrial and craft activities		
3034	08.9	Other mining and quarrying products n.e.c.		
		MANUFACTURING INDUSTRY PRODUCTS		
		Food products and other processed products of plant and animal origin; used as reproduced material		
3035	10.11 except 10.11.4 & 10.12.5	Meat (red meat, including frozen) except live animals and unprocessed and untreated products of animal origin (goes to rows 1021 - 1024); raw offal and edible fat and oils		
		Electricity, refined petroleum products for energy purposes, gas (excluding natural gas), steam, hot water, air conditioning (including energy products use for heating)	Account 513	
3115	35.11	Electricity costs		
3116	35.22	Manufactured gas for industrial purposes and for heating - gas distributed through mains (excluding natural gas, see 3030, petroleum gas from refineries, see 3119 and industrial and medical gases, see 3069)		
3117	35.30	Steam and hot water, air conditioning supply services		
3118	19.2	Refined petroleum products - motor, engine and other fuels		
3119	19.2	Petroleum gases - propane, butane etc. (excluding natural gas and industrial and medical gases)		

Figure A6.2: Extract from questionnaire covering costs of industrial and non-industrial services

No.	CPA code	Product description	Costs of services	Account code
1	2	3	4	5
4000		TOTAL		
		Support services directly linked with the production of goods and services		
4001	01.6 part	Support agricultural services to crop production		
4002	01.6 part	Support services to animal production; veterinary services excluded (row 4044)		
4003	02.10.2, 02.4	Support services to forestry		
4004	09	Mining support services; services to petroleum and natural gas extraction		
4005	13.3	Textile finishing services—bleaching, dyeing, printing etc.		530
4006	16.10.9	Drying, impregnation or chemical treatment services of timber and product of wood; support services in the processing of wood and wood products n.e.c.		
4007	25.5	Forging, pressing, stamping and roll-forming services of metal		
4008	25.6	Treatment and coating services of metals, machining		
4009	24.5	Casting services of metal and steel		
		Subcontracted services in industry and construction, trade services and other intermediation commissions. Note: enter only the value of the services, value of materials of goods excluded		
4010	14, part	Subcontracted operations in textile industry (excluding value of materials)		
4011	15, part	Subcontracted operations in footwear and leather production industry (excluding value of materials)		
4012	16, part	Subcontracted operations in production of processed wood and wood products (excluding value of materials)		
4013	25, part	Subcontracted operations as part of machine industry—processing and finishing materials services (excluding value of materials)		530, 539
4014	41, 42, 43	Subcontracted operations in construction		
4015	—	Other subcontracted operations in production of goods of other enterprises (excluding value of materials), please specify		
4016	46.1	Trade commissions		
4017	—	Other intermediation commissions, please specify		
		Transportation costs, postal and courier services		
4018	49	Land transport of freight, taxi operation services including rental services of land transport vehicles with operator		
4019	50	Water transport		
4020	51	Air transport		531
4021	52.2	Support services for transportation (loading, unloading, hauling, towing, parking services, etc., transportation excluded)		
4022	53	Postal services under universal obligation		
4023	53	Other postal and courier services		
		Repair, maintenance, installation services; conversion, reconstruction and fitting out of transport equipment		
4024	33.1	Repair and maintenance services of fabricated metal products, machinery and equipment, except motor vehicles		
4025	43	Repair and maintenance services of buildings and electrical, plumbing, heating and similar installations		
4026	45.2	Maintenance and repair services of motor vehicles		
4027	95.1	Maintenance and repair services of computers and communication equipment		
4028	95.2	Repair services of personal and household goods		532
4029	33.2	Installation services of industrial machinery and equipment		
4030	29.20.4, 29.20.5	Reconditioning, assembly, fitting out and bodywork services of motor vehicles, except installation, maintenance and repair services		
4031	30.11.9, 30.20.9, 30.30.6	Conversion, reconstruction and fitting out services of other transport equipment, except installation, maintenance and repair services		
		Rentals, rents on land, warehousing and storage services		
4032	68.2, part	Rental costs on buildings and office space owned by legal persons (except rents on land)		
4033	68.2, part	Rental costs on buildings and office space owned by natural persons (except rents on land)		
4034	68.2, part	Rents on private land		533
4035	68.2, part	Rents on public or state land		
4036	77	Rental and leasing services of motor vehicles, machinery, equipment and tangible goods (excluding real estate and financial lease)		
		Expenditure related to the intellectual property (royalties, licence fees, rights of usage, publication, reproduction, transmission, broadcasting and the like), other services n.e.c. Note: services as expenditures; if not capitalised		
4089	58.1	Royalties for the publication of books, magazines, new spapers, etc.		
4090	58.2	Royalties and similar payments for usage of software		
4091	59	Royalties for publishing (music and movies, TV series)		
4092	60	Royalties (broadcast rights, etc.) in the production and broadcast of radio and television programs		539 or 559
4093	71.2	Certification of products and processes		
4094	77.4	Royalties and fees for the use of intellectual property if it is not capitalised		
4095	96.01	Laundry and washing of textile and fur		
4096	—	Other industrial and non-industrial services not elsewhere specified please specify		

Annex B to chapter 6: Impact of capitalizing the costs of research and development in SUTs and IOTs

A. Research and development as fixed capital formation

B6.1 The 2008 SNA introduced changes relating to the treatment of research and development. Research and development is creative work undertaken on a systematic basis in order to increase the stock of knowledge, including knowledge of people, culture and society, and to enable this stock of knowledge to be used to devise new applications. The 2008 SNA does not treat research and development activity as an ancillary activity and it recommends that a separate establishment should be distinguished for research and development when possible.

B6.2 The output of research and development should be capitalized as “intellectual property products” except where it is clear that the activity does not entail any economic benefit to its producer (and hence owner), in which case it is treated as intermediate consumption.

B6.3 The 2008 SNA now includes expenditures for both bought-in and own-account research and development as gross fixed capital formation and the depreciation of these assets as consumption of fixed capital.

B6.4 Table B6.1 shows a summary of the impact of these changes using a simplified hypothetical example simply to demonstrate the impact of capitalization. The example is simplified in the sense that it shows only two years and the capitalization of research and development (R&D) is assumed to be introduced in year 1, when there is no increase in the consumption of fixed capital. In reality, the capitalization costs will be spread over several successive years and the consumption of fixed capital will occur in year 1 and throughout the lifetime of the asset.

B6.5 Under 1993 SNA, own-account R&D activity was treated as an ancillary activity and no separate output was estimated in the system, and expenditures for this purpose were not separately identified. Only in those cases, usually unimportant, where R&D services were purchased from outside specialist producers, classified in the research and development activity (ISIC Rev. 4, division 72) or imported, do R&D services appear in the SUTs as intermediate consumption.

B6.6 With the capitalization of R&D expenditures under the 2008 SNA, the output of own-account R&D is separately estimated and allocated to GCF or exports. In practice, the introduction of own-account R&D output in the system has just resulted in additional output with the intermediate inputs being left unaffected, as the intermediate inputs needed to produce the own-account R&D were already included. With the 2008 SNA change of treating research and

development not as an ancillary activity, the question for the compilation of SUTs and IOTs is whether the R&D output from own-account R&D activity should be seen as a secondary product from that particular branch or the principal product of the activity “Scientific research and development” ISIC Rev. 4, division 72. This issue of reclassification of own-account R&D activity is considered in the following section.

B6.7 The challenging methodological and practical problems related to the actual estimates of output of own-account research and development are not being dealt with at this point; reference is made to the Manual on Measuring Research and Development in ESA 2010 (Eurostat, 2014a), the Handbook on Deriving Capital Measures of Intellectual Property Products (OECD, 2010) and specific country documentations. In the following, only some problems of special interest for the compilation of SUTs and IOTs will be highlighted.

Table B6.1 Summary of the impact of the capitalization of research and development in the new 2008 SNA

Indicative impact	Change in the treatment of Research and Development			
	Market producers		Non-market producers	
	Ow n account	Bought-in	Ow n account	Bought in
	(1)	(2)	(3)	(4)
Intermediate consumption	0	-500	0	-500
Non-market consumption of fixed capital	0	0	(+200)	(+100)
Gross operating surplus	+1 000	+500	(+200)	(+100)
GVA	+1 000	+500	(+200)	(+100)
Total output	+1 000	0	(+200)	-500 (+100)
Output for ow n fnal use	+1 000	0	+1 000	0
General governeemt or NPISHs final consumption expenditures	0	0	-1 000 (+200)	-500 (+100)
Gross fixed capital formation	+1 000	+500	+1 000	*500
Impact on GDP	+1 000	+500	0 (+200)	0 (+100)

For own account value of production is 1000. For non-market producers, consumption of fixed capital in Year 2 is 200.

For bought-in, value of intermediate consumption is 500. For non-market producers, consumption of fixed capital in Year 2 is 100.

Year 1 estimates. Estimates in brackets to relate to consumption of fixed capital to impact in Year 2 for non-market producers.

B. Implications of valuation of output as sum of costs

B6.8 Output for own final use should be valued at the basic prices at which the goods and services could be sold if offered for sale on the market. When reliable market prices cannot be obtained, a second best procedure must be used, in which the value of the output of the goods or services produced for own final use is deemed to be equal to the sum of their costs of production, that is, as the sum of: intermediate consumption; compensation of employees; consumption of fixed capital; a net return to fixed capital; and other taxes (less subsidies) on production. By

convention, no net return to capital is included when own-account production is undertaken by non-market producers (2008 SNA, para. 6.125).

B6.9 The calculation of the output of own-account R&D from the sum of costs approach means that the cost structure for this particular type of output will be separately specified; this is a departure from the usual situation where intermediate and primary inputs used for various types of outputs will be indistinguishably lumped together. Based on the known cost structure, it would in principle be possible to create separate establishments for the own-account R&D activity, and if further on this activity was seen as secondary, to reclassify these establishments from the original activity to which they are classified, for example, pharmaceuticals, electronics and so forth, to the specialist activity for scientific research and development (ISIC Rev. 4, division 72).

B6.10 For a number of analytical, methodological and practical reasons, however, such a reclassification is not recommended. These reasons include, primarily, the following:

- There is an analytical interest in keeping track of those economic activities that actively involve R&D and this information would be lost through such a reclassification.
- As each R&D output is uniquely defined, the own-account research and development is usually not suitable for delivery outside the producing unit, and a reclassification would not reflect the economic reality of the activity.
- Given the lack of any other information, it is assumed that the producing unit is also the owner of the resulting R&D capital stock and the associated consumption of fixed capital – this assumption is often not valid.
- The cost structures calculated to derive the estimate of R&D output will not usually have product details corresponding to the SUTs product requirement, and in general, will only exist as internal worksheet exercises not intended for a wider audience.
- If all R&D were reclassified to ISIC Rev. 4, division 72, in many developed countries this division would increase to the same size or even larger than the agricultural sector, and seriously distort the relative proportions, in particular between manufacturing industries and service industries. It would also make industries active in R&D (such as the pharmaceutical industry) rather meaningless truncated residuals compared with the usual perception of the size and structure of these industries and thus the reclassifications would be counterproductive from the point of view of users' needs and a wide-range of analytical purposes.

C. Own-account research and development as principal or secondary output

B6.11 Research and development services, division 81 of CPC Version 2.1, also existed prior to the capitalization of R&D, and were mainly made up of the services actually sold in the market by

enterprises classified in division 72 of ISIC Rev. 4, “Scientific research and development”. This was generally small, however, when compared with the total value of own-account R&D estimated in connection with the capitalization.

B6.12 In division 81 of CPC Version 2.1, the subclasses are organized according to the type of research (for example, chemistry, biotechnology, and others), and not according to the economic activities carrying out the R&D, and all R&D services are indicated as characteristic products of ISIC Rev. 4 subclasses 7210 and 7220. This follows logically from the fact that CPC Version 2.1 was not designed for a situation where the overwhelming share of R&D services comes into existence as estimated own-account output in ISIC industries other than ISIC Rev. 4, division 72.

B6.13 Under the system of estimated own-account output of R&D services, it would be more appropriate to introduce a CPC structure for R&D similar to the structures for “Maintenance, repair and installation (except construction) services” (CPC Version 2.1, division 87) and for “Manufacturing services on physical goods owned by others” (CPC Version 2.1, division 88) where the subclasses (four digits) are made up industry specific outputs, each corresponding to a characteristic ISIC class (four digits). This means, in particular, that outputs of these services in industries which are active in R&D form principal outputs. When this approach is followed for own-account R&D services, there will be as many subclasses of CPC division 81 as there are industries with R&D activities, and own-account R&D will formally change from a secondary to a principal activity of the producing industries. The adoption of this approach will also have important implications when deriving IOTs from the SUTs, as it will prevent major structural differences between industry-by-industry IOTs and product-by-product IOTs, and avoid truncating research and development-intensive product-adjusted industries in the product-by-product IOTs.

B6.14 In practice, specialized R&D departments of enterprises with major own-account R&D activities may, for various reasons (legal, tax-related, and other), have already been classified in ISIC Rev. 4, division 72, in the business register, and thus be included in business statistics with this activity, rather than the principal activity (for example, pharmaceutical, electronic and so forth) of the parent enterprise.

B6.15 In such cases, the flows between the R&D department classified in ISIC Rev. 4, division 72, and the parent enterprise, and also the applied valuation principles, should be carefully assessed. In this connection, it is important to realize that business accounting practices will usually not follow the principle of capitalizing R&D expenditures. Thus the total output (however estimated) from an R&D department classified in ISIC Rev. 4, division 72, may, in the business accounts, reappear as intermediate consumption in the accounts of the parent enterprise. Depending on the circumstances, one solution might be to reclassify the R&D department back to the activity of the parent enterprise. In national accounts, sometimes legal structures may be overruled if they are found not to reflect economic realities. Alternatively, the intermediate consumption of R&D services could simply be removed from the parent enterprise and instead treated as gross fixed capital formation, but this may leave a truncated enterprise of little analytical

interest, as noted above, and it would still be necessary to deal with the valuation of reported output of the R&D department.

D. Balancing supply and use of research and development services

B6.16 Assuming that the output of R&D services (market, non-market and for own-use) by industry is available from the current national accounts calculations, allocation by user should, in principle, be fairly straightforward, as these services under the 2008 SNA treatment of R&D should be allocated to gross fixed capital formation. There are, however, two problem areas worth mentioning:

- Some research and development purchases are still to be treated as intermediate consumption.
- Foreign trade in research and development services must be taken into account when balancing the R&D services.

B6.17 When market R&D services are purchased by an own-account producer of R&D from a commercial R&D producer (usually, although not necessarily, classified in ISIC Rev. 4, division 72) or imported, it must be decided whether this is an acquisition of an asset or an intermediate product used as an input into the own-account production of R&D. As there is usually insufficient information available on the individual transactions for an informed decision to be made, it is recommended by the Eurostat Manual that, in the absence of any strong evidence to the contrary, all purchases by own-account R&D producers from units classified in ISIC Rev. 4, division 72, (and also purchases by other units in ISIC Rev. 4, division 72) should be treated as intermediate consumption. This assumption will also ease the distribution in cases where no product statistics for the output from ISIC Rev. 4, division 72, are available, as some output may be non-research and development services that would nonetheless usually be allocated to intermediate consumption.

B6.18 The above-mentioned case involving a specialized R&D department classified in division 72 of ISIC Rev. 4 may, however, interfere with this solution, and it is recommended that the SUTs compilers coordinate closely with the compilers of the R&D estimates made for the national accounts.

B6.19 In the 2010 Manual on Statistics of International Trade in Services (United Nations, IMF, OECD, European Union, UNCTAD, UNWTO and WTO, 2011) and Extended Balance of Payments Services Classification (EBOPS, 2010), there are special entries on R&D services, which are explicitly separated from transactions on the results of R&D (for example, royalties and license fees paid for use of patented entities).

B6.20 It would therefore appear that the balancing of the R&D services, also taking into account imports and exports, would be straightforward. The transactions registered in the balance of

payments are, however, actual economic transactions where the prices may be quite different from the cost-based valuation of the domestic own-account output of R&D services, and the delimitation of the R&D concept may also deviate. Further balance of payments transactions in R&D may include significant elements of transfer pricing and trade with subsidiaries in low-tax jurisdictions. When the aim is to achieve consistency with existing balance of payments data, the final balancing of the R&D services may be quite difficult, even though the capitalized R&D services allows trade in “used” R&D so that gross fixed capital formation may in principle (though not very realistically) become negative. More detail on these issues may be found in chapter 7 of the guidance on the impact of globalization on national accounts prepared by the Economic Commission for Europe (UNECE, 2011).

Chapter 7. Compiling the valuation matrices

A. Introduction

7.1. The compilation of valuation matrices is a fundamental step in the process of compiling SUTs. These matrices are necessary to bridge the different valuation concepts of the product flows. This chapter covers the main concepts and methods of compiling matrices for trade margins, transport margins, taxes on products and subsidies on products. In particular, in section B, the chapter starts with an overview of the valuation concepts in the 2008 SNA and of how the valuation matrices fit within the SUTs presented in chapters 5 and 6. Sections C–E elaborate on each component of the valuation matrices and describe the main compilation steps. The annex to this chapter provides further details on how to compile trade margins for the SUTs from survey data based on a country practice.

B. Valuation of product flows

7.2. Transactions are valued at the actual prices agreed upon by the purchasers and sellers. Market prices are thus the main reference for the valuation of transactions in the SUTs system, in line with 2008 SNA. In the absence of market transactions, the valuation is made according to costs incurred (for example, for non-market services produced by government) or by reference to market prices for analogous goods and services (for example, for services of owner-occupied dwellings).

1. Valuation concepts in the 2008 SNA

7.3. More than one set of prices may be used to value outputs and inputs depending on how taxes and subsidies on products, trade and transport margins are recorded. The 2008 SNA distinguishes three main valuation concepts of the flows of goods and services: the two main recommended valuations being basic prices and purchasers' prices and the lesser used producers' prices.

7.4. The basic price is the amount receivable by the producer from the purchaser for a unit of a good or service produced as output minus any tax payable, and plus any subsidy receivable, by the producer as a consequence of its production or sale. It excludes any transport charges invoiced separately by the producer (2008 SNA, para. 6.51).

7.5. The producers' price is the amount receivable by the producer from the purchaser for a unit of a good or service produced as output minus any VAT, or similar deductible tax, invoiced to the

purchaser. It excludes any transport charges invoiced separately by the producer (2008 SNA, para. 6.51).

7.6. The purchasers' price is the amount paid by the purchaser, excluding any VAT or similar tax deductible by the purchaser, in order to take delivery of a unit of a good or service at the time and place required by the purchaser. The purchasers' price of a good includes any transport charges paid separately by the purchaser to take delivery at the required time and place (2008 SNA, para. 6.64).

7.7. The difference between these valuation concepts for a product relates to trade margins, transport margins and taxes on products and subsidies on products. The relationship between the three types of prices is as follows:

Basic prices	+	Taxes on products excluding invoiced VAT
	-	Subsidies on products
	=	Producers' prices
	+	Wholesalers' trade margins
	+	Retailers' trade margins
	+	Separately invoiced transport charges
	+	VAT not deductible by the purchaser
	=	Purchasers' prices

7.8. The basic price measures the amount retained by the producer and, therefore, the price most relevant for the producers' decision-making and is often reported in business surveys. For imported products, taxes on products include import duties. When the relationship between basic prices and purchasers' prices is compiled for the total economy, the transport charges and trade margins will cancel out because they form only a reallocation of value across products.

7.9. The concept of producers' prices does not form any of the main valuations. The preferred valuation of output and GVA in the SNA is made at basic prices and for intermediate consumption at purchasers' prices. It is worth recognizing that source data from business surveys for sales may be valued at producers' prices. In these cases, data should be adjusted to a basic price valuation before they are entered into the SUTs. If this step is not completed, then a different recording of taxes on products and subsidies on products must be established, and GVA by economic activity would be partly at market prices, a practice which is not recommended by the SNA.

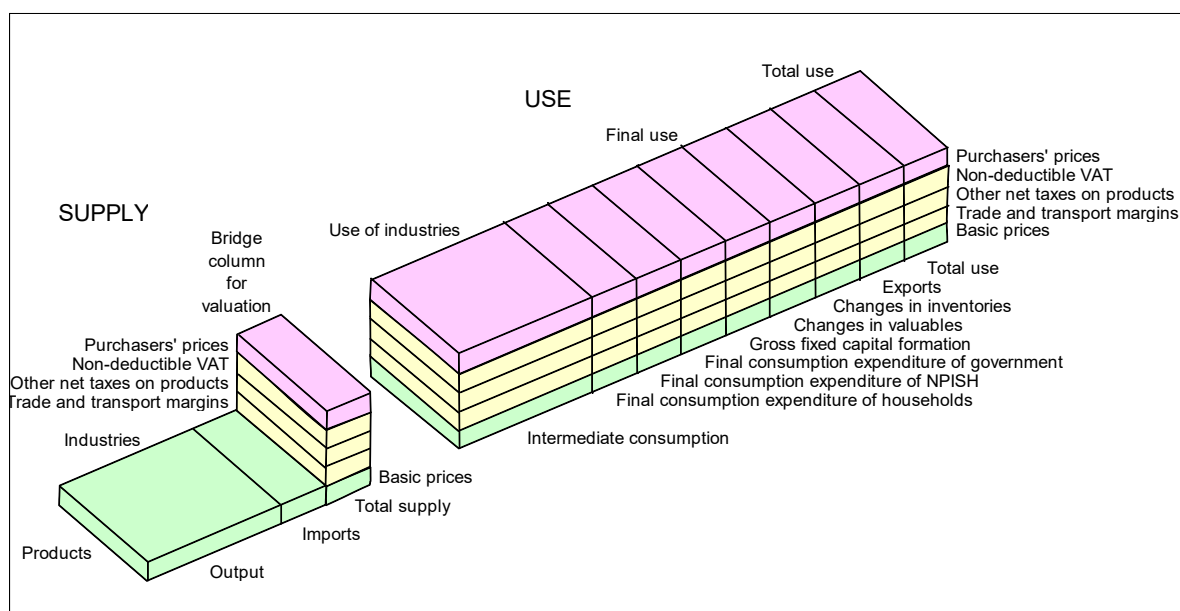
7.10. It is important to note that the relationship between basic price and purchasers' price does not describe a process over time for an identifiable product. In this case, the difference between basic prices and purchasers' prices is likely to contain an element of holding gains and losses while the product is with the producer and with wholesale and retail traders (2008 SNA, para. 3.148). The SNA value concepts are consistently defined in such a way that holding gains and losses do not become part of GVA and GDP. Hence a trade margin is relative to the replace price of the

good “at the time it is sold”, and the price of intermediate consumption relates to what the producer would have to pay to replace it “at the time it is used”.

7.11. The source data used to fill the cells in the SUTs may have different valuations:

- Production and output data are valued at basic prices.
- Intermediate consumption and final uses are valued at purchasers’ prices.
- Imports are valued at CIF prices. This is the valuation of a good delivered at the point of entry into the importing economy or the price of a service delivered to a resident, before the payment of any import duties or other taxes on imports or trade and transport margins within the country. In the SUTs framework, for national SUTs, the CIF value is taken to be the basic price of imports of goods.
- Exports are valued at FOB prices. This is the valuation of a good at the point of exit from the exporter’s economy or the price of a service delivered to a non-resident, including transport charges and trade margins up to the point of the border, and including any taxes less subsidies on the goods exported. In the SUTs framework, the FOB value is taken to be the purchasers’ price of exports of goods.

7.12. It is recommended that the different valuation components of the product flows are separated to ensure that the SUTs framework is balanced in a fully coherent and consistent manner. One of the purposes of the valuation matrices is to bridge the difference between the valuation at basic prices and the valuation at purchasers’ prices and to come up with SUTs at basic prices. Figure 7.1 illustrates how the valuation matrices link the supply table with the use table. They comprise all flows that are related to the supply and use of trade margins and transport margins and to taxes and subsidies on products.

Figure 7.1 Schematic representation of the valuation matrices in the SUTs

7.13. In the supply table, the valuation matrices consist of columns (the “bridge column” in Figure 7.1) which transform the supply of each product from basic prices to purchasers’ prices, and, in turn, match the product values in the use table compiled at purchasers’ prices.

7.14. In the use table, the valuation matrices consist of product-by-industry matrices of trade margins, transport margins, taxes on products and subsidies on products which allow for the transformation of the values of the use table from purchasers’ prices into basic prices. The availability of such matrices makes possible the balancing of SUTs at basic prices and purchasers’ prices and, as recommended in the Handbook, both valuations should be balanced simultaneously.

7.15. Although it is not strictly necessary that a balanced SUTs framework ends up with tables showing valuation both at purchasers’ and at basic prices, this is recommended for several reasons. For analytical purposes, the SUTs data must have the same valuation, and usually the basic price version is the most appropriate. This is also the case for the process of transforming the SUTs into IOTs and for the volume estimates in a consistent SUTs framework leading to the estimation of GVA in volume terms, using double deflation.

7.16. For these input-output based analytical purposes, a valuation as uniform as possible of the cells in a row of the use table is essential. The values at purchasers’ prices in the different uses will usually be affected by differences in trade and transport margins, and by differences in taxes on products and subsidies on products, according to the specific user. The uniformity requirement is therefore best fulfilled by values at basic prices, although the cells valued at basic prices may still show user specific price variation: this is the most uniform valuation concept that in practice can be achieved.

2. Valuation matrices in the SUTs framework

7.17. The valuation matrices comprise information on trade margins, transport margins, taxes on products and subsidies on products. Valuation matrices can be established for the supply table (supply-side valuation matrices) and the use table (use-side valuation matrices). In a balanced SUTs system, the use-side valuation matrices and the supply-side valuation matrices will add up to the same totals. In this section, the full set of valuation matrices is described.

(a) Supply-side valuation matrices

7.18. The supply-side valuation matrix consists of a set of columns added to the supply table at basic prices to derive the supply at purchasers' prices. These columns comprise trade margins, transport margins, VAT, taxes on products and subsidies on products. Table 7.1 shows the structure of the supply table at basic prices, including a transformation into purchasers' prices. The table corresponds to table 5.2 of chapter 5 and is reproduced here for ease of reference. The left part of this table starts with the domestic output of the various industries by products at basic prices. The inclusion of the imports valued at CIF prices by products generates the total supply by products at basic prices, as shown in column (9).

7.19. In the supply table, the output at basic prices of trade services (of which trade margins form the major part) is included in row (4) and that of transport services in row (5). To arrive at purchasers' prices for each product, the trade margin and transport margin shares of this output have to be reallocated from trade margins and transport services to the traded and transported products. Columns (10) and (11) of Table 7.1 contain the allocation of trade margins and transport margins respectively, with positive entries (+) in the rows of the traded and transported products and negative entries (-) in the rows of trade services and transport services. The totals by row of columns (10) and (11) of trade and transport margins respectively are always zero.

7.20. The columns of taxes less subsidies of products – columns (12)–(14) of Table 7.1 – are also added to total supply at basic prices in order to arrive at the total supply at purchasers' prices. Taxes on products comprise VAT-type taxes, taxes and duties on imports and other taxes on products. Similarly, subsidies on products comprise import subsidies and other subsidies on products. Taxes and subsidies on products should be compiled separately although they may be shown as a single column.

7.21. The addition of columns (10)–(14) to total supply at basic prices of column (9) gives total supply at purchasers' prices in column (16). Columns (10) and (14) thus provide the bridge needed to compare and balance total supply with total use when both sides are valued at purchasers' prices.

7.22. Both trade margins and transport margins can be produced by any industry outside the trade and transport industries. The bulk of the output of trade margins, however, is generally produced by the trade industries and the bulk of transport margins by the transport industries, as illustrated in rows (4) and (5) of Table 7.1.

Table 7.1 Supply table at basic prices, including a transformation into purchasers' prices

Millions of euros												
			INDUSTRIES					Output at basic prices	Imports	Total supply at basic prices		
			Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services				Other services	
			(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	
PRODUCTS	Agriculture	(1)	8 782	0	0	0	0	0	8 782	3 271	12 052	
	Manufacturing	(2)	796	182 982	643	1 808	133	44	186 405	124 590	310 995	
	Construction	(3)	83	961	43 060	734	255	179	45 272	563	45 835	
	Trade	(4)	1	4 773	311	54 204	640	257	60 187	600	60 787	
	Transport	(5)	13	465	66	25 538	128	125	26 335	8 150	34 485	
	Communication	(6)	160	1 781	139	43 912	1 253	982	48 228	6 234	54 463	
	Finance and business services	(7)	29	8 902	698	7 588	106 909	3 381	127 508	7 061	134 569	
	Other services	(8)	3	85	13	1 053	143	74 346	75 643	824	76 467	
Total			(9)	9 867	199 950	44 931	134 837	109 461	79 314	578 360	151 293	729 653
Adjustments	CIF/FOB adjustments on imports	(10)								- 97	- 97	
	Direct purchases abroad by residents	(11)								6 675	6 675	
Total			(12)	9 867	199 950	44 931	134 837	109 461	79 314	578 360	157 871	736 230
Total of which:												
	Market output	(13)	9 763	195 916	41 462	127 401	88 330	18 116	480 989			
	Output for own final use	(14)	104	4 029	3 468	2 134	19 890	2 670	32 295			
	Non-market output	(15)	0	4	0	5 302	1 241	58 528	65 075			

		Total supply at basic prices	VALUATION MATRICES						Total supply at purchasers' prices		
			Trade margins	Transport margins	VAT	Taxes on products	Subsidies on products	Total			
		(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)		
PRODUCTS	Agriculture	(1)	12 052	1 926	274	329	57	- 107	2 479	14 532	
	Manufacturing	(2)	310 995	48 838	2 540	13 175	7 866	- 49	72 370	383 364	
	Construction	(3)	45 835	0	0	1 529	13	0	1 542	47 377	
	Trade	(4)	60 787	- 52 341	0	575	11	0	- 51 755	9 032	
	Transport	(5)	34 485	0	- 2 800	558	71	- 448	- 2 620	31 865	
	Communication	(6)	54 463	1 493	9	3 375	217	- 34	5 059	59 522	
	Finance and business services	(7)	134 569	0	- 22	2 706	2 159	0	4 842	139 411	
	Other services	(8)	76 467	85	0	1 201	576	0	1 861	78 329	
Total		(9)	729 653	0	0	23 447	10 969	- 638	33 778	763 431	
Adjustments	CIF/FOB adjustments on imports	(10)	- 97							- 97	- 97
	Direct purchases abroad by residents	(11)	6 675							6 675	6 675
Total		(12)	736 230	0	0	23 447	10 969	- 638	40 356	770 009	
Total of which:											
	Market output	(13)									
	Output for own final use	(14)									
	Non-market output	(15)									

Table based on 2011 figures from Austria

(b) Use-side valuation matrices

7.23. The use-side valuation matrices consist of a sequence of matrices – mirroring the shape of the intermediate use and final use parts of the use table – for each component of the valuation, namely, for trade margins, transport margins, taxes on products and subsidies on products.

7.24. Table 7.2 illustrates the use table at purchasers' prices. This table corresponds to table 6.1 of chapter 6 and is reproduced here for ease of reference. It shows the structure of the table, comprising the following three sub-matrices:

- Intermediate consumption matrix showing intermediate consumption for each industry by product
- Final uses matrix showing final uses by type of final use and by product
- GVA matrix showing the components of GVA for each industry

Table 7.2 Use table at purchasers' prices

Millions of euros

		INDUSTRIES							FINAL USE							Total use at purchasers' prices		
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	Total	Final consumption expenditure		Gross fixed capital formation	Changes in valuables	Changes in inventories	Exports	Total			
									Households	NPISH							General government	
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	
PRODUCTS	Agriculture	(1)	2 583	6 570	16	371	34	49	9 623	3 595			180		- 27	1 161	4 909	14 532
	Manufacturing	(2)	2 205	107 190	12 441	16 874	6 015	8 797	153 522	71 438	3 180	26 756	2 183	3 034	123 252	229 842	383 364	
	Construction	(3)	105	2 440	9 528	2 446	3 907	1 604	20 029	1 667		25 155		- 38	563	27 348	47 377	
	Trade	(4)	33	1 883	119	2 240	259	308	4 842	3 325		67	45		753	4 189	9 032	
	Transport	(5)	14	4 386	267	8 399	822	321	14 208	5 833	3 370				8 453	17 656	31 865	
	Communication	(6)	34	2 563	299	9 359	5 919	1 833	20 008	26 444	121	5 976		67	6 905	39 514	59 522	
	Finance and business services	(7)	457	13 578	4 736	20 359	29 166	9 134	77 430	38 838	1 006	11 170		- 178	11 145	61 981	139 411	
	Other services	(8)	8	382	59	1 171	415	1 794	3 829	14 923	5 416	53 373	113	107	1	567	74 500	78 329
	Total at purchasers' prices before adjustments	(9)	5 440	138 991	27 466	61 219	46 538	23 839	303 492	166 063	5 416	61 050	69 418	2 335	2 859	152 800	459 939	763 431
Adjustments	CI/FOB adjustments on exports	(10)													- 97	- 97	- 97	
	Direct purchases abroad by residents	(11)							6 675							6 675	6 675	
	Purchases in the domestic territory by non-residents	(12)							- 12 945						12 945			
	Total at purchasers' prices	(13)	5 440	138 991	27 466	61 219	46 538	23 839	303 492	159 792	5 416	61 050	69 418	2 335	2 859	165 648	466 517	770 009
GVA	Compensation of employees	(14)	551	30 679	10 239	37 906	22 997	41 971	144 343									
	Other taxes less subsidies on production	(15)	- 1 627	1 077	546	1 755	2 004	1 103	4 858									
	Consumption of fixed capital	(16)	1 845	12 750	1 542	10 917	18 934	7 480	53 469									
	Net operating surplus/Net mixed income	(17)	3 658	16 453	5 138	23 040	18 989	4 921	72 198									
	Gross operating surplus/gross mixed income	(18)	5 503	29 203	6 680	33 957	37 923	12 401	125 667									
	GVA	(19)	4 427	60 959	17 465	73 618	62 923	55 475	274 868									
	Total input at basic prices	(20)	9 867	199 950	44 931	134 837	109 461	79 314	578 360									

Table based on 2011 figures from Austria

7.25. The matrices covering intermediate consumption and final uses are valued at purchasers' prices, thus they include trade margins and transport margins, along with taxes on products less subsidies on products. Accordingly, the sum of total intermediate consumption at purchasers' prices (column (7) of Table 7.2) and total final use (column (15) of Table 7.2) gives the total use by product at purchasers' prices, which, in a balanced system, is equal to the total supply by products at purchasers' prices in column (16) of the supply table in Table 7.1.

7.26. Table 7.3 shows the use-side valuation matrices covering trade margins (split between wholesale and retail trade margins), transport margin, VAT, taxes on products and subsidies on products. These matrices have the same structure and dimension as the intermediate consumption and final uses sub-matrices of the use table at purchasers' prices. They show the allocation of the trade margins, transport margins, taxes and subsidies on products to each element of the use table at purchasers' prices. They represent the amounts that must be deducted from each element of the use table at the purchasers' price in order to arrive at the use table at basic prices.

Table 7.3 Use-side valuation matrices

		INDUSTRIES							FINAL USE								Total use at purchasers' prices	
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	Total	Final consumption expenditure			Gross fixed capital formation	Changes in valuables	Changes in inventories	Exports	Total		
									Households	NPISH	General government							
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	
Wholesale trade margins																		
PRODUCTS	Agriculture	(1)	31	440	2	63	4	6	547	326		10		4	165	506	1 052	
	Manufacturing	(2)	164	6 450	1 415	1 879	326	1 049	11 284	5 941		2 718	15	265	8 995	18 493	29 777	
	Construction	(3)									560							
	Trade	(4)	- 196	- 6 910	- 1 420	- 1 997	- 367	- 1 082	- 11 972	- 6 464	- 569	- 2 776	- 15	- 273	- 9 232	- 19 329	- 31 301	
	Transport	(5)																
	Communication	(6)	0	20	3	55	37	27	141	197	9	48		4	72	330	472	
	Finance and business services	(7)																
	Other services	(8)																
	Total	(9)	0	0	0	0		0	0	0	0	0	0	0	0	0	0	
Retail trade margins																		
PRODUCTS	Agriculture	(1)		3	0	13	0	1	17	856						856	873	
	Manufacturing	(2)	19	56	20	339	38	90	562	16 781	477	1 142	99			18 499	19 061	
	Construction	(3)																
	Trade	(4)	- 19	- 86	- 25	- 431	- 104	- 150	- 815	- 18 363	- 511	- 1 168	- 184			- 20 226	- 21 040	
	Transport	(5)																
	Communication	(6)	0	27	5	79	66	59	236	725	34	26				785	1 021	
	Finance and business services	(7)																
	Other services	(8)											85			85	85	
	Total	(9)	0	0	0	0		0	0	0	0	0	0	0	0	0	0	
Transport margins																		
PRODUCTS	Agriculture	(1)	7	201	1	7	0	1	217	44		2		1	11	57	274	
	Manufacturing	(2)	20	1 125	191	127	30	64	1 557	303	21	144	2	26	486	983	2 540	
	Construction	(3)																
	Trade	(4)																
	Transport	(5)	- 27	- 1 321	- 191	- 135	- 31	- 65	- 1 771	- 347	- 21	- 146	- 2	- 27	- 487	- 1 030	- 2 800	
	Communication	(6)		0	0	1	1	0	4	3	0	2		0	1	5	9	
	Finance and business services	(7)	0	- 5	- 1	- 1	0	0	- 7	- 3	0	- 2	0	0	- 10	- 15	- 22	
	Other services	(8)																
	Total	(9)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Value added tax (VAT)																		
PRODUCTS	Agriculture	(1)	0	0	0	0	0	3	4	324		1				325	329	
	Manufacturing	(2)	18	71	18	68	200	942	1 317	10 624	368	734	132			11 858	13 175	
	Construction	(3)	0	1	7	4	242	185	439	265		825				1 090	1 529	
	Trade	(4)	3	16	4	20	17	39	100	467		2	7			476	575	
	Transport	(5)	0	1	0	9	8	17	36	487	34					521	558	
	Communication	(6)	0	1	0	16	207	150	374	2 888	10	103				3 001	3 375	
	Finance and business services	(7)	0	3	3	28	561	713	1 308	1 235		163				1 398	2 706	
	Other services	(8)	0	0	0	1	17	44	63	921	209		8			1 138	1 201	
	Total	(9)	22	93	32	147	1 252	2 093	3 639	17 210	621	1 830	147			19 807	23 447	
Taxes on products (excl. VAT)																		
PRODUCTS	Agriculture	(1)	0	50	0	1	0	0	51	5		1		0	0	6	57	
	Manufacturing	(2)	62	834	179	1 068	212	544	2 898	4 272	6	284	5	7	393	4 968	7 866	
	Construction	(3)	0	1	3	1	1	0	5	0		7				8	13	
	Trade	(4)	0	5	0	4	0	0	10	1		0	0			1	11	
	Transport	(5)	0	7	1	8	11	3	30	26	0				14	40	71	
	Communication	(6)	0	6	1	14	47	5	72	130	0	0		0	15	145	217	
	Finance and business services	(7)	9	88	18	142	172	37	467	936		755				1 691	2 159	
	Other services	(8)	0	0	0	0	0	0	1	574	1		0			575	576	
	Total	(9)	71	991	202	1 238	443	590	3 535	5 944	8	1 048	6	7	422	7 434	10 969	
Subsidies																		
PRODUCTS	Agriculture	(1)	0	- 89	0	0	0	0	- 89	- 2		- 5		0	- 11	- 18	- 107	
	Manufacturing	(2)	0	- 16	- 2	- 3	- 1	- 1	- 24	- 9	0	- 2	0	0	- 14	- 25	- 49	
	Construction	(3)																
	Trade	(4)																
	Transport	(5)	0	- 26	- 2	- 33	- 6	- 10	- 77	- 300	- 71				- 1	- 371	- 448	
	Communication	(6)				0			0	- 34						- 34	- 34	
	Finance and business services	(7)																
	Other services	(8)																
	Total	(9)	- 1	- 131	- 5	- 36	- 6	- 11	- 190	- 344	- 71	- 7	0	0	- 25	- 448	- 638	
Trade and transport margins																		
PRODUCTS	Agriculture	(1)	38	644	3	83	4	8	781	1 226		12		5	176	1 419	2 200	
	Manufacturing	(2)	203	7 631	1 626	2 345	393	1 204	13 403	23 025	1 058	4 004	117	291	9 481	37 975	51 378	
	Construction	(3)																
	Trade	(4)	- 215	- 6 996	- 1 445	- 2 428	- 471	- 1 233	- 12 786	- 24 827	- 1 080	- 3 944	- 199	- 273	- 9 232	- 39 555	- 52 341	
	Transport	(5)	- 27	- 1 321	- 191	- 135	- 31	- 65	- 1 771	- 347	- 21	- 146	- 2	- 27	- 487	- 1 030	- 2 800	
	Communication	(6)	1	47	8	136	104	86	381	925	44	75		4	72	1 121	1 501	
	Finance and business services	(7)	0	- 5	- 1	- 1	0	0	- 7	- 3	0	- 2	0	0	- 10	- 15	- 22	
	Other services	(8)											85			85	85	
	Total	(9)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Taxes less subsidies on products																		
PRODUCTS	Agriculture	(1)	0	- 38	0	1	0	3	- 34	327		- 3		0	- 10	313	279	
	Manufacturing	(2)	80	888	194	1 133	411	1 485	4 191	14 888	374	1 016	137	7	379	16 800	20 992	
	Construction	(3)	0	1	10	5	243	185	444	265		833				1 098	1 542	
	Trade	(4)	3	21	4	24	17	40	109	468		2	7			477	586	
	Transport	(5)	0	- 18	- 1	- 15	13	10	- 11	214	- 36				14	191	180	
	Communication	(6)	0	6	1	30	253	155	446	2 984	10	104		0	15	3 112	3 568	
	Finance and business services	(7)	9	91	21	171	733	750	1 775	2 171		918				3 090	4 865	
	Other services	(8)	0	0	0	2	17	44	63	1 495	210		8			1 713	1 777	
	Total	(9)	92	952	229	1 349	1 689	2 672	6 984	22 810	557	2 870	152	7	397	26 794	33 778	

Table based on 2011 figures from Austria

7.27. It should be noted that the use-side valuation matrices in Table 7.3 relate to the supply-side valuation matrices in Table 7.1 as follows:

- The sum of the totals in column (16) in Table 7.3 for “Wholesale trade margins” and “Retail trade margins” must be equal to “Trade margins” in the supply-side valuation matrix, namely column (10) of Table 7.1.
- The total in column (16) of Table 7.3 for “Transport margins” must be equal to the “Transport margins” in the supply-side valuation matrix, namely column (11) of Table 7.1.
- The total in column (16) of Table 7.3 for “Value added tax (VAT)” must be equal to the “Value added tax” in the supply-side valuation matrix, namely column (12) of Table 7.1.
- The total in column (16) of Table 7.3 for “Taxes on products (excl. VAT)” must be equal to the “Taxes on products” in the supply-side valuation matrix, namely column (13) of Table 7.1.
- The total in column (16) of Table 7.3 for “Subsidies” must be equal to the “Subsidies on products” in the supply-side valuation matrix, namely column (14) of Table 7.1.

7.28. In general, the information needed to construct the trade and transport margins matrices of Table 7.3 is available only to a limited extent, and some balancing between the supply and use of trade margins and transport margins is necessary. In addition, depending on available data and whether a benchmark or a current SUT is being compiled, it must be assessed whether it is best to start from the supply or the use side when estimating total trade margins and transport margins by products. In the cases when trade and transport margins are estimated first from the supply side, they will serve as a constraint when allocating the supply of trade margins and transport margins to the various use categories.

7.29. There is one type of tax on product, namely VAT, for which it is not possible to start with the supply-side estimates. In the VAT system as set out in the SNA, only non-deductible VAT is recorded as a tax on product, and there is no means by which the actual VAT payers (VAT collectors) can obtain information about the final users and their ability to deduct VAT or not. The structure of VAT by products has therefore to be estimated from the use-side by identifying all user categories not exempted from the VAT system and the appropriate effective tax rate has to be applied to all their purchases of products. One increasingly problematic area is the treatment of digital intermediation platforms in terms both of trade and transport margins and also of the taxes (in particular, VAT) paid to foreign governments.

7.30. Once all the matrices in Table 7.1 are compiled, the next step is to deduct the trade margins, transport margins and taxes less subsidies on products from the use table at purchasers’ prices to arrive at the use table at basic prices as shown in Table 7.4. In addition, it is necessary to reallocate the deducted trade margins and transport margins to the specific trade and transport service products distinguished in the product classification applied, and the taxes on products to a separate row. After these steps, the use table is transformed into a valuation at basic prices as shown in Table 7.4.

7.31. It should be noted that all the entries from rows (1)–(9) in Table 7.4 are at basic prices. Row (10) of Table 7.4 contains the net taxes on products which bridge the total use at basic prices with the total use at purchasers' prices, which is shown in row (11) of Table 7.4. This latter coincides with row (9) of the use table at purchasers' prices in Table 7.2.

Table 7.4 Use table at basic prices

Millions of euros

		INDUSTRIES							FINAL USE							Total use at basic prices
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	Total	Final consumption expenditure			Gross fixed capital formation	Changes in valuables	Changes in inventories	Exports	
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	Households	NPISH	General government	(11)	(12)	(13)	(14)	(15)
PRODUCTS	Agriculture (1)	2 545	5 964	13	287	29	38	8 877	2 042			170		- 32	996	3 176
	Manufacturing (2)	1 922	98 670	10 621	13 397	5 211	6 108	135 928	33 525		1 749	21 736	1 929	2 737	113 392	175 067
	Construction (3)	105	2 439	9 518	2 441	3 664	1 419	19 585	1 402			24 323		- 38	563	26 250
	Trade (4)	245	8 857	1 560	4 644	712	1 501	17 519	27 684			4 008	238		9 985	43 267
	Transport (5)	41	5 724	459	8 549	840	376	15 990	5 967		3 427				8 926	18 495
	Communication (6)	33	2 510	290	9 194	5 562	1 592	19 181	22 535		68	5 797		63	6 818	35 281
	Finance and business services (7)	448	13 492	4 716	20 189	28 433	8 384	75 662	36 669		1 006	10 254		- 177	11 156	58 907
	Other services (8)	8	381	59	1 169	398	1 750	3 765	13 429	5 416	53 163	113	14	1	567	72 702
	Total at basic prices (9)	5 348	138 038	27 236	59 870	44 849	21 167	296 507	143 252	5 416	60 492	66 548	2 182	2 852	152 403	433 145
	Taxes less subsidies on products (10)	92	952	229	1 349	1 689	2 672	6 984	22 810		557	2 870	152	7	397	26 794
Adjustments	Total at purchasers' prices before adjustments (11)	5 440	138 991	27 466	61 219	46 538	23 839	303 492	166 063	5 416	61 050	69 418	2 335	2 859	152 800	459 939
	Off/FOB adjustments on exports (12)														- 97	- 97
	Direct purchases abroad by residents (13)								6 675							6 675
	Purchases in the domestic territory by non-residents (14)								- 12 945						12 945	
GVA	Total at purchasers' prices (15)	5 440	138 991	27 466	61 219	46 538	23 839	303 492	159 792	5 416	61 050	69 418	2 335	2 859	165 648	466 517
	Compensation of employees (16)	551	30 679	10 239	37 906	22 997	41 971	144 343								
	Other taxes less subsidies on production (17)	- 1 627	1 077	546	1 755	2 004	1 103	4 858								
	Consumption of fixed capital (18)	1 845	12 750	1 542	10 917	18 934	7 480	53 469								
	Net operating surplus/Net mixed income (19)	3 658	16 453	5 138	23 040	18 989	4 921	72 198								
	Gross operating surplus/Gross mixed income (20)	5 503	29 203	6 680	33 957	37 923	12 401	125 667								
	GVA (21)	4 427	60 959	17 465	73 618	62 923	55 475	274 868								
	Total input at basic prices (22)	9 867	199 950	44 931	134 837	109 461	79 314	578 360								

Table based on 2011 figures from Austria

C. Trade margins

7.32. This section deals with the compilation of the valuation matrix for trade margins. The amounts involved can vary significantly by type of product and can be of great magnitude in total. In most countries, the trade margins may vary between 10 and 25 per cent of the total domestic supply of goods and services.

7.33. The data sources needed for compiling the trade margin estimates by product for the SUTs are special in the sense that they are not necessary when compiling the current national accounts GVA by industry and, for that reason, may not be readily available on an annual basis.

7.34. Whereas information on trade turnover and purchases of goods for resale by industry is usually included in the general business statistics or estimated in order to compile the annual GVA by industry, the situation is different for information on trade turnover and margins by product. These data are generally not as readily available for a number of reasons: most often they are collected at intervals of several years, if at all; or information on trade margin ratios may be available only for a limited number of products and provided by government agencies dealing with

price control or monopoly surveillance; or they are limited to what can be derived from current price statistics by, for example, comparing wholesale prices collected for the wholesale price index (WPI) with the prices collected for the consumer price index (CPI).

7.35. It is recommended that a special survey be conducted, in particular for the purpose of compiling benchmark SUTs, that covers all trading activity (both as primary and secondary output) by industry and by product at a level sufficiently detailed to match the level and classification of products applied in the SUTs. The annex to chapter 7 provides an example of a questionnaire used for a survey of this kind, along with a template for the optimal use of the collected data in compiling the SUTs trade margin matrices.

7.36. Even when less than ideal source data are available, it is still necessary to estimate the trade margin matrices. In these cases, however, the results will become more dependent on the assumptions made to populate the trade margin columns in the supply table and the trade margin matrices in the use table. In principle, the same type of trade margin data table (shown in Table 7.5 and in the annex to this chapter) must be generated, irrespective of the coverage or quality of primary source data on trade and trade margins.

1. Definition of trade margins

7.37. Even though wholesalers and retailers actually buy and sell goods, the goods purchased are not treated as part of their intermediate consumption as they are resold with only minimal processing such as grading, cleaning and packaging. Wholesalers and retailers are treated as supplying services. Their output is measured by the total value of the trade margins realized on the goods they purchase for resale and some non-margin trade services. This notwithstanding, actual trade turnover is an important supporting variable when compiling the trade margin matrices in the SUTs.

7.38. The 2008 SNA provides the following definition of a trade margin:

“A trade margin is defined as the difference between the actual or imputed price realized on a good purchased for resale and the price that would have to be paid by the distributor to replace the good at the time it is sold or otherwise disposed of” (2008 SNA, para. 6.146).

7.39. With regard to valuation, the SNA goes on to state as follows:

“Goods purchased for resale should be valued excluding any transport charges invoiced separately by the suppliers or paid to third parties by wholesalers or retailers: these transport services form part of the intermediate consumption of the wholesalers or retailers” (2008 SNA, para. 6.148).

7.40. This valuation principle means that there can be no transport margins linked to purchases of goods for resale. This follows from the fact that, in the national accounts definition of output from trade activity, the traded goods are not seen as actually flowing in and out of the trade activity.

The trade activity only adds services to the goods that are seen as flowing directly from the producer or importer to the user. As a result, there is no flow to which a transport margin could be attached.

7.41. In practice, the output of wholesalers and retailers is derived as the difference between the trading sales and the costs of goods purchased for resale, adjusted by changes in inventories (see 2008 SNA, para. 6.147):

$$\begin{aligned}
 \text{Value of output} &= \text{value of sales} \\
 &+ \text{value of goods purchased for resale but used for intermediate} \\
 &\quad \text{consumption, compensation of employees, etc.} \\
 &- \text{value of goods purchased for resale} \\
 &+ \text{value of additions to inventories of goods for resale} \\
 &- \text{value of goods withdrawn from inventories of goods for resale} \\
 &- \text{value of recurrent losses due to normal rates of wastage, theft or} \\
 &\quad \text{accidental damage}
 \end{aligned}$$

7.42. In order to derive trade margins, either for trading activities of single goods, trading activities of a statistical unit industries or total economy, data on trading sales (trade turnover), data on goods purchased for resale without further processing, and data on inventories of goods for resale at the beginning and at the end of the period must be available. As a rule, business surveys or specialized trade surveys can collect and provide data at the level of trade industries. Trading is also an important secondary activity in many non-trade industries, and trading activities in the system are measured by trade margins, regardless of whether the activity is conducted by traders as their main activity or by other industries as part of their secondary outputs.

7.43. Even though distributive trade is defined as purchases of goods for resale without any transformation, certain operations usually associated with distribution are included in the definition, such as sorting, mixing, breaking bulk and repackaging for distribution into smaller lots. Other services may also be included, if not separately invoiced, such as installation in situ.

7.44. Trade services should be separated at least into two main categories: wholesale and retail. Wholesale is the resale (sale without transformation) of new and used goods to retailers, industrial, commercial, institutional or professional users or to other wholesalers and export. Retailing is the resale (sale without transformation) of new and used goods, mainly to the general public for personal or household consumption or non-resident visitors or for a minor part to business (intermediate consumption and fixed capital formation). This separation is essential for the correct and transparent estimation of the trade channels and the cumulative trade margins for products passing through both the wholesale and the retail trade links.

7.45. The services provided by the trade industry include both margin and non-margin services. Margins services are those related to the trade activity of resale. Non-margin services are other services provided by such trade establishments as repair and installation services. Some trade

margins may, however, best be estimated and treated as non-margin services, in other words, should not be calculated as a percentage of the basic value of the underlying goods. This is the case when the underlying goods do not appear in the system, such as used household goods, in particular cars; or when the value of the underlying goods makes up a very small and fluctuating share of the total selling price like scrap materials; and when the underlying goods by convention are not specified by type (as is the case for merchanting, where the goods involved are not covered by the merchandise trade statistics but only appear as a net item in the balance of payments – with acquisitions shown as negative exports, as in SNA 14.73 and 26.21).

7.46. It should be noted that merchanting, and thus trading output, also includes the case of so-called factoryless goods production, where a principal has completely outsourced the transformation process but does not own the input materials (see para. 5.81 of the Guide to Measuring Global Production (UNECE, 2015), although, in paragraphs 2.69-113, an alternative view is explored that treats factoryless goods production units as manufacturers). The treatment as non-margin services means that this part of trading output will not be shown in the trade margin matrices. Where the above-mentioned cases are concerned, this will facilitate the use of the trade margin matrices in volume estimates and analytical uses of the resulting IOTs. With the CIF valuation of imports, there are no imports of trade margins.

7.47. Trade activity is classified in the industrial classification ISIC Rev. 4 under division 45, “Wholesale and retail trade and repair of motor vehicles and motorcycles”, division 46, “Wholesale trade, except of motor vehicles and motorcycles”, and division 47, “Retail trade, except of motor vehicles and motorcycles”. The following should be noted:

- Division 45 is a mixture of both wholesale and retail trade activities, and also of non-trade activities. In the context of SUTs and IOTs, division 45 must be subdivided into repair activities (group 452), and trade activities (groups 451 and 453).
- Group 454 “Sale, maintenance and repair of motorcycles and related parts and accessories” has to be dealt with appropriately, depending on the significance of this activity in the country. This is a heterogeneous group and if the GVA contribution is significant, then the sale, maintenance and repair activities should be separated, as the respective margin ratios are very different.
- Division 46 also covers wholesale on a fee or contract basis which is not part of the trade margin activity. It also covers some merchanting activity.
- Division 47 includes the resale (sale without transformation) of new and used goods, mainly to the general public for personal or household consumption or use, by shops, department stores, stalls, mail-order houses, hawkers and peddlers, consumer cooperatives and other outlets.

7.48. Trade services are classified in CPC Version 2.1 in division 61, “Wholesale trade services”, and division 62, “Retail trade services”. It should be noted that, even though many industries have

output of trade services as either principal or secondary products, the product classification used in the system may distinguish only a few groups such as wholesale and retail trade services.

2. Compilation of trade margin matrices

7.49. The compilation of the trade margin matrices in the SUTs may in principle be started either from the supply side or from the use side. In general, especially when compiling a benchmark SUT, the preferred approach is to start with estimating trade margins in the supply side resulting in data on the total amount of trade margins by products. These estimates will then represent the restrictions with which the use-side trade margin estimates have to comply.

7.50. Business surveys or special trade surveys usually provide data on the total output of trade margins by industries (including subdivisions of the trade industries), which then need to be transformed into margins by products if special surveys do not provide this information.

7.51. Starting the compilation of trade margins from the use-side means that estimates are made on the share of trade margins included in each element of the use table at purchasers' prices. The effective trade margin associated with each cell depends both on the typical trade channels for this particular use and on the typical product margin ratios for those parts passing through the wholesale and retail links. Normally, such information cannot be gathered by asking purchasing enterprises and establishments and other users, as they will only be aware of the last step in the distribution channel (where they have purchased the product) but clearly not of the previous steps. Even for the last step, they do not know the margin implicitly invoiced to them. Thus, plausible assumptions both on distributive channels and margin ratios have to be made and, for this purpose, it is useful to have the supply-side trade margin estimates at hand.

7.52. In the case of benchmarked estimates, it is necessary to calculate the full range of supply-side and use-side trade margin matrices by exploiting all available data sources and thus also to establish a basis for subsequent calculation, where the trade margin ratios of the previous years can be taken as the point of departure, and depending on available data, it may in this case be best to start from the use-side, as this represent the most detailed set of trade margin ratios.

7.53. The compilation of trade margins can be organized in three steps. The first two steps can be seen as dealing with estimating the supply-side trade margins by product, while the third step deals with the estimation of the use-side trade margin matrix. Box 7.1 gives a general summary of the three compilation steps for trade margins.

Box 7.1 Compilation process for trade margins

Supply-side trade margins	Step 1	Estimation of turnover and output (trade margins) by principal and secondary trade margin producers.	<i>Absolute constraint</i> for totals by producing industries <i>Output estimates</i> entered into the domestic output part of the supply table
	Step 2	(a) Estimation of trade turnover matrices (dimension: product by margin producing industry). (b) Estimation of product specific trade margin ratios. (c) Calculation of total wholesale and retail margins by product. (d) Balance step 2 (c) against step 1.	<i>Relative constraints</i> for margins by product. Result entered into the trade margin columns of the supply table
Use-side trade margins	Step 3	The starting point is step 2 (d) and the use table, either balanced or unbalanced, and either at purchasers' prices or at basic prices. Main objective: <i>If use table at purchasers' prices:</i> To compile the full matrices for trade margins to derive basic price values in each cell, and the corresponding rows for the margin producing industries. <i>If use table preliminary estimated at basic prices:</i> To compile the full matrices of trade margins in order to adjust the preliminary estimates of the rows for margin producing industries (often the case in an already established SUTs system) Estimated trade margin matrices are eventually balanced against the constraint in step 1 (absolute constraint) and in step 2 (d) (relative constraint)	Common problems: – Question of distribution channels. – No direct information of trade margin ratios by <i>user</i> category. – Extensive use of assumptions necessary.

(a) Step 1

7.54. Step 1 of the compilation process for trade margins covers the compilation of trade services by industry that must already be part of the current annual national accounts calculations of GVA by industry. The national accounts estimates of output usually do not include a distinction between primary and secondary output. In the case of trade output, however, the situation is different, since trade output must be derived separately as the difference between sales and purchase of goods for resale. The data needed to fill in the domestic production part of the supply table should therefore be readily available, and these calculations are not unique to the SUTs.

7.55. When compiling the SUTs, and in particular benchmark SUTs, there may be a need to take a closer look at existing calculations and to supplement them with more detail, such as, for example, through the introduction of the distinction between wholesale and retail output, if it has not already been done.

7.56. Trading services are an important secondary output in many industries other than trade, such as manufacturing enterprises, barber shops, museums, hotels, recreational and sporting activities, and so forth. Accordingly, as a first step, it should be assessed if current data fully cover all secondary output of trade activity, as some data sources such as industrial statistics may exclude trading, whereas data sources for service industries may not specify trade turnover separately, and some existing estimates may need adjustments. Thus, for certain industries only total sales may be known, without any distinction between trade sales and other sales. Estimates based on plausible assumptions should be made to achieve data on total trade services of the economy.

7.57. Trade turnover should be separated into wholesale trade turnover and retail sale trade turnover, and wholesale trade margins and retail trade margins recorded as different products in the domestic output part of the supply table at basic prices. In business statistics, wholesale and retail trade turnover and margins are often available for the trade industries separately (ISIC Rev. 4, divisions 45, 46 and 47). For industries with trade activity as a secondary activity this breakdown may not be available and, even in cases where trade turnover has been reported separately for wholesale and for retail sales, the value of goods purchased for resale may not be thus subdivided. In this case, wholesale and retail trade margins cannot be derived directly but have to be estimated on the basis of plausible assumptions.

7.58. If no direct separate information on the type of trade (wholesale or retail) is available from surveys, the subdivision can be based on the kind of primary economic activity. Thus all ISIC Rev. 4, division 46, may be assumed to carry out wholesale trade, and all ISIC Rev. 4, division 47, retail trade, whereas ISIC Rev. 4, division 45, must be further broken down, as indicated above. The subdivision of trade for secondary trade producers into wholesale and retail trade has in this case to be based on assumptions. For example, it can be assumed that the trade turnover of restaurants and hotels, hairdressers, cinemas and theatres will probably be retail trade turnover, whereas trade activities of advertising agents will more likely be wholesale trade. Manufacturing industries will often trade in products similar to those that they produce or in complementary products and such sales will usually be of a wholesale type, although some may be sold directly to consumers. Manufacturing industries may also be trading in similar imported goods, and such trade is again, likely to be classified as wholesale trade. There may also be industries where a grouping by size might be relevant for the correct identification of the type of trade activity performed, for example, the trade activity of small bakeries would normally be retail sale, whereas the trade activities of the larger ones would probably be wholesale trade.

7.59. For the purpose of the following calculation steps, these estimates should be made also at the most detailed level of classification of the trade industry available in source statistics. Although these more detailed estimates would not be shown in the supply table at basic prices, they will be very useful when estimating trade margins by products in cases where this information is not available from existing surveys.

7.60. In the discussion above, it has been assumed that the secondary output of trade services remains in the industries of the secondary producers. As explained in chapter 5, however, the SNA recommends the partitioning of horizontally-integrated enterprises that have production in two or more sections of the ISIC Rev. 4 (these sections are broad activity groups such as agriculture and fishing, mining, manufacturing, construction, trade and others) and the creation of new establishments to be classified together with the primary producers of the secondary product if that has not already been done in basic statistics. Such reclassification of secondary output is called “redefinition” and is typically carried out for trade activities in many countries. A redefinition implies that there will be trade activity and output of trade margins only from divisions 45–47 of ISIC Rev. 4. This will significantly simplify the calculations and will also facilitate the estimates of input structures and the calculations of industry-by-industry IOTs. It will not, however, affect the basic methodology outlined in this section and in the annex to this chapter.

7.61. Step 1 results in an estimate of total output of trade services forming an absolute constraint. This value is then disaggregated by products in step 2, as described below.

(b) Step 2

7.62. In step 2, the product dimension is in focus, in particular in the allocation of total wholesale and retail trade margins to the products to which the margins apply.

7.63. The output of both wholesale and retail trade services must be first separated into the output of margin activities (trade margins) and the output of non-margin activities (non-margin trade services), such as trade services related to used goods, waste and scrap, and to merchanting. The output of non-margin trade services do not form part of the valuation matrices (see para. 7.45).

7.64. Table 7.5 illustrates how the trade data needed in the supply table are related to the survey data (or primary data obtained in other ways, where necessary by relying on plausible estimates). The yellow shaded cells are data that will feed into the rows for wholesale and retail trade services in the domestic output part of the supply table. They are obtained as explained under step 1 above. Step 2 deals with the problem of distributing the values in the yellow shaded cells to the various products, whereupon the grey shaded column for the total trade margins will form the trade margin columns of the supply table (column (10) of Table 7.1). It should be noted that the trade turnover data appear as supporting variables only; they are not moved forward to the supply table.

7.65. In the situation where data on margins by product are already available, either from current business surveys or from special surveys conducted in connection with the SUTs, the task is limited to aligning the survey results with the product classification used in the SUTs, and to grossing up the results to make them consistent with the output data for trade dealt with in step 1 above.

7.66. In the case where no trade data by product are available, it is necessary to subdivide the margin trade turnover by each industry into turnover and trade margins by the products traded. This should be done separately for wholesale trade turnover and retail trade turnover.

Table 7.5 Trade turnover and trade margins for wholesale and retail trade margins

Activity ISIC	Ec. activity 1		Ec. activity 2		...	Trade 45		Trade 46		Trade 47		...	Ec. activity n		SUM	
	Trade turn- over	Trade margin	Trade turn- over	Trade margin		Trade turn- over	Trade margin	Trade turn- over	Trade margin	Trade turn- over	Trade margin		Trade turn- over	Trade margin	Trade turn- over	Trade margin
Product CPC	Indicate W or R		Indicate W or R		...	Indicate W or R		Wholesale (W)		Retail [R]		...	Indicate W or R			
1																
2																
:																
m																
Total wholesale																
1																
2																
:																
m																
Total retail																

7.67. The subdivision by products of trade turnover of each trade service-producing industry will result in the two trade turnover matrices, each of which as the dimensions: “Products (traded) by producing industry (output)”: one matrix for wholesale trade turnover and one matrix for retail trade turnover. These matrices are presented together in Table 7.5. The availability of source data for this subdivision may vary considerably across countries. Even in cases where no direct survey information is available, scattered information may be available, including from administrative sources and related and older surveys, although ad hoc information on specific units may not be representative for the total branch. In this case, it is of particular importance to use data from the most detailed level of subdivisions of the trade industry, as these will indicate the types of products traded.

7.68. Even with the availability of data for subdivisions of the trade industries, the estimation of trade turnover by products is not straightforward. This is the case, for example, with non-specialized trade divisions such as supermarkets and department stores where a wide range of goods are traded (even though, in some cases, it may be possible to get access to computerized cash transactions with detailed information about the goods sold and purchased data from businesses). It is generally easier to make estimates for specialized retail divisions, as their turnover by product is known from everyday life and more uniform. Estimating turnover by product for the wholesale divisions is more difficult because of the range of product mix, although there are branches with a clear concentration on one or a few product groups such as, for example, wholesale trade of motor vehicles and energy products.

7.69. For trade as secondary activity, plausible assumptions must be made about the products traded. For example, for hairdresser trade in cosmetic articles, hotel trade in souvenirs, newspapers, journals, food and beverages, and museum trade in books, multimedia products, and so on, the share of each identified product group in turnover must also be determined. In manufacturing, it can be difficult to estimate trade turnover structures by goods traded without any specific information, as the level of specialization may be very high. Information could be obtained on ad hoc basis by asking selected units with significant trade turnovers. Making these estimates

on the basis of plausible assumptions at the highest level of detail may provide acceptable results for the SUTs aggregates, even if only rudimentary information is available.

7.70. Having compiled the two trade turnover matrices in Table 7.5, it is possible to check the wholesale and retail trade turnover against the supply of the goods (domestic production and imports) from the supply table. At this stage, the comparison cannot be done at completely comparable prices, as the supply will be at basic prices and the trade turnover will be inclusive of either wholesale or retail trade margins, or both, but such checks should ensure that the trade turnover estimates are plausible in relation to the supply of the goods. For instance, there should not be much retail trade turnover of intermediate and capital goods; wholesale trade turnover should normally not be much higher than domestic production plus imports (plus wholesale margins); retail trade turnover of consumer goods should not be much higher than household expenditure for these goods. There may, however, be exceptions to these general rules for certain products, such as those that are traded twice within the same chain of distribution. This arises, for example, when one wholesaler imports a product or purchases it from many small producers (such as agriculture) and subsequently resells it to another wholesaler.

7.71. From the two trade turnover matrices, the trade margin matrices of the same dimensions must be derived. This is formally done by multiplying the trade turnover matrix by the assumed product margin ratios as described below.

7.72. The margin ratios are defined here as the share of a trade margin relative to the trade turnover. Margin ratios can be defined at the level of industries, which would show the average margin obtained by margin producing industry (the necessary information is already available from the Step 1 calculations), or at the level of products, where information is usually less readily available, although some countries may conduct regular surveys on percentage trade margins by products, classified by CPC or COICOP. It must in general be assumed that margin ratios are more closely connected with the products traded than with the industry carrying out the trading activity as either primary or secondary production.

7.73. It is obvious that even a single benchmark survey of product-specific trade margins would contribute greatly to the overall quality of the SUTs.

7.74. It is important to be aware of the basis for the calculation of percentage trade margins. When reported by enterprises, the trade margins will often appear as a percentage of the total selling price, including excise taxes and VAT, whereas the compiler of the SUTs will usually need the trade margin as either a percentage of the basic price (when estimating margins in the supply table) or a percentage of the purchasers' price excluding taxes on products (when estimating margins in the use table). The source data on percentage trade margins must therefore be adjusted to the appropriate basis before being applied in the system.

7.75. When specific survey information is missing, alternative sources for product-specific margin ratios must be explored. One possible approach might be to compare the prices observed

for the CPI and for the wholesale price index (WPI) for identical products. The same could be achieved by comparing PPIs with WPIs, which could provide proxies for wholesale margin ratios. In the case of regulated prices, the price levels in the different distribution channels may be available, and more generally price information available from the monopoly and price control agencies could be used. The margin ratios of specialized trade branches may be used as proxies for the related product margin ratios. Thus the margin ratio of the retail trade branch selling shoes could be taken as the typical retail margin for shoes. In practice, the usefulness of this approach would depend on the level of detail in the product classification applied, and the availability of data by detailed sub-branches of wholesale and retail activity, which would facilitate linking branches and products.

7.76. Having established a set of product-specific margin ratios, the multiplication of the trade turnover matrix could then be performed on the assumption that these product-specific margin ratios are valid in all industries trading in that product (as primary and as secondary). Next the resulting wholesale and retail trade margins by producing industries must be compared with the total trade margins by industries determined in step 1. The reasons for any differences could be inaccuracies in the trade turnover matrices, in the subdivision between wholesale and retail margins, and in the assumed or derived product margin ratios. These differences must be eliminated either by proportional adjustments or, if appropriate, by more refined methods.

7.77. It should be noted that difficulties in determining trade margins are attributable not only to the often weak data sources but also to the continuing changes in the structure of the trade industries, such as the following:

- Changing forms of supply of trade services
- Concentration in retail trade branches
- Growing size of shops
- Increasing importance of internet trade

7.78. These developments also affect the validity of benchmark estimates that may relatively quickly become outdated.

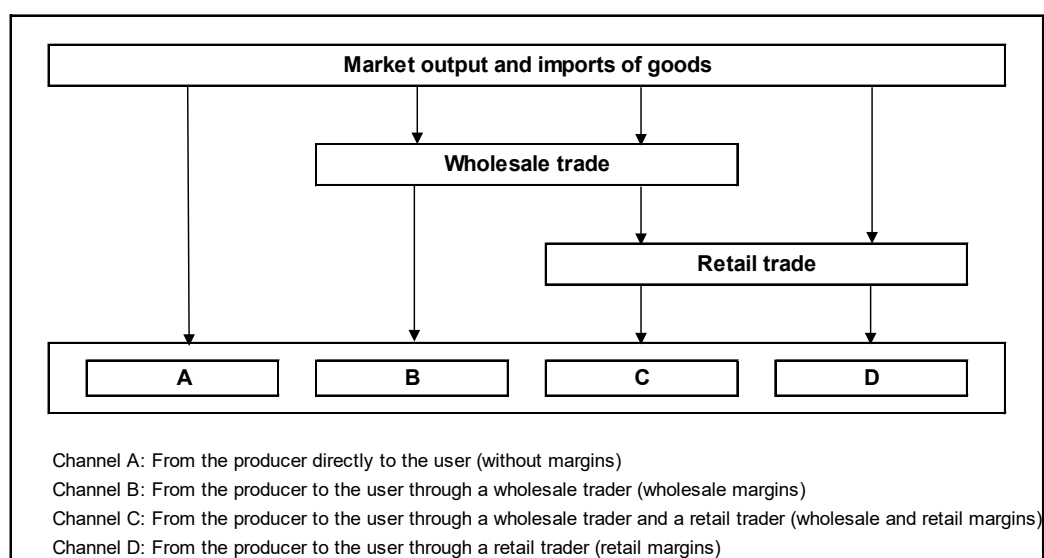
(c) Step 3

7.79. Step 3 relates to the calculation of the use-side trade margins matrices. In the previous steps, the question of trade channels has not been dealt with, as the data sources have been either survey data for the trading activities or estimates based on administrative or other indirect sources. When compiling the use table trade margin matrices, however, trade channels become important, as individual users may purchase their goods from different levels of the distribution system, or even directly from producers.

7.80. Direct data sources are more limited for the use table margins than for the supply table based margins. This is because buyers of the goods do not know what share the trade margins constitute in the price that they have paid. Sometimes they do not even know whether they have purchased the good from a trader or not. In cases where the goods have been purchased in a retail shop or from a wholesaler, the buyer will only know that the price paid includes some trade margin but not the full amount of the margin. This is because the distribution channels before the final seller are usually unknown to the buyer.

7.81. Figure 7.2 illustrates in a schematic way the possible distribution channels for goods from the producer of market output and imports of goods to the user. The distribution can go directly from the producer to the user (represented by box A); through wholesale trade (box B) in which case on wholesale margins are applicable; through wholesale and retail trade (box C) in which case both on wholesale and retail margins are applicable; or only through retail trade (box D), in which case only retail margins are applicable.

Figure 7.2 Alternative distribution channels of goods



7.82. The calculation of the use table trade margin matrices must therefore be based on plausible assumptions and eventually balanced with the estimated total supply of the trade margins by products. In principle, the following types of information are necessary:

- For each single cell of the use table, the share of total purchases that has been channelled through trade activities (for all involved steps in the distributive channel)
- The margin ratios to be applied for the products actually traded in the particular intermediate or final use part of the use table

7.83. Usually specific knowledge about the distributive channels for the goods in the individual cells is missing. The same holds true for the possible variation of the actual margin ratio across

users. For that reason, plausible assumptions have to be made. Concerning the use of wholesale and retail trade channels, the following assumptions may reasonably be made:

- For intermediate consumption, mostly wholesale trade margins and in very few cases retail trade margins are relevant. Retail trade margins for intermediate consumption can be relevant, for example, when stationery and handicrafts materials are being bought by smaller shops and small-scale enterprises.
- For household final consumption expenditure, retail trade margins are mostly relevant, but also with some exceptions when consumers have direct access to wholesale channels, thus generating wholesale trade margins, or are able to buy directly from the producer of the goods (for example, farmers, bakeries, tailors and so forth), thus not involving trading services at all.
- Wholesale services are also connected with household final consumption expenditure as some of the products bought in retail trade may have been delivered from wholesalers, and thus also include wholesale trade margins.
- For gross fixed capital formation, the wholesale channel is the most important and a very small measure of importance also attaches to the retail sale channel, for example, valuables and smaller equipment.
- In changes in inventories, it may reasonably be assumed that only wholesale margins and not retail trade margins can be involved (although the involvement of the latter is theoretically possible). Furthermore, wholesale trade margins can only be allocated to input stocks and trading stocks but not to output stocks, finished products and work-in-progress.
- For exports, it may reasonably be assumed that only wholesale margins may be involved (allowing for retail trade margins allocated to non-residents expenditure).

7.84. For much intermediate consumption, fixed capital information in equipment and machinery, and exports it is possible that no trade margins are involved at all, as bigger enterprises in particular will deal directly with one another. Imported goods are more likely to be bought through wholesalers than domestically produced goods, again depending on the size of the enterprises. Small enterprises will be more inclined to buy certain goods through retail traders.

7.85. The allocation of the trade margins to the individual cells of the use table has to be done in a step-wise manner. VAT and other net taxes on products must first be removed from the use table at purchasers' prices. The remaining value of the cells comprises only basic values and trade margins and is called the "residual" use table at purchasers' prices. It may be better first to estimate the retail trade margins for consumption of households and deduct those amounts in order to get all use data, including household final consumption expenditure, in a more uniform valuation, including only wholesale trade margins, making the allocation of these wholesale trade margins easier and of slightly higher quality. That notwithstanding, the retail margins and the wholesale margins are determined as follows:

- Based on the relationship between the total supply of each product and the trade turnover determined in step 2, it is possible to estimate the average share of the total supply of each product passing through wholesale and retail trade links.
- At this stage, it is necessary to make assumptions about the distributive pattern for each cell, on the proviso that the total trade turnover (separately for wholesale and retail trade) is known from step 2. In this way the absolute amount of each cell passing through either wholesale or retail trade is determined.
- If no specific information is available, it can be assumed that the average wholesale or retail trade margin ratio for the specific product (known from Table 7.5 in step 2) should be applied to the share of the total value passing through this trade link as determined in step 3.

7.86. When the two trade margin matrices have been compiled, they can be deducted from the “residual” use table at purchasers’ prices to obtain the use table at basic prices.

7.87. The resulting use table and trade margin matrices should be checked for overall plausibility, regarding both the relationship between allocated wholesale and retail trade margins and that between the use table data at purchasers’ prices and the allocated trade margins. In this process, the previously estimated trade margins by products of step 2 may also be reallocated.

7.88. The procedures outlined in this section for estimating the trade margin matrices is one that typically has to be applied when estimating benchmark SUTs. As mentioned above, the approach may be somewhat different when compiling annual tables on a current basis, as it may be better in this case to start from the use table side, taking as the starting point the effective trade margin ratios (in principle the proportion passing through this trade channel multiplied by the actual trade margin ratio) of the previous year as this detailed set of trade margin ratios is less prone to aggregation errors than the application of the average margin ratios by product that can be derived from the supply table.

7.89. The annex to this chapter provides a numerical example where the principles outlined in this section are illustrated in a complete template for the derivation of trade margin matrices, and which clarifies the importance of the distinction between wholesale and retail trade margins in getting the correct results.

D. Transport margins

7.90. Transport margins are another valuation component relating to the delivery chain of the products from the producer to the final user. Transport margins represent freight services of products when invoiced separately by the seller. Transport margins are transport charges paid separately by the purchaser to take delivery at the required time and place. They are included in the use of products at purchasers’ prices but not in the basic price of a manufacturer’s output or in the trade margins of wholesalers or retail traders.

7.91. Transport margins include in particular:

- Transport of goods arranged by the manufacturer, the wholesale or the retail trader in such a way that the purchaser has to pay separately for the transport costs even when the transport is done by the manufacturers, wholesale or retail traders themselves.
- Transport of goods from the place where they are manufactured or sold to the place where the purchaser takes delivery of them, in the event that the manufacturer or trader pays a third party for the transport, if this amount is invoiced separately to the purchaser.

7.92. This definition of transport margins means that the transport has to be arranged by the seller (producer or trader). This also means that transport arranged directly by the purchaser (and thus, of course, also directly paid for by the purchaser) is not included in the transport margins.

7.93. The existence of a transport margin is thus related to the way in which the transport costs are paid. This means that transport margins cannot be derived from the output of the respective transport services but that information on the payments between the two related parties of the seller and the buyer is required.

7.94. Since transport margins only occur when transport services are separately invoiced, this means that no partitioning of transactions is necessary because the transport service is already treated as a separate product and necessarily known to the purchaser. (see 2008 SNA, para. 14.130)

7.95. Thus, from a statistical point of view, transport margins should therefore be much easier to deal with and to estimate than the trade margins because they could be surveyed directly, based on information available in the bookkeeping department of the purchaser and because it is not necessary to break down the purchasers' price on the basis of assumptions. In reality, however, it may not be easy to obtain the data, for example, with large companies that keep their bookkeeping departments offshore.

7.96. Based on the definition of transport margins, which is logically connected to the definitions of basic prices, purchasers' prices and trade margins, Box 7.2 provides examples of transport costs which are not recorded as transport margins because they do not contribute to the valuation difference between basic prices and purchasers' prices.

7.97. The value concepts and the consequent transport margins are defined in the 2008 SNA in such a manner as to reflect the way in which the transport costs are treated in business accounts and thus in the source data. When the cost of transport that is arranged by the purchaser is included in the price of the intermediate consumption or final use in the source data, it should be treated as transport margins but not when it is recorded as a separate cost item. Hence the recording of transport costs in the source data will influence the actual delimitation of transport margins, in particular when source data which are not SNA-compatible are not being adapted (by the compiler of the national accounts) to bring them into line with the SNA definition. Such general adaptations

of existing source data will in general not be feasible, however, and it may even be questioned whether such adaptation should be attempted at all. Without such adaptation, the concepts of basic value and transport margins would deviate somewhat from the recommended concepts but the overall properties of the system would not be compromised.

7.98. Contrary to the treatment of trade margins, imports of transport margins can exist. This happens when a foreign carrier transports freight into, within, or out of the domestic territory. This would be the case of road, water (only inland waterways), and air (inland) transport. Pipelines within the domestic territory are normally run by a resident enterprise.

Box 7.2 Examples of transport costs which do not form transport margins

Transport margins are not the same as actual transport costs. Since these two concepts are often confused, examples of activities which are not recorded as transport margins because they do not contribute to the valuation difference between basic prices and purchasers' prices are listed below:

- If the manufacturers or traders transport the goods themselves and do not invoice the transport separately, these transport costs will be included in the basic price of the manufacturers' output or traders' output. This transport represents an ancillary activity and the individual costs of transport will not be identifiable as transport costs.
- If the manufacturer arranges for the goods to be transported by a third party without a separate invoice for the transport services, the transport costs will be included in the basic prices of the manufacturers' output. These transport costs will be identifiable and recorded as part of the manufacturers' intermediate consumption.
- If wholesale and retail traders arrange for goods to be moved from the point where they take delivery of them to that where another purchaser takes delivery, these costs will be included in the trade margin if no separate charge is made for transport to the purchaser, where these costs will be part of the intermediate consumption by the wholesale trader and retail trader.
- If a household buys goods for final consumption purposes and arranges for transport by a third party, these transport costs are recorded as household final consumption expenditure on transport services and not included in transport margins.
- If a domestic carrier transports goods from country A to country B through the domestic territory (transit transport), this will also not be considered as a transport margin as it does not relate to goods that forms part of domestic supply and use. These transport services will be recorded under exports of services.
- Transport services of domestic carriers outside the domestic territory (merchanting) are not part of the transport margins but form exports of services.
- Freight of used goods, scrap and waste, earth and similar freight connected with construction projects are also not part of transport margins as these goods are not considered as products. This also includes the transport of goods in connection with removals.

7.99. According to the specific modes of transport (such as road, railway, water, air, and pipeline), several kinds of transport margins need to be distinguished, provided they are classified as separate products in the system. In addition, the services of forwarding agencies also form part of the transport margins when paid separately by the buyer. Transport insurance services must also be considered under the same terms as the general definition of transport margins.

7.100. Compared with trade margins, transport margins are of a much lower magnitude in accordance with the definition in the 2008 SNA, which is more restricted than that used in the system based on the 1968 SNA. The transport margin is much more complex, however, not only because of the different kinds of transport margins but also because of the definition itself.

7.101. Furthermore, the data situation gives rise to numerous practical problems. The relationship between the supply of goods and the transport margins connected with them is looser than in the case of trade margins. Thus, transport costs are usually not related to the value of the goods transported; much transport is carried out as an ancillary activity and the manner in which transport costs are paid might differ from product to product and from transaction to transaction.

7.102. According to the 2008 SNA, total imports and exports of goods are to be valued FOB. For the purpose of compiling SUTs, however, total imports will be valued CIF and an appropriate adjustment item should serve for the transition between both valuation concepts. A CIF valuation means that transport costs up to the border of the importing country are included in the CIF-based value.

7.103. Transport services between the border of the importing country and the domestic location of the buyer are thus to be considered as transport margins (if paid for by the buyer and separately invoiced by the seller). By corollary, transport services between the domestic location of the seller and the border in the case of exports are also to be considered as transport margins (if paid for by the buyer and separately invoiced by the seller). Transport services delivered outside the domestic territory by resident producers will never become transport margins but are exports of trade services. Non-resident carriers can also provide transport services within the domestic territory for resident or non-resident buyers.

1. Compilation of transport margin matrices

7.104. Before embarking on the task of estimating transport margins, the compiler should carefully study the instructions provided in the business questionnaires used to collect data on sales and purchases and any other sources for these data. The instructions should be studied with a view to determining the extent to which the collected data meet the conditions for the existence of transport margins, and, if that is the case, how closely such transport margins correspond to the SNA definition.

7.105. In particular, the exact way in which the (non-margin) transport costs directly collected in the business surveys or available from other sources have been defined should be examined, in order to clarify if such cost items could possibly include those transport costs that, according to the SNA definition, are to be considered as transport margins. Only after this examination, can a decision be made on the existence or the exact delimitation of the transport margins to be estimated.

7.106. Trade margins make up the bulk of the total output of trade services (the exceptions being only the trade services on used goods, waste and scrap, and trade relating to goods in transit and merchanting), and virtually all trading activity in the economy is covered by the total output of trade services identified in the system. The situation for transport margin activity is quite different. A significant part of all transport activity in an economy takes place as ancillary activity in non-transport industries and is therefore not identified in the system. The intermediate consumption related to the ancillary activity is lumped together with the intermediate consumption related to the principal and secondary activities of the industry. Only the transport services carried out by the transport industries and, if statistically identified, a very minor output of transport services as secondary output in non-transport industries are shown explicitly in the system.

7.107. If transport margins are estimated, it is therefore not possible to assess their importance relative to the total transport activity in the economy, and the estimated transport margins should not be mistakenly perceived to reflect the actual physical freight activities carried out in the economy, comparable to the activities customarily covered by specialized transport statistics. Within the SUTs, the only way to assess total freight activity, ancillary and marketed, is by means of the distribution of those inputs typically used for transport, such as fuel, auto repair, and current taxes on motor vehicles (other taxes on production). On the other hand, these inputs will often – and ideally – have been estimated on the basis of a distribution of all motor vehicles by type and size across industry.

7.108. The size of the transport margins, and even of the total output of freight services, is usually much smaller, relatively speaking, than that of the trade margins. In some cases, the imbalance between the supply and use of a product might even be bigger than the transport margins of that specific product. It is therefore recommended that a careful review be made of those products where important transport services are involved, such as, for example, agricultural and forestry products, energy products, iron and steel products and products related to construction. This situation will vary across countries.

7.109. If no secondary transport activity is shown in the supply table at basic prices, this means that the transport of goods has been arranged by the manufacturer, the wholesale trader or the retail trader in such a way that the purchaser has to pay separately for the transport costs even when the transport is done by the seller .

7.110. As noted above, transport margins could in principle be surveyed directly, on the basis of the purchasers' bookkeeping information. In practice, however, this information is not collected, as coverage of this kind would involve not only the total transport margins paid by the enterprises being surveyed but also their distribution by product and by kind of transport. In addition, even if this information does exist, respondents may have to go back to the individual invoices and collate the data required in order to obtain it. Thus, even though it would be possible to establish such special surveys, for example related to the compilation of benchmark SUTs, it is generally not conducted as it would entail such a large burden for the businesses concerned. When such

information is available and sufficiently representative, then this is all that is needed to compile the transport margins of the system.

7.111. Box 7.3 provides four options to consider in the absence of any direct information on transport margins. The argument in support of option 4 is that the matrix of wholesale trade margins will in any event be based on inadequate information, and that it is not possible to ascertain whether the margin associated with a particular cell is the, so to speak, “pure” wholesale trade margin or if it also includes some transport margin. As trade margins will be much higher (by some 10–25 per cent) than any contribution from possible transport margins (around 0.5 per cent), the additional uncertainty introduced by choosing a joint wholesale and transport margin will therefore be rather limited. The input structure of wholesale trade will, however, be somewhat distorted.

Box 7.3 Options to consider where no data exists on transport margins

In the absence of any direct information on transport margins, there are basically four ways to proceed:

Option 1: In complete absence of any information on transport margins, decide that transport margins are insignificant, given the way in which output and intermediate consumption values are defined, and therefore need not be estimated at all.

Option 2: Concentrate on those products where important transport services are involved – normally agricultural and forestry products, energy products, iron and steel products and products related to construction – and collect ad hoc information about transport arrangement from selected enterprises.

Option 3: Decide to establish a full matrix of transport margins based on general assumptions about total transport margins and their distribution by products and uses.

Option 4: Reroute transport margins by product and by use through wholesale trade. This can be done by estimating for each type of transport output the share being transport margins. Then record this as input into wholesale trade. The output of wholesale trade should be increased by the same amount. This rerouting via wholesale trade recognizes the existence of transport margins but their actual distribution is hidden in an untraceable way in the wholesale trade margin matrix.

Where options 3 and 4 are concerned, the total transport margins by type of transport output could in principle be determined residually as the difference between total supply and the identified uses of each type of transport service. This approach would, however, require a very high degree of confidence in the preliminary estimates of transport costs entered into the use table. As previously noted, transport margins are expected to make up only a very minor share of total transport services. This residual would be highly unreliable, and probably reflect the statistical uncertainty of the estimated output and use data, rather than the actual level of any transport margins.

7.112. On the basis of the supply table alone, it is not possible to distinguish transport services paid for by the seller from those invoiced to the purchaser. Starting from the output of transport services (principal or secondary) in the industries, only total output can be calculated. From this total output, some non-margin services can be clearly deducted. These are the transport services related to transit transport and merchanting, and to used goods, waste and scrap. It will also be possible to deduct some statistically identified transport services paid for by the seller and not

invoiced separately or directly paid for by the purchaser. This could still leave a residual, however, that would be much higher than any reasonable estimate of total transport margins.

7.113. Transport costs are usually surveyed in current business statistics, at least as a cost item. By definition, these transport costs relate to the goods produced or traded. If the purchaser arranges the transport, these costs may be incorporated in intermediate consumption. Based on the structure of the output and the products traded, an estimate can be made, based on the structure of the products for which the transport costs have been paid. It is implausible to assume, however, that the transport costs are to some extent proportional to the value of the products produced or traded. Such assumptions and the subsequent estimates will be of limited use.

7.114. The various estimation steps to be followed in calculating the transport margin matrices should as far as possible be separated into the different modes of transport (for example, road, railway, water, air, pipeline, forwarding) and transport insurance; the available data might not be broken down in this way, however, and different estimation methods would need to be applied.

7.115. Given the limited availability of data, it may be better to concentrate on the products with large transport margins and to allocate margins to the remaining products according to certain plausible assumptions. As only a part of all transport services are transport margins, it is difficult to check the resulting data for plausibility. The supply and use of transport margins should of course be equal but estimation of the one side is not independent from estimation of the other side.

7.116. For the forwarding agents' services, the same estimation problem exists as for the transport itself. The forwarding agents' services are much more closely related, however, to the transport costs, and estimates could be based, if available at this stage, on the structure of the transport margins. That said, not all transport is organized by forwarding agents. Forwarding agents are usually engaged in cross-border transport rather than in domestic transport. In evaluating the practical problems connected with the correct estimates of forwarding agents' margin matrices, consideration could be given to the treatment of these services as not being part of the transport margins.

7.117. Transport insurance services are usually a very small part of the transport margins. Here too they may be more important for cross-border transport than for domestic transport. A key difference is that the insurance premiums depend on the value of the goods transported, rather than on the actual transport costs of the freight. Similarly to the forwarding agents' services, and in view of the practical implementation of such services and their limited scope, it could be decided to treat them also as ordinary services outside the margin system.

7.118. Once the use table-based transport margin matrices, whenever relevant, have been estimated, these matrices must be deducted from the use table at purchasers' prices, and the total transport margins by intermediate and final uses are then allocated to the transport service products of the applied product classification.

E. Taxes on products and subsidies on products

7.119. Taxes on products and subsidies on products are the other major valuation component in addition to the trade and transport margins. Compared with the margins matrices, the elaboration of the matrices of taxes on products and subsidies on products is less complicated because the data situation is usually more favourable and the delimitation and calculation of taxes and subsidies forms an integral part of the regular compilation of national accounts and not just an aspect of the SUTs. Thus the main task arising with regard to taxes on products and subsidies on products when compiling SUTs is the need to establish the relationship between the different kinds of taxes and subsidies and the product flows.

7.120. The matrices for taxes on products and subsidies on products are usually derived by separate calculations for each of those taxes and subsidies and related to the intermediate use and final use parts of the use table. Contrary to the trade and transport margins, no specific information on distribution channels or transport deliveries is needed here; only the relations between the product classification and the individual taxes and subsidies are needed.

7.121. A tax on a product is a tax that is payable per unit of some good or service. The tax may be a specific amount of money per unit of quantity of a good or service (the quantity units being measured either in terms of discrete units or in continuous physical variables such as volume, weight, strength, distance, time, and so forth), or it may be calculated *ad valorem* as a specified percentage of the price per unit or value of the goods or services transacted. A tax on a product usually becomes payable when it is produced, sold or imported, but it may also become payable in other circumstances, such as when a good is exported, leased, transferred, delivered, or used for own consumption or own capital formation. An enterprise may or may not itemize the amount of a tax on a product separately on the invoice or bill that it submits to its customers (2008 SNA, para. 7.88).

7.122. VAT is a special type of tax on products collected in stages by enterprises but ultimately charged in full to the final purchasers. It is described as a deductible tax, because producers are not usually required to pay to the government the full amount of the tax that they invoice to their customers, being permitted to deduct the amount of tax that they have been invoiced on their own purchases of goods or services intended for intermediate consumption or fixed capital formation. VAT is usually calculated on the price of the good or service, including any other tax on the product. VAT is also payable on imports of goods or services in addition to any import duties or other taxes on the imports (2008 SNA, para. 7.89). General sales and turnover taxes give rise to many of the same compilation problems as VAT.

7.123. A subsidy on a product is a subsidy payable per unit of a good or service. The subsidy may be a specific amount of money per unit of quantity of a good or service, or it may be calculated *ad valorem* as a specified percentage of the price per unit. A subsidy may also be calculated as the difference between a specified target price and the market price actually paid by a buyer. A subsidy

on a product usually becomes payable when the good or service is produced, sold or imported, but it may also be payable in other circumstances, such as when a good is transferred, leased, delivered or used for own consumption or own capital formation (2008 SNA, para. 7.100).

7.124. Three main categories of taxes on products are distinguished:

- VAT-type taxes
- Taxes and duties on imports excluding VAT
- Taxes on products, except VAT and import taxes

7.125. Similarly, there are three main categories of subsidies on product:

- Import subsidies
- Export subsidies
- Other subsidies on products

7.126. The 2008 SNA gives further definitions and lists typical examples for all these different types of taxes and subsidies on products. It should be noted that profits of fiscal monopolies which are transferred to the State are treated as taxes on products, and that losses of government trading organizations and subsidies to public corporations and quasi-corporations may have to be treated as subsidies on products.

7.127. Taxes on products should be recorded on an accrual basis in the national accounts, that is, when the activities, transactions or other events occur that create the liabilities to pay taxes. The amounts to be recorded in the system are determined by the amounts due for payment only when evidenced by tax assessments, declarations or other instruments which create liabilities in the form of clear obligations on the part of taxpayers. The system does not impute missing taxes not evidenced by tax assessments.

7.128. Subsidies on products are recorded when the transaction or the event (production, sale, import, and others) which gives rise to the subsidy occurs.

7.129. The recording in the SNA of transactions related to taxes on products and subsidies on products does not mirror the way in which those involved view them. The system contains no transactions between economic units that are the actual payers (collectors) of taxes on product or the actual receivers of the subsidies on products and government. In the SNA, taxes on products and subsidies on products are recorded only at the level of the total economy and are not payable out of GVA of domestic producers. They are also not split by institutional sector.

7.130. In the context of SUTs, this has the important implication that it is never necessary to consider the actual payment flows related to these taxes and subsidies but only to identify the

products to which they relate. It is therefore also irrelevant at which stage in the turnover sequence (producer, wholesaler or retailer) the tax is actually being collected or the subsidy paid out.

1. Compilation of taxes on products and subsidies on products matrices

7.131. Generally, the compilation of the taxes on products and subsidies on products matrices consists in three main steps. The first compilation step is the allocation of taxes and subsidies on products by the products of the supply table corresponding to columns (12), (13) and (14) of Table 7.1. The second compilation step is to allocate taxes and subsidies on products to the relevant entries of the use table, as shown in Table 7.3. The third compilation step covers the specific task relating to VAT necessary to calculate non-deductible VAT.

7.132. The allocation of the taxes on products and subsidies on products would be easier when SUTs are compiled at a level of product detail where there is a one-to-one relation between the product classification item and the specific tax and subsidy. Furthermore, in cases where the tax or subsidy was linked to the physical quantities, such additional information might be necessary.

7.133. In the first compilation step, the amounts of the different and specific taxes and subsidies – usually taken directly from the respective government revenue accounts – are allocated to specific products in the SUTs. If these data are not already on an accrual basis, then they must be adjusted from a cash basis to an accrual basis, which can often be done by summary time-adjustments. No further compilation steps would be needed to arrive at the required column of taxes on products (exclusive of VAT) less subsidies on products for the supply table at purchasers' prices, as in columns (13) and (14) of Table 7.1. The allocation of non-deductible VAT depends upon the user and, in general, can only be derived on the basis of the use table, and is covered later in this section.

7.134. The second compilation step refers to the allocation of taxes and subsidies on products to the entries of the use table (intermediate use and final uses) at purchasers' prices and to the separation between "other taxes on products" and "subsidies on products", as in Table 7.3. For those product categories for which the tax or subsidy has been allocated, the share of the tax or subsidy component in the purchasers' price must be calculated. This step needs to be based on the appropriate taxation basis according to tax legislation, and Table 7.3 is the result of the appropriate calculations for each single kind of taxes on products and subsidies on products.

7.135. In order adequately to allocate the taxes and subsidies to the use table elements, not only must the appropriate tax rates be explored but also the share of the use flows at which the tax rate is to be applied. A certain product classification category might include not only flows that are taxed but also other types of products not taxed, and certain products may be free of taxes for certain users. Thus, an effective rate may need to be estimated: for example, the mineral oil tax may not only have different tax rates for the different mineral oil products but some of them might also have a tax rate of zero (for example, aviation fuel) and some users, such as the agriculture

industry, may be exempt from tax. As mentioned, this problem may be alleviated by having a sufficiently detailed product classification.

7.136. There may be cases where no rates or limited data are available for the allocation of taxes and subsidies to the use table element. In these cases, a pro rata approach may need to be applied: for example, the total value of tobacco excise duty received by government may need to be prorated against all industries' purchases of tobacco except the principal industry and including components of final uses. This implicitly assumes that all purchasers of tobacco pay the same proportion of duty in relation to the value of their purchase. This is clearly a sub-optimal approach but it achieves an allocation constrained to the corresponding total value in the supply table at purchasers' prices.

7.137. Usually the taxes on products and subsidies on products are restricted to only a small number of products. Furthermore, only a small number of taxes on products cover the bulk of the total value of taxes on products. This situation is even more prevalent with subsidies on products.

7.138. The third compilation step covers a specific tax on products, VAT, which requires separate handling. According to the 2008 SNA, VAT is to be recorded net in that:

- The output of goods and services and imports are valued excluding invoiced VAT.
- Purchases of goods and services are recorded inclusive of non-deductible VAT.

7.139. VAT is recorded as being borne by the purchasers, not the sellers, and then, only by those purchasers who are not able to deduct VAT. This applies to both intermediate consumption and gross capital formation.

7.140. Accordingly, the overwhelming part of non-deductible VAT will be recorded as being levied on final uses, mainly on household final consumption expenditures. A small part of VAT, however, is levied on enterprises and institutions that are exempt from VAT.

7.141. According to the definition of purchasers' prices, only the non-deductible part of VAT is included in the purchasers' prices. Thus, the rows (products) in the use table at purchasers' prices include non-deductible VAT. In order to balance supply and use for each product, the non-deductible VAT by products has to be estimated and either included in column (13) of Table 7.1 or deducted from the use table.

7.142. In general, VAT exemptions are related to products or activities. If an industry has only exempted activities, this causes no problem. In the case of an industry that has exempted and non-exempted activities, however, additional assumptions on the estimation of non-deductible VAT are necessary. In such cases, the ratio of exempted activities and total activities must be applied to intermediate consumption in the estimation of VAT. The exemptions may differ across countries and be dependent on the respective countries' taxation policies. For estimation of VAT by type of

product, and by type of use, details should be sought from the relevant tax authorities and should be reviewed annually.

7.143. In order to calculate non-deductible VAT, it is necessary to identify those industries and final users that are exempted from VAT, and therefore not permitted to deduct VAT from their purchases, and to relate the VAT rates (explicit rate or an effective rate depending upon the product mix) to the product classification used. Both steps need to be based on the actual VAT legislation. This calculation will be further complicated where there is more than one VAT rate in operation, as some product items in the product classification applied might be mixed in terms of their VAT tax rates. In this case, additional breakdowns of those product groups would be desirable or an effective rate should be calculated, using a weighting of lower level product detail and VAT rates.

7.144. A certain part of an industry might be VAT-exempt, in which case appropriate subdivisions might be helpful. It could also be the case that certain VAT-exempt industries are normal VAT payers for their secondary outputs. The VAT legislation may also have specific rules for very small enterprises that have to be considered, for example, thresholds for being registered in the VAT system. It is important to note, for VAT exempt industries, that non-deductible VAT must be calculated both for intermediate consumption and also for gross capital formation.

7.145. Examples of exempt-type industries, dependent upon individual countries' tax legislation, tend to cover such industries as postal services, newspapers, housing, banking, insurance, some business services, education services and health services. In addition, small producers below the VAT threshold may also be exempt.

7.146. Total calculated non-deductible VAT derived by using the official tax rates and the purchasers' values of the relevant cells in the use table will generate a theoretical VAT estimate, which should exceed VAT revenue (on accrual basis) received by the government. This is because there will always be some degree of missing VAT due to evasion, cash transactions and fraud involving products that, based on other statistical sources, are included in SUTs. Using the official VAT rates may lead to an over-estimation of theoretical VAT; for example, where the tax law allows reductions for losses on debtors or in countries where there are high thresholds for registration for VAT. It can also be appropriate to lower the rate if local tax authorities are known to be inefficient.

7.147. In the balancing process of the VAT vector and matrices, the theoretical VAT must be adjusted to the revenue received (due to be paid) by government. This adjustment process should be based on information from the tax authorities, such as industries and products where VAT is or is not likely to be paid: for example, general government is likely to be fully compliant whereas households are less so.

7.148. In producing SUTs, it is also important that the rates are reviewed each year to allow for changes in VAT rates, schemes and legislation. For example, if the effective VAT rate on a product changes in mid-year, an appropriate weighted estimate for the period will need to be established.

Annex A to chapter 7. Example for deriving trade margins in SUTs based on survey data

A7.1 This annex provides an illustration of how to calculate trade margin matrices using survey data. The example is based upon the survey data obtained through the questionnaire shown in figure A7.1, which is used by the Statistical Office of Serbia. Similar information can be obtained with other forms of questionnaires.

Figure A7.1 Extract of questionnaire

No.	Code	Product description	Sales of goods produced by the enterprise (group of accounts 61)	Closing stock of products and work in progress (groups of accounts 10 and 11)	Sales of merchandise (group of account 60)	Trade margin amount or rate %	Closing stocks of goods for resale (group of accounts 13)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1072	14.3	Outerwear, knitted or crocheted; socks, sweaters, vest					
1073	15.1	Tanned or dressed leather; luggage, handbags, saddlery and harness; dressed and dyed fur					
1074	15.2	Footwear					
		Wood and products of wood and cork, articles of straw and plaiting					
1075	16.1	Cut and treated wood for further processing					
1076	16.2	Wood and products of wood and cork, except furniture (see 1147); articles of straw and plaiting					
		Paper and paper products					
1077	17.1	Pulp, paper and cardboard for further industrial processing and printing					
1078	17.2	Articles of paper and paperboard for industrial use—boxes, containers and packing material					
1079	17.2	Articles of paper and paperboard for personal use—paper towels, napkins, toilet paper, cleaning items and deletion of the pulp and paper					
1080	17.2	Paper stationary and articles of paper and paperboards (notebooks, binders, forms etc..)					
		Coke and refined petroleum products - manufacturing (columns 4–5; trade and wholesale (columns 6–8)					

Source: Structure of income and expenditure of economic subjects in the republic of Serbia 2011. Statistical office of the Republic of Serbia, 2013.

Note: Data were collected for about 250 goods and 50 services according to the Classification of Products by Activity (CPA) used in the European Union, varying from two-digit to four-digit groups. The CPA is consistent with the CPC. The same questionnaire was used for all non-financial market enterprises.

A7.2 The objective of the calculation is to populate table A7.1 with available data in order to obtain the trade margin matrices. If the available data sources are less complete, the results will of course be more dependent on the assumptions used to populate the trade margin columns (in the supply table) and the trade margin matrices (underpinning the use table). Irrespective of the coverage or quality of source data on trade and trade margins, it is important to generate the table illustrated in table A7.1 below.

A7.3 In collecting data from the above questionnaire, the following considerations are followed:

- For all economic activities, trade turnover and either purchase of goods for resale or trade margins (either absolute or as a percentage of either purchasing or selling price), data are collected by type of product.
- The product specification is at least as detailed as the product classification applied in the SUTs, and either uses the same classification or a version that is easily transformed into the SUTs product classification.
- For each product, data are sought on either opening or closing stocks of merchandise to facilitate the calculation of changes in inventories of merchandise needed in the event that the total trade margin is derived from the difference between sales and purchases.
- Surveys usually include total coverage for enterprises above a certain threshold (based on either turnover or employment) and samples for the smaller enterprises. It is assumed that the survey results have been grossed up to cover the whole population.

Based on this information it is possible to compile table A7.1.

A7.4 As shown in table A7.1, the vast majority of trade turnover and output of trade margins (trade services) originates from the three trade activities (ISIC Rev. 4, divisions 45–47), whereas many other industries generate relatively small amounts of trade output as their secondary production. In table A7.1, estimates both of grossed-up trade turnover and trade margins are shown, as this information is needed later in the compilation process.

Table A7.1 Trade data from survey: trade margins identified separately for wholesale and retail trade margins

	Ec. activity 1		Ec. activity 2		...	Trade 45		Trade 46		Trade 47		...	Ec. activity n		SUM: all W		SUM: all R	
	Trade turnover	Trade margin	Trade turnover	Trade margin		Trade turnover	Trade margin	Trade turnover	Trade margin	Trade turnover	Trade margin		Trade turnover	Trade margin	Trade turnover	Trade margin	Trade turnover	Trade margin
	Indicate W or R	Indicate W or R	Indicate W or R	Indicate W or R		Indicate W or R	Indicate W or R	Wholesale (W)	Wholesale (W)	Retail (R)	Retail (R)		Indicate W or R	Indicate W or R	W	W	R	R
Product: CPC																		
1	10	1	15	2		0	0	750	90	950	230		10	3	800	100	1 000	250
2	0	0	10	4		0	0	1 000	100	700	300		20	8	1 100	100	800	400
:																		
Total W		5		200			1 000		9 000		0			0		11 000		
Total R		0		50			2 000		0		13 000			25				17 000

A7.5 At this stage, it is important to introduce the distinction between wholesale and retail trade margins. In this example, it is assumed that this distinction is not made directly in the survey results (although it can be established from survey returns). It is therefore necessary to make decisions on the type of margin associated with the various combinations of economic production activities and products.

A7.6 First, it may be reasonably assumed that ISIC Rev. 4, division 46, wholesale, produces mainly wholesale trade margins and, similarly, that ISIC Rev. 4, division 47, retail trade, produces

mainly retail margins, whereas ISIC Rev. 4, division 45, produces a mix of margins, which must be decided on the basis of the individual products.

A7.7 For trade carried out as secondary activity, it may be assumed, for example, that trade turnover of restaurants and hotels is probably retail trade turnover; the same holds for the trade turnover of such suppliers as hairdressers, cinemas and theatres. On the other hand, trade activities of advertising agents are more likely to be wholesale trade. Manufacturing industries often trade in products similar to those that they produce or in complementary products and the majority of such trade sales are usually of the wholesale type, although some may be sold directly to consumers. These industries may also be trading in similar imported goods and such trade is again likely to be classified as wholesale trade.

A7.8 There will be some products where a trade margin may not be applicable and additional survey detail may be collected that makes it possible to improve the specific nature of these decisions.

A7.9 Once these decisions have been made, the row and column totals for wholesale margins and retail trade margins can be calculated and, for the row totals, also the turnover by product broken down by wholesale and retail turnover. Table A7.1 illustrates how the results of these decisions are fitted into the system. It should also be noted that, at this stage, only absolute and not percentage margins are being processed.

A7.10 As explained in chapter 7, the source data for trade activity and trade margins may in practice be available in a variety of forms and with a varying degree of detail. Different assumptions may therefore be needed to establish the dataset shown in table A7.1, which is essential for deriving the trade margins needed in both the supply table and the use table.

A. Supply table

A7.11 The trade activity and the trade margin entries needed in the supply table consist of rows for output of margin activities by economic activity in the “domestic output at basic prices” part of the supply table, and of the columns for trade margins needed to transform the values by product from basic prices to purchasers’ prices.

A7.12 It will become increasingly clear throughout this process, and beyond, that it is essential to retain the distinction between wholesale trade margins and retail trade margins derived in connection with table A7.1.

A7.13 Table A7.1 resembles the format (product by industry) of the supply table. All necessary information on the output of trade services and trade margins from table A7.1 can be transferred to table A7.2.

A7.14 As explained in chapter 5, many countries may choose to redefine secondary output of trade services in the supply table so that secondary output is classified together with the output of

the primary producers. Such a redefinition would mean that there will be trade activity and output of trade margins services only from ISIC Rev. 4, divisions 45–47. In this example, it is assumed that redefinition has taken place. This simplifies all the calculations in this example and facilitates the estimation of the input structures and the calculations of IOTs. The basic methodology outlined in this annex will not be affected, however.

A7.15 It should be noted that the output of trade products may also contain some non-margin items (for example, commissions, fees, margins on second-hand sales, and others), so that total supply from trade may still be positive after deduction of trade margins in the trade margin columns – this reflects actual output produced.

Table A7.2 Supply table

		Economic activity 1	Economic activity 2	...	Trade 45	Trade 46	Trade 47	...	Economic activity n	Output at basic prices	Imports	Supply at basic prices	Wholesale trade margins	Retail trade margins	VAT	Taxes on products	Subsidies on products	Supply at purchasers' price
PRODUCTS	1											1 000	100	250	150	20		1 520
	2											2 000	100	400	200		- 10	2 690
	:																	
	Wholesale	5	200		1 000	9 000			0	11 000			- 11 000					0
	Retail	0	50		2 000		13 000		25	17 000				- 17 000				0
	:																	
	Total												0	0				

A7.16 Before compiling the trade margin matrices associated with the use table, it is useful to derive a number of ratios from tables A7.1 and A7.2, which are shown in table A7.3.

Table A7.3 Distribution channels and percentage trade margins

		Total supply at purchasers' prices made comparable to trade survey turnover (from Table A7.2)		Survey turnover, grossed up (from Table A7.1)		Average percentage of supply passing through wholesale and retail trade channels		Absolute trade margins (from Table A7.1)		Average percentage trade margin for traded goods out of comparable prices	
		Basic value plus wholesale margins	Basic value plus wholesale and retail margins	Trade turnover wholesale	Trade turnover retail	Wholesale	Retail	Wholesale	Retail	Wholesale	Retail
PRODUCTS	1	1 100	1 350	800	1 000	72.7	74.1	100	250	12.5	25.0
	2	2 100	2 500	1 100	800	52.4	32.0	100	400	9.1	50.0
	:										
	Wholesale										
	Retail										
	:										
	Total										

A7.17 The question of trade channels (namely, how big a share of the supply of a given product passes through the wholesale or retail channels or both) is central when compiling the use table trade margin matrices as shown in Figure 7.2. Fortunately, there is enough combined information in tables A7.1 and A7.2 to tackle this issue.

A7.18 To calculate these shares, the total supply in the supply table must be made comparable to the turnover concept used for the survey data (assumed to be turnover exclusive of VAT and net of taxes on products). The supply table value concept comparable for the wholesale trade turnover is basic value plus wholesale trade margin (1100 for product 1). For retail trade turnover, the comparable concept is basic value plus both wholesale and retail trade margins (1350 for product 1).

A7.19 The average percentages of supply passing through each of the wholesale and retail trade channels can now be calculated as shown in table A7.3. For product 1, the shares are 72.7 (800/1100) and 74.1 (1000/1350), respectively. It is further possible to calculate the average percentage trade margin for actual traded goods from comparable purchasers' prices. For product 1, the percentage trade margins are 12.5 (100/800) and 25.0 (250/1000), respectively.

A7.20 It should be noted that the percentage trade margins are calculated as percentage of sales prices, as required for the estimates in the use table (and not the usually applied survey percentages from the traders' buying price). It should be noted that the VAT column does not need to be completed to calculate these memo items.

B. Use table

A7.21 The use table is initially valued at purchasers' prices and this table is the starting point for determining the valuation matrices that will permit the gradual transition of the use table from purchasers' prices to basic prices.

A7.22 The first step is to estimate the VAT matrix, and then to deduct it from the use table at purchasers' prices. In the next step, the matrix for other taxes on products must be determined and deducted, and the matrix for subsidies on products determined and added.

A7.23 The elements in the residual use table resulting from these procedures will consist of only basic values and trade margins, as illustrated in table A7.4, and the task is now to separate each element into its basic value and the possible wholesale and retail trade margins. In order to illustrate the restrictions and sum conditions, it is assumed that only those economic activities specified (1, 2 and n) have intermediate consumption.

Table A7.4 Use table after removal of net taxes on products

	Economic activity 1	Economic activity 2	...	Economic activity n	Total intermediate consumption	Final consumption expenditure of households	Final consumption expenditure of general government	Gross fixed capital formation	Changes in inventories	Exports	Total use at purchasers' prices
PRODUCTS	1	100	50	150	300	700	50	150	50	100	1 350
	2										
	:										
	Wholesale										
	Retail										
	:										
Total											

A7.24 The product-by-product procedure uses the information in table A7.3, as illustrated in tables A7.5, A7.6 and A7.7. If a product passes through both a wholesale channel and a retail channel, the retail trade margin comes on the top of the wholesale trade margin, corresponding to the trade margin percentages calculated in table A7.3. Accordingly, the retail trade margins must be estimated first, followed by the wholesale trade margins.

Table A7.5 Product 1: retail margins

	Economic activity 1	Economic activity 2	...	Economic activity n	Total intermediate consumption	Final consumption expenditure of households	Final consumption expenditure of general government	Gross fixed capital formation	Changes in inventories	Exports	Total use at purchasers' prices
1. Starting row (from Table A7.4)	100	50		150	300	700	50	150	50	100	1 350
2. Selected values with retail margin	50			150	200	700		100			1 000
3. Retail margin distributed	13	0		38	50	175	0	25	0	0	250
4. Average percentage retail margin	12.5			25.0		25.0	0.0	16.7	0.0	0.0	18.5

A7.25 The estimates of the retail trade margins are illustrated for product 1 in table A7.5. From table A7.3, it is known that 1,000 of the 1,350 passes through the retail trade, and the total retail trade margin on this product is 250. The knowledge of these totals provides a good starting position but it is still not known to which of the individual uses (or part thereof) the retail turnover is linked, and it is therefore necessary to decide on (or to make assumptions regarding) the figures to be entered in row (2) – based on which specific knowledge may be at hand, and on common sense, to comply with the restriction that their sum must be 1,000. For example, final consumption expenditure of households is usually assumed to include a high share of the available retail trade margins, whereas intermediate consumption and gross capital formation may have very narrow retail margin, and exports none at all, as non-resident expenditure is a summary adjustment item, and the related margins will be included in the domestic consumption concept.

A7.26 When row (2) in table A7.5 has been determined, the distribution of the retail trade margin can be determined either by distributing the 250 proportionally to the values in row (2), or by applying the percentage retail trade margin of 25 per cent from table A7.3 to the values in row (2). In row (3), the effective percentage retail trade margins relative to the elements of the use table are calculated. These are the percentages that will be used to recalculate the retail trade margin table after changes made to the original data during the balancing.

Table A7.6 Product 1: wholesale margins

	Economic activity 1	Economic activity 2	...	Economic activity n	Total intermediate consumption	Final consumption expenditure of households	Final consumption: expenditure of general government	Gross fixed capital formation	Changes in inventories	Exports	Total use at purchasers' prices
1. Table A7.5 Row (1) minus Row (3)	87.5	50.0		112.5	250.0	525.0	50.0	125.0	50.0	100.0	1 100.0
2. Values with wholesale margin	50.0			100.0	150.0	500.0		100.0		50.0	800.0
3. Wholesale margin distributed	6.3	0.0		12.5	18.8	62.5	0.0	12.5	0.0	6.3	100.0
4. Average percentage wholesale margin	7.1	0.0		11.1		11.9	0.0	10.0	0.0	6.3	9.1

A7.27 A similar procedure is used to determine the distribution of wholesale trade margins in table A7.6. The first row in this table is the first row in table A7.5 minus the estimated retail trade margins. When the estimated wholesale trade margins are deducted from row (1) in table A7.6, the row at basic prices in table A7.7 below is obtained, and thus the desired use table at basis prices has been derived. It should be noted that, in the use table at basic prices, the rows for wholesale and retail products will be made up of the column totals of the two trade margin matrices and, in addition, that they will include any non-margin trade output.

Table A7.7 Product 1: row in use table at basic prices

	Economic activity 1	Economic activity 2	...	Economic activity 2	Total intermediate consumption	Final consumption expenditure of households	Final consumption expenditure of general government	Gross fixed capital formation	Changes in inventories	Exports	Total use at purchasers' prices
Product 1 at basic prices	81.3	50.0		100.0	231.3	462.5	50.0	112.5	50.0	93.8	1 000.0

A7.28 Following the outline of these procedures, it is clear why it is essential to distinguish between wholesale and retail trade margins. If this is not done, it will not in practice be possible to manage the problem of successive trade channels. Thus the cumulative trade margin on household consumption of $(175+62.5)/463 = 51$ per cent of the basic value total could not have been derived directly from the survey results if simply aggregated.

Chapter 8. Compiling the imports use table and domestic use table

A. Introduction

8.1. This chapter describes the disaggregation of the use table into the imports use table and the domestic use table. The first table, the imports use table, contains information on the use, in the national economy, of imported products (by product) for intermediate consumption and final uses. The second table, the domestic use table, provides information on the use of domestically produced products (by product) for intermediate consumption and final use. The compilation of these two tables primarily consists in the estimation of the imports use table, since the domestic use table is obtained by subtracting the imports use table from the use table. This chapter therefore focuses mainly on the compilation of the imports use table.

8.2. The compilation of an imports use table is embedded in the SNA and it is important to balance the supply and use of products for the domestic economy, accurately to deflate components of GDP by linking imported intermediate products with appropriate import price deflators, and to ascertain the correct distribution of the changes in the volume of GVA by industry and industry contributions to GDP growth.

8.3. Historically, the compilation of the imports use table was mainly considered as an intermediate step towards the compilation of IOTs (although not an essential step). The imports use table is becoming increasingly important in its own right, however, for analytical purposes. With the globalization of economic activities, exports and imports are growing more rapidly than GDP and the GVA chains in production are becoming more complex and more international. It is therefore very important for the national accounts to provide a sectoral disaggregation of macroeconomic data for both domestic production and imports.

8.4. Over time, many domestic economies have seen significant changes in the import share of domestic supply that can be attributed to changes in international trade, and in particular, trade in goods for processing and other intermediate material inputs. In addition, many multinational enterprises, and previously large domestic businesses, have shifted their production processes around the world, taking advantage of lower production costs and thereby increasing their competitiveness and profitability.

8.5. The set of SUTs at basic prices – both in current prices and in volume terms – that should be compiled includes the following tables:

- Supply and use table at basic prices

- Domestic use table at basic prices
- Imports use table at basic prices

8.6. Since direct information for compiling the use table for imported products is generally rare and available only in exceptional cases, the recommendation is to work on a highly detailed level of product group (which means rectangular SUTs with a detailed product specification). A detailed level will help in identifying the likely users of a specific imported product.

8.7. Chapter 5 provides a detailed description of the concepts and definition of imports of goods and services, where emphasis is placed on the imports by products as part of the total supply of products. In the supply table, imports are only shown as a vector of products covering goods and services. In practice, however, it may be desirable to subdivide the import vector by regions so as separately to identify imports within and outside particular regions. Furthermore, the import vector could show separate columns, for example, displaying goods and services separately. Further elaborations that are very useful for the analysis of global value chains and globalization include splitting residents' expenditure abroad into individual products; separately identifying the transport and insurance margins included in CIF estimates of goods; and separately identifying imports of manufacturing services provided under goods for processing arrangements (ideally, also with complementary information showing the underlying value of goods processed).

8.8. This chapter focuses on the compilation of the imports use table. In particular, Section B describes the structure of the imports use table, provides a numerical example and describes how to obtain the domestic use table. Section C focuses on the compilation of the imports use table and potential issues that arise during the compilation process.

B. Structure of the imports use table and domestic use table

8.9. Imports consist of purchases of goods and services by residents from non-resident producers and suppliers. In the SNA, total imports are valued FOB. Data on detailed flows of imports by product from foreign trade statistics, however, are usually valued on a CIF basis. To reconcile the different valuations used for total imports and the product components of imports, a global CIF/FOB adjustment on imports is required, and this needs to be allocated in accordance with the type of goods involved. More details on the CIF/FOB adjustment may be found in chapter 5.

8.10. The supply of imports shown in the supply table has to be allocated in the imports use table to the different use categories of intermediate uses and final uses. The general structure of the imports use table is shown in Table 8.1. The table shows the total use of imported products, goods and services, by products and by industries and by final use categories. In the columns, the table has the same format as the use table. It distinguishes two main sub-matrices, one for the intermediate use and one for the final uses of products. The total use of imports must be equal to the total supply of imports of the supply table. This equality is given for each of the products

distinguished in the SUTs. Table 8.2 shows a numerical example of the imports use table as part of the SUTs system.

Table 8.1 Structure of the imports use table

Products \ Industries	Industries					Final uses				Total use at basic prices
	Agriculture	Manufacturing	...	Services	Total	Final Consumption	Gross capital formation	Exports	Total	
Agriculture	Imported products for intermediate consumption at CIF values					Imported products for final uses at CIF values				Imported total use
Manufacturing										
...										
Other services										
Total	Intermediate consumption by industry					Total final uses by category				

Table 8.2 Numerical example of the imports use table

Millions of euros

		INDUSTRIES							FINAL USE							Total use at basic prices		
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	Total	Final consumption expenditure			Gross fixed capital formation	Changes in valuables	Changes in inventories	Exports		Total	
									Households	NPISH	General government							
																(1)		(2)
PRODUCTS	Agriculture	(1)	191	1 680	5	170	14	17	2 077	1 079		47		9	58	1 194	3 271	
	Manufacturing	(2)	706	55 898	4 365	5 621	1 126	2 985	70 702	20 894	1 422	12 310	807	1 344	17 112	53 888	124 590	
	Construction	(3)		255	197	68	38	5	563								563	
	Trade	(4)		257	0	274	30	39	600								600	
	Transport	(5)	10	1 300	95	2 181	265	75	3 926	139	9	59	1	6	4 011	4 223	8 150	
	Communication	(6)	4	860	65	2 449	1 267	248	4 893	447	17	686		22	169	1 342	6 234	
	Finance and business service	(7)	8	1 786	106	1 566	2 654	322	6 443	145		473				618	7 061	
	Other services	(8)		14	1	110	23	127	275	384	47		118			549	824	
Total		(9)	919	62 051	4 834	12 439	5 417	3 819	89 479	23 087	0	1 495	13 575	926	1 381	21 350	61 814	151 293

Table based on 2011 figures from Austria

8.11. Once the imports use table is compiled, the domestic use table is obtained by deducting the imports use table from the use table. As shown in Table 8.3, the structure and size of the domestic use table are the same as those of the use table, except for an additional row in the primary inputs section to reflect the sum of the columns in the imports use table. The body of the domestic use table does not include direct or indirect imports of goods and services. Some countries compile and reconcile both the imports use table and domestic use table concurrently, instead of compiling the use table first, followed by compilation of the imports use table and domestic use table. This happens, for example, where there are very good quality data on both imports and domestic use, separately available.

8.12. Table 8.4 provides a numerical example of the domestic use table.

Table 8.3 Structure of the domestic use table

Industries Products	Industries					Final uses				Total use at basic prices
	Agriculture	Manufacturing	...	Services	Total	Final Consumption	Gross capital formation	Exports	Total	
Agriculture	Domestic products for intermediate consumption at basic prices					Domestic products for final uses at basic prices				Total use by product
Manufacturing										
...										
Other services										
Total at basic prices	Domestic intermediate inputs at basic prices					Final uses at basic prices				
Use of imported products, CIF	Total imported products for intermediate consumption					Total imported products for final uses				
Taxes less subsidies on products	Net taxes on products for intermediate consumption					Net taxes on products for final use				
Adjustments						Adjustments on final uses				
Total at purchasers' prices	Intermediate inputs at purchasers' prices					Final uses at purchasers' prices				
Compensation of employees	Value added by component and by industry									
Other net taxes on production										
Consumption of fixed capital										
Net operating surplus/net mixed income										
Value added at basic prices	Total value added by industry									
Total inputs at basic prices	Total input by industry									

 Empty cells
Table 8.4 Numerical example of a domestic use table

Millions of euros

		INDUSTRIES							FINAL USE								Total use at basic prices	
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	Total	Final consumption expenditure		Gross fixed capital formation	Changes in inventories	Changes in stocks	Exports	Total			
									Households	NPISH						General government		
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	
PRODUCTS	Agriculture	(1)	2 354	4 284	8	117	15	21	6 800	963		123	- 42	938	1 982	8 782		
	Manufacturing	(2)	1 216	42 772	6 256	7 776	4 085	3 123	65 227	12 631	327	9 426	1 122	1 393	96 280	121 178	186 405	
	Construction	(3)	105	2 184	9 321	2 373	3 625	1 414	19 021	1 402		24 323		- 38	563	26 250	45 272	
	Trade	(4)	245	8 601	1 560	4 370	682	1 462	16 919	27 684	1 080	4 008	238	273	9 985	43 267	60 187	
	Transport	(5)	31	4 424	364	6 368	575	301	12 063	5 828	3 418	87	2	21	4 916	14 271	26 335	
	Communication	(6)	29	1 651	226	6 745	4 295	1 343	14 289	22 088	51	5 111		40	6 649	33 940	48 228	
	Finance and business services	(7)	439	11 706	4 611	18 623	25 779	8 062	69 219	36 524	1 006	9 781	0	- 177	11 156	58 289	127 508	
	Other services	(8)	8	367	58	1 060	375	1 622	3 490	13 045	5 416	53 116	113	- 105	1	567	72 153	75 643
	Total at basic prices	(9)	4 429	75 987	22 402	47 431	39 431	17 346	207 028	120 165	5 416	58 997	52 973	1 257	1 471	131 053	371 332	578 360
	Imports	(10)	919	62 051	4 834	12 439	5 417	3 819	89 479	23 087		1 495	13 575	926	1 381	21 350	61 814	151 293
	Taxes less subsidies on products	(11)	92	952	229	1 349	1 689	2 672	6 984	22 810		557	2 870	152	7	397	26 794	33 778
	Total at purchasers' prices	(12)	5 440	138 991	27 466	61 219	46 538	23 839	303 492	166 063	5 416	61 050	69 418	2 335	2 859	152 800	459 939	763 431
Adjustments	CF/FOB adjustments on exports	(13)													- 97	- 97	- 97	- 97
	Direct purchases abroad by residents	(14)							6 675							6 675	6 675	6 675
	Purchases on the domestic territory by non-residents	(15)							- 12 945						12 945			
	Total at purchasers' prices	(16)	5 440	138 991	27 466	61 219	46 538	23 839	303 492	159 792	5 416	61 050	69 418	2 335	2 859	165 648	466 517	770 009
GVA	Compensation of employees	(17)	551	30 679	10 239	37 906	22 997	41 971	144 343									
	Other taxes less subsidies on production	(18)	- 1 627	1 077	546	1 755	2 004	1 103	4 858									
	Consumption of fixed capital	(19)	1 845	12 750	1 542	10 917	18 934	7 480	53 469									
	Net operating surplus/net mixed income	(20)	3 658	16 453	5 138	23 040	18 989	4 921	72 198									
	Gross operating surplus/gross mixed income	(21)	5 503	29 203	6 680	33 957	37 923	12 401	125 667									
	GVA	(22)	4 427	60 959	17 465	73 618	62 923	55 475	274 868									
	Total input at basic prices	(23)	9 867	199 950	44 931	134 837	109 461	79 314	578 360									

 Empty cells

Table based on 2011 figures from Austria

1. Input table for imports

8.13. Table 8.5 shows a numerical example of an input table for imports of goods and services at basic prices. This table is either a product-by-product table or industry-by-industry table but is not an IOT, as imports form an input and not an output.

8.14. It should be noted that the sub-matrices for final uses and the row totals for products are the same in the imports use table and the IOT table of imports. An input table for imports, as already noted, may also be a step in the process of compiling IOTs but not a necessary step. This is covered in more detail in chapter 12 (see box 12.3) on the transformation of SUTs into IOTs, where the transformed imports use table can be applied in two different ways.

8.15. The only way in which it differs from the imports use table in Table 8.2 is that Table 8.5 shows the intermediate use of the imports in a product-by-product format (or this could be an industry-by-industry format). The final use part is unchanged. Chapter 12 provides more details on how the imports use table, and in turn the input table for imports, can be used to produce IOTs where, for imports of goods and services, this is only an input table.

Table 8.5 Example of an input table for imports at basic prices

Millions of euros																		
		PRODUCTS							FINAL USE								Total use at basic prices	
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	Total	Final consumption expenditure			Gross fixed capital formation	Changes in valuables	Changes in inventories	Exports	Total		
									Households	NPSH	General government							
PRODUCTS	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)		
	Agriculture	(1)	176	1 722	3	148	14	15	2 077	1 079		47		9	58	1 194	3 271	
	Manufacturing	(2)	618	55 846	4 392	5 506	1 398	2 941	70 702	20 894		1 422	12 310	807	1 344	17 112	53 888	124 590
	Construction	(3)		265	204	47	44	4	563								563	
	Trade, transport and communication	(4)	9	2 095	150	5 150	1 678	337	9 419	586	26	745	1	28	4 179	5 565	14 984	
	Finance and business services	(5)	7	1 531	97	1 527	2 974	308	6 443	145		473				618	7 061	
Other services	(6)		10	0	108	29	127	275	384	47		118			549	824		
	Total	(7)	811	61 469	4 846	12 485	6 136	3 731	89 479	23 087	1 495	13 575	926	1 381	21 350	61 814	151 293	

Table based on 2011 figures from Austria

2. Input-output table for domestic output at basic prices

8.16. If the imports use table is subtracted from the use table at purchasers' prices, the corresponding domestic use table can be derived which shows only consumption of domestic produced output. A further step, however, is to subtract and reallocate the trade and transport margins and to deduct the taxes less subsidies on products, in order to achieve the SUTs at basic prices.

8.17. Table 8.6 shows the IOT for domestic output and this forms the basis for input-output analyses. It should be noted that, in this table, the use of imported goods and services is only shown in an aggregated form in one row. More detail on the transformation of SUTs into IOTs may be found in chapter 12.

Table 8.6 Input-output table for domestic output at basic prices

Millions of euros

		PRODUCTS							FINAL USE								Output at basic prices	
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	Total	Final consumption expenditure			Gross fixed capital formation	Changes in valuations	Changes in inventories	Exports	Total		
									Households	NPISH	General government							
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	
PRODUCTS	Agriculture	(1)	2 316	4 344	4	101	15	19	6 800	963		123		- 42	938	1 982	8 782	
	Manufacturing	(2)	1 091	42 919	6 362	7 534	4 369	2 951	65 227	12 631	327	9 426	1 122	1 393	96 280	121 178	186 405	
	Construction	(3)	73	1 883	9 927	1 969	3 890	1 279	19 021	1 402		24 323		- 38	563	26 250	45 272	
	Trade, transport and communication	(4)	239	13 805	2 109	18 364	5 909	2 846	43 272	55 600	4 549	9 207	239	334	21 550	91 479	134 750	
	Finance and business services	(5)	370	9 320	4 530	17 653	29 781	7 564	69 219	36 524	1 006	9 781	0	- 177	11 156	58 289	127 508	
	Other services	(6)	6	286	51	1 066	453	1 629	3 490	13 045	5 416	53 116	113	- 105	1	567	72 153	75 643
	Total	(7)	4 094	72 557	22 984	46 687	44 418	16 288	207 028	120 165	5 416	58 997	52 973	1 257	1 471	131 053	371 332	578 360
GVA	Imports	(8)	811	61 469	4 846	12 485	6 136	3 731	89 479	23 087	1 495	13 575	926	1 381	21 350	61 814	151 293	
	Taxes less subsidies on products	(9)	78	862	226	1 333	1 839	2 646	6 984	22 810	557	2 870	152	7	397	26 794	33 778	
	Total at purchasers' prices	(10)	4 983	134 889	28 056	60 506	52 393	22 665	303 492	166 063	5 416	61 050	69 418	2 335	2 859	152 800	459 939	763 431
	Compensation of employees	(11)	411	25 857	10 216	38 422	28 962	40 475	144 343									
	Other taxes less subsidies on products	(12)	- 1 446	717	545	1 762	2 267	1 014	4 858									
	Consumption of fixed capital	(13)	1 620	11 519	1 422	10 172	21 759	6 977	53 469									
	Net operating surplus/net mixed income	(14)	3 214	13 423	5 032	23 889	22 127	4 512	72 198									
	Gross operating surplus/gross mixed income	(15)	4 834	24 942	6 455	34 061	43 886	11 489	125 667									
	GVA	(16)	3 799	51 516	17 216	74 245	75 115	52 978	274 868									
	Total input at basic prices	(17)	8 782	186 405	45 272	134 750	127 508	75 643	578 360									

Table based on 2011 figures from Austria

8.18. It should be noted that table 8.6 does not contain any of the adjustment rows shown in table 8.4. The adjustment items are the following:

- CIF/FOB adjustments on exports (recorded as part of imports)
- Direct purchases abroad by residents (recorded as part of exports)
- Purchases in the domestic territory by non-residents (recorded as part of exports)

8.19. In this form, as shown in table 8.6, the IOTs always show the correct GDP; the totals for household final consumption expenditure, exports and imports in the IOTs differ, however, from the totals in the SUTs. Given that all the omitted adjustment items relate to final uses, the GDP calculated from the expenditure side (308,647) is still correct and identical to the results shown in connection with box 2.10. It is always possible to include the corresponding adjustment items in the final IOTs to arrive at the correct totals for household final consumption expenditure, exports and imports, as illustrated in Table 8.7, which is consistent with tables 8.2, 8.4 and 8.6.

8.20. Ideally, the adjustment items should be included in the IOTs for the purpose of consistency with the national accounts framework and the SUTs, and for complete coverage of the economy in analytical uses, as illustrated by table 8.7 (which, to illustrate the point, also shows a net export presentation – this presentation is covered in more detail in chapter 12). Different practices regarding these adjustment items have, however, been emerged across countries and international organizations. Thus the Austrian IOTs shown here replicate the tables contained in the Eurostat database. On the other hand, the IOTs in the OECD input-output database include all adjustment items, to ensure full consistency with national accounts data.

8.21. The problem of how to deal with the adjustment items necessarily arises when compiling empirical IOTs adjustment rows. For ease of exposition and in order not to overload the

presentation of the SUTs and IOTs, these additional rows are not included in the numerical examples in this Handbook. Their absence does not mean that they have been distributed by products and thus included in the upper part of the SUTs and IOTs, an arrangement which, for example, some analytical users of IOTs would prefer.

Table 8.7 Input-output table for domestic output at basic prices, net exports with adjustment items

Millions of euros

		INDUSTRIES							FINAL USE									Total output at basic prices	
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	Total	Total consumption expenditure			Gross fixed capital formation	Changes in val-ables	Changes in inven-tories	Exports	Less imports	Total		
									House-holds	NPISH	General government								
																			(1)
PRODUCTS	Agriculture	(1)	2 492	6 065	8	248	29	34	8 877	2 042		170		- 32	996	- 3 271	- 95	8 782	
	Manufacturing	(2)	1 708	98 765	10 754	13 040	5 768	5 893	135 928	33 525	1 749	21 736	1 929	2 737	113 392	- 124 590	50 477	186 405	
	Construction	(3)	73	2 148	10 131	2 016	3 934	1 282	19 585	1 402		24 323		- 38	563	- 563	25 687	45 272	
	Trade, transport and communication	(4)	248	15 900	2 258	23 514	7 586	3 183	52 690	56 185	4 575	9 951	240	363	25 729	- 14 984	82 060	134 750	
	Finance and business services	(5)	377	10 851	4 627	19 180	32 755	7 872	75 662	36 669	1 006	10 254		- 177	11 156	- 7 061	51 846	127 508	
	Other services	(6)	6	297	51	1 174	482	1 756	3 765	13 429	5 416	53 163	113	14	1	567	- 824	71 878	75 643
	Total at basic prices	(7)	4 905	134 027	27 830	59 173	50 554	20 019	296 507	143 252	5 416	60 492	66 548	2 182	2 852	152 403	- 151 293	281 852	578 360
ADJUSTMENTS	Taxes less subsidies on products	(8)	78	862	226	1 333	1 839	2 646	6 984	22 810		557	2 870	152	7	397		26 794	33 778
	Total at purchasers' prices	(9)	4 983	134 889	28 056	60 506	52 393	22 665	303 492	166 063	5 416	61 050	69 418	2 335	2 859	152 800	- 151 293	308 647	612 138
	CIF/FOB adjustments	(10)														- 97	97		
	Direct purchases abroad by residents	(11)								6 675							- 6 675		
	Purchases on the domestic territory by non-residents	(12)								- 12 945					12 945				
	Total	(13)	4 983	134 889	28 056	60 506	52 393	22 665	303 492	159 792	5 416	61 050	69 418	2 335	2 859	165 648	- 157 871	308 647	612 138
VALUE ADDED	Compensation of employees	(14)	411	25 857	10 216	38 422	28 962	40 475	144 343										
	Other taxes on production	(15)	- 1 446	717	545	1 762	2 267	1 014	4 858										
	Consumption of fixed capital	(16)	1 620	11 519	1 422	10 172	21 759	6 977	53 469										
	Net operating surplus	(17)	3 214	13 423	5 032	23 889	22 127	4 512	72 198										
	Gross operating surplus	(18)	4 834	24 942	6 455	34 061	43 886	11 489	125 667										
	GVA	(19)	3 799	51 516	17 216	74 245	75 115	52 978	274 868										
	Total input at basic prices	(20)	8 782	186 405	45 272	134 750	127 508	75 643	578 360										

Table based on 2011 figures from Austria

C. Compilation of the imports use table

8.22. Compilation of the imports use table may be a difficult process because direct information for the estimates of imported products by industry and by final use may not be available or only available in limited cases. As a result, direct information has to be supplemented by reasonable assumptions and indirect techniques. As noted earlier, in a large rectangular SUTs system, many homogenous products can be identified which have to be imported from abroad. Thus, the allocation of goods and services in the use table for domestic output and the use table for imports is easier if a large rectangular SUTs system is available. The two main approaches to the compilation of Imports Use tables are presented below: the first is based on the availability of directly collected data and the second on assumptions regarding the imports.

1. Using directly collected data

8.23. There are two major sources for direct information for the imports use table:

- Business surveys which could be developed further, for example, for each industry, more product detail of imports of goods and services by type of product and more information on imports of services
- Trade surveys which provide extensive details of imports of goods. Traditionally, the customs department collects foreign trade statistics.

8.24. The data linking of units' data from trade and business registers provides another source, also ensuring some degree of coherence between the two sources.

8.25. Business surveys (annual or for benchmarked years) generally collect details such as sales by type of product and purchases by type of product. These surveys could be expanded to include additional questions useful for the imports use table, such as the value of purchases of imports of goods and the value of purchases of imports of services.

8.26. For certain industries, it is important to ask specific questions on the imports of goods and services for certain specific products. For example, for the sugar refining industry it may be important to have information on the purchases of sugar beet separate from sugar cane. The economy may have little, or no, domestic production of one or both of these products and would have to rely on imports, which at the two-digit level would appear in the same product classification. This approach of asking for specific details may apply also to other products such as, for example, tobacco and tobacco leaf.

8.27. This use of the business survey data would help to provide an industry total of direct imports. These values could then be developed and matched with imports of goods from the trade data suppliers by product, to help develop the body of the imports use table in terms of intermediate use. This works for direct imports but less so for indirect imports, such as imports sold to manufacturers via resident distributors such as wholesalers. Imports by retailers could, however, be assumed in the main for final use categories. Again, this would need scrutiny, for example small items (not purchased in bulk) like stationery, may be purchased by businesses from retailers. For imports of services, indirect imports should not be an issue, because by their nature services cannot be resold – in other words, they are used only once.

8.28. International trade surveys provide much of the detail by product. Further work is often required, however, to identify the importing industry or industries. For imports of goods, for example, very detailed international trade data from the Harmonized System can be more easily used to link imports to specific products that are used by industries as intermediate consumption and those products that are components of specific categories of final use. These data could be developed with the data collectors to identify the industry to which the importer is classified and the value of imports by product.

8.29. In terms of imports of services, a product-by-industry matrix should be generated for each of the 12 components forming trade in services (applicable to both imports and exports), as

illustrated in Box 8.1, which would also highlight various improvements required to imports of services in the balance of payments. For business services, a separate matrix can be generated using imports of services data from business surveys providing details for other variables such as sales and purchases.

8.30. There are various existing sources of data used for imports of services by product, such as the following:

- International trade in services – which collects data from businesses covering their imports and exports of services by product. This is a statistical survey which has advantages over administrative data.
- International passenger survey – which collects expenditure data by product by travellers, at the point of entering or exiting the resident economy (for example, airports and ports). There is a need to separate the expenditure by business travellers (recorded as intermediate consumption) and expenditure by households (recorded as households' final consumption expenditures).
- Specific sources capturing imports of services such as shipping, air transport, financial services, and others.
- Development of non-traditional survey-type sources such as credit card data and international microdata-sharing, for example between national statistical offices.

8.31. For some of the standard services components, however, there are specific issues which need careful handling, for example, disbursements, freight costs, royalties, and so forth.

Box 8.1 Standard services components of BPM 6

- | | |
|----|--|
| 1 | Manufacturing services on physical inputs owned by others |
| 2 | Maintenance and repair services not included elsewhere (n.i.e.) |
| 3 | Transport |
| 4 | Travel |
| 5 | Construction |
| 6 | Insurance and pension services |
| 7 | Financial services |
| 8 | Charges for the use of intellectual property not included elsewhere (n.i.e.) |
| 9 | Telecommunications, computer, and information services |
| 10 | Other business services |
| 11 | Personal, cultural, and recreational services |
| 12 | Government goods and services n.i.e. |

Source: IMF (2009)

2. Alternative approach

8.32. As noted, there may be a lack of source data (unless surveys are designed to collect detail data and flows from the trade industry) to estimate so-called “indirect” import use, that is, the ultimate destination of the use of imported goods, and not, say, the wholesaler acting as an intermediary. This situation will often require strong assumptions and indirect techniques to allocate the use of imports by product for each industry and by category of final use by product.

8.33. A widely used approach in estimating import use by product across using industries and categories of final uses is the application of the import proportionality or comparability assumption. This assumes that imports are used in the same proportion across all industry intermediate inputs and final uses (except exports and allowing for imports for re-export purposes). This is often a two-step procedure in which the ratio of imports to domestic supply is first calculated and is then applied to each product that is used by industries as intermediate inputs to production and by categories of final uses (except exports). For example, if imports of semi-conductors represent 50 per cent of the domestic supply of semi-conductors, then it is assumed that each industry that purchases semi-conductors purchases 50 per cent of them from foreign sources. This procedure results in the same distribution of imported products across a given row in the use table, thus providing another reason to work at the most detailed level of products available within the SUTs system, where there are likely to be fewer users of very specific products. Thus this procedure works much better with large numbers of products (for example, 10,000) as opposed to, say, fewer than 100 products. With foreign trade data, there tends to be much more data available by products relative to other data sources. The procedure should, preferably, be applied at basic prices.

8.34. Certain products can be very straightforward to allocate. For example, there are very few users of imports of crude oil in the domestic economy, whereas imports of food are used by many industries and households, or refrigerators may be used by households or in gross fixed capital formation. Nevertheless, the main task remains, namely, to attribute allocation ratios and percentages for each category of imported products across using industries and categories of final uses.

8.35. For this procedure, BEC, or a national variant of BEC, can also be used when applying the import proportionality assumption. BEC allocates imports of goods into categories of intermediate goods, consumer goods and capital goods. The elements of BEC are the subclasses of SITC, which are defined in terms of the Harmonized System. Only broad use categories are distinguished, however, and these are of less help for the intermediate uses by specific industry. Nonetheless, BEC may be helpful in categorizing the products for the imports use table.

8.36. It is worth noting that it is time and resource-consuming, especially for the first time, to generate detailed allocation ratios and percentages covering intermediate use and final uses for each category of imported products.

8.37. In defining the allocation percentages, there is a need also to keep in mind that, as a consequence of secondary output, products are also used in industries where they might not be typical inputs. This may have already been addressed in the use table.

8.38. It is important that the import proportionality assumption or related ratio procedures be used only after direct information about import use has been compiled. Lastly, once the proportionality assumptions have been applied, it is essential to evaluate the generalized results for reasonableness, and to adjust these percentages in the light of an understanding of how the specific economy operates with regard to production chains and purchases of products by final use.

8.39. The product imbalances, and the balancing process, can often be used to correct for implausible results from an initial allocation based on proportions.

8.40. Although the proportionality approach is not time-consuming, the allocation percentages can generally also be applied for other years without any large changes. Usually a great share of total imports will fall under only a few specific product classifications (for example, manufacturing products purchased by manufacturing industries) and efforts should be concentrated on those as, to a large extent, they determine the quality of the resulting imports use table and thus the accuracy of GDP in volume terms and the distribution of GVA by industry.

8.41. A difficult category of final use in respect of the allocation of imports of goods is changes in inventories. First, it is often assumed that the import share of semi-finished and finished products is zero. Second, the category of changes in inventories is a balancing item between the inventories at the end of the period minus the inventories at the beginning of the period without knowing the inflows and outflows over the period. As a result, the sign of the estimate can be positive or negative and, in the latter case, this needs to be handled with caution, otherwise it could lead to implausible values.

8.42. Inventories of finished goods should, however, be treated separately from imports of finished goods for resale without further processing. The latter are goods likely to be held mainly by distributors.

8.43. Lastly, in the process of balancing it must be expected that the procedure may need to be repeated in order to achieve plausible results. As noted, the allocation shares might need to be corrected on the basis of implausible results, which may include negative use elements. Naturally, the allocation of imports to a use element with a zero entry is not permissible and may indicate a wider problem with the level of aggregation.

D. Specific issues in the compilation of the imports use table

8.44. There are specific issues that need careful consideration when compiling the imports use table. These include the recording of goods sent abroad for processing, investment goods repaired

abroad, imports for re-export purposes, and direct expenditures by residents abroad. These specific issues are briefly discussed in this section.

8.45. Other issues include: arrangements within multinational enterprises including transfer pricing; contract manufacturing and manufacturers; factoryless goods production and processors; foreign direct investment relationships; intellectual property products – ownership and cross-border use; international labour movement and remittances; Internet trading; limitations of national data collections; merchanting of goods and services; ownership of property abroad; special-purpose entities; and toll processing and processors. The guide to measuring global production (UNECE, 2015) provides much more detail on how handle these types of issues.

1. Goods sent abroad for processing

8.46. Sending materials or partly finished goods to another affiliate or non-affiliate enterprise for processing is an established practice which has become more common with the drop in transport costs, specialization among enterprises and the emergence of new production sources. The enterprise processing the items may be resident in the same country as its client or it may be abroad.

8.47. The procedure of sending material for processing is called “goods sent abroad for processing”. This practice is very common in industries such as wearing apparel (clothing); chemicals and the manufacture of electronic and metal goods. One variant of particular interest for the national accounts and balance of payments is goods sent abroad for processing, where the unit in country A (the principal) makes a contract with the unit in country B (the contractor) under which B transforms in a substantive way raw materials or semi-processed goods sent by country A. Throughout the process the principal maintains legal ownership of the raw materials and semi-processed goods, and also of the processed goods. The principal pays the contractor a fee for the processing.

8.48. The issue of how to record the goods sent abroad for processing in national accounts, including SUTs and balance of payments, has been the subject of extensive discussions in connection with successive versions of the SNA and the BPM. The central question has been whether to impute a value of the goods when sent abroad for processing and subsequently for the processed goods when returned to the legal owner and, in this approach, to impute transactions even though no change of ownership has taken place (the gross principle), or just to record the processing fee as a separate service delivered from the processor to the principal (the net principle).

8.49. In the following sections, the different implications of applying either the gross or the net method are illustrated by numerical examples, both for goods sent for domestic processing and for goods sent for international processing. Against this background, the international recommendations established by the 2008 SNA and BPM 6 are explained, with the inclusion of possibilities to deviate from these recommendations to reinforce desirable properties of the SUTs. The Eurostat Manual on Goods Sent Abroad for Processing (Eurostat, 2014b) provides further

details in this regard. Lastly, related measurement problems in current economic statistics are also considered.

(a) Domestic processing

8.50. In the example in Table 8.8, a principal unit classified in Industry 1 sends semi-processed goods (Product A) for further processing to a contractor unit classified in Industry 2. The contractor does not pay for the material received from the principal unit. The value of the goods sent for processing is 100. The value of the goods after processing, assumed to be finished goods requiring no further processing (Product B) is estimated at 180. Processing fees in this example are set, for the sake of simplicity, at 80. In this example, Industry 1 and Industry 2 could also be interpreted as two units belonging to the same industry but, for maximum clarity of the exposition, a two-industry case is chosen.

Table 8.8 Processing within the country

Gross recording				Net recording			
	Supply Table				Supply Table		
	Industry 1	Industry 2	Imports		Industry 1	Industry 2	Imports
Product A	100			Prod A			
Product B		180		Prod B	180		
Processing fees				Processing fees		80	
Other intermediate consumption				Other intermediate consumption			80
GVA				GVA			
Total output	100	180	0	Total output	180	80	0

Note: Industry 1 is the principal and Industry 2 is the contractor.

8.51. Under the gross treatment, transactions of the values of 100 and 180 are imputed, the 100 being the output of Industry 1, and the 180 being the output of Industry 2. As the 180 is assumed to consist of finished goods, they are not, as it were, “returned” to the owner industry but enter the product balance as an output of Product B from Industry 2, even though this industry is not the legal owner. From the point of view of Industry 1, it would be goods for resale which are therefore not recorded as a flow from Industry 1 in the system, except if held in inventories at the end of the period. The processing fees do not appear separately, as they are included in the output of Product B.

8.52. Under the net treatment, the processing fee of 80 is the only output of Industry 2 and it is used as intermediate consumption by Industry 1. The processing fee will be classified as a service and not a good. The output of Industry 1 will be 180. As processing fees can usually be found in current industrial statistics, there are no imputations associated with implementing the net treatment.

8.53. It is noted that the GVA in the two industries (30 and 50 respectively) are identical for the two alternative treatments but the input structures are quite different. For Industry 1, the gross

treatment results in a much higher GVA share in output than the net treatment, and vice versa for Industry 2.

(b) International processing

8.54. In Table 8.9, the gross and net treatments are illustrated for goods sent abroad for processing. The numerical example is essentially identical to that presented for domestic processing, except that now it is assumed that the principal is a resident of Country I, and the processor a resident of Country II, and also that the two-country case involves export and import transactions, either actual or imputed.

Table 8.9 Goods sent abroad for processing

Gross recording (1993 SNA)

Principal (Country I)

	Supply Table			Use Table		
	Industry 1	Industry 2	Imports	Industry 1	Industry 2	Exports
Product A	100					100
Product B		180				
Processing fees						
Other intermediate consumption				70		
GVA				30		
Total output	100	0	180	100	0	100

Contractor (Country II)

	Supply Table			Use Table		
	Industry 1	Industry 2	Imports	Industry 1	Industry 2	Exports
Product A			100		100	
Product B		180				180
Processing fees						
Other intermediate consumption					30	
GVA					50	
Total output	0	180	100	0	180	180

Net recording (2008 SNA)

Principal (Country I)

	Supply Table			Use Table		
	Industry 1	Industry 2	Imports	Industry 1	Industry 2	Exports
Product A						
Product B	180					
Processing fees			80	80		
Other intermediate consumption				70		
GVA				30		
Total output	180	0	80	180	0	0

Contractor (Country II)

	Supply Table			Use Table		
	Industry 1	Industry 2	Imports	Industry 1	Industry 2	Exports
Product A						
Product B						
Processing fees		80			80	
Other intermediate consumption					30	
GVA					50	
Total output	0	80	0	0	80	80

8.55. Under the gross treatment, the 100 output of semi-processed goods (Product A) from Industry 1 in Country I is the exports of goods from Country I and imports of goods into Country II, where it is used as intermediate consumption in Industry 2. The output of 180 from Industry 2 (Product B) in Country II is exported to Country I. As the 180 is assumed to consist of finished goods, they are not returned to the owner industry but enter the balance of Product B in Country I as imports of goods. As there is no change of ownership related to the 100 and the 180, the values of these transactions must be imputed, but as later noted these cross-border movements of goods will usually be included in and valued for the external merchandise trade statistics.

8.56. Under the net treatment the semi-processed goods (Product A) disappear, and the processed goods (Product B) will appear as produced in Country I, as actual output from Industry 1. Only processing fees will appear in international trade, under services. As international processing fees are usually covered both by current industrial statistics and by statistics on international trade in

services, there are no imputations associated with the implementation of the net treatment of goods sent abroad for processing.

8.57. As for domestic processing, the GVA remains the same for the two approaches but the input structure is significantly different.

8.58. It should be noted that these numerical examples are highly stylized, in order to emphasize the main characteristics of the gross and net treatments. In practice, the difference between the value of the finished goods and the semi-processed goods may not be equal to the processing fee paid, either because the prices have changed over the processing period or because part of the increase in the value of the finished goods reflects the embodiment of intellectual property or trademarks owned by the principal. It may also be that the processed goods require further processing by the principal, in which case an additional entry of intermediate consumption (180 in the example following the gross treatment) would be necessary, and output increased accordingly. It should be noted that these issues will only arise under the gross treatment. The net treatment automatically resolves them.

8.59. In practice, the situation may be much more complicated and prove a significant measurement challenge. For example, goods are often not really sent abroad; for example, they may be purchased abroad. On the other hand, goods do not necessarily return after processing; they can be shipped immediately to a third country for final use.

(c) Treatment in 2008 SNA and BPM 6

8.60. The treatment of goods sent for processing, as laid out by the 1993 SNA and BPM 5, was quite complex but the main recommendations were that domestic processing should be based on the net treatment (except when transactions take place between two establishments belonging to the same enterprise, in which case the gross treatment should be used), and that international processing should be based on the gross principle. It was indicated in the 1993 SNA that this treatment of goods for processing was designed to facilitate input-output analysis.

8.61. The question raised leading up to the discussion of 2008 SNA was whether there was still a valid reason to record goods for international processing on a gross basis or if the advent of globalization and the growing volumes of goods processed abroad suggest that a change in practice would be appropriate. In response to this discussion, the recommended treatment of goods for processing was changed in the 2008 SNA and BPM 6, in that the change in economic ownership principle was prioritized over the actual movements of goods and physical technology, so that all goods sent for processing (both domestically and internationally) should be treated according to the net principle. This much simplified recommendation is, at the same time, more in line with company accounts and principally avoids imputations but may still require adjustments to ensure that the domestic activity is consistent with that recorded in the imports and exports data.

8.62. The new treatment of goods for processing potentially leads to larger variation in input-output coefficients and it is important to put this change into perspective. In the 1993 SNA, the net treatment was already applied if the goods were sent for processing to a non-affiliated domestic processor. Moreover, input structures change for a number of reasons: for example, because of changes in product mix, more or less use of semi-finished products, changes in capital and labour intensities, and outsourcing of services. The change in goods sent abroad for processing merely adds to the other changes in the 2008 SNA, which in turn change the input-output coefficients.

8.63. The effects on the input structure of the alternative treatments of goods sent for processing are illustrated in Table 8.10 for Industry 1. The input structures when using the gross or net treatment are taken from Table 8.8. To get an idea of what the input structure would have looked like if no goods had been sent for processing, the processing fee has been decomposed into “other intermediates” and “GVA”, using the input structure of Industry 2, and the components added to the “net treatment” column to obtain the result in the last column. When looking at the GVA to output ratios of the three input structures, they are obviously quite different. From the standpoint of input-output technology, the structure in the last column would be preferred as neither of the first two columns would be a good approximation. In practice, when following the 2008 SNA, the input structure of an industry sending goods for processing would probably represent a weighted average of the structures in the second and the last column, as the industry will probably also have some primary output that has not been sent for processing. When the share of goods sent for processing changes, the overall input structure of the industry will change even though no technological change has taken place.

Table 8.10 Industry 1 – Alternative input structures

	Gross treatment	Net treatment	Processing fee breakdown	Structure when no processing
Processing fees		80	-80	
Other intermediates	70	70	30	100
GVA	30	30	50	80
Output	100	180		180
GVA/Output	0.30	0.17		0.44

8.64. As already emphasized, Table 8.10 provides a stylized example. There are various business activity models, and mixed models, not just within an industry but within an individual business – with the impact of globalization, this is becoming more prevalent. For example, the same oil company can cover the following three activities:

- Extract the crude oil, refine it and then sell the refined petroleum. This activity has the full input structure as indicated in the final column of Table 8.10.
- Extract the crude oil, sell the crude oil for processing to another company and then purchase the refined petroleum from that company for resale without any further

processing. Note, in this case, that there is a change in economic ownership and the oil company undertakes no refining.

- Extract the crude oil, pay another company a processing fee for refining the crude oil (note, no change of economic ownership), and then sell the refined petroleum. Again, the oil company undertakes no refining. This activity is indicated in the second column of Table 8.10.

8.65. If the three activities are separable, then they would reflect different input structures. This illustrates, however, that an individual company, and an industry, can reflect a mix of activities and input structures from both the point of view of the outsourcing industry and the point of view of the processing industry. This makes the data collection, measurement and interpretation of the industry input structures much harder.

8.66. In paragraphs 28.18 and 28.19, the 2008 SNA outlines two ways to proceed that would retain the technical interpretation:

- Option 1: split the economic activity into two: one processing on own account, and one for goods sent for processing.
- Option 2: use the gross recording approach.

8.67. It should be emphasized that option 1 is the recommendation of the 2008 SNA and, for goods sent abroad for processing, BPM 6. Option 2 is shown as a supplementary presentation that may be adopted for reasons of continuity with past practices. Option 1 more accurately reflects the economic processes taking place, while option 2 focuses on the physical transformation process (2008 SNA, para. 28.20). The idea – when seen in the perspective of a published IOT – is rather theoretical, however, as only in very exceptional cases would it be realistic to have the tables include such dual branches, and furthermore this way of reasoning could be extended to cover all other reasons for differences in input structures and thus lead to an expansion of the number of economic activities ad infinitum. In these cases, where the individual producer units represent blends of traditional production and contracting-out, this approach would anyway not be feasible.

8.68. Even in gross recording, the input structures of principals and contractors would probably be quite different from the input structure when processing on own account. Specific adjustments may, however, be appropriate in certain cases. If, for example, the share of oil refining made on contract basis (as a contractor) varies drastically over time, it might be justified to apply the gross treatment in this case, while on the other hand an increasing trend in goods sent abroad for processing should not be counteracted. Practical data problems will also limit the options. Whereas it may be perfectly realistic to implement the gross treatment in the above oil refinery case, it would require industrial insight and data beyond any realistic possibility to adjust for the latter general trend.

8.69. The increasing activity by businesses involving the sending of goods for processing, whether domestically or internationally, will in any event affect the input-output coefficients (both for the principal and the contractor), and imputing sets of data that are completely detached from the actual economic transactions and their statistical recording is not a viable way of dealing with the complications arising from the institutional changes taking place in the economy. The growth of outsourcing under the globalization of markets means that these inherent institutional changes are more rapid and more significant, and this is a phenomenon with which the input-output compilers and analytical users will have to accommodate. This will not be done by making an artificial world of their own that denies these structural changes, rather than exposing them.

(d) Data and balancing issues

8.70. There are three main data sources involved in preparing the industry and product estimates for goods sent for processing in the SUTs:

- Industrial statistics in which manufacturers provide information on receipts for doing work to the orders of others (as contractor) and subcontracting expenses (as principal) but in which manufacturers are not asked to estimate values for the materials received for processing or for the processed goods when returned to the principal
- Merchandise trade statistics in which estimated values for the goods sent abroad for processing – and returned from processing abroad – are included as a border crossing principle, rather than following a change of ownership principle
- Statistics on international trade in services in which in-going and out-going processing fees are recorded

8.71. If data from all three data sources are just taken at face value and entered into the SUTs, major imbalances will obviously result.

8.72. Under the net treatment, the merchandise trade related to goods sent abroad for processing will have no counterpart in the data recorded by domestic industrial statistics and must therefore be removed from the merchandise trade data. This will often be possible, where these types of goods have been given a special code in the customs procedure. If, however, they are indistinguishably included, a more comprehensive approach may be needed, based on specific information on industrial practices and processes. Under the net approach, the processing fees recorded in the statistics on international trade in services should be directly applicable, and in principle consistent with (but not equal to) the processing fees recorded in domestic industrial statistics, as these also include payments for domestic processing.

8.73. In order to obtain qualitative data on the fees paid and received for international processing, detailed comparisons of different data sources at the firm level may be required. Survey data on the import and export of services may provide the basis for the fee paid and received but may be

incomplete in terms both of respondents and of product detail. These data must be supplemented, and reconciled, with data on industrial production and international goods flows under processing. They must also be consistent with firm data on turnover and costs, ensuring that the domestic account and rest of the world account are consistent.

8.74. Under the gross treatment, the imbalance between supply and use of goods would be removed by imputing additional inputs and outputs for domestic industries corresponding to the merchandise imports and exports related to processing. As the processing fees are in this case embedded in the value of the processed goods, they should be removed from the statistics on international trade in services before these data are entered into the SUTs.

8.75. In order to keep track of the conceptual and data-linked complexities of goods sent for processing, both domestically and internationally, it is recommended that a subsystem be established in which all the related goods and services balances are separately set out and analysed, so as to secure full coverage and consistency in the SUTs before the data are entered into the full system.

2. Investment goods repaired abroad

8.76. Investment goods which are sent abroad for major repair result in substantial amounts of value being created in the reconstruction. Both the export and the re-import are part of the import and export flows. In the case of minor repair, maintenance or servicing, however, the flows concerned are not to be recorded under imports and exports. In the case of major repairs, similar problems of recording in the SUTs framework occur. Thus, for practical reasons, it may be assumed that it usually would be only a minor repair. Furthermore, it could be assumed that the cross-border transport of investment goods for repair is quite rare and thus negligible (with the exception of products like aircraft and ships, where the activity and values involved are often very significant).

3. Imports for re-export purposes

8.77. Another challenge in the compilation of the imports use table is posed by re-exports. Re-exports are transactions of goods which were previously imported with a change in economic ownership and then exported without any substantial transformation. These re-exports are included as exports in foreign trade statistics. In the case of products that are not produced domestically, any exports of these products could easily be identified as re-exports. It should be noted that – with the exception of transport margins – re-exports of services are, by their very nature, not possible, thus the identification of re-exports is only a problem for goods.

4. Direct expenditures by residents abroad

8.78. A further problem which may also be of some importance concerning the data involved is posed by direct purchases abroad by residents in connection with tourism. These direct purchases

abroad by residents should cover all purchases of goods and services made by residents while travelling abroad for business or pleasure. Such purchases are part of the import flows and need to be estimated on a product basis. As a result, these purchases must be allocated to intermediate use in the case of business travellers and to household final consumption in the case of private travellers.

5. Transit trade

8.79. These are goods admitted under special customs procedures that allow the goods to pass through the territory. They are excluded from the general merchandise of the territory of transit. The issue of transit trade takes various forms, including quasi-transit trade and others, and poses challenges in terms both of statistics and of measurement. For some countries, these may be significant, in particular with large ports. Transit trade, also referred to as “simple transit”, and quasi-transit trade flows do appear in merchandise trade statistics and should be excluded from national accounts and balance of payments.

E. Enhancements to the imports use table for analytical uses

8.80. In addition to those enhancements described above, the analytical potential of the imports use table for users is considerably enhanced if the table also shows supplementary classifications such as the distinction between “competitive imports” and “complementary imports”, or imports subdivided by regions, such as country or region of origin.

8.81. Competitive imports are products that are also domestically produced and thus are consequential in estimating an accurate domestic use table. Complementary imports (also referred to as “non-competitive imports”) are products that are not domestically produced.

8.82. This distinction is of analytical interest as both types of imports can be expected to have a different relationship with and importance for the national economy. Competitive imports may be the subject of economic analysis concerning substitution policies and effects. Complementary imports, as products not produced in the national economy, are sometimes vital and analyses may focus on the impact of changes in their prices or volume.

8.83. In theory, the distinction between competitive and complementary imports seems to be clear. In practice, however, a number of borderline cases need to be solved. For the validity of this distinction, the product level of disaggregation is of utmost importance. Even at a very detailed product level, it is sometimes difficult to classify the products as competitive or complementary. Furthermore, this classification may not be stable over time.

8.84. Compared to the distinction between competitive and complementary imports, a geographical breakdown of the use table of imports is easier to compile as there may be data problems but no basic conceptual problems. Where goods are concerned, information on the

geographical origin of the imports is usually available in foreign trade statistics. For services, the data situation concerning the geographical breakdown of imports is less favourable.

8.85. The main problem in compiling use tables on imports with a geographical breakdown is how to allocate a single product imported from two geographical regions to the respective use categories: should it be allocated proportionally to the assumed users? Similar questions already arise when compiling the use table of imports without geographical breakdowns: is the import share of an imported product the same in all use categories? In consequence, a geographical breakdown might need such additional assumptions to be made, in order to allocate the imports.

Chapter 9. Compiling SUTs in volume terms

A. Introduction

9.1. One of the major objectives of national accounts is to provide comprehensive and coherent data which can be used for analysing and evaluating the performance of an economy. Data on the real growth of major economic flows such as production, household consumption, capital formation and exports serve as inputs for formulating economic policy. Furthermore, the national accounts data play a key role in helping to investigate the causal mechanisms within an economy.

9.2. Estimation of the parameters for macroeconomic models by applying econometric methods requires consistent time series of national accounts data with a focus on annual changes. The decomposition of annual current price changes into price changes and volume changes is therefore an important aspect of the compilation of national accounts.

9.3. Contrary to data in current prices, much of the data in volume terms cannot be directly observed. They must be derived from current price data combined with appropriate price and volume indicators, meaning that estimates in volume terms are derived more through modelling or based on proxies rather than estimates used when formulating data in current prices. In addition, the choice of index formulae influences the result of the estimates in volume terms.

9.4. The calculation of price and volume changes for the transactions of goods and services in the national accounts is ideally supported through the use of the SUTs framework. When established in an accounting framework, the volume indices and deflators of several variables at different levels of aggregation are interrelated in a systematic way. By applying an appropriate combination of price and volume index formulae, all the identities and relationships of SUTs in current prices are maintained in the SUTs in volume terms.

9.5. When balanced both in current prices and in volume terms, the SUTs ensure coherent and consistent deflation of the components of the production and expenditure approaches to measuring GDP, together with coherent and consistent estimates of price and volume indices. Another advantage of compiling price and volume measures within the SUTs framework is that price and volume measures can be derived for such important balancing items as GVA and GDP through applying the so-called “double deflation” approach – this is the recommended SNA approach to estimating GVA in volume terms.

9.6. This chapter focuses on the compilation of SUTs in volume terms. In section B it is acknowledged that there are alternative approaches to compiling SUTs in volume terms. Section

C describes the compilation of SUTs in volume terms using the H-Approach and sections D and E present more details on how to deflate the various components of the SUTs. Lastly, section F presents some considerations for compilation of IOTs in volume terms. The balancing of SUTs in current prices and in volume terms (and also at basic prices and at purchasers' prices) is covered in chapter 11.

B. Recognition of alternative approaches

9.7. Based on issues like the availability of data, resources and time, it is important to note that there are different ways of deflating SUTs, and also different sequences in so doing. In turn, these variations will also generate different balancing schemes and, most important, different degrees of quality for the detail and the aggregates.

9.8. This chapter focuses on a recommended approach, although alternative approaches are available. One such alternative is the deflation of uses at purchasers' prices as inputs to generate the SUTs in volume terms and balance the rest of SUTs, which may be deflated at basic prices (or producer prices), for example, output. This approach may be easier to implement but has in-built incoherence and inconsistency reflected through balancing different valuations. Albeit balanced out, the impact on the quality of the aggregates is implicit (and not explicit), and, in addition, for some areas like gross fixed capital formation is clearly sub-optimal, as appropriate price indices at purchasers' prices are not usually available.

9.9. In addition, SUTs in volume terms for one period can be compiled using SUTs in current prices for one period and deflators. The recommended approach, however, includes a time-series dimension.

9.10. Another alternative approach is to remove taxes, subsidies, trade and transport margins and imports, to deflate domestic output at basic prices (with weighting for exports using export price indices) and to deflate imports using import price indices. Trade and transport margins and taxes and subsidies in volume terms are estimates using rates of the previous year and the volume change at basic prices.

9.11. The underlying principles of this approach form the recommended approach covered in detail in this chapter (referred to as the H-Approach), an overview of which is given in Figure 9.1. This is a transparent, coherent and consistent approach compiling SUTs both in current prices and in previous years' prices and ensuring the better matching of the valuation of current price values with appropriate price indices.

9.12. There are many benefits of the H-Approach as, for example, it allows for the data confrontation to take place in both current prices and in volume terms and also ensures consistent deflation across the national accounts. In addition, the incorporation of high-quality alternatives helps to improve the quality of the SUTs in volume terms, for example, by deflating household final consumption expenditure using CPIs. Using either a dual approach or all the information

available, the result is conducive to the production of better quality estimates of household final consumption expenditure but these will be different from those purely based on the CPIs. There is a variety of reasons for the use of CPIs but it does have some drawbacks: for example, CPIs may not allow for discounts or bulk purchases, or fully meet national accounts requirements. Thus, the implicit household final consumption expenditure deflator from the SUTs would be, and should be, different from the CPI. For such areas as capital formation, however, it is unlikely that there is any collection of direct prices at purchasers' prices which can be used to deflate components like gross fixed capital formation, thus alternative proxies, such as producer prices, may be used instead. In these cases, the use of the H-Approach provides a much better basis and results in a higher quality volume estimate.

9.13. The H-Approach presumes SUTs in current prices for the year t , SUTs in current prices for the year $t-1$ and appropriate deflators are available. This is not, however, a necessary precondition, as SUTs in volume terms can be created with just SUTs in current prices for the year t and appropriate deflators. At the same time, this approach does not reflect the time series dimension and the use of margin and tax rates in the previous year.

C. Overview of the steps in the H-Approach with a focus on volumes

9.14. This section provides an overview of the compilation process, where Figure 9.1 shows a framework for estimating SUTs simultaneously at purchasers' prices and at basic prices and in current prices and volume terms. This section supplements the description of the H-Approach provided in chapters 2 and 3.

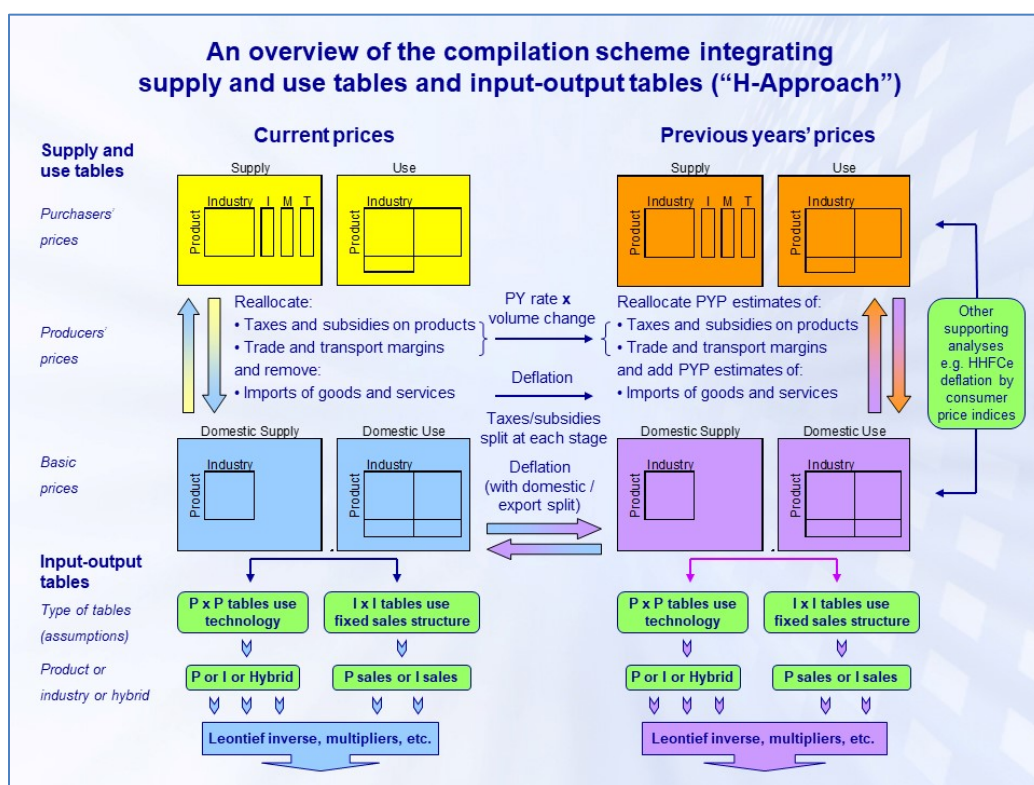
9.15. On the left-hand side of the "H", the current price data are presented, while the right-hand side of the "H" presents the volume data in previous years' prices. In the middle, the links between SUTs in current prices and in volume term (through the process of deflation) are shown, where the SUTs on both sides are valued at basic prices. The two legs below the middle cover the IOTs in current prices and in volume terms. The compilation flow can be from left-top to left-middle then through deflation to the right-middle and then to right-top. As explained in chapter 3, if better final use-based purchasers' price deflators exist, such as CPIs, then a compilation flow starting from the right-top can be incorporated, reversing the process and ensuring a balanced approach at each stage. The H-Approach allows for the use of such deflators and volume-based indicators. In terms of system design, however, there are different options of how better quality price deflators can be used but these need to be clearly set out at the outset. For example, it may be agreed that the deflation of household final consumption expenditure using CPIs gives a better measure of the volume of such expenditure in practice than deflation at basic prices, thus these deflators could be incorporated, with the objective of retaining these results and balancing back to the basic price valuation of household final consumption expenditure.

9.16. This framework is shown as a summary in figures 3.4 and 3.5 covering a simplified version of the outputs in the compilation schematic layout for SUTs and IOTs in current prices and in previous years' prices. Application of this compilation scheme will generate consistent and coherent estimates of volume and price indices for all entries of the SUTs linking the different valuations and the valuation matrices, together with the IOTs.

9.17. Figure 9.1 provides an overview that underpins the annexes covering the sequence and the stage of production processes compiling SUTs in both current prices and in volume terms, provided in chapter 4.

9.18. As stated in the introduction, many of the above transactions in volume terms cannot be directly observed. As a result, volume estimates must be derived from current price data combined with information on price or volume changes. As a consequence, the starting point for volume estimates are the SUTs in current prices as shown in the top left corner of Figure 9.1.

Figure 9.1 Overview of the compilation schematic layout linking SUTs in current prices and in volume terms



Compiled by Sanjiv Mahajan June 2009

1. Step 1

9.19. Once current price SUTs at purchasers' prices (i.e. top left-hand side of Figure 9.1) have been established, the separation of the valuation matrices covering taxes on products, subsidies on products, trade margins and transport margins, and the separation of the imports use table are used to derive a domestic use table at basic prices (i.e. middle of the left-hand side of Figure 9.1). The compilation of each of these matrices and tables is covered in chapters 7 and 8 of this Handbook.

9.20. At this stage (i.e. middle of the left-hand side of Figure 9.1), the tables for imported goods and services and domestically produced goods and services form the starting point for the compilation of the IOTs in current prices (i.e. bottom of the left-hand side of Figure 9.1) but also the first step in the deflation phase for compiling SUTs in volume terms.

2. Step 2

9.21. For intermediate consumption at purchasers' prices in the use table, appropriate price indices are mostly not available. The output and import price indices might be used as an approximation (see below); the optimal process, however, is to compile the use table at basic prices by deduction of the valuation matrices from the use table at purchasers' prices, shown on the left-hand side of Figure 9.1. In order to apply the most appropriate price indices, the use table at basic prices should be split between the uses of imported goods and services (imports use table), separate from the uses of domestically produced goods and services (domestic use table) shown in the middle of the left-hand side of Figure 9.1.

9.22. The domestic use table at basic prices is deflated using appropriate price deflators (or use of volume indicators) applied across each product in both the supply table and use table, allowing for the separation of domestically consumed products and exported products, deflated using export price indices (EPs). This assumes that the sale price charged by the seller is the same as the purchase price paid by the purchaser. Implicitly, this may not, for example, allow for bulk purchases made at a discounted price. If, however, the deflation is carried out at a very detailed level, then the impact will be insignificant.

9.23. In the ideal case, producer price indices (PPIs) are a correctly weighted average of domestic sales and exports. In practice, however, often a weighted average has to be constructed by applying weights of the previous year. The preferred option is to split produced goods and services between those consumed domestically and those exported, and then to deflate using PPIs and EPs, respectively.

9.24. The same approach holds for the imports of goods and services when applying import price indices (IPIs).

9.25. Within this step, the use of volume indicators, if appropriate, may form better quality than deflating current prices with inappropriate deflators. Further areas where a different approach for

deflation may be considered is handling self-balanced concepts such as FISIM. In current prices, FISIM is balanced across the production, income and expenditure approaches and can be shown in the form of SUTs and there is no need for any balancing adjustments. Therefore, if the SUTs in current prices are balanced at this step, you can remove FISIM as a balanced change leaving the SUTs in current prices (excluding FISIM) still balanced. FISIM can then be deflated separately using an alternative, or more appropriate, deflation approach generating a balanced FISIM in volume terms. Balanced FISIM in volume terms can be added to balanced SUTs in volume terms (excluding FISIM) to give balanced SUTs in volume terms. This ensures good quality, appropriate, consistent and coherent deflation of self-balanced concepts. There are several examples that may be addressed using a self-balanced approach such as insurance, consumption of fixed capital for non-market units, imputed rental of owner-occupied dwellings, etc.

9.26. The above step results in SUTs at basic prices in previous years' prices. These data in volume terms sit in the middle of the right-hand side of Figure 9.1.

9.27. In the case when the SUTs in current prices and at purchasers' prices are already balanced, the above step should result in balanced SUTs at basic prices in previous years' prices, assuming the requirements on prices mentioned above are fulfilled. In practice, however, some balancing will be required.

3. Step 3

9.28. Step 3 consists in the deflation of the valuation matrices for taxes, subsidies and margins (trade and transport) by applying the previous year rates to the volumes at basic prices (or volume change as appropriate).

9.29. In combination with the SUTs at basic prices, in previous years' prices, the volume changes can be calculated, and these form the basis for the volume estimates for all the individual entries of the valuation matrices. In this compilation step, there are two key issues:

- First, it is important to check the plausibility of the volume estimates, in particular trade and transport margins for the goods and services which have large changes in quality, as the quality change forms part of the volume changes in the national accounts. The consequences for GVA in the trade and transport industries might be unacceptable.
- Second, separate independent PPIs might be available for transport services. Confrontation with the implicit price indices resulting from the process indicated by Figure 9.1 might lead to unacceptable differences with the observed price indices. This, in turn, may necessitate a re-evaluation of earlier estimates.

4. Step 4

9.30. Step 4 consists in the compilation of SUTs at purchasers' prices in previous years' prices – as shown in the top-right corner of Figure 9.1 – by adding the SUTs at basic prices and the valuation matrices in previous years' prices obtained in the previous steps.

9.31. At this stage, additional plausibility checks are available, and in some cases, essential. These may include, as described earlier, the use of household final consumption expenditure in volume terms using CPIs. Confrontation of the resulting implicit price indices from the SUTs at purchasers' prices with observed purchasers' price indices like the CPI may reveal implausible results, leading to the need to re-evaluate and adjust earlier estimates.

9.32. The reassessment and allocation of adjustments may ultimately mean, in some cases, that the current price SUTs at purchasers' prices may need to change or that one or several of the intermediate steps may need to be altered.

9.33. In the case of starting with balanced SUTs in current prices and at purchasers' prices, each transitional step thereafter creates a balanced matrix. As a consequence, the resulting SUTs in previous years' prices both at basic prices and at purchasers' prices will also be balanced. Any adjustments resulting from the additional plausibility checks will be incorporated in a balanced manner and improve plausibility and, in turn, quality.

5. Step 5

9.34. Step 5 consists in the compilation of IOTs in previous years' prices using the same assumptions as applied for the compilation of IOTs in current prices. This is shown in the bottom right-hand side of Figure 9.1.

6. Other points to note

9.35. Options on the starting point, in other words, whether to use the current price SUTs at purchasers' prices and at basic prices balanced or unbalanced as a starting point, is discussed in chapter 11 of this Handbook. This choice does not alter the steps or the processes of deflation but it will have different impacts on the processes, resources, systems and schedules needed to compile these tables.

9.36. Figure 9.2 shows the links between the table in Figure 9.1 at the current price estimates and in volume estimates for two successive years. In particular:

- The link between the SUTs of year $t-1$ in current prices of $t-1$ and the SUTs year t in prices of $t-1$ are the SUTs with the corresponding volume indices
- The connecting link between the SUTs in current prices of year t and the SUTs of year t in prices of $t-1$ are the SUTs with the information on price indices

From the SUTs in current prices of year t and year $t-1$, the SUTs with value indices can be derived.

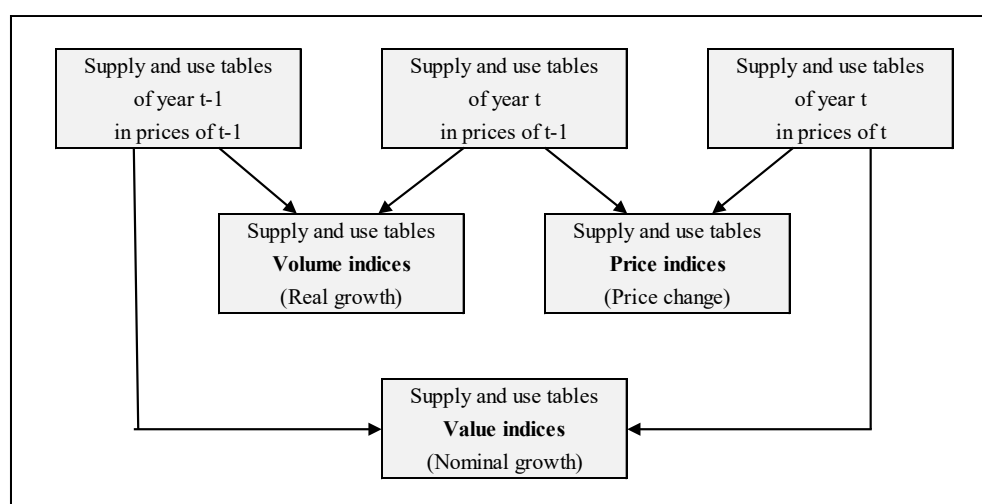
9.37. At the end of this estimation process, a complete picture is available as laid out in Figure 9.2, for every column of the SUTs, not only in current prices but also in previous years' prices for outputs, intermediate uses, final uses and imports of goods and services. It should be noted the SUTs may be unbalanced at this stage.

9.38. This set of data makes it possible to check the consistency of the data, in which process, even if the results in current prices look plausible, the analysis of volume and price data may reveal significant problems. Such analysis may include, for example, a comparison of changes in the volume of output by industry with the corresponding changes in the volume of intermediate consumption and the volume of GVA.

9.39. The analysis in volume terms is far superior, in particular when prices are changing rapidly. In several cases, these data can be checked with actual data in volumes, for example, the use of energy products or the volume of sales by product as in agriculture.

9.40. The value-price-volume analysis can lead to amendments on either of the estimated variables before the balancing process has commenced.

Figure 9.2 Link between SUTs in current prices and in volume terms



9.41. An empirical example of SUTs reflecting the components of Figure 9.2 (including prices indices, volume indices and value indices) is shown in Table 9.1 and Table 9.2. From these tables, information on inflation, real growth and nominal growth can be extracted at a detailed level, along with nominal GDP, real GDP growth and the GDP deflator. Table 9.3 also shows that the growth rates for GDP can be directly derived from the SUTs by applying the formulae for the production, income and expenditure approaches.

9.42. It should be noted that the estimates of “direct purchases abroad by residents” and “purchases on the domestic territory by non-residents” in row (11) of Table 9.1 and rows (11) and

(12) of Table 9.2 are not shown separately in these tables but incorporated in the products of the table. This is an alternative presentation of direct purchases abroad by residents and domestic purchases by non-residents: these adjustments form the difference between the national concept and domestic concept of household final consumption expenditure. The breakdown by product allows for more appropriate deflation by product, whereas the two aggregates form a heterogeneous suite of products.

Table 9.1 Supply table in current prices and in volume terms

		OUTPUT OF INDUSTRIES							Imports	Total supply at basic prices	VALUATION			Total supply at purchasers' prices	
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	Output at basic prices			Trade and transport margins	Taxes less subsidies on products	Total		
		(1)	(2)	(3)	(4)	(5)	(6)	(7)			(8)	(9)	(10)	(11)	(12)
Supply table at current prices in year t															
PRODUCTS	Agriculture	(1)	25 773	153			20	16	25 962	15 384	41 346	10 903	535	11 438	52 784
	Manufacturing	(2)	1 262	316 757	1 757	18 741	5 157	11 821	355 495	336 807	692 302	112 979	39 419	152 398	844 700
	Construction	(3)	92	1 104	87 896	296	2 062	2 134	93 584	1 564	95 148		8 260	8 260	103 408
	Trade	(4)	144	9 919	356	116 902	2 893	672	130 886	3 429	134 315	- 109 321	829	- 108 492	25 823
	Transport	(5)	385	17	69	63 645	1 348	977	66 441	10 671	77 112	- 14 864	- 996	- 15 860	61 252
	Communication	(6)		4 722		41 679	2 895	585	49 881	6 277	56 158	198	3 160	3 358	59 516
	Finance and business services	(7)	489	6 992	1 053	13 015	195 536	73 409	290 494	55 242	345 736		9 124	9 124	354 860
	Other services	(8)	304	2 512		2 887	3 540	216 628	225 671	16 638	242 309	105	3 008	3 113	245 422
Adjustment	Total	(9)	28 449	342 176	91 131	256 965	213 451	306 242	1 238 414	446 012	1 684 426		63 339	63 339	1 747 765
Adjustment	CI/FOB adjustments on imports	(10)								- 3 569	- 3 569				- 3 569
	Direct purchases abroad by residents	(11)													
	Total	(12)	28 449	342 176	91 131	256 965	213 451	306 242	1 238 414	442 443	1 680 857		63 339	63 339	1 744 196
Price index (t-1 = 100)															
PRODUCTS	Agriculture	(1)	102.0	104.1			100.0	100.0	102.0	106.9	103.8	97.5	107.6	97.9	102.4
	Manufacturing	(2)	107.5	108.0	101.3	99.9	100.5	101.3	107.2	108.0	107.6	100.2	101.9	100.7	106.2
	Construction	(3)	103.4	100.8	100.5	102.1	100.9	99.8	100.5	101.6	100.5		100.1	100.1	100.5
	Trade	(4)	98.0	100.1	99.2	99.7	99.9	99.9	99.7	100.0	99.7	99.5	102.3	99.5	100.4
	Transport	(5)	102.4	100.0	101.5	102.0	101.6	101.6	102.0	102.2	102.0	103.5	99.4	103.2	101.7
	Communication	(6)		97.5		99.6	97.4	98.3	99.2	98.3	99.1	101.5	100.5	100.6	99.2
	Finance and business services	(7)	102.9	101.0	100.5	100.7	100.2	101.0	100.4	101.0	100.5		93.6	93.6	100.3
	Other services	(8)	102.4	100.2		102.2	115.9	100.8	101.0	103.4	101.1	99.1	110.2	109.8	101.2
Adjustment	Total	(9)	102.2	107.4	100.5	100.3	100.4	100.8	102.4	106.5	103.4		100.8	100.8	103.3
Adjustment	CI/FOB adjustments on imports	(10)								108.1	108.1				108.1
	Direct purchases abroad by residents	(11)													
	Total	(12)	102.2	107.4	100.5	100.3	100.4	100.8	102.4	106.5	103.4		100.8	100.8	103.3
Supply table of year t at previous years' prices															
PRODUCTS	Agriculture	(1)	25 274	147			20	16	25 457	14 386	39 843	11 183	497	11 680	51 523
	Manufacturing	(2)	1 174	293 265	1 734	18 755	5 129	11 673	331 730	311 934	643 664	112 711	38 690	151 401	795 065
	Construction	(3)	89	1 095	87 486	290	2 043	2 138	93 141	1 540	94 681		8 253	8 253	102 934
	Trade	(4)	147	9 913	359	117 309	2 897	673	131 298	3 428	134 726	- 109 828	810	- 109 018	25 708
	Transport	(5)	376	17	68	62 377	1 327	962	65 127	10 444	75 571	- 14 367	- 1 002	- 15 369	60 202
	Communication	(6)		4 845		41 862	2 973	595	50 275	6 385	56 660	195	3 144	3 339	59 999
	Finance and business services	(7)	475	6 925	1 048	12 924	195 168	72 709	289 249	54 690	343 939		9 745	9 745	353 684
	Other services	(8)	297	2 507		2 628	3 055	214 997	223 484	16 094	239 578	106	2 729	2 835	242 413
Adjustment	Total	(9)	27 832	318 714	90 695	256 145	212 612	303 763	1 209 761	418 901	1 628 662		62 866	62 866	1 691 528
Adjustment	CI/FOB adjustments on imports	(10)								- 3 303	- 3 303				- 3 303
	Direct purchases abroad by residents	(11)													
	Total	(12)	27 832	318 714	90 695	256 145	212 612	303 763	1 209 761	415 598	1 625 359		62 866	62 866	1 688 225
Volume index (t-1 = 100)															
PRODUCTS	Agriculture	(1)	99.9	96.1			87.0	88.9	99.9	103.5	101.1	103.6	111.9	103.9	101.8
	Manufacturing	(2)	95.4	104.5	89.4	98.5	94.4	103.1	103.8	103.3	103.6	103.6	98.7	102.3	103.3
	Construction	(3)	127.1	129.4	104.1	117.9	89.3	99.9	103.9	101.2	103.8		98.2	98.2	103.4
	Trade	(4)	79.5	94.4	104.1	104.8	99.0	90.5	103.6	101.4	103.6	103.4	98.8	103.5	104.1
	Transport	(5)	98.9	100.0	82.9	103.2	129.2	101.9	103.6	100.3	103.1	104.9	106.5	105.0	102.7
	Communication	(6)		109.4		101.6	109.6	97.2	102.7	97.8	102.1	97.0	97.5	97.5	101.9
	Finance and business services	(7)	99.6	103.0	97.1	103.3	102.7	100.4	102.1	107.4	102.9		98.2	98.2	102.8
	Other services	(8)	95.2	99.1		99.4	100.3	100.5	100.5	98.3	100.3	94.6	101.9	101.6	100.4
Adjustment	Total	(9)	99.6	104.2	103.6	103.3	102.4	100.6	102.6	103.5	102.8		98.6	98.6	102.7
Adjustment	CI/FOB adjustments on imports	(10)								100.9	100.9				100.9
	Direct purchases abroad by residents	(11)													
	Total	(12)	99.6	104.2	103.6	103.3	102.4	100.6	102.6	103.5	102.8		98.6	98.6	102.7
Supply table of t-1 at current prices															
PRODUCTS	Agriculture	(1)	25 299	153			23	18	25 493	13 900	39 393	10 799	444	11 243	50 636
	Manufacturing	(2)	1 230	280 614	1 939	19 032	5 431	11 317	319 563	301 843	621 406	108 774	39 201	147 975	769 381
	Construction	(3)	70	846	84 076	246	2 289	2 141	89 668	1 521	91 189		8 404	8 404	99 593
	Trade	(4)	185	10 503	345	111 978	2 927	744	126 682	3 381	130 063	- 106 185	820	- 105 365	24 698
	Transport	(5)	380	17	82	60 429	1 027	944	62 879	10 408	73 287	- 13 701	- 941	- 14 642	58 645
	Communication	(6)		4 428		41 199	2 713	612	48 952	6 530	55 482	201	3 224	3 425	58 907
	Finance and business services	(7)	477	6 724	1 079	12 509	190 079	72 409	283 277	50 908	334 185		9 925	9 925	344 110
	Other services	(8)	312	2 531		2 644	3 045	213 878	222 410	16 366	238 776	112	2 678	2 790	241 566
Adjustment	Total	(9)	27 953	305 816	87 521	248 037	207 534	302 063	1 178 924	404 857	1 583 781		63 755	63 755	1 647 536
Adjustment	CI/FOB adjustments on imports	(10)								- 3 272	- 3 272				- 3 272
	Direct purchases abroad by residents	(11)													
	Total	(12)	27 953	305 816	87 521	248 037	207 534	302 063	1 178 924	401 585	1 580 509		63 755	63 755	1 644 264
Value index (t-1 = 100)															
PRODUCTS	Agriculture	(1)	101.9	100.0			87.0	88.9	101.8	110.7	105.0	101.0	120.5	101.7	104.2
	Manufacturing	(2)	102.6	112.9	90.6	98.5	95.0	104.5	111.2	111.6	111.4	103.9	100.6	103.0	109.8
	Construction	(3)	131.4	130.5	104.5	120.3	90.1	99.7	104.4	102.8	104.3		98.3	98.3	103.8
	Trade	(4)	77.8	94.4	103.2	104.4	98.8	90.3	103.3	101.4	103.3	103.0	101.1	103.0	104.6
	Transport	(5)	101.3	100.0	84.1	105.3	131.3	103.5	105.7	102.5	105.2	108.5	105.8	108.3	104.4
	Communication	(6)		106.6		101.2	106.7	95.6	101.9	96.1	101.2	98.5	98.0	98.0	101.0
	Finance and business services	(7)	102.5	104.0	97.6	104.0	102.9	101.4	102.5	108.5	103.5		91.9	91.9	103.1
	Other services	(8)	97.4	99.2		101.6	116.3	101.3	101.5	101.7	101.5	93.8	112.3	111.6	101.6
Adjustment	Total	(9)	101.8	111.9	104.1	103.6	102.9	101.4	105.0	110.2	106.4		99.3	99.3	106.1
Adjustment	CI/FOB adjustments on imports	(10)								109.1	109.1				109.1
	Direct purchases abroad by residents	(11)													
	Total	(12)	101.8	111.9	104.1	103.6	102.9	101.4	105.0	110.2	106.3		99.3	99.3	106.1

Netherlands 2011

Table 9.2 Use table in current prices and in volume terms

		INPUT OF INDUSTRIES						Total	FINAL USE						Total use at purchasers' prices		
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services		Final consumption expenditure			Gross fixed capital formation	Changes in inventories	Exports		Total	
		(1)	(2)	(3)	(4)	(5)	(6)		Households	NPISH	General government	(11)	(12)	(13)		(14)	(15)
Use table at current in year t																	
PRODUCTS	Agriculture	(1)	5 514	17 236	114	205	248	1 091	24 408	5 937			143	- 91	22 387	28 376	52 784
	Manufacturing	(2)	10 133	175 862	26 285	26 880	7 573	30 984	277 717	122 814		9 119	51 545	1 409	382 096	566 983	844 700
	Construction	(3)	298	1 718	24 504	1 608	2 212	15 934	46 274	453		566	54 015	2 100	57 134	103 408	
	Trade	(4)	107	2 975	336	7 421	2 115	576	13 530	4 750				7 543	12 293	25 823	
	Transport	(5)	291	3 230	162	23 568	2 738	1 903	31 890	5 679		492		23 191	29 362	61 252	
	Communication	(6)	170	3 027	1 068	13 842	7 911	5 089	32 107	10 794	868		6 717	2	9 028	27 409	59 516
	Finance and business services	(7)	2 067	36 341	7 131	46 730	61 900	52 851	207 020	77 968	5	3 585	16 820	49 462	147 840	354 860	
	Other services	(8)	172	2 306	1 236	4 214	4 569	13 381	25 878	55 061	4 610	153 396	1 162	206	5 109	219 544	245 422
	Total	(9)	18 752	242 695	60 836	124 466	89 266	122 809	658 824	283 456	5 483	167 158	130 402	1 526	500 916	1 088 941	1 747 765
Adjustment	CF/FOB adjustments on exports	(10)													- 3 569	- 3 569	- 3 569
	Direct purchases abroad by residents	(11)															
	Purchases in the domestic territory by non-residents	(12)															
	Total	(13)	18 752	242 695	60 836	124 466	89 266	122 809	658 824	283 456	5 483	167 158	130 402	1 526	497 347	1 085 372	1 744 196
	Compensation of employees	(14)	2 677	42 141	18 786	71 831	70 666	111 939	318 040								318 040
	Other taxes less subsidies on production	(15)	- 722	176	5	18	- 930	1 190	- 263								- 263
	Consumption of fixed capital	(16)	3 683	18 241	2 234	18 024	13 259	51 627	107 068								107 068
	Net operating surplus	(17)	4 059	38 923	9 270	42 626	41 190	18 677	154 745								154 745
	GVA	(18)	9 697	99 481	30 295	132 499	124 185	183 433	579 590								579 590
Total input at basic prices	(19)	28 449	342 176	91 131	256 965	213 451	306 242	1 238 414	283 456	5 483	167 158	130 402	1 526	497 347	1 085 372		
Price index (t-1 = 100)																	
PRODUCTS	Agriculture	(1)	105.8	109.9	102.7	108.5	102.5	96.2	108.2	100.4			94.7	568.8	97.7	98.0	102.4
	Manufacturing	(2)	114.1	111.5	103.3	107.0	103.9	103.5	109.2	102.1		99.5	99.6	85.7	106.8	104.9	106.2
	Construction	(3)	101.4	103.1	100.3	102.4	101.1	101.8	101.0	97.4		97.1	100.0		102.0	100.0	100.5
	Trade	(4)	100.0	101.2	101.2	101.7	102.4	101.8	101.7	102.2					97.3	99.1	100.4
	Transport	(5)	101.7	101.4	101.9	102.4	100.4	102.1	102.1	101.8		103.1			101.3	101.4	101.7
	Communication	(6)	100.0	98.3	97.9	98.7	99.6	99.2	99.0	102.4	100.9		96.7	100.0	98.1	99.5	99.2
	Finance and business services	(7)	101.4	100.8	100.4	100.7	101.7	98.7	100.5	101.7	100.0	101.8	95.7		99.0	100.1	100.3
	Other services	(8)	102.4	101.4	102.1	102.5	102.4	105.6	103.9	102.7	103.8	100.2	101.7		99.5	101.8	100.9
	Total	(9)	109.3	109.0	101.6	102.2	101.7	101.1	104.5	102.1	103.3	100.2	99.1		83.0	104.8	102.6
Adjustment	CF/FOB adjustments on exports	(10)													108.1	108.1	108.1
	Direct purchases abroad by residents	(11)															
	Purchases in the domestic territory by non-residents	(12)															
	Total	(13)	109.3	109.0	101.6	102.2	101.7	101.1	104.5	102.1	103.3	100.2	99.1		83.0	104.8	102.6
	Compensation of employees	(14)	102.7	102.1	101.3	101.8	101.5	101.4	101.6								101.6
	Other taxes less subsidies on production	(15)	98.0	193.4	- 500.0	- 1 800.0	98.8	113.9	48.3								48.3
	Consumption of fixed capital	(16)	99.1	100.7	99.4	99.9	98.8	98.8	97.7								98.8
	Net operating surplus	(17)	79.8	106.1	92.4	93.1	96.3	103.8	97.7								97.7
	GVA	(18)	98.8	103.5	98.3	98.6	99.5	100.6	100.1								100.1
Total input at basic prices	(19)	102.2	107.4	100.5	100.3	100.4	100.8	102.4	102.1	103.3	100.2	99.1		83.0	104.8	102.6	
Use table at previous years' prices																	
PRODUCTS	Agriculture	(1)	5 212	15 677	111	189	242	1 134	22 565	5 912			151	- 16	22 911	28 958	51 523
	Manufacturing	(2)	8 882	157 696	25 450	25 123	7 291	29 932	254 374	120 255		9 167	51 739	1 645	357 885	540 691	795 065
	Construction	(3)	294	1 667	24 422	1 570	2 188	15 654	45 795	465		583	54 033	2 058	57 139	102 934	
	Trade	(4)	107	2 940	332	7 299	2 065	596	13 309	4 646				7 753	12 399	25 708	
	Transport	(5)	286	3 186	159	23 022	2 728	1 863	31 244	5 579		477		22 902	28 958	60 202	
	Communication	(6)	170	3 079	1 091	14 028	7 939	6 139	32 446	10 538	860		6 947	2	9 206	27 553	59 999
	Finance and business services	(7)	2 039	36 056	7 102	46 387	60 856	53 336	205 976	76 632	5	3 523	17 570	49 978	147 708	353 684	
	Other services	(8)	168	2 274	1 210	4 112	4 461	12 675	24 900	53 632	4 443	153 068	1 143	207	5 020	217 513	242 413
	Total	(9)	17 158	222 575	59 877	121 730	87 770	121 499	630 609	277 659	5 308	166 818	131 583	1 838	477 713	1 060 919	1 691 528
Adjustment	CF/FOB adjustments on exports	(10)													- 3 303	- 3 303	- 3 303
	Direct purchases abroad by residents	(11)															
	Purchases in the domestic territory by non-residents	(12)															
	Total	(13)	17 158	222 575	59 877	121 730	87 770	121 499	630 609	277 659	5 308	166 818	131 583	1 838	474 410	1 057 616	1 688 225
	Compensation of employees	(14)	2 606	41 261	18 537	70 563	69 589	110 413	312 969								312 969
	Other taxes less subsidies on production	(15)	- 737	91	- 1	- 1	- 941	1 045	- 544								- 544
	Consumption of fixed capital	(16)	3 718	18 116	2 247	18 045	13 424	52 821	108 371								108 371
	Net operating surplus	(17)	5 087	36 671	10 035	45 808	42 770	17 985	158 356								158 356
	GVA	(18)	10 674	96 139	30 818	134 415	124 842	182 264	579 152								579 152
Total input at basic prices	(19)	27 832	318 714	90 695	256 145	212 612	303 763	1 209 761	277 659	5 308	166 818	131 583	1 838	474 410	1 057 616		
Volume index (t-1 = 100)																	
PRODUCTS	Agriculture	(1)	102.0	103.3	119.4	71.3	103.4	101.1	102.6	101.5			84.8	- 9.1	102.0	101.1	101.8
	Manufacturing	(2)	99.5	106.3	101.6	99.3	99.9	97.5	103.6	99.4		102.8	112.5	43.1	104.0	103.2	103.3
	Construction	(3)	100.7	126.0	109.8	102.3	101.2	99.8	105.9	103.3		96.0	101.9		91.6	101.4	103.4
	Trade	(4)	97.3	99.0	108.1	99.3	99.2	95.0	99.2	99.2					117.4	109.9	104.1
	Transport	(5)	100.0	97.7	100.0	101.7	108.4	96.4	101.5	100.8		97.3			104.9	103.9	102.7
	Communication	(6)	98.8	102.5	100.6	104.7	101.3	95.3	101.6	97.5	100.0		102.9	7.4	107.9	102.2	101.9
	Finance and business services	(7)	98.9	104.9	104.0	104.8	104.7	99.1	103.2	100.5	100.0	96.2	100.2		106.2	102.2	102.8
	Other services	(8)	99.4	103.6	98.5	107.8	102.7	99.1	101.5	101.8	99.8	99.7	98.2		72.6	103.1	100.2
	Total	(9)	100.2	105.7	105.1	102.7	103.8	98.6	103.2	100.2	99.8	99.8	105.6		42.7	104.4	102.4
Adjustment	CF/FOB adjustments on exports	(10)													100.9	100.9	100.9
	Direct purchases abroad by residents	(11)															
	Purchases in the domestic territory by non-residents	(12)															
	Total	(13)	100.2	105.7	105.1	102.7	103.8	98.6	103.2	100.2	99.8	99.8	105.6		42.7	104.4	102.4
	Compensation of employees	(14)	100.1	100.5	98.8	101.4	101.6	100.4	100.8								100.8
	Other taxes less subsidies on production	(15)	137.2	- 104.6	5.3	.9	94.4	136.4	55.5								55.5
	Consumption of fixed capital	(16)	101.6	100.8	101.5	100.3	100.4	102.0	101.3								101.3
	Net operating surplus	(17)	99.7	101.2	104.8	109.0	101.5	110.7	104.7								104.7
	GVA	(18)	98.6	101.0	100.9	103.8	101.5	101.9	102.0								102.0
Total input at basic prices	(19)	99.6	104.2	103.6	103.3	102.4	100.6	102.6	100.2	99.8	99.8	105.6		42.7	104.4	102.4	

Handbook on Supply and Use Tables and Input-Output Tables with Extensions and Applications

		INPUT OF INDUSTRIES							FINAL USE							Total use at purchasers' prices		
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	Total	Final consumption expenditure			Gross fixed capital formation	Changes in values in inventories	Exports	Total			
									Households	NPISH	General government							
																	(1)	(2)
Use Table of T-1 at current prices																		
PRODUCTS	Agriculture	(1)	5 108	15 176	93	265	234	1 122	21 998	5 823	178	176	22 461	28 638	50 636			
	Manufacturing	(2)	8 926	148 356	25 043	25 305	7 297	30 707	245 634	120 960	8 919	45 974	3 820	344 074	769 381			
	Construction	(3)	292	1 323	22 247	1 534	2 161	15 686	43 243	450	607	53 047	2 246	56 350	99 381			
	Trade	(4)	110	2 970	307	7 347	2 082	596	13 412	4 683			6 603	11 286	24 698			
	Transport	(5)	286	3 260	159	22 630	2 517	1 933	30 785	5 535	490		21 835	27 860	58 645			
	Communication	(6)	172	3 004	1 084	13 401	7 835	6 439	31 935	10 806	860	6 748	27	8 531	26 972	58 907		
	Finance and business services	(7)	2 062	34 384	6 828	44 243	58 107	53 998	199 622	76 231	5	3 664	17 538	47 050	144 488	344 110		
	Other services	(8)	169	2 194	1 229	3 815	4 345	12 786	24 538	52 706	4 451	153 552	1 164	285	480 270	241 566		
	Total	(9)	17 125	210 667	56 990	118 540	84 578	123 267	611 167	277 194	5 316	167 232	124 649	4 308	457 670	1 036 369	1 647 536	
Adjustments	CF/FOB adjustments on exports	(10)													- 3 272	- 3 272	- 3 272	
	Direct purchases abroad by residents	(11)																
	Purchases in the domestic territory by non-residents	(12)																
GVA	Total	(13)	17 125	210 667	56 990	118 540	84 578	123 267	611 167	277 194	5 316	167 232	124 649	4 308	454 398	1 033 097	1 644 264	
	Compensation of employees	(14)	2 603	41 042	18 765	69 595	68 466	110 000	310 471								310 471	
	Other taxes less subsidies on production	(15)	- 537	- 87	- 19	- 106	- 997	766	- 980								- 980	
	Consumption of fixed capital	(16)	3 660	17 973	2 213	17 989	13 366	51 781	106 982								106 982	
	Net operating surplus	(17)	5 102	36 221	9 572	42 019	42 121	16 249	151 284								151 284	
	GVA	(18)	10 828	95 149	30 531	129 497	122 956	178 796	567 757								567 757	
	Total input at basic prices	(19)	27 953	305 816	87 521	248 037	207 534	302 063	1 178 924	277 194	5 316	167 232	124 649	4 308	454 398	1 033 097		
	Value index (T-1 = 100)																	
	PRODUCTS	Agriculture	(1)	107.9	113.6	122.6	77.4	106.0	97.2	111.0	102.0		80.3	- 51.7	99.7	99.1	104.2	
Manufacturing		(2)	113.5	118.5	105.0	106.2	103.8	100.9	113.1	101.5		102.2	112.1	111.1	108.3	109.8		
Construction		(3)	102.1	129.9	110.1	104.8	102.4	101.6	107.0	100.7		93.2	101.8	93.5	101.4	103.8		
Trade		(4)	97.3	100.2	109.4	101.0	101.6	96.6	100.9	101.4				114.2	108.9	104.6		
Transport		(5)	101.7	99.1	101.9	104.1	108.8	98.4	103.6	102.6		100.4		106.2	105.4	104.4		
Communication		(6)	98.8	100.8	98.5	103.3	101.0	94.6	100.5	99.9	100.9		99.5	7.4	105.8	101.6		
Finance and business services		(7)	100.2	105.7	104.4	105.6	106.5	97.9	103.7	102.3	100.0	97.8	95.9	105.1	102.3	103.1		
Other services		(8)	101.8	105.1	100.6	110.5	105.2	104.7	105.5	104.5	103.6	99.9	99.8	72.3	104.9	101.2		
Total		(9)	109.5	115.2	106.7	105.0	105.5	99.6	107.8	102.3	103.1	100.0	104.6	35.4	109.4	105.1	106.1	
Adjustments	CF/FOB adjustments on exports	(10)													109.1	109.1	109.1	
	Direct purchases abroad by residents	(11)																
	Purchases in the domestic territory by non-residents	(12)																
GVA	Total	(13)	109.5	115.2	106.7	105.0	105.5	99.6	107.8	102.3	103.1	100.0	104.6	35.4	109.5	105.1	106.1	
	Compensation of employees	(14)	102.8	102.7	100.1	103.2	103.2	101.8	102.4								102.4	
	Other taxes less subsidies on production	(15)	134.5	- 202.3	- 26.3	- 17.0	93.3	155.4	26.8								26.8	
	Consumption of fixed capital	(16)	100.6	101.5	100.9	100.2	99.2	99.7	100.1								100.1	
	Net operating surplus	(17)	79.6	107.5	96.8	101.4	97.8	114.9	102.3								102.3	
	GVA	(18)	89.6	104.6	99.2	102.3	101.0	102.6	102.1								102.1	
	Total input at basic prices	(19)	101.8	111.9	104.1	103.6	102.9	101.4	105.0	102.3	103.1	100.0	104.6	35.4	109.5	105.1	106.1	

Netherlands 2011

Table 9.3 Gross domestic product in current prices and in volume terms

Production approach		Income approach		Expenditure approach	
GROSS DOMESTIC PRODUCT					
t at current prices					
Total output at basic prices	1 238 414	Compensation of employees	318 040	Household final consumption expenditure	283 456
- Intermediate consumption	- 658 824	+ Other net taxes on production	- 263	+ NPISH final consumption expenditure	5 483
		+ Capital consumption	107 068	+ Government consumption expenditure	167 158
		+ Net operating surplus	154 745	+ Gross fixed capital formation	130 402
				+ Acquisitions less disposals of valuables	
= Value added at basic prices	579 590	= Value added at basic prices	579 590	+ Changes in inventories	1 526
				+ Exports of goods and services	497 347
+ Taxes less subsidies on products	63 339	+ Taxes less subsidies on products	63 339	- Imports of goods and services	- 442 443
= Gross domestic product	642 929	= Gross domestic product	642 929	= Gross domestic product	642 929
INFLATION					
Annual change of prices in percent					
Total output at basic prices	2.4	Compensation of employees	1.6	Household final consumption expenditure	2.1
- Intermediate consumption	4.5	+ Other net taxes on production	-51.7	+ NPISH final consumption expenditure	3.3
		+ Capital consumption	-1.2	+ Government consumption expenditure	0.2
		+ Net operating surplus	-2.3	+ Gross fixed capital formation	-0.9
				+ Acquisitions less disposals of valuables	0.0
= Value added at basic prices	0.1	= Value added at basic prices	0.1	+ Changes in inventories	-17.0
				+ Exports of goods and services	4.8
+ Taxes less subsidies on products	0.8	+ Taxes less subsidies on products	0.8	- Imports of goods and services	6.5
= Gross domestic product	0.1	= Gross domestic product	0.1	= Gross domestic product	0.1
GROSS DOMESTIC PRODUCT					
t at prices of previous year					
Total output at basic prices	1 209 761	Compensation of employees	312 969	Household final consumption expenditure	277 659
- Intermediate consumption	- 630 609	+ Other net taxes on production	- 544	+ NPISH final consumption expenditure	5 308
		+ Capital consumption	108 371	+ Government consumption expenditure	166 818
		+ Net operating surplus	158 356	+ Gross fixed capital formation	131 583
				+ Acquisitions less disposals of valuables	
= Value added at basic prices	579 152	= Value added at basic prices	579 152	+ Changes in inventories	1 838
				+ Exports of goods and services	474 410
+ Taxes less subsidies on products	62 866	+ Taxes less subsidies on products	62 866	- Imports of goods and services	- 415 598
= Gross domestic product	642 018	= Gross domestic product	642 018	= Gross domestic product	642 018
REAL GROWTH					
Annual real growth rates in percent					
Total output at basic prices	2.6	Compensation of employees	0.8	Household final consumption expenditure	0.2
- Intermediate consumption	3.2	+ Other net taxes on production	-44.5	+ NPISH final consumption expenditure	-0.2
		+ Capital consumption	1.3	+ Government consumption expenditure	-0.2
		+ Net operating surplus	4.7	+ Gross fixed capital formation	5.6
				+ Acquisitions less disposals of valuables	0.0
= Value added at basic prices	2.0	= Value added at basic prices	2.0	+ Changes in inventories	-57.3
				+ Exports of goods and services	4.4
+ Taxes less subsidies on products	-1.4	+ Taxes less subsidies on products	-1.4	- Imports of goods and services	3.5
= Gross domestic product	1.7	= Gross domestic product	1.7	= Gross domestic product	1.7
GROSS DOMESTIC PRODUCT					
t-1 at current prices					
Total output at basic prices	1 178 924	Compensation of employees	310 471	Household final consumption expenditure	277 194
- Intermediate consumption	- 611 167	+ Other net taxes on production	- 980	+ NPISH final consumption expenditure	5 316
		+ Capital consumption	106 982	+ Government consumption expenditure	167 232
		+ Net operating surplus	151 284	+ Gross fixed capital formation	124 649
				+ Acquisitions less disposals of valuables	
= Value added at basic prices	567 757	= Value added at basic prices	567 757	+ Changes in inventories	4 308
				+ Exports of goods and services	454 398
+ Taxes less subsidies on products	63 755	+ Taxes less subsidies on products	63 755	- Imports of goods and services	- 401 585
= Gross domestic product	631 512	= Gross domestic product	631 512	= Gross domestic product	631 512
NOMINAL GROWTH					
Annual nominal growth rates in percent					
Total output at basic prices	5.0	Compensation of employees	2.4	Household final consumption expenditure	2.3
- Intermediate consumption	7.8	+ Other net taxes on production	-73.2	+ NPISH final consumption expenditure	3.1
		+ Capital consumption	0.1	+ Government consumption expenditure	0.0
		+ Net operating surplus	2.3	+ Gross fixed capital formation	4.6
				+ Acquisitions less disposals of valuables	0.0
= Value added at basic prices	2.1	= Value added at basic prices	2.1	+ Changes in inventories	-64.6
				+ Exports of goods and services	9.5
+ Taxes less subsidies on products	-0.7	+ Taxes less subsidies on products	-0.7	- Imports of goods and services	10.2
= Gross domestic product	1.8	= Gross domestic product	1.8	= Gross domestic product	1.8

Netherlands 2011

D. Price and volume indicators in theory

9.43. The price and volume changes for aggregates are derived from price and volume changes of separate products, preferably at a low level of aggregation. Ideally, price changes on the product level should refer to specific products; for example, every year the prices of exactly the same product can change (in accordance with brand type, weight, quality, and other properties) and should be observed. Generally speaking, in terms of quality, the greater the number of products in the SUTs, the better the required matching of appropriate values and prices.

9.44. The price and volume indicators must meet a number of requirements in order to be appropriate for the estimation of price and volume indices within the SUTs framework. These requirements are discussed in relation to the concept of output. They also apply, however, to all other transactions in goods and services. Examples of the requirements include:

- The prices and quantities should relate directly to output. This means that they should refer to complete end-products and not to contributory activities or to contributory intermediate or primary inputs. In the case of prices, they must also refer to the right valuation, for example, output at basic prices.
- The prices and quantities should have sufficient stratification, meaning that different prices and quantities should be available for all different product groups making up the output.
- The product classification of prices and quantities should have sufficient and detailed matching. This requirement will be fully met, for example, if there is only one product in a product group. If there is more than one product within a product group, an additional requirement is that the composition of the product group does not change over time, which is a weak assumption and not very likely.
- The prices and quantities should be representative for the product group. Usually, prices and quantities available do not cover all products of the product group or are based on a sample survey. Changes in the prices and quantities that are observed should be representative of changes in the prices and quantities that are not observed.
- If prices differ among users for the same products, then separate price indices should be collected and used; this is very important when, for example, distinguishing price changes between domestic users and for export.
- The changes in values resulting from changes of quality should be excluded from the price index and included in the volume index.

9.45. The requirement of matching for volume estimation means that the compilation of SUTs in volume terms will require much more detail in terms of products and prices than is necessary

for the compilation of SUTs in current prices. It is preferable, however, that the classification of the SUTs in volume terms should be similar to the level of detail of the SUTs in current prices.

9.46. Considerations relating to the compilation of SUTs in volume terms constitute one of the many criteria to be borne in mind when determining the size of the SUTs – this covered in more detail in chapter 4. The balance between, on the one hand, the assumptions to be made in order to obtain sufficient detail to ensure homogeneity of price and values and, on the other, the gain in quality must be taken into account when determining the classifications in the SUTs.

9.47. For the most part, comprehensive information is not available, either on prices or on quantities. Accordingly, estimates will be based on limited information. It is recognized that limited price data and limited quantity data do not provide the same possibilities. Price information from a sample with a certain size may well be more representative than quantity information from a sample of the same size. This observation is based on the consideration that, if there is a competitive market for a specific product grouping, there will be a tendency to use one price for the total supply of that product. In that case, a relatively small sample will be sufficient for observing the price and price changes of the total supply of that product.

9.48. For their part, however, changes in quantities are less liable to such equalizing tendencies. Thus it is true that, in an expanding market, all producers will try to increase their supply but the realization will depend on restrictive factors, such as production capacity and financing facilities. Alongside fast-growing producers, there will be slow-growing producers and possibly even shrinking producers. This means that, in order to obtain reliable estimates for quantities, the samples will have to be larger – even much larger. As a consequence, it is common practice to derive price indices from price samples and, afterwards, to compile data in volume terms by combining current price data and price indices. In many cases, this approach is efficient and cost-saving.

9.49. Although the standard price method can be applied for many goods and services, there are still a number of transactions for which observation of prices has not been realized or is even not possible. The latter condition applies to certain cases in which, owing to the nature of the definition and measurement of output in current prices, the direct observation of appropriate prices is not possible, such as with non-market services, FISIM and insurance services.

E. Price and volume indicators in practice

9.50. This section briefly covers a range of price and volume indicators that may be available as the building blocks for the preparation of SUTs in volume terms and are not meant to be exhaustive. These indicators should be considered applicable, as appropriate, to the various parts of the H-Approach, for example, in ensuring consistency as far as possible between the supply table components and the use tables components. The section covers:

- Supply table at basic prices

- Use table at basic prices
- GVA by industry
- Valuation matrices

9.51. More information on the choice of index formulae may be found in chapter 15 of the 2008 SNA and the Handbook on Price and Volume Measures in National Accounts (Eurostat, 2016).

1. Supply table at basic prices

9.52. Under this heading, the domestic production of goods and services and imports of goods and services are separately considered.

(a) Domestic production: producers' price indices

9.53. The PPIs usually fulfil the general requirements for price indicators, such as valuation, adjustment for quality change and level of detail, and separate prices should preferably be available for domestic sales and for exports. For total output, a weighted average should be used. For this reason, PPIs are the best indicators for the deflation of output of goods and services (either by product as required or by industry).

9.54. Most PPIs are collected as producers' prices, and these will suffice for many industries. Industries paying large amounts of excise duties, however, such as oil, alcohol and tobacco, will require adjustments when, for example, rates have changed. One disadvantage of PPIs is that they are mostly Laspeyres-type indices and might use fixed weighting schemes, generally updated only once every five years. This militates in favour of applying PPIs at the lowest possible level of detail when deflating the domestic supply of products for domestic consumption.

9.55. It is not always possible to observe prices directly because the products of concern are not the same over time, as is the case, for example, with unique goods and services and goods that change rapidly in quality, such as computers, mobile phones, tablets, and so forth.

9.56. If the products in the domestic output part of the supply table which are sold for domestic consumption are deflated using PPIs (or adjusted to form basic price indices), then the same indices need to be used for the corresponding products in the domestic use table on the right-hand side of the H-Approach, thereby ensuring consistency together with balance. The domestic output part of the supply table that is sold for export should be deflated using export price indices for consistency. Export price indices are covered later in this chapter.

9.57. For some products, price indicators other than direct observation may need to be considered. These include tariff indices, model pricing, hedonic price indices, unit value indices, consumer price indices, extrapolation by quantity indicators, input methods, and non-market production.

(i) Tariff indices

9.58. Certain types of services (for example, commercial services and services of general medical practitioners) are paid for on a tariff basis, for example, with a fee per time unit. One problem arising with indices of these types is that, in this approach, no account is taken of changes in the quality of the services provided and of changes in the productivity per time unit. Thus, tariff-based price indices are only appropriate deflators if adjustments can be made for changes of quality and productivity or when it is known for sure that such changes are within acceptable limits.

(ii) Model pricing

9.59. In the model pricing approach, the producers are asked to provide price estimates for typical products. Model price indices are candidate approximate deflators when there are significant changes in product specification from one year to the next and, in particular, in areas where products are unique. An important advantage of these indices is that, since the same product or project is priced, the quality is unchanged. They do have some disadvantages, however, in such areas as rapid product change and the degree of representation of the observed price change for total supply is questionable.

(iii) Hedonic price indices

9.60. Hedonic price indices are candidate deflators when product specification and quality change significantly. This method is based on an assessment of certain measurable characteristics of products. For example, in personal computers, memory and processing speed are two such characteristics. The main advantage is that quality changes are explicitly captured, enabling productivity changes also to be taken into account. A serious drawback is the complexity of the method. Furthermore, the resulting quality adjustment factor seems to be highly dependent on the choice of the characteristics and the choice of the regression model.

(iv) Unit value indices

9.61. Unit value indices of a product may be derived when, for both the current year and the base year, information on value and quantity is available for domestic supply (for example, from production surveys). Dividing the values by the corresponding quantities gives the so-called “unit values”. Under certain conditions, a unit value index can be applied as a price index, for example, when a PPI is not available. Unit value indices often cover a heterogeneous set of products, and, therefore, their usefulness is generally limited when they are applied to homogeneous products where the quality does not change rapidly over time.

9.62. Unit values are quantity weighted as opposed to the regular price indices, which will be time weighted (such as the annual average of monthly indices). In some cases, where quantities do not move smoothly over the year and prices vary considerably, unit values may be the only approach to get the correct relationship between current price value and volume. Thus we may

often choose to rely on unit values for energy products and some agricultural products, with changes in inventories in current prices adjusted accordingly.

(v) Consumer price indices

9.63. For some products, CPIs may be used as approximate deflators for domestic supply. This is only possible, however, in cases where private households buy a considerable part, or all, of the supply of a product, and trade and transport margins and taxes and subsidies play a very small part in the value at purchasers' prices. Special attention must be given to changes in tax rates, in particular such taxes as VAT. As only non-deductible VAT should be taken into account, any modification in the legal entitlements to VAT deduction must be treated in the same way as a modification of the rates of invoiced VAT and therefore as a variation of the price of the tax. This effect would not be fully detected if CPI were used to deflate taxes because only the products of final consumption are considered in CPI, and not, for example, those of intermediate consumption of exempt industries.

9.64. The CPIs are serious candidate deflators for service products mainly provided to private households. An advantage of CPIs is that they take due account of changes in quality. On the other hand, most CPIs are of the Laspeyres type and might use fixed weighting schemes, generally updated only once every five years, and this is a disadvantage. This argues in favour of applying CPIs at the lowest possible level of detail when deflating the domestic supply of products for domestic consumption.

(vi) Extrapolation by quantity indicators

9.65. Although the standard price method can be applied for many goods and services, there are still some transactions for which observation of prices has not been realized or is even not possible. The latter category includes those cases where, owing to the nature of the definition and measurement of output in current prices, direct observation of appropriate prices is not possible: for example, non-market services, FISIM and insurance companies. If price observation is not possible, then the use of quantity information is an alternative.

9.66. As mentioned above, for some products, it is impossible to collect price data and, in order to decompose a value change in a price change and a volume change, quantity indicators must be used. For some industries, mostly the object of government involvement (for example, public transport, medical services and cultural services) or government supervision (for example, banking and insurance), a considerable volume of detailed quantity data are already collected by the national statistical offices or government agencies. Examples may be found in the medical sector (for example, number of inpatients of short-stay hospitals classified by diagnosis-related groups), the cultural sector (for example, the number of visitors attending theatrical performances), and the banking sector (for example, the number of saving accounts, number of credits granted to commercial and private customers, number of payments on bank accounts). Of course, quantity indicators must fulfil the general requirements concerning quality adjustment.

(vii) Input methods

9.67. Input methods use the weighted price or volume changes of intermediate and primary inputs as a proxy for the price or volume change of the output of an industry. The advantage of input methods for deflation within a SUTs framework is that the necessary data are readily available, as all inputs exist in the SUTs in current prices and can be deflated on the use table side in a manner consistent with the supply table side. It should be noted that the input method is recommended for capitalized research and development, although it is not the total production of an industry. On the other hand, a considerable disadvantage is that the price and volume indicators are not directly related to output. As a result, the change in GVA in volume terms, and also productivity changes of an industry, cannot be properly calculated. For that reason, input methods must be avoided as far as possible. Another disadvantage is that input methods can only be applied for the total production of an industry and a separate deflation of the different products of an industry is impossible. It should be noted that, although the output method is recommended by the SNA, it should only be used when the method and the results have been tested carefully for an appropriate number of years.

(viii) Non-market production

9.68. Special attention must be paid to non-market production by general government and NPISHs. By definition, the output of non-market producers in current prices equals the sum of the costs of inputs. As the SUTs accounting rules are valid both in current prices and in volume terms, it can be argued that the output of non-market producers in volume terms equals the sum of inputs in volume terms. This means that, in fact, an input method is applied. This approach, however, places a considerable restraint on the estimation of volume and price indices for the non-market services. Independent estimates of GVA in volume terms and productivity changes are not possible if the input method is used. If non-market services contribute a significant amount in an economy, estimates of the volume growth of such macroeconomic variables as GDP are liable to be biased if input methods are applied. Applying quantity methods can improve the estimates of output and GVA in volume terms. The quantity indicators should preferably be adjusted for changes in quality.

(b) Imports of goods and services

(i) Import price indices

9.69. Import price indices usually meet the general requirements for price indicators, such as valuation, adjustment for quality change and detail. Accordingly, import prices are the best indicators for the deflation of import of goods and services. One disadvantage of most import price indices are that they tend to be Laspeyres-type indices and may use fixed weighting schemes, generally updated only once every five years. This militates in favour of the application of import price indices at the lowest possible level of detail when deflating the imports of goods and services.

One problem here is that import price data covering services are of limited availability, necessitating the search for alternatives.

(ii) Unit value indices

9.70. Foreign trade statistics often provide the value of imports along with the corresponding quantities at a detailed level. Using this information, unit value indices can be derived. One problem arising with unit value indices when used for deflation purposes is that they often cover a heterogeneous set of product groups. Sometimes the unit of measurement is the kilogram or the unit is simply the number of items. This means that, in many cases, unit value indices have heterogeneity problems, limiting the scope for their use as deflators. If, however, no appropriate information from price statistics is available, and the unit values refer to similar mass products for which the quality does not change rapidly over time, they may be applied as useful proxy deflators.

(iii) Other proxies

9.71. Generally, directly observed deflators for services are limited in terms of availability, which means that the deflation of services within SUTs may need to use proxies based on rough assumptions – nonetheless, deflation through SUTs ensures consistency. A good assumption may be that, for every product, the price generating conditions at the domestic market tend to bring about one price, which may hold for both domestic supply and imported services. In terms of the strength of this assumption, the price index of the domestic supply of a service is an acceptable proxy for the price index of the imports of that service. The validity of the assumption depends on whether imports are a large part of the domestic market and on whether imported services are of same quality as domestically produced services.

2. Use table at basic prices

9.72. In the ideal scenario, the use table at basic prices can be derived as the sum of the domestic use table at basic prices and the imports use table at basic prices both in volume terms.

(a) Domestic use table at basic prices

9.73. Assuming a competitive economy, for the deflation of the use table with goods and services from domestic production, PPIs are appropriate price indices, as (in the main, except in such areas as duty-related industries) they are valued at basic prices. In cases where volume indicators are used for the compilation of the volume estimates in the supply table, the residually derived price indices (in other words, the implied price index derived from the difference between the value index and the volume index) can be used. If available, it is better to apply dedicated volume or price indices for specific transactions.

(b) Imports use table at basic prices

9.74. Assuming a competitive economy, for the deflation of the imports use table, import prices are appropriate price indices. In cases where volume indicators are used for the compilation of the volume estimates in the supply table, the residually derived price indices can be used. If available, again, dedicated volume or price indices should preferably be applied for specific transactions.

3. GVA by industry

9.75. Although GVA is not deflated directly, this section covers how GVA is established in volume terms and its constituents.

(a) Double deflation approach

9.76. Total GVA in current prices by industry is compiled as the difference between output and the intermediate consumption of goods and services. For the estimates in volume terms, the same method is applied. As a result, the following condition holds:

$$\begin{array}{lcl} \text{GVA in volume terms} & \text{equals} & \text{deflated output} \\ & \text{less} & \text{deflated intermediate consumption} \end{array}$$

9.77. The corresponding price and volume indices are derived afterwards. This approach is also known as the “double deflation” approach.

9.78. From a theoretical perspective, this approach is superior to the so-called “single deflation” methods, since it takes into account changes in both the composition of outputs and composition of inputs in order to derive GVA as a residual. With a double deflation approach, the volume index of GVA is the result of independent estimates of the volume indices of output and intermediate consumption, and the results are pre-eminently appropriate for productivity analysis. It is important to note that additional quality assurance checks are often needed when establishing plausible volume growth rates, in particular, where intermediate consumption forms a large proportion of output. There are three variations of the double deflation approach.

9.79. *Double deflation:* As described, “double deflation” covers the deflation of current price estimates for output and of intermediate consumption separately using appropriate price indices. The volume estimate of GVA is derived by subtracting the volume of intermediate consumption from the volume of output.

9.80. *Double extrapolation:* In double extrapolation, the previous year values of output and intermediate consumption are extrapolated using appropriate volume indices, and then the volume estimate of GVA is derived by subtracting the volume of intermediate consumption from the volume of output.

9.81. *Extrapolation and deflation*: This is a combination of the extrapolation of output of the previous year by a volume index and the deflation of intermediate consumption of the current year by a price index; then the volume estimate of GVA is derived by subtracting the volume of intermediate consumption from the volume of output.

(b) Compensation of employees

9.82. The compensation of employees is part of total GVA and it is useful to estimate it in volume terms as it increases the range of options for economic analysis using SUTs: for example, the results can be used in the analysis of labour productivity. Another application is in price analysis, where, for example, the price change of the output of an industry is linked to, and explained by, the price changes of the inputs, including compensation of employees.

9.83. The compensation of employees consists of two parts, wages and salaries, both in cash and in kind, and employers' social contributions. The deflation of both parts should be closely connected, since both relate to the same labour input. Thus, both volume indices must be the same, rendering it unnecessary to estimate price and volume indices for both parts separately.

9.84. Since employers' social contributions are liable to complex legislation, it is difficult to observe their price index. This means that, in practice, wages and salaries will be deflated, and the resulting volume index will also be applied in the calculation employers' social contributions in volume terms.

9.85. An important question relates to the appropriate unit of the volume of labour. Many candidate units suffer from heterogeneity; for example, the numbers of employed persons do not give an indication of the number of hours worked per person. Even full-time equivalent jobs are not really an adequate unit since they do not carry information about reductions in working hours and differences in the education level, skill and other attributes of the employees. For this reason, for the purpose of measuring the volume of the input of labour in an industry, the most appropriate quantity unit may be the actual number of hours worked, classified by education level, skill, and other properties. The corresponding price is the value of this unit.

(c) Other taxes and subsidies on production

9.86. The payment of other taxes on production is related to the use of certain inputs in the production process or to socially unwelcome results of production processes. Examples of the former include taxes on real estate property, taxes on motor vehicles owned by producers. Levies on pollution caused by production processes represent an example of the latter.

9.87. Taxes can be based on values (for example, the value of a building) or quantities (for example, tons of pollutants), which means that the deflation of other taxes on production is in principle comparable with the deflation of taxes on products, and that the same formulae are

applicable. In practice, however, the deflation of other taxes on production is more difficult to achieve because of a serious lack of appropriate indicators for price and volume.

9.88. In principle, price or quantity indicators can be used to derive other taxes on production in volume terms. Because of the complexity of the tariff structure of most taxes and the lack of appropriate data, quantity methods will prevail. The use of quantity indicators requires a direct link between them and the tax. For instance, the indicator for the tax on real estate property needs a direct relation to the amount of real estate property owned by producers. A candidate proxy indicator is the volume index of the total stock real estate property. The index of the total tons of emitted pollutants per kind of pollution tax could serve as an appropriate indicator for taxes on pollution. The price indices are derived afterwards from the combination of the value index and the volume index, and they can be applied for the deflation of the tax payments by industry.

9.89. The practical elaboration of the volume estimation of taxes on production presented above can be similarly applied to subsidies on production.

(d) Gross operating surplus

9.90. Gross operating surplus in volume terms is a residual item calculated as GVA minus compensation of employees and minus other taxes on production plus subsidies on production. Direct deflation of gross operating surplus is not possible because no appropriate price or volume indices are available. Furthermore, the economic interpretation of gross operating surplus in volume terms is questionable, and many view it as a meaningless concept.

4. Valuation matrices

(a) Trade margins

9.91. Trade margins are the remuneration for the services mainly provided by the trade industry to producers, consumers and exports in the distribution of goods. Trade margins can also be generated by industries other than the trade industry. As with other services, the appropriate deflation of trade services requires price or volume indicators directly related to the service provided. Alongside the difficulty of defining the services provided by the trade industry precisely, numerous aspects influence the quality of the services of the trade industry, such as the amount of information given to the customers, after-sales services, delivery time, assortment, competence of shop assistants and availability of parking lots. As a result, methods to observe or derive price and volume indices based on direct price and quantity indicators are not available.

9.92. As trade margins in current prices are defined as the difference between the value of goods sold and the value of the same goods purchased for resale by trade industry, double deflation would offer an alternative, and theoretically sound solution. This would comprise the independent deflation of sales and purchases for resale, and the subsequent calculation of trade margins in volume terms as the difference. This approach requires high quality price indices for both

purchases for resale and sales of products by the trade industry (and other industries as appropriate).

9.93. A third option is to apply a proxy for the estimation of the volume index of the trade margin on a product, based on the assumption that the volume change of trade margins equals the volume change of the underlying product flow. An alternative means of formulating this proxy is to take as the percentage of trade margins in volume terms to be applied on the product flow in volume terms, the percentage of the current prices of period $t-1$. The percentages of trade margins are defined as the ratio of trade margins and the relevant product flow valued at basic prices. In this option, the price change is a residual item derived from the current price trade margins and the trade margins in volume terms. This method provides better quality results when applied at a detailed product and industry level. Application of the margin rate in the previous year assumes no change in quality on the margin. Some countries do this by taking the mid-point rate between the two years, but other approaches can also be applied.

9.94. For every entry of the use table, if applicable, the trade margins in volume terms can be estimated as:

$$TR_{t/t-1} = TR_{t-1/t-1} \times VI_{flow}$$

where

$TR_{t/t-1}$ = trade margins of t in prices of $t - 1$

$TR_{t-1/t-1}$ = trade margins of $t - 1$ in prices of $t - 1$

VI_{flow} = volume change of the underlying product

9.95. The underlying assumption is more valid, if the degree to which trade involved in the concerning transactions does not change from one year to another. The position of trade in a market, however, is reflected in what might be termed the “involvement rate”, which can be defined as the ratio between turnover of trade and the relevant product flow which can differ from year to year. These changes influence the estimates in volume terms.

9.96. The relation of the flow and the turnover of trade can be written as:

$$VI_{trt} = F \times VI_{flow}$$

where

VI_{trt} = volume index of turnover trade

F = rate of trade in the product flow

Trade margins in volume terms can be written as:

$$TR_{t/t-1} = TR_{t-1/t-1} \times VI_{trt}$$

9.97. If $F = 1$, then the involvement rate of trade in the product flow has not been changed from period $t - 1$ to period t , and the volume index of the turnover of trade equals the product flow.

9.98. If $F \neq 1$, then the product flow assumption is not valid. In order to refine the estimates, data must be collected on involvement rates by product (and preferably industry).

9.99. A further major improvement can be achieved by collecting a detailed breakdown of trade margins, by type of product, and by type of outlet, assuming that different outlets provide different qualities of services. In this way, the quality changes due to turnover shifts between outlets can be addressed.

(b) Transport margins

9.100. For transport margins, there is more than one way of estimating the volume estimates.

9.101. The first approach is similar to the method for compiling trade margins in volume terms – that is using the rates of the previous year. This implicitly assumes that transport costs are proportional to the value of the product, something which may not be universally true.

9.102. An alternative option for the deflation of transport margins is the use of price indices for the output of transport industries. A necessary condition is the existence of a matrix of transport margins, by type of transport (column) and by type of product (row). By column, the price index of the relevant type of transport can be applied. The resulting volume change of the transport margins can be checked for plausibility with the volume changes resulting from the so-called “margin method”. Generally, it is expected that these two volume changes should be similar.

9.103. A further approach is first to derive volume estimates, by applying the volume changes of the transported products on the previous years’ results, and, second, to inflate these with the appropriate price indices in order to arrive at current price estimates. Consequently, the initial current price estimates will then be overruled.

(c) Taxes on products

9.104. Taxes on products are taxes that are payable per unit of a certain good or service purchased. The tax may be a specific amount of money per unit of quantity of a good or service, or it may be calculated as a specified percentage of the price per unit or value of the goods and services purchased.

9.105. Taxes on products affect the price of a product and not the volume. This means that, for deflation, for a specific product, it is a requirement that the volume index including any taxes on products equals the volume index excluding any taxes on products. As a result, the volume index of the tax must also equal the volume index at basic prices of the product on which the tax is

applied. It should be noted, however, that the volume index of GVA for the whole economy will not necessarily move in line with the volume index of GDP as there is no direct link between the volume of taxes on products (or subsidies on products) and the volume of GVA. The taxes on products are directly linked to the sales of goods and services and therefore relate to the volume of output and not to GVA. The volume change of GVA is not necessarily the same as the volume change of output, because the volume change of intermediate consumption might – or will – be different as a consequence of more efficient production, outsourcing and other effects.

9.106. In the case of taxes on products on “quantities”, for every entry of the use table, if applicable, taxes on products in volume terms can be estimated as:

$$T_{t/t-1} = T_{t-1/t-1} \times QI_{flow}$$

where

$T_{t/t-1}$ =tax on products t in prices of $t - 1$

$T_{t-1/t-1}$ =tax on products $t - 1$ in prices of $t - 1$

9.107. Examples of this application cover excise duties on tobacco, alcoholic drinks and fuel.

9.108. In the case of taxes on products on “values”, for every entry of the use table, if applicable, taxes on products in volume terms can be estimated as:

$$T_{t/t-1} = T_{t-1/t-1} \times VI_{flow}$$

where

$T_{t/t-1}$ =tax on products t in prices of $t - 1$

$T_{t-1/t-1}$ =tax on products $t - 1$ in prices of $t - 1$

9.109. An example of taxes levied on prices is VAT.

9.110. In the case of taxes on products on “value”, the price index of the tax is usually different for different transactions. The reason is that they depend on the price index of the value at basic prices of the transactions. Furthermore, different tariffs exist for different products.

9.111. This approach to the calculation of taxes on products can lead to odd-looking results when a new tax appears or an existing tax disappears, as shown in Box 9.1.

(d) Subsidies on products

9.112. The practical elaboration of the estimation in volume terms of the taxes on products presented above also applies in the same way to subsidies on products – thus the equations in that section also apply to subsidies on products (with the replacement of T by S).

9.113. It must be recognized that the assumption that the volume change of trade (and transport) margins and taxes and subsidies on products equals the volume change of the transactions at purchasers' prices can lead to unacceptable results – the focus is the volume change at basic prices. For products with a rapid increase of quality (for example, computers, mobile phones, and so forth) and also the volume changes of the relevant valuation layers, this change in quality should be included. This may lead to unacceptable growth rates of GVA and labour productivity for specific branches in wholesale and retail, necessitating ad hoc adjustments.

9.114. Similar to taxes, new subsidies can appear or an existing subsidy disappears, and this is covered in Box 9.1.

Box 9.1 Treatment of newly introduced and disappearing taxes and subsidies

Newly introduced and disappearing taxes and subsidies

When using Laspeyres volume indices and Paasche price indices, taxes on products and subsidies on products affect the price of a product and not the volume, implying that the volume index of the value including tax (or subsidy) of a product equals the volume index of the value excluding tax (or subsidy).

Therefore the volume index of the value including tax (or subsidy) also equals the volume index of the tax (or subsidy) value. In the case of newly introduced or disappearing taxes (or subsidies), these conditions give rise to remarkable results. In the example used to demonstrate the impact, trade and transport margins are omitted for convenience. However, these results are in conformity with the registration of changes in taxes on products as a price change.

Newly introduced taxes on products

Applying the guidelines, the volume change at purchasers' prices equals the volume change at basic prices, which means that taxes on products in volume terms equal zero, while the current price amount is not zero, as shown in the table below.

	Year Current prices	Price index	Year t Volume terms	Volume index	Year t-1 Current prices
Output at basic prices	1 000	100	1 000	100	1 000
Taxes on products	100	0	0	0	0
Output at purchasers' prices	1 100	110	1 000	100	1 000

As expected, the introduction of a tax on products results in an increase in the purchasers' prices.

Disappearing taxes on products

Applying the guidelines, the volume change at purchasers' prices equals the volume change at basic prices, which means that taxes on products in volume terms are not zero, while in current prices the amount equals zero, as shown in the table below.

	Year Current prices	Price index	Year t Volume terms	Volume index	Year t-1 Current prices
Output at basic prices	1 000	100	1 000	100	1 000
Taxes on products	0	0	100	100	100
Output at purchasers' prices	1 000	91	1 100	100	1 100

As expected, the disappearance of a tax on products results in a decrease in the purchasers' prices.

5. Use table at purchasers' prices

9.115. The use table at purchasers' prices can be derived from the use table at basic prices and the valuation matrices. In order to ensure consistency in the system, the bridge columns between the supply table at basic prices and the use table at purchasers' prices are derived from the valuation matrices (row totals).

9.116. For the use table at purchasers' prices, alternative options for price and volume estimation are available using indicators appropriate for this valuation. This option entails additional plausibility checks on volumes and prices, in particular for the valuation matrices.

(a) Intermediate consumption by industries

9.117. Intermediate consumption price indices (ICPIs) usually meet the general requirements such as valuation, adjustment for quality change and detail. For that reason, ICPIs are the best indicators for the deflation of intermediate consumption of goods and services. A key problem, however, is that ICPIs are very rarely collected by national statistics offices and, if available, do not cover intermediate consumption of services. Thus, in a sense, the H-Approach removes this problem by deflating at basic prices (or producers' prices) and at a very disaggregated level by product, in which process a single price can be used for both output and intermediate consumption by product, thereby matching the price paid by the purchaser with the price received by the seller.

9.118. In some cases, where ICPIs are not available, CPIs can be used as proxy deflators for intermediate consumption of products. An important requirement is that market conditions for intermediate use and such areas as household final consumption expenditure are comparable. This means, for example, that the share of wholesale and retail margins in the purchasers' price should be the same. One example of goods where intermediate use and household final consumption expenditure often show comparable price changes is fuel for motor-vehicles.

9.119. In a number of cases, the error in the estimation of total GDP due to the use of less appropriate price indices will be limited. When intermediate consumption accounts for the bulk of the turnover of a domestically produced product, the under-estimation of intermediate consumption and, thus, over-estimation of GVA in one industry will be counter-balanced by an under-estimation of output (product, trade or transport margins), thus an under-estimation of GVA in another industry.

(b) Exports of goods and services

9.120. Export price indices usually fulfil the general requirements such as valuation, adjustment for quality change and detail. For that reason, export price indices are the best indicators for the deflation of exports of goods and services. One problem here is that the availability of export price statistics covering services tends to be limited. In addition, they often have the added disadvantage

of being Laspeyres-type indices and use fixed weighting schemes generally updated only once every five years. This militates in favour of their application at the lowest possible level of detail.

9.121. Where deflation by unit value indices is concerned, foreign trade statistics often provide the value of exports along with the corresponding quantities at a detailed level. From this information, unit value indices can be derived. A problem with unit value indices when used for deflation purposes is that they often refer to heterogeneous product groups. The unit of measurement can be kilos or simply the number of products. That means that, in many cases, unit value indices suffer from heterogeneity issues, limiting the possibilities for their use as deflators. If, however, no appropriate information from producer's price statistics is available and the unit values refer to similar mass products where the quality does not change rapidly over time, then they can be applied as useful proxies of deflators.

9.122. Currently, deflation using exports price data and unit value indices is only possible for the export of goods. Direct deflators for services are limited in availability. A general problem consists in the exact observation of the exports by product group according to the classification in the SUTs. The second, and for deflation most important, problem is that the price observation of exported services is not well developed in many countries. For that reason, in the national accounts, there is a tendency for the deflation of the exports of services to resort to proxies based on rough assumptions.

9.123. A simple but rough assumption is that, for every product, the price index for exports equals the price index of domestic production. Another possibility would be to collect information on the price changes of that service in the customer countries (see imports of services).

(c) Household final consumption expenditure

9.124. CPIs usually fulfil the general requirements such as valuation, adjustment for quality change and detail. For this reason, CPIs are the best indicators for the deflation of household final consumption expenditure (for both goods and services). Most CPIs are Laspeyres-type, which militates in favour of the application of CPIs at the lowest possible level of detail. Balancing the SUTs at basic prices is complicated when CPIs are used, owing to the differences in valuation. Thus household final consumption expenditure deflated using CPIs should be used to validate the household final consumption expenditure deflated at basic prices and transformed into household final consumption expenditure at purchasers' prices, which is the right-hand side of the H-Approach.

(d) Government consumption

9.125. Collective and individual government consumption equals government production minus sale of market production by government and own account fixed capital formation. Estimates in volume terms can be derived following the same approach. For social benefits in kind, similar indicators may be used as for consumption of households.

(e) Gross fixed capital formation

9.126. Specific price indices for fixed capital goods usually fulfil the general requirements such as valuation, adjustment for quality change and detail. For that reason, directly collected specific price indices for capital goods form the best indicators for deflation of gross fixed capital formation in goods. A major problem is that price indices for capital goods are hardly ever collected as part of the collection of prices by national statistics offices, and proxy producer price-type indices are used instead. Again, the disadvantages are that price indices for capital goods are often Laspeyres-type and that they use fixed weighting schemes generally updated only once every five years. This militates in favour of the application of price indices for capital goods at the lowest possible level of detail when deflating the domestic supply of products. For gross fixed capital formation, more weight will be given to deflation through the basic price valuation to purchasers' price valuation on the right-hand side of the H-Approach, when compared with the results using proxy price indices for capital goods.

(f) Changes in inventories

9.127. The calculations of changes in inventories in current prices and in volume terms are often closely interlinked. If high quality current price estimates can be made because reliable and appropriate data are available, then it is often possible to make high quality estimates in volume terms as well, since the same data are used.

9.128. In the ideal case, information is available on the exact times and quantities of additions to and withdrawals from the inventory and the price of the product at those times. It is then in principle straightforward to calculate the changes in inventories in current prices and in volume terms. Additions and withdrawals have to be valued at the prices prevailing at the times at which they take place. The changes in inventories in volume terms can be calculated by valuing the quantities of additions and withdrawals at the average prices of the previous year.

9.129. In practice, the data available for the calculation of changes in inventories are not sufficient for a perfect estimation. Assumptions and approximations have to be made. The estimation methodology for changes in inventories (both in current prices and in volume terms) is highly dependent on the kind of information on inventories that is available. In general, enterprises will not provide data on quantities but only on the value of their inventories at the beginning and end of the year according to their own bookkeeping system. This makes it difficult to calculate the current price value from the volume change, as the adjustment for holding gains and losses is incorrect or missing.

9.130. These bookkeeping systems also do not generally value inventories according to SNA rules but follow other systems, such as historic cost systems, the LIFO system, and others. In consequence, these values cannot be used directly in the national accounts. In order to calculate correctly the change in volume of inventories, information is needed on the bookkeeping system

used in the enterprise. The first step is to estimate the change in volume, and the result should then be multiplied with an appropriate price index to arrive at changes in inventories in current prices. As a result, the quality of this process, and the quality of the subsequent estimates, provides some scope for adjusting the current price changes in inventories in the balancing process.

F. Input-output tables in volume terms

9.131. As with SUTs, IOTs can be compiled in volume terms. The reference year can in particular be a problem for IOTs, however, because they are often compiled at irregular, non-annual intervals, such as once every five years. This does not line up with other national accounts data which are required by international standards to make use of the previous year as base year in the calculation of volume measures and the chaining of annual data at prices of a fixed reference year.

9.132. The SUTs in volume terms will generally be compiled at prices of the previous year. This makes possible the calculation of growth rates by comparing the volume measures with the current price values of the previous year. Results of the chaining of complete SUTs will not be additive: that is to say, in the resulting SUTs the elements of a given row will not add up to the row total, and the same will be true for the columns. The resulting tables can only be used to analyse the time path of one particular element at a time. They cannot be used to any real effect for such purposes as the analysis of the time path of the total input structure of an industry or the market shares for a product.

9.133. As for IOTs, if they are only compiled once every five years, it is of course possible to use the previous years' prices as well but the results cannot be used to calculate growth rates, and this clearly limits the usefulness of such tables.

9.134. The alternative would be to compile IOTs in prices of the year five years prior to the current year, for example, the year 2010 in prices of 2005. This could be done by performing the same transformation process as for the current price IOTs, but it would require the availability of coherent SUTs in the same valuation, which is a problem when chaining is used, as already mentioned.

9.135. Another possibility for the derivation of such IOTs is to deflate directly the IOTs in current prices by finding appropriate price or volume indices for the products. These should be indices of the price or volume change in the five years between the base year and the current year. This procedure will entail the use of a different base year than for the SUTs in volume terms, introducing the risk of inconsistencies.

9.136. The recommended approach would be to compile IOTs annually, and to apply the same price and volume methodology as used for SUTs.

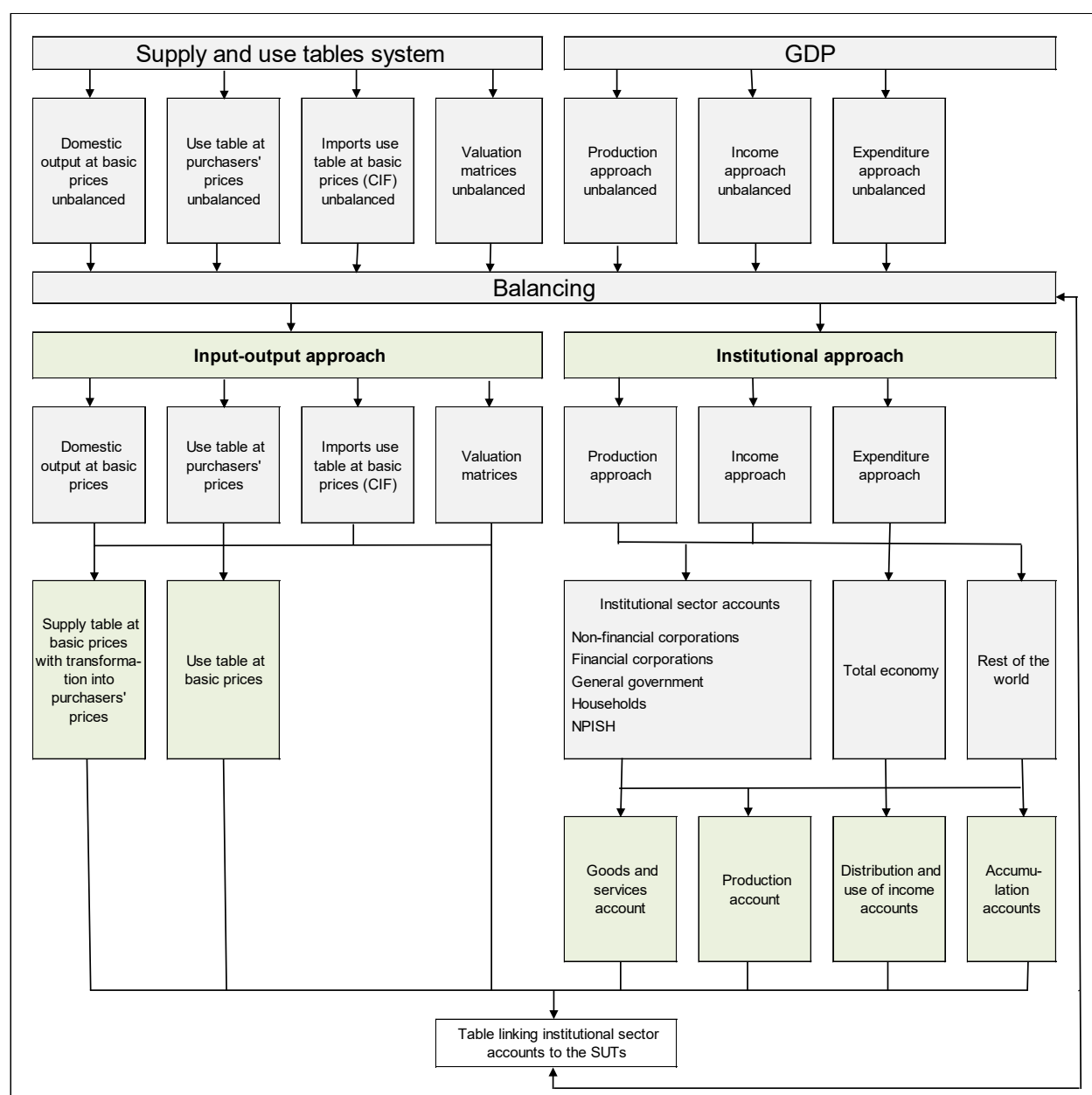
Chapter 10. Linking the supply and use tables to the institutional sector accounts

A. Introduction

10.1. When compiling SUTs, it is important to ensure that they are linked to, and consistent with, the institutional sector accounts of the SNA. This is fundamental to ensuring a full integration of the accounts and also a full integration of the SUTs with the regular annual compilation of the national accounts. In this way data in the SUTs, such as GVA and GDP, are made to be consistent and coherent with the institutional sector accounts, and vice versa. This is achieved through the compilation and balancing process of SUTs incorporating a table that cross-classifies data by industry, by type of factor incomes and by institutional sector.

10.2. Linking SUTs to the institutional sector accounts extends the role of SUTs, enabling them to increase the quality, consistency and coherence of the national accounts, where the SUTs have specific links bringing together parts of the national accounts. Figure 10.1 shows the links between the industry accounts (for example, the SUTs system) and the institutional sector accounts as part of the balancing framework within the national accounts. Figure 10.1 provides a different perspective from that illustrated in figure 1.1 but contains a more detailed diagram linking the two parts of the national accounts framework.

10.3. This chapter describes the links between the SUTs and the institutional sector accounts. Section B provides a description of the institutional sectors in the national accounts and the differences in perspective between the institutional accounts and SUTs. Section C provides a description of how the SUTs are linked to the institutional sector accounts and gives the layout of the table linking these accounts. Section C also provides a numerical example of the linking table and how the SUTs are linked to the goods and services accounts, production accounts and generation of income accounts. Section D presents various approaches to establishing the link between the SUTs and the institutional accounts and describes some issues that may arise in the compilation of the linking table.

Figure 10.1 Links between the industry accounts and the institutional sector accounts

B. Institutional sectors and subsectors

10.4. The institutional sectors of the 2008 SNA group together institutional units on the basis of their principal functions, behaviour and objectives. The following institutional sectors are distinguished in the 2008 SNA:

- *Non-financial corporations*: these are institutional units that are principally engaged in the production of market goods and non-financial services (2008 SNA, para. 4.94).

- *Financial corporations*: these consist of all resident corporations that are principally engaged in providing financial services, including insurance and pension funding services, to other institutional units (2008 SNA, para. 4.98).
- *General government*: this consists of the following groups of resident institutional units: first, all units of central, state or local government; second, all non-market non-profit institutions that are controlled by government units (2008 SNA, para. 4.127). It consists of institutional units that, in addition to fulfilling their political responsibilities and their role of economic regulation, produce services (and possibly goods) for individual or collective consumption mainly on a non-market basis and redistribute income and wealth.
- *Households*: these consist of all resident households (2008 SNA, para. 4.158). Households are institutional units consisting of one individual or a group of individuals. All physical persons in the economy must belong to one and only one household. The principal functions of households are to supply labour, to undertake final consumption and, as entrepreneurs, to produce market goods and non-financial (and possibly financial) services. The entrepreneurial activities of a household consist of unincorporated enterprises that remain within the household except under certain specific conditions.
- *NPISHs*: these consist of all resident non-profit institutions, except those controlled by government, that provide non-market goods or services to households or to the community at large (2008 SNA, para. 4.31).

10.5. Table 10.1 provides a summary of institutional sectors and subsectors with a link to the concepts of market and non-market producers. Often there is a misconception that the distinction between market and non-market producers corresponds to the distinction between the private and public sectors. This is not the case in the national accounts. The public sector includes all resident institutional units controlled directly or indirectly by resident government units. In other words, the public sector consists of all units of the general government sector plus all resident public corporations (2008 SNA, para. 22.164). Accordingly, the public sector includes both market and non-market producers so long as they are controlled directly or indirectly by resident government units.

10.6. It should be noted that the SUTs and the institutional sector accounts reflect different ways of looking at and measuring the economy. In the SUTs, the analysis by products and industries emphasizes the production processes, the flows of goods and services, and the use of primary inputs (for example capital, labour, and others). In this way the units are chosen to reflect technical-economic relations, for example, in units of production such as establishments. As a result, economic activities are studied from the viewpoint of the specific units that carry out the production. The balance between supply (resources) and uses of products constitutes the central element of this type of functional analysis.

10.7. In the institutional approach, the analysis focuses on the generation and distribution of income, and the investment and financing of capital by institutional sectors. In this case, the units that are chosen reflect the general economic behaviour of so-called “institutional units” according to their economic objectives, functions and behaviour.

Table 10.1 Summary of institutional sectors and subsectors

Market / non-market producers	Institutional sectors	Sub-sectors (summary)
Market	Non-financial corporations	
Market	Financial corporations	Central Bank Deposit-taking corporations except the Central Bank Money market funds (MMF) Non-MMF investment funds Other financial intermediaries except insurance corporations and pension funds Financial auxiliaries Captive financial institutions and money lenders Insurance corporations Pension funds
Non-market	General government	Central government State government Local government Social security funds*
Market	Households	
Non-market	Non-profit institutions serving households	
n/a	Rest of the world	

Social security funds may also be grouped at the various government levels.

10.8. Table 10.2 provides a summary of the key features of the SUTs approach and institutional approach. The two types of approaches are linked and should be integrated via the linking table

10.9. Ideally, if a single common unit could meet the needs of both SUTs and the institutional sector accounts, this would further improve the coherence, consistency and compilation of the two areas of the national accounts framework (see also section D on the compilation methods). In the 2008 SNA, however, the recommended unit for the SUTs is the establishment, while for the institutional sector account it is the institutional unit. Information on the legal status and ownership of the establishments is important for the cross-classification of establishments into institutional sectors.

Table 10.2 Main features of the SUT approach and the institutional sector approach

	SUT approach	Institutional sector approach
Objectives	Production relationship Goods and services flows (equilibrium of resources and uses)	To record the economic data of institutional units grouped in terms of their economic objectives, functions and behaviour
Accounts	Goods and services account Production account Generation of income account (integrated in the SUT framework)	A complete system of accounts: goods and services account; production account; distribution and use of income account; accumulation accounts, balance sheets
Types of units: Elementary Aggregates	Production units (establishments, etc.) Industries (type of economic activity)	Institutional units (households, corporations, etc.) Institutional sectors

10.10. As an establishment always belongs to an institutional unit, it is possible to link the production activities of industries and institutional sectors. The output of an institutional unit is equal to the sum of the outputs of the individual establishments of which the institutional unit is composed, thus including deliveries between establishments within the institutional unit.

10.11. To clarify relationships and contents of industries and institutional sectors, the 2008 SNA proposes the cross-classification of GVA and its components (and if possible, also for output and intermediate consumption) by both industry and by institutional sector. This is essentially the GVA part of the use table broken down also by sectors, in addition, to become the table linking SUTs to institutional sectors.

10.12. In order to implement the table linking SUTs and institutional sector accounts, it would be a great advantage to have clear links between units and institutional sectors on the business register, and as a feature of business survey results. The split by institutional sector of units classified by industry would meet the compilation requirements. This would also facilitate similar cross-classifications for output, intermediate consumption and variables such as gross fixed capital formation and the compensation of employees.

C. Table linking SUTs and institutional sector accounts

10.13. Figure 10.2 shows how the SUTs are linked to the sequence of accounts by institutional sector through a linking table. The rows of the linking table contain information by institutional sector on the following:

- Transaction of production accounts: total output and intermediate consumption
- Transactions on the generation of income account: GVA, compensation of employees, other taxes less subsidies on production and imports
- Transaction of the accumulation accounts: gross fixed capital formation

10.14. The linking table thus records complete data of three specific sector accounts of the system as a whole: the production account, the generation of income account and the accumulation

accounts, broken down simultaneously by industries (by column) and by institutional sector (by row). In this way, the interrelations between the systems become clear and their coherence is ensured in both the institutional sector accounts and the SUTs.

10.15. The starting point for linking the SUTs to the institutional sector accounts is the supply table at basic prices, including a transformation at purchasers' prices and the use table at purchasers' prices. Table 10.3 shows a numerical example of the linking table linking the SUTs in table 5.2 of chapter 5 and table 6.1 of chapter 6 to the institutional sector accounts.

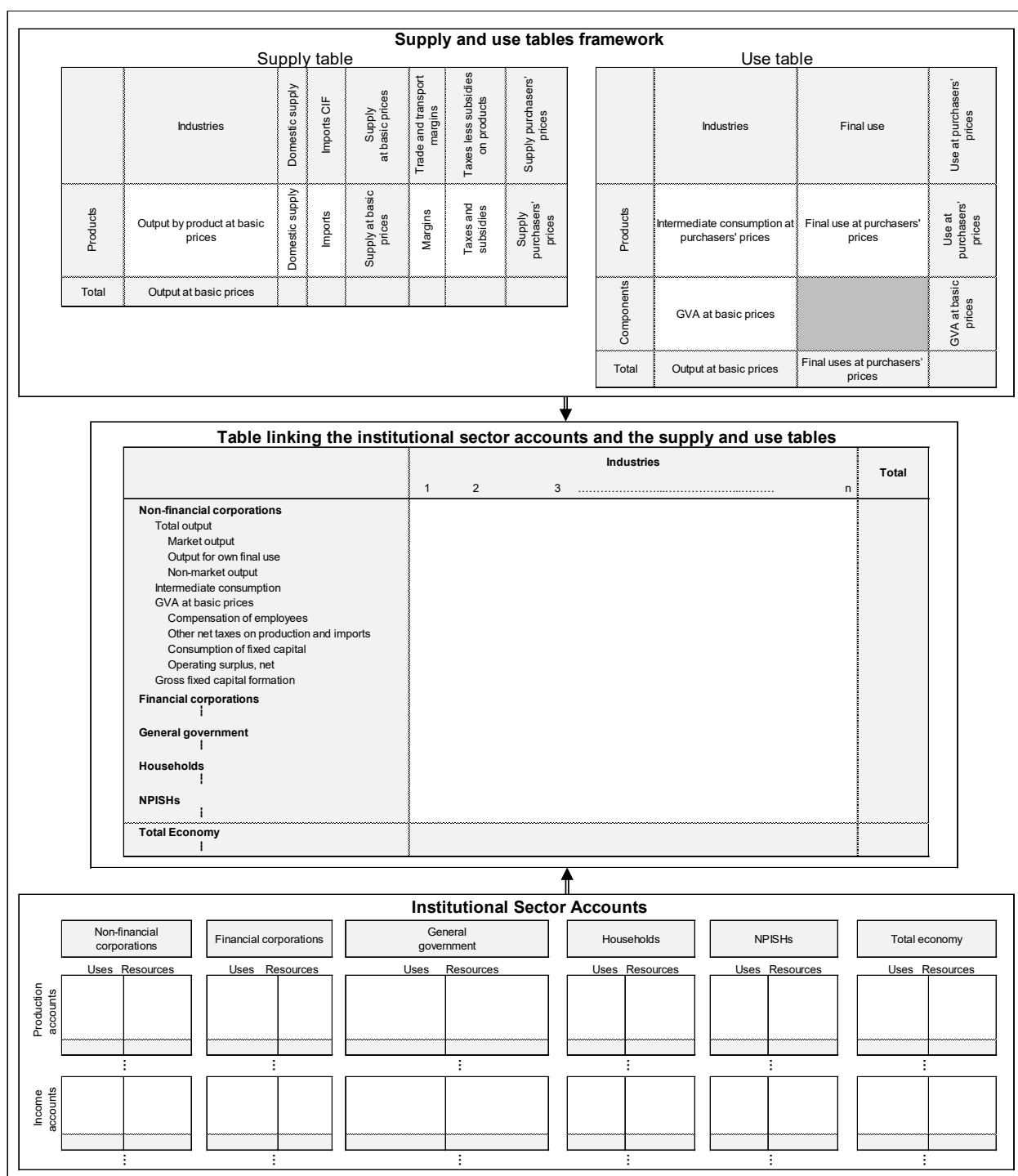
Figure 10.2 Link between the SUTs and institutional sector accounts

Table 10.3 Numerical example showing the table linking the SUTs and institutional sector accounts

Millions of euros

INSTITUTIONAL SECTORS	INDUSTRIES						Total
	Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	
	(1)	(2)	(3)	(4)	(5)	(6)	
1. Non-financial corporations							
Total output	1 837	196 180	37 517	112 204	53 204	8 809	409 751
Market output	1 828	192 193	37 244	110 468	52 633	8 685	403 050
Output for own final use	9	3 988	273	1 736	571	124	6 701
Other non-market output							
Intermediate consumption	972	136 868	24 907	52 394	24 513	3 129	242 784
Gross value added at basic prices	864	59 313	12 611	59 810	28 691	5 679	166 967
Compensation of employees	349	29 839	9 247	32 573	13 008	4 912	89 928
Other net taxes on production and imports	- 133	1 017	479	1 360	955	- 340	3 337
Consumption of fixed capital	165	12 533	1 370	8 598	9 208	526	32 401
Operating surplus, net	482	15 923	1 515	17 279	5 519	582	41 300
Gross fixed capital formation	277	14 376	1 058	10 419	14 605	931	41 665
2. Financial corporations							
Total output					25 058		25 058
Market output					24 802		24 802
Output for own final use					256		256
Other non-market output							
Intermediate consumption					12 351		12 351
Gross value added at basic prices					12 706		12 706
Compensation of employees					8 125		8 125
Other net taxes on production and imports					926		926
Consumption of fixed capital					1 810		1 810
Operating surplus, net					1 846		1 846
Gross fixed capital formation					1 923		1 923
3. General government							
Total output	18	225		5 753	3 045	53 382	62 423
Market output	18	221		151	1 489	578	2 457
Output for own final use				300	314	2 223	2 837
Other non-market output		4		5 302	1 241	50 581	57 129
Intermediate consumption	13	149		2 555	1 366	15 609	19 692
Gross value added at basic prices	5	77		3 198	1 678	37 773	42 731
Compensation of employees	3	12		1 494	782	30 725	33 017
Other net taxes on production and imports	1	1		127	34	1 260	1 423
Consumption of fixed capital	1	20		1 582	625	5 890	8 118
Operating surplus, net	0	44		- 5	237	- 103	173
Gross fixed capital formation	2	11		2 318	1 251	5 661	9 243
4. Households							
Total output	8 012	3 544	7 413	16 880	28 141	9 096	73 086
Market output	7 918	3 503	4 218	16 782	9 391	8 847	50 659
Output for own final use	95	41	3 195	98	18 749	249	22 428
Other non-market output							
Intermediate consumption	4 455	1 975	2 559	6 270	8 307	2 746	26 311
Gross value added at basic prices	3 558	1 569	4 854	10 610	19 834	6 350	46 775
Compensation of employees	198	828	992	3 839	1 081	1 390	8 329
Other net taxes on production and imports	- 1 495	60	67	267	89	- 70	- 1 081
Consumption of fixed capital	1 622	249	183	777	7 264	580	10 675
Operating surplus, net	3 233	432	3 612	5 726	11 399	4 450	28 852
Gross fixed capital formation	2 036	129	101	586	12 114	887	15 853
5. Non-profit institutions serving households							
Total output						8 029	8 029
Market output						7	7
Output for own final use						74	74
Other non-market output						7 948	7 948
Intermediate consumption						2 356	2 356
Gross value added at basic prices						5 672	5 672
Compensation of employees						4 944	4 944
Other net taxes on production and imports						253	253
Consumption of fixed capital						475	475
Operating surplus, net						0	0
Gross fixed capital formation						734	734
6. Total							
Total output	9 867	199 950	44 931	134 837	109 447	79 315	578 347
Market output	9 763	195 916	41 462	127 401	88 315	18 116	480 975
Output for own final use	104	4 029	3 468	2 134	19 890	2 670	32 295
Other non-market output		4		5 302	1 241	58 529	65 077
Intermediate consumption	5 440	138 991	27 466	61 219	46 538	23 841	303 495
Gross value added at basic prices	4 428	60 958	17 465	73 618	62 909	55 475	274 852
Compensation of employees	551	30 679	10 239	37 906	22 997	41 971	144 343
Other net taxes on production and imports	- 1 627	1 077	546	1 755	2 004	1 103	4 858
Consumption of fixed capital	1 788	12 803	1 553	10 958	18 908	7 472	53 480
Operating surplus, net	3 715	16 400	5 128	22 999	19 001	4 929	72 171
Gross fixed capital formation	2 314	14 516	1 160	13 323	29 892	8 212	69 418

Table based on 2011 figures from Austria

10.16. Among the institutional sector accounts as a whole, it is the goods and services accounts, the production accounts and the generation of income accounts that are important for balancing the SUTs and the institutional sector accounts.

10.17. If the goods and services accounts, production accounts, and generation of income accounts are compiled and balanced in an integrated manner as part of the SUTs compilation and balancing process, then all the components of the first three accounts of the national accounts framework become available from the balanced SUTs. This ensures a high degree of consistency and coherence between the SUTs and the institutional sector accounts. In addition, there is a powerful data quality feedback loop from the institutional sector accounts affecting SUTs, and vice versa.

10.18. Furthermore, when the data for gross fixed capital formation is compiled by industry, by product and by institutional sector and as an integrated input to the use table, this also provides a key link between the SUTs and part of the accumulation accounts. Although these could be compiled as a satellite system, they should be integrated as an input to the SUTs process and are available on a consistent basis after the SUTs are balanced.

10.19. Although this type of approach is recommended, in many countries the SUTs are compiled separately from the institutional sector accounts. See also section D on compilation methods.

1. Goods and services accounts

10.20. The goods and services accounts show the total supply of a product for the whole economy, together with how it has been used. The main components for the whole economy balance are:

Output + imports + taxes on products – subsidies on products (Total resources)
equals

Intermediate consumption + final consumption + gross capital formation + exports (Total uses)

10.21. The goods and services are traced through the economy from their original producers (either resident producers or producers abroad) to their users (either resident users or users abroad). With output being valued at basic prices and uses at purchasers' prices, then taxes on products less subsidies on products must be included in the resources part to ensure that a purchasers' prices balance can be struck.

10.22. It is important to note that the goods and services account is by definition in balance and therefore has no balancing item, and all the components are available from the SUTs. In essence, all these are totals of variables available in the SUTs. Table 10.4 shows a numerical example covering goods and services.

Table 10.4 Goods and services for the whole economy

Millions of euros

Uses	Total	Resources	Total
Intermediate consumption	303 492	Output	578 360
Final consumption expenditure	226 258	Imports of goods and services	157 871
Final consumption by households	159 792	Taxes less subsidies on products	33 778
Final consumption by non-profit organisations	5 416	Taxes on products	34 416
Final consumption by government	61 050	Subsidies on products (-)	- 638
GCF	74 612		
GFCF	69 418		
Changes in inventories	2 335		
Acquisition less disposal of valuables	2 859		
Exports of goods and services	165 648		
Total	770 009	Total	770 009

Table based on 2011 figures from Austria

2. Production account

10.23. The production account shows the transactions relating to the production process and is drawn up for institutional sectors and for industries. For the whole economy, and for each institutional sector, the resources include output and the uses include intermediate consumption.

10.24. The production account generates one of the most important balancing items in the system, namely, GVA, the value generated by any unit engaged in production activity and in turn, the link to the major aggregate, GDP. GVA is economically significant for both the institutional sectors and the industries.

10.25. As with balancing items for all the accounts, value added may be calculated before or after allowance is made for consumption of fixed capital, and is therefore available on a gross or net basis. Given that output is valued at basic prices and intermediate consumption at purchasers' prices, GVA will not include taxes on products and will include subsidies on products.

10.26. The production account at the whole economy level includes among resources, in addition to the output of goods and services, the taxes on products less subsidies on products. This enables GDP at market prices to be obtained as a balancing item.

10.27. Again, all the components are available from the SUTs and are totals for the variables available in the SUTs. Table 10.5 shows a numerical example covering the production account for the whole economy. The same variables underpin the whole economy by the institutional sectors, except that GVA is shown instead of GDP as the balancing item on the uses side.

Table 10.5 Production account for the whole economy

Millions of euros			
Uses	Total	Resources	Total
Intermediate consumption	303 492	Output	578 360
		Market output	480 989
		Output for own final use	32 295
		Non-market output	65 075
GDP	308 647	Taxes less subsidies on products	33 778
Consumption of fixed capital	53 469	Taxes on products	34 416
NDP	255 177	Subsidies on products (-)	- 638
Total	612 138	Total	612 138

Table based on 2011 figures from Austria

10.28. All the estimates for the production account for the whole economy in Table 10.5 can be derived from the SUTs in order to derive GDP. Alongside this, GVA by industry can also be linked and similarly derived from the same SUTs, as shown in Table 10.6.

Table 10.6 Link between GDP and industry GVA

Millions of euros						
Industries	Output	Intermediate consumption	Gross value added at basic prices	Taxes on products	Subsidies on products	GDP at market prices
Agriculture	9 867	5 440	4 427			
Manufacturing	199 950	138 991	60 959			
Construction	44 931	27 466	17 465			
Trade, transport and communication	134 837	61 219	73 618			
Finance and business services	109 461	46 538	62 923			
Other services	79 314	23 839	55 475			
Total	578 360	303 492	274 868	34 416	638	308 647

Table based on 2011 figures from Austria

3. Generation of income account

10.29. The generation of income account analyses the extent to which GVA can cover compensation of employees and other taxes less subsidies on production. It measures the gross operating surplus, which is the surplus (or deficit) on production activities before account has been taken of the interest, rents or charges which the production unit must pay on financial assets or on tangible non-produced assets which it has borrowed or rented and must receive on financial assets or on tangible non-produced assets of which it is the owner.

10.30. The gross operating surplus corresponds to the income which the units obtain from their own use of their production facilities. Although the institutional sector accounts have balancing items in each of the accounts, gross operating surplus is the last balancing item in the national accounts framework that can be calculated linking industries, institutional sectors and subsectors.

10.31. In the case of unincorporated enterprises in the household sector, the balancing item of the generation of income account implicitly includes an element corresponding to remuneration for

work carried out by the owners or members of their family which cannot be distinguished from their profits as entrepreneur. This is referred to as “mixed income”.

10.32. In the case of own account production of accommodation services by owner-occupier households, the balancing item of the generation of income account is an operating surplus, and not mixed income.

10.33. The generation of income account can also be presented by industries and is usually published with the main national accounts releases. These can be shown as the industry columns of the use table and presented as sectors, subsectors and industries which are the source, rather than the destination, of primary income.

10.34. It is essential that these industries (and underlying units) are the same as those applied in the SUTs, the industry-by-industry IOTs and the institutional sector accounts. Where this is not possible at best a clear bridge should be provided, to explain the differences.

10.35. All the components could be available from the SUTs and would consist of totals of variables available in the GVA part of the use table if the SUTs incorporated these components as part of the SUTs compilation and balancing process.

10.36. Table 10.7 shows a numerical example of the generation of income account for the whole economy. It is underpinned by a similar breakdown by institutional sector (and by industry), except that GVA is shown instead of GDP as the starting point, in line with the function of GVA as the balancing item of the production account for each institutional sector.

Table 10.7 Generation of income account for the whole economy

Millions of euros

Uses	Total	Resources	Total
Compensation of employees	144 343	GDP	308 647
Wages and salaries			
Employers' social contributions			
Taxes on production and imports			
Taxes on products	34 416		
VAT type taxes			
Taxes and duties on imports excluding VAT			
Taxes on products except VAT and import taxes			
Other taxes on production	4 858		
Subsidies	- 638		
Subsidies on products			
Other subsidies on production			
Gross operating surplus	125 667		
Gross mixed income			
Total	308 647	Total	308 647

Table based on 2011 figures from Austria

10.37. The income approach to measuring GDP is obtained by summing together:

- Gross operating surplus
- Compensation of employees

- Taxes on production and imports less any subsidies on production
- Taxes on products and imports less any subsidies on products

D. Compilation methods

10.38. Using the tables linking the SUTs to the institutional sector accounts, it is possible to make a direct comparison with information from the SUTs and the institutional sector accounts for each period. This at least guarantees that, after the balancing process, consistency is ensured between the SUTs and the sector accounts. Even after independently compiling the SUTs and institutional sector accounts, the linking table may be established to check the consistency of results.

10.39. The compilation procedure is designed in such a way that, in the first stage, the SUTs on the one hand and the institutional sector accounts on the other are independently compiled. In the second stage, the comparison between the two types of information is made in the linking table. In the event of data incompatibilities, a revision process will start on both approaches until a new assessment can be reached.

10.40. There are, however, other possible compilation methods (see Eurostat, 2008) as shown in Box 10.1, used to link the SUTs to the institutional sector accounts.

Box 10.1 Compilation methods used to link SUTs to the institutional sector accounts

General structure of the national accounts compilation procedure	Role of the linking matrix in the compilation process
Method A Independent compilation of SUTs and institutional sector account	Ex post reconciliation of both approaches
Method B Compilation based on instruments and statistical sources of SUTs as a core element with a secondary role assigned to institutional sector accounts	The linking matrix as the first stage in the compilation of institutional sectors accounts
Method C Compilation based on instruments and statistical sources of institutional sectors accounts with a secondary role assigned to SUTs	The linking matrix as the first stage in the compilation of SUTs
Method D Simultaneous compilation of SUTs and institutional sector accounts	The linking matrix as a central instrument in the compilation of the system of national accounts

10.41. In method A, the two approaches are elaborated in an independent way. The particular role of the linking matrix is to contrast them and to facilitate the reconciliation process. Method B and method C represent opposite alternatives in the compilation of national accounts. Method B builds on the production SUTs methods and corresponding sources of information. Method C starts from the institutional sectors from which SUTs elements will be derived. In both alternatives the role of the linking matrix is similar: it represents the missing link between the two approaches. When the SUTs approach is the starting point, the linking matrix gives data for the first two accounts of institutional sectors – the production and generation of income accounts. In the event that institutional sector accounts are the main and initial stage, the linking matrix helps to distribute data over the different industries as the first stage in the compilation of SUTs.

10.42. The recommended approach would be to start simultaneously from an institutional and SUTs standpoint, as indicated in method D. The advantage of this method is that the two different standpoints are wholly compatible from the very beginning of the national accounts compilation process. In terms of the relevance of the linking table, such a method would mean that it would form the core of the whole compilation process.

10.43. The statistical requirements for this method are significant but, by that token, so are the benefits. The main aspect is that the databases should be structured according to the institutional sector with which the units are associated. As indicated in the linking table, five basic types of information are required (broken down by institutional sectors) to prepare SUTs:

- Production data broken down into matrices by products and industries and valued at basic prices, or at least the total for each industry.
- Intermediate consumption data, broken down by products and industries and valued at purchasers' prices, or at least the total for each industry.
- Data on the cost of primary inputs, particularly wage earners' compensation (with a breakdown of wages and salaries and employers' social contributions), and consumption of fixed capital: these data should be disaggregated by industries.
- Data on the gross fixed capital formation and stock variations broken down by types of products and industries. In the case of gross fixed capital formation, the data are valued at purchasers' prices, while in the case of changes in inventories (stock variations) the data are shown at basic prices.
- Data on labour input broken down by the employers' industries and by employment category (wage and salary earners, self-employed) – also defined by the numbers of people employed and hours worked.

10.44. The availability of such a cross-classified database makes possible the simultaneous compilation of SUTs and institutional sector accounts and increases the overall improvement of all requirements to compile the national accounts as follows:

- One of the main advantages of the linking table is that it allows for the possibility of stating and analysing the different types of production (market, non-market, for own final use) which depend on the institutional approach. Definition of the concepts of market output, output for own final use and other non-market output can only be understood by looking, in addition, at features of the institutional unit and the establishment that produce that output. The distinctions are defined in a top-down way: that is to say, the distinction is first defined for institutional units, then for establishment and then for their output.
- The simultaneous compilation of aspects of institutional sectors and industries is a prerequisite for the estimation of taxes of the value-added type. If some details of intermediate consumption and gross fixed capital formation are available in the suggested approach, then it is possible to achieve a more accurate compilation of VAT.
- The linking table allows for a clear identification of non-market household production activities: the (imputed) production of rental services of owner-occupied dwellings; output of household services produced by employing paid staff; own-account construction, and so forth.

10.45. When the compilation of the linking table is at the core of the compilation of national accounts (as in method D in Box 10.1), it is important to cross-classify the original data by industry and sector. In this regard, it is important to keep the link between national accounting, on the one side, and business accounting and public finance, on the other side, as close as possible.

10.46. From the perspective of compiling the linking table, major problems arise from vertically integrated enterprises. A vertically integrated enterprise is one in which different stages of production, which are usually carried out by different enterprises, are carried out in succession by different parts of the same enterprise (2008 SNA, para. 5.23). Business accounting data will be consolidated, without specific detail on the stages and intra-enterprise transactions involved among the different units. This causes difficulties in distinguishing intermediate consumption and other current costs as output of one stage which is, for example, intermediate consumption of another stage. Moreover, gross operating surplus may not be differentiated among the different parts of the enterprise, thus appropriate adjustments would be required; more details of the type of redefinition required may be found in chapter 5.

10.47. The 2008 SNA recommends (see 2008 SNA, para. 5.26) that, when a vertically integrated enterprise spans two or more sections of ISIC, at least one establishment must be distinguished within each section. With such a treatment, the activities of units engaged in vertically integrated activities will not cross section boundaries of ISIC.

Chapter 11. Balancing the supply and use tables

A. Introduction

11.1 The balancing of SUTs is a fundamental step in the compilation process of SUTs. The usefulness of the SUTs is underpinned by the set of identities between elements of the tables which enable the consistent integration of the components of the three approaches to measuring GDP.

11.2 In fact, since the SUTs are populated in the first stage with data derived from many sources, each of which has its own sample and reliability margins, definitions and peculiarities, the basic identities of the SUTs are not satisfied when the tables are first put together (as described in the previous chapters) and the resulting estimates of GDP emerging from the three approaches are likely to be very different, and to differ from year to year. In order to achieve a single, coherent and consistent estimate, all the identities and plausibility relations in the SUTs have to be satisfied, and the initial unbalanced SUTs need therefore to be balanced, preferably using a time series perspective.

11.3 The ideal scenario, linked to the H-Approach to the compilation of SUTs (as shown in figure 9.1), is for the full set of SUTs to be balanced simultaneously at basic prices and at purchasers' prices, as well as in current prices and volume terms. In addition, if the balancing also takes into account the institutional sector accounts, IOTs, PSUTs and EE-IOTs, balanced as a single package or sequentially, the integration and reliability of the system are greatly enhanced.

11.4 This approach, however, is demanding in terms of data, resources and computer systems. In practice, balancing will often be less extensive and sequential procedures will be applied. The sequential theme may, for example, concern volume estimates, valuation matrices or the import matrix. The choice of a variant for the balancing process in practice depends upon criteria such as the availability of data. In the estimation of volume data, the application of appropriate price indices is a key factor to consider.

11.5 Whatever choice is made with regard to the set-up of the balancing, it is important to recognize that any version of the SUTs or a particular stage of the process is not finished until all the subsequent estimates are made and checked for plausibility. The balancing phase is then an iterative process and feedback loops to earlier stages in the process will improve the quality of the final result and can also indicate future improvements to source data.

11.6 The main objective of this chapter is to provide an overview of the balancing of SUTs. Section B provides an overview of the basic identities that need to be satisfied in the SUTs system.

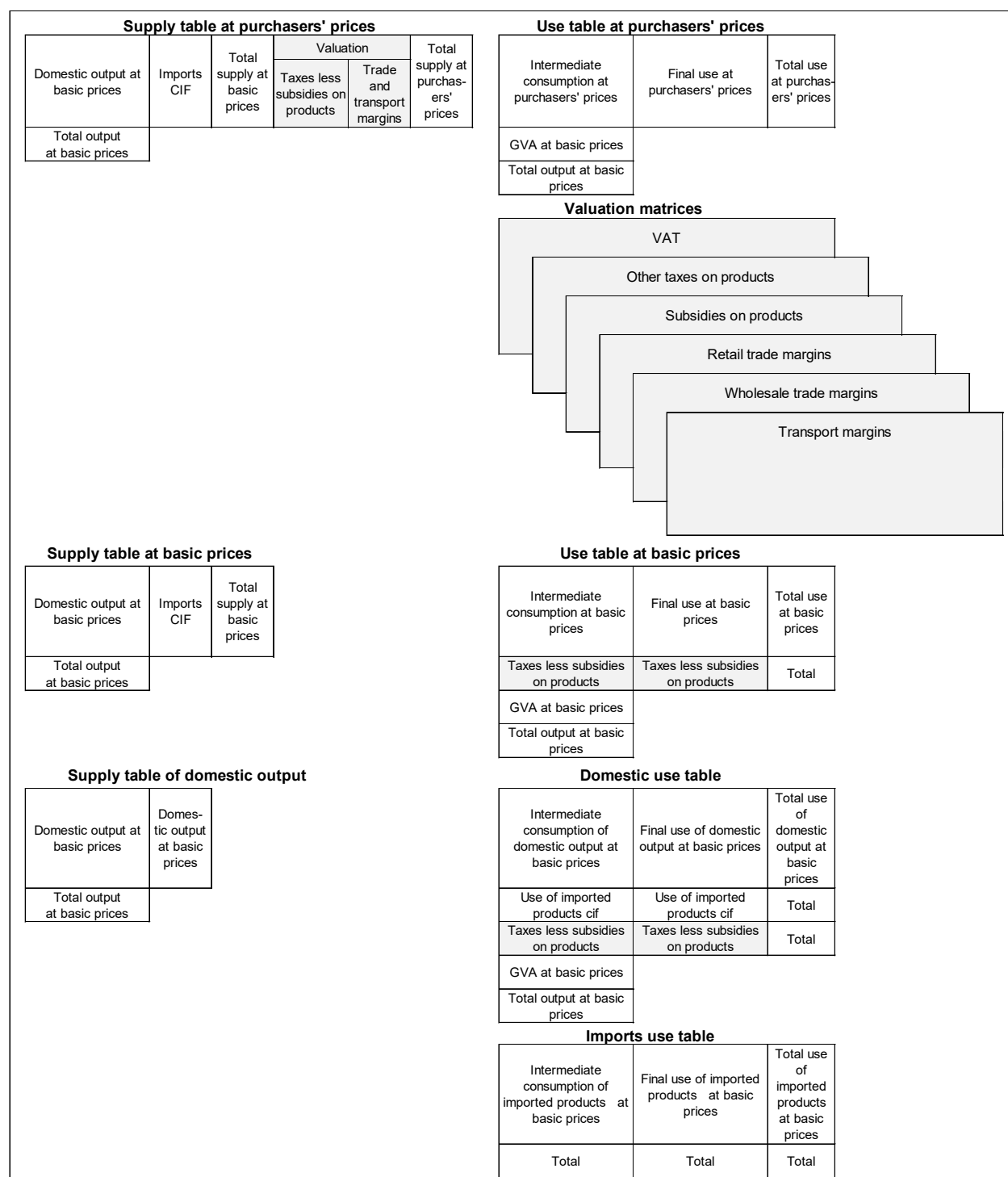
Section C describes different methods of balancing – sequential and simultaneous methods – and a general approach to the investigation and resolution of inconsistencies. Section D describes a step-by-step procedure for simultaneous balancing in current prices and volume terms. Section E describes the benefits of extending the balancing to include the institutional sector accounts, IOTs, PSUTs and EE-IOTs. Lastly, section F provides a list of practical considerations for balancing. These include, for example, the use of automated and manual balancing procedures, the role of balancing in benchmark years and the importance of documenting adjustments to the data. Annex A to this chapter provides a numerical example of how the unbalanced initial SUTs are balanced through a simultaneous balancing process.

B. Overview of the system and basic identities

11.7 The balancing starts with a set of tables which consists of the following (in current prices and previous years' prices):

- SUTs at purchasers' prices
- Valuation matrices
- SUTs at basic prices
- Use table at basic prices with a split between the domestic use table and the imports use table

11.8 The full system of SUTs, as presented in figure 2.2 in chapter 2, therefore consists of SUTs both at purchasers' prices and basic prices and a set of valuation matrices bridging the valuation gap between the supply table and the use table, together with the corresponding dimension covering previous years' prices. The use table at basic prices is also split between a table showing uses of domestically produced goods and services (domestic use table) and a table showing the imports of goods and services (imports use table). Although not shown here, the domestic output table at basic prices needs to be split between that for domestic consumption and that for export for the purposes of deflation, as covered in chapter 9. Figure 11.1 shows the full set of tables and matrices irrespective of the price basis. In addition, the IOTs at basic prices, IOTs of domestic output at basic prices and the input tables of imports also play a key feedback role in terms of quality, coherence and consistency (whether through sequential or simultaneous balancing).

Figure 11.1 Simplified SUTs system

11.9 In the ideal case, all tables and matrices presented in Figure 11.1 are balanced simultaneously both in current prices (top-down in the scheme and the top left-hand side of the H-Approach) and in volume terms (bottom-up in the scheme and the top right-hand side of the H-Approach). By so doing, it is possible to avoid implausible results such as those showing imports

for a product smaller than re-exports (apart from cases when the re-exports have come from inventories) and negative values at basic prices. In addition, the volume and price changes can be judged on plausibility with possible implications for the current price estimates.

11.10 Being an accounting framework, the SUTs have basic identities which are directly linked to the three approaches to measuring GDP. With the inclusion of taxes and subsidies in the SUTs, differences will exist in the identities at the macro (total economy) and the meso (product or industry) level.

1. Basic identities of SUTs

(a) Supply = use

11.11 The “total supply equals total use” identity must be satisfied for the whole economy (macro level) and also for each product (product level). In the first case, at the macro level, this identity has to be satisfied at purchasers’ prices. Total supply at purchasers’ prices consists of domestically produced, and imported, goods and services plus taxes on products less subsidies on products. At the macro level, trade and transport margins do not appear separately in this identity because they are part of the output of goods and services at basic prices. Total use consists of intermediate consumption, final consumption of households and government, gross capital formation and exports, which are all valued at purchasers’ prices.

11.12 At the product level, the “total supply equals total use” identity is defined both at purchasers’ prices and at basic prices. In the first case, the total supply consists of domestically produced and imported goods and services, trade and transport margins plus taxes on products less subsidies on products. The total use consists of intermediate consumption, final consumption, gross capital formation and exports, which are all valued at purchasers’ prices.

11.13 In the case of basic prices, the total supply consists only of domestically produced and imported goods and services. The total use consists of intermediate consumption, final consumption, gross capital formation and exports, which are all valued at basis prices. In the basic price case, the trade and transport margins are treated as ordinary services.

(b) Output = input

11.14 The “total output equals total input” identity is also defined at different levels: for the whole economy (macro level) and by industry (industry level). For the whole economy, the output is at basic prices and the input consists of intermediate consumption at purchasers’ prices and GVA at basic prices.

11.15 At the industry level, this identity is also defined at basic prices. The output is defined at basic prices and the input consists of intermediate consumption at purchasers’ prices and GVA at basic prices. The column totals of the valuation matrices appear as separate rows in the SUTs system at basic prices.

(c) Trade and transport margins “used” = trade and transport margins produced and imported

11.16 In the supply table at purchasers’ prices, the trade and transport margins appear in separate columns. These columns provide the total constraint for the relevant valuation matrices. This means that, for each product, the trade and transport margins in the supply table have to be equal to the sum by columns of the relevant valuation matrices (namely, retail trade, wholesale trade and transport margins).

(d) Value change = volume change x price change

11.17 When the above identities are not satisfied, it is not always easy to discover the causes and, for that reason, it is always helpful to have additional information. For example, incorporating price and volume information helps considerably with identifying and analysing inconsistencies within the SUTs. Preferably, the basic identities referred to above should apply both in current prices and in volume terms. This requirement depends upon the choice of the index formulae. In this case, the combination of the Laspeyres volume index and Paasche price index formula ensures that the identities in this section also hold for the volume terms.

11.18 With the inclusion of volume estimates, the SUTs identities in current prices and in volume terms have to be satisfied, and the less strict relations between variables based on price and volume changes can be judged on plausibility.

11.19 Looking at the industries, it can be seen that the volume change of production is very similar to the volume change of intermediate consumption. This relation is stronger for the output goods and input of raw materials than for services. Nevertheless, when there is a large difference between the two volume changes, this indicates that there may be something wrong in the data and further investigation is advisable.

11.20 When combined with labour data, the volume changes of GVA can be used to calculate changes in labour productivity. It is important to note that the labour data should be calculated on the same basis (for example, using the same statistical unit) as the economic data. This being so, labour productivity is expected to rise gradually every year (except for periods such as the start of a recession). A decrease or a high growth of productivity can also indicate possible mistakes in the data.

11.21 When looking at products in a competitive economy, it is expected that price changes should be more or less the same for all economic agents (except for areas like foreign trade). If the price change of a certain user deviates significantly from the average, this may indicate that something is wrong and further investigation is advisable.

11.22 For household final consumption expenditure, the plausibility of volume changes of products for general use, such as food, can be evaluated comparing them with other indicators such as population growth.

11.23 One optional check would be to view the time series of variables. Sudden breaks in time series can indicate a signal of implausible data in the SUTs. Again, further investigation would be necessary before concluding that adjustments are necessary. For example, the impact of globalization and rapidly changing ways of organizing production processes by enterprises, can lead to justified breaks in time series.

C. Balancing

11.24 Balancing of the SUTs refers to the iterative process of reconciling differences between the different parts of the SUTs. For balancing, there is no general theory or mathematical programme available enabling the entire process to be automated. There is a clear, controlled role for automated balancing techniques but after, and only after, all the significant imbalances have been resolved manually. In balancing, however, it is very important to follow a systematic approach to problem solving involving basic identities, checks on plausibility and credibility, and the investigation of possible causes of inconsistencies. This section reviews the two main approaches to balancing (sequential and simultaneous balancing) and provides a general guide on how to investigate sources of inconsistency.

1. Simultaneous or sequential balancing

11.25 The compilation of SUTs in current prices and in volume terms can be organized in two ways: through a sequential approach whereby the SUTs are balanced first in current prices, subsequently deflated and finally balanced in volume terms; or through a simultaneous approach whereby the SUTs in current prices and in volume terms are balanced at the same time. At the end of the balancing process, the tables in current prices and in volume terms are available and balanced. There are advantages and disadvantages to each approach but, in general, the simultaneous balancing approach is recommended.

11.26 The main advantage of sequential balancing is that it is, in general, less complicated because it is only necessary to deal with values in current prices during balancing and also because there may be a lack of reliable price data at a sufficiently detailed level. The major disadvantage of the sequential approach, however, is that problems encountered while compiling SUTs in volume terms sometimes make it necessary to amend the current price tables that have already been finished, and perhaps even published.

11.27 In general, in the sequential approach, it is preferable to employ an iterative procedure with feedback loops to the SUTs in current prices. Moreover, the SUTs in current prices should not be considered as final until all tables of the SUTs system (including SUTs in volume terms) are checked for coherence and plausibility.

11.28 The main advantage of the simultaneous approach is that it makes it possible to analyse value, price and volume indices in relation to one another. The outcome of the analysis may affect data in volume terms and also current price data. In other words, all three indices must give a plausible picture. This clearly improves the quality of the outcome of the balancing process. It should also be noted that the simultaneous approach can be useful not only in the balancing phase but also in the phase in which basic data are prepared for national accounts purposes. This approach affords an opportunity to check the data by comparing price and volume indices before they are entered in the SUTs system. Simultaneous balancing in current prices and in volume terms may result in a different allocation of adjustments than balancing in current prices only.

11.29 The simultaneous approach requires every transaction of the SUTs to be available, including current prices, deflation detail and prices of the previous year. In order to calculate indices, the system also requires values in current prices of the previous year. For every entry in the SUTs, three values must be available:

- Value for year t in prices of $t-1$
- Value in current prices for year $t-1$
- Value in current prices for year t

11.30 Figure 11.2 illustrates the above configuration, presented in the form of what might be termed a “six-pack”.

Figure 11.2 Six-pack

Description	Data	Description	Data
t at current prices	525	Price index	102.9
t in prices of $t-1$	510	Volume index	102.0
$t-1$ at current prices	500	Value index	105.0

11.31 The six-pack table can be used by compilers of national accounts and SUTs to cross-check consistency of data – analytical tools should ensure that such analyses are readily available to aid validation and balancing. Although the results in current prices may at first sight look plausible, an analysis of the volume and price data can show implausible results and lead to adjustments in the current price data. It is important to check the comparison between changes in the volume of output by industry, its intermediate consumption and GVA. When prices are changing rapidly it is particularly evident that analysis in volume terms is to be preferred, for example, in the oil and chemical industries.

11.32 One major advantage of the simultaneous approach is that it provides an opportunity to analyse value, price and volume indices in relation to one another, and the impact of any

adjustments on all items of the six-pack immediately, in terms of plausibility, including the impact on macro and meso economic aggregates, such as GDP and GVA by industry.

2. Balancing: investigative dimension

11.33 In general, an effective way to approach the balancing process is to investigate the inconsistencies in the SUTs in a systematic manner. The first step would be to select the major inconsistencies. The second step would be to carry out a critical search for results of data processed in compiling the national accounts. With regard to the use table in particular, the main items are the result of partitioning source data into product groups. The allocation may be changed without altering the original aggregates. In practice, it is apparent that not all problems can be solved in this way. The third step would be to consult the expert knowledge of the statistician who is compiling the source statistics. If major inconsistencies still remain, the fourth step would be to contact the reporting company and have a critical discussion regarding the data that it has provided.

11.34 The balancing is driven by two linked underlying themes: the reconciliation of estimates of industry GVA between the income-based and production-based approaches; and the reconciliation of supply and use for each product, essentially through the matching of production and expenditure. As all the components of production, income and expenditure are integrated within a single framework, when the identities are reconciled, the estimates based on the three approaches will be equal.

11.35 It should be noted that these reconciliations must also ensure that consistency and coherence over time is achieved. For example, consistency over time of individual series, both within the SUTs and in suppliers' own detailed series; consistency over time of aggregated series; consistency of estimates in current prices, estimates in volume terms and the implied deflators, both at the aggregate and component level; and consistency in terms of growth rates and levels.

11.36 When assessing these aspects, the impact of revisions to earlier years and the quality of the relative data sources should also be taken into account.

11.37 It should be noted that, during the balancing, the basic identities of SUTs in current prices and, if applicable, in volume terms, must be satisfied and that the values in the SUTs are consistent and plausible, providing a coherent set of price and volume changes. In a set of balanced SUTs, the identities of the framework are satisfied as well as less strict plausibility relationships, such as the volume change of output of goods resembling the volume change of intermediate consumption. Through the process of balancing, the detection of inconsistencies and implausibilities, on the one hand, and the identification of their causes, on the other, forms the most important part of the exercise. With this knowledge, the resolution of any inconsistencies is much more straightforward.

11.38 Any difference between the total supply and total use of any product points to an inconsistency in the system, and forms the start for a balancing procedure in the cause should be sought by:

- Analysing the transformation of source data and the validity of assumptions made (see the various compilation chapters)
- Analysing the underlying source data, if necessary, at the unit level
- Discussing the data with experts in the respective areas or even survey respondents
- Analysing the data in the form of time series
- Carrying out a number of credibility checks, for example, of the following:
 - GVA to total output ratios, while recognizing that activities such as processing require careful consideration
 - Changes in the composition of GVA weights
 - Taxes on products, trade and transport margins as a proportion of the supply and use of products
 - Search for outliers in price and volume ratios (if applicable)
- Comparing data with other data sources (which are not from the statistical office or central bank), for example qualitative and quantitative sources covering specific industries or products such as company reports, regulatory reports, trade association analysis, and others
- Comparing and reconciling inconsistencies between different survey data sources providing different estimates for the same or similar variables (for example, turnover from monthly sources compared with audited annual sources)
- Using other proxy indicators to facilitate the identification of plausible SUTs variables, for example, VAT-based indicators to compare with GVA and turnover
- Analysing related volume ratios for variables such as output and intermediate consumption

11.39 Working with statistical data based on sample surveys and questionnaires, and influenced by, among other things, non-response type issues, involves working with reliability margins (for each cell), so there will inevitably be inconsistencies. The cause would then be a statistical measurement issue. In such a case, balancing could be automated using the reliability margins of the statistics concerned as weights. Some of the methods of automated balancing described later in this chapter are based on this principle.

11.40 Statistics, however, are never ideal and inconsistencies are not only caused by sampling and other such activities, but may be caused by issues that are of a non-statistical nature. It is the sources of these inconsistencies that make manual balancing essential, as a preliminary step towards any form of automated balancing.

3. Examples of causes of data inconsistencies

11.41 There could be several reasons for data inconsistencies and they can arise at various stages of the collection and processing of data. Some of the inconsistencies that are frequently encountered in the compilation of SUTs are presented below.

(a) Inconsistencies in data at the unit level

11.42 For the collection of data on sales and purchases, most statistical units such as enterprises, establishments or kind-of-activity units are defined. These consist of sets of legal units. In the simplest case, the statistical unit is the same as the legal unit but often the statistical unit consists of more than one legal unit. Having a well-defined statistical unit does not necessarily mean that it corresponds, for example, to tax units used by the company concerned for its tax declaration or to the level of consolidation in the bookkeeping. Where the reporting unit uses their bookkeeping or tax records, the reporting unit is not likely to be the same as the statistical unit. This can lead to data missing from certain legal units or even double-counting. This risk increases when data are collected by different agencies, for example, the national statistical offices, national central banks and the tax authorities.

11.43 Another ever-increasing and widespread cause of inconsistencies is the impact of globalization, reflecting issues such as production abroad and the trade flows associated with intellectual property products, together with the impact of any change – or lack of change – of economic ownership. When the unit in a country is the economic owner of all goods and services purchased and sold, it will report its worldwide activity in business statistics, even when the goods concerned never enter the country of residence of the unit. On the other hand, foreign trade statistics on goods are based on goods crossing borders, so data on goods that never enter the country of residence of the unit will be missing. In this case, there is an inconsistency between business statistics and foreign trade statistics, both of which serve as a source for the SUTs system. The Guide to Measuring Global Production (UNECE, 2015) provides extensive detail on how to handle these types of issues.

11.44 Examples of other causes of inconsistencies at the unit level are mismatches and mistakes. One example of a mismatch is the difference between the calendar year and the bookkeeping year, where for a significant number of units the bookkeeping year differs from the calendar year used in the national accounts (and other annual statistics). Entering the bookkeeping data in the questionnaire causes inconsistencies in the SUTs when these data are confronted with other statistics.

11.45 The survey questionnaires for business statistics are designed in such a way that data covering different branches can be compared and added together. The needs of users like national accounts require specific definitions of variables in the survey questionnaires, which cannot always be derived directly from bookkeeping records. When respondents use their own definitions of variables, this may also cause inconsistencies in the SUTs.

11.46 When detailed information on such variables as output and intermediate consumption is sought via survey questionnaires, it is possible that respondents may allocate products to the wrong CPC product code, with the result that the contents of product codes in the SUTs are no longer comparable.

11.47 Last but not least, a business can provide incomplete data. If, for example, data on changes in inventories are lacking, the transformation from either sales to output or purchases to intermediate consumption cannot be made. This will in turn also affect GVA and GDP.

(b) Inconsistencies in processing survey data

11.48 The processing of collected micro data into subject matter statistics can create inconsistencies. Although procedures for grossing up are routine, the target population is less straightforward. An important issue in this regard is linked to the updating of the business register and the consequence of correctly identifying active or non-active units during the reporting period. A further related issue is the detection and treatment of outliers.

11.49 Small enterprises are often given less detailed survey questionnaires, making it necessary to break down the aggregated variables to the level of detail required of large enterprises. The assumptions made for this calculation may be incorrect. The same holds for the breakdown of variables from business statistics to the product classification used in the SUTs. For the compilation of valuation matrices, the trade and transport margins and taxes and subsidies on production must be allocated to the various users (industries and final consumption categories). If scant detailed information is available, then a range of assumptions is applied, and these may also lead to inconsistencies, in particular in the SUTs at basic prices.

11.50 Another potential cause of inconsistencies is the coverage of the hidden and informal economy. When no estimates for the hidden and informal economy are included in the SUTs, or where those estimates are insufficient, then inconsistencies will arise. When, for example, consumers buy a beer in a bar, they usually do not know whether it is, economically speaking, an “illegal” (for example, smuggled) or a “legal” beer, meaning that it is reported in household consumption, while the “illegal” beer will not be recorded in business statistics.

(c) Inconsistencies in volume data

11.51 Deflating SUTs data can itself generate inconsistencies in the SUTs in volume terms. As most price index numbers based on observation are Laspeyres indices, inconsistencies result when work is carried out at a level of aggregation above the observation of the price data. The observed price data often do not keep account of discounts, bulk purchases and negotiated prices (in particular in business-to-business sales or business-to-households sales), with the result that they do not always match the actual value of the transactions. The impact is due less to the different prices paid than to the changing weights on price changes. In addition, if the discount bulk purchases and other such items are stable and form a constant share over time, there is little impact.

In the process of balancing the full SUTs system, the implicit price indices resulting from the SUTs system must be reconciled with the observed indices, such as the CPIs and PPIs by specific products.

4. Reliability of data in the unbalanced SUTs

11.52 One important and very useful step to be taken before starting the balancing process is the assessment of the reliability and quality of the data in the unbalanced SUTs. In general, less reliable data will, and should, be adjusted to a relatively higher degree. It should, however, be borne in mind that even weaker estimates cannot endlessly absorb inconsistencies; for example, positive changes in inventories for a product for a large number of consecutive years are implausible or generate implausible ratios or movements in ratios, for GVA as a proportion of output or trade margin as a proportion of domestic output at basic prices.

11.53 The quality of the estimates will influence the role that the variable will play in the balancing process of the SUTs. Some variables are predetermined when entered directly into the system and kept at their original value throughout the entire balancing process; for example, data on taxes and subsidies which are directly derived from government administrative data sources and data derived from exhaustive sources (such as regulatory sources).

11.54 A perfect sample with a 100 per cent response rate can still generate inconsistencies. Although such source statistics can be judged as very reliable, they may still be adjusted in the balancing process. Estimates using models, for example fixed input structures based on the previous period, expert guesses, use of data for the previous period, and others, are likely to be adjusted earlier in the balancing process.

11.55 A ranking of the reliability of estimates for entries and aggregates in the SUTs should always be borne in mind, in particular in the manual balancing phase. This ranking information is an essential input for any automated balancing procedures and is covered later in this chapter.

11.56 Box 11.1 and Box 11.2 provide two examples illustrating the simultaneous balancing process. With each example it is very important to have details on the reliability of the data before starting to look for a solution or implementing any adjustments.

11.57 Box 11.1 illustrates a situation where the discrepancies are balanced in current prices and in volume terms. The value, price and volume analysis can lead to adjustment of any of the estimated variables. Sometimes the results can be checked using observed quantity data, which are also available for the supply and use of energy products. Another possible check in the simultaneous approach is the ratio of the volume of GVA to the input of labour.

11.58 The example in Box 11.2 shows that the comparison of volume indices of the main supplier and the main user offers one solution for a balancing problem.

Box 11.1 Example of discrepancies balanced in current prices and in volume terms

The price and volume changes of domestic production and exports can be compared in the simplified example below (there is need to accept possible inconsistencies between the price indices from the supply and use sides in the use of a simplified example). This example, for demonstration purposes only, excludes margins, taxes, subsidies and imports. The supply minus use shows the discrepancies between supply (domestic production) and use (exports and by other users) in current prices and in volume terms.

	Supply minus use	Domestic production	Exports	Other uses		Domestic production	Exports	Other uses
Value t in current prices	-10	525	420	115	Price index	102.9	100.0	103.6
Value t in prices of $t-1$	-21	510	420	111	Volume index	102.0	105.0	111.0
Value $t-1$ in current prices	0	500	400	100	Value index	105.0	105.0	115.0

In this example, there is a discrepancy both in current prices and in volume terms. The first step is to get an idea about the reliability of the data. In this case, data on both domestic production and exports in current prices are considered to be very reliable. Thus, a sensible solution would be to adjust “other uses”. If the price index (102.9) is considered to be correct, the adjustment should be made both in current prices and in volume terms. This results with the following situation.

	Supply minus use	Domestic production	Exports	Other uses		Domestic production	Exports	Other uses
Value t in current prices	0	525	420	105	Price index	102.9	100.0	102.9
Value t in prices of $t-1$	-12	510	420	102	Volume index	102.0	105.0	102.0
Value $t-1$ in current prices	0	500	400	100	Value index	105.0	105.0	105.0

The discrepancy in current prices has been eliminated but, in volume terms, a discrepancy remains. Assuming that the figures for the price of domestic production are reliable, and that the difference between the volume index of domestic production and exports should not be too large, then the balancing results in an adjustment of the price of the export and a minor adjustment of other uses.

	Supply minus use	Domestic production	Exports	Other uses		Domestic production	Exports	Other uses
Value t in current prices	0	525	420	105	Price index	102.9	102.7	104.0
Value t in prices of $t-1$	0	510	409	101	Volume index	102.0	102.3	101.0
Value $t-1$ in current prices	0	500	400	100	Value index	105.0	105.0	105.0

Box 11.2 Example of simultaneous balancing comparing volume indices

Large discrepancies between volume changes of the main user of important raw materials and volume changes of the main supplier (for instance imports) are an indication for inconsistent data.

	Supply minus use	Domestic production	Imports	Main user	Other uses		Domestic production	Imports	Main user	Other uses
Value t in current prices	0	50	468	426	92	Price index	100.0	104.0	100.0	100.0
Value t in prices of $t-1$	-18	50	450	426	92	Volume index	100.0	100.0	103.9	102.2
Value $t-1$ in current prices	0	50	450	410	90	Value index	100.0	104.0	103.9	102.2

In this example, no discrepancy between supply and use in current prices is assumed. The value indices of imports and the main user are both plausible: 104.0 and 103.9 respectively. Analysis reveals, however, that volume indices of imports and the main user differ: 100.0 versus 103.9, which is not plausible. Further analysis is necessary to find the solution for this balancing problem. It is not inconceivable that the value in current prices also needs to be adjusted.

5. Documentation

11.59 Many decisions leading to corrections, adjustments and subjective estimates are entered by the balancers, and these may provoke a struggle when referred to other statistical and available sources or when common sense considerations are taken into account. Thus, it is important that the considerations and rationale behind the solutions implemented are visible to other balancers, and the solutions are sustainable and can be reproduced if the same problems are encountered in subsequent years. Such corrections should be recorded in a systematic way.

11.60 It is also important to record separately the steps and links between the source data through to the balanced data so that they can be reviewed in subsequent balancing exercises to investigate source data incoherence, bias and other such factors (Mahajan and Penneck, 1999). For example:

National accounts source data (covering business survey data, household survey data, census data, administrative based data, extrapolations and models (for example, PIM, FISIM), company accounts based data, etc.)

plus	coverage (including exhaustiveness) adjustments
plus	conceptual adjustments
plus	quality (data validation) adjustments
plus	balancing and coherence adjustments
equals	national accounts final estimates on 2008 SNA basis

11.61 Balancing adjustments can, and should, be part of a process table describing the steps taken from the source statistics to the final estimates in the balanced SUTs. If the balancing adjustments are recorded in a systematic manner, they can point to flaws in source statistics or even a bias in the balancing process itself. Again, the feedback loop can be powerful in that suppliers of source data can improve survey questionnaires, data collection, data processing, and other processes, thus cumulatively improving the quality of the national accounts estimates.

D. Step-by-step procedure for simultaneous balancing

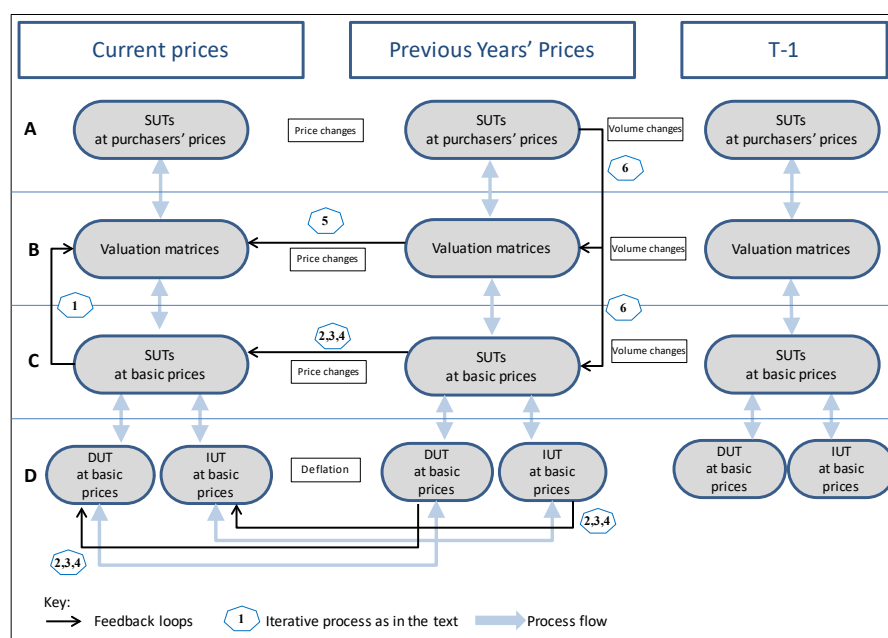
11.62 This section sets out a step-by-step process for the simultaneous balancing of SUTs in current prices and in volume terms. The process presented below relies on a sequence of tables which starts from the SUTs at purchasers' prices, valuation matrices, SUTs at basic prices, and the domestic use table and imports use table at basic prices. An alternative sequence could be to split the SUTs at purchasers' prices into the domestic use table and imports use table at purchasers' prices. Nevertheless the latter is not a commonly used sequencing of tables and the first is the recommended approach to the compilation of SUTs.

11.63 Figure 11.3 provides a flowchart showing how to carry out the balancing of SUTs indicating what types of balancing are performed at each step. Note that balancing is an iterative

process, so the figures shows a number of feedback loops that need to be carried out in order to arrive at a final set of balanced SUTs in current prices and volume terms.

11.64 At the start of the balancing process, an estimate is available for all entries in the full SUTs system both in current prices and in volume terms. In combination with balanced and fixed data of the previous year, volume changes can then be compiled. The balancing effort starts by checking all the inconsistencies and implausible estimates in the system. This is summarized in the sequence of steps below.

Figure 11.3 Overview of the SUTs balancing framework for simultaneous balancing



(a) Differences between supply and use of products at purchasers' prices in current prices

11.65 These types of checks are represented in current prices in part A of Figure 11.3. Differences between the supply and use of products at purchasers' prices in current prices point towards inconsistencies, possibly caused by data processing with national accounts (for example, transformation to national accounts definitions and requirements) or by inconsistencies in observed data (for example, as a result of the impact of globalization).

(b) Unwanted negative entries in the SUTs at basic prices: parts B and C in current prices

11.66 These types of checks are represented in current prices in parts B and C of Figure 11.3. The unwanted negative entries in the SUTs at basic prices can be caused by mistakes in the calculation of the valuation matrices, which would lead to a recalculation of these matrices. This is a process that should be continued until all unwanted negatives are eliminated and the valuation matrices look plausible. There are cases, however, where negative entries are plausible in areas such as

changes in inventories or exports of goods (for example, merchanting). This step forms the first key iterative process of balancing the SUTs system (heptagon 1 in the figure).

(c) Differences between supply and use of products at basic prices in current prices

11.67 These checks are represented in current prices in part C of Figure 11.3. These differences will point towards inconsistencies of the type covered in subsection (a) above, due to issues such as data processing inconsistencies in observed data.

(d) Inconsistencies between domestic supply and use of products from domestic origin and use of imported products in current prices

11.68 These checks are represented in current prices in part D of Figure 11.3. The use table is initially split into the domestic use table and imports use table independently of the supply table. Thus there may be inconsistencies at the product level such as, for example, the value of exports being larger than the value of domestic supply or the value of re-exports larger than the value of imports. In these cases, values need to be adjusted. The breakdown of the supply of domestic and imports in the supply table can also be used to inform the split between the domestic use table and imports use table.

(e) Differences between the supply and use of products at basic prices in previous years' prices

11.69 These checks are represented in previous years' prices in parts C and D of Figure 11.3. When starting with balanced SUTs in current prices, the differences between supply and use of products in volume terms point towards inconsistencies in the applied price indices; for example, failure to deflate domestic output and exports separately or failure to use an appropriate weighted average. Furthermore, there may be weaknesses in the PPIs regarding details on discounts, bulk purchases and negotiated prices which can also cause inconsistencies. In addition, the use of a Laspeyres-type index can also play a role in generating inconsistencies.

11.70 These checks can also reveal errors in the SUTs in current prices. In this case, the SUTs in current prices need to be rebalanced. This forms the second key iterative process in balancing the full SUTs system (heptagon 2 in the figure).

(f) Plausibility of volume changes of output and intermediate consumption

11.71 These checks are represented in current prices in part C of Figure 11.3, comparing the SUTs in previous years' prices and the SUTs for period $t-1$. When combined with previous years data, the deflated SUTs at basic prices in previous years' prices provide a framework for judging the volume changes of output, intermediate consumption and GVA at the industry level. Implausible results will need adjustment of the estimates in volume terms, and if necessary, the

current price estimates in the SUTs. This forms the third iterative process in balancing the SUTs system (heptagon 3 in the figure).

(g) Plausibility of changes in labour productivity

11.72 These checks are represented in part C comparing the SUTs in previous years' prices and the SUTs for period t-1 in current prices in Figure 11.3. As with the changes in volume of output and intermediate consumption, the changes in labour productivity can be used to assess the plausibility of the resulting GVA in volume terms at both the macro level and the industry level. Implausible results will require adjustment of the estimates in volume terms, and if necessary, the estimates in current prices in the SUTs. This forms the fourth iterative process in balancing the SUTs system (heptagon 4 in the figure).

(h) *Confrontation of implicit price indices of valuation matrices and observed PPIs and changes in tariffs*

11.73 These checks are represented in part (B) in Figure 11.3, confronting the valuation matrices for period t in current and previous years' prices. The volumes for the valuation layers are calculated by applying the rates of the previous year to the estimates in volume terms. Therefore, for all entries of the valuation matrices, implicit prices can be compiled. If available, observed producer prices indices can be compared with these implicit prices indices. It is likely that there will be possibilities for a data confrontation for specific areas, for example, transport services. For taxes and subsidies linked to the value of the transaction concerned (for example, VAT), the changes in tariffs can be used to assess the plausibility of the implicit prices. Implausible results will need adjustment of the estimates in volume terms, and if necessary, the current price estimates in the SUTs. This forms the fifth iterative process in balancing the SUTs system (heptagon 5 in the figure).

(i) Confrontation of implicit purchasers' price indices resulting from calculation and observed purchasers' price indices such as the CPIs

11.74 These checks are represented in in current prices and in previous years' prices in part A of Figure 11.3, confronting the SUTs for period t. The SUTs in volume terms at purchasers' prices are compiled as the sum of the SUTs at basic prices and the valuation matrices (all in volume terms). When the SUTs at basic prices are balanced, the SUTs at purchasers' prices are by definition also balanced. At this point in the balancing process, a confrontation of observed purchasers' price-type indices such as the CPI and the calculated implicit purchasers' price indices may show that the latter may be implausible, while accepting that they should not be the same. If there are significant differences, then the estimates in volume terms of all underlying component tables (the valuation matrices, and the SUTs at basic prices) may need to be reconsidered. This forms the sixth iterative process in balancing the SUTs system (heptagon 6 in the figure).

(j) Overall assessment of the second order effects of balancing steps (a)–(i)

11.75 Through the balancing procedure, the trade and transport margins are very likely to be adjusted as a result of manual and automated corrections. As a consequence, the total use of trade and transport margins will probably not equal the total supply, even if they were in balance in the initial version of the system, and will need to be constrained.

11.76 Similarly, VAT should be recalculated based on the adjusted results in the use table. The total of non-deductible VAT which is the result of the balancing procedure cannot be expected to be an exact match for the VAT receipts based on government accounts. If official rates and tax legislation alone are used in the calculations, the computed VAT total will normally exceed the target; this is closer to the concept of theoretical VAT as opposed to cash-collected VAT (on an accrued basis). To be realistic, however, the model used to estimate VAT should take into account the expected patterns of tax evasion by keeping account of various issues like the hidden economy. Nevertheless, the total estimated VAT will not necessarily equal the government data, so final corrections will be needed. It may be preferable to adjust VAT proportionally in specific columns, where the exact share of VAT liable is uncertain. A final proportional adjustment of VAT on many products, most likely to be household final consumption expenditure, can be used to eliminate the remaining difference.

11.77 One important final check is to ensure that the resulting effective (and implied) tax rates do not exceed the legal rates, for example, the standard rate of VAT.

E. Alternative balancing methods

11.78 The ideal balancing scenario covered in the previous section based on the H-Approach consists in simultaneously balancing SUTs at basic prices and at purchasers' prices both in current prices and in volume terms. This balancing is data demanding and the choice between simultaneous balancing and any other variation of balancing methods depends heavily upon the availability of data, human resources and information technology systems. If the ideal scenario is not possible, then alternatives can be considered such as, for example, balancing the SUTs at purchasers' prices and balancing at basic prices or prioritizing them or launching an iterative process with feedback loops.

11.79 The choice made among the alternative scenarios will have various consequences, in particular for the use of price indicators. In the ideal scenario, the price indicators optimally match the SUTs being deflated in terms of underlying flows and valuations. Diversion from the ideal scenario will require additional compilation, assumptions and approximations in the use of price indicators.

1. Balancing SUTs at basic prices

11.80 Assuming that the balancing process is not ended before all components of the full SUTs are checked for plausibility, the balancing at basic prices only provides a close approximation to the ideal scenario.

11.81 Balancing at basic prices requires stripping out the trade margins, transport margins, taxes on products and subsidies on products from the initial current price use table at purchasers' prices. The deflation of the SUTs then takes place at basic prices, with the application of PPIs and import prices for the supply table and a weighted average of those indicators for the use table. Weights could be derived from the domestic use table and imports use table of the previous year. When the SUTs are balanced both in current prices and in volume terms, the volume changes of the valuation matrices can be compiled, applying the volume changes of the corresponding entries of the use table.

11.82 Subsequently, the SUTs at purchasers' prices including non-deductible VAT, can be derived both in current prices and in volume terms. The resulting price indices can be checked for plausibility with observed indices on household consumption (for example, the CPIs) and export price indices as in the ideal scenario. The price indices resulting from the sequentially compiled domestic use table and imports use table can also be checked for plausibility with the observed PPIs and import price indices.

2. Balancing SUTs at purchasers' prices

11.83 Balancing at purchasers' prices requires a very different approach for a number of entries in the SUTs, and in general, more approximations and assumptions because of the lack of appropriate price indices, in particular for those cases where trade and transport margins play a substantial role. As a first step, non-deductible VAT may have to be stripped out from the initial use table, which includes VAT (this step may often be carried out in the preprocessing of source data for the SUTs).

11.84 As the supply table is valued at basic prices, the deflation for this part will be similar to the ideal scenario applying PPIs, import price indices or other appropriate indicators.

11.85 The compilation of the volumes of trade margins, transport margins, taxes on products and subsidies on products using this balancing approach will be performed at an aggregate level. If applicable, at this stage, only the total trade margins, transport margins, taxes on products and subsidies on products for each product are included in the system as part of the bridge columns between the supply table and the use table. In order to compile the volume changes, the volume change of the underlying aggregated flows must be determined. For an accurate estimate of the volume index of the valuation layers, it is therefore very important to determine which part of the supply or use of a product is liable to this valuation layer. For example, the retail trade margins are mainly linked to household final consumption expenditure.

11.86 For deflation of the use table at purchasers' prices (excluding the deflation of VAT), price indicators are required other than those applied in the ideal scenario in particular for intermediate consumption of goods and services and gross fixed capital formation, where the ideal price indices are often not available and must be replaced by proxies. More details may be found in chapter 9.

F. Extending the balancing of SUTs to include institutional sector accounts, IOTs, PSUTs and EE-IOTs

11.87 The previous section describes a process for a simultaneous balancing of SUTs at basic prices and purchasers' prices both in current prices and in volume terms. It therefore focuses on balancing within a SUTs system, even though the compilation of SUTs is not seen here as a separate and isolated exercise from the compilation of the national accounts and from the compilation of IOTs, PSUTs, EE-IOTs or other satellite accounts that may also be compiled. This implies that the balancing process has to be extended to ensure a coherent and consistent integration of SUTs with the national accounts (namely, institutional sector accounts) and related products (for example, IOTs, PSUTs and EE-IOTs).

11.88 The balancing of SUTs can be extended to include additional accounts either in a simultaneous or sequential manner. There are clear benefits in this extended balancing due to the additional feedback loops which would eventually lead to further improvements in the quality of the SUTs, and also the other products in terms of consistency, coherence and integration. Thus, in general, it is recommended that the balancing be extended to include the features outlined in the following subsections.

1. Institutional sector accounts

11.89 Together with the SUTs, the institutional sector accounts lie at the core of the national accounts. The sector accounts provide an overview of the various economic activities covering production, consumption, generation of income and distribution of income, accumulation of wealth and relations with the rest of the world. The SUTs and institutional sector accounts thus have several variables in common such as output, intermediate consumption, GVA and its components linked by industry and by institutional sector. Analysing and balancing SUTs and the institutional sector accounts can point to implausible data in SUTs, suggesting the need for a rebalancing of the SUTs, for example highlighting classification issues or showing where cells should have zero or non-zero entries.

11.90 Like the SUTs system, the institutional sector accounts constitute a balancing framework consisting of a set of well-defined variables and a number of basic identities. For the total economy, the production account and the generation of income account are in fact an aggregate of the domestic production part of the SUTs without the dimensions products and industries. One-to-one links exist for production, intermediate consumption and GVA. In addition, compensation of employees is directly linked to the SUTs system. Other macroeconomic variables with a strong link to SUTs and institutional sector accounts include household and government consumption

(use of disposable income account), and fixed capital formation (capital account). Lastly, taxes and subsidies on products and other taxes on production appear in both systems.

11.91 From a conceptual point of view, the links between SUTs and institutional sector accounts are strong. In statistical practice, however, it is not always easy to transform industry data on production into institutional sector data and vice versa. For that purpose, a set of tables is constructed with a dual classification. In this table the transactions are classified by industry (SUTs) and by institutional sector (sector accounts) (see the linking table in chapter 10).

11.92 The SUTs are the most developed and detailed framework use for the estimation of GDP and other macroeconomic variables in the scope of production, consumption, gross capital formation, exports, import, and income. The three approaches to measuring GDP are combined in one system based on a great variety of source data which are confronted and compared with one another in order to find possible causes of inconsistencies. The high reliability, strengths, and quality of SUTs estimates ensure that they have a strong influence on the sector accounts. It could be said that, in the main, there is a one-way traffic between SUTs and institutional sector accounts, but the use of dual classification provides possibilities for feedback in both directions. For the time being, feedback is limited because the level of aggregation in the institutional sector accounts is very high, making it difficult to trace back inconsistencies and implausible results on a sector level to specific industries in the SUTs system.

11.93 When GVA by industry from the production approach is available, it should be balanced against the GVA from the income approach for the corresponding industry, linking the factor incomes and the institutional sectors. This link is extremely important between the industry accounts and the institutional sector accounts, as illustrated below:

(a) For each industry, using the production approach:

Total output at basic prices

less total intermediate consumption at purchasers' prices
equals GVA at basic prices (production approach)

(b) For the corresponding industry, using the income approach reflecting the different factor incomes:

Self-employment income (mixed income and quasi-corporations)

plus gross trading profits of private financial corporations
plus gross trading profits of private non-financial corporations
plus gross trading surplus of public corporations (financial and non-financial)
plus rental income
plus non-market consumption of fixed capital
less holding gains and losses
less intermediate consumption of FISIM
plus other taxes on production and imports

less other subsidies on production
 equals GVA at basic prices (income approach)

11.94 Note that, for each of the factor income components shown in (b), there is an institutional sector breakdown.

11.95 The approaches shown in (a) and (b) bring together within the SUTs framework a range of data from sources such as administrative data and business surveys, and this, when balanced, ensures a better quality estimate first of GVA by industry, then for the whole economy GVA, and hence GDP and GNI.

11.96 The full breakdown in (b) in terms of data may not be available by detailed industry (and by institutional sector) for a range of reasons. The minimum that should be incorporated in the SUTs compilation and balancing process is covered in (c) below:

(c) Gross operating surplus
 plus compensation of employees
 plus other taxes on production and imports
 less other subsidies on production
 equals GVA at basic prices (income approach)

11.97 It should be noted again that, for each component shown in (c), there is an institutional sector breakdown.

2. IOTs

11.98 The links between the SUTs and IOTs in the bottom left-hand side and right-hand side of the H-Approach have been covered through the links to the separation of imports of goods and services and valuation matrices needed to produce the SUTs at basic prices and the SUTs in volume terms, and then in turn, the IOTs.

11.99 Integrating the production of IOTs with the production and balancing of SUTs makes possible the effective and timely use of powerful feedback loops in indicating data problems within the SUTs or with the steps transforming SUTs to IOTs. For example, addressing negative cell entries in the IOTs can improve the quality of the SUTs. It is recommended that IOTs should be produced (simultaneously or sequentially) alongside SUTs rather than at a much later stage or even less frequently than the SUTs.

3. PSUTs and EE-IOTs

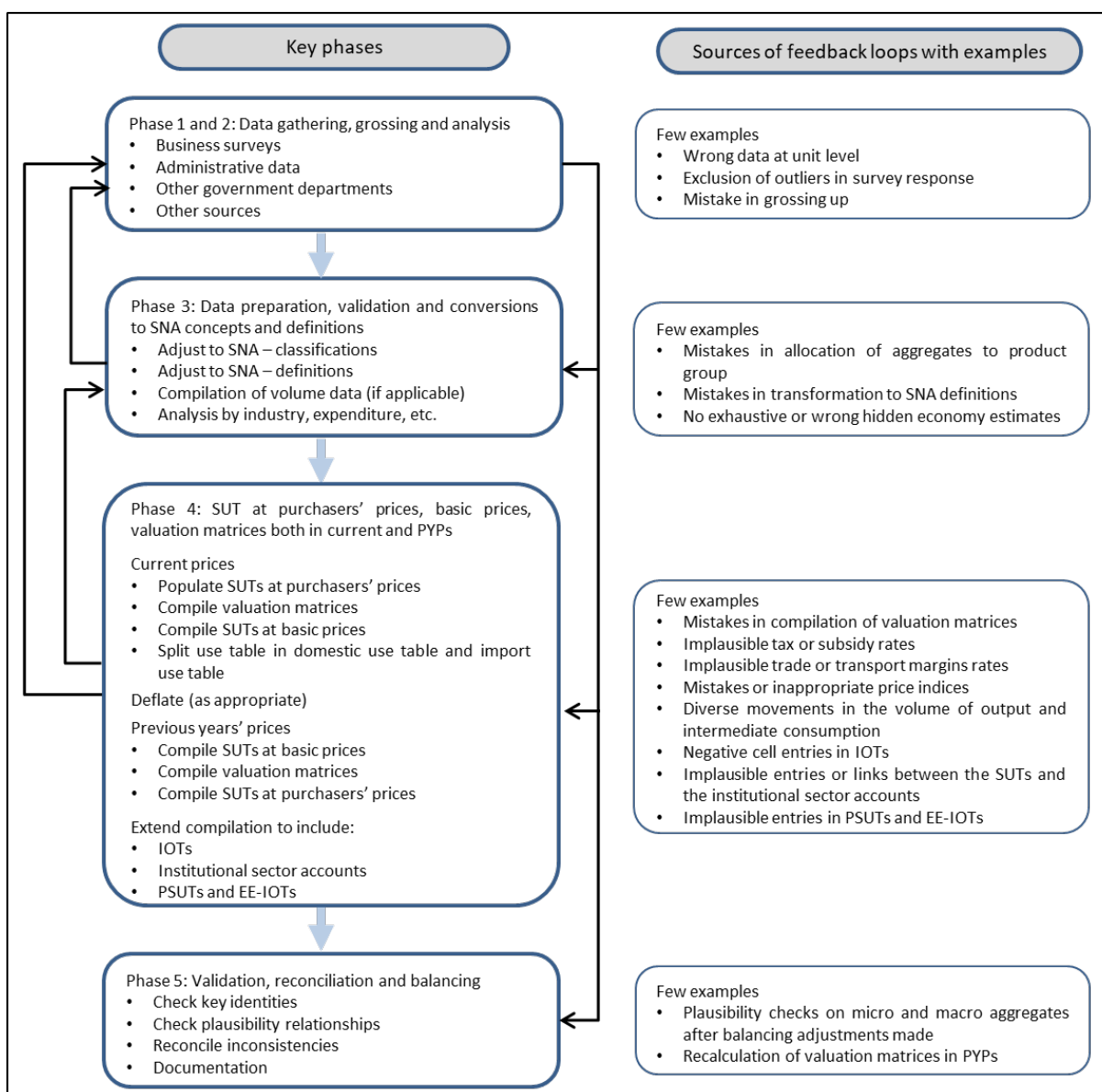
11.100 The links with the PSUTs and EE-IOTs are of a different nature because the transactions in the physical SUTs are expressed in other units (for example, kilograms, terajoules, and others) or are more detailed in terms of industries and products. The balancing of PSUTs and EE-IOTs in combination with SUTs and IOTs is described in chapter 13.

4. Key feedback loops generated by the balancing process

11.101 In the ideal scenario, all balancing is performed simultaneously, implying that all feedback loops are part of an integrated process. This includes the feedback loops within the SUTs process together with the loops going back to earlier steps in the full statistical process chain.

11.102 Figure 11.4 illustrates key feedback loops that can be generated from within the balancing process as well as examples of the sources of the feedback loops.

11.103 When the balancing is extended to cover IOTs, the links to the institutional sector accounts, PSUTs, EE-IOTs, and other satellite accounts, there will be more feedback loops that become available than are shown in Figure 11.4, and, in turn, each adds a further data quality improvement dimension.

Figure 11.4 Sources of feedback loops emanating from the balancing process

G. Practical aspects of balancing

11.104 The balancing of SUTs is not a simple task, as it requires priorities to be set because of time and resources constraints. Below are some practical considerations to be considered in the balancing process.

1. Automated and manual balancing

11.105 The balancing process covering the full SUTs system incorporates automated and manual balancing. The first step is to separate the inconsistencies between those needing further research and those which can be resolved using automated procedures. In general, large inconsistencies

require more attention than smaller ones but such indicators as time series, revision analyses, input-output ratios and labour productivity can also point to serious problems in the data. For the smaller inconsistencies, automated procedures may be used, although they will still require assessment in terms of the quality and plausibility of the results.

11.106 For the inconsistencies selected for manual balancing, a more or less reversed version of the initial full statistical process from surveys to national accounts can be applied. It would start with a critical investigation of the national accounts process, transforming source data into data for use in SUTs. In the use table in particular there are many entries which are the result of splitting aggregated source data to the product level as required in the SUTs. The allocation may be changed without altering the initial aggregate total, and thus GVA. In practice, many of the problems will not be resolved through this approach. The next step is to have recourse to expert knowledge on the specific subject matter and, if major inconsistencies persist, it may prove necessary, as a final step, to contact the survey respondent or source supplier to instigate a critical discussion of the data that they have provided.

11.107 The adjustments to entries in the SUTs will obviously affect other flows and ratios. To systematize the balancing process in order, for example, to take into account inter-industrial relationships, it is helpful to distinguish separable blocks of industries in which the main producers and users of products are represented and to assign to these blocks separate groups of input-output specialist statisticians responsible for balancing. For example, cement is definitely required by the construction industry and cement supplies can therefore be used to cross-check the plausibility of construction estimates. The SUTs can be divided into various groupings of related industries; one section of groupings, for example, could include the following, although this is not an exhaustive list:

- Agriculture, and fishing (ISIC Rev. 4, divisions 01 and 03), manufacture of food products and beverages (ISIC Rev. 4, divisions 10–12), hotels and restaurants (ISIC Rev. 4, divisions 55 and 56)
- Manufacture of metals and metal products (ISIC Rev. 4, divisions 24 and 25) and manufacture of machinery and means of transport industries (ISIC Rev. 4, divisions 26, 29 and 30)
- Forestry (ISIC Rev. 4, division 02) and industries producing wood and wood products (ISIC Rev. 4, divisions 16 and 31)
- Quarrying and non-metallic mineral products (ISIC Rev. 4, divisions 08 and 23) and construction (ISIC Rev. 4, divisions 41 and 43)
- Manufacture of textile (ISIC Rev. 4, division 13) and textile products, footwear (ISIC Rev. 4, divisions 14 and 15)
- Chemical industries, including plastic products (ISIC Rev. 4, divisions 19–22)

- Energy sectors (ISIC Rev. 4, divisions 05, 06 and 35)

11.108 Automation is essential for the preparation and management of the SUTs system. The SUTs information technology system (including the supporting modules and analytical tools and function) will play various roles in the compilation process so it will need to be designed to a high standard and with an eye to the whole range of functions that it has to carry out.

11.109 Many of the calculations in the preparation stage are carried out by automated procedures. At every stage of the process, the information technology system can provide quick and clear overviews of the data in every chosen configuration. The information technology system produces the first parts of the SUTs, which are essential for the detection of major integration problems and, down the line, it enables investigation of the finest details, so that the causes of balancing problems can be detected in the most efficient manner possible. Lastly, the information technology system can help to develop appropriate solutions along with professional and orderly documentation of all the adjustments made during the compilation and balancing processes. The information technology system is a powerful instrument which is indispensable for all operations from the source data through to the final set of balanced SUTs.

11.110 Modules can be designed to eliminate small discrepancies between supply and use at the product level in an adequate and efficient way. This may imply one-dimensional proportional distribution of the discrepancies over a selected set of users.

11.111 Balancing the whole SUTs system requires a multidimensional approach to the reconciliation of inconsistencies and a range of human inputs which cannot be automated. The experiences to date of balancing SUTs in an automated manner have shown that full automated balancing does not yet work; the quality generated is poor, many issues require further investigation and, overall, the savings gained are tenuous at best. Most experiences have shown that a combination of automated and manual statistical techniques and procedures is the best workable solution to the underpinning of a SUTs system.

11.112 Working with statistical data based on sample surveys, survey questionnaires and influenced by such issues as non-response, means working with sample margins of error. Even when samples are perfect and there is a 100 per cent response rate there will be inconsistencies. Statistics therefore constitute a major cause of differences and it could be argued that balancing could be carried out in an automated manner using the sample margins of the statistics concerned as weights. Statistics, however, are never ideal and inconsistencies are not only caused by sampling but also have various causes of a non-statistical nature. It is these causes of inconsistencies that make manual balancing necessary as a preliminary step to be taken prior to using automated balancing.

11.113 The decision as to what can be balanced using automated procedures and what should be done manually depends not only on the nature of the inconsistencies but also on the context of where the balancing takes place.

11.114 If the automated balancing of SUTs is carried out independently for consecutive years, and is applicable to current prices and volume estimates, then the initial discrepancies must be small. Experience has shown that even small differences in initial data can lead to totally different results in RAS (covered in box 11.3) and optimization procedures, which indicates that the balanced SUTs are not comparable over time. As long as the discrepancies are small, the chances that the results will no longer be comparable should also be small.

11.115 One major step forward is to include linkages (for example, growth rate expectations) between consecutive years, preferably including price and volume indicators. In setting such a scenario, the basic identities of the SUTs can be used as a restrictive measure in the automated balancing process and also in the criteria relating to plausibility referred to above. Including price and volume ratios and defined restrictions attributable to them would help to assure the plausibility of growth rates and comparability over time. This also allows for the option of leaving major discrepancies to the automated system.

11.116 Box 11.3 provides a short description of automated methods that are often used to remove some of the inconsistencies in the SUTs.

Box 11.3 Methods used for automated balancing SUTs

RAS method

The RAS method (the abbreviation is derived from the phrase “raking and scaling”) is a well-known and widely used method for data reconciliation. Its aim is to achieve consistency between the entries of some non-negative matrix and pre-specified column totals and row totals. It is very easy to apply and to understand. RAS has a narrow range of applicability, however; for example, it can only be applied to non-negative matrices.

It is used to revise the internal entries in a matrix so that they agree with the margin totals. RAS is used when the margin totals – total supply and use of products, or total output by kind of activity, for example – are believed to be correct but the breakdown inside the matrix is not consistent with the margin totals. Over the years, many extensions, variations and improvements of the RAS method have been developed. Some examples include:

- GRAS (generalized RAS) allows for matrices in which some of the elements are predefined, in addition to the row totals and the column totals. The GRAS method allows for matrices with negative entries (see Lenzen and others, 2007).
- TRAS (three-stage RAS) extends RAS by including constraints on arbitrary subsets of the matrix elements, instead of only fixing row totals and column totals (see Cole, 1992, and Gilchrist and St. Louis, 1999).
- KRAS, as developed by Lenzen and others (2009), includes the aforementioned features of GRAS and TRAS and further generalises RAS for the case of conflicting source data. The simplest case is when two data sources prescribe two different values for the same matrix entry. In order to converge, the original RAS method can use only conflicting values, whereas the KRAS method will use both values and allow for different reliabilities of the data sources.

More details on different updating methods may be found in chapter 18 of this Handbook and in United Nations (1999).

Stone method

The Stone method is another method of data reconciliation whereby data are adjusted in order to satisfy a set of linear constraints. In adjusting the initial data, the Stone method uses information on the relative reliabilities of

the initial data given in a covariance matrix. Data that are considered to be the most reliable are modified least, and vice versa. The Stone method yields a set of fully reconciled data, with minimum variance. It translates the reconciliation problem into a mathematical (weighted quadratic) optimization problem under linear constraints.

In practical applications of the method, a covariance matrix of the initial data is often unavailable. Accordingly, applications generally use estimates of relative variances. There are several ad hoc methods for estimating relative variances. One method is to have a specialist estimate of 95 per cent confidence intervals and to use the interval sizes as an approximation for variances. Another method may be to distinguish several categories such as “relatively unreliable”, “normally reliable” and “relatively reliable”, and all variables within the same group are assigned the same variance.

It is often desirable in practice for reconciliation to affect large values more than small values in an absolute sense. If this is the intention, then the following variances may be chosen:

$\text{Var}(x_i) = \theta_i^2 x_i^2$ where, θ_i is a parameter that depends on the reliability, or reliability category, of x_i .

In practice, determining the correct ratios between the various variances is a process of trial and error, which means that one particular ratio is chosen based on a degree of prior knowledge and simple assumptions (for example, that variances are equal in the absence of prior knowledge), and then judging whether the results are acceptable. If not, the variances are then modified.

Convex quadratic constrained optimization

An option for balancing SUTs both in current prices and in volume terms simultaneously is the application of loss function, which includes current price and volume estimates and also linear price and volume ratios. The loss function must be minimized under a set of linear constraints, and the loss is defined as the difference between the initial data and the balanced data. For the price and volume ratios, linear constraints are applied. The constraints are either strong or weak.

Strong constraints include, for example, identities of the SUTs and upper and lower boundaries (subsidies should be less than or equal to zero). Weaker constraints include, for example: volume change of output related to volume change of input; ratio of taxes and subsidies to the basic price estimate (in the form of a percentage of the variable-like tax); ratio of margins to the basic price estimate (in the form of a percentage of the margins); and re-exports are smaller than the corresponding imports (weak because of differences in valuation).

More specific constraints at a product level can be specified in this optimization problem. It is also possible to extend the optimization problem and to include, for example, the transformation to industry-by-industry IOTs, including the estimation of valuation matrices.

2. Balancing benchmark and consecutive years

11.117 In the ideal situation, the annual SUTs and IOTs present the current state of the art when it comes to balancing of the basic statistics. The data from the annual SUTs and IOTs are further improved, however, with every new benchmarking exercise, leading to an entire new time series. Benchmarking is a regular process in economic statistics, whereby data sources for the same target variable with different frequencies are reconciled and the inconsistencies between the different estimates are corrected. Benchmarking leads to revisions of earlier estimates of the target variables. This section deals with two types of benchmark revisions for SUTs and IOTs – periodic benchmarking and annual benchmarking – and considerations about balancing of SUTs and IOTs in these two cases.

11.118 Periodic benchmarking refers to significant revisions, for example, conceptual changes, new or changed basic data sources that originate from incorporating data from periodic benchmark censuses (which are carried out every five to ten years), revised international guidelines like the 2008 SNA and BPM 6 and other changes that cannot be incorporated on a continuous basis because of resource constraints. Backward revisions to the time series affecting several previous years are also carried out based on the benchmark exercise.

11.119 In general, planning for period benchmark revisions should seek to coordinate all major changes to be synchronized for a common year for implementation, thereby achieving, at a minimum of about once every five years, a maximum degree of consistency within national accounts, balance of payments and other related domains and the statistics concerned are based on the best possible data.

11.120 In this context annual benchmarking mainly refers to the regular revisions to annual accounts made possible by the availability of new and more complete data sources. Annual benchmarking, however, also includes revisions due to the alignment of short-term survey results (say, turnover variable) based on small survey samples with much larger sample-based annual structural surveys. Following the reconciliation of short-term survey sources before the benchmarking process with more complete and detailed annual data sources such as structural statistics, these sources are then fed through the SUTs framework. Combining annual benchmarking and annual chain-linking also ensures that improved accuracy of the levels and growth rates of the economy is reached more quickly, and again, achieved through the SUTs framework.

11.121 Consistency is one of the key elements of national accounts. Theoretically, the whole time series and every level of detail should be consistent. In practice, it may not be possible to publish all results at the same time, although the SUTs and the main aggregates such as GDP should be fully consistent. More and more years of SUTs are being produced and this poses a growing challenge as to how to maintain their consistency as a long-run dataset. Statistics Denmark, for example, as part of its implementation of the 2008 SNA and BPM 6, ensured that the Danish IOTs were retained on a consistent basis going back to 1966 – a significant exercise in its own right.

11.122 Annual benchmark revisions, as carried out in Ireland and the United Kingdom, entail less of a planning burden in synchronizing a common year, as the changes form an integral process to compiling annual SUTs.

11.123 In theory, the balancing procedure for period benchmark SUTs is the same as for SUTs compiled annually and also for investigations into the causes of inconsistencies and the search for solutions. For non-benchmark years, however, much information can be derived from looking at the previous years. When balanced product-flow systems already exist for the previous year, it can safely be assumed that the general structure of the system will be more or less similar to the preceding year, unless of course specific information is available showing that there are major

structural changes in some industries. Together with price and volume estimates, the T-1 data provide a valuable source in detecting inconsistencies in the SUTs.

11.124 For periodic benchmark SUTs, all inconsistencies must be investigated thoroughly and exhaustively. This is even more important as the level estimates of the subsequent years are based on this benchmark. Consequently, the balancing of benchmark SUTs should be carried out manually to a great degree, so that a good quality base can be achieved and the use of automated procedures should be limited to small discrepancies.

3. Organization of the balancing function

11.125 The organization of the balancing function can be set up in different ways across teams. The following are possible examples:

- Centralized balancing team – a single person or a very small team of people form the central balancing team whose designated role is to take all the validated and investigated data from the compilation teams and balance the SUTs using the tools (manual and automated) at their disposal. They will lead and coordinate the implementation of balancing adjustments across the components and record the adjustments as “pure” balancing adjustments. They may generate the areas for investigation in liaison with the compilers. The team may also be organized in such a way that it does not simply produce a balanced SUTs dataset but also provides feedback to the compilers on the balancing adjustments as appropriate, for example, household final consumption expenditure or gross fixed capital formation, to enable the compilers to generate publications consistent with the final dataset. Alternatively, the compilers could use the final dataset to generate the publications with no additional data flows in the system.
- Decentralized balancing – here the balancing is devolved to industry, product or topic specialists (for example, energy, household final consumption expenditure or the compilers). They will undertake a row and column balancing process related to their allocated row, column and topic. After so many iterations, the central team may simply use automated tools to achieve a final balance. The role of the central team is different in that it focuses solely on the automated part, whereas the manual balancing is left to the specialists and is also coordinated across the balancers.

11.126 There are pros and cons to either approach:

- In centralized balancing, the control and order assessment and also the overall quality control sit with the central team and form an efficient way of achieving a coherent and high-quality balance. The specialist experts’ knowledge can be used for quality assurance of the inputs and the balanced picture.

- In decentralized balancing, the specialist experts' knowledge is more extensively used but requires much more communication and coordination to be effective, and the balancing adjustments and quality adjustments may be less clear.

11.127 In both cases, such issues as documentation, communication and coordination are crucial, and it is also essential to ensure that staff have appropriate skills and knowledge. It is preferable, however, to have a centralized rather than a decentralized balancing arrangement.

Annex A to chapter 11. Balancing supply and use tables

A11.1 This annex presents a numerical example of the balancing of SUTs in current prices and in previous years' prices in line with the SUTs balancing framework shown in figure 11.3 (fully consistent with the H-Approach). This provides an example of the type of thinking and issue resolution that goes to achieve a balanced SUTs system. The numbers in the tables in this annex have been divided by 1,000 for readability and presentation purposes; in reality the differences are larger than the small numbers shown. This annex consists of three sections.

A11.2 Section 1 of the annex shows the following (unbalanced) tables:

- Table A11.1: Supply and use tables 2011 in current price (SUTs at purchasers' prices, Taxes less subsidies on products, trade and transport margins, SUTs at basic prices, imports use table and domestic use table).
- Table A11.2: Price indices for supply and use tables 2011 (SUTs at purchasers' prices, taxes less subsidies on products, trade and transport margins, SUTs at basic prices, imports use table and domestic use table). It should be noted that some prices may be based on actual source data or derived from independent volume estimates (using the valuation matrices) or implicit prices, but a complete set of prices is shown for the purposes of this balancing example.
- Table A11.3: Supply and use tables 2011 in previous years' prices (SUTs at purchasers' prices, taxes less subsidies on products, trade and transport margins, SUTs at basic prices, imports use table and domestic use table).
- Table A11.4: Volume indices tables for supply and use tables 2011 (SUTs at purchasers' prices, taxes less subsidies on products, trade and transport margins, SUTs at basic prices, imports use table and domestic use table).
- Table A11.5: Supply and use tables 2010 in current price (SUTs at purchasers' prices, taxes less subsidies on products, trade and transport margins, SUTs at basic prices, imports use table and domestic use table).

A11.3 Section 2 of the annex follows the same sequence of tables but shows the corresponding balanced SUTs system for both price bases. The cells highlighted in yellow are those changed in order to achieve balanced SUTs.

A11.4 Section 3 of the annex provides an overview of the numerical adjustments required to achieve the balanced system.

A11.5 The inconsistencies in the system are revealed by differences between the supply and use at the product level and discrepancies between trade and transport margins in the SUTs at purchasers' prices and the row totals of the respective valuation matrices. The same holds for taxes less subsidies on products. In general, the net operating surplus is used as a balancing item in achieving the identity that the total output equals the total input.

A11.6 If the inconsistencies are relatively small, then automated procedures can be applied to balance the SUTs system simultaneously in the different valuations. In the case of large discrepancies or implausible input-output ratios in volume terms or implausible movements in the price changes on a row, then further research would have to be undertaken before any automated balancing could be applied.

A11.7 In the numerical example, the estimates for agriculture products in current prices show a discrepancy both at purchasers' prices and at basic prices. Conversely, the estimates in previous years' prices are balanced. The latter does not mean that the previous years' price-based data are plausible although, having examined the data on volumes and prices at a more detailed level than presented in the example, it was concluded that the prices were too high and the volumes looked plausible. As a consequence, the estimates in current prices for the supply of agriculture products by the agriculture industry needed to be adjusted.

A11.8 The estimates for manufacturing products show a discrepancy in current and previous years' prices both at purchasers' prices and at basic prices. The difference in the previous years' prices based data at purchasers' prices is somewhat larger than in current prices because the initial estimate of taxes less subsidies on products and trade and transport margins in previous years' prices seems to be very low. The separate use table based on domestic output (namely, domestic use table) and imports (namely, imports use table) shows a big discrepancy for this product group in the domestic use table, which is counterbalanced in the imports use table. In addition to the difference between supply and use of products in current prices and in previous years' prices, there is an implausible price index for exports. In this case, the estimates in previous years' prices can be adjusted in order to get plausible price indices. A second reason for the inconsistency is the delineation of exports and re-exports. At the point when products are imported, it is not always known whether or not the products will be re-exported in the same form. As a consequence, the data available for re-exports from source statistics should be a minimum and the actual estimate in the SUTs will be much higher, as shown in this example. The delineation problem caused a small inconsistency which can be solved by adjusting total exports.

A11.9 The balancing adjustments and approaches applied to the valuation matrices differ between estimates in current prices and estimates in previous years' prices.

A11.10 The current price estimates of trade and transport margins (TTM) do not show an inconsistency in the purchasers' prices table; the TTM column is consistent with the output of trade. In the basic price table, however, there is an inconsistency caused by the difference between

the TTM matrix and the TTM column in the supply table at purchasers' prices. The gap is 680, which is found on the manufacturing product row.

A11.11 Similarly, for taxes less subsidies on products (TLS), there were some differences between the TLS column of the supply table and the total column of the TLS matrix. These inconsistencies appear on the product rows for manufacturing (MAN) and finance and business services (FBS).

A11.12 In both cases, the data in the current price valuation matrices need to be adjusted, as in most cases the current price TLS on products is derived from government data and therefore fixed. The same approach is less strong for TTM, in which the output control totals generally prevail over the estimates of the valuation matrices.

A11.13 For the previous years' prices-based estimates of TLS and TTM, the opposite holds because, while the estimates for TTM and TLS on products are compiled using the volume change of the relevant transactions, the estimates in the valuation matrices determine TTM and TLS in previous years' prices, so the TLS and TTM columns of the SUTs at purchasers' prices have to be adjusted.

A11.14 Both in current prices and in previous years' prices, the transport products show a discrepancy. In this case the inconsistency was not caused by the valuation matrices but, on looking at the imports use table and through additional research, it was identified that the estimate of import of transport services was too high.

A11.15 The discrepancy in communication is shown in both valuation matrices in current prices and in previous years' prices and can also be seen in the domestic use table. On examining the data, it was decided that consumption of households had to be adjusted.

A11.16 A similar procedure was followed for financial and business services and other services, in which cases it was decided that the intermediate consumption of TIC and consumption of households should be adjusted respectively. It should be noted that the discrepancy for the FBS product in the basic price table is somewhat larger than in the purchasers' price table because of the wrong estimate in the TLS matrix.

1. Unbalanced SUTs system (table A11.1–table A11.5)

Table A11.1: Supply and use tables 2011 in current prices

Supply Table at basic prices, transf. to purchasers' prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	26.0	0.2	0.0	26.2	15.4	41.6	10.9	0.5	53.0
MAN	1.3	318.5	35.7	355.5	336.8	692.3	113.0	39.4	844.7
CON	0.1	88.0	4.5	92.6	1.6	94.1	8.3		102.4
TTC	0.5	15.1	231.6	247.2	73.4	320.6	-124.0	3.0	199.6
FBS	0.5	8.0	282.0	290.5	55.2	345.7		9.1	354.9
OSE	0.3	2.5	222.9	225.7	16.6	242.3	0.1	3.0	245.4
TOT	28.7	432.3	776.7	1237.7	499.1	1736.7	0.0	63.3	1800.1

Use Table at purchasers' prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTIn	TOTpp	Check
AGR	5.5	17.4	1.5	24.4	5.9			0.1	-0.1	22.4	28.4	52.8	0.2
MAN	10.1	202.1	65.4	277.7	122.8		9.1	51.5	1.4	383.1	568.0	845.7	-1.0
CON	0.3	26.2	19.8	46.3	0.5		0.6	53.7		2.1	56.8	103.1	-0.7
TTC	0.6	10.8	66.2	77.5	21.4	0.9	0.5	6.7	0.0	92.3	121.8	199.3	0.3
FBS	2.1	43.5	161.8	207.3	78.0	0.0	3.6	16.8		49.5	147.8	355.2	-0.3
OSE	0.2	3.5	22.2	25.9	54.7	4.6	153.4	1.2	0.2	5.1	219.1	245.0	0.4
TIC	18.8	303.5	336.8	659.1	283.3	5.5	167.2	130.1	1.5	554.5	1142.0	1801.1	-1.0
OTLS	-0.7	0.2	0.3	-0.3								-0.3	
COE	2.7	60.9	254.4	318.0								318.0	
CFC	3.7	20.5	82.9	107.1								107.1	
NOS	4.3	47.2	102.2	153.7								153.7	
GVA	9.9	128.8	439.8	578.5								578.5	
TOT	28.7	432.3	776.7	1237.7	283.3	5.5	167.2	130.1	1.5	554.5	1142.0		

Check

Taxes less subsidies on products

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR							0.5		0.5
MAN							39.4		39.4
CON							8.3		8.3
TTC							3.0		3.0
FBS							9.1		9.1
OSE							3.0		3.0
TOT							63.3		63.3

Use Table for taxes less subsidies on products

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTIn	TOTpp	Check
AGR	0.0	0.0	0.0	0.1	0.4					0.1	0.5	0.5	
MAN	0.2	1.7	6.4	8.3	26.8		0.4	2.2	0.0	1.7	31.1	39.4	0.0
CON	0.0	0.0	2.0	2.0	0.0		0.1	6.1		0.0	6.2	8.3	
TTC	0.0	-0.1	1.1	1.0	1.7		-0.1	0.4		-0.1	2.0	3.0	
FBS	0.0	0.0	4.4	4.4	1.6		0.1	3.0		0.0	4.7	9.1	0.0
OSE	0.0	0.0	0.1	0.1	2.3		-0.4	0.7		0.3	2.9	3.0	
TOT	0.2	1.6	14.0	15.8	32.9		0.1	12.5	0.0	2.0	47.5	63.3	0.0

Check

Trade and transport margins

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR							10.9		10.9
MAN							113.0		113.0
CON									
TTC							-124.0		-124.0
FBS									
OSE							0.1		0.1
TOT							0.0		0.0

Use Table for trade and transport margins

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTIn	TOTpp	Check
AGR	0.4	1.6	0.4	2.5	3.2			0.0	0.0	5.2	8.4	10.9	
MAN	0.9	20.5	9.2	30.6	39.1		3.1	5.1	-0.1	34.5	81.7	112.3	0.7
CON													
TTC	-1.3	-22.2	-9.6	-33.1	-42.3		-3.1	-5.1	0.1	-39.7	-90.2	-123.3	-0.7
FBS													
OSE					0.1					0.0	0.1	0.1	
TOT					0.0			0.0	0.0	0.0	0.0	0.0	

Check

Supply Table at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	26.0	0.2	0.0	26.2	15.4	41.6			
MAN	1.3	318.5	35.7	355.5	336.8	692.3			
CON	0.1	88.0	4.5	92.6	1.6	94.1			
TTC	0.5	15.1	231.6	247.2	73.4	320.6			
FBS	0.5	8.0	282.0	290.5	55.2	345.7			
OSE	0.3	2.5	222.9	225.7	16.6	242.3			
TOT	28.7	432.3	776.7	1237.7	499.1	1736.7			

Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTIn	TOTpp	Check
AGR	5.1	15.7	1.1	21.9	2.4			0.1	-0.1	17.1	19.5	41.3	0.3
MAN	9.0	179.9	49.8	238.8	56.9		5.6	44.3	1.5	346.9	455.2	694.0	-1.7
CON	0.3	26.2	17.7	44.3	0.4		0.5	47.6		2.1	50.6	94.8	-0.7
TTC	1.8	33.0	74.8	109.6	62.0	0.9	3.7	11.4	-0.1	132.1	210.0	319.7	1.0
FBS	2.1	43.5	157.4	202.9	76.4	0.0	3.5	13.8		49.5	143.1	346.1	-0.3
OSE	0.2	3.5	22.1	25.8	52.3	4.6	153.8	0.4	0.2	4.8	216.1	241.9	0.4
TLN	0.2	1.6	14.0	15.8	32.9		0.1	12.5	0.0	2.0	47.5	63.3	
TIC	18.8	303.5	336.8	659.1	283.3	5.5	167.2	130.1	1.5	554.5	1142.0	1801.1	-1.1
OTLS	-0.7	0.2	0.3	-0.3								-0.3	
COE	2.7	60.9	254.4	318.0								318.0	
CFC	3.7	20.5	82.9	107.1								107.1	
NOS	4.3	47.2	102.2	153.7								153.7	
GVA	9.9	128.8	439.8	578.5								578.5	
TOT	28.7	432.3	776.7	1237.7	283.3	5.5	167.2	130.1	1.5	554.5	1142.0		

Check

Imports CIF

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR					15.4				
MAN					336.8				
CON					1.6				
TTC					73.4				
FBS					55.2				
OSE					16.6				
TOT					499.1				

Imports Use Table

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTIn	TOTpp	Check
AGR	0.8	6.4	0.5	7.7	1.2			0.0	-0.1	6.6	7.7	15.4	
MAN	1.3	98.0	20.9	120.2	24.5		1.6	19.2	-0.2	121.3	166.5	286.6	50.2
CON	0.0	0.5	0.4	0.9				0.7			0.7	1.6	
TTC	0.0	2.7	14.0	16.8	0.7			2.7	0.0	52.7	56.1	72.9	0.5
FBS	0.2	14.0	24.5	38.8	0.2			2.2	14.1		16.5	55.2	
OSE	0.0	0.5	3.9	4.5	12.0		0.0	0.1	0.0		12.2	16.6	
TLN													
TIC	2.3	122.1	64.2	188.7	38.7		1.6	24.8	-0.1	194.7	259.7	448.4	50.7

Check

Supply Table for domestic output at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	26.0	0.2	0.0	26.2					
MAN	1.3	318.5	35.7	355.5					
CON	0.1	88.0	4.5	92.6					
TTC	0.5	15.1	231.6	247.2					
FBS	0.5	8.0	282.0	290.5					
OSE	0.3	2.5	222.9	225.7					
TOT	28.7	432.3	776.7	1237.7					

Domestic Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTIn	TOTpp	Check
AGR	4.3	9.3	0.6	14.2	1.2			0.1	0.0	10.5	11.7	26.0	0.2
MAN	7.8	81.9	28.9	118.6	32.4		4.0	25.1	1.6	225.6	288.7	407.4	-51.9
CON	0.3	25.7	17.4	43.4	0.4		0.5	46.9		2.1	49.9	93.3	-0.7
TTC	1.8	30.3	60.7	92.8	61.3	0.9	3.7	8.6	-0.1	79.4	153.9	246.7	0.5
FBS	1.8	29.5	132.9	164.2	76.2	0.0	3.5	11.6		35.4	126.6	290.8	-0.3
OSE	0.2	3.0	18.1	21.3	40.2	4.6	153.8	0.4	0.1	4.8	203.9	225.3	0.4
TLN	0.2	1.6	14.0	15.8	32.9		0.1	12.5	0.0	2.0	47.5	63.3	
TIC	16.4	181.4	272.6	470.5	244.6	5.5	165.5	105.3	1.6	359.7	882.3	1352.8	-51.8
IMP	2.3	122.1	64.2	188.7	38.7		1.6	24.8	-0.1	194.7	259.7	448.4	
TOT	18.8	303.5	336.8	659.1	283.3	5.5	167.2	130.1	1.5	554.5	1142.0	1801.1	-51.8
OTLS	-0.7	0.2	0.3	-0.3								-0.3	
COE	2.7	60.9	254.4	318.0								318.0	
CFC	3.7	20.5	82.9	107.1								107.1	
NOS	4.3	47.2	102.2	153.7								153.7	
GVA	9.9	128.8	439.8	578.5								578.5	
TOT	28.7	432.3	776.7	1237.7	283.3	5.5	167.2	130.1	1.5	554.5	1142.0		

Check

Table A11.2: Price indices for supply and use tables 2011

Supply Table at basic prices, transf. to purchasers' prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	102.0	104.1		102.0	106.9	103.8	97.5	107.6	102.5
MAN	107.5	108.0	100.5	107.2	108.0	107.6	101.1	103.2	106.4
CON	103.4	100.5	100.5	100.5	101.6	100.5		100.1	100.4
TTC	101.1	99.2	100.3	100.2	110.7	102.4	100.8	101.4	103.4
FBS	102.9	100.9	100.4	100.4	101.0	100.5		93.6	100.3
OSE		100.2		101.0	103.4	101.1	99.1	110.2	101.2
TOT	102.2	105.8	100.5	102.4	107.3	103.7		101.6	103.7

Use Table at purchasers' prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	105.8	109.9	98.7	108.2	100.4			94.7	568.8	97.7	98.0	102.4	0.0
MAN	114.1	110.4	105.0	109.2	102.1		99.5	99.6	85.7	106.7	104.9	106.2	0.2
CON	101.4	100.5	101.8	101.0	97.4		97.1	100.0		102.0	100.0	100.5	0.0
TTC	100.9	100.1	100.8	100.7	102.2					108.0	106.2	104.0	-0.5
FBS	101.4	100.7	100.4	100.5	101.7		101.8			99.0	100.1	100.3	0.0
OSE	102.4	101.7	104.3	103.9	102.7	103.8		101.7	99.5	101.8	100.9	101.2	0.0
TIC	109.3	107.5	101.7	104.5	102.1	103.3	100.2	99.1	83.0	105.7	103.2	103.6	-0.3
OTLS	98.0	201.1	269.9	48.3								48.3	
COE	102.7	101.9	101.5	101.6								101.6	
CFC	99.1	100.6	98.4	98.8								98.8	
NOS	80.9	103.2	96.2	97.7								97.7	
GVA	91.1	102.2	99.7	100.1								100.1	
TOT	102.2	105.8	100.5	102.4	102.1	103.3	100.2	99.1	83.0	105.7	103.2		

Check

Taxes less subsidies on products

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR								107.6	
MAN								103.2	
CON								100.1	
TTC								101.4	
FBS								93.6	
OSE								110.2	
TOT								101.6	

Use Table for taxes less subsidies on products

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	92.3	612.5	91.4	230.0	100.8					96.2	99.8	107.6	
MAN	103.7	107.3	103.9	104.6	101.7		98.7	93.8	-100.0	104.4	101.3	101.9	1.3
CON			102.2	102.1	90.7		95.1	99.6			100.0	99.4	
TTC				99.4	103.3						102.4	101.4	
FBS	123.1	100.0	102.0	102.1	105.7		103.8			44.4	86.4	93.3	0.3
OSE				241.7	109.2			101.1			108.5	110.2	
TOT	105.1	110.1	103.1	103.8	102.5		102.3	92.5	-100.0	105.0	99.8	100.7	0.8

Check

Trade and transport margins

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR							97.5		
MAN							101.1		
CON									
TTC							100.8		
FBS									
OSE							99.1		
TOT									

Use Table for trade and transport margins

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	90.4	96.7	100.7	96.3	104.9			85.7	100.0	94.0	97.9	97.5	
MAN	113.7	102.6	102.3	102.8	97.8		97.9	97.5	93.5	99.5	98.5	99.6	1.5
CON													
TTC	105.0	102.1	102.2	102.3	98.3		97.9	97.5	93.8	98.8	98.4	99.4	1.4
FBS													
OSE					97.6					104.2	99.1	99.1	
TOT													

Check

Supply Table at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	102.0	104.1		102.0	106.9	103.8			
MAN	107.5	108.0	100.5	107.2	108.0	107.6			
CON	103.4	100.5	100.5	100.5	101.6	100.5			
TTC	101.1	99.2	100.3	100.2	110.7	102.4			
FBS	102.9	100.9	100.4	100.4	101.0	100.5			
OSE		100.2		101.0	103.4	101.1			
TOT	102.2	105.8	100.5	102.4	107.3	103.7			

Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	107.2	111.2	98.1	109.5	94.9			95.6	725.0	98.9	98.0	103.8	0.0
MAN	114.4	111.4	105.6	110.2	105.5		100.5	100.2	85.7	107.5	106.3	107.6	-0.1
CON	101.4	100.5	101.7	101.0	98.3		97.4	100.0		102.0	100.1	100.5	0.0
TTC	103.8	101.5	101.0	101.2	99.5		98.5	97.0	93.7	105.0	102.8	102.2	0.2
FBS	101.2	100.7	100.4	100.5	101.7		101.7	100.6		99.0	100.6	100.5	0.0
OSE	102.4	101.8	104.1	103.7	102.4	103.8		102.5	99.5	101.5	100.8	101.1	0.0
TLS	105.1	110.1	103.1	103.8	102.5		102.3	92.5	-100.0	105.0	99.8	100.7	
TIC	109.3	107.5	101.7	104.5	102.1	103.3	100.2	99.1	83.0	105.7	103.2	103.6	0.1
OTLS	98.0	201.1	269.9	48.3								48.3	
COE	102.7	101.9	101.5	101.6								101.6	
CFC	99.1	100.6	98.4	98.8								98.8	
NOS	80.9	103.2	96.2	97.7								97.7	
GVA	91.1	102.2	99.7	100.1								100.1	
TOT	102.2	105.8	100.5	102.4	102.1	103.3	100.2	99.1	83.0	105.7	103.2		

Check

Imports cif

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR					106.9				
MAN					108.0				
CON					101.6				
TTC					110.7				
FBS					101.0				
OSE					103.4				
TOT					107.3				

Imports Use Table

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	113.0	114.5	98.0	113.2	100.1			100.0	207.1	102.1	101.4	106.9	
MAN	112.6	114.4	107.2	113.1	102.0		99.6	99.5	-311.5	89.4	92.1	99.9	8.1
CON	100.0	101.0	102.0	101.4				101.8			101.8	101.6	
TTC	100.0	100.5		101.2						115.2	113.9	110.7	-0.1
FBS	97.0	101.4	100.4	100.8	106.7						101.6	101.0	
OSE	100.0	100.8	101.4	101.4	104.2				99.0	97.1	104.1	103.4	
TLS													
TIC	110.7	112.3	102.8	108.9	102.6		99.6	99.4	-100.8	96.5	97.6	102.0	5.3

Check

Supply Table for domestic output at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	102.0	104.1		102.0					
MAN	107.5	108.0	100.5	107.2					
CON	103.4	100.5	100.5	100.5					
TTC	101.1	99.2	100.3	100.2					
FBS	102.9	100.9	100.4	100.4					
OSE		100.2		101.0					
TOT	102.2	105.8	100.5	102.4					

Domestic Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	106.2	109.0	98.2	107.6	90.0			94.6	-181.3	97.0	95.9	102.0	0.0
MAN	114.7	107.9	104.5	107.5	108.4		100.8	100.7	98.0	120.7	116.7	113.9	-6.7
CON	101.4	100.5	101.7	101.0	98.3		97.4	100.0		102.0	100.0	100.5	0.0
TTC	103.9	101.5	100.9	101.2	99.5		98.5	97.2	93.5	99.2	99.2	99.9	0.3
FBS	101.8	100.4	100.4	100.4	101.7		101.7	100.5		98.0	100.5	100.4	0.0
OSE	102.5	101.9	104.7	104.2	101.9	103.8		102.5	100.0	101.5	100.7	101.0	0.0
TLS	105.1	110.1	103.1	103.8	102.5		102.3	92.5	-100.0	105.0	99.8	100.7	
TIC	109.1	104.4	101.4	102.8	102.0	103.3	100.2	99.0	96.1	111.5	104.9	104.2	-6.4
IMP	110.7	112.3	102.8	108.9	102.6		99.6	99.4	-100.8	96.5	97.6	102.0	
TOT	109.3	107.5	101.7	104.5	102.1	103.3	100.2	99.1	83.0	105.7	103.2	103.6	-6.4
OTLS	98.0	201.1	269.9	48.3								48.3	
COE	102.7	101.9	101.5	101.6								101.6	
CFC	99.1	100.6	98.4	98.8								98.8	
NOS	80.9	103.2	96.2	97.7								97.7	
GVA	91.1	102.2	99.7	100.1								100.1	
TOT	102.2	105.8	100.5	102.4	102.1	103.3	100.2	99.1	83.0	105.7	103.2		

Check

Table A11.3: Supply and use tables 2011 at previous years' prices

Supply Table at basic prices, transf. to purchasers' prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	25.5	0.1	0.0	25.7	14.4	40.1	11.2	0.5	51.8
MAN	1.2	295.0	35.6	331.7	311.9	643.7	111.7	38.2	793.6
CON	0.1	87.6	4.5	92.2	1.5	93.7	8.3		102.0
TTC	0.5	15.2	231.0	246.7	66.4	313.1	-123.0	3.0	193.0
FBS	0.5	8.0	280.8	289.2	54.7	343.9		9.7	353.7
OSE	0.3	2.5	220.7	223.5	16.1	239.6	0.1	2.7	242.4
TOT	28.1	408.4	772.5	1209.0	465.0	1674.0	0.0	62.4	1736.4

Use Table at purchasers' prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR	5.2	15.8	1.6	22.6	5.9			0.2	0.0	22.9	29.0	51.5	0.2
MAN	8.9	183.1	62.3	254.4	120.3			9.2	51.7	1.6	358.9	541.7	-2.5
CON	0.3	26.1	19.4	45.8	0.5			0.6	53.7		2.1	56.8	-0.7
TTC	0.6	10.8	65.6	77.0	21.0	0.9	0.5	6.9	0.0	85.5	114.7	191.7	1.3
FBS	2.0	43.2	161.1	206.3	76.6	0.0	3.5	17.6		50.0	147.7	354.0	-0.3
OSE	0.2	3.5	21.2	24.9	53.2	4.4	153.1	1.1	0.2	5.0	217.1	242.0	0.4
TIC	17.2	282.5	331.3	630.9	277.5	5.3	166.8	131.3	1.8	524.3	1107.0	1737.9	-1.5
OTLS	-0.7	0.1	0.1	-0.5								-0.5	
COE	2.6	59.8	250.6	313.0								313.0	
CFC	3.7	20.4	84.3	108.4								108.4	
NOS	5.3	45.7	106.3	157.3								157.3	
GVA	10.9	126.0	441.2	578.1								578.1	
TOT	28.1	408.4	772.5	1209.0	277.5	5.3	166.8	131.3	1.8	524.3	1107.0		

Check

Taxes less subsidies on products

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR							0.5		
MAN							38.2		
CON							8.3		
TTC							3.0		
FBS							9.7		
OSE							2.7		
TOT							62.4		

Use Table for taxes less subsidies on products

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR	0.0	0.0	0.0	0.0	0.4					0.1	0.5	0.5	
MAN	0.2	1.6	6.2	8.0	26.4		0.4	2.3	0.0	1.6	30.7	38.7	-0.5
CON	0.0	0.0	2.0	2.0	0.1		0.1	6.1		0.0	6.3	8.3	
TTC	0.0	-0.1	1.1	1.0	1.7		-0.1	0.5	-0.1	2.0	2.0	3.0	
FBS	0.0	0.0	4.3	4.3	1.5		0.1	3.9		0.0	5.5	9.7	
OSE	0.0	0.0	0.0	0.0	2.1		-0.4	0.7		0.3	2.7	2.7	
TOT	0.2	1.5	13.6	15.3	32.1		0.1	13.5	0.0	1.9	47.6	62.9	-0.5

Use Table for trade and transport margins

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR							11.2		
MAN							111.7		
CON									
TTC							-123.0		
FBS									
OSE							0.1		
TOT							0.0		

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR	0.4	1.7	0.4	2.6	3.1			0.0	0.0	5.6	8.6	11.2	
MAN	0.8	20.0	9.0	29.8	39.9		3.2	5.2	-0.1	34.7	82.9	112.7	-1.0
CON													
TTC	-1.2	-21.7	-9.4	-32.3	-43.1		-3.2	-5.2	0.1	-40.2	-91.7	-124.0	1.0
FBS													
OSE					0.1					0.0	0.1	0.1	
TOT	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0

Supply Table at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	25.5	0.1	0.0	25.7	14.4	40.1			
MAN	1.2	295.0	35.6	331.7	311.9	643.7			
CON	0.1	87.6	4.5	92.2	1.5	93.7			
TTC	0.5	15.2	231.0	246.7	66.4	313.1			
FBS	0.5	8.0	280.8	289.2	54.7	343.9			
OSE	0.3	2.5	220.7	223.5	16.1	239.6			
TOT	28.1	408.4	772.5	1209.0	465.0	1674.0			

Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR	4.8	14.1	1.1	20.0	2.5			0.1	0.0	17.2	19.9	39.8	0.2
MAN	7.9	161.6	47.1	216.6	53.9		5.6	44.2	1.7	322.6	428.0	644.7	-1.0
CON	0.3	26.1	17.4	43.8	0.4		0.5	47.6		2.1	50.6	94.4	-0.7
TTC	1.8	32.5	74.0	108.3	62.4	0.9	3.8	11.7	-0.1	125.8	204.4	312.8	0.3
FBS	2.0	43.2	156.8	202.0	75.1	0.0	3.4	13.7		50.0	142.2	344.2	-0.3
OSE	0.2	3.5	21.2	24.9	51.0	4.4	153.5	0.4	0.2	4.7	214.3	239.2	0.4
TLS	0.2	1.5	13.6	15.3	32.1		0.1	13.5	0.0	1.9	47.6	62.9	
TIC	17.2	282.5	331.3	630.9	277.5	5.3	166.8	131.3	1.8	524.3	1107.0	1737.9	-1.0
OTLS	-0.7	0.1	0.1	-0.5								-0.5	
COE	2.6	59.8	250.6	313.0								313.0	
CFC	3.7	20.4	84.3	108.4								108.4	
NOS	5.3	45.7	106.3	157.3								157.3	
GVA	10.9	126.0	441.2	578.1								578.1	
TOT	28.1	408.4	772.5	1209.0	277.5	5.3	166.8	131.3	1.8	524.3	1107.0		

Check

Imports CIF

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR					14.4				
MAN					311.9				
CON					1.5				
TTC					66.4				
FBS					54.7				
OSE					16.1				
TOT					465.0				

Imports Use Table

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR	0.7	5.6	0.5	6.8	1.2			0.0	0.0	6.4	7.6	14.4	
MAN	1.1	85.6	19.5	106.2	24.0		1.6	19.3	0.1	135.7	180.7	286.9	25.0
CON	0.0	0.5	0.3	0.9				0.7			0.7	1.5	
TTC	0.0	2.7	13.9	16.6	0.7			2.8	0.0	45.8	49.3	65.9	0.5
FBS	0.2	13.8	24.4	38.5	0.2			2.1		13.9	16.2	54.7	
OSE	0.0	0.5	3.9	4.4	11.6		0.0	0.1	0.0		11.7	16.1	
TLS													
TIC	2.1	108.7	62.5	173.3	37.7		1.6	24.9	0.1	201.8	266.2	439.5	25.5

Supply Table for domestic output at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	25.5	0.1	0.0	25.7					
MAN	1.2	295.0	35.6	331.7					
CON	0.1	87.6	4.5	92.2					
TRA	0.5	15.2	231.0	246.7					
TRN	0.5	8.0	280.8	289.2					
COM	0.3	2.5	220.7	223.5					
TOT	28.1	408.4	772.5	1209.0					

Domestic Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR	4.1	8.5	0.6	13.2	1.3			0.1	0.0	10.8	12.2	25.5	0.2
MAN	6.8	75.9	27.7	110.4	29.9		3.9	24.9	1.7	186.9	247.3	357.7	-26.0
CON	0.3	25.6	17.1	43.0	0.4		0.5	46.9		2.1	49.9	92.8	-0.7
TTC	1.7	29.9	60.2	91.8	61.7	0.9	3.8	8.9	-0.1	80.0	155.1	246.9	-0.2
FBS	1.8	29.4	132.4	163.5	74.9	0.0	3.4	11.6		36.1	126.0	289.5	-0.3
OSE	0.2	3.0	17.3	20.5	39.5	4.4	153.5	0.4	0.1	4.7	202.6	223.1	0.4
TLS	0.2	1.5	13.6	15.3	32.1		0.1	13.5	0.0	1.9	47.6	62.9	
TIC	15.1	173.7	268.8	457.6	239.8	5.3	165.2	106.4	1.7	322.5	840.8	1298.4	-26.5
IMP	2.1	108.7	62.5	173.3	37.7		1.6	24.9	0.1	201.8	266.2	439.5	
TOT	17.2	282.5	331.3	630.9	277.5	5.3	166.8	131.3	1.8	524.3	1107.0	1737.9	-26.5
OTLS	-0.7	0.1	0.1	-0.5								-0.5	
COE	2.6	59.8	250.6	313.0								313.0	
CFC	3.7	20.4	84.3	108.4								108.4	
NOS	5.3	45.7	106.3	157.3								157.3	
GVA	10.9	126.0	441.2	578.1								578.1	
TOT	28.1	408.4	772.5	1209.0	277.5	5.3	166.8	131.3	1.8	524.3	1107.0		

Check

Table A11.4: Volume indices for supply and use tables 2011

Supply Table at basic prices, transf. to purchasers' prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	100.8	96.1	87.8	100.8	103.5	101.8	103.6	111.9	102.2
MAN	95.4	104.4	99.4	103.8	103.3	103.6	102.7	97.4	103.1
CON	127.1	103.2	95.6	102.8	101.2	102.8		98.2	102.4
TTC	92.6	98.9	103.8	103.4	101.9	103.1	102.8	95.1	103.2
FBS	99.6	102.2	102.1	102.1	107.4	102.9		98.2	102.8
OSE	95.2	99.1	100.5	100.5	98.3	100.3	94.6	101.9	100.4
TOT	100.4	103.8	102.0	102.6	103.4	102.8	104.2	97.8	102.6

Use Table at purchasers' prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	102.0	103.4	96.5	102.6	101.5			84.8	-9.1	102.0	101.1	101.8	0.5
MAN	99.5	105.6	98.5	103.6	99.4			102.8	112.5	43.1	104.3	103.4	-0.3
CON	100.7	110.7	100.2	105.9	103.3			96.0	101.3	91.6	100.9	103.1	-0.7
TTC	99.1	100.0	101.3	101.1	99.7					104.5	103.4	102.5	0.7
FBS	98.9	104.7	103.0	103.3	100.5			96.2		106.2	102.2	102.9	-0.1
OSE	99.4	101.8	101.4	101.5	101.0	99.8		98.2	72.6	103.1	100.0	100.2	0.2
TIC	100.2	105.5	101.5	103.2	100.1	99.8	99.8	105.3	42.7	104.3	102.4	102.7	0.2
OTLS	137.2	-84.9	-30.6	55.5								55.5	
COE	100.1	100.0	101.0	100.8								100.8	
CFC	101.6	100.9	101.4	101.3								101.3	
NOS	104.4	99.9	105.9	104.0								104.0	
GVA	100.8	100.2	102.3	101.8								101.8	
TOT	100.4	103.8	102.0	102.6	100.1	99.8	99.8	105.3	42.7	104.3	102.4		

Check

Taxes less subsidies on products

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR								111.9	
MAN								97.4	
CON								98.2	
TTC								95.1	
FBS								98.2	
OSE								101.9	
TOT								97.8	

Use Table for taxes less subsidies on products

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	92.9	-40.0	134.6	-375.0	101.4					110.5	103.3	111.9	
MAN	94.3	97.2	97.6	97.5	98.2			100.8	105.1	-100.0	104.0	99.0	-1.3
CON			98.1	98.1	105.9			96.5	98.2		100.0	98.2	
TTC				92.2	98.2							96.7	
FBS	130.0	-14.3	100.4	100.7	102.9			89.0		225.0		96.3	
OSE				240.0	101.0			101.1				101.1	
TOT	96.4	99.7	98.3	98.4	98.7			79.8	98.2	-100.0	103.5	98.7	-0.8

Use Table for trade and transport margins

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR								103.6	
MAN								102.7	
CON									
TTC								102.8	
FBS									
OSE								94.6	
TOT									

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	101.4	104.6	95.9	102.5	101.7			82.4	-50.0	105.4	103.9	103.6	
MAN	97.9	107.5	97.8	104.1	100.7			101.5	118.6	-183.3	105.2	103.4	-0.9
CON													
TTC	99.2	107.3	97.7	104.0	100.8			101.5	118.5	-162.0	105.2	103.5	-0.8
FBS													
OSE					95.3					92.3	94.6	94.6	
TOT													

Supply Table at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	100.8	96.1		100.8	103.5	101.8			
MAN	95.4	104.4	99.4	103.8	103.3	103.6			
CON	127.1	103.2	95.6	102.8	101.2	102.8			
TTC	92.6	98.9	103.8	103.4	101.9	103.1			
FBS	99.6	102.2	102.1	102.1	107.4	102.9			
OSE		99.1		100.5	98.3	100.3			
TOT	100.4	103.8	102.0	102.6	103.4	102.8			

Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	102.1	103.1	95.9	102.4	101.3			85.1	-7.1	100.9	99.9	101.1	0.6
MAN	99.8	105.5	98.7	103.7	99.0			103.7	112.3	45.7	104.2	103.8	-0.2
CON	100.7	110.7	100.4	106.3	103.0			96.0	101.7	91.6	101.2	103.5	-0.7
TTC	99.1	104.7	101.0	102.1	100.5			101.0	109.7	-102.6	104.7	103.5	0.1
FBS	98.7	104.7	103.1	103.4	100.5			96.4	102.0		106.2	102.5	-0.1
OSE	99.4	101.6	101.4	101.4	101.0	99.8		93.7	72.6	103.2	100.0	100.2	0.2
TLN	96.4	99.7	98.3	98.4	98.7			79.8	98.2	-100.0	103.5	98.7	
TIC	100.2	105.5	101.5	103.2	100.1	99.8	99.8	105.3	42.7	104.3	102.4	102.7	-0.1
OTLS	137.2	-84.9	-30.6	55.5								55.5	
COE	100.1	100.0	101.0	100.8								100.8	
CFC	101.6	100.9	101.4	101.3								101.3	
NOS	104.4	99.9	105.9	104.0								104.0	
GVA	100.8	100.2	102.3	101.8								101.8	
TOT	100.4	103.8	102.0	102.6	100.1	99.8	99.8	105.3	42.7	104.3	102.4		

Check

Imports CIF

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR					103.5				
MAN					103.3				
CON					101.2				
TTC					101.9				
FBS					107.4				
OSE					98.3				
TOT					103.4				

Imports Use Table

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	100.9	106.3	88.2	104.2	105.0			78.8	-32.2	104.5	102.9	103.5	
MAN	104.1	106.6	99.7	105.2	100.4			92.0	117.4	3.1	86.4	90.0	8.3
CON	100.0	110.6	96.6	104.5				97.4				97.4	
TTC	102.3	99.0		100.2						101.7	101.4	101.1	0.8
FBS	98.8	103.5	107.7	106.1				92.1			110.8	107.4	
OSE	85.7	104.5	101.8	102.1	97.6				57.8	97.2	97.0	98.3	
TLN													
TIC	102.3	105.9	102.9	104.8	99.7			92.2	114.7	6.3	91.3	93.6	5.7

Supply Table for domestic output at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	100.8	96.1		100.8					
MAN	95.4	104.4	99.4	103.8					
CON	127.1	103.2	95.6	102.8					
TTC	92.6	98.9	103.8	103.4					
FBS	99.6	102.2	102.1	102.1					
OSE		99.1		100.5					
TOT	100.4	103.8	102.0	102.6					

Domestic Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	102.3	101.0	103.0	101.5	98.0			86.7	19.8	98.9	98.1	99.9	0.9
MAN	99.1	104.3	98.0	102.3	98.0			109.4	108.6	79.6	122.5	116.8	-8.1
CON	100.7	110.7	100.5	106.3	103.0			96.0	101.8	91.6	101.3	103.5	-0.8
TTC	99.0	105.3	101.1	102.4	100.5			101.0	114.6	-110.0	106.6	104.2	-0.1
FBS	98.7	105.3	102.3	102.8	100.5			96.4	98.3		105.3	101.5	-0.1
OSE	100.0	101.2	101.3	101.2	102.1	99.8		93.7	95.5	103.2	100.2	100.3	0.2
TLN	96.4	99.7	98.3	98.4	98.7			79.8	98.2	-100.0	103.5	98.7	
TIC	99.9	105.3	101.2	102.7	100.2	99.8	99.8	103.3	72.5	114.6	105.5	104.5	-8.0
IMP	102.3	105.9	102.9	104.8	99.7			92.2	114.7	6.3	91.3	93.6	97.7
TOT	100.2	105.5	101.5	103.2	100.1	99.8	99.8	105.3	42.7	104.3	102.4	102.7	-8.0
OTLS	137.2	-84.9	-30.6	55.5								55.5	
COE	100.1	100.0	101.0	100.8								100.8	
CFC	101.6	100.9	101.4	101.3								101.3	
NOS	104.4	99.9	105.9	104.0								104.0	
GVA	100.8	100.2	102.3	101.8								101.8	
TOT	100.4	103.8	102.0	102.6	100.1	99.8	99.8	105.3	42.7	104.3	102.4		

Table A11.5: Supply and use tables 2010 in current prices

Supply Table at basic prices, transf. to purchasers' prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	25.3	0.2	0.0	25.5	13.9	39.4	10.8	0.4	50.6
MAN	1.2	282.6	35.8	319.6	301.8	621.4	108.8	39.2	769.4
CON	0.1	84.9	4.7	89.7	1.5	91.2		8.4	99.6
TTC	0.6	15.4	222.6	238.5	65.1	303.6	-119.7	3.1	187.1
FBS	0.5	7.8	275.0	283.3	50.9	334.2		9.9	344.1
OSE	0.3	2.5	219.6	222.4	16.4	238.8	0.1	2.7	241.6
TOT	28.0	393.3	757.6	1178.9	449.7	1628.6	0.0	63.8	1692.3

Use Table at purchasers' prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	5.1	15.3	1.6	22.0	5.8				0.2	0.2	22.5	28.6	50.6
MAN	8.9	173.4	63.3	245.6	121.0			8.9	46.0	3.8	344.1	523.7	769.4
CON	0.3	23.6	19.4	43.2	0.5			0.6	53.0		2.2	56.4	99.6
TTC	0.6	10.8	64.8	76.1	21.0	0.9	0.5	6.7	0.0	81.8	110.9	187.1	
FBS	2.1	41.2	156.3	199.6	76.2	0.0	3.7	17.5	47.1		47.1	144.5	344.1
OSE	0.2	3.4	20.9	24.5	52.7	4.5	153.6	1.2	0.3	4.9	217.0	241.6	
TIC	17.1	267.7	326.4	611.2	277.2	5.3	167.2	124.6	4.3	502.5	1081.2	1692.3	
OTLS	-0.5	-0.1	-0.3	-1.0								-1.0	
COE	2.6	59.8	248.1	310.5								310.5	
CFC	3.7	20.2	83.1	107.0								107.0	
NOS	5.1	45.8	100.4	151.3								151.3	
GVA	10.8	125.7	431.2	567.8								567.8	
TOT	28.0	393.3	757.6	1178.9	277.2	5.3	167.2	124.6	4.3	502.5	1081.2		

Check

Taxes less subsidies on products

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR								0.4	
MAN								39.2	
CON								8.4	
TTC								3.1	
FBS								9.9	
OSE								2.7	
TOT								63.8	

Use Table for taxes less subsidies on products

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	0.0	0.0	0.0	0.0	0.4					0.1	0.5	0.4	
MAN	0.2	1.6	6.3	8.2	26.9		0.4	2.2	0.0	1.6	31.0	39.2	
CON	0.0	2.0	2.0	2.0	0.1		0.1	6.3		0.0	6.4	8.4	
TTC	0.0	-0.1	1.1	1.1	1.7	-0.1	0.5	-0.1		2.0	2.0	3.1	
FBS	0.0	0.0	4.3	4.3	1.5		0.1	4.1		0.0	5.7	9.9	
OSE	0.0	0.0	0.0	0.0	2.1	-0.4	0.7	0.3		2.7	2.7		
TOT	0.2	1.5	13.8	15.5	32.5	0.1	13.8	0.0	1.8	48.2	63.8		

Check

Trade and transport margins

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR							10.8		
MAN							108.8		
CON									
TTC							-119.7		
FBS									
OSE							0.1		
TOT							0.0		

Use Table for trade and transport margins

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	0.4	1.6	0.4	2.5	3.0				0.0	0.0	5.3	8.3	10.8
MAN	0.8	18.6	9.2	28.6	39.6		3.2	4.4	0.0	32.9	80.2	108.8	
CON													
TTC	-1.2	-20.2	-9.7	-31.1	-42.7		-3.2	-4.4	-0.1	-38.2	-88.6	-119.7	
FBS													
OSE					0.1					0.0	0.1	0.1	
TOT					0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0

Check

Supply Table at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	25.3	0.2	0.0	25.5	13.9	39.4			
MAN	1.2	282.6	35.8	319.6	301.8	621.4			
CON	0.1	84.9	4.7	89.7	1.5	91.2			
TTC	0.6	15.4	222.6	238.5	65.1	303.6			
FBS	0.5	7.8	275.0	283.3	50.9	334.2			
OSE	0.3	2.5	219.6	222.4	16.4	238.8			
TOT	28.0	393.3	757.6	1178.9	449.7	1628.6			

Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	4.7	13.7	1.2	19.5	2.5				0.2	0.2	17.1	19.9	39.4
MAN	7.9	153.2	47.8	208.9	54.5		5.4	39.4	3.8	309.6	412.5	621.4	
CON	0.3	23.6	17.4	41.2	0.4		0.5	46.8		2.2	50.0	91.2	
TTC	1.8	31.1	73.3	106.2	62.1	0.9	3.7	10.7	0.1	120.1	197.5	303.6	
FBS	2.1	41.2	152.1	195.4	74.8	0.0	3.5	13.4		47.1	138.8	334.2	
OSE	0.2	3.4	20.9	24.5	50.5	4.5	154.0	0.5	0.3	4.6	214.3	238.8	
TLS	0.2	1.5	13.8	15.5	32.5		0.1	13.8	0.0	1.8	48.2	63.8	
TIC	17.1	267.7	326.4	611.2	277.2	5.3	167.2	124.6	4.3	502.5	1081.2	1692.3	
OTLS	-0.5	-0.1	-0.3	-1.0								-1.0	
COE	2.6	59.8	248.1	310.5								310.5	
CFC	3.7	20.2	83.1	107.0								107.0	
NOS	5.1	45.8	100.4	151.3								151.3	
GVA	10.8	125.7	431.2	567.8								567.8	
TOT	28.0	393.3	757.6	1178.9	277.2	5.3	167.2	124.6	4.3	502.5	1081.2		

Check

Imports CIF

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR					13.9				
MAN					301.8				
CON					1.5				
TTC					65.1				
FBS					50.9				
OSE					16.4				
TOT					449.7				

Imports Use Table

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	0.7	5.3	0.6	6.5	1.2				0.0	0.1	6.1	7.4	13.9
MAN	1.1	80.4	19.5	101.0	23.9		1.8	16.4	1.7	157.1	200.9	301.8	
CON	0.0	0.5	0.4	0.8				0.7			0.7	1.5	
TTC	0.0	2.7	13.8	16.6	0.7			2.9	0.0	45.0	48.6	65.1	
FBS	0.2	13.3	22.7	36.3	0.2			1.7		12.8	14.6	50.9	
OSE	0.0	0.5	3.8	4.3	11.8				0.2	0.0	12.1	16.4	
TLS													
TIC	2.0	102.6	60.8	165.4	37.8		1.8	21.7	1.9	221.0	284.2	449.7	

Check

Supply Table for domestic output at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	25.3	0.2	0.0	25.5					
MAN	1.2	282.6	35.8	319.6					
CON	0.1	84.9	4.7	89.7					
TTC	0.6	15.4	222.6	238.5					
FBS	0.5	7.8	275.0	283.3					
OSE	0.3	2.5	219.6	222.4					
TOT	28.0	393.3	757.6	1178.9					

Domestic Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	4.0	8.4	0.6	13.0	1.3				0.1	0.1	10.9	12.5	25.5
MAN	6.8	72.8	28.2	107.9	30.6		3.6	22.9	2.1	152.5	211.7	319.6	
CON	0.3	23.1	17.0	40.4	0.4		0.5	46.1		2.2	49.3	89.7	
TTC	1.7	28.4	59.5	89.6	61.4	0.9	3.7	7.8	0.1	75.1	148.9	238.5	
FBS	1.8	27.9	129.4	159.1	74.5	0.0	3.5	11.8		34.3	124.2	283.3	
OSE	0.2	2.9	17.1	20.2	38.7	4.5	154.0	0.5	0.1	4.5	202.2	222.4	
TLS	0.2	1.5	13.8	15.5	32.5		0.1	13.8	0.0	1.8	48.2	63.8	
TIC	15.1	165.0	265.6	445.7	239.4	5.3	165.5	102.9	2.4	281.5	796.9	1242.7	
IMP	2.0	102.6	60.8	165.4	37.8		1.8	21.7	1.9	221.0	284.2	449.7	
TOT	17.1	267.7	326.4	611.2	277.2	5.3	167.2	124.6	4.3	502.5	1081.2	1692.3	
OTLS	-0.5	-0.1	-0.3	-1.0								-1.0	
COE	2.6	59.8	248.1	310.5								310.5	
CFC	3.7	20.2	83.1	107.0								107.0	
NOS	5.1	45.8	100.4	151.3								151.3	
GVA	10.8	125.7	431.2	567.8								567.8	
TOT	28.0	393.3	757.6	1178.9	277.2	5.3	167.2	124.6	4.3	502.5	1081.2		

Check

2. Balanced supply and use tables (tables A11.6–A11.10)

Table A11.6: Supply and use tables 2011 in current prices

Supply Table at basic prices, transf. to purchasers' prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	25.8	0.2	0.0	26.0	15.4	41.3	10.9	0.5	52.8
MAN	1.3	318.5	35.7	355.5	336.8	692.3	113.0	39.4	844.7
CON	0.1	89.0	4.5	93.6	1.6	95.1		8.3	103.4
TTC	0.5	15.1	231.6	247.2	72.9	320.1	-124.0	3.0	199.1
FBS	0.5	8.0	282.0	290.5	55.2	345.7		9.1	354.9
OSE	0.3	2.5	222.9	225.7	16.6	242.3		0.1	245.4
TOT	28.4	433.3	776.7	1238.4	498.6	1737.0	0.0	63.3	1800.3

Use Table at purchasers' prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	5.5	17.4	1.5	24.4	5.9				0.1	-0.1	22.4	28.4	52.8
MAN	10.1	202.1	65.4	277.7	122.8			9.1	51.5	1.4	382.1	567.0	844.7
CON	0.3	26.2	19.8	46.3	0.5			0.6	54.0		2.1	57.1	103.4
TTC	0.6	10.8	66.2	77.5	21.2	0.9	0.5	6.7	0.0	92.3	121.6	199.1	
FBS	2.1	43.5	161.5	207.0	78.0	0.0	3.6	16.8		49.5	147.8	354.9	
OSE	0.2	3.5	22.2	25.9	55.1	4.6	153.4	1.2	0.2	5.1	219.5	245.4	
TIC	18.8	303.5	336.5	658.8	283.5	5.5	167.2	130.4	1.5	553.5	1141.5	1800.3	
OTLS	-0.7	0.2	0.3	-0.3								-0.3	
COE	2.7	60.9	254.4	318.0								318.0	
CFC	3.7	20.5	82.9	107.1								107.1	
NOS	4.1	48.2	102.5	154.7								154.7	
GVA	9.7	129.8	440.1	579.6								579.6	
TOT	28.4	433.3	776.7	1238.4	283.5	5.5	167.2	130.4	1.5	553.5	1141.5		

Check

Taxes less subsidies on products

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR								0.5	0.5
MAN								39.4	39.4
CON								8.3	8.3
TTC								3.0	3.0
FBS								9.1	9.1
OSE								3.0	3.0
TOT								63.3	63.3

Use Table for taxes less subsidies on products

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	0.0	0.0	0.0	0.1	0.4					0.1	0.5	0.5	
MAN	0.2	1.6	6.4	8.3	26.8			0.4	2.2	0.0	1.7	31.1	39.4
CON	0.0	0.0	2.0	2.0	0.0			0.1	6.1	0.0	6.2	8.3	
TTC	0.0	-0.1	1.1	1.0	1.7			-0.1	0.4	-0.1	2.0	3.0	
FBS	0.0	0.0	4.4	4.4	1.6			0.1	3.0	0.0	4.7	9.1	
OSE	0.0	0.0	0.1	0.1	2.3			-0.4	0.7	0.3	2.9	3.0	
TOT	0.2	1.6	14.0	15.9	32.9			0.1	12.5	0.0	2.0	47.5	63.3

Use Table for trade and transport margins

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR							10.9		10.9
MAN							113.0		113.0
CON									
TTC							-124.0		-124.0
FBS									
OSE							0.1		0.1
TOT							0.0		0.0

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	0.4	1.6	0.4	2.5	3.2				0.0	0.0	5.2	8.4	10.9
MAN	0.9	20.4	9.2	30.5	39.9			3.1	5.1	-0.1	34.5	82.5	113.0
CON													
TTC	-1.3	-22.0	-9.6	-33.0	-43.1			-3.1	-5.1	0.1	-39.7	-91.0	-124.0
FBS													
OSE					0.1						0.1	0.1	
TOT					0.0				0.0	0.0	0.0	0.0	0.0

Supply Table at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	25.8	0.2	0.0	26.0	15.4	41.3			
MAN	1.3	318.5	35.7	355.5	336.8	692.3			
CON	0.1	89.0	4.5	93.6	1.6	95.1			
TTC	0.5	15.1	231.6	247.2	72.9	320.1			
FBS	0.5	8.0	282.0	290.5	55.2	345.7			
OSE	0.3	2.5	222.9	225.7	16.6	242.3			
TOT	28.4	433.3	776.7	1238.4	498.6	1737.0			

Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	5.1	15.7	1.1	21.9	2.4				0.1	-0.1	17.1	19.5	41.3
MAN	9.0	180.1	49.8	238.9	56.1			5.6	44.3	1.5	345.9	453.4	692.3
CON	0.3	26.2	17.7	44.3	0.4			0.5	47.9		2.1	50.9	95.1
TTC	1.8	32.9	74.8	109.5	62.6	0.9	3.7	11.4	-0.1	132.1	210.6	320.1	
FBS	2.1	43.5	157.1	202.6	76.4	0.0	3.5	13.8		49.5	143.1	345.7	
OSE	0.2	3.5	22.1	25.8	52.7	4.6	153.8	0.4	0.2	4.8	216.5	242.3	
TLS	0.2	1.6	14.0	15.9	32.9			0.1	12.5	0.0	2.0	47.5	63.3
TIC	18.8	303.5	336.5	658.8	283.5	5.5	167.2	130.4	1.5	553.5	1141.5	1800.3	
OTLS	-0.7	0.2	0.3	-0.3								-0.3	
COE	2.7	60.9	254.4	318.0								318.0	
CFC	3.7	20.5	82.9	107.1								107.1	
NOS	4.1	48.2	102.5	154.7								154.7	
GVA	9.7	129.8	440.1	579.6								579.6	
TOT	28.4	433.3	776.7	1238.4	283.5	5.5	167.2	130.4	1.5	553.5	1141.5		

Check

Imports CIF

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR					15.4				
MAN					336.8				
CON					1.6				
TTC					72.9				
FBS					55.2				
OSE					16.6				
TOT					498.6				

Imports Use Table

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	0.8	6.4	0.5	7.7	1.2				0.0	-0.1	6.6	7.7	15.4
MAN	1.3	98.0	20.9	120.2	24.7			1.6	19.2	-0.2	171.3	216.7	336.8
CON	0.0	0.5	0.4	0.9					0.7		0.7	1.6	
TTC	0.0	2.7	14.0	16.8	0.7				2.7	0.0	52.7	56.1	72.9
FBS	0.2	14.0	24.5	38.8	0.2				2.2	14.1	16.5	55.2	
OSE	0.0	0.5	3.9	4.5	12.0			0.0	0.1	0.0	12.2	16.6	
TLS													
TIC	2.3	122.1	64.2	188.7	38.9			1.6	24.8	-0.1	244.7	309.9	498.6

Supply Table for domestic output at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	25.8	0.2	0.0	26.0					
MAN	1.3	318.5	35.7	355.5					
CON	0.1	89.0	4.5	93.6					
TTC	0.5	15.1	231.6	247.2					
FBS	0.5	8.0	282.0	290.5					
OSE	0.3	2.5	222.9	225.7					
TOT	28.4	433.3	776.7	1238.4					

Domestic Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	4.3	9.3	0.6	14.2	1.2				0.1	0.0	10.5	11.7	26.0
MAN	7.8	82.1	28.9	118.8	31.4			4.0	25.1	1.6	174.6	236.7	355.5
CON	0.3	25.7	17.4	43.4	0.4			0.5	47.2		2.1	50.2	93.6
TTC	1.8	30.2	60.7	92.7	61.9	0.9	3.7	8.6	-0.1	79.4	154.5	247.2	
FBS	1.8	29.5	132.6	163.9	76.2	0.0	3.5	11.6		35.4	126.6	290.5	
OSE	0.2	3.0	18.1	21.3	40.6	4.6	153.8	0.4	0.1	4.8	204.3	225.7	
TLS	0.2	1.6	14.0	15.9	32.9			0.1	12.5	0.0	2.0	47.5	63.3
TIC	16.4	181.4	272.3	470.2	244.6	5.5	165.5	105.6	1.6	308.7	831.6	1301.8	
IMP	2.3	122.1	64.2	188.7	38.9			1.6	24.8	-0.1	244.7	309.9	498.6
TOT	18.8	303.5	336.5	658.8	283.5	5.5	167.2	130.4	1.5	553.5	1141.5	1800.3	
OTLS	-0.7	0.2	0.3	-0.3								-0.3	
COE	2.7	60.9	254.4	318.0								318.0	
CFC	3.7	20.5	82.9	107.1								107.1	
NOS	4.1	48.2	102.5	154.7								154.7	
GVA	9.7	129.8	440.1	579.6								579.6	
TOT	28.4	433.3	776.7	1238.4	283.5	5.5	167.2	130.4	1.5	553.5	1141.5		

Check

Table A11.7: Price indices for supply and use tables 2011

Supply Table at basic prices, transf. to purchasers' prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	102.0	104.1		102.0	106.9	103.8	97.5	107.6	102.4
MAN	107.5	108.0	100.5	107.2	108.0	107.6	100.2	101.9	106.2
CON	103.4	100.5	100.5	100.5	101.6	100.5		100.1	100.5
TTC	101.1	99.2	100.3	100.2	110.7	102.4	100.0	101.4	104.0
FBS	102.9	100.9	100.4	100.4	101.0	100.5		93.6	100.3
OSE		100.2		101.0	103.4	101.1	99.1	110.2	101.2
TOT	102.2	105.8	100.5	102.4	107.3	103.7		100.8	103.6

Use Table at purchasers' prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	105.8	109.9	98.7	108.2	100.4			94.7	568.8	97.7	98.0	102.4	
MAN	114.1	110.4	105.0	109.2	102.1		99.5	99.6	85.7	106.8	104.9	106.2	
CON	101.4	100.5	101.8	101.0	97.4		97.1	100.0		102.0	100.0	100.5	
TTC	100.9	100.1	100.8	100.7	102.2					108.0	106.2	104.0	
FBS	101.4	100.7	100.4	100.5	101.7		101.8			99.0	100.1	100.3	
OSE	102.4	101.7	104.3	103.9	102.7	103.8		101.7	99.5	101.8	100.9	101.2	
TIC	109.3	107.5	101.7	104.5	102.1	103.3	100.2	99.1	83.0	105.8	103.2	103.6	
OTLS	98.0	201.1	269.9	48.3								48.3	
COE	102.7	101.9	101.5	101.6								101.6	
CFC	99.1	100.6	98.4	98.8								98.8	
NOS	79.8	103.2	96.2	97.7								97.7	
GVA	90.8	102.2	99.7	100.1								100.1	
TOT	102.2	105.8	100.5	102.4	102.1	103.3	100.2	99.1	83.0	105.8	103.2		

Check

Taxes less subsidies on products

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR							107.6		
MAN							101.9		
CON							100.1		
TTC							101.4		
FBS							93.6		
OSE							110.2		
TOT							100.8		

Use Table for taxes less subsidies on products

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	92.3	612.5	91.4	230.0	100.8					96.2	99.8	107.6	
MAN	103.7	106.1	103.9	104.3	101.7		98.7	93.8	-100.0	104.4	101.3	101.9	
CON			102.2	102.1	90.7		95.1	99.6		100.0	99.4	100.1	
TTC				99.4	103.3						102.4	101.4	
FBS	123.1	100.0	102.7	102.8	105.7		103.8			44.4	86.4	93.6	
OSE				241.7	109.2			101.1			108.5	110.2	
TOT	105.1	108.7	103.3	103.9	102.5		102.3	92.5	-100.0	105.0	99.8	100.8	

Check

Trade and transport margins

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR							97.5		
MAN							100.2		
CON									
TTC							100.0		
FBS									
OSE							99.1		
TOT									

Use Table for trade and transport margins

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	90.4	96.7	100.7	96.3	104.9				85.7	100.0	94.0	97.9	97.5
MAN	113.7	102.0	102.3	102.4	99.8			97.9	97.5	93.5	99.5	100.2	
CON													
TTC	105.0	101.6	102.2	101.9	100.2			97.9	97.5	93.8	98.8	99.3	100.0
FBS													
OSE					97.6					104.2	99.1	99.1	
TOT													

Check

Supply Table at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	102.0	104.1		102.0	106.9	103.8			103.8
MAN	107.5	108.0	100.5	107.2	108.0	107.6			107.6
CON	103.4	100.5	100.5	100.5	101.6	100.5			100.5
TTC	101.1	99.2	100.3	100.2	110.7	102.4			102.4
FBS	102.9	100.9	100.4	100.4	101.0	100.5			100.5
OSE		100.2		101.0	103.4	101.1			101.1
TOT	102.2	105.8	100.5	102.4	107.3	103.7			103.7

Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	107.2	111.2	98.1	109.5	94.9				95.6	725.0	98.9	98.0	103.8
MAN	114.4	111.5	105.6	110.3	104.0		100.5	100.2	85.7	107.6	106.2	107.6	
CON	101.4	100.5	101.7	101.0	98.3		97.4	100.0		102.0	100.1	100.5	
TTC	103.8	101.1	101.0	101.1	100.8		98.5	97.0	93.7	105.0	103.1	102.4	
FBS	101.2	100.7	100.4	100.5	101.7		101.7	100.6		99.0	100.6	100.5	
OSE	102.4	101.8	104.1	103.7	102.4	103.8		102.5	99.5	101.5	100.8	101.1	
TLS	105.1	108.7	103.3	103.9	102.5		102.3	92.5	-100.0	105.0	99.8	100.8	
TIC	109.3	107.5	101.7	104.5	102.1	103.3	100.2	99.1	83.0	105.8	103.2	103.6	
OTLS	98.0	201.1	269.9	48.3								48.3	
COE	102.7	101.9	101.5	101.6								101.6	
CFC	99.1	100.6	98.4	98.8								98.8	
NOS	79.8	103.2	96.2	97.7								97.7	
GVA	90.8	102.2	99.7	100.1								100.1	
TOT	102.2	105.8	100.5	102.4	102.1	103.3	100.2	99.1	83.0	105.8	103.2		

Check

Imports CIF

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR					106.9				
MAN					108.0				
CON					101.6				
TTC					110.7				
FBS					101.0				
OSE					103.4				
TOT					107.3				

Imports Use Table

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	113.0	114.5	98.0	113.2	100.1				100.0	207.1	102.1	101.4	106.9
MAN	112.6	114.4	107.2	113.1	102.8		99.6	99.5	-311.5	106.6	105.3	108.0	
CON	100.0	101.0	102.0	101.4				101.8			101.8	101.6	
TTC	100.0	100.5		101.2						115.2	113.9	110.7	
FBS	97.0	101.4	100.4	100.8	106.7						101.6	101.0	
OSE	100.0	100.8	101.4	101.4	104.2				99.0	97.1	104.1	103.4	
TLS													
TIC	110.7	112.3	102.8	108.9	103.2		99.6	99.4	-100.8	107.9	106.4	107.3	

Check

Supply Table for domestic output at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	102.0	104.1		102.0					
MAN	107.5	108.0	100.5	107.2					
CON	103.4	100.5	100.5	100.5					
TTC	101.1	99.2	100.3	100.2					
FBS	102.9	100.9	100.4	100.4					
OSE		100.2		101.0					
TOT	102.2	105.8	100.5	102.4					

Domestic Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	106.2	109.0	98.2	107.6	90.0				94.6	-181.3	97.0	95.9	102.0
MAN	114.7	108.1	104.5	107.6	105.0		100.8	100.7	98.0	108.5	106.9	107.2	
CON	101.4	100.5	101.7	101.0	98.3		97.4	100.0		102.0	100.0	100.5	
TTC	103.9	101.1	100.9	101.0	100.8		98.5	97.2	93.5	99.2	99.7	100.2	
FBS	101.8	100.4	100.4	100.4	101.7		101.7	100.5		98.0	100.5	100.4	
OSE	102.5	101.9	104.7	104.2	101.9	103.8		102.5	100.0	101.5	100.7	101.0	
TLS	105.1	108.7	103.3	103.9	102.5		102.3	92.5	-100.0	105.0	99.8	100.8	
TIC	109.1	104.4	101.4	102.8	101.9	103.3	100.2	99.0	96.1	104.1	102.0	102.3	
IMP	110.7	112.3	102.8	108.9	103.2		99.6	99.4	-100.8	107.9	106.4	107.3	
TOT	109.3	107.5	101.7	104.5	102.1	103.3	100.2	99.1	83.0	105.8	103.2	103.6	
OTLS	98.0	201.1	269.9	48.3								48.3	
COE	102.7	101.9	101.5	101.6								101.6	
CFC	99.1	100.6	98.4	98.8								98.8	
NOS	79.8	103.2	96.2	97.7								97.7	
GVA	90.8	102.2	99.7	100.1								100.1	
TOT	102.2	105.8	100.5	102.4	102.1	103.3	100.2	99.1	83.0	105.8	103.2		

Check

Table A11.8: Supply and use tables 2011 at previous years' prices

Supply Table at basic prices, transf. to purchasers' prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	25.3	0.1	0.0	25.5	14.4	39.8	11.2	0.5	51.5
MAN	1.2	295.0	35.6	331.7	311.9	643.7	112.7	38.7	795.1
CON	0.1	88.6	4.5	93.1	1.5	94.7		8.3	102.9
TTC	0.5	15.2	231.0	246.7	65.9	312.6	-124.0	3.0	191.5
FBS	0.5	8.0	280.8	289.2	54.7	343.9		9.7	353.7
OSE	0.3	2.5	220.7	223.5	16.1	239.6		0.1	242.4
TOT	27.8	409.4	772.5	1209.8	464.5	1674.3	0.0	62.9	1737.1

Use Table at purchasers' prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTin	TOTpp	Check
AGR	5.2	15.8	1.6	22.6	5.9				0.2	0.0	22.9	29.0	51.5
MAN	8.9	183.1	62.3	254.4	120.3			9.2	51.7	1.6	357.9	540.7	795.1
CON	0.3	26.1	19.4	45.8	0.5		0.6	54.0		2.1	57.1	102.9	
TTC	0.6	10.8	65.6	77.0	20.8	0.9	0.5	6.9	0.0	85.5	114.5	191.5	
FBS	2.0	43.2	160.8	206.0	76.6	0.0	3.5	17.6		50.0	147.7	353.7	
OSE	0.2	3.5	21.2	24.9	53.6	4.4	153.1	1.1	0.2	5.0	217.5	242.4	
TIC	17.2	282.5	331.0	630.6	277.7	5.3	166.8	131.6	1.8	523.3	1106.5	1737.1	
OTLS	-0.7	0.1	0.1	-0.5								-0.5	
COE	2.6	59.8	250.6	313.0								313.0	
CFC	3.7	20.4	84.3	108.4								108.4	
NOS	5.1	46.7	106.6	158.4								158.4	
GVA	10.7	127.0	441.5	579.2								579.2	
TOT	27.8	409.4	772.5	1209.8	277.7	5.3	166.8	131.6	1.8	523.3	1106.5		

Check

Taxes less subsidies on products

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR								0.5	
MAN								38.7	
CON								8.3	
TTC								3.0	
FBS								9.7	
OSE								2.7	
TOT								62.9	

Use Table for taxes less subsidies on products

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTin	TOTpp	Check
AGR	0.0	0.0	0.0	0.0	0.4					0.1	0.5	0.5	
MAN	0.2	1.6	6.2	8.0	26.4		0.4	2.3	0.0	1.6	30.7	38.7	
CON	0.0	2.0	2.0	2.0	0.1		0.1	6.1		0.0	6.3	8.3	
TTC	0.0	-0.1	1.1	1.0	1.7		-0.1	0.5		-0.1	2.0	3.0	
FBS	0.0	0.0	4.3	4.3	1.5		0.1	3.9		0.0	5.5	9.7	
OSE	0.0	0.0	0.0	0.0	2.1		-0.4	0.7		0.3	2.7	2.7	
TIC	0.2	1.5	13.6	15.3	32.1		0.1	13.5	0.0	1.9	47.6	62.9	

Check

Trade and transport margins

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR							11.2		
MAN							112.7		
CON									
TTC							-124.0		
FBS									
OSE							0.1		
TOT							0.0		

Use Table for trade and transport margins

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTin	TOTpp	Check
AGR	0.4	1.7	0.4	2.6	3.1				0.0	0.0	5.6	8.6	11.2
MAN	0.8	20.0	9.0	29.8	39.9			3.2	5.2	-0.1	34.7	82.9	112.7
CON													
TTC	-1.2	-21.7	-9.4	-32.3	-43.1		-3.2	-5.2	0.1	-40.2	-91.7	-124.0	
FBS													
OSE										0.0	0.1	0.1	
TOT	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0

Check

Supply Table at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	25.3	0.1	0.0	25.5	14.4	39.8			
MAN	1.2	295.0	35.6	331.7	311.9	643.7			
CON	0.1	88.6	4.5	93.1	1.5	94.7			
TTC	0.5	15.2	231.0	246.7	65.9	312.6			
FBS	0.5	8.0	280.8	289.2	54.7	343.9			
OSE	0.3	2.5	220.7	223.5	16.1	239.6			
TOT	27.8	409.4	772.5	1209.8	464.5	1674.3			

Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTin	TOTpp	Check
AGR	4.8	14.1	1.1	20.0	2.5				0.1	0.0	17.2	19.9	39.8
MAN	7.9	161.6	47.1	216.6	53.9		5.6	44.2	1.7	321.6	427.0	643.7	
CON	0.3	26.1	17.4	43.8	0.4		0.5	47.9		2.1	50.9	94.7	
TTC	1.8	32.5	74.0	108.3	62.2	0.9	3.8	11.7	-0.1	125.8	204.2	312.6	
FBS	2.0	43.2	156.5	201.7	75.1	0.0	3.4	13.7		50.0	142.2	343.9	
OSE	0.2	3.5	21.2	24.9	51.4	4.4	153.5	0.4	0.2	4.7	214.7	239.6	
TLS	0.2	1.5	13.6	15.3	32.1		0.1	13.5	0.0	1.9	47.6	62.9	
TIC	17.2	282.5	331.0	630.6	277.7	5.3	166.8	131.6	1.8	523.3	1106.5	1737.1	
OTLS	-0.7	0.1	0.1	-0.5								-0.5	
COE	2.6	59.8	250.6	313.0								313.0	
CFC	3.7	20.4	84.3	108.4								108.4	
NOS	5.1	46.7	106.6	158.4								158.4	
GVA	10.7	127.0	441.5	579.2								579.2	
TOT	27.8	409.4	772.5	1209.8	277.7	5.3	166.8	131.6	1.8	523.3	1106.5		

Check

Imports CIF

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR					14.4				
MAN					311.9				
CON					1.5				
TTC					65.9				
FBS					54.7				
OSE					16.1				
TOT					464.5				

Imports Use Table

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTin	TOTpp	Check
AGR	0.7	5.6	0.5	6.8	1.2				0.0	0.0	6.4	7.6	14.4
MAN	1.1	85.6	19.5	106.2	24.0		1.6	19.3	0.1	160.7	205.7	311.9	
CON	0.0	0.5	0.3	0.9				0.7			0.7	1.5	
TTC	0.0	2.7	13.9	16.6	0.7			2.8	0.0	45.8	49.3	65.9	
FBS	0.2	13.8	24.4	38.5	0.2			2.1		13.9	16.2	54.7	
OSE	0.0	0.5	3.9	4.4	11.6			0.0	0.1	0.0	11.7	16.1	
TLS													
TIC	2.1	108.7	62.5	173.3	37.7		1.6	24.9	0.1	226.8	291.2	464.5	

Check

Supply Table for domestic output at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	25.3	0.1	0.0	25.5					
MAN	1.2	295.0	35.6	331.7					
CON	0.1	88.6	4.5	93.1					
TRA	0.5	15.2	231.0	246.7					
TRN	0.5	8.0	280.8	289.2					
COM	0.3	2.5	220.7	223.5					
TOT	27.8	409.4	772.5	1209.8					

Domestic Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTin	TOTpp	Check
AGR	4.1	8.5	0.6	13.2	1.3				0.1	0.0	10.8	12.2	25.5
MAN	6.8	75.9	27.7	110.4	29.9		3.9	24.9	1.7	160.9	221.3	331.7	
CON	0.3	25.6	17.1	43.0	0.4		0.5	47.2		2.1	50.2	93.1	
TTC	1.7	29.9	60.2	91.8	61.5	0.9	3.8	8.9	-0.1	80.0	154.9	246.7	
FBS	1.8	29.4	132.1	163.2	74.9	0.0	3.4	11.6		36.1	126.0	289.2	
OSE	0.2	3.0	17.3	20.5	39.9	4.4	153.5	0.4	0.1	4.7	203.0	223.5	
TLS	0.2	1.5	13.6	15.3	32.1		0.1	13.5	0.0	1.9	47.6	62.9	
TIC	15.1	173.7	268.5	457.3	240.0	5.3	165.2	106.7	1.7	296.5	815.3	1272.6	
IMP	2.1	108.7	62.5	173.3	37.7		1.6	24.9	0.1	226.8	291.2	464.5	
TOT	17.2	282.5	331.0	630.6	277.7	5.3	166.8	131.6	1.8	523.3	1106.5	1737.1	
OTLS	-0.7	0.1	0.1	-0.5								-0.5	
COE	2.6	59.8	250.6	313.0								313.0	
CFC	3.7	20.4	84.3	108.4								108.4	
NOS	5.1	46.7	106.6	158.4								158.4	
GVA	10.7	127.0	441.5	579.2								579.2	
TOT	27.8	409.4	772.5	1209.8	277.7	5.3	166.8	131.6	1.8	523.3	1106.5		

Check

Table A11.9: Volume indices for supply and use tables 2011

Supply Table at basic prices, transf. to purchasers' prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	99.9	96.1	87.8	99.9	103.5	101.1	103.6	111.9	101.8
MAN	95.4	104.4	99.4	103.8	103.3	103.6	103.6	98.7	103.3
CON	127.1	104.3	95.6	103.9	101.2	103.8		98.2	103.4
TTC	92.6	98.9	103.8	103.4	101.1	102.9	103.6	95.1	102.4
FBS	99.6	102.2	102.1	102.1	107.4	102.9		98.2	102.8
OSE	95.2	99.1	100.5	100.5	98.3	100.3	94.6	101.9	100.4
TOT	99.6	104.1	102.0	102.6	103.3	102.8	104.2	98.6	102.6

Use Table at purchasers' prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	102.0	103.4	96.5	102.6	101.5			84.8	-9.1	102.0	101.1	101.8	
MAN	99.5	105.6	98.5	103.6	99.4		102.8	112.5	43.1	104.0	103.2	103.3	
CON	100.7	110.7	100.2	105.9	103.3		96.0	101.9		91.6	101.4	103.4	
TTC	99.1	100.0	101.3	101.1	98.8					104.5	103.2	102.4	
FBS	98.9	104.7	102.8	103.2	100.5		96.2			106.2	102.2	102.8	
OSE	99.4	101.8	101.4	101.5	101.8	99.8		98.2	72.6	103.1	100.2	100.4	
TIC	100.2	105.5	101.4	103.2	100.2	99.8	99.8	105.6	42.7	104.1	102.3	102.6	
OTLS	137.2	-84.9	-30.6	55.5								55.5	
COE	100.1	100.0	101.0	100.8								100.8	
CFC	101.6	100.9	101.4	101.3								101.3	
NOS	99.7	102.0	106.2	104.7								104.7	
GVA	98.6	101.0	102.4	102.0								102.0	
TOT	99.6	104.1	102.0	102.6	100.2	99.8	99.8	105.6	42.7	104.1	102.3		

Check

Taxes less subsidies on products

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR								111.9	
MAN								98.7	
CON								98.2	
TTC								95.1	
FBS								98.2	
OSE								101.9	
TOT								98.6	

Use Table for taxes less subsidies on products

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	92.9	-40.0	134.6	-375.0	101.4					110.5	103.3	111.9	
MAN	94.3	97.2	97.6	97.5	98.2		100.8	105.1	-100.0	104.0	99.0	98.7	
CON		98.1	98.1	98.1	105.9		96.5	98.2		100.0	98.2	98.2	
TTC				92.2	98.2						96.7	95.1	
FBS	130.0	-14.3	100.4	100.7	102.9		89.0			225.0	96.3	98.2	
OSE				240.0	101.0			101.1			101.1	101.9	
TOT	96.4	99.7	98.3	98.4	98.7		79.8	98.2	-100.0	103.5	98.7	98.6	

Use Table for trade and transport margins

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR							103.6		
MAN							103.6		
CON									
TTC							103.6		
FBS									
OSE							94.6		
TOT									

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	101.4	104.6	95.9	102.5	101.7			82.4	-50.0	105.4	103.9	103.6	
MAN	97.9	107.5	97.8	104.1	100.7		101.5	118.6	-183.3	105.2	103.4	103.6	
CON													
TTC	99.2	107.3	97.7	104.0	100.8		101.5	118.5	-162.0	105.2	103.5	103.6	
FBS													
OSE					95.3					92.3	94.6	94.6	
TOT													

Supply Table at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	99.9	96.1	87.8	99.9	103.5	101.1			
MAN	95.4	104.4	99.4	103.8	103.3	103.6			
CON	127.1	104.3	95.6	103.9	101.2	103.8			
TTC	92.6	98.9	103.8	103.4	101.1	102.9			
FBS	99.6	102.2	102.1	102.1	107.4	102.9			
OSE	99.1			100.5	98.3	100.3			
TOT	99.6	104.1	102.0	102.6	103.3	102.8			

Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	102.1	103.1	95.9	102.4	101.3			85.1	-7.1	100.9	99.9	101.1	
MAN	99.8	105.5	98.7	103.7	99.0		103.7	112.3	45.7	103.9	103.5	103.6	
CON	100.7	110.7	100.4	106.3	103.0		96.0	102.3		91.6	101.8	103.8	
TTC	99.1	104.7	101.0	102.1	100.2		101.0	109.7	-102.6	104.7	103.4	102.9	
FBS	98.7	104.7	102.9	103.2	100.5		96.4	102.0		106.2	102.5	102.9	
OSE	99.4	101.6	101.4	101.4	101.8	99.8		93.7	72.6	103.2	100.2	100.3	
TLS	96.4	99.7	98.3	98.4	98.7		79.8	98.2	-100.0	103.5	98.7	98.6	
TIC	100.2	105.5	101.4	103.2	100.2	99.8	99.8	105.6	42.7	104.1	102.3	102.6	
OTLS	137.2	-84.9	-30.6	55.5								55.5	
COE	100.1	100.0	101.0	100.8								100.8	
CFC	101.6	100.9	101.4	101.3								101.3	
NOS	99.7	102.0	106.2	104.7								104.7	
GVA	98.6	101.0	102.4	102.0								102.0	
TOT	99.6	104.1	102.0	102.6	100.2	99.8	99.8	105.6	42.7	104.1	102.3		

Check

Imports CIF

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR					103.5				
MAN					103.3				
CON					101.2				
TTC					101.1				
FBS					107.4				
OSE					98.3				
TOT					103.3				

Imports Use Table

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	100.9	106.3	88.2	104.2	105.0			78.8	-32.2	104.5	102.9	103.5	
MAN	104.1	106.6	99.7	105.2	100.4		92.0	117.4	3.1	102.3	102.4	103.3	
CON	100.0	110.6	96.6	104.5				97.4			97.4	101.2	
TTC	102.3	99.0		100.2						101.7	101.4	101.1	
FBS	98.8	103.5	107.7	106.1	92.1						110.8	107.4	
OSE	85.7	104.5	101.8	102.1	97.6				57.8	97.2	97.0	98.3	
TLS													
TIC	102.3	105.9	102.9	104.8	99.7		92.2	114.7	6.3	102.6	102.4	103.3	

Supply Table for domestic output at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	99.9	96.1	87.8	99.9					
MAN	95.4	104.4	99.4	103.8					
CON	127.1	104.3	95.6	103.9					
TTC	92.6	98.9	103.8	103.4					
FBS	99.6	102.2	102.1	102.1					
OSE	99.1			100.5					
TOT	99.6	104.1	102.0	102.6					

Domestic Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	102.3	101.0	103.0	101.5	98.0			86.7	19.8	98.9	98.1	99.9	
MAN	99.1	104.3	98.0	102.3	98.0		109.4	108.6	79.6	105.5	104.6	103.8	
CON	100.7	110.7	100.5	106.3	103.0		96.0	102.4		91.6	101.9	103.9	
TTC	99.0	105.3	101.1	102.4	100.1		101.0	114.6	-110.0	106.6	104.0	103.4	
FBS	98.7	105.3	102.1	102.6	100.5		96.4	98.3		105.3	101.5	102.1	
OSE	100.0	101.2	101.3	101.2	103.1	99.8		93.7	95.5	103.2	100.4	100.5	
TLS	96.4	99.7	98.3	98.4	98.7		79.8	98.2	-100.0	103.5	98.7	98.6	
TIC	99.9	105.3	101.1	102.8	100.2	99.8	99.8	105.6	42.7	104.1	102.3	102.6	
IMP	102.3	105.9	102.9	104.8	99.7		92.2	114.7	6.3	102.6	102.4	103.3	
TOT	100.2	105.5	101.4	103.2	100.2	99.8	99.8	105.6	42.7	104.1	102.3	102.6	
OTLS	137.2	-84.9	-30.6	55.5								55.5	
COE	100.1	100.0	101.0	100.8								100.8	
CFC	101.6	100.9	101.4	101.3								101.3	
NOS	99.7	102.0	106.2	104.7								104.7	
GVA	98.6	101.0	102.4	102.0								102.0	
TOT	99.6	104.1	102.0	102.6	100.2	99.8	99.8	105.6	42.7	104.1	102.3		

Table A11.10: Supply and use tables 2010 in current prices

Supply Table at basic prices, transf. to purchasers' prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	25.3	0.2	0.0	25.5	13.9	39.4	10.8	0.4	50.6
MAN	1.2	282.6	35.8	319.6	301.8	621.4	108.8	39.2	769.4
CON	0.1	84.9	4.7	89.7	1.5	91.2		8.4	99.6
TTC	0.6	15.4	222.6	238.5	65.1	303.6	-119.7	3.1	187.1
FBS	0.5	7.8	275.0	283.3	50.9	334.2		9.9	344.1
OSE	0.3	2.5	219.6	222.4	16.4	238.8	0.1	2.7	241.6
TOT	28.0	393.3	757.6	1178.9	449.7	1628.6	0.0	63.8	1692.3

Use Table at purchasers' prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	5.1	15.3	1.6	22.0	5.8				0.2	0.2	22.5	28.6	50.6
MAN	8.9	173.4	63.3	245.6	121.0		8.9	46.0	3.8	344.1	523.7	769.4	
CON	0.3	23.6	19.4	43.2	0.5		0.6	53.0		2.2	56.4	99.6	
TTC	0.6	10.8	64.8	76.1	21.0	0.9	0.5	6.7	0.0	81.8	110.9	187.1	
FBS	2.1	41.2	156.3	199.6	76.2	0.0	3.7	17.5		47.1	144.5	344.1	
OSE	0.2	3.4	20.9	24.5	52.7	4.5	153.6	1.2	0.3	4.9	217.0	241.6	
TIC	17.1	267.7	326.4	611.2	277.2	5.3	167.2	124.6	4.3	502.5	1081.2	1692.3	
OTLS	-0.5	-0.1	-0.3	-1.0								-1.0	
COE	2.6	59.8	248.1	310.5								310.5	
CFC	3.7	20.2	83.1	107.0								107.0	
NOS	5.1	45.8	100.4	151.3								151.3	
GVA	10.8	125.7	431.2	567.8								567.8	
TOT	28.0	393.3	757.6	1178.9	277.2	5.3	167.2	124.6	4.3	502.5	1081.2		

Check

Taxes less subsidies on products

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR								0.4	
MAN							10.8	39.2	
CON								8.4	
TTC							-119.7	3.1	
FBS								9.9	
OSE							0.1	2.7	
TOT							0.0	63.8	

Use Table for taxes less subsidies on products

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	0.0	0.0	0.0	0.0	0.4					0.1	0.5	0.4	
MAN	0.2	1.6	6.3	8.2	26.9		0.4	2.2	0.0	1.6	31.0	39.2	
CON	0.0	2.0	2.0	2.0	0.1		0.1	6.3		0.0	6.4	8.4	
TTC	0.0	-0.1	1.1	1.1	1.7		-0.1	0.5	-0.1	2.0	2.0	3.1	
FBS	0.0	0.0	4.3	4.3	1.5		0.1	4.1		0.0	5.7	9.9	
OSE	0.0	0.0	0.0	0.0	2.1		-0.4	0.7		0.3	2.7	2.7	
TOT	0.2	1.5	13.8	15.5	32.5		0.1	13.8	0.0	1.8	48.2	63.8	

Trade and transport margins

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR								10.8	
MAN							108.8		
CON									
TTC							-119.7		
FBS									
OSE							0.1		
TOT							0.0		

Use Table for trade and transport margins

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	0.4	1.6	0.4	2.5	3.0				0.0	0.0	5.3	8.3	10.8
MAN	0.8	18.6	9.2	28.6	39.6		3.2	4.4	0.0	32.9	80.2	108.8	
CON													
TTC	-1.2	-20.2	-9.7	-31.1	-42.7		-3.2	-4.4	-0.1	-38.2	-88.6	-119.7	
FBS													
OSE					0.1					0.0	0.1	0.1	
TOT					0.0			0.0	0.0	0.0	0.0	0.0	0.0

Supply Table at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	25.3	0.2	0.0	25.5	13.9	39.4			
MAN	1.2	282.6	35.8	319.6	301.8	621.4			
CON	0.1	84.9	4.7	89.7	1.5	91.2			
TTC	0.6	15.4	222.6	238.5	65.1	303.6			
FBS	0.5	7.8	275.0	283.3	50.9	334.2			
OSE	0.3	2.5	219.6	222.4	16.4	238.8			
TOT	28.0	393.3	757.6	1178.9	449.7	1628.6			

Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	4.7	13.7	1.2	19.5	2.5				0.2	0.2	17.1	19.9	39.4
MAN	7.9	153.2	47.8	208.9	54.5		5.4	39.4	3.8	309.6	412.5	621.4	
CON	0.3	23.6	17.4	41.2	0.4		0.5	46.8		2.2	50.0	91.2	
TTC	1.8	31.1	73.3	106.2	62.1	0.9	3.7	10.7	0.1	120.1	197.5	303.6	
FBS	2.1	41.2	152.1	195.4	74.8	0.0	3.5	13.4		47.1	138.8	334.2	
OSE	0.2	3.4	20.9	24.5	50.5	4.5	154.0	0.5	0.3	4.6	214.3	238.8	
TLS	0.2	1.5	13.8	15.5	32.5		0.1	13.8	0.0	1.8	48.2	63.8	
TIC	17.1	267.7	326.4	611.2	277.2	5.3	167.2	124.6	4.3	502.5	1081.2	1692.3	
OTLS	-0.5	-0.1	-0.3	-1.0								-1.0	
COE	2.6	59.8	248.1	310.5								310.5	
CFC	3.7	20.2	83.1	107.0								107.0	
NOS	5.1	45.8	100.4	151.3								151.3	
GVA	10.8	125.7	431.2	567.8								567.8	
TOT	28.0	393.3	757.6	1178.9	277.2	5.3	167.2	124.6	4.3	502.5	1081.2		

Check

Imports CIF

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR					13.9				
MAN					301.8				
CON					1.5				
TTC					65.1				
FBS					50.9				
OSE					16.4				
TOT					449.7				

Imports Use Table

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	0.7	5.3	0.6	6.5	1.2				0.0	0.1	6.1	7.4	13.9
MAN	1.1	80.4	19.5	101.0	23.9		1.8	16.4	1.7	157.1	200.9	301.8	
CON	0.0	0.5	0.4	0.8				0.7			0.7	1.5	
TTC	0.0	2.7	13.8	16.6	0.7			2.9	0.0	45.0	48.6	65.1	
FBS	0.2	13.3	22.7	36.3	0.2			1.7	12.8	14.6	50.9		
OSE	0.0	0.5	3.8	4.3	11.8				0.2	0.0	12.1	16.4	
TLS													
TIC	2.0	102.6	60.8	165.4	37.8		1.8	21.7	1.9	221.0	284.2	449.7	

Supply Table for domestic output at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	25.3	0.2	0.0	25.5					
MAN	1.2	282.6	35.8	319.6					
CON	0.1	84.9	4.7	89.7					
TTC	0.6	15.4	222.6	238.5					
FBS	0.5	7.8	275.0	283.3					
OSE	0.3	2.5	219.6	222.4					
TOT	28.0	393.3	757.6	1178.9					

Domestic Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR	4.0	8.4	0.6	13.0	1.3				0.1	0.1	10.9	12.5	25.5
MAN	6.8	72.8	28.2	107.9	30.6		3.6	22.9	2.1	152.5	211.7	319.6	
CON	0.3	23.1	17.0	40.4	0.4		0.5	46.1		2.2	49.3	89.7	
TTC	1.7	28.4	59.5	89.6	61.4	0.9	3.7	7.8	0.1	75.1	148.9	238.5	
FBS	1.8	27.9	129.4	159.1	74.5	0.0	3.5	11.8		34.3	124.2	283.3	
OSE	0.2	2.9	17.1	20.2	38.7	4.5	154.0	0.5	0.1	4.5	202.2	222.4	
TLS	0.2	1.5	13.8	15.5	32.5		0.1	13.8	0.0	1.8	48.2	63.8	
TIC	15.1	165.0	265.6	445.7	239.4	5.3	165.5	102.9	2.4	281.5	796.9	1242.7	
IMP	2.0	102.6	60.8	165.4	37.8		1.8	21.7	1.9	221.0	284.2	449.7	
TOT	17.1	267.7	326.4	611.2	277.2	5.3	167.2	124.6	4.3	502.5	1081.2	1692.3	
OTLS	-0.5	-0.1	-0.3	-1.0								-1.0	
COE	2.6	59.8	248.1	310.5								310.5	
CFC	3.7	20.2	83.1	107.0								107.0	
NOS	5.1	45.8	100.4	151.3								151.3	
GVA	10.8	125.7	431.2	567.8								567.8	
TOT	28.0	393.3	757.6	1178.9	277.2	5.3	167.2	124.6	4.3	502.5	1081.2		

Check

3. Balancing adjustments incorporated in the supply and use tables (tables A11.11–A11.14)

No change made in table A11.10, thus no table A11.15.

Table A11.11 Supply and use tables 2011 in current prices

Supply Table at basic prices, transf. to purchasers' prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	0.3					0.3			0.3
MAN									
CON		-1.0		-1.0		-1.0			-1.0
TTC					0.5	0.5			0.5
FBS									
OSE									
TOT	0.3	-1.0		-0.8	0.5	-0.3			-0.3

Use Table at purchasers' prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR													0.2
MAN										1.0	1.0	1.0	-1.0
CON								-0.3			-0.3	-0.3	-0.7
TTC					0.2						0.2	0.2	0.3
FBS			0.3	0.3									-0.3
OSE					-0.4						-0.4	-0.4	0.4
TIC			0.3	0.3	-0.2		-0.3			1.0	0.5	0.8	-1.0
OTLS													
COE													
CFC													
NOS	0.3	-1.0	-0.3	-1.1									-1.1
GVA	0.3	-1.0	-0.3	-1.1									-1.1
TOT	0.3	-1.0		-0.8	-0.2		-0.3			1.0	0.5		

Check

Taxes less subsidies on products

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR									
MAN									
CON									
TTC									
FBS									
OSE									
TOT									

Use Table for taxes less subsidies on products

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR													
MAN		0.0		0.0								0.0	0.0
CON													
TTC												0.0	0.0
FBS			0.0	0.0									
OSE													
TOT		0.0	0.0	0.0								0.0	0.0

Trade and transport margins

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR									
MAN									
CON									
TTC									
FBS									
OSE									
TOT									

Use Table for trade and transport margins

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR													
MAN		0.1		0.1	-0.8						-0.8	-0.7	0.7
CON													
TTC		-0.1		-0.1	0.8						0.8	0.7	-0.7
FBS													
OSE													
TOT											0.0	0.0	0.0

Supply Table at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	0.3					0.3			
MAN									
CON		-1.0		-1.0		-1.0			
TTC					0.5	0.5			
FBS									
OSE									
TOT	0.3	-1.0		-0.8	0.5	-0.3			

Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR													0.3
MAN		-0.1		-0.1	0.8					1.0	1.8	1.7	-1.7
CON								-0.3			-0.3	-0.3	-0.7
TTC		0.1		0.1	-0.6						-0.6	-0.5	1.0
FBS			0.3	0.3									-0.3
OSE					-0.4						-0.4	-0.4	0.4
TLS		0.0	0.0	0.0									0.0
TIC			0.3	0.3	-0.2		-0.3			1.0	0.5	0.8	-1.1
OTLS													
COE													
CFC													
NOS	0.3	-1.0	-0.3	-1.1									-1.1
GVA	0.3	-1.0	-0.3	-1.1									-1.0
TOT	0.3	-1.0		-0.7	-0.2		-0.3			1.0	0.5		

Check

Imports CIF

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR									
MAN									
CON									
TTC					0.5				
FBS									
OSE									
TOT					0.5				

Imports Use Table

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR													
MAN					-0.2					-50.0	-50.2	-50.2	50.2
CON													
TTC													0.5
FBS													
OSE													
TLS													
TIC					-0.2					-50.0	-50.2	-50.2	50.7

Supply Table for domestic output at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	0.3					0.3			
MAN									
CON		-1.0		-1.0		-1.0			
TTC									
FBS									
OSE									
TOT	0.3	-1.0		-0.8					

Domestic Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR													0.2
MAN		-0.1		-0.1	1.0					51.0	52.0	51.9	-51.9
CON								-0.3			-0.3	-0.3	-0.7
TTC		0.1		0.1	-0.6						-0.6	-0.5	0.5
FBS			0.3	0.3									-0.3
OSE					-0.4						-0.4	-0.4	0.4
TLS		0.0	0.0	0.0									0.0
TIC			0.3	0.3	-0.2		-0.3			51.0	50.7	51.0	-51.8
IMP					-0.2					-50.0	-50.2	-50.2	51.8
TOT			0.3	0.3	-0.2		-0.3			1.0	0.5	0.8	
OTLS													
COE													
CFC													
NOS	0.3	-1.0	-0.3	-1.1									-1.1
GVA	0.3	-1.0	-0.3	-1.1									-1.0
TOT	0.3	-1.0		-0.8	-0.2		-0.3			1.0	0.5		

Check

Table A11.12: Price indices for supply and use tables 2011

Supply Table at basic prices, transf. to purchasers' prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	0.0			0.0		0.0			0.0
MAN							0.9	1.3	0.2
CON		0.0		0.0		0.0			0.0
TTC					-0.1	0.0	0.8		-0.5
FBS									0.0
OSE									0.0
TOT	0.0	0.0		0.0	0.0	0.0		0.8	0.0

Use Table at purchasers' prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR													0.0
MAN										0.0	0.0	0.0	0.2
CON									0.0		0.0	0.0	0.0
TTC					0.0						0.0	0.0	-0.5
FBS			0.0	0.0								0.0	0.0
OSE					0.0						0.0	0.0	0.0
TIC			0.0	0.0	0.0			0.0		0.0	0.0	0.0	-0.3
OTLS													
COE													
CFC													
NOS	1.1	0.0	0.0	0.0								0.0	
GVA	0.3	0.0	0.0	0.0								0.0	
TOT	0.0	0.0		0.0	0.0			0.0		0.0	0.0		

Check

Taxes less subsidies on products

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR									
MAN								1.3	
CON									
TTC									
FBS									
OSE									
TOT								0.8	

Use Table for taxes less subsidies on products

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR													
MAN		1.3		0.3								0.1	1.3
CON													
TTC													
FBS			-0.7	-0.7								-0.3	0.3
OSE													
TOT		1.3	-0.2	-0.1								0.0	0.8

Trade and transport margins

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR									
MAN							0.9		
CON									
TTC							0.8		
FBS									
OSE									
TOT									

Use Table for trade and transport margins

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR													
MAN		0.6		0.4	-2.0						-1.0	-0.6	1.5
CON													
TTC		0.6		0.4	-1.9						-0.9	-0.5	1.4
FBS													
OSE													
TOT													

Supply Table at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	0.0			0.0		0.0			104.2
MAN									
CON		0.0		0.0		0.0			102.0
TTC					-0.1	0.0			102.0
FBS									
OSE									
TOT	0.0	0.0		0.0	0.0	0.0			-103.7

Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR													0.0
MAN		-0.1		-0.1	1.5					0.0	0.2	0.1	-0.1
CON								0.0			0.0	0.0	0.0
TTC		0.4		0.1	-1.3						-0.4	-0.2	0.2
FBS			0.0	0.0								0.0	0.0
OSE					0.0						0.0	0.0	0.0
TLS		1.3	-0.2	-0.1								0.0	
TIC			0.0	0.0	0.0			0.0		0.0	0.0	0.0	0.1
OTLS													
COE													
CFC													
NOS	1.1	0.0	0.0	0.0								0.0	
GVA	0.3	0.0	0.0	0.0								0.0	
TOT	0.0	0.0		0.0	0.0			0.0		0.0	0.0		

Check

Imports CIF

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR									
MAN									
CON									
TTC					-0.1				
FBS									
OSE									
TOT					0.0				

Imports Use Table

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR													
MAN					-0.8					-17.2	-13.2	-8.1	8.1
CON													
TTC													-0.1
FBS													
OSE													
TLS													
TIC					-0.5					-11.4	-8.9	-5.3	5.3

Supply Table for domestic output at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	0.0			0.0					
MAN									
CON		0.0		0.0					
TTC									
FBS									
OSE									
TOT	0.0	0.0		0.0					

Domestic Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR													0.0
MAN		-0.2		-0.1	3.3					12.2	9.8	6.7	-6.7
CON								0.0			0.0	0.0	0.0
TTC		0.4		0.1	-1.3						-0.5	-0.3	0.3
FBS			0.0	0.0								0.0	0.0
OSE					0.0						0.0	0.0	0.0
TLS		1.3	-0.2	-0.1								0.0	
TIC			0.0	0.0	0.1			0.0		7.4	2.9	1.9	-6.4
IMP					-0.5					-11.4	-8.9	-5.3	
TOT		0.0	0.0	0.0	0.0			0.0		0.0	0.0	0.0	-6.4
OTLS													
COE													
CFC													
NOS	1.1	0.0	0.0	0.0								0.0	
GVA	0.3	0.0	0.0	0.0								0.0	
TOT	0.0	0.0		0.0	0.0			0.0		0.0	0.0		

Table A11.13: Supply and use tables 2011 at previous years' prices

Supply Table at basic prices, transf. to purchasers' prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	0.2			0.2		0.2			0.2
MAN							-1.0	-0.5	-1.5
CON		-1.0		-1.0		-1.0			-1.0
TTC					0.5		1.0		1.5
FBS									
OSE									
TOT	0.2	-1.0		-0.7	0.5	-0.3		-0.5	-0.8

Use Table at purchasers' prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR													0.2
MAN										1.0	1.0	1.0	-2.5
CON											-0.3	-0.3	-0.7
TTC					0.2						0.2	0.2	1.3
FBS			0.3	0.3								0.3	-0.3
OSE					-0.4						-0.4	-0.4	0.4
TIC			0.3	0.3	-0.2			-0.3		1.0	0.5	0.8	-1.5
OTLS													
COE													
CFC													
NOS	0.2	-1.0	-0.3	-1.0									-1.0
GVA	0.2	-1.0	-0.3	-1.0									-1.0
TOT	0.2	-1.0		-0.7	-0.2			-0.3		1.0	0.5		

Check

Taxes less subsidies on products

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR									
MAN								-0.5	
CON									
TTC									
FBS									
OSE									
TOT								-0.5	

Use Table for taxes less subsidies on products

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR													
MAN													-0.5
CON													
TTC													
FBS													
OSE													
TOT													-0.5

Trade and transport margins

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR									
MAN									
CON								-1.0	
TTC									
FBS								1.0	
OSE									
TOT									

Use Table for trade and transport margins

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR													
MAN													-1.0
CON													
TTC													1.0
FBS													
OSE													
TOT													

Supply Table at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	0.2			0.2		0.2			
MAN									
CON		-1.0		-1.0		-1.0			
TTC					0.5		1.0		
FBS									
OSE									
TOT	0.2	-1.0		-0.7	0.5	-0.3			

Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR													0.2
MAN										1.0	1.0	1.0	-1.0
CON											-0.3	-0.3	-0.7
TTC					0.2						0.2	0.2	0.3
FBS			0.3	0.3								0.3	-0.3
OSE					-0.4						-0.4	-0.4	0.4
TLS													
TIC			0.3	0.3	-0.2			-0.3		1.0	0.5	0.8	-1.0
OTLS													
COE													
CFC													
NOS	0.2	-1.0	-0.3	-1.0									-1.0
GVA	0.2	-1.0	-0.3	-1.0									-1.0
TOT	0.2	-1.0		-0.7	-0.2			-0.3		1.0	0.5		

Check

Imports CIF

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR									
MAN									
CON									
TTC					0.5				
FBS									
OSE									
TOT					0.5				

Imports Use Table

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR													
MAN										-25.0	-25.0	-25.0	25.0
CON													
TTC													0.5
FBS													
OSE													
TLS													
TIC											-25.0	-25.0	25.5

Supply Table for domestic output at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	0.2			0.2					
MAN									
CON		-1.0		-1.0					
TRA									
TRN									
COM									
TOT	0.2	-1.0		-0.7					

Domestic Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTfin	TOTpp	Check
AGR													0.2
MAN										26.0	26.0	26.0	-26.0
CON											-0.3	-0.3	-0.7
TTC					0.2						0.2	0.2	-0.2
FBS			0.3	0.3								0.3	-0.3
OSE					-0.4						-0.4	-0.4	0.4
TLS													
TIC			0.3	0.3	-0.2			-0.3		26.0	25.5	25.8	-26.5
IMP										-25.0	-25.0	-25.0	
TOT			0.3	0.3	-0.2			-0.3		1.0	0.5	0.8	-26.5
OTLS													
COE													
CFC													
NOS	0.2	-1.0	-0.3	-1.0									-1.0
GVA	0.2	-1.0	-0.3	-1.0									-1.0
TOT	0.2	-1.0		-0.7	-0.2			-0.3		1.0	0.5		

Check

Table A11.14: Volume indices for supply and use tables 2011

Supply Table at basic prices, transf. to purchasers' prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	0.9			0.9		0.6			0.5
MAN							-0.9	-1.3	-0.2
CON		-1.2		-1.1		-1.1			-1.0
TTC					0.8	0.2	-0.8		0.8
FBS									
OSE									
TOT	0.9	-0.2		-0.1	0.1	0.0		-0.8	0.0

Use Table at purchasers' prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR													0.5
MAN										0.3	0.2	0.1	-0.3
CON								-0.6			-0.5	-0.3	-0.7
TTC					1.0						0.2	0.1	0.7
FBS			0.2	0.1								0.1	-0.1
OSE					-0.8						-0.2	-0.2	0.2
TIC			0.1	0.0	-0.1			-0.2	0.2	0.0	0.0	0.0	0.2
OTLS													
COE													
CFC													
NOS	4.7	-2.1	-0.3	-0.7									-0.7
GVA	2.2	-0.8	-0.1	-0.2									-0.2
TOT	0.9	-0.2		-0.1	-0.1			-0.2	0.2	0.0			

Check

Taxes less subsidies on products

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR									
MAN								-1.3	
CON									
TTC									
FBS									
OSE									
TOT								-0.8	

Use Table for taxes less subsidies on products

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR													
MAN													-1.3
CON													
TTC													
FBS													
OSE													
TOT													-0.8

Trade and transport margins

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR									
MAN									
CON								-0.9	
TTC								-0.8	
FBS									
OSE									
TOT									

Use Table for trade and transport margins

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR													
MAN													-0.9
CON													
TTC													-0.8
FBS													
OSE													
TOT													

Supply Table at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	0.9			0.9		0.6			
MAN									
CON		-1.2		-1.1		-1.1			
TTC					0.8	0.2			
FBS									
OSE									
TOT	0.9	-0.2		-0.1	0.1	0.0			

Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR													0.6
MAN										0.3	0.2	0.2	-0.2
CON								-0.6			-0.6	-0.3	-0.7
TTC					0.3						0.1	0.1	0.1
FBS			0.2	0.1								0.1	-0.1
OSE					-0.8						-0.2	-0.2	0.2
TLS													
TIC			0.1	0.0	-0.1			-0.2	0.2	0.0	0.0	0.0	-0.1
OTLS													
COE													
CFC													
NOS	4.7	-2.1	-0.3	-0.7									-0.7
GVA	2.2	-0.8	-0.1	-0.2									-0.2
TOT	0.9	-0.2		-0.1	-0.1			-0.2	0.2	0.0			

Check

Imports CIF

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR									
MAN									
CON					0.8				
TTC									
FBS									
OSE									
TOT					0.1				

Imports Use Table

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR													
MAN										-15.9	-12.4	-8.3	8.3
CON													0.8
TTC													
FBS													
OSE													
TLS													
TIC										-11.3	-8.8	-5.6	5.7

Supply Table for domestic output at basic prices

	AGR	MMC	SER	DP	IMP	SUPbp	TTM	TLS	SUPpp
AGR	0.9			0.9					
MAN									
CON		-1.2		-1.1					
TTC									
FBS									
OSE									
TOT	0.9	-0.2		-0.1					

Domestic Use Table at basic prices

	AGR	MMC	SER	TIC	FCH	FCN	FCG	GFCF	INV	EXP	TOTlin	TOTpp	Check
AGR													0.9
MAN										17.1	12.3	8.1	-8.1
CON								-0.7			-0.6	-0.3	-0.8
TTC					0.3						0.1	0.1	-0.1
FBS			0.2	0.2								0.1	-0.1
OSE					-1.0						-0.2	-0.2	0.2
TLS													
TIC			0.1	0.1	-0.1			-0.3	9.2	3.2	2.1	2.1	-8.0
IMP									-11.3	-8.8	-5.6	-5.6	-8.0
TOT			0.1	0.0	-0.1			-0.2	0.2	0.0	0.0	0.0	
OTLS													
COE													
CFC													
NOS	4.7	-2.1	-0.3	-0.7									-0.7
GVA	2.2	-0.8	-0.1	-0.2									-0.2
TOT	0.9	-0.2		-0.1	-0.1			-0.2	0.2	0.0	0.0	0.0	

<i>Key:</i>	AGR	agriculture
	CFC	consumption of fixed capital
	COE	compensation of employees
	CON	construction
	DP	domestic product
	EXP	exports
	FBS	finance and business services
	FCG	final consumption expenditure of general government
	FCH	final consumption expenditure of households
	FCN	final consumption expenditure of non-profit institutions serving households
	GFCF	gross fixed capital formation
	GVA	gross value added
	IMP	imports
	INV	inventories (changes in)
	MAN	manufacturing
	MMC	manufacturing, mining and construction
	NOS	net operating surplus
	OSE	other services
	OTLS	other taxes less subsidies on production
	SER	services
	SUPbp	supply at basic prices
	SUPpp	supply at purchasers' prices
	TIC	total intermediate consumption
	TLS	taxes less subsidies on products
	TOT	total
	TOTfin	total at final uses
	TOTpp	total at purchasers' prices
	TTC	trade, transport and communication
	TTM	trade and transport margins

Chapter 12. Transforming the supply and use tables into input-output tables

A. Introduction

12.1 The present chapter describes the methods for transforming SUTs into IOTs (product-by-product and industry-by-industry). The compilation of IOTs is quite different in nature from the compilation of SUTs and is better described as an analytical step or transformation rather than a compilation process.

12.2 Section B below provides an overview of IOTs and a description of the terminology relating to them. Section C focuses on the transformation of SUTs into IOTs. In particular, it describes the tables that form the starting point for the compilation of IOTs, and covers some issues related to when it is necessary to use square rather than rectangular SUTs for the compilation of IOTs, and how to deal with secondary production in IOTs. Section D describes the input-output framework; it outlines the theoretical models for the compilation of IOTs; and it provides numerical examples of transformation of SUTs into IOTs based on the different theoretical models. Section E provides an empirical application of the transformation models. It also discusses general issues on IOTs, such as the relationship between types of tables, technology and share markets; the link between IOTs and official statistics; and the requirements of IOTs. Annexes A and B to this chapter elaborate further on the mathematical derivation of different IOTs and discuss how to handle negatives in IOTs. Annex C provides a list of references for the treatment of secondary production.

B. Overview of the relationship between IOTs and SUTs

12.3 The SUTs form a central part of the national accounts, providing a framework to bring together a range of data and, through balancing, to ensure the coherency and consistency of various parts of the national accounts. The SUTs thus serve many purposes, in particular, statistical and analytical, not just for producers but also for a range of different users, and their analytical dimension is especially enhanced when the SUTs are transformed into IOTs. For analytical purposes, the assumptions about the relationships between inputs and outputs are required irrespective of whether the products have been produced by the primary industry or by other industries as their secondary output.

12.4 The SUTs constitute the basis for compiling IOTs and it is recommended that they are compiled first, and then the IOTs. This approach makes the best use of data collected from businesses and other sources in order to compile SUTs and then – via a range of assumptions –

move to the basis of the IOTs. The domestic output part of the supply table and the intermediate use part of the use table are always product-by-industry tables and often rectangular, meaning that many more products than industries are distinguished. IOTs, on the other hand, always reflect the same number of industries (industry-by-industry IOTs) or the same number of products (product-by-product IOTs).

12.5 Table 12.1 shows the general structure of a product-by-product IOT at basic prices. A similar table can be compiled for an industry-by-industry IOT where the first quadrant contains an industry-by-industry rather than a product-by-product matrix. In practice, the IOTs can have different presentations, such as input-output tables with net exports. The fundamental elements of the general structure remain the same, however. The different presentations of IOTs and the components are covered later in this chapter. In addition, as noted earlier, for ease of exposition and not to overload the presentation of the SUTs and IOTs, the adjustment items are not included in the numerical examples in this Handbook. This issue is explained further in chapter 8, paragraphs 8.18–8.21.

Table 12.1 Product-by-product IOT at basic prices

		PRODUCTS							FINAL USE							Output at basic prices	
		Agriculture	Manufacturing	Construction	Trade, transport and communication	Finance and business services	Other services	Total	Final consumption expenditure			Gross fixed capital formation	Changes in valuables	Changes in inventories	Exports		Total
									Households	NPISH	General government						
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
PRODUCTS	Agriculture	(1)	Intermediate consumption of domestic products at basic prices					Final use of domestic products at basic prices									
	Manufacturing	(2)															
	Construction	(3)															
	Trade, transport and communication	(4)															
	Finance and business services	(5)															
	Other services	(6)															
Total at basic prices		(7)						Final use of domestic products at basic prices									
Imports		(8)	Intermediate consumption of imported products					Final use of imported products									
Taxes less subsidies on products		(9)	Taxes less subsidies on products					Taxes less subsidies on products									
Total at purchasers' prices		(10)															
GVA	Compensation of employees	(11)	GVA at basic prices														
	Other taxes less subsidies on production	(12)															
	Consumption of fixed capital	(13)															
	Net operating surplus/Net mixed income	(14)															
GVA		(15)															
Input at basic prices		(16)															

 = empty

12.6 In classical input-output theory, the impact analysis (for example, the effect on GVA of an increase in household final consumption) requires that the output side and the input side are classified in identical ways (either by products or by industries). This enables the direct and indirect effects to be traced through the system: any output will require inputs, which in turn, require further outputs, and so forth. As the basis of the use table is a product-by-industry matrix, it is not possible to directly link the required outputs to the required inputs, and thus it is necessary to transform either the product dimension into an industry dimension or vice versa, which can be achieved by applying the information available in the supply table. Although the impact analysis can also be undertaken by iterative procedures directly based on the supply table and use table datasets, this approach also involves relying on one of the transformation models discussed later in this chapter.

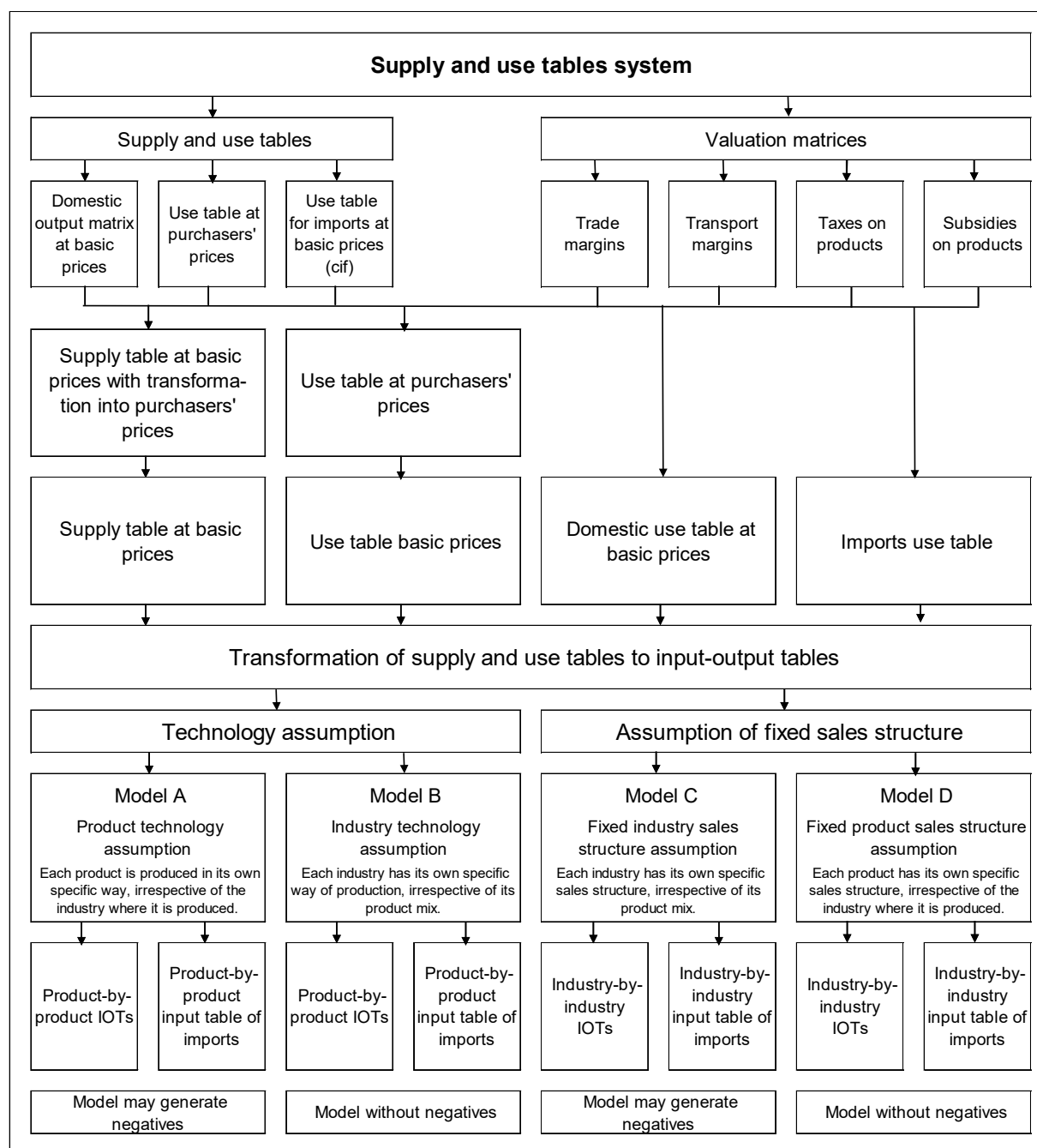
Thus, compiling IOTs is an analytical step and various assumptions have to be made and sometimes adjustments are required for the transformation of SUTs into IOTs.

12.7 In the main, the construction of IOTs depends on the treatment of secondary products. Many units may produce only one group of products, which are the primary products of the industry to which they are classified. Some units, however, produce products that are not among the primary products of the industry to which they belong. As a result, there will be many non-diagonal entries in the supply table. The treatment of secondary products rests upon the separation of outputs or inputs associated with secondary products, so that they can be added to the outputs or inputs of the industry in which the secondary product is the characteristic or principal output or, alternatively, to create industry-adjusted product groups.

12.8 As an analytical tool, input-output-based data are conveniently integrated into macroeconomic models in order to analyse the links between final uses and levels of industrial output. Input-output analysis can also serve various analytical purposes such as impact analysis, productivity analysis, employment effects, analysis of the interdependence of structures and analysis of price change. Analytical uses of IOTs are illustrated in chapters 19 and 20 of this Handbook.

12.9 The SUTs at basic prices with a split of the use table between the use of domestically produced products and the use of imported products constitute the starting point for the transformation of SUTs to IOTs. In essence, therefore, this provides the starting point, as shown in figure 2.2, for the bottom left-hand side of the H-Approach for IOTs in current prices and for the bottom right-hand side of the H-Approach for IOTs in previous years' prices.

12.10 Figure 12.1 provides an overview of the various tables that feed into the transformation of SUTs into IOTs and the various types of IOTs linked to the different model assumptions. It is worth noting that, for industry-by-industry IOTs, the corresponding import flow tables does not necessarily need to be transformed to an industry-by-industry format but can be retained in the SUTs format as product-by-industry.

Figure 12.1 Transformation of SUTs into IOTs

1. Terminology used with reference to IOTs

12.11 Over many years, the terminology associated with symmetric IOTs and SUTs, and the understanding of these tables, have evolved. In particular, the use of the term “symmetric” is often misunderstood. In terms of its lexical meaning, the use of “symmetric” is correct here, in that the

IOTs are square tables (thus allowing, for example, for matrix inversion and the generation of multipliers) and in the way in which the industry-by-industry tables or product-by-product tables are presented. From a conceptual point of view, however, it is incorrect to use the term “symmetric”, in that the transformations reflect industry-adjusted product-by-industry IOTs or product-by-product-adjusted industry IOTs, which means, in essence, that there is no symmetry in the dimension of the matrix (more details on this may be found in Box 12.1).

12.12 In addition, in mathematical terms, IOTs are not symmetric matrices, in that the table element (i,j) is not equal to element (j,i) . In other words, the use of steel by the manufacturing industry of basic metals is not the same as the use of basic metals by the steel industry. Thus in this Handbook, and for future guidance, it is recommended that the term “IOTs” be used, and not “symmetric IOTs”.

12.13 It is also worth noting that, in the transformations used in model A and model B shown in Figure 12.1, where technology assumptions are applied, no technology is involved in the physical production processes, but only economic transactions measured in monetary terms. With the transformations, the institutional characteristics of the industries remain unaffected.

12.14 Box 12.1 provides further clarification on some key misunderstandings regarding input-output-related terminology.

Box 12.1 Clarification of IOTs terminology

The terminology used to date for IOTs can be traced back to chapter 3 of the 1968 SNA. As the understanding of IOTs has evolved, however, some clarifications need to be made in two specific areas to ensure the correct understanding and correct use of specific terminology.

The first area relates to the use of the term “technology assumptions”. This term was previously also applied to the assumptions used for compiling industry-by-industry IOTs from SUTs. Based on the work of Konijn and Steenge (1995), it was clarified that only the much weaker sales structure assumptions were necessary and those actually used in this case. This aspect was further elaborated in Thage (2002) and the new terminology, in which there is a clear distinction between technology assumptions and sales structure assumptions. These contributed to the understanding of the real differences between product-by-product IOTs and industry-by-industry IOTs, and became the new standard terminology with the introduction of the Eurostat Manual on Supply, Use and Input-Output Tables (2008), the 2008 SNA and the ESA 2010 (Eurostat, 2013c).

The second area relates to the use of the terms “product-by-product” IOTs and “industry-by-industry” IOTs. The same term is used to characterize both rows and columns, although their implications are very different. Starting with the use table for intermediate consumption, the products are indicated in the rows and industries indicated in the columns.

- “Product-by-product” IOTs are compiled by adjusting the columns (using technology assumptions) but in this process the industries remain as industries with all the characteristics of the producing units, transforming intermediate and primary inputs into output, including all the institutional characteristics. The industries are not transformed into products.
- “Industry-by-industry” IOTs are compiled by adjusting the rows (using sales structure assumptions) but in this process the products remain as products, just composed in a different way. The products are not transformed into industries.

Accordingly, a more precise terminology reflecting the actual procedures would require the use of different terms reflecting the meaning of the rows and the columns of the IOTs. This may be summarized as follows:

- References to “product-by-product industries” should be replaced by “product-by-product-adjusted industries”, retaining the word “industry”.
- References to “industry-by-industry” should be replaced by “industry-adjusted product-by-industry”, retaining the word “product”.

Even though these terms are conceptually correct, they are currently not used. In general, it is important to remember that “product-by-product” and “industry-by-industry” references are short-hand versions of a more descriptive terminology, and to be aware of how this affects the understanding of the contents of the tables.

C. Conversion of SUTs to IOTs

1. Starting point for the transformation

12.15 The starting point for the transformation of SUTs into IOTs consists of the set of the following tables:

- Supply table at basic prices
- Use table at basic prices
- Domestic use table at basic prices
- Imports use table at basic prices

12.16 Although the use table is initially compiled at purchasers’ prices, the transformation of the use table into basic prices is viewed as a step towards the compilation of IOTs, which are usually compiled at basic prices and not purchasers’ prices. Intermediate uses and final uses calculated at basic prices are one step further removed from the basic statistics collected and actual observations in the economy thus needing the compilation of valuation matrices, as described in chapter 7 of this Handbook.

12.17 The structure and data contents of unbalanced SUTs in current prices, including the domestic use table and imports use table, are covered in detail in chapters 3–8. These chapters cover all the building blocks required in producing the set of tables necessary to compile IOTs. Various issues of particular relevance when compiling IOTs from SUTs, such as statistical units, redefinitions and the relationship between products and industries, are further elaborated in this chapter.

12.18 The recommended approach to the compilation of IOTs is thus the preparation of separate IOTs for domestic output and imported products, which are derived from the domestic use table and the imports use table of the SUTs system. The statistical requirements for such a separation are considerable but the results allow considerable flexibility in the treatment of imports and enable a clear analysis of the changes in the impact of the use of supplies from resident producers and from non-resident suppliers.

12.19 A simple numerical example of IOTs obtained using different models is presented in this chapter. The starting point for the various IOTs is the SUTs in Table 12.2, which include a supply table at basic prices, a domestic use table at basic prices and an imports use table. It should be noted that, in the domestic use table, the imports are still included in the table but shown in a single row so that the resulting table still adds up to the supply table.

Table 12.2 Numerical example of rectangular SUTs for the transformation

Supply table at basic prices							
	Industries			Output	Imports	Total supply	
	Agriculture	Manufacturing and construction	Services				
Products	Agriculture	25.77	5.15	7.04	37.96	15.38	53.35
	Manufacturing	1.26	313.51	35.72	350.50	399.47	749.97
	Construction	0.09	89.00	4.49	93.58	1.56	95.15
	Trade, transport and communication	0.53	15.08	231.60	247.21	72.93	320.13
	Finance and business services	12.49	8.05	262.96	283.49	55.24	338.74
	Other services	0.30	2.51	222.86	225.67	16.64	242.31
Total		40.45	433.31	764.66	1 238.41	561.23	1 799.64

Domestic use table at basic prices								
	Industries			Final use			Total use	
	Agriculture	Manufacturing and construction	Services	Final consumption expenditure	Gross capital formation	Exports		
Products	Agriculture	4.35	9.28	0.61	13.18	0.08	10.47	37.96
	Manufacturing	7.78	82.09	28.90	31.15	26.74	173.83	350.50
	Construction	0.30	25.71	17.39	0.89	47.20	2.10	93.58
	Trade, transport and communication	1.80	30.22	60.71	68.96	8.57	76.95	247.21
	Finance and business services	1.82	29.48	132.56	73.04	11.63	34.97	283.49
	Other services	0.17	3.03	18.14	202.77	0.55	1.02	225.67
Imports		2.54	123.74	78.29	75.33	36.50	244.82	561.23
	GVA	21.70	129.78	428.07				579.54
Total		40.45	433.31	764.66	465.33	131.27	544.17	

Imports use table at basic prices								
	Industries			Final use			Total use	
	Agriculture	Manufacturing and construction	Services	Final consumption expenditure	Gross capital formation	Exports		
Products	Agriculture	0.77	6.40	0.48	1.25	0.03	6.45	15.38
	Manufacturing	1.49	99.62	34.95	61.07	30.80	171.55	399.47
	Construction	0.00	0.52	0.35		0.69		1.56
	Trade, transport and communication	0.04	2.69	14.05	0.73	2.72	52.70	72.93
	Finance and business services	0.23	13.99	24.54	0.22	2.16	14.10	55.24
	Other services	0.01	0.52	3.93	12.06	0.10	0.02	16.64
Total		2.54	123.74	78.29	75.33	36.50	244.82	561.23

2. Square versus rectangular SUTs

12.20 A square supply table is required when the assumptions for the transformation into IOTs are based on a product technology assumption (model A) and a fixed industry sales structure model (model C), which require the calculation of an inverse matrix based on the supply table. In most countries, the SUTs are rectangular, with many more products than industries, and this requires an aggregation of the product dimension with the result that the number of industries will determine the dimension of the resulting square SUTs.

12.21 In the case of the industry technology assumption (model B), square matrices are not required but the application of the formula directly to the existing dimensions of the SUTs will result in square IOTs with as many rows and columns as the number of products, which will usually not make much sense, and the aggregation loss of information will not depend on whether the aggregation is made before or after application of the formula.

12.22 Only in the case of the fixed product sales structure assumption (model D), is there a clear gain relative to minimizing the aggregation loss of information by using the formula directly on the SUTs rectangular matrices, resulting in industry-by-industry IOTs with as many rows and columns as the number of industries. On this point, see Thage (2005) and (2011).

12.23 In the cases where aggregation by product is necessary to obtain a square SUTs configuration, each product must be assigned to a primary producer. The existing link between CPC and ISIC provides a guide for the assignment of products to industries. In general, several products will need to be assigned to the same industry. This means that, in the product technology assumption (model A), these products are assumed to share the same input structure. The assignment of products to industries can then be used to aggregate the supply table to obtain a square table. For the product technology assumption (model A), it is not strictly required to aggregate the use table at this stage. In order ultimately to obtain square IOTs, however, an aggregation of the use table will be necessary at some point. It may therefore also be performed before the calculation of the IOTs.

12.24 Table 12.3 shows a numerical example of a square SUTs constructed from the rectangular SUTs shown in Table 12.2.

Table 12.3 Numerical example of square SUTs for the transformation

Supply table at basic prices						
		Industries			Output	Imports
		Agriculture	Manufacturing and construction	Services		
Products	Agriculture	25.77	5.15	7.04	37.96	15.38
	Manufacturing and construction	1.35	402.51	40.21	444.08	401.04
	Services	13.32	25.64	717.41	756.37	144.81
	Total	40.45	433.31	764.66	1 238.41	561.23
Domestic use table at basic prices						
		Industries			Final use	
		Agriculture	Manufacturing and construction	Services	Final consumption expenditure	Gross capital formation
Products	Agriculture	4.35	9.28	0.61	13.18	0.08
	Manufacturing and construction	8.08	107.80	46.29	32.04	73.94
	Services	3.78	62.72	211.40	344.77	20.75
	Imports	2.54	123.74	78.29	75.33	36.50
GVA		21.70	129.78	428.07		
Total		40.45	433.31	764.66	465.33	131.27
Imports use table						
		Industries			Final use	
		Agriculture	Manufacturing and construction	Services	Final consumption expenditure	Gross capital formation
Products	Agriculture	0.77	6.40	0.48	1.25	0.03
	Manufacturing and construction	1.49	100.14	35.30	61.07	31.49
	Services	0.28	17.20	42.52	13.02	4.98
	Total	2.54	123.74	78.29	75.33	36.50

3. Secondary production

12.25 The existence of cell entries reflecting secondary production in the square-based supply table is the only reason for the difference between product-by-product and industry-by-industry IOTs. Secondary production thereby creates the need to choose between the alternative product technology assumptions and market share assumptions. In the case of no secondary production, the domestic use table would represent an IOT.

12.26 The 2008 SNA (para. 28.46) distinguishes between three types of secondary products:

- *Subsidiary products* are those products that are technologically unrelated to the primary product.
- *By-products* are those products that are produced simultaneously with another product but which can be regarded as secondary to that product, for example, gas produced by blast furnaces.

- *Joint products* are those products that are produced simultaneously with another product but which cannot be said to be secondary, for example, beef and hide produced by slaughtering animals.

12.27 The importance of secondary production is closely connected to the type of economic unit (for example, establishments versus enterprises) used when collecting data from businesses and compiling the SUTs. Most often, the compiler may not be in a position to distinguish between the three types of secondary products. In consequence, the four standard models for compiling IOTs from SUTs do not make such a distinction, whereas it may play a role when compiling product-by-product IOTs based on the assumption of hybrid technology.

12.28 In most economies, there are probably limited cases of pure subsidiary products, by-products or joint products. In most cases, there are some joint costs and some costs that can be attributed to the distinctive outputs.

12.29 In practice, where certain kinds of secondary production would potentially create problems in the resulting IOTs no matter which transformation formula is used, a priori redefinitions (for example, breakdowns of vertically integrated economic units) may offer possible solutions. Such very special features of production structure will usually be well known to the compilers and could be taken into account on an ad hoc basis. The handling of secondary production in SUTs is covered in more detail in chapter 5.

4. Main theoretical models used for the derivation of IOTs

12.30 There are four main transformation methods used to derive IOTs from SUTs. As shown in Figure 12.1 and summarized in Figure 12.2, the four basic transformation models are based on the following assumptions:

- Product technology assumption (model A):
Each product is produced in its own specific way, irrespective of the industry where it is produced.
- Industry technology assumption (model B):
Each industry has its own specific way of production, irrespective of its product mix.
- Fixed industry sales structure assumption (model C):
Each industry has its own specific sales structure, irrespective of its product mix.
- Fixed product sales structure assumption (model D):
Each product has its own specific sales structure, irrespective of the industry where it is produced.